

Höhere Mathematik

RĂZVAN GHEORGHE GURĂU

INSTITUT FÜR THEORETISCHE PHYSIK DER UNIVERSITÄT HEIDELBERG,
PHILOSOPHENWEG 19, 69120 HEIDELBERG, GERMANY

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Chapter 1

Introduction

Axioms are statements which are accepted to be true. *Theorems* (lemmas, propositions, etc.) are statements that can be proven using logical reasoning starting from axioms. We know since the work of Gödel that no system of axioms is complete, that is for any system of axioms which includes the natural numbers it is possible to formulate statements that are undecidable, that is can neither be proved true nor be proven false. This is because, essentially, there are more possible statements that can be formulated than there are possible proofs.

Reductio ad absurdum is a strategy of proving a statement is true by first assuming the statement is false and proving that this implies a contradiction (i.e. some other false statement). As a true statement can not imply a false one, one concludes that the original statement is true.

Mathematical induction is a strategy of proving a statement indexed by a natural number n is true by first proving it is true for $n = 0$ and then proving that if the statement for n is true then the one for $n + 1$ is also true. Transfinite induction is indexed by ordinal numbers instead of natural ones.

The following logic symbols are useful $\exists, \forall, \Rightarrow, \Leftrightarrow, =$. We denote $\equiv$ when the left hand side and the right hand side are equal by definition.

Part I

HöMa I

Chapter 2

Sets and numbers

We review some elementary notions which will be useful for the sequel. In the beginning we will denote the multiplication of numbers by $\times$, but drop this notation later on.

2.1 Sets (Mengen)

The notion of *set* is a primitive notion in mathematics, therefore it can not be defined in terms of other more fundamental notions. Instead of providing a definition, one lists the axioms sets obey. Informally, a set is the mathematical model for a collection of different things.

A set X contains elements x , which we denote $x \in X$. The elements are mathematical objects of any kind. The only question one can ask about a set and an element is whether the element belongs to the set. This has the consequence that, when listing the elements of a set, every element is written only once: $\{2\} = \{2, 2\} = \{2, 2, 2\}$ and so on, as they all represent the set which contains only the element 2.¹

We follow the Zermelo–Fraenkel axioms of set theory plus the axiom of choice. The axiom of choice is sometimes contested, but not by us. Most axioms are intuitive.

Axiom 2.1 (*Zermelo–Fraenkel axioms (ZF-Axiome)*). We denote the empty set $\emptyset$ the set with no elements. We have

- (Extension) A set is determined by its elements, that is, if $x \in A \Leftrightarrow x \in B \Rightarrow$ then $A = B$. We denote $B \subset A$ if $x \in B \Rightarrow x \in A$.
- (Pairing) If A and B are sets, then $\{A, B\}$ is a set.
- (Subsets or specification) If A is a set and $P(x)$ is a property of some of the elements x , then $\{x \in A \mid P(x)\}$ is a set.
- (Union, or sum set) For any set of sets² A there exist a set $\bigcup A = \{x \mid x \in D \in A\}$ which is the union of all the elements of the elements of A . In particular this allows us to unpack the pairs of sets:

$$\bigcup \{A, B\} = \{x \mid x \in A \text{ or } x \in B\} \equiv A \cup B.$$

For instance $\bigcup \{\{1, 2\}, \{2, 3\}\} = \{1, 2, 3\}$.

- (Power set) For any set A there exists the power set $\mathcal{P}(A)$ consisting in all the subsets of A , $\mathcal{P}(A) = \{B \mid B \subset A\}$. As the empty set is a subset of any set, $\emptyset \in \mathcal{P}(A)$ for any A .

¹Multi sets which can contain the same element more than once can be defined.

²In ZF theory everything is a set, so any set is a set of sets.

- (Infinity). Let w be some set. Then, by the axiom of pairing $\{w\}$ is a set (obtained by pairing w with itself $\{w, w\} \equiv \{w\}$) and so is $S(w) = w \cup \{w\} = \bigcup\{w, \{w\}\}$. Note that $\bigcup\{w\} = w$, but $\{w\} \neq w$.

There exist a set A such that $\emptyset \in A$ and $w \in A \Rightarrow S(w) \in A$.

- (Replacement) The image of a set under a function is a set, that is if f is a function and A is a set, $f(A) = \{f(x) \mid x \in A\}$ is a set.
- (Foundation, or regularity) Every non empty set A contains an element that is disjoint from A .

We denote the intersection of the sets A and B the set consisting of the elements which belong to both A and B , that is $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$. Two sets are disjoint if their intersection is the empty set. The axiom of foundation states that for all $A \neq \emptyset$, there exist $x \in A$ such that $x \cap A = \emptyset$.

This implies that no set is an element of itself: for any set A , $\{A\}$ is a set by the axiom of pairing and by the axiom of regularity $\{A\}$ must contain an element which is disjoint from $\{A\}$. As $\{A\}$ contains only A , it follows that $A \cap \{A\} = \emptyset$. Reading this from right to left, as $A \in \{A\}$ it follows that $A \notin A$.

The set difference of X and Y consists in the element of X which do not belong to Y , $X \setminus Y = \{x \mid x \in X \text{ and } x \notin Y\}$. The Cartesian product of two non empty sets X and Y is the set of ordered pairs $X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}$.

2.1.1 Functions (Funktionen)

Definition 2.1 (**Functions**). A function from a set X to a set Y is an assignment of an unique element of Y to each element of X . The set X is called the domain of the function and the set Y is called the codomain of the function.

A function, its domain, and its codomain, are declared by the notation $f : X \rightarrow Y$, and the value of a function f at an element $x \in X$, denoted by $f(x)$, is called the image of x under f , or the value of f applied to the argument x . Functions are sometimes called maps or mappings. For example, the *identity function* $\text{id} : X \rightarrow X$ is the function which assigns to each element of X itself, that is $\forall x \in X, \text{id}_X(x) = x$.

Two functions f and g are equal if their domain and codomain sets are the same and their output values agree on the whole domain, that is for $f, g : X \rightarrow Y$ we denote $f = g$ if $f(x) = g(x), \forall x \in X$. The *composition* of two functions $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ is the function $g \circ f : X \rightarrow Z$ defined by $\forall x \in X, g \circ f(x) = g(f(x)) \in Z$. The *image* of a set $U \subset X$ under the function $f : X \rightarrow Y$ is the set $f(U) = \{y \in Y \mid \exists x \in U \text{ s.t. } f(x) = y\} \subset f(X)$. The *preimage* of a set $V \subset f(X)$ is the set $f^{-1}(V) = \{x \in X \mid f(x) \in V\} \subset X$.

A function $f : X \rightarrow Y$ is called *injective*, or *one to one*, if each element in Y is the image of at most one element in X , that is $\forall x_1, x_2 \in X, f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$. A function $f : X \rightarrow Y$ is called *surjective*, or *onto*, if $f(X) = Y$ that is each element in Y is the image of some element in X : $\forall y \in Y, \exists x \in X$ such that $f(x) = y$. A function which is both injective and surjective (one to one and onto) is called *bijective*. A bijection $f : X \rightarrow Y$ has a well defined *inverses*, that there exists a function $f^{-1} : Y \rightarrow X$ such that $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$. Indeed, as f is a bijection, then $\forall y \in Y, \exists x \in X$ (due to surjectivity) and x is unique (due to injectivity) such that $f(x) = y$. Let us denote this element x_y . The *inverse* of f is the function $f^{-1} : Y \rightarrow X$ defined by $f^{-1}(y) = x_y$.

2.1.2 The axiom of choice (Das Auswahlaxiom)

The most disputed axiom of set theory is the axiom of choice.

Axiom 2.2 (**The axiom of choice**). For any set $\mathcal{X}$ of nonempty sets, there exists a choice function $f : \mathcal{X} \rightarrow \bigcup \mathcal{X}$ that maps each set $X \in \mathcal{X}$ to an element $x \in X$.

Reformulated, this means that for any set of nonempty sets $\{X_\alpha \neq \emptyset \mid \alpha \in A\}$, there exists a set that contains exactly one element per X_α , that is there exists the set $\{x_\alpha \in X_\alpha \mid \alpha \in A\}$. What is non obvious about this axiom is that the index set A can be anything, for instance the real numbers, a set of ordinal numbers (see below) etc. In such cases it is often non trivial to exhibit a choice function and one needs to rely on the axiom of choice to conclude that such a function exists.

2.1.3 Binary relations (Zweistellige Relationen)

Elements of sets can be in binary relations of several types.

Definition 2.2 (**Binary relations**). *Several binary relations among the elements of a set X important for the sequel. We say that:*

- $\sim$ is an equivalence relation on X if:
 - (reflexive) $\forall x \in X$, then $x \sim x$,
 - (symmetric) $\forall x, y \in X$, if $x \sim y$ then $y \sim x$,
 - (transitive) $\forall x, y, z \in X$, if $x \sim y$ and $y \sim z$ then $x \sim z$.
- $\leq$ is a partial order relation on X if:
 - (reflexive) $\forall x \in X$, then $x \leq x$,
 - (antisymmetric) $\forall x, y \in X$, if $x \leq y$ and $y \leq x$ then $y = x$,
 - (transitive) $\forall x, y, z \in X$, if $x \leq y$ and $y \leq z$ then $x \leq z$.

A set with a partial order relation is called a partially ordered set, or a poset. A poset can contain pairs of elements that are not ordered, that is there can exist $x, y \in X$ such that $x \not\leq y$ and $y \not\leq x$. A partial order is a total order if any two elements are ordered:

- (connected) $\forall x, y \in X$, either $x \leq y$ or $y \leq x$.
- $<$ is a strict partial order relation on X if:
 - (irreflexive) $\forall x \in X$, then $x \not< x$,
 - (asymmetric) $\forall x, y \in X$, if $x < y$ then $y \not< x$,
 - (transitive) $\forall x, y, z \in X$, if $x < y$ and $y < z$ then $x < z$.

For every partial order $\leq$ there exists an associated strict partial order $<$ defined by $x < y$ if $x \leq y$ and $x \neq y$.

A strict partial order is a strict total order relation on X if:

- (connected) $\forall x, y \in X$, if $x \neq y$ then either $x < y$ or $y < x$,
- $\leq_w$ is a well-ordering relation on X if
 - (total order) $\leq_w$ is a total order on X ,
 - (least element) every non empty subset of X has a least element, that is $\forall Y \subset X, Y \neq \emptyset$, there exists $y \in Y$ such that $\forall z \in Y, y \leq_w z$.

A finite poset can be represented by a Hasse diagram with nodes the elements $x \in X$ and links between x and y if $x < y$ but there exists no z with $x < z < y$.

2.1.4 Zorn's lemma (Das Lemma von Zorn)

Lemma 2.1 (**Zorn's lemma**). A chain T in a partially order set X is a subset $T \subset X$ that is totally ordered, that is $\forall x, y \in T$ either $x \leq y$ or $y \leq x$. If any chain T in X has an upper bound, that is $\forall T$ chain, $\exists x_T \in X$ such that $\forall x \in T$, $x \leq x_T$, then X has at least one maximal element, that is an element that is not smaller than any other element in X . Formally, $\exists x_X \in X$ such that $\forall x \in X \setminus \{x_X\}$, $x_X \not\leq x$.

This lemma is in fact equivalent to the axiom of choice. In proving this lemma, the axiom of choice is used as follows. Assume there exists a partially ordered set X such that every chain has an upper bound and also for every element in X there is another element bigger than it. For every chain T we then define $b(T) \in X$, an element *strictly* larger than any element in T (this exists because T has an upper bound, and that upper bound, being an element in X , has a bigger element). In order to actually chose $b(T)$ we need to use the axiom of choice. The contradiction comes from building, using the function b , a strictly totally ordered sequence in X which is “too long”, that is it has more elements than there can be in any set as it is indexed by the ordinal numbers which are a proper class, not a set (see below).

Exercise 2.1 (The poset of partitions). A partition Π of a set X is a set of subsets of X , $\Pi = \{X_a \mid a \in A \neq \emptyset\}$ such that $X = \bigcup_{a \in A} X_a$ and $X_a \cap X_b = \emptyset$ if $a \neq b$. A partition Π_1 is finer than Π_2 (and Π_2 is coarser than Π_1) if $\forall X \in \Pi_1$, $\exists Y \in \Pi_2$ such that $X \subset Y$. We denote $\Pi_1 \leq \Pi_2$.

Prove that $\leq$ is a partial order on the set of partitions. Check that this poset verifies Zorn's lemma. Show that this poset is a lattice, that is a poset in which every pair of elements x, y has a unique supremum $x \vee y$ (smallest element larger than both, called the join) and a unique infimum $x \wedge y$ (largest element smaller than both, called the meet). Show that this lattice has a greatest lower bound as well as a smallest upper bound. For Zorn's lemma - the upper bound on any chain in the poset is the join of all the partitions in the chain; the poset has a maximal element, the one set partition.

Exercise 2.2 (Equivalence classes, quotient). Let X be a set with an equivalence relation. For any $x \in X$, we define the equivalence class $[x] = \{y \in X \mid y \sim x\}$. Prove that an equivalence relation partitions a set into the disjoint union of equivalence classes. First $X = \bigcup_{x \in X} [x]$, that is X is the union of equivalence classes of its elements. Second, two equivalence classes are either equal or disjoint: assume that two equivalence classes are not disjoint $[x] \cap [y] \neq \emptyset$. Then $\exists z \in [x] \cap [y]$ hence $z \sim x$ and $z \sim y$ and (by transitivity) $x \sim y$. But then, again by transitivity $\forall z' \in [x]$, $z' \sim x \sim y$ hence $z' \in [y]$ and $[x] \subset [y]$. Similarly, $[y] \subset [x]$ hence $[x] = [y]$.

We denote $\{[x] \mid x \in X\} \equiv X / \sim$ and we call it the quotient of X by $\sim$.

Exercise 2.3 (Equivalence classes of permutations.). The permutations of n elements are the bijections on the finite set with n elements $\sigma : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$. We denote the set of all the permutations over n elements by S_n . The composition of permutations is sometimes denoted as a product $\sigma_2 \circ \sigma_1 \equiv \sigma_2 \sigma_1$. For $\sigma_1, \sigma_2 \in S_n$, we denote $\sigma_1 \sim \sigma_2$ if there exists $\pi \in S_n$ such that $\sigma_1 = \pi \sigma_2 \pi^{-1}$.

How many elements are there in S_n ? Prove that $\sim$ is an equivalence relation. Characterize the quotient set $S_n / \sim$. How many elements are there in the equivalence class of a permutation? Hint: any permutation can be decomposed into a product of disjoint cycles.

2.2 Numbers (Zahlen)

We now discuss the various sets of numbers we will encounter in these notes. Different numbers are introduced in order to solve equations: solving $x + n = 0$ with n natural leads to integers, solving $mx + n = 0$ for m, n integers leads to rational numbers, $x^2 = r > 0$ with r rational leads to real numbers and finally $P[x] = 0$ for some polynomial P leads to complex numbers.

2.2.1 Natural numbers $\mathbb{N}$ (Natürliche Zahlen)

The natural numbers are a primitive notion. Informally they are obtained by starting from 0 and iteratively adding 1 an arbitrary number of times. One can construct the natural numbers starting from sets. This construction is then generalized to ordinal numbers.

Construction 2.1 (Natural numbers $\mathbb{N}$). *By the axiom of infinity, there exist sets which contain the empty set $\emptyset$ and are closed under the successor function (such sets are said to be inductive). The smallest such set is the set of natural numbers $\mathbb{N}$ and is constructed in detail as follows:*

- zero is the empty set $0 = \emptyset$.
- the successor $S(a)$ of any set a is $S(a) = a \cup \{a\}$ and we label:

$$\begin{aligned} 0 &= \emptyset, \\ 1 &= 0 \cup \{0\} = \{0\} = \emptyset \cup \{\emptyset\} = \{\emptyset\}, \\ 2 &= 1 \cup \{1\} = \{0, 1\} = \{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\}, \\ 3 &= 2 \cup \{2\} = \{0, 1, 2\} = \{\emptyset, \{\emptyset\}\} \cup \left\{ \{\emptyset, \{\emptyset\}\} \right\} = \left\{ \emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \right\}. \end{aligned}$$

- $\mathbb{N}$ is the set consisting in $\emptyset$ and all its successors.

We endow $\mathbb{N}$ with further structure:

- (addition) the *addition* of natural numbers is the internal binary operation $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ defined as $\forall a, b \in \mathbb{N}, a + 0 = a$ and $a + S(b) = S(a + b)$,
- (multiplication) the *multiplication* is the internal binary operation $\times: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ defined as $\forall a, b \in \mathbb{N}, a \times 0 = 0$, $a \times S(b) = a + a \times b$.
- (total order) the natural numbers are *totally ordered* by the relation $a \leq b$ if $a \in b$ or $a = b$. Equivalently, there exists $c \in \mathbb{N}$ such that $a + c = b$. This total order is a well order.

The operation $+$ substitutes the successor function by denoting $S(0) = 1$, $S(a) = a + S(0) = a + 1$. We use the notation $1, 1+1=2, 2+1=3$ and so on and write the natural numbers as $\mathbb{N} = \{0, 1, 2, \dots, n, \dots\}$. We also denote $n \times n \equiv n^2$, $n \times n^2 = n^3$ and so on.

Exercise 2.4. *Prove that $+$ is commutative $a+b = b+a$ and associative $a+(b+c) = (a+b)+c$. We have $0+0=0$. For $a \neq 0$ we proceed by induction:*

we show that the statement for $S^{-1}(a)$ implies the statement for a . We first show that 0 commutes with every a . It is enough to show that $0+a = a$. Indeed, $0+a = S(0+S^{-1}(a)) = S(S^{-1}(a)) = a$. Now we want to show that $S(0)$ commutes with any a . First $a+S(0) = S(a+0) = S(a)$. Second, for $a \neq 0$ we have $a = S(S^{-1}(a)) = S(S^{-1}(a)+0) = S^{-1}(a)+S(0)$. Assuming $S(0)+S^{-1}(a) = S^{-1}(a)+S(0)$, then $S(0)+a = S(S(0)+S^{-1}(a)) = S(S^{-1}(a)+S(0)) = S(a)$. We then proceeded to $S(S(0))$ and so on.

Prove that $\times$ is commutative, associative and distributive with respect to the addition, that is $a \times (b+c) = a \times b + a \times c$. Prove that $\leq$ is a total order on $\mathbb{N}$ and that it is compatible with $+$ and $\times$, that is $\forall a, b, c \in \mathbb{N}, a \leq b \Rightarrow a+c \leq b+c$ and $a \times c \leq b \times c$. Finally, prove that $n \times p = n + n + \dots + n$ where the addition is performed p times.

2.2.2 Ordinals and cardinal numbers (Ordinal und Kardinal Zahlen)

Ordinal numbers generalise the notion of numerals (first, second, third and so on) to infinite sets. Roughly speaking, ordinals are labels which include the natural numbers, are totally ordered and are such that every set of ordinals is well ordered.

Definition 2.3 (von Neumann ordinals). An ordinal λ is the set having as elements precisely the ordinals strictly smaller than itself, which we denote by $\lambda = [0, \lambda)$. We stress that λ itself is not an element of λ .

More precisely, an ordinal λ is a transitive strictly well ordered set by set membership:

- $\in$ is a strict well order relation among the elements in the set λ :
 - that is for all $x \in \lambda$, $x \notin x$; for all $x, y \in \lambda$, if $x \neq y$ then either $x \in y$, or $y \in x$; for all $x, y, z \in \lambda$, if $x \in y$ and $y \in z$ then $x \in z$,
 - for all $A \subset \lambda$, $A \neq \emptyset$, $\exists x \in A$ such that $\forall y \in A$ either $x \in y$ or $x = y$.
- λ is transitive, that is for all $x \in \lambda$ then $x \subset \lambda$.

For every ordinal $\lambda = [0, \lambda)$, we denote $\lambda + 1 = \lambda \cup \{\lambda\}$ the successor of λ . It is immediate that $\lambda \cup \{\lambda\}$ is also an ordinal, and it is in fact the least element of all the ordinals strictly larger than λ . We always have $\bigcup \lambda \subset \lambda$. The ordinals themselves are strictly well ordered by set membership: for β and λ two ordinals, $\beta < \lambda$ if $\beta \in \lambda$. Two ordinals either coincide or one is strictly large than the other.

It can be proven that the union of any set of ordinals is an ordinal and every set of ordinals has a least upper bound (a supremum) obtained by taking the union of all the ordinals in the set. A *limit ordinal* is an ordinal that is neither 0 nor the successor of an ordinal³ and limit ordinals arise as least upper bounds on sets of all smaller ordinals:⁴ λ is a limit ordinal if $\forall \beta < \lambda$, $\exists \gamma$ s.t. $\beta < \gamma < \lambda$. Every ordinal is either $0 = \emptyset$, or a successor ordinal or a limit ordinal and in the latter case, $\lambda = \bigcup \lambda$.

By construction, the natural numbers are ordinals. The smallest ordinal greater than every natural number, denoted ω , is a limit ordinal: this is nothing but the well ordered infinite set of all the natural numbers. To check that ω is a limit ordinal we observe that, by definition, $n < \omega \Rightarrow n \in \mathbb{N}$ but $n < n + 1 \in \mathbb{N}$ and, because $n + 1$ is itself a natural number, $n + 1 < \omega$. The successor of ω is the ordinal $\omega + 1 = \omega \cup \{\omega\}$. A construction of ω is provided by the two letter words with alphabet $\mathbb{N}$ arranged in lexicographical order $(0, 0), (0, 1), (0, 2), \dots, (1, 0), (1, 1), \dots$. Then ω is the position of $(1, 0)$, $\omega + 1$ the one of $(1, 1)$, $\omega \cdot 2$ the one of $(2, 0)$ and so on. The smallest ordinal larger than any $\omega \cdot n$ is denoted ω^2 .

The class of all ordinals is *not a set*, but a proper class. If it were a set, this class would be an ordinal λ (obtained by taking the supremum of all the ordinals) hence a set. But, as λ contains all the ordinals, $\lambda \in \lambda$, which is a contradiction.

The axiom of choice implies that every set can be well ordered (Zermelo's theorem) by choosing the element given by the choice function on the subset of elements not yet chosen. In detail, take f a choice function for the non empty subsets of A and define for every ordinal α the element $a_\alpha = f(A \setminus \{a_\xi \mid \xi < \alpha\})$ if $A \setminus \{a_\xi \mid \xi < \alpha\} \neq \emptyset$. Then the listing of elements of A in the order $\{a_\alpha\}_\alpha$ is a well ordering of A . For every well ordered set S , there exists an order preserving bijection between S and exactly one ordinal.

Cardinal numbers are a generalization of the natural numbers used to measure the cardinality (size) of sets. Two sets have the same cardinality (the same cardinal number) if and only if there exist a bijection between them. The cardinal of a set X is the least ordinal number such that there exists a not necessarily order preserving bijection between X and that ordinal.

The cardinality of a finite set is the number of elements in that set hence a natural number. The “transfinite” cardinal numbers, often denoted using the Hebrew Aleph symbol $\aleph$ followed by a subscript, describe the sizes of infinite sets. There exists a transfinite sequence of cardinal numbers:

$$0, 1, 2, \dots, n, \dots \aleph_0, \aleph_1, \dots \aleph_\alpha \dots,$$

³A limit ordinal is a limit point of the class of ordinal numbers, with respect to the order topology defined in the next chapter. The other ordinals are isolated points in the same topology.

⁴Of course, there are sets of ordinals whose supremum is *not* a limit ordinal.

where the transfinite cardinal numbers are indexed by ordinals. The cardinal of $\mathbb{N}$ is the first transfinite cardinal number, Aleph zero $\aleph_0$, which is the smallest ordinal greater than every natural number, ω . A set with cardinality $\aleph_0$ is called *countable*. Cardinals and ordinals are distinct objects: the ordinal $\omega + 1 = \omega \cup \{\omega\}$ is the successor of ω , hence a distinct ordinal, but the cardinal of $\omega + 1$ is still $\aleph_0$. The same holds for $\omega + 2, \omega \cdot 2, \omega^2, \omega^\omega$.

We denote the cardinal of a set X by $|X|$. The power set of the set $P(X) = \{Y \mid Y \subset X\}$ has cardinal $|P(X)| = 2^{|X|}$, and is a cardinal number strictly larger than $|X|$.

Theorem 2.1 (Cantor's theorem). *Let f be a map from a set X to its power set $\mathcal{P}(X)$. Then f is not surjective and $|X| < |\mathcal{P}(X)|$.*

Proof. Consider the set $B = \{x \in X \mid x \notin f(x)\} \subset X$ hence $B \in \mathcal{P}(X)$. Assume f is surjective. Then there exists $y \in X$ such that $f(y) = B$. But by construction $y \in B \Leftrightarrow y \notin f(y) = B$, which is a contradiction. The last assertion follows by noting that $f(x) = \{x\}$ is an injective function from X to $\mathcal{P}(X)$. $\square$

2.2.3 Integer numbers $\mathbb{Z}$ (Ganze Zahlen)

Informally, the integer numbers are constructed as $\mathbb{Z} = \mathbb{N} \cup \{-1, -2, \dots\}$, where $-n$ is the inverse of n with respect to the addition. Integers can be modeled as differences or natural numbers $a - b$.

Construction 2.2 (Integer numbers $\mathbb{Z}$). *The integer numbers can be constructed as the quotient set $\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim$ with $\sim$ the equivalence relation $(a, b) \sim (c, d)$ if $a + d = b + c$. The addition, multiplication and total order on the integers are defined as:*

- $[(a, b)] + [(c, d)] = [(a + c, b + d)],$
- $[(a, b)] \times [(c, d)] = [(a \times c + b \times d, a \times d + b \times c)],$
- $[(a, b)] < [(c, d)]$ if $a + d < b + c$.

The natural numbers $\mathbb{N}$ are embedded in $\mathbb{Z}$ by $n \rightarrow [(n, 0)]$.

Note that $[(a, b)] + [(b, a)] = [(a + b, a + b)] = [(0, 0)]$ hence the additive inverse of $[(a, b)]$ is $[(b, a)]$ which we denote $[(b, a)] \equiv -[(a, b)]$. If $a \geq b$ then $a = b + c$ and $[(a, b)] = [(c, 0)] \equiv c$; if on the contrary $a < b$ then $b = a + c$ and $[(a, b)] = -[(b, a)] \equiv -c$ which leads to the usual notation $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$. We observe that $[(b, a)] = [(a, b)] \times [(0, 1)] = -[(a, b)]$, that is $(-1) \times c = -c$ and $(-1) \times (-c) = c$.

Exercise 2.5. *Prove that $\sim$ is an equivalence relation. Prove that the addition of integers is commutative, associative and compatible with the total order. Prove that the multiplication on integers is associative, distributive with respect to the addition and the multiplication by positive elements is compatible with the total order while the multiplication by negative elements reverses the order.*

Definition 2.4 (Group (Gruppe)). *A group $(G, \circ, e)$ is a set G with an internal binary operation $\circ : G \times G \rightarrow G$ such that:*

- (identity element) there exist $e \in G$ such that $e \circ g = g \circ e = g$,
- (associative) $g_1 \circ (g_2 \circ g_3) = (g_1 \circ g_2) \circ g_3$, $\forall g_1, g_2, g_3 \in G$,
- (inverse) for any $g \in G$, there exists $g^{-1} \in G$ such that $g \circ g^{-1} = g^{-1} \circ g = e$.

A group is Abelian, or commutative, if $\forall g_1, g_2 \in G$, $g_1 \circ g_2 = g_2 \circ g_1$.

Definition 2.5 (Axiomatic definition of $\mathbb{Z}$). *$(\mathbb{Z}, +, 0)$ is the only non trivial totally ordered Abelian group whose positive elements (i.e. elements strictly greater than 0) are well-ordered.*

2.2.4 Rational numbers $\mathbb{Q}$ (Rationale Zahlen)

Informally, rational numbers are modeled as ratios p/q with p and q integers.

Construction 2.3 (**Rational numbers $\mathbb{Q}$**). *The rational numbers can be constructed as the quotient set $\mathbb{Q} = \mathbb{Z} \times \mathbb{Z}_{\neq 0} / \sim$ with $\sim$ the equivalence relation $(p_1, q_1) \sim (p_2, q_2)$ if $p_1 \times q_2 = q_1 \times p_2$. Clearly $[(p, q)] = [(a \times p, a \times q)]$ for any $a \in \mathbb{Z}_{\neq 0}$, in particular $[(p, q)] = [(-p, -q)]$. The addition, multiplication and total order on the rational numbers are:*

- $[(p_1, q_1)] + [(p_2, q_2)] = [(p_1 \times q_2 + q_1 \times p_2, q_1 \times q_2)]$,
- $[(p_1, q_1)] \times [(p_2, q_2)] = [(p_1 \times p_2, q_1 \times q_2)]$,
- $[(p_1, q_1)] < [(p_2, q_2)]$ if $q_1, q_2 > 0$ and $p_1 \times q_2 < q_1 \times p_2$.

The integer numbers $\mathbb{Z}$ are embedded in $\mathbb{Q}$ by $p \rightarrow [(p, 1)]$. We use the notation $[(p, q)] \equiv \frac{p}{q}$.

The additive and multiplicative inverses of $x = [(p, q)]$ are $-x = [(-p, q)]$ and $x^{-1} = [(q, p)]$ (where we need to assume $p \neq 0$ for the multiplicative inverse to exist). We can define the ratio of two rational numbers as the multiplication of the first by the multiplicative inverse of the second: $x/y = x \times y^{-1}$.

Theorem 2.2 (**$\mathbb{Q}$ is dense in itself**). *The set $\mathbb{Q}$ is dense in itself, that is for any $x, y \in \mathbb{Q}$ and $x < y$ there exists $z \in \mathbb{Q}$ such that $x < z < y$. More precisely (see Sec. 3) each point in $\mathbb{Q}$ is a limit point of $\mathbb{Q}$ in the standard topology.*

Proof. We have $x = [(p_1, q_1)]$ and $y = [(p_2, q_2)]$ with $q_1, q_2 > 0$ and $p_1 \times q_2 < q_1 \times p_2$. We denote $z = (x + y)/2 = [(p_1 \times q_2 + q_1 \times p_2, 2 \times q_1 \times q_2)]$, where $2 \times q_1 \times q_2 = q_1 \times q_2 + q_1 \times q_2$ as the multiplication is distributive. Taking into account that the multiplication is compatible with the total order on $\mathbb{Z}$ we have:

$$\begin{aligned} q_1 \times (p_1 \times q_2) &< q_1 \times (q_1 \times p_2) \Rightarrow 2 \times q_1 \times q_2 \times p_1 < q_1 \times (p_1 \times q_2 + q_1 \times p_2) \Rightarrow x < z, \\ (p_1 \times q_2) \times q_2 &< (q_1 \times p_2) \times q_2 \Rightarrow (p_1 \times q_2 + q_1 \times p_2) \times q_2 < 2 \times q_1 \times q_2 \times p_2 \Rightarrow z < y. \end{aligned}$$

□

Theorem 2.3 (**$\mathbb{Q}$ is countable**). *There exists a bijection between $\mathbb{N}$ and $\mathbb{Q}$. More generally, a countable union of countable sets is countable, that is there exists a bijection between $\mathbb{N}$ and the set $\bigcup_{n \in \mathbb{N}} S_n$, with $S_n = \{s_{np} | p \in \mathbb{N}\}$.*

Proof. Let us arrange the sets $(S_n = \{s_{np} | p \in \mathbb{N}\})_{n \in \mathbb{N}}$ on a square grid in the plane putting s_{np} at coordinates (n, p) . We list the sets S_n along the horizontal and pile them S_1, S_2 and so on along the vertical. The bijection is obtained by stepping along the axes and along the diagonals:

$$s_{11} \rightarrow s_{12} \rightarrow s_{21} \rightarrow s_{31} \rightarrow s_{22} \rightarrow s_{13} \rightarrow s_{14} \rightarrow s_{23} \rightarrow s_{32} \rightarrow s_{41} \rightarrow \dots$$

It follows that $\mathbb{Z} \subset \bigcup_{a \in \mathbb{N}} \{(a, b) | b \in \mathbb{N}\}$ is countable and $\mathbb{Q} \subset \bigcup_{p \in \mathbb{Z}} \{(p, q) | q \in \mathbb{Z}_{\neq 0}\}$ is also.

□

Exercise 2.6. *Prove that $\sim$ is an equivalence relation. Prove that the addition of rational numbers is commutative, associative and compatible with the total order. Prove that the multiplication of rational numbers is commutative, associative, distributive with respect to the addition and compatible with the total order.*

Exercise 2.7. *Prove that the decimal representation of any rational number either stops or ends up with an infinitely repeated (periodic) sequence of decimals. Consider $p < q$ and recall the algorithm of long division (we denote $(..0)$ a possibly empty sequences of the decimal zero):*

```

p.000000..... | q
p0(..0)         | 0.(.0)c1(..0)c2(..0)c3(..0) ****
-c1 x q
-----
r10(..0)
-c2 x q
-----
r20(..0)
-c3 x q
-----
r30(..0)
*****

```

The rests $r_1, r_2, \dots$ can only take a finite number of values $0, 1, \dots, q-1$. If one of the rests is 0 the decimal representation stops. If not, there is a first repeat of the rests $r_i = r_j$ and the decimal representation ends up in a periodically repeating sequence.

2.2.5 Real numbers $\mathbb{R}$ (Reelle Zahlen)

Perhaps the most useful numbers we will use are the real numbers. Informally the real numbers sit “in between” the rational ones, that is they are numbers that are not rational but can be arbitrarily well approximated by rational ones. An example of a number that is not rational is $\sqrt{2}$.⁵

Real numbers can be constructed using Dedekind cuts. Morally, a Dedekind cut represents the real number b by the set $(-\infty, b) \cap \mathbb{Q}$.

Construction 2.4 (Real numbers $\mathbb{R}$). *The real numbers $\mathbb{R}$ can be constructed as the set of Dedekind cuts of the rational numbers. A Dedekind cut (A, B) of the rational numbers is a partition of $\mathbb{Q}$ into two non empty disjoint sets $\mathbb{Q} = A \cup B$, $A, B \neq \emptyset$, $A \cap B = \emptyset$ such that:*

- (*A is closed downwards*) if $x, y \in \mathbb{Q}$ with $x < y$ and $y \in A$ then $x \in A$,
- (*A does not contain a greatest element*) if $x \in A$ then there exists $y \in A$ such that $x < y$.

As $B = \mathbb{Q} \setminus A$, we can identify the cut (A, B) by its first set only, namely A , which is a nonempty, downwards closed strict subset of $\mathbb{Q}$ (hence bounded from above) but with no greatest element. $\mathbb{Q}$ is embedded in the set of Dedekind cuts by $r \rightarrow A = \{a \in \mathbb{Q} \mid a < r\}$, which does not have a greatest element in $\mathbb{Q}$ since $\mathbb{Q}$ is dense in itself. For example the Dedekind cut corresponding to 0 is $\mathbb{Q}_{<0}$, the set of strictly negative rational numbers. The arithmetic operations and total order on the Dedekind cuts are defined as:

- (*addition*) $A_1 + A_2 = \{a_1 + a_2 \mid a_1 \in A_1 \text{ and } a_2 \in A_2\}$,
- (*subtraction*) $A_1 - A_2 = \{a_1 - b_2 \mid a_1 \in A_1 \text{ and } b_2 \in \mathbb{Q} \setminus A_2\}$,
- (*negation*) $-A_2 = 0 - A_2 = \{a_1 - b_2 \mid a_1 \in \mathbb{Q}_{<0} \text{ and } b_2 \in \mathbb{Q} \setminus A_2\}$,
- (*total order*) $A_1 < A_2$ if A_1 is a strict subset of A_2 , that is $A_1 \subset A_2$ and $A_2 \setminus A_1 \neq \emptyset$, also denoted $A_1 \subsetneq A_2$, and this is a total order as the Dedekind cuts are closed downwards,
- (*multiplication*) we have two cases:
 - for $A_1, A_2 \geq 0$ we define $A_1 \times A_2 = \{a_1 \times a_2 \mid 0 \leq a_1 \in A_1 \text{ and } 0 \leq a_2 \in A_2\} \cup \mathbb{Q}_{<0}$,
 - otherwise we use one of the equalities $A_1 \times A_2 = -(A_1 \times (-A_2)) = -((-A_1) \times A_2) = ((-A_1) \times (-A_2))$,
- (*division*) we again have two cases:

⁵Euclid, Elements (proposition 117, Book X) Alexandria, Ptolemaic Egypt, year ≈ 300 BCE. Assume $\sqrt{2} = p/q$ with p and q co primes. Then $p^2 = 2q^2$ hence p^2 divides by 2 hence p divides by 2. Then $q^2 = 2(p/2)^2$ hence q^2 divides by 2 hence q divides by 2 hence p and q are not co primes.

- if $A_1 \geq 0$ and $A_2 > 0$ then $A_1/A_2 = \{a_1/b_2 \mid 0 \leq a_1 \in A_1 \text{ and } b_2 \in \mathbb{Q} \setminus A_2\} \cup \mathbb{Q}_{<0}$,
- otherwise we use one of the equalities $A_1/A_2 = -(A_1/(-A_2)) = -((-A_1)/A_2) = (-A_1)/(-A_2)$.

The important remark is that any nonempty subset of the real numbers thus constructed which is bounded from above has a *supremum*, that is a least upper bound.

Theorem 2.4 (**Every nonempty, bounded subset of $\mathbb{R}$ has a supremum in $\mathbb{R}$**). *The supremum, that is the least upper bound, of a bounded above subset $S = \{A \mid A \in X \text{ and } A < B\}$ of a totally ordered set X is an element $I \in X$ such that $\forall A \in S, A \leq I$ and $\forall I' < I, \exists A \in S$ with $A \not\leq I'$.⁶ The supremum of the set of Dedekind cuts $S = \{A \mid A \in \mathbb{R} \text{ and } A < B\}$ is the Dedekind cut $I = \bigcup_{A \in S} A$, that is any nonempty subset of $\mathbb{R}$ bounded above has a supremum in $\mathbb{R}$.*

Proof. We consider $S = \{A \mid A \in \mathbb{R} \text{ and } A < B\}$. The set $I = \bigcup_{A \in S} A$ is a Dedekind cut:

- as $A \subset \mathbb{Q}$ then $I \subset \mathbb{Q}$; as $I \supset A \neq \emptyset$ then $I \neq \emptyset$; as $I < B$, then $I \neq \mathbb{Q}$,
- I is closed downwards, if $y \in I$ then $y \in A$ for some A and as A is closed downwards, $x < y \Rightarrow x \in A \subset I$.
- if I has a greatest element x , then $x \in A$ for some A and, as $A \subset I$, x is the greatest element of A , which is impossible.

Moreover, I is an upper bound for S because $\forall A \in S, A \subset I$ hence $A \leq I$. Finally, any $I' < I$ is not an upper bound for S because, if $I' < I$ then $\exists x \in I \setminus I'$. As $x \in I$, then $\exists A \in S$ such that $x \in A$ hence (as $x \notin I'$) $A \not\leq I'$. $\square$

Definition 2.6 (**Algebraic field (Körper)**). *An algebraic field $(K, +, 0, \times, 1)$ is a set such that:*

- $(K, +, 0)$ is an Abelian group, $(K \setminus \{0\}, \times, 1)$ is an Abelian group and $\times$ is distributive with respect to $+$ that is $a \times (b + c) = a \times b + a \times c$.

If moreover K is totally ordered by $\leq$ we say that $\leq$ is compatible with $+$ and $\times$ if for all $x, y, z \in K$, if $x \leq y$ then $x + z \leq y + z$, and if $0 \leq x, y$ then $0 \leq x \times y$. We say that $\leq$ is Dedekind complete if every non-empty subset of K bounded from above has a least upper bound, that is $\forall X \subset K, X \neq \emptyset$, if $B = \{b \in K \mid \forall x \in X, x \leq b\} \neq \emptyset$ then $\exists u \in B$ such that $\forall b \in B, u \leq b$. In this case u is called the supremum of X .

The rational numbers $\mathbb{Q}$ are an algebraic field with a compatible order $\leq$ which is not Dedekind complete: the set $A = \{x \in \mathbb{Q}, x^2 < 2\}$ has upper bounds in $\mathbb{Q}$, e.g. 1.416, but it has no least upper bound as $\sqrt{2} \notin \mathbb{Q}$.

Definition 2.7 (**Axiomatic definition of $\mathbb{R}$**). *$(\mathbb{R}, +, 0, \times, 1)$ is the unique Dedekind complete compatibly totally ordered algebraic field.*

The real numbers can be represented as points on an infinite straight line, called the real line, depicted in Fig. 2.1.

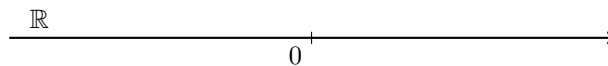


Figure 2.1: The real axis.

Theorem 2.5 (**$\mathbb{R}$ has the Archimedean property (Archimedisches Axiom)**). *The real numbers have the Archimedean property, that is for any $x, y \in \mathbb{R}$ with $0 < x < y$, there exists $n \in \mathbb{N}$ such that $y < nx$.*

Proof. Without loss of generality we can assume $y = 1$. Assume there exists $0 < x < 1$ such that $nx \leq 1$ for any $n \in \mathbb{N}$. Denote S the set of all such x and observe that S is bounded from above by 1. Then S has a supremum, which we denote c . As c is the supremum of S , $n(c/2) \leq 1$ for all $n \in \mathbb{N}$ ⁷ but there exists $p \in \mathbb{N}$ such that $1 < p(2c)$, which is impossible. $\square$

⁶Equivalently, if B is such that $\forall A \in S, A \leq B$ then $I \leq B$.

⁷As there exists $c/2 < x \leq c$ with $x \in S$, and as the order is compatible $n(c/2) < nx \leq 1$

Theorem 2.6 ($\mathbb{Q}$ is dense in $\mathbb{R}$). For any $x, y \in \mathbb{R}$ with $x < y$ there exists a rational number $q \in \mathbb{Q}$ such that $x < q < y$.

Proof. We denote $\tilde{x}$ the cut associated to the real number x , that is $\tilde{x}$ is a subset of $\mathbb{Q}$ that is closed downwards and is bounded from above but has no greatest element. The Dedekind cut of $q \in \mathbb{Q}$ is denoted $\tilde{q} = \{p \in \mathbb{Q} \mid p < q\}$. We denote $\tilde{x} \subsetneq \tilde{y}$ the relation of strict inclusion $\tilde{x} \subset \tilde{y}$ and $\tilde{y} \setminus \tilde{x} \neq \emptyset$. As $x < y$ we have $\tilde{x} \subsetneq \tilde{y}$ and:

If $x \in \mathbb{Q}$ then $x \in \tilde{y}$. This follows if $\exists s \in \tilde{y}$, $x < s$. Assume the contrary $\forall s \in \tilde{y}$, $s \leq x$. But then the inequality needs to be strict, otherwise, as $\tilde{y}$ has no greatest element, $\exists s' \in \tilde{y}$, $s' > s = x$. Finally, if $\forall s \in \tilde{y}$, $s < x$ then $\tilde{y} \subset \tilde{x}$ and $y \leq x$, contradicting the hypothesis $x < y$.

There exist $q \in \mathbb{Q}$, $q \in \tilde{y} \setminus \tilde{x}$ and if $x \in \mathbb{Q}$ we can always chose $q > x$. The first part follows from $\tilde{x} \subsetneq \tilde{y}$. If $x \in \mathbb{Q}$ then by the previous item $x \in \tilde{y}$ and as $\tilde{y}$ has no greatest element, there exists $q \in \mathbb{Q}$, $x < q$ and $q \in \tilde{y}$.

We have $\tilde{x} \subsetneq \tilde{q}$. Any $r \in \tilde{x}$, $r \neq q$. As $\mathbb{Q}$ is totally ordered then either $r < q$ or $q < r$. If $q < r$, as $\tilde{x}$ is downwards closed, $q \in \tilde{x}$ contradicting $q \in \tilde{y} \setminus \tilde{x}$. We conclude that $r < q$ hence $r \in \tilde{q}$ and $\tilde{x} \subset \tilde{q}$. If $x \notin \mathbb{Q}$ then $x \neq q$ and the inclusion is strict. If $x \in \mathbb{Q}$, as we can chose $q > x$, the inclusion is again strict.

We have $\tilde{q} \subsetneq \tilde{y}$. This is trivial as $q \in \tilde{y}$ and $\tilde{y}$ is closed downwards.

□

Theorem 2.7 ($\mathbb{R}$ is uncountable). There is no surjective function $f : \mathbb{N} \rightarrow (0, 1) \subset \mathbb{R}$, that is the set of real numbers is not countable.

Proof. Every $a \in (0, 1)$ admits a decimal representation as $0.a_1a_2\dots a_n\dots$ obtained by choosing inductively a_k to be the largest digit 0, 1, 2 up to 9 such that $\sum_{p=1}^k a_p 10^{-p} \leq x$ and $x = \sup_{k \geq 1} \{\sum_{p=1}^k a_p 10^{-p}\}$. This definition avoids the “problem of trailing 9s”, i.e. 0.49999... is not the decimal representation of any real number, as the decimal representation of $1/2$ is 0.50000.... The price to pay is that not any list of digits is an admissible list of decimals of a real number: lists ending in trailing 9s, $a_k = 9$ for $k > n$ are not admissible and need to be converted in lists of decimals ending in trailing 0s, as in $0.a_1\dots a_n9999 \rightarrow 0.a_1\dots(a_n+1)0000\dots$. The gain is that there is a one to one correspondence between real numbers and lists of decimals that is $x = y$ if and only if all their decimals are equal.

Consider now any function $f : \mathbb{N} \rightarrow (0, 1)$ and consider for all $i \in \mathbb{N}$ the decimal representation $f(i) = 0.i_1i_2\dots i_n\dots$. We now build a real number $x \in (0, 1)$ such that $\forall i, f(i) \neq x$. We take x with decimal representation $x = 0.x_1x_2\dots x_n\dots$ and we ensure that $x \neq f(i)$ by choosing the i 'th decimal x_i of x to be different from the i 'th decimal i_i of $f(i)$ (if we make a table with rows the decimals of $f(i)$, we modify the diagonal decimals and collect them to form x). For instance we can chose $x_i = i_i + 1$ if $0 \leq i_i \leq 7$ and $x_i = 0$ for $i_i = 8, 9$. In particular x can not contain any decimal 9, so it is an admissible decimal representation of a real number. As $x \neq f(i)$ for any i , $x \notin f(\mathbb{N})$ and f is not surjective. □

It can be shown that $|\mathbb{R}| = 2^{\aleph_0}$, that is the cardinal of the real numbers is equal to the cardinal of the power set of the natural numbers.⁸ The continuum hypothesis states that there is no other cardinal number in between $\aleph_0$ and $2^{\aleph_0}$, that is $2^{\aleph_0} = \aleph_1$. This statement is undecidable using the standard Zermelo–Fraenkel set theory axioms.

Some simple functions of real numbers are:

- *Inverses.* For $x \in \mathbb{R}$ the additive inverse of x is $-x$ and for $x \in \mathbb{R}_{\neq 0}$ the multiplicative inverse is $1/x = x^{-1}$. The multiplicative inverses of 0 is undetermined.
- *Absolute value.* For $x \in \mathbb{R}$ the absolute value $|x|$ of x is x if $x \geq 0$ and $-x$ otherwise.

⁸Think of the decimal representation in base 2.

- *Integer powers, polynomials.* For $x \in \mathbb{R}_{\neq 0}$, $x^0 = 1$ and $x^{n+1} = x \times x^n$; $0^1 = 0$ but 0^0 is undermined. Polynomials are linear combinations of integer powers. A polynomial of degree n is $P(x) = a_n \times x^n + a_{n-1} \times x^{n-1} + \dots + a_0$, with $a_0, \dots, a_n \in \mathbb{R}$ and $a_n \neq 0$.
- *n -roots.* For $x \in \mathbb{R}_{\geq 0}$, the root of order n for $n \in \mathbb{Z}_{\neq 0}$ is:
 - $n > 0$, it is the Dedekind cut $x^{\frac{1}{n}} = \{0 \leq q \in \mathbb{Q} \mid q^n < x\} \cup \mathbb{Q}_{<0}$.
 - for $n < 0$ it is defined as $x^{\frac{1}{n}} = (x^{-\frac{1}{n}})^{-1}$.

For $n \neq 0$, $0^{\frac{1}{n}} = 1$ and $0^{\frac{1}{0}}$ is undetermined.

- *Rational powers.* For $x \in \mathbb{R}_{\geq 0}$, and $p, q \in \mathbb{Z}$, $x^{\frac{p}{q}} = (x^{\frac{1}{q}})^p$

Exercise 2.8. *Prove that the arithmetic operations defined for Dedekind cuts are commutative, associative, compatible with the ordering and that the multiplication is distributive with respect to the addition. Prove that if $0 < x, y$ then $0 < x \times y$. Prove that this implies that if $a > 0$ and $x < y$ then $a \times x < a \times y$.*

Exercise 2.9. *Prove that $A = \{a \in \mathbb{Q} \mid a^2 = a \times a < 2 \text{ or } a < 0\}$ is a valid Dedekind cut which does not correspond to a rational number. What real number does this cut represent?*

Exercise 2.10. *Prove, using Dedekind cuts, that the sequence $0.999\dots$ of trailing 9 is the real number 1. We denote the truncated sequence of n decimals $9, 0.(9)_n = 1 - 10^{-n} < 1$, and $0.999\dots$ the supremum of this subset of real numbers. The Dedekind cut of $0.999\dots$ consists in the rational numbers r such that there exists some n with $r < 0.(9)_n < 1$, hence r belongs to the cut of 1. Consider now any element in the cut of 1, $\frac{a}{b} < 1$ hence $0 < a < b$. Then $1 - \frac{a}{b} = \frac{b-a}{b} \geq \frac{1}{b} > \frac{1}{10^6}$ hence $0.(9)_6 > \frac{a}{b}$, therefore $\frac{a}{b}$ belongs to the cut of $0.(9)_6$ and consequently to the one of $0.999\dots$*

2.2.6 Complex numbers $\mathbb{C}$ (Komplexe Zahlen)

The last class of numbers we will be using are the complex numbers.

Construction 2.5 (**Complex numbers $\mathbb{C}$**). *The complex numbers $\mathbb{C}$ are the Cartesian product $\mathbb{R} \times \mathbb{R}$ endowed with the operations:*

- (addition) $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$,
- (multiplication) $(x_1, y_1) \times (x_2, y_2) = (x_1 \times x_2 - y_1 \times y_2, x_1 \times y_2 + y_1 \times x_2)$,

and form an algebraic field. The real numbers $\mathbb{R}$ are embedded in $\mathbb{C}$ by $x \rightarrow (x, 0)$

In order to simplify the notation, whenever no confusion can arise, we suppress the symbol $\times$ for multiplication.

As $\mathbb{C} = \mathbb{R} \times \mathbb{R}$, the complex numbers are represented as an infinite plane, called the complex plane. The real numbers $\mathbb{R}$ are embedded into $\mathbb{C}$ by the mapping $x \rightarrow (x, 0)$. In particular $(1, 0) \equiv 1$ and denoting $(0, 1) \equiv i$ we can write any complex number as:

$$z = (x, y) = (x, 0) + (0, 1) \times (y, 0) = x + i \times y, \quad i^2 = i \times i = -1.$$

Let $z = x + iy$ be a complex number. We call x the *real part* of z which we denote $\text{Re}(z)$, and y the *imaginary part* of z , which we denote by $\text{Im}(z)$. The *complex conjugated* of z , denoted $\bar{z}$, is obtained by keeping the real part fixed, but taking the additive inverse of the imaginary part $z = x + iy \rightarrow \bar{z} = x - iy$.

The complex numbers can be represented as points in a plane, called the complex plane spanned by two orthogonal real axes: $z = x + iy = (x, y)$ represents a point with coordinates x on the horizontal and y on the vertical, as depicted in Fig. 2.2.

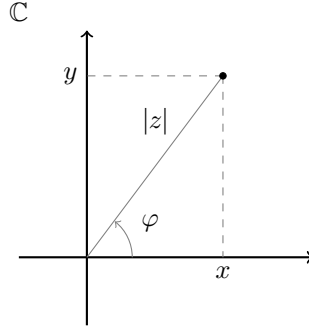


Figure 2.2: The complex plane.

The *absolute value* (or modulus) of z is the non negative real number⁹ $|z| = (x^2 + y^2)^{1/2} = (z\bar{z})^{1/2}$, $|z| > 0$ for $z \neq 0$. It represents the distance from the origin $0 = (0, 0)$ and $z = (x, y)$. The *argument* of z is the angle φ between the horizontal axis and the line $(0, 0), (x, y)$ and:

$$x = |z| \cos \varphi, \quad y = |z| \sin \varphi \quad \Rightarrow \quad z = |z|(\cos \varphi + i \sin \varphi).$$

A priori the angle φ belongs to the interval $(-\pi, \pi)$. This is the *principal determination* of the argument, denoted $\text{Arg}(z)$. However, the argument is not uniquely defined: it is a periodic function with period 2π , that is:

$$z = |z|(\cos \theta + i \sin \theta), \quad \theta = \text{Arg}(z) + 2k\pi, \quad k \in \mathbb{Z}.$$

Like for real numbers, simple functions of complete numbers are:

- *Inverses.* For $z \in \mathbb{C}$ the additive inverse of z is $-z$ and for $z \in \mathbb{C}_{\neq 0}$ the multiplicative inverse is $\frac{1}{z} = \frac{\bar{z}}{z \times \bar{z}}$. The multiplicative inverses of 0 is undetermined.
- *Integer powers, polynomials.* For $z \in \mathbb{R}_{\neq 0}$, $z^0 = 1$ and $z^{n+1} = z \times z^n$; $0^1 = 0$ but 0^0 is undermined. Polynomials are linear combinations of integer powers $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0$, with $a_0, \dots, a_n \in \mathbb{R}$ and $a_n \neq 0$.
- *n-roots.* For $z = |z|(\cos \varphi + i \sin \varphi) \in \mathbb{C}_{\neq 0}$ and for any $n \in \mathbb{N}_{\neq 0}$ we have n distinct roots of order n of z :

$$\xi^n = z \Rightarrow \xi \in \{\xi_0, \dots, \xi_{n-1}\}, \quad \xi_k = |z|^{1/n} \left(\cos\left(\frac{\varphi + 2k\pi}{n}\right) + i \sin\left(\frac{\varphi + 2k\pi}{n}\right) \right).$$

- *Rational powers.* For $x \in \mathbb{C}_{\neq 0}$, and $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_+$, $z^{\frac{p}{q}} = (z^{\frac{1}{q}})^p$

The most important property of the complex numbers, whose proof is postponed to Chapter 10, states that $\mathbb{C}$ is an algebraically closed field, that is any polynomial with complex coefficients admits a root in $\mathbb{C}$. In detail, for any polynomial with complex coefficients $P(z)$ there exist $r \in \mathbb{C}$ such that $P(r) = 0$. Denoting r a root of the polynomial $P(z)$, we note that $z^n - r^n = (z - r) \sum_{q=0}^{n-1} z^q r^{n-q}$ hence $P(z) = P(z) - P(r) = (z - r)Q(z)$ for Q some polynomial of degree $n - 1$, and iterating we get $P(z) = a_n (z - r_1) \times \dots (z - r_n)$ with $r_1, \dots, r_n$ the n (not necessarily distinct) roots of $P(z)$ and a_n the coefficient of the monomial of highest degree.

Exercise 2.11. Prove that the addition and multiplication of complex numbers are commutative and associative and the multiplication is distributive with respect to the addition, that is the complex numbers are an unordered algebraic field.

Exercise 2.12. Prove that $i^2 = i \times i = -1$

⁹Strictly speaking the non negative determination of the square root is constructed as a Dedekind cut.

Exercise 2.13. *Prove using the trigonometric circle that $\cos(\varphi_1 + \varphi_2) = \cos(\varphi_1)\cos(\varphi_2) - \sin(\varphi_1)\sin(\varphi_2)$ and $\sin(\varphi_1 + \varphi_2) = \sin(\varphi_1)\cos(\varphi_2) + \cos(\varphi_1)\sin(\varphi_2)$. Conclude that for $z_1 = |z_1|(\cos \varphi_1 + \imath \sin \varphi_1)$ and $z_2 = |z_2|(\cos \varphi_2 + \imath \sin \varphi_2)$ then $z_1 z_2 = |z_1||z_2|(\cos(\varphi_1 + \varphi_2) + \imath \sin(\varphi_1 + \varphi_2))$. Show that for any n -root of z , $(\xi_k)^n = z$. Prove that any complex number such that $\xi^n = z$ is one of the n -roots of z .*

Exercise 2.14. *What is the image of the line $\operatorname{Re}(z) = \text{const.}$ under the function $f(z) = 1/z$?*

Chapter 3

Topological spaces

Topology formalizes the notion of being close in the absence of a way to measure distances.

3.1 Topology

Definition 3.1 (**Topology** (Topologie)). A topological space is a set X endowed with a topology $\mathcal{T}$. A topology $\mathcal{T}$ on X is a set of subsets of X , $\mathcal{T} = \{U \mid U \subset X\} \subset \mathcal{P}(X)$ such that:

- $\emptyset, X \in \mathcal{T}$,
- (closed under arbitrary union) for any index set¹ A if $(U_\alpha)_{\alpha \in A} \in \mathcal{T}$ then $\bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}$ (or, stated fancier, $\mathcal{A} \subset \mathcal{T} \Rightarrow \bigcup \mathcal{A} \subset \mathcal{T}$),
- (closed under finite intersection) for any finite set I , if $(U_i)_{i \in I} \in \mathcal{T}$ then $\bigcap_{i \in I} U_i \in \mathcal{T}$.

A subset Y of X is endowed with the subspace (or induced) topology $\mathcal{T}_Y = \{U \cap Y \mid U \in \mathcal{T}\}$.

The topologies of a set X form a poset, in fact a lattice, with the inclusion. For $\mathcal{T}_1$ and $\mathcal{T}_2$ two topologies on X , if $\mathcal{T}_1 \subset \mathcal{T}_2$ we say that $\mathcal{T}_2$ is finer than $\mathcal{T}_1$ (resp. $\mathcal{T}_1$ is coarser than $\mathcal{T}_2$).

Any set X has two trivial topologies: the *discrete topology* consisting in all the subsets of X (the finest topology on X) and the *indiscrete topology* consisting in only the empty set and the set itself (the coarsest topology in X). Topologies that sit in between these two are usually more interesting.

Definition 3.2 (**Open sets and neighborhoods** (Offene Menge, Umgebungen)). The elements U in a topology $U \in \mathcal{T}$ are called open sets. We reserve U as a notation for open sets. A set is called closed if its complement is open.² Sets that are both open and closed (for instance $\emptyset$ and X) are sometimes called clopen.

For any $x \in X$, a neighborhood of x in X is an set $V \subset X$ which includes an open set containing x , that is $x \in U \subset V$, with $U \in \mathcal{T}$. Without loss of generality we restrict to open neighborhoods, that is a neighborhood of a point is any open set containing that point.

A topology is called Hausdorff if for any two distinct points $x_1, x_2 \in X$, $x_1 \neq x_2$ one can find two disjoint neighborhoods, one containing x_1 and the other one x_2 , that is $x_1 \in V_1$, $x_2 \in V_2$ and $V_1 \cap V_2 = \emptyset$.

In a Hausdorff topology the points are not indistinguishably close to each other. The discrete topology is Hausdorff, while the indiscrete one is not.

¹Strictly speaking, any ordinal A , finite, countable or uncountable.

²As $\bigcap_\alpha (X \setminus U_\alpha) = X \setminus (\bigcup_\alpha U_\alpha)$ and $\bigcup_i (X \setminus U_i) = X \setminus (\bigcap_i U_i)$, arbitrary intersection and finite unions of closed sets are closed.

A *subbase* in a topology is a subset of open sets $U \in \mathcal{T}$ such that any other open set is obtained by taking arbitrary unions of finite intersections of them. A *base* is a subset of open sets such that any other open set can be obtained by taking only unions of them. For example, every totally ordered set X can be endowed with a topology, called the order topology, generated by the subbase $\{x|x < b\}$ and $\{x|a < x\}$ for all $a, b \in X$. A base in the order topology consists in these sets together with $(a, b) = \{x|a < x < b\}$. The ordinal numbers $\lambda = [0, \lambda)$, $\lambda + 1 = \lambda \cup \{\lambda\}$ are totally ordered and the space of ordinal numbers can be endowed with the order topology.

Definition 3.3 (**Continuous and open functions**). A map (function) $f : X \rightarrow Y$ from a topological space $(X, \mathcal{T}_1)$ to a topological space $(Y, \mathcal{T}_2)$ is called

- open if for all $U \in \mathcal{T}_1$, $f(U) \in \mathcal{T}_2$,
- continuous if for all $V \in \mathcal{T}_2$, $f^{-1}(V) \in \mathcal{T}_1$.

If $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on X with $\mathcal{T}_1 \subset \mathcal{T}_2$ then the map $\text{id}_X : (X, \mathcal{T}_1) \rightarrow (X, \mathcal{T}_2)$ is open and $\text{id}_X : (X, \mathcal{T}_2) \rightarrow (X, \mathcal{T}_1)$ is continuous.

A topological space is called *connected* if it can not be divided into two disjoint nonempty open sets, $X \neq X_1 \cup X_2$ with $X_1 \cap X_2 = \emptyset$. For $x, y \in X$, a *path* from x to y is a continuous function³ $\gamma : [0, 1] \rightarrow X$ with $f(0) = x$ and $f(1) = y$. A topological space is *path connected* if any two points in it are connected by a path.

3.1.1 Compact sets (Kompakter Menge)

A class of sets that will play an important role below is that of compact sets.

Definition 3.4 (**Compact sets**). A set $K \subset X$ is called compact if from any open cover of K , $\bigcup_{\alpha \in A} U_\alpha \supset K$ with $U_\alpha \in \mathcal{T}$, one can select a finite open cover of K , $\bigcup_{\alpha \in I \subset A} U_\alpha \supset K$ for some finite set I .

Theorem 3.1 (**Properties of compact sets**). We have:

- let K be a compact subset of X and $U \subset K$ an open set. Then $K \setminus U$ is compact.

Proof. Let any open cover $(U_\alpha)_{\alpha \in A}$ of $K \setminus U$, then $U, (U_\alpha)_{\alpha \in A}$ is an open cover of K and it possesses a finite subcover $(U_i)_{i \in I}$ (where one of U_i can be U). As U contains no point in $K \setminus U$, $(U_i)_{i \in I} \setminus \{U\}$ is a finite cover of $K \setminus U$. $\square$

- if K is a compact subset of $(X, \mathcal{T})$ then K is compact in the induced topology $(K, \mathcal{T}_K)$.

Proof. For any cover of K by open sets in the induced topology $(K \cap U_\alpha)_{\alpha \in A}$ with $U_\alpha \in \mathcal{T}$, then $(U_\alpha)_{\alpha \in A}$ is an open cover of K in $\mathcal{T}$ and admits a finite subcover $(U_i)_{i \in I}$ which is such that $(K \cap U_i)_{i \in I}$ covers K . $\square$

- the image of a compact set under a continuous function is a compact set.

Proof. Let $f : X \rightarrow Y$ a continuous function and consider $K \subset X$ compact. For any open cover of $f(K)$, $f(K) \subset \bigcup_{\alpha \in A} V_\alpha \subset f(X)$, the sets $(f^{-1}(V_\alpha))_{\alpha \in A}$ are open (as f is continuous) and form a cover of K , $\bigcup_{\alpha \in A} f^{-1}(V_\alpha) \supset K$. Hence one can select a finite cover of K , $(f^{-1}(V_i))_{i \in I}$ leading to a finite open cover of $(V_i)_{i \in I}$ of $f(K)$. $\square$

³We endow $[0, 1] \subset \mathbb{R}$ with the subspace topology inherited from the order topology (which coincides with the standard topology, see below) on $\mathbb{R}$.

3.1.2 Topological boundary, closure and interior (Topologischer Rand)

Definition 3.5 (**Topological boundary, closure and interior**). Consider a subset of a topological space $Y \subset X$. For any point in $x \in X$, we denote $U(x)$ an open neighborhood of x , that is an open set $U(x) \in \mathcal{T}$ containing x , $x \in U(x)$. The point x is:

- an accumulation (or limit) point of Y if every neighborhood of x contains at least one other point in Y , that is $\forall U(x), (U(x) \cap Y) \setminus \{x\} \neq \emptyset$. A limit point of Y does not necessarily belong to Y .
- an interior point of Y if it has a neighborhood entirely contained in Y , that is $\exists U(x) \subset Y$.
- a isolated point of Y if it belongs to Y but is not a limit point of Y , that is $x \in Y$ and $\exists U(x)$ such that $(U(x) \cap Y) \setminus \{x\} = \emptyset$.⁴

We stress that the isolated points of a set, while by definition are not limit points, can be interior points: if X is endowed with the discrete topology and $Y \subset X$, every point of $y \in Y$ is at the same time an isolated and an interior point of Y as $\{y\}$ is open.

Exercise 3.1. The interior Y° of a subset Y of a topological space X is the union of all the open sets contained in Y , that is $Y^\circ = \bigcup(\mathcal{P}(Y) \cap \mathcal{T})$. Prove that:

- $x \in Y^\circ$ if and only if x is an interior point of Y . If x is interior to Y then $\exists U(x) \subset Y$ hence $U(x) \in \mathcal{P}(Y) \cap \mathcal{T}$ thus $U(x) \subset Y^\circ$. Conversely, any point in Y° must belong to at least one of the open sets in $\mathcal{P}(Y) \cap \mathcal{T}$, all of which are entirely included in Y .
- the interior $Y^\circ \subset Y$ is an open set in $\mathcal{T}$. By definition.
- a set U is open if and only if $U = U^\circ$. If $U = U^\circ$ then U is open (as U° is open). Conversely, consider an open U ; first, $U^\circ \subset U$; second, as U is open then both $U \in \mathcal{P}(U)$ and $U \in \mathcal{T}$, hence $U \subset \bigcup(\mathcal{P}(U) \cap \mathcal{T}) = U^\circ$.

The closure $\bar{Y}$ of Y is the union of Y with all its limit points. The boundary of Y is the set difference between its closure and its interior, $\partial Y = \bar{Y} \setminus Y^\circ$. We have that:

- $x \in \bar{Y}$ if and only if $\forall U(x), U(x) \cap Y \neq \emptyset$. If $x \in \bar{Y}$ then either $x \in Y$ or x is a limit point of Y and in both cases $\forall U(x), U(x) \cap Y \neq \emptyset$. Conversely, if $x \in Y$ then $x \in \bar{Y}$ and if $x \notin Y$ then $\forall U(x), x \notin U(x) \cap Y$ hence $U(x) \cap Y \setminus \{x\} = U(x) \cap Y \neq \emptyset$ and x is a limit point of Y .
- the complement of the closure is the interior of the complement $X \setminus \bar{Y} = (X \setminus Y)^\circ$. Indeed, if $x \in X \setminus \bar{Y}$ then on the one hand $\exists U(x)$ such that $U(x) \cap Y \setminus \{x\} = \emptyset$ and on the other $x \notin Y$ (as $Y \subset \bar{Y}$) hence $U(x) \cap Y = \emptyset$. It follows $U(x) \subset X \setminus Y$, that is $x \in (X \setminus Y)^\circ$. Conversely, if $x \in (X \setminus Y)^\circ$ then $\exists U(x), U(x) \subset X \setminus Y$, which implies $U(x) \cap Y = \emptyset$ hence x can not belong to $\bar{Y}$.
- C is closed if and only if $C = \bar{C}$. By the previous item, $\bar{C}$ is closed hence $C = \bar{C}$ implies C is closed. Conversely, if C is closed then $X \setminus C$ is open and $X \setminus C = (X \setminus C)^\circ = X \setminus \bar{C}$ by the previous item.
- X splits into the disjoint partition $X = \bar{Y} \cup (X \setminus \bar{Y}) = Y^\circ \cup \partial Y \cup (X \setminus Y)^\circ$. From the definition of $\bar{Y}$ and item 2.
- the boundaries of a set and of its complement are equal $\partial Y = \partial(X \setminus Y)$ and $\partial Y = \bar{Y} \cap \overline{X \setminus Y}$. By applying the previous item to Y and $Y \rightarrow X \setminus Y$.
- ∂Y is a closed set. As its complement is open.
- ∂Y consists of all the points $x \in X$ such that $\forall U(x), U(x) \cap Y \neq \emptyset$ and $U(x) \cap (X \setminus Y) \neq \emptyset$. Indeed, any point $x \in \partial Y$ is either a point in Y or a limit point of Y (or both) and any $U(x)$ will contain a point in Y ; the same goes for $X \setminus Y$. Conversely, by the first item above, $x \in \bar{Y}$ and $x \in \overline{X \setminus Y}$ hence $x \in \bar{Y} \cap \overline{X \setminus Y} = \partial Y$.

⁴ $\{x\}$ is an open set in the induced topology $\mathcal{T}_Y$.

- for any set Y , $\partial Y \supset \partial\partial Y$ and $\partial Y = \partial\partial Y$ if and only if ∂Y has no interior points. *Indeed, as ∂Y is closed, $\overline{\partial Y} = \partial Y$ and $\partial\partial Y = \partial Y \setminus (\partial Y)^\circ$.*
- the boundary ∂V of any open set V has empty interior. The same holds for closed sets. *As V is open, $V = V^\circ$ hence $V \cap \partial V = \emptyset$ as the boundary and the interior of any set are disjoint. Take any point $x \in \partial V$; for any $U(x)$, $U(x) \cap V \neq \emptyset$ hence $U(x)$ can not be a subset of ∂V . The same result follows for close sets as they have the same boundary as their complements, which are open.*
- every closed set with empty interior is the boundary of its complement. *Indeed, as $B = \bar{B}$ and $B^\circ = \emptyset$ then on the one hand $\partial B = \bar{B} = B$ and on the other $\partial B = \partial(X \setminus B) = \partial(X \setminus B)^\circ$.*
- we stress that:
 - * the isolated points of Y are not necessarily limit points of $X \setminus Y$, as in the discrete topology.
 - * for any Y the interior of Y° is Y° as Y° is open and the closure of $\bar{Y}$ is $\bar{Y}$ as $\bar{Y}$ is closed.
 - * a set Y is dense in X if $\bar{Y} = X$. If U is dense in X and open then $\partial U = X \setminus U$.
 - * in general $\partial\bar{Y} \neq \partial Y \neq \partial Y^\circ$ hence in general $\bar{Y}^\circ \neq \bar{Y}$ and $(\bar{Y})^\circ \neq Y^\circ$! The isolated points of Y belong to $\bar{Y}$ but are not accumulation points of Y . Such points usually do not belong to Y° (unless the topology is discrete and they belong as one element sets to the topology), hence usually they belong to ∂Y but not to ∂Y° . If U is open dense in X , then $\partial(\bar{U}) = \partial X = \emptyset$ as X is both closed and open but $\partial U = X \setminus U \neq \emptyset$.

3.2 Metric spaces and the standard topology

In many cases of interest, the topology of a space comes from a notion of metric. Formally, a topological space X is called *metrizable* if it is homeomorphic⁵ to a metric space. A metric space is a space in which one can measure distances.

We use ϵ as a notation for a (possibly small) strictly positive real number.

Definition 3.6 (**Metric space (Metrischer Raum)**). A metric space X is a set endowed with a distance function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ (the non negative reals) such that:

- $d(x, x) = 0$,
- (symmetry and positivity) if $x \neq y$ then $d(x, y) = d(y, x) > 0$,
- (triangle inequality) for all x, y, z , $d(x, y) \leq d(x, z) + d(z, y)$.

The most interesting property of a distance is the triangle inequality. Morally it means that the distance between x and y is measured along the shortest path, and passing through another point z will always represent a detour which lengthens the distance.

A distance function of a metric space induces a topology.

Definition 3.7 (**The standard topology**). The standard topology on a metric space (X, d) is the topology generated by the subbase of all open balls $B_r(x) = \{y \mid d(x, y) < r\} \subset X$, for all $x \in X$, $r \in \mathbb{R}_{>0}$.

The closure of a ball in the standard topology is $\bar{B}_r(x) = \{y \mid d(x, y) \leq r\}$. Indeed, if y is an accumulation point of $B_r(x)$, then $\forall \epsilon, \exists z \in B_\epsilon(y) \cap B_r(x)$ hence $d(x, y) \leq d(x, z) + d(z, y) < r + \epsilon$. Assuming $d(x, y) = r' > r$, $\epsilon < r' - r$ leads to a contradiction. The boundary of the ball is $\partial B_r(x) = \{y \mid d(x, y) = r\}$.

⁵That is mapped by a *bicontinuous bijection*, a continuous bijection whose inverse is also continuous.

Theorem 3.2 (**In the standard topology, every open set contains a ball around any of its points**). *For any open set U in the standard topology on some metric space (X, d) , and for any $y \in U$, there exists a small enough $\epsilon > 0$ such that $B_\epsilon(y) \subset U$. The converse (if $\forall y \in U, \exists B_\epsilon(y) \subset U$ then U is open) is trivially true.*

A topological space X is called first countable if $\forall x \in X$, there exists a countable set of open neighborhoods of x , $\{U_n(x) | n \in \mathbb{N}\}$ such that for any open set U , $x \in U \Rightarrow U_i(x) \subset U(x)$ for some $i \in \mathbb{N}$. All metric spaces are first countable.

Proof. We first show that this property holds for any ball, then for an arbitrary union of balls and finally for a finite intersection of balls.

Consider a ball $B_r(x)$ centered at x for some x and some other point $y \in B_r(x)$. As $d(x, y) < r$, there exists ϵ such that $0 < \epsilon < r - d(x, y)$. By the triangle inequality, $\forall z \in B_\epsilon(y)$, $d(z, x) \leq d(z, y) + d(y, x) \leq \epsilon + d(y, x) < r$ therefore $B_\epsilon(y) \subset B_r(x)$.

Let $y \in \bigcup_{\alpha \in A} B_{r_\alpha}(x_\alpha)$ and denote $B_{r_1}(x_1)$ (one of) the ball(s) to which y belongs; then there exists ϵ such that $B_\epsilon(y) \subset B_{r_1}(x_1)$ and the same ball is contained in the union $B_\epsilon(y) \subset \bigcup_{\alpha \in A} B_{r_\alpha}(x_\alpha)$. Finally, if $y \in B_{\epsilon_1}(y) \subset B_{r_1}(x_1)$ and $y \in B_{\epsilon_2}(y) \subset B_{r_2}(x_2)$, then $y \in B_{\min(\epsilon_1, \epsilon_2)}(y) \subset B_{\epsilon_1}(y) \cap B_{\epsilon_2}(y)$. The lemma follows by noting that any open set can be obtained by taking arbitrary unions and finite intersection balls.

The countable set $B_{\frac{1}{n}}(x)$ fulfills the requirement of first countability at x . $\square$

From this lemma it follows that the balls form in fact a base, not only a subbase in the standard topology, that is one can obtain any open set U by taking unions of balls, one for each of its points, which are fully contained in U .

An important result is that metrizable topologies are Hausdorff.

Theorem 3.3 (**The standard topology is Hausdorff**). *The standard topology on a metric space is Hausdorff; the same holds for all metrizable topologies.*

Proof. Consider any two points $x \neq y$, hence such that $d(x, y) > 0$. Then the balls $B_{d(x, y)/3}(x)$ and $B_{d(x, y)/3}(y)$ are totally disjoint. Indeed, assume there exists $z \in B_{d(x, y)/3}(x) \cap B_{d(x, y)/3}(y)$. Then $d(z, x), d(z, y) < d(x, y)/3$, and from the triangle inequality $d(x, y) \leq d(x, z) + d(z, y) < 2d(x, y)/3$ which is impossible. $\square$

Theorem 3.4 (**Any distance function is continuous**). *Any distance function is continuous in either of its arguments.*

Proof. Let $f(x) = d(x_0, x)$ for some fixed x_0 , and let $V \subset \mathbb{R}_{\geq 0}$ open. For any x in $f^{-1}(V)$, $f(x) \in V$ and, as V is open, there exists some ϵ_x such that $(f(x) - \epsilon_x, f(x) + \epsilon_x) \subset V$. Any $y \in B_{\epsilon_x/2}(x)$ is such that $d(x_0, x) - d(x, y) \leq d(x_0, y) \leq d(x_0, x) + d(x, y)$, that is $|d(x_0, y) - d(x_0, x)| \leq d(x, y) = \epsilon_x/2 < \epsilon_x$ hence $f(y) \in (f(x) - \epsilon_x, f(x) + \epsilon_x)$ thus $f(y) \in V$ and $y \in f^{-1}(V)$. We conclude that $B_{\epsilon_x/2}(x) \subset f^{-1}(V)$ hence x is an interior point of $f^{-1}(V)$ and as x is arbitrary, V is open. $\square$

Sorgenfrey, or lower limit topology (Sorgenfrey-Gerade)

Although in many cases topologies are metrizable, there exist also non metrizable topologies. An example is provided for instance by the Sorgenfrey topology of $\mathbb{R}$ generated by the basis of all half open intervals $[a, b)$. This is a non metrizable topology on $\mathbb{R}$.

Exercise 3.2. *Let us consider the real numbers $\mathbb{R}$. Prove that:*

$$d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}, \quad d(x, y) = |x - y| = \begin{cases} x - y, & x - y \geq 0 \\ -x + y, & x - y < 0 \end{cases}.$$

is a distance function on $\mathbb{R}$. Any $x, y, z \in \mathbb{R}$ with $x < y$, we have $z < x < y$ or $x < z < y$ or $x < y < z$. Consider now the complex numbers $\mathbb{C}$. Prove that $d : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{R}_{\geq 0}$, $d(z_1, z_2) = |z_1 - z_2|$ is a distance function. This follows from $|z_1 + z_2| \leq |z_1| + |z_2|$ which follows from $(x_1 + x_2)^2 + (y_1 + y_2)^2 \leq x_1^2 + y_1^2 + x_2^2 + y_2^2 + 2\sqrt{(x_1^2 + y_1^2)(x_2^2 + y_2^2)} \Leftrightarrow x_1^2 x_2^2 + y_1^2 y_2^2 + 2x_1 x_2 y_1 y_2 \leq x_1^2 x_2^2 + x_1^2 y_2^2 + y_1^2 x_2^2 + y_1^2 y_2^2$ which is true as $0 \leq (x_1 y_2 - x_2 y_1)^2$.

3.2.1 The $\mathbb{R}^D$ metric space

The D dimensional Euclidean space is the set $\mathbb{R}^D = \{(x^1, \dots, x^D) | x^\mu \in \mathbb{R}\}$. We denote $0 = (0, \dots, 0)$ the origin in $\mathbb{R}^D$, that is the point with coordinates $x^\mu = 0$, $\mu = 1 \dots D$. In most cases $\mathbb{R}^D$ is regarded as a topological space endowed with the standard topology with respect to a distance function which we now discuss. Starting from the next chapter, unless otherwise specified, we will always assume that the $\mathbb{R}^D$ is endowed with this topology.

The space $\mathbb{R}^D$ is a metric space. In fact several standard distance functions are usually defined on it:

$$d_p(x, y) = \|x - y\|_p = \left(\sum_{\mu=1}^D |x^\mu - y^\mu|^p \right)^{1/p}, \quad p \in [1, \infty),$$

and $d_\infty(x, y) = \|x - y\|_\infty = \max_{\mu=1, \dots, D} |x^\mu - y^\mu|$. Observe that for $D = 1$ all these distance functions are equal and reproduce the absolute value of the difference.

Lemma 3.1. *The distance functions d_p are equivalent on $\mathbb{R}^D$, that is for any $p, q \geq 1$ there exist two constants c_{pq} and C_{pq} such that*

$$c_{pq} d_p(x, y) \leq d_q(x, y) \leq C_{pq} d_p(x, y).$$

Proof. It is enough to prove that d_p is equivalent to d_∞ . We have $\max_\mu |x^\mu - y^\mu| \leq (\sum_{\mu=1}^D |x^\mu - y^\mu|^p)^{1/p} \leq D^{1/p} \max_\mu |x^\mu - y^\mu|$. $\square$

This lemma implies that the standard topology on the D dimensional Euclidean space $(\mathbb{R}^D, d_p)$ is independent on which distance function d_p is chosen (i.e. a set is open with respect to one of these distance functions if and only if it is open with respect to any other). For simplicity the common choice is to use d_2 :

$$d_2(x, y) = \|x - y\|_2 = \sqrt{\sum_{\mu=1}^D |x^\mu - y^\mu|^2} \equiv \|x - y\| = d(x, y),$$

that is $\|\cdot\|$ and $d(\cdot, \cdot)$ with no index p are understood to correspond to $p = 2$. This is just the familiar Euclidean distance between points in dimension n . Note that, from the topological point of view, $\mathbb{R}^2$ endowed with the Euclidean $\|x - y\|$ is identical to the complex plane $\mathbb{C}$ with distance given by the absolute value $|z_1 - z_2|$ on complex numbers.

Theorem 3.5 (In the standard topology $\overline{\mathbb{Q}} = \mathbb{R}$). *In the standard topology the closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ is $\mathbb{R}$.*

Proof. The balls in the standard topology of $\mathbb{R}$ (which incidentally is also the order topology) with distance function $d(x, y) = |x - y|$ have the form $B_\epsilon(x) = (x - \epsilon, x + \epsilon)$. From Theorem 2.6 we have that $\mathbb{Q}$ is dense in $\mathbb{R}$, that is for any $x, y \in \mathbb{R}$ with $x < y$ there exists a rational number $q \in \mathbb{Q}$ such that $x < q < y$. Consider any $x \in \mathbb{R}$. From Theorem 3.2 any neighborhood of x contains a ball $B_\epsilon(x) = (x - \epsilon, x + \epsilon)$ and as $\mathbb{Q} \cap B_\epsilon(x) \setminus \{x\} = \mathbb{Q} \cap ((x - \epsilon, x) \cup (x, x + \epsilon)) \neq \emptyset$ we conclude that x is a limit point of $\mathbb{Q}$. $\square$

Exercise 3.3. *Give an example of an open dense subset of $\mathbb{R}$ in the standard topology. $\mathbb{R} \setminus \{0\}$ or more generally $\mathbb{R} \setminus M$ for any finite set M .*

Exercise 3.4 (Hölder's inequality (Hölder-Ungleichung)). *Let $p, q \in \mathbb{R}_{\geq 1}$ such that $1/p + 1/q = 1$. Then, for any $(a^\mu, b^\mu)_{\mu=1, \dots, D} \in \mathbb{R}_{\geq 0}$ we have the inequality:*

$$\sum_{\mu=1}^D a^\mu b^\mu \leq \left[\sum_{\mu=1}^D (a^\mu)^p \right]^{1/p} \left[\sum_{\mu=1}^D (b^\mu)^q \right]^{1/q}.$$

We first note that, for any $p, q > 1$ with $1/p + 1/q = 1$ and any $x, y > 0$, as the logarithm (see Section 7.2.1 for a reminder on the definition and properties of the logarithm) is a concave function we have:

$$\ln \left(\frac{1}{p} x^p + \left(1 - \frac{1}{p}\right) y^q \right) \geq \frac{1}{p} \ln(x^p) + \frac{1}{q} \ln(y^q) = \ln(xy),$$

hence, as the logarithm is strictly increasing, $xy \leq \frac{x^p}{p} + \frac{y^q}{q}$, which is also known as the Young inequality. Thus: $\sum_{\mu=1}^n x^\mu y^\mu \leq \frac{\sum_{\mu=1}^D (x^\mu)^p}{p} + \frac{\sum_{\mu=1}^n (y^\mu)^q}{q}$, and the lemma follows by setting

$$x^\mu = \frac{a^\mu}{(\sum_{\mu=1}^D (a^\mu)^p)^{1/p}} \text{ and } y^\mu = \frac{b^\mu}{(\sum_{\mu=1}^D (b^\mu)^q)^{1/q}}.$$

Prove the Minkowski inequality $\|x+y\|_p \leq \|x\|_p + \|y\|_p$. Conclude that that all the functions $d_p(x, y) = \|x-y\|_p$ are distance functions. We have, with $x+y = (x^1+y^1, \dots, x^D+y^D)$

$$\|x+y\|_p^p = \sum_{\mu} |x^\mu + y^\mu|^p = \sum_{\mu} |x^\mu + y^\mu| |x^\mu + y^\mu|^{p-1} \leq \sum_{\mu} (|x^\mu| + |y^\mu|) |x^\mu + y^\mu|^{p-1} = \sum_{\mu} |x^\mu| |x^\mu + y^\mu|^{p-1} + \sum_{\mu} |y^\mu| |x^\mu + y^\mu|^{p-1} \leq (\|x\|_p + \|y\|_p) \|x+y\|_p^{p-1},$$

where we used Hölder's inequality. We conclude by observing that $\|x+y\|_p^{p-1} = \left(\sum_{\mu} (|x^\mu| + |y^\mu|)^{p-1} \right)^{\frac{p-1}{p}} = \|x+y\|_p^{p-1}$.

Exercise 3.5 (One point compactification (Alexandroff-Kompaktifizierung)). To any topological space $(X, \mathcal{T})$ one can add a point, usually denoted $\{\infty\}$, and build the space $X^* = X \cup \{\infty\}$ endowed with the topology $\mathcal{T}^*$ consisting in all the open sets $U \in \mathcal{T}$ together with the sets $V = (X \setminus K) \cup \{\infty\}$ for K compact in X .

Prove that $(X^*, \mathcal{T}^*)$ is compact. Any open cover of X^* will need to cover $\{\infty\}$, hence will contain a set of the form $(X \setminus K) \cup \{\infty\}$ whose complement in X is compact. What are the open sets in $\mathbb{N} \cup \{\infty\}$? In the discrete topology any subset of $\mathbb{N}$ and a set is compact if and only if it is finite. The neighborhoods of $\{\infty\}$ are $\{\infty\} \cup P$ for P any infinite subset of $\mathbb{N}$. Give a base in the topology of $\mathbb{R} \cup \{\infty\}$. Besides the balls $B_\epsilon(x) = (x-\epsilon, x+\epsilon)$ we add $B_\epsilon(\infty) = (-\infty, -1/\epsilon) \cup \{\infty\} \cup (1/\epsilon, \infty)$. Prove that $\mathbb{R} \cup \{\infty\}$ is homeomorphic to a circle. $\mathbb{R} \cup \{\infty\}$ is mapped on a circle by the inverse stereographic projection: draw a circle tangent at the south pole to $0 \in \mathbb{R}$ and map any point $x \in \mathbb{R}$ to the intersection of the line (x, N) (where N is the north pole of the circle) with the circle then identify $\{\infty\}$ with the north pole N on the circle. Prove that the one point compactification of $\mathbb{C}$ is homeomorphic to a 2 dimensional sphere, the Riemann sphere.

Chapter 4

Sequences

In this chapter we discuss sequences. One can consider sequences in generic topological spaces, metric spaces in $\mathbb{R}^D$ endowed with the standard topology. Sequences in non metrizable topological spaces are more complicated and will not be discussed here. However, it is important to discuss as much as possible general metric spaces: one does in fact encounter in physics such spaces, for instance in quantum mechanics where a central role is played by spaces of functions. Such spaces are metric spaces in their own right: one does talk about distances between functions, sequences of functions, and so on.

4.1 Sequences in metric spaces

From now on X is always understood to designate a metric space (X, d) endowed with the standard topology. Unless otherwise specified, when discussing the space $\mathbb{R}^D$, we always consider it endowed with some distance function $d(x, y) = \|x - y\|$.

A sequence in a metric space X is a list of elements $x_0, x_1, x_2, \dots, x_n, \dots$ in X in which repetitions are allowed and the order matters. Formally sequences are defined as follows.

Definition 4.1 (**Sequence (Folgen)**). A sequence in a metric space (X, d) is a function with domain $\mathbb{N}$ and co domain X , $x : \mathbb{N} \rightarrow X$. We denote $x(n) = x_n$ and we identify a sequence by its image $(x_0, x_1, x_2, \dots) \equiv (x_n)_{n \in \mathbb{N}}$. In order to simplify the notation we sometimes denote a sequence simply (x_n) .

A subsequence of a sequence is obtained by deleting some elements without changing the order of the remaining ones, that is a subsequence of $(x_n)_{n \in \mathbb{N}}$ is the sequence $(x_{n_k})_{k \in \mathbb{N}}$ with n_k a strictly increasing sequence of natural numbers $n_k > n_{k'}$ if $k > k'$. In particular, for any subsequence, $n_k \geq k$.

A sequence converges if x_n approaches some fixed value when $n \rightarrow \infty$.

Definition 4.2 (**Convergence (Konvergenz)**). A sequence $(x_n)_{n \in \mathbb{N}}$ in (X, d) is said to converge to x if for any $\epsilon > 0$ there exists $N_\epsilon \in \mathbb{N}$ such that $n > N_\epsilon \Rightarrow d(x, x_n) < \epsilon$. Equivalently, for any ball $B_\epsilon(x)$, there exists N_ϵ such that $n > N_\epsilon \Rightarrow x_n \in B_\epsilon(x)$. We denote this by $\lim_{n \rightarrow \infty} x_n = x$, $\lim x_n = x$ or simply $x_n \rightarrow x$. The point x is called the limit of (x_n) .

This definition can be extended with zero effort to general topological spaces.

Lemma 4.1. A sequence in (X, d) converges if and only if all of its subsequences converge. A convergent sequence and its subsequences converge to the same limit.

Proof. As any sequence is one of its own subsequences, one implication is trivial. For the other, assume (x_n) converges to x , that is for any ϵ , $\exists N_\epsilon$ such that $n > N_\epsilon \Rightarrow d(x_n, x) < \epsilon$. As $n_k \geq k$ for any subsequence, $d(x_{n_k}, x) < \epsilon$ also, and the lemma follows. $\square$

Limits of sequences and limit points of subspaces of metric spaces are intimately related.

Theorem 4.1 (**Limit points in metric spaces are limits of sequences**). *A point x is an accumulation (limit) point of a subset Y of a metric space (X, d) if and only if there exists a sequence (y_n) with $y_n \in Y$, $y_n \neq x$ such that $\lim y_n = x$.*

In particular, if $(y_n)_n \in Y$ is a convergent sequence, $\lim y_n = x$, then $x \in \bar{Y}$.

Proof. If $\lim y_n = x$ then every ball $B_\epsilon(x)$ will contain all the elements $(y_n)_{n > N_\epsilon}$ which belong to Y and are different from x . Conversely, take the sequence of balls of radius $1/n$ centered at x . As x is an accumulation point $B_{\frac{1}{n}}(x) \cap Y \setminus \{x\} \neq \emptyset$. From the axiom of choice (!) we can choose an element y_n per set $B_{\frac{1}{n}}(x) \cap Y \setminus \{x\} \neq \emptyset$ which yields a sequence such that $\forall N > 0, n > N \Rightarrow d(y_n, x) < 1/N$ and we conclude from the Archimedean property of $\mathbb{R}$. $\square$

In metric spaces, limit of sequences go through continuous functions. For X and Y topological spaces, a function $f : X \rightarrow Y$ is called *sequentially continuous* if for all $(x_n) \subset X$ such that $x_n \rightarrow x \in X$, the sequence $(f(x_n))_n \subset Y$ is a sequence which converges to $f(x)$, $f(x_n) \rightarrow f(x)$. It can be shown that if X and Y are arbitrary topological spaces, a continuous function is sequentially continuous. The reciprocal does not hold for general topological spaces but holds if X is first countable.

Theorem 4.2 (**In metric spaces continuous is sequentially continuous**). *For X and Y metric spaces, a function $f : X \rightarrow Y$ is sequentially continuous if and only if it is continuous.*

Proof. Let f a continuous function. For any ϵ , let $B_\epsilon(f(x))$ be the open ball of radius ϵ around $f(x)$. Then $x \in f^{-1}(B_\epsilon(f(x)))$ (note that $f^{-1}(B_\epsilon(f(x)))$ includes other points in X if f is not injective) thus $f^{-1}(B_\epsilon(f(x)))$ is an open set in X (due to continuity) that includes x . It follows that there exists some δ such that $B_\delta(x) \subset f^{-1}(B_\epsilon(f(x)))$. As $x_n \rightarrow x$, there exists some N such that $x_n \in B_\delta(x)$ for $n > N$ and then $n > N \Rightarrow f(x_n) \in B_\epsilon(f(x))$.

Suppose now that f is sequentially continuous, but not continuous at some point $x \in X$, which means there exists an open neighborhood V of $f(x)$ such that $f^{-1}(V)$ is not an open neighborhood of x . As any open neighborhood of x contains some ball $B_\delta(x)$, it follows that for any δ , $B_\delta(x) \setminus f^{-1}(V) \neq \emptyset$. Choosing a sequence (x_n) with $x_n \in B_{\frac{1}{n}}(x) \setminus f^{-1}(V)$ we have that $x_n \rightarrow x$ but $f(x_n) \notin V$. But then $f(x_n) \not\rightarrow f(x)$, because V is open and contains a ball around $f(x)$. Hence f is not sequentially continuous, leading to a contradiction. $\square$

Theorem 4.3 (**Any distance function is continuous**). *Any distance function is continuous in either of its arguments.*

Proof. It is enough to prove sequential continuity. Fix a point z and consider $d(x, z)$ as a function of its first argument. For any x and x_n we have $d(x_n, z) \leq d(x_n, x) + d(x, z)$ and $-d(x_n, x) \leq d(x_n, z) - d(x, z)$ hence $|d(x_n, z) - d(x, z)| \leq d(x_n, x)$. We conclude that for any sequence $x_n \rightarrow x \Rightarrow d(x_n, z) \rightarrow d(x, z)$. $\square$

4.1.1 Cauchy sequences (Cauchy Folgen)

A Cauchy sequence is a sequence such that its terms become increasingly closer for $n \rightarrow \infty$.

Definition 4.3 (**Cauchy sequence**). *A sequence $(x_n)_{n \in \mathbb{N}}$ in (X, d) is called a Cauchy sequence if for any $\epsilon > 0$ there exists $N_\epsilon \in \mathbb{N}$ such that $m, n > N_\epsilon \Rightarrow d(x_m, x_n) < \epsilon$.*

Every convergent sequence is Cauchy. The converse (reciprocal) of this statement is *not true*. For instance the real sequence defined recursively by $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$, $x_0 = 1$ is a Cauchy sequence in $\mathbb{Q}$ but does not converge in $\mathbb{Q}$.

Lemma 4.2. *A Cauchy sequence in (X, d) converges if and only if it has a convergent subsequence.*

Proof. One implication is trivial. Assume now we have a Cauchy sequence $(x_n)_{n \in \mathbb{N}}$ with a convergent subsequence $(x_{n_k})_{k \in \mathbb{N}}$. For any ϵ choose N_ϵ such that for $k, l > N_\epsilon$ both $d(x_k, x_l) < \epsilon/2$ and $d(x_{n_k}, x) < \epsilon/2$. as $n_k \geq k$, it follows that $d(x_k, x) \leq d(x_k, x_{n_k}) + d(x_{n_k}, x) < \epsilon$. $\square$

Exercise 4.1. *Prove that any convergent sequence is Cauchy.* Use $d(x_m, x_n) < d(x_m, x) + d(x, x_n)$. *Prove that $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$, $x_0 = 1$ is a Cauchy sequence in $\mathbb{Q}$.* The fact that the sequence stays in $\mathbb{Q}$ is trivial. Then denote $x_n = \sqrt{2} - \epsilon_n$, the recursion becomes $\sqrt{2} - \epsilon_{n+1} = \frac{2+2+\epsilon_n^2-2\sqrt{2}\epsilon_n}{2\sqrt{2}-2\epsilon_n} \Rightarrow \epsilon_{n+1} = \frac{4-2\sqrt{2}\epsilon_n-4-\epsilon_n^2+2\sqrt{2}\epsilon_n}{2\sqrt{2}-2\epsilon_n} = -\frac{\epsilon_n^2}{2(\sqrt{2}-\epsilon_n)}$. By induction we show that $|\epsilon_{n+1}| < \epsilon_n^2$ and $\epsilon_0 < 1/2$. Thus $|x_{n+p} - x_n| \leq |\epsilon_{n+1}| + |\epsilon_{n+2}| \dots |\epsilon_{n+p+1}| \leq \epsilon_n^2 + \epsilon_n^4 + \dots + \epsilon_n^{2^n} < \frac{\epsilon_n^2}{1-\epsilon_n^2} < \frac{1}{1-1/4} (\frac{1}{2})^n = 2^{2-n}$ which becomes arbitrary small with n .

Theorem 4.4 (**In metric spaces compact is sequentially compact**). *A subset $K \subset X$ of a topological space X is called sequentially compact if every sequence in K admits a subsequence that converges to some point in K . If (X, d) is a metric space then $K \subset X$ is compact if and only if it is sequentially compact.*

Proof. We have:

Sequentially compact $\Rightarrow$ compact. Assume K is sequentially compact and take $\{U_\alpha\}_\alpha$ an open cover of K .

First, there exists $\delta > 0$ (independent of x) such that $\forall x \in K$, $B_\delta(x) \subset U_\alpha$ for some α . Assume no such δ exists. Then for any n there exists $x_n \in K$ such that $\forall \alpha$, $B_{1/n}(x_n) \not\subset U_\alpha$. Chose a convergent subsequence of $(x_n)_n$, say $x_{n_i} \rightarrow x \in K$. Now, $x \in U_{\alpha_1}$ for some α_1 and there exists $\epsilon > 0$ such that $B_\epsilon(x) \subset U_{\alpha_1}$. Pick n_i such that $d(x, x_{n_i}) < \frac{\epsilon}{2}$ and $\frac{1}{n_i} < \frac{\epsilon}{2}$, thus $B_{1/n_i}(x_{n_i}) \subset U_{\alpha_1}$ which is a contradiction.

Second, there are finitely many balls of radius δ centered at points in K whose union covers K . Suppose this is false. Pick $x_1 \in K$ and $B_\delta(x_1) \not\subset K$, pick $x_2 \in K \setminus B_\delta(x_1)$ and $B_\delta(x_1) \cup B_\delta(x_2) \not\subset K$, iterate. We build a sequence x_n such that $j \neq j' \Rightarrow d(x_j, x_{j'}) > \delta$, hence x_n does not contain any Cauchy subsequence hence can not have any convergent subsequence.

As each $B_\delta(x)$ is a subset of some U_α , the finite cover of K by balls of radius δ selects a finite cover of K by sets U_α .

Compact $\Rightarrow$ sequentially compact. Assume K is compact and take a sequence $(x_n)_n$ in K . If x_n has a constant subsequence we conclude. If not, we can always pick a subsequence consisting in pairwise distinct elements which we label $(x_p)_{p \in \mathbb{N}}$.

Assume that $\forall y \in K$, $\exists \delta_y > 0$ such that $B_{\delta_y}(y)$ contains only finitely many terms in the sequence $(x_p)_p$. As K is compact, the open cover $\bigcup_y B_{\delta_y}(y) \supset K$ has a finite subcover $\bigcup_{i=1}^l B_{\delta_{y_i}}(y_i) \supset K$. But this finite union can only contains finitely many of the x_p 's, a contradiction.

Then $\exists y \in K$ such that $\forall \delta > 0$, $B_\delta(y)$ contains infinitely many terms in the sequence (x_p) . Then the subsequence obtained by picking one element $x_{p_k} \in \{x_p, p \in \mathbb{N}\} \cap B_{1/k}(y)$ for each $k \in \mathbb{N}$ is a subsequence of x_p which converges to $y \in K$.

□

Definition 4.4 (**Cauchy complete spaces**). *A metric space such that every Cauchy sequence converges is called complete or Cauchy complete.*

Not all metric spaces are Cauchy complete: for instance $\mathbb{Q}$ is not, as can be seen by considering the Cauchy sequence consisting in the finite truncations of the decimal representation of $\sqrt{2}$ which does not converge in $\mathbb{Q}$. Starting from a Cauchy incomplete space X one can build a Cauchy complete one (called the completion of X) by adding to X the limits of all the Cauchy sequences¹ of elements in X . This procedure provides an alternative construction of the real numbers starting from the rational ones.

¹Strictly speaking one must take equivalence classes of Cauchy sequences with the same limit.

4.1.2 Banach fixed-point theorem (Fixpunktsatz von Banach)

A function (map) $T : X \rightarrow X$ is called a contracting map if there exists $q \in [0, 1)$ such that $d(T(x), T(y)) \leq q d(x, y)$ for all $x, y \in X$. The constant q is called a *Lipschitz constant* for T .

Theorem 4.5 (**Banach fixed-point theorem**). *A contracting map T on a (non empty) Cauchy complete metric space X admits a fixed point x^* , that is $T(x^*) = x^*$. Furthermore, for any x_0 the recursively defined sequence $x_{n+1} = T(x_n)$ for $n \geq 0$ is such that $d(x^*, x_n) < \frac{q^n}{1-q} d(x_1, x_0)$ hence converges and $\lim x_n = x^*$.*

The fixed point exist under the weaker condition $d(T(x), T(y)) \leq d(x, y)$ if X is compact.

Proof. Starting from x_0 , we build $x_{n+1} = T(x_n)$ for $n \geq 0$. Then $d(x_{n+1}, x_n) \leq q^n d(x_1, x_0)$ hence for $p \geq 1$:

$$d(x_{n+p}, x_n) \leq \sum_{i=0}^{p-1} d(x_{n+1+i}, x_{n+i}) \leq \sum_{i=0}^{p-1} q^{n+i} d(x_1, x_0) = \frac{1-q^p}{1-q} q^n d(x_1, x_0) < \frac{1}{1-q} q^n d(x_1, x_0),$$

hence, as $q^n \rightarrow 0$, x_n is Cauchy and, as X is complete, it converges $\lim x_n = x^* \in X$. Taking $p \rightarrow \infty$ above we find $d(x^*, x_n) \leq q^n / (1-q) d(x_1, x_0)$. We have that $\forall \epsilon$, there exists N_ϵ such that $d(x^*, x_k) < \epsilon/2$ for all $k > N_\epsilon$. Picking any $k > N_\epsilon$ we have:

$$d(x^*, T(x^*)) \leq d(x^*, x_{k+1}) + d(T(x_k), T(x^*)) < \frac{\epsilon}{2} + q \frac{\epsilon}{2} < \epsilon,$$

hence $x^* = T(x^*)$. Note that the fixed point is unique as T is contracting. $\square$

Note that in this proof we used the fact that $q^n \rightarrow 0$ for $0 \leq q < 1$. We will see below that this is a direct consequence of the monotone convergence theorem.

Exercise 4.2. *We will prove later that $\mathbb{R}$ is Cauchy complete. Consider the function $T(x) = \frac{x}{2} + \frac{1}{x}$. Prove that $T([1, \infty)) \subset [1, \infty)$. $\frac{x}{2} + \frac{1}{x} \geq 1 \Leftrightarrow x^2 + 1 \geq 2x \Leftrightarrow (x-1)^2 \geq 0$. Prove T is a contraction on $[1, \infty)$. $T(x) - T(y) = (x-y)(\frac{1}{2} - \frac{1}{xy})$ and for all $x, y > 1$, we have $0 < (xy)^{-1} < 1 \Rightarrow -2^{-1} < (xy)^{-1} - 2^{-1} < 2^{-1} \Rightarrow |2^{-1} - (xy)^{-1}| < 2^{-1}$. Find the fixed point of this contraction. $x^* = \sqrt{2}$.*

Propose an algorithm to compute $\sqrt{2}$ as the limit of a rational sequence. $x_{n+1} = T(x_n)$, $x_0 = 1$.

4.2 Sequences in $\mathbb{R}$ and $\mathbb{R}^D$

A particularly rich case is the one of real sequence $(x_i)_{i \in \mathbb{N}}$, $x_i \in \mathbb{R}$.

4.2.1 Homogeneous linear recurrence (Homogener lineare Differenzengleichung)

Often one encounters recursively defined sequences in which a term is fixed by a finite number of its predecessors. $x_{n+q} = F(x_n, x_{n+1}, \dots, x_{n+q-1})$. If F is continuous, then the limit of a recursively defined sequence, if it exists, is a solution of the equation $x^* = F(x^*, \dots, x^*)$.

Let us consider real or complex valued sequences. If a recursively defined sequence is not convergent, one is interested in finding a closed formula for the general term x_n (this is the equivalent in the world of sequences of solving differential equations). A particular case or recurrence that can always be solved is that of a homogeneous linear recursion with constant coefficients:²

$$x_{n+q} = \sum_{p=0}^{q-1} c_p x_{n+p}, \quad x_0, x_1, x_2, \dots, x_{q-1},$$

²One can consider more general recursions with non constant, n dependent, coefficients $c_p(n)$ or inhomogeneous obtained by adding a known sequence b_n to the right hand side of the recurrence relation. If the homogeneous recurrence is solved, the inhomogeneous case can also be solved by the so called "variation of constants" method. We will discuss this later in the context of differential equations.

where c_p are fixed. In order to start the recursive computation of all the x_n s, the first q elements in the sequence $x_0, x_1, x_2, \dots, x_{q-1}$ need to be given. Linear recursions can always be translated in *finite difference equations* using the finite difference operator:

$$(\Delta x)_n = x_{n+1} - x_n, \quad (\Delta^{k+1} x)_n = \left(\Delta(\Delta^k x) \right)_n, \quad x_{n+p} = \sum_{k=0}^p \binom{p}{k} (\Delta^k x)_n.$$

This provides a straightforward parallel between such recursions and differential equations which we will discuss later.

We note that if $(y_n^{(1)})$ and $(y_n^{(2)})$ are two sequences respecting a homogeneous recursion, so is any linear combination of them. In order to solve the recursion it is then enough to:

- identify q linearly independent solutions $y_n^{(i)}$ for $i = 1, \dots, q$,
- posit the general solution:

$$x_n = \sum_{i=1}^q d_i y_n^{(i)},$$

- fix the constants d_i by solving the boundary conditions $x_j = \sum_{i=1}^q d_i y_j^{(i)}$ for $j = 0, \dots, q-1$. Indeed, as long as $y_n^{(i)}$ are linearly independent, this linear system of equations will admit a unique solution.

The q linearly independent solutions $y_n^{(i)}$ are found by solving the *characteristic equation* of the recurrence. Posit that $x_n = t^n$ with some fixed t , then the homogeneous linear recursion becomes the characteristic equation:

$$P(t) = t^q - \sum_{p=0}^{q-1} c_p t^p = 0.$$

Being a polynomial equation of degree q , this equation has q not necessarily distinct complex roots. Clearly any root r of this polynomial is such that $x_n = r^n$ will respect the recurrence, hence if all the roots r_i are distinct the general solution of the recursion relation is:

$$x_n = \sum_{i=1}^q d_i r_i^n.$$

In particular, as advertised, the linear system of equations determining the coefficients d_i admits a unique solution as:

$$\det(y_j^{(i)}) = \begin{vmatrix} 1 & \dots & 1 \\ r_1 & \dots & r_q \\ \vdots & & \vdots \\ r_1^{p-1} & \dots & r_q^{q-1} \end{vmatrix} = \prod_{1 \leq i < j \leq q} (r_i - r_j) \neq 0.$$

If the root r has multiplicity ν , then the linearly independent solutions corresponding to it are $r^n, n r^n, n^2 r^n$ and so on up to $n^{\nu-1} r^n$.

Exercise 4.3. Solve for the Fibonacci numbers $x_0 = 0, x_1 = 1, x_{n+2} = x_{n+1} + x_n$. The characteristic equation is $t^2 = t + 1$ with roots $t = \frac{1 \pm \sqrt{1+4}}{2}$ hence $x_n = d_+ \left(\frac{1+\sqrt{5}}{2} \right)^n + d_- \left(\frac{1-\sqrt{5}}{2} \right)^n$ with $d_+ + d_- = 0$, $(d_+ - d_-)\sqrt{5} = 2$ hence $x_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$.

Exercise 4.4. Solve and compare the general term of the sequences $a_0 = 0$ and $a_1 = 1$ and recursive definitions:

$$a_{n+2} - 2a_{n+1} + a_n + \omega^2 a_n = 0, \quad a_{n+2} - 2a_{n+1} + a_n - \lambda^2 a_n = 0$$

Exercise 4.5. *First order non-homogeneous linear recursions can always be solved. Given two known sequences (f_n) with $f_n \neq 0$ and (g_n) prove that the general term in the recursive sequence $a_0, a_{n+1} = f_n a_n + g_n$ is:*

$$a_n = \left(\prod_{k=0}^{n-1} f_k \right) \left(a_0 + \sum_{m=0}^{n-1} \frac{g_m}{\prod_{k=0}^m f_k} \right).$$

We verify that $a_1 = f_0(a_0 + \frac{g_0}{f_0})$ and $f_n a_n + g_n = f_n \left(\prod_{k=0}^{n-1} f_k \right) \left(a_0 + \sum_{m=0}^{n-1} \frac{g_m}{\prod_{k=0}^m f_k} \right) + g_n = \left(\prod_{k=0}^n f_k \right) \left(a_0 + \sum_{m=0}^{n-1} \frac{g_m}{\prod_{k=0}^m f_k} + \frac{g_n}{\prod_{k=0}^n f_k} \right).$

4.2.2 Limits of real sequences, lim sup and lim inf

The distance on $\mathbb{R}$ is the absolute value $d(x, y) = |x - y|$. Convergent real sequence can be manipulated much like numbers: they can be added, subtracted, multiplied, divided ordered and so on.

Theorem 4.6 (Properties of limits). *If (a_n) and (b_n) are two convergent sequences in $\mathbb{R}$, $\lim a_n = a$ and $\lim b_n = b$, then:*

- $\lim |a_n| = |a|$.
- $\lim(a_n \pm b_n) = a \pm b$.
- $\forall c \in \mathbb{R}, \lim(ca_n) = ca$.
- $\lim(a_n b_n) = ab$.
- if $b \neq 0$ then $\lim \frac{a_n}{b_n} = \frac{a}{b}$.
- if $\forall n > N, a_n \leq b_n$ then $a \leq b$. This implies the so called sandwich theorem (Einschnürungssatz): if c_n is a sequence such that $a_n \leq c_n \leq b_n$ and $\lim a_n = \lim b_n = L$ then $\lim c_n = L$.

Proof. The proof is straightforward. For instance, as $|a| \leq |a - a_n| + |a_n|$ and $|a_n| \leq |a_n - a| + |a|$ we conclude that $||a| - |a_n|| \leq |a - a_n|$ and $\lim |a_n| = |a|$; also:

$$\left| \frac{a_n}{b_n} - \frac{a}{b} \right| = \left| \frac{a_n}{b_n} - \frac{a_n}{b} + \frac{a_n}{b} - \frac{a}{b} \right| \leq |a_n| \left| \frac{b - b_n}{|b_n| |b|} \right| + \frac{|a_n - a|}{|b|},$$

and for n large enough $|b| - \epsilon < |b_n|, |a_n| < |a| + \epsilon$.

For the last point, assuming that $b < a$ the ball $B_{\frac{|b-a|}{3}}(a)$ contains all the $a_n, n > N_1$ and the ball $B_{\frac{|b-a|}{3}}(b)$ contains all the $b_n, n > N_2$, hence for $N > \max(N_1, N_2), b_n \leq a_n$ contradicting the hypothesis. $\square$

Any real sequence $(x_n)_n \in \mathbb{R}$ admits an asymptotic upper (lower) bound:

$$\limsup x_n = \lim_{n \rightarrow \infty} \left(\sup_{m \geq n} x_m \right), \quad \liminf x_n = \lim_{n \rightarrow \infty} \left(\inf_{m \geq n} x_m \right),$$

where we allow $\limsup x_n = \pm\infty, \liminf x_n = \pm\infty$. We have that:

- we denote $\limsup x_n = \infty$ if for any $y \in \mathbb{R}$, there exist infinitely many terms in (x_n) such that $x_n > y$, that is $\forall y, \forall N, \exists n > N$ such that $x_n > y$.
- if $\limsup x_n = L < \infty$, then for any $\epsilon > 0$ only a finite number of terms in (x_n) are such that $x_n > L + \epsilon$, but infinitely many terms are such that $x_n > L - \epsilon$, that is $\forall \epsilon, \exists N, \forall n > N, L + \epsilon \geq x_n$ but $\forall \epsilon, \forall N, \exists n > N$ such that $x_n > L - \epsilon$.

Clearly, the sequence (x_n) converges to x if and only if $\limsup x_n = \liminf x_n = x \neq \pm\infty$.³

³One sometimes encounters the notation $\lim x_n = \infty$ if $\forall y, \exists N_y$ such that $n > N_y \Rightarrow x_n > y$. This is however a dangerous notations because such a sequence x_n does not converge.

4.2.3 Monotone convergence theorem (Satz von der monotonen Konvergenz)

An interesting case of sequences in $\mathbb{R}$ are the monotonic, bounded ones.

Definition 4.5 (**Monotonicity**). A sequence in $\mathbb{R}$ is called:

- monotonically increasing if $a_n \geq a_p$ for $n > p$.
- monotonically decreasing if $a_n \leq a_p$ for $n > p$.
- bounded if $|a_n| \leq M$ for some fixed M .

Theorem 4.7 (**Monotone convergence theorem**). A sequence (x_n) , $x_n \in \mathbb{R}$ that is monotonic and bounded is convergent. If x_n is monotonically increasing (resp. decreasing) then it converges to its supremum (resp. infimum).

Proof. Let (x_n) monotonically increasing and bounded. $\{x_n\}$ is a bounded subset of $\mathbb{R}$ hence has a supremum (least upper bound) which we denote x . Now for all ϵ , there exists N such that $x_N > x - \epsilon$ (otherwise $x - \epsilon$ would be an upper bound on (x_n)) and $\forall n > N$, $x - \epsilon < x_N \leq x_n \leq x$. $\square$

Exercise 4.6 (Limits of classical sequences.). Prove Bernoulli's inequality $(1+x)^n \geq 1+nx$ for x real, $x > -1$ and $n \in \mathbb{N}$.

By induction, $(1+x)^{n+1} = (1+x)^n(1+x) \geq (1+nx)(1+x) = 1+(n+1)x+nx^2 \geq 1+(n+1)x$. Prove that:

- $\forall s \in \mathbb{Q}_{>0} \lim_{n \rightarrow \infty} \frac{1}{n^s} = 0$. Chose $N_\epsilon > 1/(\epsilon^{1/s})$.
- $\forall a > 0 \lim_{n \rightarrow \infty} a^{1/n} = 1$. It is enough to prove this for $a > 1$. We have $a = (1+a^{1/n}-1)^n \geq 1+n(a^{1/n}-1) \Rightarrow 0 < |a^{1/n}-1| \leq \frac{a-1}{n}$ hence by the sandwich criterion $\lim a^{1/n} = 1$.
- $\lim_{n \rightarrow \infty} n^{1/n} = 1$. Note that if $n > 1$ then $n^{1/q} > 1$ for any q . Now $n^{1/2} = [1+(n^{1/(2n)}-1)]^n \geq 1+n(n^{1/(2n)}-1) \Rightarrow 0 < n^{1/(2n)}-1 < n^{-1/2}-n^{-1} \rightarrow 0$ and $\lim(n^{1/n}) = (\lim n^{1/(2n)})(\lim n^{1/(2n)}) = 1$.
- $0 < q < 1$, $\lim_{n \rightarrow \infty} q^n = 0$. As $0 < q^{n+1} < q^n$ this is a monotonic bounded sequence hence converges at its infimum. Assume this infimum is $c > 0$. Then $c < q^n$ for all n , but $q^p \leq \frac{1}{q}c$ for some p , as $\frac{1}{q}c$ is not the infimum, which is a contradiction.
- for $n \in \mathbb{N}$, $x > 1$, $\lim_{n \rightarrow \infty} \frac{n^k}{x^n} = 0$. We have $x^n \geq 1+n(x-1) \geq n^{1/2}n^{1/2}(x-1) \Rightarrow \frac{n^{1/2}}{x^n} < \frac{1}{n^{1/2}(x-1)} \rightarrow 0$. Then $\frac{n^k}{x^n} = \left(\frac{n^{1/2}}{(x^{1/2})^n}\right)^{2k} \rightarrow 0$.
- denoting $n! = 1 \cdot 2 \cdot \dots \cdot n$, prove that $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$ for any real x . It is enough to check for $x > 0$. We denote $a_n = \frac{x^n}{n!} > 0$ and note that $a_{n+1} = \frac{x}{n+1}a_n$ hence a_n is monotonic decreasing for $n > x$ hence converges to its infimum. Assuming this infimum is $c > 0$, then $c < \frac{x^n}{n!}$ for all n , but there exists some $n > x$ such that $\frac{x^n}{n!} < c \frac{n+1}{x}$, as $c \frac{n+1}{x}$ is not an upper bound on a_n for $n > x$.
- prove that $x_n = (1+n^{-1})^n$ converges. We denote $\lim(1+n^{-1})^n = e$ and we call it Euler's number. $(1+n^{-1})^n = \sum_{p=0}^n \binom{n}{p} \frac{1}{n^p} = 1 + \sum_{p=1}^n \frac{1}{p!} \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{p-1}{n}\right)$. Hence $x_{n+1} - x_n = \sum_{p=1}^n \frac{1}{p!} \left[\left(1 - \frac{1}{n+1}\right)\left(1 - \frac{2}{n+1}\right) \dots \left(1 - \frac{p-1}{n+1}\right) - \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{p-1}{n}\right)\right] + \frac{1}{(n+1)!} \left(1 - \frac{1}{n+1}\right)\left(1 - \frac{2}{n+1}\right) \dots \left(1 - \frac{n}{n+1}\right)$, and as $1 - \frac{1}{n+1} > 1 - \frac{1}{n} > 0$ it follows that $x_{n+1} \geq x_n$ hence x_n is monotonically increasing. Furthermore, as $x_n \leq \sum_{p \geq 0} \frac{1}{p!}$ and $p! > \frac{1}{8} 2^p$ for $p = 0, 1, \dots$ then $x_n \leq 8 \sum_{p \geq 0} \frac{1}{2^p} = 8 \frac{1 - \lim_{N \rightarrow \infty} (1/2)^N}{1 - 1/2} < 16$ hence it is bounded from above.

4.2.4 The Bolzano-Weierstrass theorem (Satz von Bolzano-Weierstrass)

We have seen that sequences that are monotonic and bounded converge. In fact, boundedness suffices to ensure convergence of a subsequence.

Theorem 4.8 (**Nested intervals theorem (Intervallschachtelung)**). Let $[a_n, b_n] \in \mathbb{R}$ be a family of nested closed intervals $[a_{n+1}, b_{n+1}] \subset [a_n, b_n]$ indexed by $n \in \mathbb{N}$. If $\lim (a_n - b_n) = 0$ then there exists exactly one real number $\{x\} = \bigcap_{n \in \mathbb{N}} [a_n, b_n]$.

Proof. Due to nesting, both (a_n) and (b_n) are monotonic and bounded, hence convergent. If they converge to different limits then $\lim(a_n - b_n) \neq 0$ contradicting the hypothesis. It follows that $\lim a_n = \lim b_n = x$ and, due to monotonicity, $a_n \leq x \leq b_n$ for all n hence $\{x\} \in \bigcap_{n \in \mathbb{N}} [a_n, b_n]$. Finally, assuming there exist some $x' \neq x$ with $\{x'\} \in \bigcap_{n \in \mathbb{N}} [a_n, b_n]$, x' can not belong to the intervals with diameter $b_n - a_n < |x - x'|$, which leads to a contradiction. $\square$

Theorem 4.9 (Bolzano-Weierstrass theorem). *A sequence in $\mathbb{R}^D$ is called bounded if $\|x_n\| = d(x_n, 0) \leq M$. Any bounded sequence in $\mathbb{R}^D$ has a convergent subsequence.*

Proof. It is enough to prove this for $D = 1$, as the general D case can be obtained by going component by component.⁴ A sequence in $\mathbb{R}^D$ is bounded if and only if the sequence of any of its components $(x_n^\mu)_{n \in \mathbb{N}}$ are bounded. We first select a convergent subsequence for the first component, then we select a subsequence of the subsequence such that the second component converges also and so on.

We denote a_0 and b_0 an upper and a lower bound for (x_n) and we declare the interval $I_0 = [a_0, b_0]$. We chose the midpoint $\frac{a_0+b_0}{2}$ and note that at least one of the intervals $[a_0, \frac{a_0+b_0}{2}]$ and $[\frac{a_0+b_0}{2}, b_0]$ has an infinity of elements of (x_n) . We declare this interval as $I_1 = [a_1, b_1]$ and we iterate. As the diameter of the interval halves at every step we have built a family of nested intervals. We denote $\{x\} = \bigcap_{k \in \mathbb{N}} [a_k, b_k]$.

We pick a subsequence (x_{n_k}) of (x_n) as follows. We first pick x_{n_0} one element in the sequence (x_n) . At each step, among the successors of $x_{n_{k-1}}$, that is x_n with $n > n_{k-1}$, an infinity belongs to I_k (as an infinity of elements in the sequence belongs to I_k and $x_{n_{k-1}}$ has only finitely many predecessors). We pick one of them and declare it x_{n_k} . Then (x_{n_k}) is a subsequence of (x_n) and $d(x_{n_k}, x) < \frac{|a_0-b_0|}{2^k}$. We have $\lim_k x_{n_k} = x$. Indeed, for any ϵ , there exists k_ϵ such that $\frac{|a_0-b_0|}{2^{k_\epsilon}} < \epsilon$ hence for any $k > k_\epsilon$, as $I_k \subset I_{k_\epsilon}$, we have that $d(x_{n_k}, x) < \epsilon$.⁵ $\square$

Whenever the splitting in half of an interval I_n leads to two intervals containing an infinity of elements of (x_n) , each of these subsets of (x_n) has its own convergent subsequence: a bounded sequence can have subsequences which converge to different limits.

4.2.5 $\mathbb{R}^D$ is Cauchy complete

An example of a Cauchy complete metric space is provided by the real numbers.

Theorem 4.10 ($\mathbb{R}^D$ is Cauchy complete). *The real numbers $\mathbb{R}$ endowed with the standard topology are a Cauchy complete metric space. The same holds for $\mathbb{R}^D$.*

Proof. Any Cauchy sequence is bounded. Indeed, pick some ϵ , and for $n > N_\epsilon$, $|x_{N_\epsilon} - x_n| < \epsilon \Rightarrow |x_n| < |x_{N_\epsilon}| + \epsilon$. Then, by the Bolzano-Weierstrass theorem any Cauchy sequence in $\mathbb{R}$ has a convergent subsequence we conclude by Lemma 4.2.

This generalizes to $\mathbb{R}^D$ by noting that the sequence corresponding to the first component x_n^1 of any Cauchy sequence $x_n = (x_n^1, \dots, x_n^D) \in \mathbb{R}^D$ is Cauchy. $\square$

Corollary 4.1. *$\mathbb{R}$ is the Cauchy completion of $\mathbb{Q}$.*

Proof. As $\mathbb{Q} \subset \mathbb{R}$ and $\mathbb{R}$ is Cauchy complete, any Cauchy sequence of rational numbers converges to some number in $\mathbb{R}$.

In order to prove that $\mathbb{R}$ is the Cauchy completion of $\mathbb{Q}$ we need to prove that $\forall x \in \mathbb{R}$, there exists a Cauchy sequence of rational numbers converging to x . From Theorem 2.6, for any real number x and for any n there exists a rational number q_n such that $x - n^{-1} < q_n < x$, hence the sequence $(q_n)_{n \in \mathbb{N}}$ is a sequence of rational numbers converging to x . As any sequence in $\mathbb{R}$ is convergent if and only if it is Cauchy, $(q_n)_{n \in \mathbb{N}}$ is Cauchy. $\square$

⁴We note that for all the distances we have defined $\|x - y\| < \sum_{i=1}^D |x^i - y^i|$.

⁵Alternatively, if (x_n) has a constant subsequence $(x_{n_k})_{k \in \mathbb{N}}$, we conclude. If not, then x is an accumulation point of the set $\{x_n | n \in \mathbb{N}\}$ and we conclude using Lemma 4.1. Indeed, for any ϵ consider the bubble $B_\epsilon(x) = (x - \epsilon, x + \epsilon)$. As $\lim(a_n - b_n) \rightarrow 0$ there exist N_ϵ such that $n \geq N_\epsilon$ $|a_n - b_n| < \epsilon$ hence, as $a_n \leq x \leq b_n$, $I_n \subset B_\epsilon(x)$. In particular $I_{N_\epsilon} \subset B_\epsilon(x)$ and I_{N_ϵ} contains an infinity of elements of (x_n) , at least one of which is different from x as (x_n) does not have a constant subsequence.

4.2.6 Heine–Borel theorem (Satz von Heine–Borel)

An important result is the characterisation of compact sets in the standard topology on $\mathbb{R}^D$. A set in $(\mathbb{R}^D, d)$ is called *bounded* if it can be entirely covered by one (big enough) ball centered at the origin.

Theorem 4.11 (**The Heine–Borel theorem**). *In the standard topology $(\mathbb{R}^D, d)$, a set is compact if and only if it is closed and bounded.*

Proof. The theorem follows from the following assertions:

If K is compact then K is closed. This holds in any Hausdorff topology. The standard topology on $\mathbb{R}^D$ Hausdorff by Theorem 3.3.

Take x a limit point of K . Any finite collection of open sets $U_i, i \in I$ such that $\forall i, \exists V_i(x)$ some open neighborhood of x such that $U_i \cap V_i(x) = \emptyset$ can not cover K . Indeed, $\bigcap_{i \in I} V_i(x)$ is an open set, thus a neighborhood of x hence it contains a point in $x' \in K$ and $x' \notin \bigcup_{i \in I} U_i$.

If K is compact but not close, then $\exists x$ limit point of K such that $x \notin K$. For every $y \in K$, pick an open set $U(y)$ containing y such that it is disjoint from some neighborhood $V_y(x)$ of x . This exists as the topology is Hausdorff. The family $U(y), y \in K$ is an open cover of K , $\bigcup_{y \in K} U(y) \supset K$, but any finite subfamily will be of the type discussed previously, hence can not cover K .

Note that if $x \in K$, such $V_y(x)$ exist only for $y \in K, y \neq x$ and $\bigcup_{y \in K, y \neq x} U(y)$ does not cover K .

If K is compact then K is bounded. This hold for any metric space in the standard topology.

Consider a compact K . The family $\{U_1(x) | x \in K\}$ of bubbles of radius 1 is an open cover of K hence admits an open subcover $\{U_1(x_i) | x_i \in I\}$ for some finite set I . The set of pairwise distances between the centers of the bubbles $\{d(x_i, x_j) | x_i, x_j \in I\}$ is a finite set hence its maximum $M = \max_{x_i, x_j \in I} d(x_i, x_j)$ is a finite number. As any point belongs to one of the bubbles in this family, by the triangle inequality, any two points are at a distance of at most $d(x, y) \leq 1 + M + 1$, hence K has finite diameter.

A closed subset of a compact set is compact. This holds for any topological space.

Let $S \subset K$ a closed subset of the compact set K , and let $\{U_\alpha | \alpha \in A\}$ an open cover of S . Then $U = \mathbb{R}^D \setminus S$ is open, $\{U\} \cup \{U_\alpha | \alpha \in A\}$ is an open cover of K , hence contains a finite cover $\{U_i | i \in I\}$ (where one of the U_i can be U) of K which also covers S . As U does not contain any element in S , $\{U_i | i \in I\} \setminus \{U\}$ is an open cover of S .

If a set is closed and bounded, then it is compact. This holds for finite dimensional Cauchy complete metric space (we will define later the dimension of a space).

By the property above, it is enough to prove that the set $T_0 = [-a, a]^D$ with $a > 0$ is compact. Assume T_0 is not compact. Then there exists an infinite open cover $\bigcup_{\alpha \in A} U_\alpha \supset K$ which admits no finite subcover for K . Similarly to the nested intervals construction, we subdivide $T_0 = \times_{i=1}^D ([-a, 0] \cup [0, a])$ into 2^n closed boxes of the type $[0, a]^p \times [-a, 0]^{n-p}$. At least one of these sub boxed requires an infinite subcover of $(U_\alpha)_{\alpha \in A}$. Call this box T_1 and iterate. We obtain a sequence of nested closed boxes $T_0 \supset T_1 \supset \dots$, where the diameter of T_k is $(2a)2^{-k}$, each of which requires an infinite subcover of $(U_\alpha)_{\alpha \in A}$. We pick a sequence $x_k \in T_k$ and x_k is Cauchy as $\forall p \geq k, x_p \in T_k$.

As $\mathbb{R}^D$ is Cauchy complete, x_k converges and $x = \lim x_k \in T_k$ for all k as T_k are closed. There exists U , one of the U_α such that $x \in U$, and since U is open there exists some ϵ such that $B_\epsilon(x) \subset U$, hence for k larger than some N_ϵ , $T_k \subset B_\epsilon(x) \subset U$, thus only one set in $(U_\alpha)_{\alpha \in A}$ suffices to cover all these T_k , leading to a contradiction.

Hence T_0 is compact.

□

Note that this property uses the completeness of $\mathbb{R}^D$, hence fails for incomplete metric spaces like $\mathbb{Q}$. It also fails in complete infinite dimensional spaces (more on this later) in which the subdivision of boxes into boxes fails.

Corollary 4.2. *The inverse of a continuous bijection between two compact Hausdorff spaces is continuous.*

Proof. Consider $f : X \rightarrow Y$ a continuous bijection with X compact, hence $Y = f(X)$ is compact. Take any $U \subset X$ open set, then $X \setminus U$ is compact (always), $f(X \setminus U)$ is compact (f continuous), $Y \setminus f(X \setminus U)$ is open (Y Hausdorff plus the first item in the proof of Heine-Borel), and using $f(U) \cap f(X \setminus U) = \emptyset$ (f bijection) we conclude that $f(U) = Y \setminus f(X \setminus U)$ is open. We thus established that f is an open map. Denote $g : Y \rightarrow X$ the inverse of f and take any open set $U \subset X$. Then $g^{-1}(U) = f(U)$ which is open. $\square$

4.3 Series

Series are infinite sums over sequences.

Definition 4.6 (**Series (Reihen)**). *Starting from any sequence (a_n) valued in $\mathbb{R}$ or $\mathbb{C}$ one can build a series, which is a formal infinite sum denoted:*

$$\sum_{k=0}^{\infty} a_k = a_0 + a_1 + a_2 + \dots$$

The elements a_n of the sequence (a_n) are called the general term of the series. A series is said to be:

- convergent if the sequence of partial sums $s_n = \sum_{k=0}^n a_k$ converges $s_n \rightarrow s$. In this case we say that $\sum_{k=0}^{\infty} a_k$ converges to s .
- absolutely convergent if the series of the absolute values $\sum_{k=0}^{\infty} |a_k|$ is convergent.
- divergent if it is not convergent.

Note that if a series is convergent then the general term $a_n = s_n - s_{n-1}$ must converge to 0, $a_n \rightarrow 0$ (otherwise the sequence of partial sums is not Cauchy).⁶ Not only this, but it must in fact converge fast enough. Some examples of series are:

$$\sum_{k \geq 1} \frac{1}{k^2}, \quad \sum_{k \geq 1} \frac{(-1)^k}{k}, \quad \sum_{k \geq 1} \frac{1}{k}.$$

Also, if a series is absolutely convergent then it is convergent: indeed, $|\sum_{k=n}^m a_k| \leq \sum_{k=n}^m |a_k|$ and one concludes that if the sequence of partial sums of $\sum_k |a_k|$ is Cauchy, then the sequence of partial sums of $\sum_k a_k$ is also Cauchy.

Exercise 4.7. *Prove that the three series above are respectively absolutely convergent, convergent and divergent. The first series is absolutely convergent as $\frac{1}{k^2} < \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$ hence $s_n < 1 + 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n-1} - \frac{1}{n} = 2 - \frac{1}{n}$ thus s_n is monotonic and bounded. The second series is convergent as $|s_m - s_n| \leq \frac{1}{n+1}$ for $m > n$ hence s_n is Cauchy, but it is not absolutely convergent. To see this, consider as $s_n = \sum_{k \geq 1} \frac{1}{k}$ and grouping the terms we have:*

$$s_{2^n} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \dots > 1 + 2^0 \times \frac{1}{2} + 2^1 \times \frac{1}{4} + 2^2 \times \frac{1}{8} + \dots 2^{n-1} \times \frac{1}{2^n} = 1 + \frac{n}{2},$$

hence s_{2^n} is a subsequence of s_n which is not convergent.

The main question one faces when encountering a series is to determine whether the series is absolutely convergent, convergent, or divergent. Two important series are encountered often.

- *The geometric series.* The geometric series $\sum_{k \geq 0} q^k$ is absolutely convergent and converges to $(1 - q)^{-1}$ for $0 < q < 1$; it is divergent for $q \geq 1$. This follows by noting that $s_n = \frac{1 - q^{n+1}}{1 - q}$ for $q \neq 1$ and $s_n = n + 1$ for $q = 1$.

⁶If s_n converges then it is Cauchy then $\forall \epsilon > 0, \exists N_\epsilon$ such that $p > N_\epsilon \Rightarrow |s_{p+1} - s_p| = |a_{p+1}| < \epsilon$ hence $a_p \rightarrow 0$.

- *The generalized harmonic series.* The series $\zeta(\alpha) = \sum_{k \geq 1} \frac{1}{k^\alpha}$ is absolutely convergent for $\alpha > 1$ and divergent for $\alpha \leq 1$. By the Cauchy condensation test (see below) the series has the same nature as:

$$\sum_{k \geq 0} 2^k \frac{1}{2^{k\alpha}} = \sum_{k \geq 0} \frac{1}{2^{k(\alpha-1)}}.$$

Remark that if a series with positive terms converges then it converges absolutely.

Theorem 4.12 (**Convergence tests (Konvergenzkriterium)**). *Given a generic series $\sum_{k \geq 0} a_k$:*

- **Cauchy condensation test.** *If $a_k \geq 0$ and a_k is monotonically decreasing then $\sum_{k \geq 1} a_k$ converges if and only if $\sum_{k \geq 1} 2^k a_{2^k}$ converges and moreover $\sum_{k \geq 1} a_k \leq \sum_{k \geq 0} 2^k a_{2^k} \leq 2 \sum_{k \geq 1} a_k$. This follows by noting that:*

$$\begin{aligned} \sum_{k \geq 1} a_k &= a_1 + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots \leq \sum_{k \geq 0} 2^k a_{2^k}, \\ \sum_{k \geq 0} 2^k a_{2^k} &= (a_1 + a_2) + (a_2 + a_4) + (a_4 + a_4) + (a_4 + \cdots) \leq 2 \sum_{k \geq 1} a_k. \end{aligned}$$

- **Direct comparison test.** *If $|a_k| \leq b_k$ (hence $b_k \geq 0$) and $\sum_{k \geq 0} b_k$ is convergent then $\sum_{k \geq 0} a_k$ is absolutely convergent and $|\sum_{k \geq 0} a_k| \leq \sum_{k \geq 0} b_k$. Indeed, as $\sum_{k \geq 0} b_k$ is convergent, the sequence $u_n = \sum_{k \geq 0}^n b_k$ is convergent hence Cauchy; but then $|s_m - s_n| \leq \sum_{k=n+1}^m |a_k| \leq \sum_{k=n+1}^m b_k \leq u_m - u_n$, hence s_m is Cauchy thus convergent and $|s_n| \leq u_n$.*
- **Alternating series.** *If a_k is monotonically decreasing and $a_k \rightarrow 0$ then $\sum_{k \geq 0} (-1)^k a_k$ is convergent. This follows by noting that for $m > n$, $|s_m - s_n| \leq |a_{n+1}|$.⁷*
- **Cauchy-Hadamard.** *Denote $\lim_{n \rightarrow \infty} (\sup_{m \geq n} |a_m|^{1/m}) = r$. If $r < 1$ then the series is absolutely convergent. If $r > 1$ it is divergent.*
- *if $\lim \left| \frac{a_{n+1}}{a_n} \right| = r$ then, if $r < 1$ the series is absolutely convergent whereas if $r > 1$ it is divergent.*
- **Integral criterion.** *This criterion can be revisited after the reader familiarizes themselves with integrals, discussed in Chapter 11. Assume a_k is monotone decreasing and $a_k \rightarrow 0$. If there exists monotonically decreasing function $f: [1, \infty) \rightarrow \mathbb{R}_+$ such that $f(n) = a_n$ then the series $\sum_{n \geq 1} a_n$ is convergent if and only if $\int_1^\infty f(t) dt < \infty$.*
- **Abel's test.** *If $\sum_{k \geq 1} a_k$ is convergent and (b_k) is a monotonic bounded sequence then $\sum_{k \geq 1} a_k b_k$ is convergent. The importance of this test is that it applies when $\sum_{k \geq 1} a_k$ is convergent, but not absolutely convergent. Call $A_n = \sum_{k=1}^n a_k$ and note⁸ that $\sum_{k=1}^n a_k b_k = A_n b_{n+1} - \sum_{k=1}^n A_k (b_{k+1} - b_k)$. Both (b_n) and (A_n) converge, then on the one hand $A_n b_{n+1}$ converges and on the other $\sum_{k=n}^m A_k (b_{k+1} - b_k) \leq (\sup_k |A_k|) \sum_{k=n}^m |b_{k+1} - b_k|$. As (b_n) is monotonic $\sum_{k=n}^m |b_{k+1} - b_k| = |b_m - b_n|$ and, as b_m is convergent hence it is Cauchy, $\sum_{k=n}^m A_k (b_{k+1} - b_k)$ is also Cauchy and consequently convergent.*

Exercise 4.8. *Prove the unproven statements above.*

Exercise 4.9. *We have defined $e = \lim(1 + n^{-1})^n$. Prove that $e = \sum_{k \geq 0} \frac{1}{k!}$.*

We have $s_n = \sum_{k=1}^n \frac{1}{k!}$ and $t_n = (1 + n^{-1})^n = \sum_{p=0}^n \binom{n}{p} \frac{1}{n^p} = \sum_{p=1}^n \frac{1}{p!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{p-1}{n}\right) \leq s_n$. We know that $t_n \rightarrow e$ and it is trivial to check that s_n converges. We have $\lim t_n = e \leq \lim s_n$. Now define $v_m = \sum_{p \geq 0}^m \frac{1}{p!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{p-1}{n}\right)$ for $m < n$. We have $v_m \leq t_n$ hence $\lim_{n \rightarrow \infty} v_m \leq \lim t_n = e$, but $\lim_{n \rightarrow \infty} v_m = s_m$ hence $s_m \leq e$ and $\lim s_m \leq e$.

⁷Alternatively $[s_{2m+1}, s_{2m+2}] \subset [s_{2m-1}, s_{2m}]$ and $s_{2m} - s_{2m-1} = a_{2m} \rightarrow 0$ hence by the nested interval theorem $\bigcap_m [s_{2m-1}, s_{2m}] = \lim s_{2m} = \lim s_{2m-1} = \sum_{k \geq 0} (-1)^k a_k$.

⁸More generally we have $\sum_{k=m}^n b_k (A_k - A_{k-1}) = b_{n+1} A_n - b_m A_{m-1} - \sum_{k=m}^n (b_{k+1} - b_k) A_k$.

4.3.1 Absolute convergent series (absolut konvergente Reihen)

Absolute convergent series have some additional, more constraining properties.

Lemma 4.3. *The sum of an absolutely convergent series is invariant under rearranging the terms in the series. That is, let an absolutely convergent series $\sum_k |a_k| < \infty$, $\sum_k a_k = A$. For any permutation $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ (bijection), $\sum_k a_{\sigma(k)} = A$.*

Proof. For any ϵ let N_ϵ such that for $N > N_\epsilon$ both $\sum_{k=N}^\infty |a_k| < \epsilon/2$ and $|\sum_{k=1}^N a_k - A| < \epsilon/2$. Fix some σ and take N also larger than $\max\{\sigma^{-1}(1) \dots \sigma^{-1}(N_\epsilon)\} = \max\sigma^{-1}(\{1, \dots, N_\epsilon\})$, that is the list $a_{\sigma(0)}, a_{\sigma(1)}, \dots, a_{\sigma(N)}$ includes all the terms $a_0, a_1, \dots, a_{N_\epsilon}$, and possibly others. Then:

$$\left| \sum_{i=1}^N a_{\sigma(i)} - A \right| \leq \left| \sum_{j=1}^{N_\epsilon} a_j - A \right| + \left| \sum_{i=1, \sigma(i) > N_\epsilon}^N a_{\sigma(i)} \right| \leq \left| \sum_{j=1}^{N_\epsilon} a_j - A \right| + \sum_{j=N_\epsilon}^\infty |a_j| \leq \epsilon.$$

□

Theorem 4.13 (Cauchy product). *The Cauchy product of the two series $\sum_{k \geq 0} a_k$ and $\sum_{k \geq 0} b_k$ is the series $\sum_{p \geq 0} c_p$ with $c_p = \sum_{k=0}^p a_k b_{p-k}$. If both $\sum_{k \geq 0} a_k$ and $\sum_{k \geq 0} b_k$ converge and at least one of them converges absolutely then $\sum_{p \geq 0} c_p$ converges to $(\sum_{k \geq 0} a_k)(\sum_{k \geq 0} b_k)$. If both series are absolutely convergent then their Cauchy product is also absolutely convergent.*

Proof. Denote the partial sums $A_n = \sum_{k=0}^n a_k$, $B_n = \sum_{k=0}^n b_k$ and $C_n = \sum_{k=0}^n c_k$ and assume $\sum_k a_k$ converges absolutely. We have:

$$C_n = \sum_{i=0}^n \sum_{k=0}^i a_k b_{i-k} = \sum_{k=0}^n \sum_{i=k}^n a_k b_{i-k} = \sum_{k=0}^n a_k B_{n-k} = \sum_{k=0}^n a_{n-k} B_k = \sum_{k=0}^n a_{n-k} (B_k - B) + A_n B.$$

Since $B_n \rightarrow B$ and $\sum_k |a_k| < \infty$ then there exists some N_ϵ such that for $n > N_\epsilon$, $|B_n - B| \leq \frac{\epsilon/3}{\sum_k |a_k| + 1}$. On the other hand as A_n converges then $a_k \rightarrow 0$ hence there exists an M_ϵ such that for $n > M_\epsilon$, $|a_n| \leq \frac{\epsilon/3}{N_\epsilon(\max_{i \leq N_\epsilon-1} |B_i - B| + 1)}$. Finally, there exists an L_ϵ such that $n > L_\epsilon$, $|A - A_n| \leq \frac{\epsilon/3}{|B| + 1}$, thus for $n \geq \{L_\epsilon, M_\epsilon + N_\epsilon\}$:

$$\begin{aligned} |C_n - AB| &= \left| \sum_{k=0}^n a_{n-k} (B_k - B) + (A_n - A)B \right| \leq \sum_{k=0}^{N_\epsilon-1} \underbrace{|a_{n-k}|}_{\geq M_\epsilon} |B_k - B| + \sum_{k=N_\epsilon}^n |a_{n-k}| |B_k - B| + |A_n - A| |B| \\ &\leq \frac{\epsilon/3}{N_\epsilon(\max_{i \leq N_\epsilon-1} |B_i - B| + 1)} N_\epsilon \max_{i \leq N_\epsilon-1} |B_i - B| + \left(\sum_k |a_k| \right) \frac{\epsilon/3}{\sum_k |a_k| + 1} + \frac{\epsilon/3}{|B| + 1} |B| \leq \epsilon. \end{aligned}$$

Now, if both $\sum_k a_k$ and $\sum_k b_k$ converge absolutely, then:

$$\sum_{i=0}^n |c_i| \leq \sum_{i=0}^n \sum_{k=0}^i |a_k| |b_{i-k}| \leq \sum_{k=0}^n |a_k| \sum_{i=k}^n |b_{i-k}| \leq \left(\sum_k |a_k| \right) \left(\sum_i |b_i| \right),$$

hence $\sum_{i \geq 0} c_i$ is absolutely convergent. □

Theorem 4.14 (Fubini-Tonelli theorem for series). *Consider the double series $\sum_{i \geq 0} \sum_{j \geq 0} a_{ij}$ with a_{ij} complex numbers. If $\sum_{i \geq 0} \sum_{j \geq 0} |a_{ij}| < \infty$ then the series is convergent and the two sums commute $\sum_{i \geq 0} \sum_{j \geq 0} a_{ij} = \sum_{j \geq 0} \sum_{i \geq 0} a_{ij}$.*

Proof. The hypothesis $\sum_{i \geq 0} \sum_{j \geq 0} |a_{ij}| < \infty$ implies $\forall i$, $M_i = \sum_{j=0}^\infty |a_{ij}| < \infty$ and $\sum_{i=0}^\infty M_i < \infty$.

Let us denote $f_n(m) = \sum_{i=0}^n \sum_{j=0}^m a_{ij}$ and $f(m) = \sum_{i=0}^\infty \sum_{j=0}^m a_{ij}$ and this series converges as $|\sum_{j=0}^m a_{ij}| < M_i$. Observe that for any m , $|f_n(m) - f(m)| \leq \sum_{i=k+1}^\infty |\sum_{j=0}^m a_{ij}| \leq \sum_{i=k+1}^\infty M_i$ and taking into account that $\sum_{i=0}^\infty M_i$ converges it follows that $f_n \rightarrow f$ uniformly in m , that is

$$\forall \epsilon, \forall m \quad \exists N_\epsilon \text{ such that } n > N_\epsilon \Rightarrow |f_n(m) - f(m)| < \epsilon.$$

As $|f(k) - f(m)| \leq |f(k) - f_n(k)| + |f_n(k) - f_n(m)| + |f_n(m) - f(m)|$, this uniform convergence and the convergence of $\sum M_i$ imply that $f(k)$ is a Cauchy sequence hence convergent, $\lim_{k \rightarrow \infty} f(k) = f$,

On the other hand, for any i , $\sum_{j=0}^{\infty} a_{ij}$ converges absolutely, therefore $\lim_{m \rightarrow \infty} f_n(m) = \sum_{i=0}^n \lim_{m \rightarrow \infty} \sum_{j=0}^m a_{ij} = A_n$ exists and, as $|A_n - A_k| \leq \sum_{i=k+1}^n M_i$, A_n is a Cauchy sequence, hence convergent $\lim_{n \rightarrow \infty} A_n = A$.

Finally:

$$|f - A| \leq |f - f(k)| + |f(k) - f_n(k)| + |f_n(k) - A_n| + |A_n - A|,$$

which is arbitrarily small for k and n large enough, therefore $f = A$. $\square$

4.3.2 Power series (Potenzreihen)

An important subclass of series are the power series, generalizing polynomials, of the form $P(z) = \sum_{n \geq 0} c_n (z - z_0)^n$, where z_0 is some fixed complex number (called the center of the disk of convergence) and z is a complex variable.

Lemma 4.4. *Let $P(z)$ a power series. If $P(z)$ converges at some z , then $P(z')$ converges absolutely for all z' such that $|z' - z_0| < |z - z_0|$.*

Proof. As $P(z)$ converges then $c_n (z - z_0)^n \rightarrow 0$ hence there exists some constant such that $\forall n, |c_n (z - z_0)^n| < S$. For any z' with $|z' - z_0| < |z - z_0|$, pick u such that $|z' - z_0| < u < |z - z_0|$. Consider the partial sum:

$$S_N = \sum_{n=0}^N |c_n (z' - z_0)^n| \leq \sup_n \{|c_n (z - z_0)^n|\} \sum_{n=0}^N \left| \frac{z' - z_0}{z - z_0} \right|^n \leq S \frac{1}{1 - \frac{u}{|z - z_0|}}.$$

We have that S_N is monotonic and bounded hence convergent. $\square$

Definition 4.7 (Radius of converge). *The radius of convergence of the power series $P(z) = \sum_{n \geq 0} c_n (z - z_0)^n$ is the nonnegative real number $r \in \mathbb{R}_{\geq 0}$ such that the series converges (absolutely) if $|z - z_0| < r$ and diverges if $|z - z_0| > r$, that is:*

$$r = \sup \left\{ |z - z_0| \mid |P(z)| < \infty \right\}.$$

The domain (disk) of convergence of the series is the open disk $B_r(z_0) = \{z \in \mathbb{C} \mid |z - z_0| < r\}$, and the power series converges absolutely inside its domain of convergence. The radius of convergence of a power series is:

$$\frac{1}{r} = \limsup |c_n|^{1/n}.$$

The radius of convergence of a power series can also be computed as $r = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ if the limit exists. For instance, the radius of convergence of the geometric series $\sum_{n \geq 0} z^n$ is 1.

Lemma 4.5. *Consider a power series and pick z such that $|z - z_0| \leq \rho < r$. Then there exists a constant C such that $|P(z) - \sum_{n=0}^N c_n (z - z_0)^n| \leq C |z - z_0|^{N+1}$.*

Proof. There exists u with $\rho < u < r$, thus $|c_n u^n|$ is bounded by some S hence:

$$\left| \sum_{n=N+1}^P c_n (z - z_0)^n \right| \leq S \sum_{n=N+1}^P \frac{|z - z_0|^n}{u^n} \leq |z - z_0|^{N+1} \frac{S}{u^{N+1}} \frac{1}{1 - \frac{\rho}{u}}.$$

$\square$

Exercise 4.10 (Cauchy–Hadamard theorem). *Prove that the series $P(z) = \sum_n c_n z^n$ converges absolutely for $|z| < r$ and diverges for $|z| > r$ where $r^{-1} = \limsup |c_n|^{1/n}$. Denote $t = \frac{1}{r} = \limsup |c_n|^{1/n}$. Then for any ϵ only a finite number of terms are such that $|c_n| > (t + \epsilon)^n$ hence $\sum_n |c_n z^n| \leq \sum_n |z|^n (t + \epsilon)^n$ which converges for $|z| < \frac{1}{t + \epsilon}$. For any ϵ , $|c_n| > (t - \epsilon)^n$ for an infinity of terms, hence for $|z| = \frac{1}{t - \epsilon}$, $|c_n z^n| \not\rightarrow 0$ and the series can not converge.*

4.3.3 The exponential function (Exponentialfunktion)

Many simple functions are defined as the sum of power series. We now discuss some examples. The *exponential function* is defined as the sum of the power series:

$$\exp(z) \equiv e^z = \sum_{n \geq 0} \frac{1}{n!} z^n \quad r = \lim_{n \rightarrow \infty} \frac{(n+1)!}{n!} = \infty,$$

where we stress that the radius of convergence of this series is ∞ , that is the series converges for any $z \in \mathbb{C}$. The exponential has a number of interesting properties.

Proposition 4.1. *We have $e^{z_1} e^{z_2} = e^{z_1 + z_2}$, in particular $e^z \neq 0$. Furthermore, for any $z \in \mathbb{C}$ we have:*

$$e^z = \lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n.$$

Proof. The Cauchy product of e^{z_1} and e^{z_2} is absolutely convergent to:

$$e^{z_1} e^{z_2} = \sum_{n \geq 0} \frac{1}{n!} \sum_{p=0}^n \frac{n!}{p!(n-p)!} z_1^p z_2^{n-p} = \sum_{n \geq 0} \frac{1}{n!} (z_1 + z_2)^n = e^{z_1 + z_2},$$

and we note that $e^z e^{-z} = e^0 = 1$.

In order to prove the last equality, for any ϵ , let N such that $\sum_{k \geq N+1} \frac{|z|^k}{k!} < \frac{\epsilon}{4}$, which is possible as the exponential has infinite radius of convergence). As $|(1 + \frac{z}{n})^n| \leq \frac{1}{k!} |z|^k$, for any $n > N$ we have that:

$$\begin{aligned} \left| \left(1 + \frac{z}{n}\right)^n - \sum_{k \geq 0} \frac{1}{k!} z^k \right| &\leq \sum_{k=1}^N \frac{|z|^k}{k!} \left| \left(1 - \frac{1}{n}\right) \dots \left(1 - \frac{k-1}{n}\right) - 1 \right| + \sum_{k=N+1}^n \frac{|z|^k}{k!} + \sum_{k \geq N+1} \frac{|z|^k}{k!} \\ &\leq \sum_{k=1}^N \frac{|z|^k}{k!} \left| \left(1 - \frac{1}{n}\right) \dots \left(1 - \frac{k-1}{n}\right) - 1 \right| + \frac{\epsilon}{2} < \epsilon, \end{aligned}$$

for n large enough. □

Exercise 4.11. *We call $f(z) = \sum_{k \geq 0} a_k z^k$ the generating function of the coefficients a_k . It might happen that the series is divergent, in which this is just a formal equality, or it is convergent. Compute the generating function of the coefficients defined recursively as $a_{n+1} = \sum_{k=0}^n a_n a_{n-k}$, $a_0 = 1$. What is the radius of converge of this generaing function?*

Exercise 4.12 (Ramanujan's trick). *One can sometimes attribute a finite value to the sum of a divergent series. For instance:*

$$\sum_{k \geq 1} k = 1 + 2 + \dots + n + \dots = -\frac{1}{12}.$$

Denote $1 + 2 + \dots + n + \dots = c$. Show that $c - 4c = \sum_{p \geq 1} (-1)^{p+1} p$. We have by subtracting 4 from the second term, 8 from the fourth, 12 from the sixth and so on that $c - 4c = 1 + (2 - 4) + 3 + (4 - 8) + 5 + (6 - 12) + \dots = 1 - 2 + 3 - 4 + 5 - 6 + \dots = \sum_{p \geq 1} (-1)^{p+1} p$. Use $\frac{1}{1+x} = \sum_{k \geq 0} (-1)^k x^k$ to conclude that $\sum_{p \geq 1} (-1)^{p+1} p = \frac{1}{4}$. We have $\frac{1}{(1+x)^2} = \left[\sum_{k \geq 0} (-1)^k x^k \right] \left[\sum_{k' \geq 0} (-1)^{k'} x^{k'} \right] = \sum_{p \geq 0} \sum_{k=0}^p (-1)^{p-k} x^{p-k} (-1)^k x^k = \sum_{p \geq 0} (-1)^p (p+1) x^p \Rightarrow \sum_{p \geq 0} (-1)^p (p+1) = \frac{1}{4}$. We will see later that this resummation can be more rigorously justified.

Chapter 5

Vector spaces, first contact

Vector spaces are ubiquitous in mathematics.

5.1 Vector spaces

The simplest examples of vector spaces are $\mathbb{R}^D$ and $\mathbb{C}^D$, but many other vector spaces are relevant.

Definition 5.1 (**Vector space (Vektorraum)**). A vector space V over the algebraic field $\mathbb{F} = \mathbb{R}, \mathbb{C}$ is a set endowed with two operations:

- the addition $+: V \times V \rightarrow V$, $(x, y) \rightarrow x + y$ such that $(V, +, 0)$ is an Abelian group (that is $+$ is associative and commutative, 0 is the identity element for $+$ and every element $x \in V$ has an additive inverse, or negation, $-x$).
- multiplication by scalars $\cdot: \mathbb{F} \times V \rightarrow V$, $(\alpha, x) \rightarrow \alpha \cdot x$ such that:
 - $\cdot$ is compatible with the multiplication $\times$ in $\mathbb{F}$:

$$\alpha \cdot (\beta \cdot x) = (\alpha \times \beta) \cdot x .$$

- the identity of $\cdot$ is the multiplicative identity in $\mathbb{F}$, denoted 1 :

$$1 \cdot x = x .$$

- $\cdot$ is distributive with respect to both the addition in $\mathbb{F}$ and the addition in V :

$$\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y , \quad (\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x .$$

$\mathbb{F}$ is called the field of scalars. When no confusion can arise, we suppress the $\cdot$ and $\times$ notation for the multiplication.

The D dimensional Euclidean space $\mathbb{R}^D = \{x = (x^1, \dots, x^D) \mid x^\mu \in \mathbb{R}\}$ is an example of vector space over $\mathbb{R}$ with:

- the addition of $x = (x^1, \dots, x^D)$ and $y = (y^1, \dots, y^D)$ is $x + y = (x^1 + y^1, \dots, x^D + y^D)$.
- the multiplication of x by $\alpha \in \mathbb{R}$ is $\alpha x = (\alpha x^1, \dots, \alpha x^D)$.

Similarly, $\mathbb{C}^D$ is a vector space over $\mathbb{C}$. We typically represent vectors as column vectors in these cases.

Exercise 5.1. Prove that $0 \cdot x = 0$. $\alpha \cdot x = (\alpha + 0) \cdot x = \alpha \cdot x + 0 \cdot x$.

We often encounter vector spaces with more structure.

Definition 5.2 (**Norm**). A norm on the vector space $(V, \mathbb{F})$ is a function $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ such that:

- $\|v\| \geq 0$ and $\|v\| = 0$ if and only if $v = 0$.¹
- $\|\alpha x\| = |\alpha| \|x\|$, $\forall \alpha \in \mathbb{F}$ and $\forall x \in V$.
- $\|x + y\| \leq \|x\| + \|y\|$, $\forall x, y \in V$.

Any normed vector space is a metric space with distance function $d(x, y) = \|x - y\|$. A Banach space is a Cauchy complete normed vector space.

For instance $\mathbb{R}^D$ and $\mathbb{C}^D$ are normed spaces (and in fact Banach spaces) with norm any of the p -norms $\|x\|_p = (\sum_{\mu=1}^D |x^\mu|^p)^{1/p}$. Remark that $\mathbb{Q}^D$ with $\|\cdot\|_p$ is a finite dimensional normed vector space over $\mathbb{Q}$ which is not Banach as it is not Cauchy complete. In fact all the finite dimensional vector spaces (see below) over Cauchy complete fields of scalars are Banach spaces.

Note that if V is a vector space and $Y \subset V$ is a vector subspace then $V/Y = \{[x] \mid x_1 \sim x_2 \text{ if } x_1 - x_2 \in Y\}$ is a vector space. An important lemma is the following.

Lemma 5.1 (**Direct sum decomposition**). Consider Y_1 and Y_2 are subspaces of X such that $Y_1 + Y_2 = \{y_1 + y_2 \mid y_1 \in Y_1, y_2 \in Y_2\} = X$, that is any $x \in X$ can be decomposed as a sum $x = y_1 + y_2$ with $y_1 \in Y_1$ and $y_2 \in Y_2$. If $Y_1 \cap Y_2 = \{0\}$ (or equivalently Y_1 and Y_2 are linearly independent)², then the decomposition is unique. In this case we denote $X = Y_1 \oplus Y_2$ and say that X is the direct sum of Y_1 and Y_2 .

Proof. If x admits two decompositions $x = y_1 + y_2 = y'_1 + y'_2$, then $y_1 - y'_1 = y'_2 - y_2$ hence, as $y_1 - y'_1 \in Y_1$ and $y'_2 - y_2 \in Y_2$ we conclude that $y_1 = y'_1$ and $y_2 = y'_2$. $\square$

Given two vector spaces one can build new vector spaces.

Definition 5.3 (**Direct sum of vector spaces**). Let Y_1 and Y_2 be two vector spaces over $\mathbb{F}$. The direct sum $Y_1 \oplus Y_2$ is the set of pairs (y_1, y_2) with $y_1 \in Y_1$ and $y_2 \in Y_2$ endowed with the vector space operations $\alpha(y_1, y_2) = (\alpha y_1, \alpha y_2)$ and $(y_1, y_2) + (y'_1, y'_2) = (y_1 + y'_1, y_2 + y'_2)$.

5.2 Hamel bases, dimension of a vector space

Vector spaces are closed under taking linear combinations. Thus given a vector space, an important question is to identify a subset of vectors such that any other vector can be obtained as a linear combination of them.

Definition 5.4 (**Hamel basis**). A Hamel basis in a vector space $(V, \mathbb{F})$ is a set of vectors $B \subset V$ such that:

- Linearly independent. For every finite subset $\{e_1, \dots, e_n\} \subset B$, $\alpha_1 e_1 + \dots + \alpha_n e_n = 0 \Rightarrow \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$.
- Spanning. Every vector writes as a finite linear combination of vectors in the basis, that is $\forall v \in V$ there exist $\beta_1, \beta_2, \dots, \beta_m \in \mathbb{F}$ and $\{e_1, \dots, e_m\} \subset B$ such that $v = \beta_1 e_1 + \dots + \beta_m e_m$.

The scalars $\beta_1, \dots, \beta_m$ are called the components of v in the basis B .

¹It is in fact sufficient to require $\|v\| > 0$ if $v \neq 0$.

²For any $y_1 \in Y_1$ and $y_2 \in Y_2$, $\alpha_1 y_1 + \alpha_2 y_2 = 0$ if and only if $\alpha_1 = \alpha_2 = 0$.

The span of the vectors S is the set of all finite linear combinations of these vectors $\text{Span } S = \{\alpha_1 e_1 + \dots + \alpha_n e_n \mid \alpha_i \in \mathbb{F}, e_i \in S\}$. If a vector $w \neq 0$ is not in the span of the linearly independent vectors S then $\{w\} \cup S$ are linearly independent. Indeed, assume

$$\alpha w + \alpha_1 e_1 + \dots + \alpha_n e_n = 0,$$

such that not all $\alpha, \alpha_1, \dots, \alpha_n$ are zero. Then $\alpha \neq 0$ (otherwise e_i would not be linearly independent) and

$$w = -\frac{\alpha_1}{\alpha} e_1 - \dots - \frac{\alpha_n}{\alpha} e_n,$$

and at least one of the α_i s is not zero (as $w \neq 0$) hence $w \in \text{Span } S$. The span of any set of vectors is a vector spaces.

We stress that even though a set B might be infinite, both linear independence and span involve only a *finite* number of vectors in B . On the positive side, every vector space has a Hamel basis.

Theorem 5.1 (**Every vector space has a Hamel basis**). *Every vector space has a Hamel basis. All Hamel basis of a vector space have the same cardinality. The cardinal number of a basis is called the dimension of the vector space.*

Proof. Let $(V, \mathbb{F})$ a vector space and let $\mathcal{X}$ the set of linearly independent subsets of V . $\mathcal{X}$ is non empty (as it contains the empty set) and is partially ordered by inclusion $\subset$.

For any subset $\mathcal{Y} \subset \mathcal{X}$ which is totally ordered (i.e. $\forall Y_1, Y_2 \in \mathcal{Y}$, either $Y_1 \subset Y_2$ or $Y_2 \subset Y_1$) denote $S_{\mathcal{Y}} = \bigcup_{Y \in \mathcal{Y}} Y$. $S_{\mathcal{Y}}$ is linearly independent: every finite subset of $S_{\mathcal{Y}}$ is a subset of some element in $\mathcal{Y}$. Indeed, consider a finite subset $\{s_1, \dots, s_n\} \in S_{\mathcal{Y}}$. Then there exist $Y_1 \ni s_1, \dots, Y_n \ni s_n$, $Y_1, \dots, Y_n \in \mathcal{Y}$ and, as $\mathcal{Y}$ is totally ordered, one of $Y_1, \dots, Y_n$ is the largest of them and contains $s_1, \dots, s_n$. It follows that $S_{\mathcal{Y}} \in \mathcal{X}$ and is an upper bound for $\mathcal{Y}$ with respect to the order relation $\subset$.

By Zorn's lemma, $\mathcal{X}$ has a maximal element, i.e. an element that is not smaller than any other element. Denote $S_{\max} \in \mathcal{X}$ a maximal element. Then $S_{\max}$ is linearly independent, as it belongs to $\mathcal{X}$. Moreover, $S_{\max}$ spans V . Indeed, assume it does not, then $\exists w \in V$ such that w is not in the span of $S_{\max}$ (i.e. no finite linear combination of elements in $S_{\max}$ equals w). Then $S_{\max} \cup \{w\}$ is linearly independent (as any finite subset of $S_{\max} \cup \{w\}$ is linearly independent) and strictly larger than $S_{\max}$, contradicting the fact that $S_{\max}$ is maximal.

The fact that all basiss have the same cardinality can be proven abstractly (using again Zorn's lemma). As we comment below, the infinite dimensional case is not of practical use and mostly academic. In finite dimensions it follows from lemma 5.2 below. $\square$

The drawback of Hamel basis is that they are useful only if the dimension of the space is finite, that is the Hamel bases have a finite number of elements. In this case a Hamel basis is called simply a basis. If the cardinal of the Hamel basis is infinite, things are more complicated.

5.3 Finite dimensional vector spaces

Theorem 5.2 (**Steinitz exchange lemma**). *Let U and W be finite subsets of a vector space V . If U is a set of linearly independent vectors and W spans V , then:*

- $|U| \leq |W|$,
- *there is a set $W' \subset W$ with $|W'| = |W| - |U|$ such that $U \cup W'$ spans V .*

Proof. We denote $U = \{u_1, \dots, u_m\}$ and $W = \{w_1, \dots, w_n\}$. We will show that “we can exchange k elements in W for elements in U ”, that is $\forall k, 0 \leq k \leq m$ the set $\{u_1, \dots, u_k, w_{k+1}, \dots, w_n\}$, where the w 's have been reordered in a k dependent manner, spans V .

We proceed by induction. $k = 0$ is trivial. Assume that for $k < m$, $\{u_1, \dots, u_k, w_{k+1}, \dots, w_n\}$ spans V then $\exists \mu_1, \dots, \mu_n$ such that:

$$u_{k+1} = \sum_{j=1}^k \mu_j u_j + \sum_{j=k+1}^n \mu_j w_j,$$

and there must exist $j \geq k+1$ with $\mu_j \neq 0$ (as the u s are linearly independent). In particular, this implies that $k+1 \leq n$. We relabel this $j = k+1$ and write:

$$w_{k+1} = \frac{1}{\mu_{k+1}} \left(u_{k+1} - \sum_{j=1}^k \mu_j u_j - \sum_{j=k+2}^n \mu_j w_j \right).$$

Thus $w_{k+1} \in \text{Span}\{u_1, \dots, u_k, u_{k+1}, w_{k+2}, \dots, w_n\}$. As, in addition, $\text{Span}\{u_1, \dots, u_k, u_{k+1}, w_{k+2}, \dots, w_n\}$ contains all $u_1, \dots, u_k, w_{k+2}, \dots, w_n$, then $\text{Span}\{u_1, \dots, u_k, u_{k+1}, w_{k+2}, \dots, w_n\} \supset \text{Span}\{u_1, \dots, u_k, w_{k+1}, w_{k+2}, \dots, w_n\} = V$. $\square$

Although an analog of this lemma holds for V an infinite dimensional vector space, its true force comes into play in the finite dimensional case. In particular, in finite dimensions, we can build a basis in a vector space by completing a basis in a subspace.

Corollary 5.1 (**Vector subspace**). *Let X be a vector space with a finite Hamel basis. Then:*

- *if a basis in X has a finite number of elements, any other basis in X will have the same number of elements. We denote $\dim X$ this common number and call it the dimension of X .*

Proof. As any two basis B_1 and B_2 are both spanning and linearly independent, we have both $|B_1| \leq |B_2|$ and $|B_2| \leq |B_1|$. $\square$

- *any set of $\dim X$ linearly independent vectors in X spans X (hence is a basis in X); any set of $\dim X$ vectors that spans X is linearly independent (hence a basis in X).*

Proof. We denote $n = \dim X$. Take $U = \{u_1 \dots u_n\} \in X$ a set of n linearly independent vectors. Choosing a basis $W = \{w_1, \dots, w_n\}$ in X and exchanging its elements with the elements of U , as $|U| = |W|$ we conclude that U spans X . For the second assertion, assume that the $\dim X$ vectors in W span X but are not linearly independent. Then one can choose a subset $W' \subset W$ with $\dim X - 1$ elements which spans X . Exchanging W' for $\dim X - 1$ vectors in a basis we obtain $\dim X - 1$ vectors which are both linearly independent and span X , which is impossible. $\square$

Let $Y \subset X$ be a vector subspace of X , then:

- *any basis in Y can be completed to a basis in X ; if $\dim Y = \dim X$ then $Y = X$ and $Y \subsetneq X$ if and only if $\dim Y < \dim X$.*

Proof. We denote $\dim Y = m$. The assertion follows by exchanging the elements of a basis $\{w_1, \dots, w_n\}$ in X for the elements of a basis $\{u_1, \dots, u_m\}$ in Y to obtain the set of $n = \dim X$ vectors $\{u_1, \dots, u_m, w_{m+1}, \dots, w_n\}$ that span X , hence are a basis in X . $\square$

If $Y_1, Y_2 \subset X$ are vector subspaces of X , then:

- *$Y_1 \cap Y_2$ is a vector subspace of X and $\dim X + \dim(Y_1 \cap Y_2) \geq \dim Y_1 + \dim Y_2$.*

Proof. This follows by choosing a basis $\{e_1, \dots, e_{\dim(Y_1 \cap Y_2)}\}$ in $Y_1 \cap Y_2$ and completing it to a basis in Y_1 by adding $\{f_{\dim(Y_1 \cap Y_2)+1}, \dots, f_{\dim Y_1}\}$ and respectively to a basis in Y_2 by adding $\{h_{\dim(Y_1 \cap Y_2)+1}, \dots, h_{\dim Y_2}\}$. As the set of all vectors e, f, h is linearly independent³ in X , we have $\dim X \geq \dim Y_1 + \dim Y_2 - \dim(Y_1 \cap Y_2)$. $\square$

³As $\alpha e + \beta f \in Y_1$ and $h \in Y_2 \setminus Y_1$, if $\alpha e + \beta f = -\gamma h$ then $\gamma = 0$ and $\alpha e + \beta f$ implies $\alpha = \beta = 0$.

Definition 5.5 (**Tensor product of vector spaces**). Let X and Y be two vector spaces over $\mathbb{F}$, with $(x_i)_{i \in I}$ and $(y_j)_{j \in J}$ bases in X and respectively Y . The tensor product space $X \otimes Y$ is the linear span of all the pairs $(x_i \otimes y_j)_{i \in I, j \in J}$, that is:

$$X \otimes Y = \text{Span}\{x_i \otimes y_j, i \in I, j \in J\} = \left\{ \sum_{i,j=1}^{\text{finite}} \alpha_{ij} x_i \otimes y_j, \alpha_{ij} \in \mathbb{F}, i \in I, j \in J \right\},$$

where the tensor product $\otimes$ is linear in each of its arguments. We have $\dim(X \otimes Y) = \dim(X) \dim(Y)$.

We stress that $X \otimes Y$ contains the tensor products $v \otimes w$ with $v \in X$ and $w \in Y$, that is $v \otimes w = \sum_{i,j} v_i w_j x_i \otimes y_j$, but contains also more general linear combinations $\sum_{i,j} \alpha_{ij} x_i \otimes y_j$.

5.4 Linear operators

Definition 5.6 (**Linear operators**). An operator $\mathcal{A} : X \rightarrow Y$ from a vector space to a vector space is called linear if $\mathcal{A}(\alpha x + \beta y) = \alpha \mathcal{A}(x) + \beta \mathcal{A}(y)$. We sometimes denote $\mathcal{A}(x) = Ax$ in order to simplify the notation. The operators $\mathcal{A}$ also form a vector space over $\mathbb{F}$, with $(\alpha \mathcal{A} + \beta \mathcal{B})(x) = \alpha \mathcal{A}(x) + \beta \mathcal{B}(x)$. The kernel of $\mathcal{A}$ is the set $\ker \mathcal{A} = \{x \in X \mid Ax = 0\}$, and the image of $\mathcal{A}$ is the set $\text{Im } \mathcal{A} = \{y \in Y \mid \exists x \in X, Ax = y\}$. Both $\ker \mathcal{A}$ and $\text{Im } \mathcal{A}$ are vector spaces.

A linear operator is also called a linear homomorphism. If $\mathcal{A} : X \rightarrow X$, then $\mathcal{A}$ is called a linear endomorphism.

Theorem 5.3 (**The rank nullity theorem**). Let $\mathcal{A} : X \rightarrow Y$ a linear operator on a finite dimensional vector space X . Then $\dim(\ker \mathcal{A}) + \dim(\text{Im } \mathcal{A}) = \dim(X)$.

Proof. We denote $\dim X = n$. The kernel $\ker \mathcal{A}$ is a vector subspace of X . We denote $k = \dim(\ker \mathcal{A})$, we chose a basis $\{v_1, \dots, v_k\}$ in the kernel and we extend it to a basis $= \{v_1, \dots, v_k, w_{k+1}, \dots, w_n\}$ in X . Then:

$$\text{Im } \mathcal{A} = \text{Span}\{\mathcal{A}v_1, \dots, \mathcal{A}v_k, \mathcal{A}w_{k+1}, \dots, \mathcal{A}w_n\} = \text{Span}\{\mathcal{A}w_{k+1}, \dots, \mathcal{A}w_n\},$$

that is the vectors $\mathcal{A}w_{k+1}, \dots, \mathcal{A}w_n$ span $\text{Im } \mathcal{A}$. These vectors are linearly independent as:

$$\sum_{i=k+1}^n \alpha_i \mathcal{A}w_i = 0 \Rightarrow \mathcal{A}\left(\sum_{i=k+1}^n \alpha_i w_i\right) = 0 \Rightarrow \sum_{i=k+1}^n \alpha_i w_i \in \ker \mathcal{A} \Rightarrow \sum_{i=k+1}^n \alpha_i w_i + \sum_{i=1}^k \alpha_i v_i = 0 \Rightarrow \alpha_i = 0, \quad i = 1, \dots, n,$$

hence form a basis in $\text{Im } \mathcal{A}$. □

Definition 5.7 (**Operator norm**). The operator norm of $\mathcal{A}$ is:

$$\|\mathcal{A}\|_{\text{op.}} = \inf \left\{ c \geq 0 \mid \|\mathcal{A}v\| \leq c\|v\|, \forall v \in V \right\} = \sup_{v \in V} \frac{\|\mathcal{A}v\|}{\|v\|} = \sup_{v \in V, \|v\|=1} \|\mathcal{A}v\|.$$

We will prove later that any linear operator on a finite dimensional vector space is bounded, that is $\|\mathcal{A}\|_{\text{op.}} \leq \infty$.

Exercise 5.2. Prove that $\|\cdot\|_{\text{op.}}$ is a norm. Prove it is submultiplicative $\|AB\|_{\text{op.}} \leq \|A\|_{\text{op.}} \|B\|_{\text{op.}}$.

The linear operators $\mathcal{A} : X \rightarrow Y$, with point wise addition and multiplication by scalars form a vector space, and in fact a Banach space with respect to $\|\cdot\|_{\text{op.}}$, which we denote $B_{\text{op.}}(X, Y)$.

5.4.1 The dual space

Definition 5.8 (**The dual**). A linear form or linear functional on the vector space V over $\mathbb{F}$ is a linear function $\varphi : V \rightarrow \mathbb{F}$, $\varphi(\alpha x + \beta y) = \alpha \varphi(x) + \beta \varphi(y)$, for $x, y \in V$, $\alpha, \beta \in \mathbb{F}$

The dual space V^* is the vector space of linear functionals on V endowed with the vector space operations:

- $(\varphi_1 + \varphi_2)(x) = \varphi_1(x) + \varphi_2(x)$, $\varphi_1, \varphi_2 \in V^*$, $x \in V$,
- $(\alpha \cdot \varphi)(x) = \alpha \cdot \varphi(x)$, $\varphi \in V^*$, $\alpha \in \mathbb{F}$, $x \in V$.

The elements of the dual are also called covectors. There exists a natural homomorphic injection of V into $V^{**} = \{\Phi : V^* \rightarrow \mathbb{F} \mid \Phi \text{ linear}\}$ (which can be promoted to an isomorphic bijection in finite dimension) $v \rightarrow \Phi(v)$ defined by $(\Phi(v))(\varphi) = \varphi(v)$, $\forall v \in V$ and $\forall \varphi \in V^*$.

For every basis in a finite dimensional vector space, one can build a basis in the dual space.

Construction 5.1 (Dual basis). If the dimension of V is finite, V^* has the same dimension. Given a basis $\{e_1, \dots, e_N\}$ in V one can build a basis in V^* called the dual basis by specifying the linear functionals θ^j defined by $\theta^j(e_i) = \delta_i^j$, that is $\theta^j(\sum_{i=1}^N v^i e_i) = v^j$. By definition θ^j are linear functionals and they are linearly independent as:

$$\sum_{i=1}^N \alpha_i \theta^i(v) = 0, \quad \forall v \Rightarrow \sum_{i=1}^N \alpha_i \theta^i(e_k) = \alpha_k = 0,$$

and span V^* as any functional φ writes as:

$$\forall v, \quad \varphi(v) = \varphi\left(\sum_{i=1}^N v^i e_i\right) = \sum_{i=1}^N \varphi(e_i) v^i = \sum_{i=1}^N \varphi(e_i) \theta^i(v) \Rightarrow \varphi = \sum_{i=1}^N \varphi(e_i) \theta^i.$$

Lemma 5.2 (Basis in $B_{\text{op.}}(X, Y)$). If X and Y are finite dimensional vector spaces, then the vector space of the linear operators is the tensor product $B_{\text{op.}}(X, Y) = Y \otimes X^*$. Given $\{e_j, \theta^j\}_{j \in J}$ a basis in X and its dual basis in X^* , respectively $\{f_i, \mu^i\}_{i \in I}$ dual bases in Y and Y^* , any linear operator $\mathcal{A} : X \rightarrow Y$ decomposes in the tensor product basis $f_i \otimes \theta^j$ as $\mathcal{A} = \sum_{i \in I, j \in J} [\mu^i(\mathcal{A}e_j)] f_i \otimes \theta^j$. For instance, the identity operator on $\mathcal{I} : V \rightarrow V$, $\mathcal{I}(v) = v$ writes as $\mathcal{I} = \sum_{i=1}^{\dim V} e_i \otimes \theta^i$.

Proof. It is enough to note that any vector $y \in Y$ decomposes as $y = \sum_{i \in I} y^i f_i = \sum_{i \in I} f_i \mu^i(y)$ hence:

$$\mathcal{A}(x) = \sum_{i \in I} f_i \mu^i(\mathcal{A}x) = \sum_{i \in I, j \in J} f_i [\mu^i(\mathcal{A}e_j)] x^j = \sum_{i \in I, j \in J} [\mu^i(\mathcal{A}e_j)] f_i \theta^j(x) = \sum_{i \in I, j \in J} [\mu^i(\mathcal{A}e_j)] (f_i \otimes \theta^j)(x).$$

We denote $\mu^i(\mathcal{A}e_j) = A_{ij}^i$ and call $(A_{ij}^i)_{i \in I, j \in J}$ the matrix elements of $\mathcal{A}$ in the given bases. □

5.5 Affine spaces

An affine space is morally what is left from a vector space when one forgets all the information that depends on the origin. For instance, while the sum of two vector depends on the origin and is not captured in the affine structure, the difference (or any affine combination $\lambda \vec{v} + (1 - \lambda) \vec{w}$ does not and is encoded in the affine structure.

Definition 5.9 (Affine space). An affine space is a set V and a vector space $\vec{V}$ and an operation $+ : V \times \vec{V} \rightarrow V$, $(a, \vec{v}) \mapsto a + \vec{v}$ such that (denoting with an arrow the elements in $\vec{V}$):

- (right identity) $a + \vec{0} = a$, $\forall a \in V$, where $\vec{0}$ is the vector zero in $\vec{V}$,
- (associativity) $\forall \vec{v}, \vec{w} \in \vec{V}$ and $\forall a \in V$ we have $(a + \vec{v}) + \vec{w} = a + (\vec{v} + \vec{w})$, where $\vec{v} + \vec{w}$ is the addition in $\vec{V}$,
- (free and transitive action) $\forall a \in V$, the mapping from $\vec{V}$ to V , $\vec{v} \mapsto a + \vec{v}$ is a bijection.

The dimension of an affine space is the dimension of the associated vector space.

For every $\vec{v} \in \vec{V}$, the translation by $\vec{v}$, that is the map $T_{\vec{v}} : V \rightarrow V$, $T_{\vec{v}}(a) = a + \vec{v}$, is a bijection. Note that $T_{\vec{0}}$ is the identity and $T_{\vec{v}+\vec{w}} = T_{\vec{w}}T_{\vec{v}}$. If $T_{\vec{v}}(a) = T_{\vec{v}}(b)$ then $T_{-\vec{v}}T_{\vec{v}}(a) = T_{-\vec{v}}T_{\vec{v}}(b)$ hence $a = b$ and for any b , $T_{\vec{v}}(T_{-\vec{v}}(b)) = b$.

The third property in the definition is equivalent to saying that for every $a, b \in V$, there exist a unique $\vec{v}$ such that $b = a + \vec{v}$.⁴ We sometimes denote this vector $\vec{ab}$ or $b - a$. The subtraction is such that:

- $\forall a \in V, \forall \vec{v} \in \vec{V}$, there exists a unique $b \in V$ such that $b - a = \vec{v}$, namely $a + \vec{v}$,
- $\forall a, b, c \in V, (c - b) + (b - a) = c - a$, as $(a + (b - a)) + (c - b) = c = a + ((b - a) + (c - b))$.

An *affine subspace* of V is a subset of the form $a + \vec{W} = \{a + \vec{w}, \vec{w} \in \vec{W}\}$ with a a point in V and $\vec{W}$ a vector subspace of $\vec{V}$. The vector subspace $\vec{W}$ is called the *direction* of the affine subspace and two affine subspaces with the same direction are called *parallel*. Given any direction $\vec{W}$ for any point $a \in V$, there is only one affine subspace of direction $\vec{W}$, namely $a + \vec{W}$, which is known as Euclid's fifth postulate. Any translation of the base point maps an affine subspace to a parallel subspace.

Exercise 5.3. Consider a linear operator on a finite dimensional vector space $\mathcal{H} : X \rightarrow Y$. Prove that for any fixed $y \in \text{Im } \mathcal{H}$, the space of solutions of the equation $\mathcal{H}x = y$ is an affine space over $\ker \mathcal{H}$. We denote $W_y = \{x \in X \mid \mathcal{H}x = y\}$ and define $+$: $W_y \times \ker \mathcal{H} \rightarrow W_y$ by $(x, v) \rightarrow x + \vec{v}$, with $+$ the addition in X . Clearly, for any $x \in W_y$, $\mathcal{H}(x + \vec{v}) = \mathcal{H}(x) + \mathcal{H}(\vec{v}) = y$ hence $x + \vec{v} \in W_y$ and the associativity is trivial. It remains to check that for any x , the function $\vec{v} \rightarrow x + \vec{v}$ is a bijection. As x is fixed, this is clearly an injection. For any $x' \in W_y$ we have $\mathcal{H}(x' - x) = y - y = 0$ hence $x' - x \in \ker \mathcal{H}$ and $x' = x + (x' - x)$.

Definition 5.10 (Affine map). An affine map (affine homomorphism) between two affine spaces $(X, \vec{X})$ and $(Y, \vec{Y})$ is a map $\mathcal{A} : X \rightarrow Y$ such that the map $\vec{\mathcal{A}} : \vec{X} \rightarrow \vec{Y}$, $\vec{\mathcal{A}}(b - a) = \mathcal{A}(b) - \mathcal{A}(a)$ is a well defined linear map, that is if $b - a = d - c$ then $\mathcal{A}(b) - \mathcal{A}(a) = \mathcal{A}(d) - \mathcal{A}(c)$. It follows that $\mathcal{A}(a + \vec{v}) = \mathcal{A}(a) + \vec{\mathcal{A}}(\vec{v})$, hence $\mathcal{A}$ is defined by its value at a point and the associated linear map $\vec{\mathcal{A}}$.

An affine transformation, or endomorphisms is an affine map from $\mathcal{A} : V \rightarrow V$. The translation $T_{\vec{v}}$ is an endomorphisms. Given a point b and a linear map $\vec{\mathcal{A}} : \vec{V} \rightarrow \vec{V}$ we can build the affine transformation $L_{b, \vec{\mathcal{A}}} : V \rightarrow V$, $L_{b, \vec{\mathcal{A}}}(a) = b + \vec{\mathcal{A}}(b - a)$, that is the linear map $\vec{\mathcal{A}}$ centered at “the origin” b .

For two choices of an origin o, o' , if $\lambda_1 + \dots + \lambda_n = 0$, then for any $a_1, \dots, a_n$:

$$\lambda_1(a_1 - o) + \dots + \lambda_n(a_n - o) = \lambda_1(a_1 - o') + \dots + \lambda_n(a_n - o').$$

Indeed, using $\lambda_n = -\lambda_1 - \dots - \lambda_{n-1}$, both sides equal the vector $\lambda_1(a_1 - a_n) + \dots + \lambda_{n-1}(a_{n-1} - a_n)$. As this vector is independent of the choice of origin o , we denote it $\lambda_1 a_1 + \dots + \lambda_n a_n$. Note that the subtraction is just the case with two terms.

Exercise 5.4. Prove that $g = o + \lambda_1(a_1 - o) + \dots + \lambda_n(a_n - o)$ is independent of o if $\lambda_1 + \dots + \lambda_n = 1$. If $g = o + \vec{v}$ and $g' = o' + \vec{v}'$ then $g - o = \vec{v}$, $g' - o' = \vec{v}'$ hence $(g - o) - (g' - o') = \vec{v} - \vec{v}'$, but the right hand side is also $(g - o) + (o - g') - [(o - g') + (g' - o')] = (g - g') - (o - o')$, therefore $g - g' = o - o' + \vec{v} - \vec{v}'$. Denoting $g = o + \lambda_1(a_1 - o) + \dots + \lambda_n(a_n - o)$ and $g' = o' + \lambda_1(a_1 - o') + \dots + \lambda_n(a_n - o')$, and taking into account that $a - b = -(b - a)$ (if $a = b + \vec{v}$ then $b = a + (-\vec{v})$), we get $g - g' = (o - o') - \lambda_1(o - o') - \dots - \lambda_n(o - o') = \vec{0}$.

For $\lambda_1 + \dots + \lambda_n = 1$ and for some choice of the origin o , we denote $g = o + \lambda_1(a_1 - o) + \dots + \lambda_n(a_n - o)$. This is independent of o and we simplify the notation to $g = \lambda_1 a_1 + \dots + \lambda_n a_n$ and say that g is the affine combination (or the barycenter) of the a_i s with weights λ_i . The affine span of $X \subset A$ is the set of all finite affine combinations of elements in X and its direction is the linear span of $x - y$ with $x, y \in X$. Fixing $x_0 \in X$, the direction of the affine span is the linear span of $x - x_0$ for all $x \in X$.⁵

⁴As $\vec{v} \mapsto a + \vec{v}$ is surjective, there exist some $\vec{v}$ such that $a + \vec{v} = b$. As $\vec{v} \mapsto a + \vec{v}$ is injective, if $a + \vec{v}_1 = a + \vec{v}_2$ then $a + (\vec{v}_1 - \vec{v}_2) = a$ but at the same time $a + \vec{0} = a$ hence $\vec{v}_1 - \vec{v}_2 = \vec{0}$.

⁵Equivalently, the affine span of X is the smallest affine space containing X and it is the intersection of all the affine subspaces containing X . Its direction is the intersection of the directions of the affine subspaces containing it.

Definition 5.11 (**Affine basis, affine and barycentric coordinates**). An affine basis in a affine space $V, \vec{V}$ of dimension n is a set of $n+1$ elements in $\{e_0, \dots, e_n\}$ such that $\{e_1 - e_0, \dots, e_n - e_0\}$ is a basis in $\vec{V}$. Every $x \in V$ has a unique decomposition in an affine basis $x = e_0 + \lambda_1(e_1 - e_0) + \dots + \lambda_n(e_n - e_0)$, or equivalently:

$$x = \lambda_0 e_0 + \lambda_1 e_1 + \dots + \lambda_n e_n, \quad \lambda_0 + \dots + \lambda_n = 1.$$

The coordinates $\lambda_0, \dots, \lambda_n$ are called the barycenter coordinates of x .

An affine coordinate frame in an affine space $V, \vec{V}$ of dimension n is a point $o \in V$, called the origin and a basis $\{\vec{v}_1, \dots, \vec{v}_n\}$ in $\vec{V}$. For each point $x \in V$, there exists a unique n -tuple $\lambda_1, \dots, \lambda_n$ such that $x = o + \lambda_1 \vec{v}_1 + \dots + \lambda_n \vec{v}_n$. The numbers λ_i are called the affine coordinates of x in the frame $\{o, \vec{v}_1, \dots, \vec{v}_n\}$.

The point with barycentric coordinates $\lambda_0, \dots, \lambda_n, \lambda_0 + \dots + \lambda_n = 1$ in the affine basis $\{e_0, \dots, e_n\}$ corresponds to the point with affine coordinates $\lambda_1, \dots, \lambda_n$ in the affine frame $\{o = e_0, \vec{v}_1 = e_1 - e_0, \dots, \vec{v}_n = e_n - e_0\}$. Conversely, the point with affine coordinates $\lambda_1, \dots, \lambda_n$ in the affine frame $\{o, \vec{v}_1, \dots, \vec{v}_n\}$ corresponds to the point with barycentric coordinates $1 - \lambda_1 - \dots - \lambda_n, \lambda_1, \dots, \lambda_n$ in the affine basis $\{e_0 = o, e_1 = o + \vec{v}_1, \dots, e_n = o + \vec{v}_n\}$.

Exercise 5.5. Prove that $x = e_0 + \lambda_1(e_1 - e_0) + \dots + \lambda_n(e_n - e_0)$ if and only if $x = (1 - \lambda_1 - \dots - \lambda_n)e_0 + \lambda_1 e_1 + \dots + \lambda_n e_n$.

If $x = (1 - \lambda_1 - \dots - \lambda_n)e_0 + \lambda_1 e_1 + \dots + \lambda_n e_n$ then, choosing e_0 as origin, we have $x = e_0 + (1 - \lambda_1 - \dots - \lambda_n)(e_0 - e_0) + \lambda_1(e_1 - e_0) + \dots + \lambda_n(e_n - e_0)$ and we conclude. Conversely, if $x = e_0 + \lambda_1(e_1 - e_0) + \dots + \lambda_n(e_n - e_0)$ then $x - e_0 = (1 - \lambda_1 - \dots - \lambda_n)(e_0 - e_0) + \lambda_1(e_1 - e_0) + \dots + \lambda_n(e_n - e_0)$ and we conclude.

Exercise 5.6. Any vector space is an affine space over itself. We denote the vectors in $\mathbb{R}^3$ as (x, y, z) . Prove that the planes defined as $P_a = \{(x, y, z) | x + y + z = a, a \in \mathbb{R}\}$ are parallel affine subsets of $\mathbb{R}^3$. Identify barycentric and affine coordinates on P_a . We denote $e_0 = (1, 0, 0)$, $e_1 = (0, 1, 0)$ and $e_2 = (0, 0, 1)$. The planes P_a can be parametrized as $P_a = \{(a - y - z, y, z) | y, z \in \mathbb{R}\}$. Denoting $a = (a, 0, 0)$ and $\vec{v}_1 = e_1 - e_0$ and $\vec{v}_2 = e_2 - e_0$ we have $P_a = a + y\vec{v}_1 + z\vec{v}_2$, with + the addition in $\mathbb{R}^3$, hence they are all parallel affine subspaces in $\mathbb{R}^3$. An affine basis in P_a is $e_0^a = (a, 0, 0)$, $e_1^a = (0, a, 0)$ and $e_2^a = (0, 0, a)$: any element in P_a decomposes in this basis as $(1 - \lambda_1 - \lambda_2)e_0^a + \lambda_1 e_1^a + \lambda_2 e_2^a = (1 - a\lambda_1 - a\lambda_2, a\lambda_1, a\lambda_2)$ and an affine frame is $(o = e_0^a, \vec{v}_1 = e_1^a - e_0^a, \vec{v}_2 = e_2^a - e_0^a)$.

Chapter 6

Matrices

We will now be more precise with the notation and distinguish between vectors, forms and operators and their expressions in various bases. We will denote $\mathbf{x}$ the vectors in the (finite dimensional) vector space X , $\varphi \in X^*$ the linear forms and $\mathcal{A}: X \rightarrow Y$ the linear operators from X to Y .

Once we pick a basis $\{\mathbf{e}_i\}_{i=1,\dots,N}$, any vector will decompose uniquely as $\mathbf{x} = \sum_{i=1}^N x^i \mathbf{e}_i$, with x^i the components of the vector in the basis. These components can be organized in a *column vector*:

$$x = \begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^N \end{pmatrix} = x^1 e_1 + \dots + x^N e_N, \quad e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots \quad e_N = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

where the entry i in the column vector x corresponding to $\mathbf{x} \in X$ is the component x^i of the vector in the direction $\mathbf{e}_i$. In particular, the basis vectors are columns with only one entry 1 and the other 0. In a basis, any vector space X is identified with the vector space $\mathbb{R}^N$ or $\mathbb{C}^N$ of column vectors.

Choosing a basis in the N dimensional vector space X and a basis in the M dimensional vector space Y allows us to represent the linear operators $\mathcal{A}: X \rightarrow Y$ as $M \times N$ *matrices*, that is a table of numbers with M rows and N columns. In detail, let $\{\mathbf{e}_j\}_{j=1\dots N}$ be a basis in X and $\{\theta^j\}_{j=1\dots N}$ its dual basis in X^* , that is $\theta^j(\mathbf{e}_k) = \delta_k^j$, $\theta^j(\sum_{k=1}^N x^k \mathbf{e}_k) = x^j$ and respectively let $\{\mathbf{f}_i, \mu^i\}_{i=1,\dots,M}$ be a pair of dual bases in Y and Y^* . For any vector in $\mathbf{y} \in Y$ we have $\mathbf{y} = \sum_{i=1}^M y^i \mathbf{f}_i = \sum_{i=1}^M \mu^i(\mathbf{y}) \mathbf{f}_i$, and the linear operator $\mathcal{A}: X \rightarrow Y$ decomposes in the tensor product basis $\mathbf{f}_i \otimes \theta^j$ as :

$$\begin{aligned} \mathcal{A}\mathbf{x} &= \sum_{i=1}^N \mathbf{f}_i \mu^i(\mathcal{A}(\mathbf{x})) = \sum_{i=1}^N \mathbf{f}_i \mu^i \left(\mathcal{A} \left(\sum_{j=1}^M x^j \mathbf{e}_j \right) \right) = \sum_{i=1}^N \sum_{j=1}^M \mathbf{f}_i [\mu^i(\mathcal{A}\mathbf{e}_j)] x^j = \sum_{i=1}^N \sum_{j=1}^M [\mu^i(\mathcal{A}\mathbf{e}_j)] \mathbf{f}_i \theta^j(\mathbf{x}) \\ \Rightarrow \mathcal{A} &= \sum_{i=1}^N \sum_{j=1}^M [\mu^i(\mathcal{A}\mathbf{e}_j)] \mathbf{f}_i \otimes \theta^j. \end{aligned}$$

We denote $\mu^i(\mathcal{A}\mathbf{e}_j) = A_{ij}^i = a_{ij}$ and call $(a_{ij})_{i=1\dots N, j=1\dots M}$ the *matrix elements* of $\mathcal{A}$ in the given bases. For instance, the identity operator on a vector space V , $\mathcal{I}: V \rightarrow V$, $\mathcal{I}(v) = v$ writes as $\mathcal{I} = \sum_{i=1}^N \mathbf{e}_i \otimes \theta^i$ and has matrix elements $I_{ij}^i = \delta_{ij}$.

As a point of notation, we use calligraphic letters $\mathcal{A}, \mathcal{B}$ and so on for operators, capital letters A, B and so on for the matrices of these operators in some bases, capital letters A_{ij}^i, B_{ij}^i for the matrix elements if we track the position of the indices and lower case letters $A_{ij}^i = a_{ij}, B_{ij}^i = b_{ij}$ if we write all the indices in subscript.

The numbers $(a_{ij})_{i=1\dots N, j=1\dots M}$ form a matrix, that is a table with M rows and N columns, having thus $M \times N$ entries a_{ij} , which are just (real or complex) numbers. We denote $\mathfrak{M}_{M \times N}(\mathbb{F})$ the space of $M \times N$ matrices whose entries a_{ij} belong

to $\mathbb{F} = \mathbb{R}, \mathbb{C}$:

$$A \in \mathfrak{M}_{M \times N}(\mathbb{F}), \quad A = \begin{pmatrix} a_{11} & \dots & a_{1N} \\ \vdots & & \\ a_{M1} & \dots & a_{MN} \end{pmatrix}.$$

The action of the operator $\mathcal{A} : X \rightarrow Y$ on a vector $\mathbf{x}$ is the *row-column multiplication* of the corresponding matrix $A \in \mathfrak{M}_{M \times N}(\mathbb{F})$ with column vector $x \in \mathbb{F}^N$, that is the component i in the basis $\mathbf{f}_i$ of the vector $\mathcal{A}\mathbf{x}$ is:

$$\mu^i(\mathcal{A}\mathbf{x}) \equiv (Ax)^i = \sum_{j=1}^N A_{ij}^i x^j = \sum_{j=1}^N a_{ij} x^j,$$

which writes in terms of tables of numbers:

$$A = \begin{pmatrix} a_{11} & \dots & a_{1N} \\ \vdots & & \\ a_{M1} & \dots & a_{MN} \end{pmatrix}, \quad x = \begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^N \end{pmatrix}, \quad Ax = \begin{pmatrix} a_{11} & \dots & a_{1N} \\ \vdots & & \\ a_{M1} & \dots & a_{MN} \end{pmatrix} \begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^N \end{pmatrix} = \begin{pmatrix} a_{11}x^1 + \dots + a_{1N}x^N \\ \vdots \\ a_{M1}x^1 + \dots + a_{MN}x^N \end{pmatrix}.$$

In words, one multiplies term by term the entries in the rows in A with the entries in the column vector and adds up the results.

The linear forms $\varphi : X \rightarrow \mathbb{F}$ are represented as $1 \times N$ matrices, that is line vectors. The form $\varphi = \sum_{i=1}^N \varphi_i \theta^i$ acts on vectors as $\varphi(\mathbf{x}) = \sum_{j=1}^N \varphi_j \theta^j(\mathbf{x}) = \sum_{j=1}^N \varphi_j x^j$ and it is represented by the line vector:

$$\varphi = (\varphi_1 \quad \varphi_2 \quad \dots \quad \varphi_N), \quad \theta^1 = (1 \quad 0 \quad \dots \quad 0), \quad \theta^2 = (0 \quad 1 \quad \dots \quad 0), \quad \theta^N = (0 \quad 0 \quad \dots \quad 1).$$

Note that, while the components $x^j = \theta^j(\mathbf{x})$ of the vector $\mathbf{x}$ and the matrix elements $A_{ij}^i = \mu^i(\mathcal{A}\mathbf{e}_j)$ of the operator $\mathcal{A}$ depend the bases we have chosen, the vectors $\mathbf{x}$ and $\mathcal{A}\mathbf{x}$ are basis independent. We will see later that, in order to reflect this, the components of vectors and the matrix elements of operators change in a precise way when changing basis. The identity operator is special: its matrix elements in any basis are $I_j^i = \delta_{ij}$.

6.1 The algebra of matrices

Once we fixed bases, the set of operators is $B_{\text{op.}}(X, Y)$ with $\dim X = N$ and $\dim Y = M$ is represented by the set of matrices $\mathfrak{M}_{M \times N}(\mathbb{F})$ and consequently the matrices form a vector space with $(A+B)_{ij}^i = a_{ij} + b_{ij}$ and $(\alpha A)_{ij}^i = \alpha a_{ij}$, that is entry-wise addition and multiplication by scalars. The dimension of this vector space is $M \times N$, and a basis consists in the matrices associated to $\mathbf{f}_i \otimes \theta^j$:

$$(\mathbf{f}_i \otimes \theta^j)_l^k \equiv (E_{(ij)})_l^k = \delta_i^k \delta_l^j \quad E_{(ij)} = \begin{matrix} & & j \\ & & \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & & & & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & & & & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} \end{matrix}.$$

In order to simplify the notation we often suppress the tensor product symbol.

Linearity operators can be composed, that is if $\mathcal{B} : X \rightarrow Y$ and $\mathcal{A} : Y \rightarrow Z$ with $\dim(X) = N$, $\dim Y = M$, and $\dim(Z) = P$ are linear operators, then $\mathcal{A} \circ \mathcal{B} : X \rightarrow Z$ is a linear operator. Fixing bases $\{\mathbf{e}_j, \theta^j\}$, $\{\mathbf{f}_i, \mu^i\}$, $\{\mathbf{h}_k, \rho^k\}$ in X, Y respectively

Z , and taking into account that the identity in Y is $\sum_i \mathbf{f}_i \theta^i$, the matrix element of the composition is:

$$\rho^k[\mathcal{A} \circ \mathcal{B}(\mathbf{e}_j)] \equiv (A \cdot B)^k_j = \sum_{i=1}^{M=\dim Y} \rho^k[\mathcal{A}(\mathbf{f}_i)] \mu^i[\mathcal{B}(\mathbf{e}_j)],$$

corresponding to the the *row-column product* of then two matrices:

$$A \in \mathfrak{M}_{P \times M}(\mathbb{F}), \quad B \in \mathfrak{M}_{M \times N}(\mathbb{F}), \quad A \cdot B \in \mathfrak{M}_{P \times N}(\mathbb{F}), \quad (A \cdot B)^k_j = \sum_{i=1}^M A^k_i B^i_j = \sum_{i=1}^M a_{ki} b_{ij}.$$

In terms of tables of numbers one multiplies term by term the rows in A with columns in B and adds up the results:

$$\begin{pmatrix} a_{11} & \dots & a_{1M} \\ \vdots & & \\ a_{P1} & \dots & a_{PM} \end{pmatrix} \begin{pmatrix} b_{11} & \dots & b_{1N} \\ \vdots & & \\ b_{M1} & \dots & b_{MN} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + \dots + a_{1M}b_{M1} & \dots & a_{11}b_{1N} + \dots + a_{1M}b_{MN} \\ \vdots & & \\ a_{P1}b_{11} + \dots + a_{PM}b_{M1} & \dots & a_{P1}b_{1N} + \dots + a_{PM}b_{MN} \end{pmatrix}.$$

In particular the basis matrices are matrix products $E_{(ij)} = \mathbf{f}_i \cdot \theta^j$. Note that the identity matrix with matrix elements $I^i_j = \delta_{ij}$ is an identity for the matrix product $A \cdot I = I \cdot A = A$.

Definition 6.1 (**Algebra over a field**). *An algebra $\mathfrak{M}$ over a field of scalars $\mathbb{F}$ is a vector space over $\mathbb{F}$ endowed with a bilinear product $\cdot : \mathfrak{M} \times \mathfrak{M} \rightarrow \mathfrak{M}$. The bilinearity of the product can be states equivalently as:*

- (right distributive) $(A + B) \cdot C = A \cdot C + B \cdot C$ for any $A, B, C \in \mathfrak{M}$,
- (left distributive) $A \cdot (B + C) = A \cdot B + A \cdot C$ for any $A, B, C \in \mathfrak{M}$,
- (compatible with scalars) $(\alpha A) \cdot (\beta B) = (\alpha\beta)(A \cdot B)$, for any $\alpha, \beta \in \mathbb{F}$ and $A, B \in \mathfrak{M}$.

The algebra is associative if $\cdot$ is associative $A \cdot (B \cdot C) = (A \cdot B) \cdot C$, commutative if $A \cdot B = B \cdot A$, and unital if $\cdot$ has a unit element $I \cdot A = A \cdot I = A$.

The set of $N \times N$ matrices, $\mathfrak{M}_{N \times N}(\mathbb{F})$ with the matrix product is an associative unital but non commutative algebra over $\mathbb{F}$. The same holds for the set of operators $B_{\text{op.}}(X, X)$ with the composition. For convenience, we supress from now on the $\cdot$ in the matrix product.

Theorem 6.1 (**The left and right inverses of a matrix are equal**). *Let A be a $N \times N$ matrix. The left inverse of A , if it exist, is the matrix L such that $LA = I$, and the right inverse is the matrix R such that $AR = I$. If the left inverse of a matrix exists then the right inverse of the matrix exists and the two are equal.*

In any associative unital algebra, if both the left and the right inverses exist then they are equal.

Proof. Consider a finite dimensional vector space V , and $\Phi : V \rightarrow V$ injective ($\Phi(v_1) = \Phi(v_2) \Rightarrow v_1 = v_2 \Rightarrow \ker(\Phi) = 0$). By the rank nullity theorem, $\dim(\text{Im}(\Phi)) = \dim V$ hence $\text{Im}(\Phi) = V$ and Φ is surjective.

Consider A a matrix with left inverse L . We define the linear operator $\Phi : \mathfrak{M}_{N \times N}(\mathbb{F}) \rightarrow \mathfrak{M}_{N \times N}(\mathbb{F})$, $\Phi(X) = AX$. Then $\Phi(X) = 0 \Leftrightarrow AX = 0 \Leftrightarrow LAX = X = 0$, hence $\dim(\ker(\Phi)) = 0$ and Φ is surjective and there exist R such that $AR = I$.

If both the left and right inverses exists then $L = L(AR) = (LA)R = R$. $\square$

Note that this is trivial when recalling that a matrix represents an operator in some basis. By the rank nullity theorem an operator $\mathcal{A} : X \rightarrow X$ is surjective if and only if it is injective (it has zero kernel) and in this case it is invertible.

6.2 Change of basis

Vectors and linear operators are intrinsic notions and do not depend on a particular basis, but their representations (components of vectors, matrix elements and so on) do. For instance the matrix associated to an operator $\mathcal{A} : X \rightarrow Y$ in some bases has entries $a_{ij} = \boldsymbol{\mu}^i(\mathcal{A}\mathbf{e}_j)$ hence depends on the basis vector $\mathbf{e}_i$ and forms $\boldsymbol{\mu}^i$.

Let us assume we have a basis $\mathbf{e}_i$ in X and we select a new basis $\tilde{\mathbf{e}}_i$. As both $\mathbf{e}_i$ and $\tilde{\mathbf{e}}_i$ are bases, we can decompose the vectors in the new basis as linear combinations of the vectors in the old one:

$$\tilde{\mathbf{e}}_i = \sum_{j=1}^N \mathbf{e}_j S_{ij}^j, \quad \tilde{\mathbf{e}}_i = \begin{pmatrix} S_{i1}^1 \\ \vdots \\ S_{iN}^N \end{pmatrix}, \quad S = \begin{pmatrix} \tilde{\mathbf{e}}_1 & \dots & \tilde{\mathbf{e}}_N \end{pmatrix},$$

where the last notation signifies that the matrix S is just the matrix obtained by juxtaposing the column vectors $\tilde{\mathbf{e}}_j$ corresponding to the vector $\tilde{\mathbf{e}}_j$ decomposed in the basis $\mathbf{e}_i$. We can of course expand $\mathbf{e}_i$ on the basis $\tilde{\mathbf{e}}_i$, $\mathbf{e}_i = \sum_j \tilde{\mathbf{e}}_j Q_{ji}^j = \sum_{j,l} \mathbf{e}_l S_{jl}^l Q_{ji}^j$ and, as the $\mathbf{e}_i$ s are linearly independent, we conclude that $Q = S^{-1}$, hence in particular S is invertible.

The dual basis changes accordingly (where all the sums are understood to run from 1 to N):

$$\tilde{\boldsymbol{\theta}}^i = \sum_j (S^{-1})^i_j \boldsymbol{\theta}^j, \quad \tilde{\boldsymbol{\theta}}^i(\tilde{\mathbf{e}}_k) = \sum_j (S^{-1})^i_j \boldsymbol{\theta}^j \left(\sum_l \mathbf{e}_l S_{lk}^l \right) = \sum_{j,l} (S^{-1})^i_j \delta_l^j S_{lk}^l = \delta_k^i.$$

The components of a vector $\sum_i v^i \mathbf{e}_i$ and of a form $\sum_i \varphi_i \boldsymbol{\theta}^i$, with $v^i = \boldsymbol{\theta}^i(v)$ and $\varphi_i = \varphi(\mathbf{e}_i)$ transform as:

$$\tilde{v}^i = \tilde{\boldsymbol{\theta}}^i(v) = \sum_j (S^{-1})^i_j \boldsymbol{\theta}^j(v) = \sum_j (S^{-1})^i_j v^j, \quad \tilde{\varphi}_i = \varphi(\tilde{\mathbf{e}}_i) = \sum_j \varphi(\mathbf{e}_j) S_{ij}^j = \sum_j \varphi_j S_{ij}^j,$$

and we say that φ_i transforms *covariantly*, as it transforms like the basis elements $\mathbf{e}_i$, while $\boldsymbol{\theta}^i$ and v^i transform *contravariantly*. The matrix elements of an operator $\mathcal{A} : X \rightarrow X$ transform under a change of basis as:

$$\tilde{A}_{ij}^i = \tilde{\boldsymbol{\theta}}^i(\mathcal{A}\tilde{\mathbf{e}}_j) = \sum_l (S^{-1})^i_l \boldsymbol{\theta}^l \left(\mathcal{A} \sum_q \mathbf{e}_q S_{jq}^q \right) = \sum_{l,a} (S^{-1})^i_l A^l_a S_{jq}^q = [S^{-1}AS]^i_j.$$

Note that, consistently, $\tilde{A}\tilde{v} = (S^{-1}AS)S^{-1}v = S^{-1}(Av)$.

Two matrices related by a transformation $\tilde{A} = S^{-1}AS$, for S some invertible matrix are called similar, and similar matrices correspond to the same operator expressed in different bases. The matrix of an operator $\mathcal{A} : X \rightarrow Y$ changes under a change of bases in both X and Y as $\tilde{A} = T^{-1}AS$.

Definition 6.2 (The trace). The trace of a square matrix is the sum of its diagonal elements, $\text{Tr}(A) = \sum_i A^i_i = \sum_i a_{ii}$. The trace is cyclical $\text{Tr}(AB) = \text{Tr}(BA)$ and it is invariant under a change of basis, $\text{Tr}(S^{-1}AS) = \text{Tr}(A)$.

As it is common in the literature, when no risk of confusion arises, we will conflate the operators with their matrix representation in some basis. We will then talk about the kernel of a matrix consisting in vectors in the kernel of the corresponding operator expressed in the appropriate basis, and so on, of course keeping in mind that the matrix is basis dependent.

In order to simplify the notation, from now on we write the matrix indices in subscript.

6.3 Gaussian elimination

Matrices can be brought into a standard form by linear manipulations. The algorithm we present below is used to determine the kernel and image of a matrix, to solve system of linear equations with constant coefficients and to compute the inverse of a matrix (if it exists).

Consider $A \in \mathfrak{M}_{M \times N}(\mathbb{F})$ a $M \times N$ matrix. The process of Gaussian elimination simplifies the matrix thorough a sequence of the following steps:

- swapping two rows, say i and j of the matrix A , that is $A \rightarrow SA$ with $S_{nn} = 1$ if $n \neq i, j$, $S_{ij} = S_{ji} = 1$ and all the other entries set to 0.
- multiplying one row by a non zero constant, $a_{i\bullet} \rightarrow \alpha a_{i\bullet}$ (the dot signifies that we apply the operations along the entire row), that is $A \rightarrow D_{\alpha}^i A$ with $(D_{\alpha}^i)_{kl} = \delta_{kl} + (\alpha - 1)\delta_{ik}\delta_{il}$:

$$(D_{\alpha}^i A)_{kq} = \sum_l (\delta_{kl} + (\alpha - 1)\delta_{ik}\delta_{il})a_{lq} = a_{kq} + (\alpha - 1)\delta_{ik}a_{iq}.$$

- taking a linear combination of two rows, $a_{i\bullet} \rightarrow a_{i\bullet} + \alpha a_{j\bullet}$, that is $A \rightarrow L_{\alpha}^{i,j} A$ with $(L_{\alpha}^{i,j})_{kl} = \delta_{kl} + \alpha\delta_{ik}\delta_{jl}$:

$$(L_{\alpha}^{i,j} A)_{kq} = \sum_l (\delta_{kl} + \alpha\delta_{ik}\delta_{jl})a_{lq} = a_{kq} + \alpha\delta_{ik}a_{jq}.$$

Exercise 6.1. Prove that $S^2 = D_{\alpha}^i D_{-\alpha}^i = L_{\alpha}^{i,j} L_{-\alpha}^{i,j} = I$.

As all the elementary operations are invertible, the matrix obtain after any sequence of such operations can be written as $B = FA$ for F some invertible $M \times M$ matrix.

Construction 6.1 (**Gaussian elimination**). Using a combination of these operations one reduces the matrix as follows:

- if an entry on the first column is non zero, by permuting the rows we can always bring it in the position a_{11} . For all the rows with $i > 1$, we take the row linear combination $a_{i\bullet} \rightarrow a_{i\bullet} - \frac{a_{i1}}{a_{11}}a_{1\bullet}$. This sets all the entries on the first column a_{i1} with $i > 1$ to 0 and we pass to the next column.
- when we reached the column k , we only look at the rows with $i \geq k$. If any of these rows has a non zero entry in the column k , by row permutations we can bring it in the position a_{kk} . For all the rows below, that is with $i > k$, we take the linear combination $a_{i\bullet} \rightarrow a_{i\bullet} - \frac{a_{ik}}{a_{kk}}a_{k\bullet}$, which sets all $a_{ik} = 0$ for $i > k$ and we pass to the next column.
- at the end of these sequence of steps we obtain the row echelon form of the matrix, which is such that:
 - all the rows with only zero entries are at the bottom.
 - the left-most nonzero entry in every nonzero row (called the leading entry, or the pivot) is on the right of the leading entry of every row above.
- we use rescalings to set all the pivot entries to 1, and linear combinations of the rows to set to zero all the elements above the pivots. This leads to the reduced row echelon form (RREF) of the matrix, which is unique.

We call dominant the columns in the RREF which contain the pivots, and non dominant the other ones.

An example of Gaussian elimination applied to a matrix goes as follows:

$$\begin{pmatrix} 1 & 3 & 1 & 9 \\ 1 & 1 & -1 & 1 \\ 3 & 11 & 5 & 35 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 1 & 9 \\ 0 & -2 & -2 & -8 \\ 0 & 2 & 2 & 8 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 1 & 9 \\ 0 & -2 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & -2f_1 + f_2 & -3f_1 + 4f_2 \end{pmatrix},$$

where we denoted f_1, f_2, f_3 the standard basis in $\mathbb{R}^3$.

Theorem 6.2 (**The RREF of a matrix is unique**). *The RREF of a matrix A is unique.*

Proof. The dominant columns in a RREF are the elements in the standard basis $f_1, \dots, f_M$ in $\mathbb{R}^M$ encountered in this order. The non dominant columns are linear combinations of the dominant columns which situated at their left, with coefficients exactly the entries in the column. It follows that a RREF writes on columns like:

$$\begin{pmatrix} f_1 & \dots & f_k & \sum_{i=1}^k \alpha_{ik+1} f_i & \sum_{i=1}^k \alpha_{ik+2} f_i & \dots & f_l & \dots \end{pmatrix}.$$

If a column is identically zero, we see it as a linear combination with zero coefficients of the columns at its left. In particular, any matrix which can be obtain from B as a product FB with F some invertible matrix will have the form:

$$\begin{pmatrix} Ff_1 & \dots & Ff_k & \sum_{i=1}^k \alpha_{ik+1} Ff_i & \sum_{i=1}^k \alpha_{ik+2} Ff_i & \dots & Ff_l & \dots \end{pmatrix}.$$

If B' is a RREF of A , then $B' = FB$ and its dominant columns must reproduce $f_1, \dots, f_M$ in $\mathbb{R}^M$ in this order. Thus the first k dominant columns must respect $Ff_1 = f_1, Ff_2 = f_2$, up to $f_k = Ff_k$. Then the first non dominant column is $\sum_{i=1}^k \alpha_{ik+1} Ff_i$ which equals $\sum_{i=1}^k \alpha_{ik+1} f_i$ and iterating we conclude that $B' = B$. $\square$

6.3.1 Finding bases in the kernel and the image of a matrix

The columns of the matrix A can also be interpreted as the column vectors Ae_i , with $e_1, \dots, e_N$ the standard basis in $\mathbb{R}^N$. As $A = FB$ with F invertible and B the RREF of A , it follows that:

$$\begin{pmatrix} Ae_1 & \dots & Ae_N \end{pmatrix} = \begin{pmatrix} Ff_1 & \dots & Ff_k & \sum_{i=1}^k \alpha_{ik+1} Ff_i & \sum_{i=1}^k \alpha_{ik+2} Ff_i & \dots & Ff_l & \dots \end{pmatrix},$$

and we conclude that:

- a basis in the image of A , which is the span of its columns $Ae_1, \dots, Ae_N$, is $Ff_1, Ff_2, \dots, Ff_l, \dots$ (from the RREF it is clear these vector are spanning, and they are linear independent as F is invertible). A basis in the image is given by the dominant columns of A , that is the columns which contain the pivots after reduction to the RREF.
- each non dominant column of A yields a relation $Ae_j = \sum_{i < j} \alpha_{ij} Ae_i$, hence a non trivial vector $v_j = e_j - \sum_{i < j} \alpha_{ij} e_i \in \ker(A)$. All such vectors are linearly independent. Indeed, denoting J the set of indices of the non dominant columns, let us assume that:

$$\sum_{j \in J} \gamma_j v_j = \sum_{j \in J} \gamma_j \left(e_j - \sum_{i < j} \alpha_{ij} e_i \right) = 0.$$

Denoting $j_{\max}$ the maximal index in J , $e_{j_{\max}}$ is multiplied only by $\gamma_{j_{\max}}$ in this linear combination, hence $\gamma_{j_{\max}} = 0$ as the e_j s are linearly independent. Iterating we conclude that $\gamma_j = 0$. As we have built $N - \dim(\text{Im } A)$ linearly independent vectors in $\ker(A)$, they form a basis by the rank-nullity theorem.

Consider for instance the matrix A with RREF B as before:

$$A = \begin{pmatrix} 1 & 3 & 1 & 9 \\ 1 & 1 & -1 & 1 \\ 3 & 11 & 5 & 35 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & -2f_1 + f_2 & -3f_1 + 4f_2 \end{pmatrix},$$

where we denoted f_1, f_2, f_3 the standard basis in $\mathbb{C}^3$. We have the bases:

$$\text{Im}(A): \quad Ae_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, \quad Ae_2 = \begin{pmatrix} 3 \\ 1 \\ 11 \end{pmatrix}, \quad \ker(A): \quad e_3 + 2e_1 - e_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad e_4 + 3e_1 - 4e_2 = \begin{pmatrix} 3 \\ -4 \\ 0 \\ 1 \end{pmatrix}.$$

Indeed, $e_3 + 2e_1 - e_2$ and $e_4 + 3e_1 - 4e_2$ are linearly independent and:

$$\begin{pmatrix} 1 & 3 & 1 & 9 \\ 1 & 1 & -1 & 1 \\ 3 & 11 & 5 & 35 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 - 3 + 1 \\ 2 - 1 - 1 \\ 6 - 11 + 5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 & 1 & 9 \\ 1 & 1 & -1 & 1 \\ 3 & 11 & 5 & 35 \end{pmatrix} \begin{pmatrix} 3 \\ -4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 - 12 + 9 \\ 3 - 4 + 1 \\ 9 - 44 + 35 \end{pmatrix}.$$

One can reconstruct $B = L_{-3}^{1;2} D_{-\frac{1}{2}}^2 L_1^{3;2} L_{-3}^{3;1} L_{-1}^{2;1} A$.

6.3.2 Solving systems of linear equations with constant coefficients

Any system of M linear equations with N unknowns and constant coefficients:

$$a_{11}x^1 + \cdots + a_{1N}x^N = y^1, \quad a_{21}x^1 + \cdots + a_{2N}x^N = y^2, \quad \dots, \quad a_{M1}x^1 + \cdots + a_{MN}x^N = y^M,$$

can be written as a matrix equation $Ax = y$ with:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1N} \\ \vdots & & \\ a_{M1} & \cdots & a_{MN} \end{pmatrix}, \quad x = \begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^N \end{pmatrix}, \quad y = \begin{pmatrix} y^1 \\ y^2 \\ \vdots \\ y^N \end{pmatrix}.$$

There are three cases:

- $y \notin \text{Im}(A)$. In this case the system has no solutions (the system is incompatible).
- $y \in \text{Im}(A)$ and $\ker(A) = 0$. In this case the system has a unique solution (the system is compatible determined).
- $y \in \text{Im}(A)$ and $\ker(A) \neq 0$. In this case the system has an infinity of solutions (the system is compatible undermined).
The set of solutions $\{x, Ax = y\}$ is an affine space over $\ker(A)$, that is given x_0 a particular solution of $Ax_0 = y$, $\{x, Ax = y\}$ is the affine subspace $x_0 + \ker(A)$ of $\mathbb{C}^N$.

We define the *augmented matrix* of the system $[A|y]$ as the matrix consisting in the matrix A with an additional column y . In order to determine in which of the three cases we are (and solve the system if possible), we observe that x is a solution of $Ax = y$ if and only if x is a solution of $RAx = Ry$ for any invertible R , that is x is a solution of the system with augmented matrix $[RA|Ay]$. The system becomes trivial if $[A|y]$ is brought in RREF, $[B|Y]$ namely:

- if the last column, Y , is dominant, then in the system of equations corresponding to the RREF one of the equations reads $0 = 1$, hence the system is incompatible.
- if the last column, Y , is non dominant, then:
 - either $M = N$ and the RREF of A is the identity, $B = I$. In this case the system has a unique solution $x = Y$.

- or the non dominant columns in B lead to a basis in $\ker(A)$ and a particular solution is $x_0^{j(i)} = Y^i$, where $j(i)$ is the label of the i 'th dominant column, and we complete x_0 with 0 on all other entries.

As examples, we consider the systems of equations with augmented matrices (we eliminate the vertical line separating y in order to simplify the notation) below:

$$\begin{pmatrix} 1 & 1 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 3 & 2 & -4 & 7 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -3 & 4 \\ 0 & -1 & 5 & -6 \\ 0 & -1 & 5 & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -3 & 4 \\ 0 & -1 & 5 & -6 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The last column is dominant and the system incompatible (no solution).

$$\begin{pmatrix} 1 & 1 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 3 & 2 & -4 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -3 & 4 \\ 0 & -1 & 5 & -6 \\ 0 & -1 & 5 & -6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -3 & 4 \\ 0 & -1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 & -2 \\ 0 & 1 & -5 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

A basis in $\ker(A)$ is $e_3 - 2e_1 + 5e_2$, and the general solution writes:

$$x = \begin{pmatrix} -2 \\ 6 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 2 & 4 & 23 \\ 1 & 1 & 1 & 7 \\ 4 & 2 & 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 4 & 23 \\ 0 & -1 & -3 & -16 \\ 0 & -6 & -15 & -90 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 4 & 23 \\ 0 & 1 & 3 & 16 \\ 0 & 0 & 3 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 4 & 23 \\ 0 & 1 & 3 & 16 \\ 0 & 0 & 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -2 & -9 \\ 0 & 1 & 3 & 16 \\ 0 & 0 & 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

The solution of the system is unique:

$$x = \begin{pmatrix} -5 \\ 10 \\ 2 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & -1 & 1 & -1 & 1 \\ 2 & 1 & -3 & 0 & 2 \\ 5 & -2 & 0 & -3 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 & -1 & 1 \\ 0 & 3 & -5 & 2 & 0 \\ 0 & 3 & -5 & 2 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -\frac{2}{3} & -\frac{1}{3} & 1 \\ 0 & 1 & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

The solution of the system is:

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} \frac{2}{3} \\ \frac{5}{3} \\ 1 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ 0 \\ 1 \end{pmatrix}.$$

6.3.3 Computing the matrix inverse

Let A be a $N \times N$ matrix. The inverse of A is such that $AC = I$ or, looking at the columns, the column vectors of the inverse solve the equations $AC_i = e_i$, with e_i the i 'th basis vector in the standard basis. The augmented matrix of these N

systems of N equations is $[A|I]$, where the i 'th column of the identity matrix is e_i . All these systems can be simultaneously solved if and only if the RREF of A is I and in this case the RREF of $[A|I]$ is $[I|C]$. For instance:

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 4 & 2 & 1 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 4 & 1 & 0 & 0 \\ 0 & -1 & -3 & -1 & 1 & 0 \\ 0 & -6 & -15 & -4 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -2 & -1 & 2 & 0 \\ 0 & 1 & 3 & 1 & -1 & 0 \\ 0 & 0 & 3 & 2 & -6 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -2 & -1 & 2 & 0 \\ 0 & 1 & 3 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{2}{3} & -2 & \frac{1}{3} \end{pmatrix} \\ \rightarrow \begin{pmatrix} 1 & 0 & 0 & \frac{1}{3} & -2 & \frac{2}{3} \\ 0 & 1 & 0 & -1 & 5 & -1 \\ 0 & 0 & 1 & \frac{2}{3} & -2 & \frac{1}{3} \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 4 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & -2 & \frac{2}{3} \\ -1 & 5 & -1 \\ \frac{2}{3} & -2 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Note that A is invertible if and only if $\ker(A) = 0$, that is the only solution of $Ax = 0$ is $x = 0$. If A is not invertible, then $Ax = 0$ admits a non trivial solution.

6.4 The determinant

The determinant of a $N \times N$ matrix is a degree N homogeneous polynomial in the matrix entries with some special properties.

We denote S_n the symmetric group, that is the group of permutations of n elements with group operation the composition:

$$S_n = \{ \sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}, \sigma \text{ bijection} \}, \quad |S_n| = n!.$$

The composition of permutations is also called the product of permutations. A permutation is sometimes denoted in a two line notation:

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & N \\ \sigma(1) & \sigma(2) & \dots & \sigma(N) \end{pmatrix},$$

and a rearrangement (permutation) of the columns denotes the same permutation.

Any permutation can be decomposed into disjoint cycles $\sigma = (p, \sigma(p), \sigma^2(p) \dots)(q, \sigma(q) \dots) \dots$, where each cycle collects all the iterated images $p, \sigma(p), \dots$ in the order in which they are encountered. A fixed point $\sigma(p) = p$ is denoted (p) . A cyclical permutation of the elements in a cycle leads to the same cycle $(p, \sigma(p) \dots \sigma^{-1}(p)) = (\sigma^{-1}(p)p\sigma(p) \dots)$. For instance

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} = (1)(2)(34), \quad \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = (12)(34), \quad \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix} = (123)(4), \quad \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = (1234).$$

We denote d_i the number of cycles of length i in the cycle decomposition of σ , and the cycle type of σ is the collection $\{d_i\}_{i \geq 1}$. Two permutations have the same cycle type $\{d_i\}_{i \geq 1}$ if and only if they are conjugated $\sigma_1 = \pi \sigma_2 \pi^{-1}$ for some $\pi \in S_n$.

Exercise 6.2. *Prove that two permutations have the same cycle type if and only if they are conjugated. If $\sigma_2(i) = j$ and $\sigma_1 = \pi \sigma_2 \pi^{-1}$ then $\sigma_1(\pi(i)) = \pi(j)$. Listing the cycles of σ_2 and the replacing all the a by $\pi(a)$ we obtain the cycles of σ_1 . Conversely, we list the cycles of σ_2 on a line and the cycles of σ_1 immediately below, making sure that the cycles match in length top and bottom, interpreting this as a two line notation for a permutation π , $\sigma_1 = \pi \sigma_2 \pi^{-1}$.*

Exercise 6.3. *Prove that there are $\frac{n!}{\prod_i d_i! i^{d_i}}$ distinct permutations with a cycle type $\{d_i\}_{i \geq 1}$. Prove that $\sum_{d_i \geq 0}^{\sum i d_i = n} \frac{n!}{\prod_i d_i! i^{d_i}} = n!$. We list the permutation τ in cycle notation. The $n!$ permutations π of the labels (balls) while keeping the cycles of τ (boxes) fixed generates all the permutations of the same cycle type. However, the permutations are over counted: a permutation π whose only effect is to permute circularly the labels in a cycle of τ lead to the same τ , and a permutation π whose only effect is to permute all the labels in a cycle with all the labels in another cycle of the same length preserving their order also leads to the same τ , hence τ is generated $\prod_i d_i! i^{d_i}$ times by the $n!$ permutations of labels π . For the second part:*

$$\sum_{d_i \geq 0}^{\sum i d_i = n} \frac{n!}{\prod_i d_i! i^{d_i}} = n! \left[\sum_{d_i \geq 0} \prod_{i \geq 1} \frac{1}{d_i! i^{d_i}} x^{d_i} \right]_{x^n} = n! \left[e^{-\ln(1-x)} \right]_{x^n} = n!,$$

where $[\dots]_{x^n}$ denotes the coefficient of x^n in a formal power series.

Exercise 6.4. Prove that there are $\frac{n!}{\prod_i d_i!(i!)^{d_i}}$ ways to distribute N labelled balls into d_1 boxes of size 1, d_2 boxes of size 2 and so on. As before, with the exception that any permutation of the labels of all the balls in the same box leads to the same assignment of balls to boxes.

A transposition is a permutation with only one cycle of length two, and $n-2$ cycles of length 1. Any permutation can be written as a product, that is a composition, of transpositions. Indeed, any permutation is a product of its cycles completed by fixed points and any cycle writes as a product of transpositions (where the product is read left to right and we do not write the fixed points in the transpositions):

$$(a_1 \ a_2 \ a_3 \ \dots \ a_i) = (a_1 \ a_2) \cdot (a_2 \ a_3) \cdot \dots \cdot (a_{i-1} \ a_i).$$

A cycle of length i needs $i-1$ transpositions, and a permutation of a give cycle type needs $\sum_i d_i(i-1)$. The signature of the permutation $\epsilon(\sigma) = (-1)^{\sum_i d_i(i-1)}$ tracks the parity of the number of transpositions in such a decomposition. Equivalently, a permutation is odd (signature -1) if the number of cycles of even length of the permutation is odd and it is even (signature $+1$) otherwise. The same permutation can be decomposed in transpositions in many different ways, but the parity of the number of transposition is the same in all such decompositions.¹

The signature of a product of permutation is the product of signatures of the permutations (as we can first decompose both permutations in transpositions and then multiply to obtain a product of transpositions).

Exercise 6.5. Prove that $\epsilon(\tau) = \epsilon(\tau^{-1})$. τ and τ^{-1} have the same cycle type.

Exercise 6.6. Prove that a permutation can be written as the product of its cycles, completed by fixed points, for instance $(12)(34) = (12)(3)(4) \cdot (1)(2)(34)$.

Definition 6.3 (**The determinant**). The determinant of a square $N \times N$ matrix $A \in \mathfrak{M}_{N \times N}(\mathbb{F})$ with matrix elements a_{ij} is the polynomial:

$$\det(A) = \begin{vmatrix} a_{11} & \dots & a_{1N} \\ \vdots & & \\ a_{N1} & \dots & a_{NN} \end{vmatrix} = \sum_{\sigma \in S_N} \epsilon(\sigma) \prod_{i=1}^n a_{i\sigma(i)}.$$

The determinant is the unique function $\det : \mathfrak{M}_{N \times N}(\mathbb{F}) \rightarrow \mathbb{F}$ such that $\det(I) = 1$, is antisymmetric under permutations of the rows of A , linear in each row and zero if two rows coincide.² In detail, denoting $r_1, \dots, r_N$ the rows of the matrix:

$$\begin{vmatrix} r_1 + \tilde{r}_1 \\ r_2 \\ \vdots \\ r_N \end{vmatrix} = \begin{vmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{vmatrix} + \begin{vmatrix} \tilde{r}_1 \\ r_2 \\ \vdots \\ r_N \end{vmatrix}, \quad \begin{vmatrix} \alpha r_1 \\ \alpha r_2 \\ \vdots \\ \alpha r_N \end{vmatrix} = \alpha \begin{vmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{vmatrix}, \quad \begin{vmatrix} r_1 \\ \vdots \\ r_i \\ \vdots \\ r_N \end{vmatrix} = (-1)^{i+1} \begin{vmatrix} r_i \\ r_1 \\ \vdots \\ r_N \end{vmatrix}.$$

The same properties hold for columns, as $\det(A) = \det(A^t)$ with the transposed matrix defined as $(A^t)_{ij} = a_{ji}$.³

The minor $\det(A_{\hat{K}\hat{J}})$ of A is the determinant of the matrix obtained from A by erasing the rows with indices in K and columns with indices in J (this is defined only for $|K| = |J|$). Minors can also be defined for $M \times N$ matrices and correspond to determinants of square sub matrices.

¹The group S_n is the group of words with generators $\tau_i = (i, i+1)$ and relations $\tau_i^2 = 1$, $\tau_i \tau_{i+1} \tau_i = \tau_{i+1} \tau_i \tau_{i+1}$ for $i < n$ and $\tau_i \tau_j = \tau_j \tau_i$ if $|i-j| \geq 2$. All the relations preserve the parity of the number of letters of the words.

²Antisymmetry follows from noting that for any τ fixed permutation, $\sum_{\sigma \in S_N} \epsilon(\sigma) \prod_{i=1}^n a_{\tau(i)\sigma(i)} = \epsilon(\tau) \sum_{\sigma \in S_N} \epsilon(\sigma) \prod_{i=1}^n a_{i\sigma(i)}$ and trivially implies the determinant is zero if two rows coincide. Linearity is obvious as the determinant is a homogeneous polynomial of degree 1 in the entries in each row.

³We have $\sum_{\sigma \in S_N} \epsilon(\sigma) \prod_{i=1}^n a_{\sigma(i)i} = \sum_{\sigma \in S_N} \epsilon(\sigma) \prod_{j=1}^n a_{\sigma\sigma^{-1}(j)\sigma^{-1}(j)}$, and we change summation variable to σ^{-1} .

The determinant of a diagonal matrix is just the product of the diagonal entries:

$$\begin{vmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N \end{vmatrix} = \lambda_1 \lambda_2 \dots \lambda_N ,$$

as in this case only the identity permutation contributes. The same holds if the matrix is upper (or lower) triangular:

$$a_{ij} = 0, \quad i > j \Rightarrow \det(A) = \prod_{i=1}^N a_{ii} .$$

Exercise 6.7. Prove that $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$ and $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{32}a_{21} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{31}a_{22} .$

An important property of the determinant is that it respects the matrix product.

Proposition 6.1 (**The determinant respects the matrix product**). For any two $N \times N$ matrices A and B we have $\det(AB) = \det(A) \det(B)$.

Proof. By linearity of the determinant in the columns we have:

$$\det(AB) = \begin{vmatrix} \sum_{j_1=1}^N a_{1j_1} b_{j_1 1} & \dots & \sum_{j_N=1}^N a_{1j_N} b_{j_N N} \\ \vdots & \ddots & \vdots \\ \sum_{j_1=1}^N a_{Nj_1} b_{j_1 1} & \dots & \sum_{j_N=1}^N a_{Nj_N} b_{j_N N} \end{vmatrix} = \sum_{j_1=1}^N \dots \sum_{j_N=1}^N b_{j_1 1} \dots b_{j_N N} \begin{vmatrix} a_{1j_1} & \dots & a_{1j_N} \\ \vdots & \ddots & \vdots \\ a_{Nj_1} & \dots & a_{Nj_N} \end{vmatrix} .$$

If two of the j 's are equal, the last determinant has two identical columns and is zero. Ordering the j 's and taking into account that the determinant is antisymmetric by permutation of the columns we get:

$$\det(AB) = \det(A) \sum_{\sigma \in S_N} \epsilon(\sigma) b_{j_{\sigma(1)} 1} \dots b_{j_{\sigma(N)} N} = \det(A) \det(B) .$$

□

This leads to the following crucial remark.

Corollary 6.1 (**The determinant is basis independent**). Under a change of basis a matrix changes like $\tilde{A} = S^{-1}AS$, and $\det(\tilde{A}) = \det(S^{-1}) \det(A) \det(S) = \det(A) \det(SS^{-1}) = \det(A)$, that is the determinant is invariant under a change of basis.

This might seem surprising at first sight as the matrix elements change under a change of basis, and the determinant is a complicated polynomial in the matrix entries.

6.4.1 The determinant and the inverse of a matrix

Lemma 6.1 (**A is invertible if and only if $\det(A) \neq 0$**). Let us denote B the row echelon form of A . Then

$$\pm \det(A) = \det(B) = \prod_{i=1}^N b_{ii} .$$

We have seen that a matrix is invertible if and only if its RREF is I , or equivalently all the diagonal entries in its row echelon form are non zero $b_{ii} \neq 0$ for $i = 1, \dots, N$, therefore A is invertible if and only if $\det(A) \neq 0$.

Proof. Up to a sign, the determinant is invariant under the linear manipulations which bring A to its row echelon form. This can be checked by observing that the matrices S which encode row permutations and $L_\alpha^{i,j}$ which take linear combinations of rows have determinant ± 1 ($L_\alpha^{i,j}$ is either upper or lower triangular). $\square$

Lemma 6.2 ($\det(A) = 0$ if and only if the columns of A are linearly dependent). Let us denote B the row echelon form of A . $\det(A) = 0$ if and only if B has at least a row of 0, hence at least one non dominant column and a linear relation between the columns.

The determinant gives a criterion of invertibility for matrices. A recursive way to compute the determinant is given by the following theorem.

Theorem 6.3 (Cofactor expansion of determinant). The determinant of a matrix can be computed by expanding along the rows:

$$\det(A) = \sum_{j=1}^N (-1)^{i+j} a_{ij} \det(A_{i\hat{j}}), \quad \forall i = 1, \dots, N,$$

where the cofactor $\det(A_{i\hat{j}})$ is the determinant of the matrix obtained by deleting the row i and the column j in A . The same holds when expanding along columns, as the determinant is invariant under taking the transposed.

Proof. Without loss of generality we fix $i = 1$ and we classify the permutations σ according to $\sigma(1)$:

$$\det(A) = \sum_{j=1}^N \sum_{\sigma \in S_n}^{\sigma(1)=j} \epsilon(\sigma) a_{1j} \prod_{i' \neq 1} a_{i'\sigma(i')}.$$

Each term above is the determinant of a matrix with only one non zero entry on the first, row a_{1j} and all the others set to 0 (that is the determinant is linear in its first row):

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ \vdots & & & \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & \dots & 0 \\ \vdots & & & \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & \dots & 0 \\ \vdots & & & \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{vmatrix} + \dots + \begin{vmatrix} 0 & 0 & \dots & a_{1N} \\ \vdots & & & \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{vmatrix}$$

and by permuting the columns, the j 'th term is:

$$(-1)^{j+1} \begin{vmatrix} a_{1j} & 0 \\ \vdots & A_{1\hat{j}} \end{vmatrix} = (-1)^{j+1} \sum_{\sigma \in S_n} \epsilon(\sigma) \prod_i b_{i\sigma(i)},$$

where we labelled the elements of the new matrix generically by b_{ij} . Now, the only non trivial element on the first row is $b_{11} = a_{1j}$, hence the only permutations contributing must have $\sigma(1) = 1$. Such a permutation consists in a cycle (1) and a permutation τ on $\{2, \dots, n-2\}$, with $\epsilon(\tau) = \epsilon(\sigma)$, hence

$$(-1)^{j+1} \sum_{\sigma \in S_n} \epsilon(\sigma) \prod_i b_{i\sigma(i)} = (-1)^{j+1} a_{1j} \sum_{\tau \in S_{n-1}} \epsilon(\tau) \prod_{i=2}^n b_{i\tau(i)} = (-1)^{j+1} a_{1j} \det(A_{1\hat{j}}).$$

$\square$

This cofactor expansion allows us to give an explicit formula for the inverse A^{-1} of any matrix A with $\det A \neq 0$.

Corollary 6.2 (**Inverse of a matrix**). If A is a $N \times N$ matrix with $\det(A) \neq 0$, then A is invertible, that is there exist a matrix A^{-1} such that $(AA^{-1})_{ik} = (A^{-1}A)_{ik} = \delta_{ik}$. The inverse matrix has matrix elements:

$$(A^{-1})_{ij} = \frac{1}{\det(A)} (-1)^{i+j} \det(A_{\hat{j}\hat{i}}),$$

where we stress that the indices of the cofactor are permuted. If A represents the operator $\mathcal{A}$ in some basis, the inverse matrix A^{-1} represents the inverse operator $\mathcal{A}^{-1}$ in the same basis.

Proof. We have:

$$(AA^{-1})_{ik} = \sum_{j=1}^N a_{ij} (A^{-1})_{jk} = \frac{1}{\det(A)} \sum_{j=1}^N a_{ij} (-1)^{k+j} \det(A_{\hat{k}\hat{j}}).$$

By the cofactor expansion, if $j = k$, then the right hand side is 1. If $j \neq k$, then

$$\sum_{j=1}^N a_{ij} (-1)^{k+j} \det(A_{\hat{k}\hat{j}}),$$

is nothing but the determinant of the matrix A , where the k 'th row has been replaced by a copy of the i 'th row. As this matrix has two identical rows, its determinant is zero. $\square$

A matrix is invertible if and only if it has zero kernel, that is $\ker A = 0 \Leftrightarrow \det A \neq 0$.

Corollary 6.3 (**Cramer's rule**). Let A be an invertible $N \times N$ matrix. Then the system of equations $Ax = y$ has a unique solution $x = A^{-1}y$ and, using the formula for the inverse and the cofactor expansion of the determinant we have:

$$x^i = \frac{1}{\det A} \sum_{j=1}^N (-1)^{i+j} \det(A_{\hat{j}\hat{i}}) y^j = \frac{\det(A^i)}{\det A},$$

where the matrix A^i is obtained from A by replacing the column i with the column vector y .

6.4.2 Block matrices

Often one deals which are partitioned by some criterion into blocks. Consider four matrices: A of sizes $M \times M$, B of size $M \times N$, C of size $N \times M$ and D of size $N \times N$ which are organized in a $(M+N) \times (M+N)$ matrix:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Proposition 6.2 (**Determinant and inverse of a block matrix**). If $B = 0$ then the determinant of the block matrix is:

$$\begin{vmatrix} A & 0 \\ C & D \end{vmatrix} = \det(A) \det(D),$$

and similarly if $C = 0$. If A is invertible then:

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = \det(A) \det(D - CA^{-1}B),$$

and if D is invertible it is $\det(D)\det(A - BD^{-1}C)$. If both A and D are invertible, then the inverse of the block matrix is:

$$\begin{pmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{pmatrix} \\ \begin{pmatrix} (A - BD^{-1}C)^{-1} & -(A - BD^{-1}C)^{-1}BD^{-1} \\ -D^{-1}C(A - BD^{-1}C)^{-1} & D^{-1} + D^{-1}C(A - BD^{-1}C)^{-1}BD^{-1} \end{pmatrix}.$$

Proof. Assume the block $B = 0$. Then, as the entries i, j for $i \in \{1, \dots, M\}$ and $j \in \{M + 1, \dots, M + N\}$ in the block matrix are zero, the permutations which contribute to the evaluation of the determinant of the block matrix must respect $\sigma(\{1, \dots, M\}) \subset \{1, \dots, M\}$ and consequently $\sigma(\{M + 1, \dots, M + N\}) \subset \{M + 1, \dots, M + N\}$. It follows that σ factors into two permutations each acting on one of the subsets of indices leading to:

$$\begin{vmatrix} A & 0 \\ C & D \end{vmatrix} = \det(A)\det(D),$$

and similarly if $C = 0$.

If A is invertible, then, denoting I_N and I_M the $N \times N$ respectively $M \times M$ identity matrix we have:

$$\begin{aligned} \begin{vmatrix} A & B \\ C & D \end{vmatrix} &= \det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \det \begin{pmatrix} I_M & -B \\ 0 & I_N \end{pmatrix} = \det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \det \begin{pmatrix} A & 0 \\ 0 & I_N \end{pmatrix} \det \begin{pmatrix} A^{-1} & -A^{-1}B \\ 0 & I_N \end{pmatrix}, \\ &= \det(A)\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \det \begin{pmatrix} A^{-1} & -A^{-1}B \\ 0 & I_N \end{pmatrix} = \det(A)\det \begin{pmatrix} I_M & 0 \\ CA^{-1} & D - CA^{-1}B \end{pmatrix} = \det(A)\det(D - CA^{-1}B) \end{aligned}$$

and the corresponding manipulation works if D invertible. The last part is just a direct check. $\square$

6.5 Minors of a matrix

Let us consider a $M \times N$ matrix A , where M can equal N or not. We denote $A_{IJ} = (a_{ij})_{i \in I, j \in J}$, where I is a set of row indices and J a set of column indices the sub-matrix consisting in the rows with indices in I and the columns with indices in J . A_{IJ} is a square matrix if $|I| = |J|$, and in this case the determinant $\det(A_{IJ}) \equiv [A_{IJ}]$ is called a minor of the matrix A . If A is a square matrix, A_{II} are called the *principal minors* of A . A matrix has many minors and some of them can be zero.

Proposition 6.3 (Minors and rank). *The rank of A is the dimension of the image of A , $\text{rank}(A) = \dim(\text{Im}(A))$. The matrix A has rank r if and only if it has a nonzero $r \times r$ minor.*

Proof. If r columns of A are linearly dependent then all the $r \times r$ minors of A containing these columns are zero (as their columns are linearly dependent). Conversely (up to a permutation of the columns) if all the $r \times r$ minors of A containing the first r columns are zero, bringing A in its RREF:

$$(Ae_1 \quad \dots \quad Ae_N) = (Ff_1 \quad \dots \quad Ff_k \quad \sum_{i=1}^k \alpha_{ik+1}Ff_i \quad \sum_{i=1}^k \alpha_{ik+2}Ff_i \quad \dots \quad Ff_l \quad \dots),$$

with F is invertible we conclude that $r > k$ hence the r columns are linearly dependent. $\square$

The determinant of a product of rectangular matrices can be computed in terms of the minors.

Theorem 6.4 (Cauchy-Binet formula). *Let A be an $M \times N$ matrix and B be a $N \times M$ matrix with $M \leq N$. Let $1 \leq j_1 < \dots < j_M \leq N$ and denote $A_{j_1 \dots j_M}$ the $M \times M$ sub-matrix of A with column indices $j_1, \dots, j_M$ and $B_{j_1 \dots j_M}$ the $M \times M$*

sub-matrix of B with row indices $j_1 \dots j_M$. Then:

$$\det(AB) = \sum_{1 \leq j_1 < \dots < j_M \leq N} \det(A_{j_1 \dots j_M}) \det(B_{j_1 \dots j_M}).$$

If $M = N$ this reduces to $\det(AB) = \det(A) \det(B)$.

Proof. The proof parallels the case $N = M$. By linearity of the determinant in the columns we have:

$$\det(AB) = \begin{vmatrix} \sum_{j_1=1}^N a_{1j_1} b_{j_1 1} & \dots & \sum_{j_M=1}^N a_{1j_M} b_{j_M M} \\ \vdots & \ddots & \vdots \\ \sum_{j_1=1}^N a_{Mj_1} b_{j_1 1} & \dots & \sum_{j_M=1}^N a_{Mj_M} b_{j_M M} \end{vmatrix} = \sum_{j_1=1}^N \dots \sum_{j_M=1}^N b_{j_1 1} \dots b_{j_M M} \begin{vmatrix} a_{1j_1} & \dots & a_{1j_M} \\ \vdots & \ddots & \vdots \\ a_{Mj_1} & \dots & a_{Mj_M} \end{vmatrix}.$$

If two of the j s are equal, the last determinant has two identical columns and is zero. Ordering the j 's and taking into account that the determinant is antisymmetric by permutation of the columns we get:

$$\det(AB) = \sum_{1 \leq j_1 < \dots < j_M \leq N} \det(A_{j_1 \dots j_M}) \sum_{\sigma \in S_M} \epsilon(\sigma) b_{j_{\sigma(1)} 1} \dots b_{j_{\sigma(M)} M} = \sum_{1 \leq j_1 < \dots < j_M \leq N} \det(A_{j_1 \dots j_M}) \det(B_{j_1 \dots j_M}).$$

□

As a direct consequence, the minors of a product of rectangular matrices can also be computed.

Corollary 6.4. *Let A be an $M \times N$ matrix and B be a $N \times P$ matrix with $M \leq N$ and let $I \subset \{1, \dots, M\}$ respectively $J \subset \{1, \dots, P\}$ with $|I| = |J|$. Then:*

$$\det[(AB)_{IJ}] = \sum_K \det(A_{IK}) \det(B_{KJ}),$$

where K runs over all the subsets of $\{1, \dots, N\}$ with $|K| = |I| = |J|$.

Classifying the permutations by their fixed points one can write a diagonal expansion of the determinant.

Lemma 6.3 (**Diagonal expansion of the determinant**). *Let A be a $N \times N$ matrix and let $I \subset \{1, \dots, N\}$. We denote $A_{\hat{I}\hat{I}}$ the sub matrix of A obtained from A by deleting the rows and columns with indices in the set I . Then:*

$$\det(A) = \sum_{I \subset \{1, \dots, N\}} \det(A_{\hat{I}\hat{I}}) \prod_{i \in I} a_{ii}.$$

6.5.1 The matrix tree theorem

Definition 6.4 (**Graph**). *A graph $G = (V, E)$ is a finite set of vertices $V = \{1, \dots, N\}$ and a set $E = \{(i, j) | i, j \in V\}$ of edges, that is pairs of vertices.⁴ A subgraph of G is a graph having the same vertices as G , but only some of the edges. A root of the graph is a marked vertex, say the vertex 1.*

Examples of graphs are:

$$\begin{aligned} G_1 : N = 3, \quad E &= \{(1, 2), (1, 3), (2, 3)\} \\ G_2 : N = 4, \quad E &= \{(1, 2), (1, 3), (1, 4), (2, 3), (3, 4)\}. \end{aligned}$$

A *path* in a graph is a sequence of edges $e_1, \dots, e_n$ such that any two consecutive edges share a vertex $e_l \cap e_{l+1} \neq \emptyset$. A path can also be identified by the ordered list of vertices $i_0, i_1, \dots, i_n$ it visits, $e_0 = (i_0, i_1), e_1 = (i_1, i_2), \dots, e_n = (i_{n-1}, i_n)$. A path is *simple* if all the intermediate vertices $v_1, \dots, v_n$ are different and a simple path is a *loop* (or *cycle*) if $i_0 = i_n$. Any path can be decomposed into simple paths and cycles.

⁴Self loops and multiple edges are excluded.

Definition 6.5 (**Connected graph, tree**). A graph is connected if any two vertices of the graph can be connected by a path. A graph is a tree if it is connected and has no loops. In a tree any two vertices are connected by only one simple path (if two different simple paths exist, their union contains a loop). We denote $\kappa(G)$ the number of subgraphs of the connected graph G which are trees.

Lemma 6.4. A graph over N vertices is a tree if and only if has no cycle and has exactly $N - 1$ edges.

Proof. Deleting any edge in a tree disconnects the graph, that is it increases the number of connected components of the graph by 1. Indeed, any edge (i, j) partitions the vertices in the tree into two sets

- the vertices in the subset I which are connected to the vertex i by simple paths which do not pass through (i, j) . They are connected to j by adding (i, j) to the paths connecting them to i .
- the vertices in the subset J which are connected to the vertex j by simple paths which do not pass through (i, j) . They are connected to i by adding (i, j) to the paths connecting them to j .

All the simple paths connecting vertices in I to vertices in J pass through (i, j) . Deleting (i, j) cuts all the paths that pass through (i, j) , hence splits the graph into two connected components containing the vertices i and I respectively j and J .

Iterating the edge deletions, the process stops when all the edges in the tree have been deleted and the graph has one vertex per connected component, thus at the end one has exactly N connected components. As T had one connected component, it follows that exactly $N - 1$ edges deletions have been performed.

Conversely, if T has N vertices, $N - 1$ edges and no cycle then T must be a tree. Indeed, assume T is not connected and denote $|E_i|$ and $|V_i|$ the number of edges and vertices in the connected component i , and $\sum_i |E_i| = N - 1$, $\sum_i |V_i| = N$. Then for at least one i , $|E_i| \geq |V_i|$. One can always chose a tree in a connected graph by a greedy algorithm: one iteratively tests at step l one of the untested edges, say e_l . If e_l does not form a loop with the edges already chosen at previous steps, then one ads e_l to the set of chosen edges and iterates. If e_l does form a loop with the edges already chosen at previous steps, one does not add it to the set of chosen edges, marks it as tested and iterates. A tree over the $|V_i|$ vertices uses only $|V_i| - 1$ edges and adding edges to a tree leads to a cycles (the two end vertices of the excess edge are connected by the tree and also by the edge). $\square$

There are several matrices associated to a graph:

- the $N \times N$ *adjacency matrix* A with entries $A_{ij} = 1$ if $(i, j) \in E$ and 0 if not. For the two graphs examples above we have:

$$A_1 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

- the $N \times |E|$ *incidence matrix* B . We orient the edges, say from the smaller to the larger label, and we define:

$$B_{ie} = \begin{cases} 1, & \text{if } e \text{ enters } i \\ -1, & \text{if } e \text{ exits } i \\ 0, & \text{otherwise} \end{cases}.$$

For the two examples before, ordering the edges lexicographically, we have

$$B_1 = \begin{pmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}, \quad B_2 = \begin{pmatrix} -1 & -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

- the $N \times N$ diagonal *degree* matrix D with entries $D_{ii} = d_i$, with d_i the number of edges incident to the vertex i (the number of edges to which i belongs). For the examples before:

$$D_1 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad D_2 = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

- the $N \times N$ *Laplacian* matrix $L = D - A$, which is for our two examples:

$$L_1 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

Lemma 6.5. *The product of the incidence matrix and its transposed is the Laplacian matrix $BB^t = L$.*

Proof. For our second example for instance we have:

$$\begin{pmatrix} -1 & -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

In general, $(BB^t)_{ij} = \sum_e B_{ie}B_{je}$. If i and $j \neq i$ are connected by an edge (ij) then the sum reduces to one term and $B_{i(ij)}B_{j(ij)} = -1$, as one of the entries is 1 and the other -1 ; if i and $j \neq i$ are not connected then the sum is zero; if $i = j$ then the sum is $\sum_e B_{ie}B_{ie}$ and each of the edges incident to i contribute a 1. $\square$

Note that the columns of L are linearly dependent, hence $\det(L) = 0$. We consider the matrix $\tilde{L}$ obtained from L by erasing the line and column corresponding to one vertex, chosen as root. We have $\tilde{L} = \tilde{B}\tilde{B}^t$, with $\tilde{B}$ the incidence matrix with the row corresponding to the root vertex erased.

Lemma 6.6. *The $(N-1) \times (N-1)$ minors of $\tilde{B}$ are the determinants of the matrices $\tilde{B}_I$, where I is a subsets of $N-1$ edges $I \subset E$. The minors are:*

$$\det(\tilde{B}_I) = \begin{cases} \pm 1 & \text{if } I \text{ is a tree} \\ 0 & \text{if not} \end{cases}.$$

Proof. Assume I has a simple cycle⁵ formed by the edges $e_1 = (i_0, i_1) \dots e_k = (i_{k-1}, i_0)$. By factoring a some global sign we can reorient all the edges along the cycle such that e_1 is oriented from i_0 to i_1 , e_2 from i_1 to i_2 and so on up to e_k oriented from i_{k-1} to i_0 . Then $B_{ie_1} + B_{ie_2} + \dots B_{ie_k} = 0$ for any i as:

- either i is not a vertex in the cycle and the all the terms are 0,
- or i is a vertex in the cycle, but then the consecutive terms B_{ie_s} and $B_{ie_{s+1}}$ to which i belongs are respectively $+1$ and -1 and cancel.

Thus in order for $\det(\tilde{B}_I) \neq 0$, I must be a graph with N vertices, $N - 1$ edges and no cycles, therefore I must be a tree. We consider one of the edges incident to the root, say (r, i) . This column has only one non zero entry, and expanding the minor along this column we obtain a minor corresponding to a graph in which the edge e has been deleted, and vertex i marked as root in the connected component generated by the deletion of e (that is the row corresponding to i is eliminated in B). Iterating, the process stops when all the edges have been exhausted. As at each step we multiply the result by ± 1 , the result follows. $\square$

Theorem 6.5 (**The matrix tree theorem**). *The determinant of $\tilde{L}$ equals the number of trees of G , $\det(\tilde{L}) = \kappa(G)$.*

Proof. As $\tilde{L} = \tilde{B}\tilde{B}^t$, from Cauchy-Binet $\det(\tilde{L}) = \sum_I |B_I|^2$ where the sum runs over all the subsets $I \subset E$ with $|I| = N - 1$. The only non zero contributions correspond to sets I which are trees, and each tree contributes exactly 1. $\square$

For the two examples we discussed:

$$\det(\tilde{L}_1) = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3, \quad \det(\tilde{L}_2) = \begin{vmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 8.$$

For the second graph we have the trees depicted in Fig. 6.1 below.

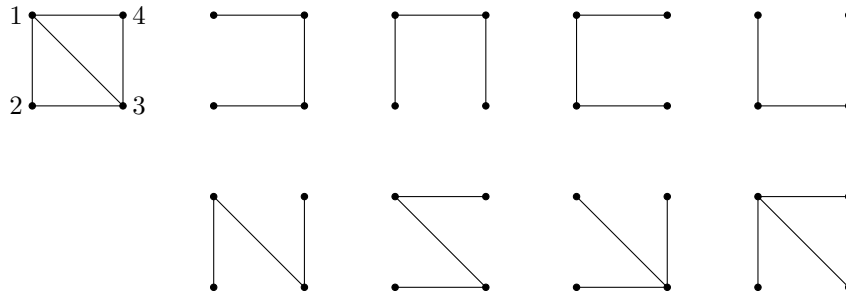


Figure 6.1: Trees in a graph.

⁵If I has a cycle it has a simple cycle.

Chapter 7

Functions, first contact

An important role in physics and mathematics is played by the continuous functions.

7.1 Continuity

Definition 7.1 (**Pointwise continuity**). For X and Y metric spaces, a function $f : X \rightarrow Y$ is called pointwise continuous if $\forall x, \forall \epsilon, \exists \delta_{\epsilon, x}$ such that $d(x, x') < \delta \Rightarrow d(f(x), f(x')) < \epsilon$. A function can be continued at a point x if there exists a function $\bar{f}$ whose domain includes x , is continuous at x and is such that $f = \bar{f}$ on the domain of f .

As we have seen in Chapter 3, a function is continuous if the preimage of any open set is open. A function between metric spaces is sequentially continuous (see Chapter 4) if for any $x_n \rightarrow x$, $f(x_n) \rightarrow f(x)$ and in metric spaces a function is continuous if and only if it is sequentially continuous.

Theorem 7.1 (**Sequential and pointwise continuity are equivalent**). A function $f : X \rightarrow Y$ with X and Y metric spaces is continuous if and only if it is sequentially continuous.

Proof. Consider a pointwise continuous function $f : X \rightarrow Y$ and $x_n \rightarrow x$. For any ϵ , there exist δ such that $d(x_n, x) < \delta \Rightarrow d(f(x_n), f(x)) < \epsilon$ and (for this δ but also for any other δ) there exist N_δ such that $n > N_\delta \Rightarrow d(x_n, x) < \delta$, hence $f(x_n) \rightarrow f(x)$.

Conversely, if f is not sequentially continuous at x , then there exist a sequence $x_n \rightarrow x$ such that $f(x_n)$ does not converge to $f(x)$. Then there exist some ϵ such that for any N_ϵ , there exist some $n > N_\epsilon$ with $d(f(x_n), f(x)) > \epsilon$. Consequently, for this ϵ , for any δ there exists $n > N_\delta$ with $d(x_n, x) < \delta$ but such that $d(f(x_n), f(x)) > \epsilon$ and f is not pointwise continuous at x . $\square$

It follows that a function is continuous if and only if it is pointwise continuous. A direct proof of this fact goes as follows.

Theorem 7.2 (**Continuous is pointwise continuous**). A function between two metric spaces is continuous if and only if it is pointwise continuous.

Proof. The proof follows the one establishing that, in metric spaces, sequential continuity and continuity are equivalent.

Let f a continuous function. For any ϵ , let $B_\epsilon(f(x))$ be the open ball of radius ϵ around $f(x)$. Then $x \in f^{-1}(B_\epsilon(f(x)))$ (note that $f^{-1}(f(x))$ includes other points in X if f is not injective) thus $f^{-1}(B_\epsilon(f(x)))$ is an open set in X (due to continuity) that includes x . It follows that there exists some $\delta_{\epsilon, x}$ such that $B_{\delta_{\epsilon, x}}(x) \subset f^{-1}(B_\epsilon(f(x)))$.

Suppose now that f is pointwise continuous but not continuous. This means that there exists an open set V such that $f^{-1}(V)$ is not open. Then there exist some $x \in f^{-1}(V)$ such that $\forall \delta, B_\delta(x) \setminus f^{-1}(V) \neq \emptyset$. As $f(x) \in V$ and V is

open, $\exists \epsilon$ with $B_\epsilon(f(x)) \subset V$. It follows that $\exists \epsilon$ such that $\forall \delta$, picking $x' \in B_\delta(x) \setminus f^{-1}(V)$ we have both $d(x, x') < \delta$ and $d(f(x), f(y)) \geq \epsilon$. $\square$

Exercise 7.1. Prove that the composition of continuous functions is continuous.

Exercise 7.2. Prove sums, products and ratios (when defined) of continuous functions are continuous functions. Conclude that any polynomial is a continuous function. $|z^n - w^n| = |z - w| \sum_{p=0}^{n-1} |z|^p |w|^{n-p} \leq |z - w|(s+1)(\max\{|z|, |w|\})^n$.

Exercise 7.3. Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$f(x) = \begin{cases} \frac{1}{q}, & \text{for } x = \frac{p}{q} \text{ with } p, q \text{ co primes and } q > 0 \\ 1, & \text{for } x = 0 \\ 0, & \text{for } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases},$$

is continuous at all $x \in \mathbb{R} \setminus \mathbb{Q}$ but discontinuous at all $x \in \mathbb{Q}$. For any $x \in \mathbb{Q}$, $x + \frac{\sqrt{2}}{N}$ is irrational and arbitrarily close to x . For any $x \in \mathbb{R} \setminus \mathbb{Q}$ and for any q there exists an integer such that $\frac{p}{q} < x < \frac{p+1}{q}$ chose $\delta < \min\{|x - \frac{p}{q}|, |x - \frac{p+1}{q}|\} < \frac{1}{q} \leq \frac{1}{\epsilon} \neq 0$. All the rational numbers in $(x - \delta, x + \delta)$ are of the form $\frac{p'}{q'}$ with denominator $q' > \frac{1}{\epsilon}$, hence $f(\frac{p'}{q'}) < \epsilon$.

Exercise 7.4. Prove that a power series defines a continuous function at all the points inside its convergence domain. Without loss of generality, let us consider a power series centered at zero, $P(z) = \sum_{n \geq 0} a_n z^n$. Note that $P(z)$ and $P'(z) = \sum_{n \geq 0} a_{n+1} (n+1) z^n$ have the same radius of convergence $r^{-1} = \limsup |a_n|^{1/n} = \limsup (n+1)^{1/n} |a_{n+1}|^{1/n}$ as $(n+1)^{1/n} \rightarrow 1$. Now for any two points in the convergence disk, denote $\rho = \max\{|z|, |w|\} < r$ then:

$$|P(z) - P(w)| \leq |z - w| \sum_{n \geq 0} |a_{n+1}| \sum_{p=0}^n z^p w^{n-p} \leq |z - w| \sum_{n \geq 0} a_{n+1} (n+1) \rho^n.$$

We will now focus on the case of $f : \mathbb{R} \rightarrow \mathbb{R}$. Note that a function f is continuous on a closed interval $[a, b]$ if it is continuous at $x \in (a, b)$ and the one sided limits exist at the end points $\lim_{x \searrow a} f(x) = f(a)$, $\lim_{x \nearrow b} f(x) = f(b)$.

7.2 The intermediate value theorem (Zwischenwertsatz)

Continuous functions do not jump between values.

Theorem 7.3 (The intermediate value theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ a continuous function. Then $f([a, b])$ is an interval and $f([a, b]) \supset [\min\{f(a), f(b)\}, \max\{f(a), f(b)\}]$. That is any intermediate value between $f(a)$ and $f(b)$ is attained by some point in $[a, b]$. Equivalently, for any $t \in [0, 1]$ there exists $c \in [a, b]$ such that $f(c) = tf(a) + (1-t)f(b)$.

Proof. We assume $f(a) < u < f(b)$, as the other case is similar. Let $S = \{x \in [a, b] \mid f(x) \leq u\}$. S is nonempty as $a \in S$ and bounded from above (by b) hence it has a supremum $c = \sup S$. As f is continuous at c , for any ϵ there exists δ such that $|x - c| < \delta \Rightarrow |f(x) - f(c)| < \epsilon$, that is $f(x) - \epsilon < f(c) < f(x) + \epsilon$. As c is the supremum of S , there exists $x \in S \cap (c - \delta, c]$, hence such that $f(x) \leq u$ and $f(c) < u + \epsilon$. On the other hand $\forall x' \in (c, c + \delta)$, $x' \notin S$ hence $u < f(x')$ and $u - \epsilon < f(c)$. Thus $f(c) = u$.

Finally, $f([a, b])$ is an interval because, for any $c \in [a, b]$, all the values between $f(c)$ and $f(a)$ are attained. $\square$

Theorem 7.4 (Path connected is connected). Recall that a topological space is called connected if it can not be divided into two disjoint nonempty open sets, $X \neq X_1 \cup X_2$ with $X_1 \cap X_2 = \emptyset$. For $x, y \in X$, a path from x to y is a continuous function $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$. A topological space is path connected if any two points in it are connected by a path.

Every path connected topological space is connected. The converse is not true.

Proof. We first prove that the interval $[0, 1]$ endowed with the subspace topology inherited from $\mathbb{R}$ is connected. Suppose there exist non empty open sets $U, V \in [0, 1]$ which are disjoint $U \cap V = \emptyset$ and such that $U \cup V = [0, 1]$. We define the function:

$$F : [0, 1] \rightarrow \mathbb{R}, \quad F(x) = \begin{cases} 0, & x \in U \\ 1, & x \in V \end{cases},$$

which is continuous because for any open set $Q \in \mathbb{R}$, $F^{-1}(Q)$ is

- $\emptyset$, if $0, 1 \notin Q$,
- U is $0 \in Q$, $1 \notin Q$,
- V if $0 \notin Q$, $1 \in Q$,
- $U \cup V = [0, 1]$ if $0, 1 \in Q$,

which are all open. By the intermediate value theorem F takes all the values between 0 and 1, which is impossible.

Assume now X is a path connected topological space and assume that it splits as the disjoint union of two open sets $X = X_1 \cup X_2$ with $X_1 \cap X_2 = \emptyset$. Chose a point $x_1 \in X_1$, a point $x_2 \in X_2$ and a path γ from x_1 to x_2 . Then $\gamma^{-1}(X_1) \ni x_1$ and $\gamma^{-1}(X_2) \ni x_2$ are open (as γ is continuous), disjoint and such that $[0, 1] = \gamma^{-1}(X_1) \cup \gamma^{-1}(X_2)$ which is impossible. $\square$

Lemma 7.1. *Let $I \subset \mathbb{R}$ an interval (could be open or unbounded), if $f : I \rightarrow J = f(I) \subset \mathbb{R}$ is strictly monotonic and continuous on I , then f is a bijection between I and J , J is an interval and f^{-1} is strictly monotonic and continuous on J .*

Proof. Let us assume f is strictly increasing, that is $x_1 > x_2 \Leftrightarrow f(x_1) > f(x_2)$. Then f is injective (as $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$) and it is also surjective on J , hence f is bijective. By the intermediate value theorem, J is an interval. As f is strictly increases, f^{-1} is also strictly increasing (as $y_1 > y_2 \Rightarrow f^{-1}(y_1) > f^{-1}(y_2)$).

It remains to prove that f^{-1} is continuous. Combining the fact that f is strictly increasing with the intermediate value theorem we conclude that the image of any open interval (x_1, x_2) is the open interval $(f(x_1), f(x_2))$ (as $x_1 < x < x_2 \Rightarrow f(x_1) < f(x) < f(x_2)$, it follows that $f((x_1, x_2)) \subset (f(x_1), f(x_2))$ and $f((x_1, x_2)) \supset (f(x_1), f(x_2))$ by the intermediate value theorem). It follows that for any interval $(y_1, y_2) \in J$, there exist $x_1, x_2 \in I$ with $f(x_1) = y_1$ and $f(x_2) = y_2$ hence $f^{-1}((y_1, y_2)) = f^{-1}((f(x_1), f(x_2))) = (x_1, x_2) = (f^{-1}(y_1), f^{-1}(y_2))$ and we conclude using the fact that the standard topology is generated by the open intervals. $\square$

For $f : \mathbb{R} \rightarrow \mathbb{R}$ we denote:

- $\lim_{x \rightarrow \infty} f(x) = L$ if $\forall \epsilon, \exists y$ such that $x > y \Rightarrow |f(x) - L| < \epsilon$
- $\lim_{x \rightarrow x_0} f(x) = \infty$ if $\forall L, \exists \delta$ such that $|x - x_0| < \delta \Rightarrow f(x) > L$.
- $\lim_{x \rightarrow \infty} f(x) = \infty$ if $\forall L, \exists y$ such that $x > y \Rightarrow f(x) > L$. In the same spirit, for a sequence $(x_n) \in \mathbb{R}$ we denote $\lim x_n = \infty$ if $\forall L, \exists N$ such that $n > N \Rightarrow x_n > L$.

7.2.1 The exponential function revisited

We revisited the exponential function restricted to real arguments, $\exp : \mathbb{R} \rightarrow \mathbb{R}_{>0}$, $\exp(x) \equiv e^x = \sum_{n \geq 0} \frac{1}{n!} x^n$.

Proposition 7.1. *Restricted to real arguments the exponential is monotonic, continuous and surjective on $\mathbb{R}_{>0} = (0, \infty)$.*

Proof. We have $e^0 = 1$ and for any positive x , $e^x > 1$. As $e^{x+y} = e^x e^y$, $\exp(x)$ is strictly increasing and $e^{-x} = (e^x)^{-1} > 0$, that is the exponential is always positive.

As $e^{\delta x} = 1 + \delta x \sum_{k \geq 1} \frac{1}{k!} \delta x^{k-1}$, for $0 < \delta x < 1$ we have $|e^{\delta x} - 1| \leq \frac{\delta x}{1 - \delta x}$, hence $|e^x - e^y| \leq e^x \frac{|x-y|}{1-|x-y|}$ for $|x-y| < 1$ that is e^x is continuous at any $x \in \mathbb{R}$ (by a similar argument we obtain continuity at any $z \in \mathbb{C}$). For $x > 0$, $e^x > 1 + x$ hence $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow -\infty} e^x = 0$ and by the intermediate value Theorem e^x takes all the values in the interval $(0, \infty)$. $\square$

It follows that the exponential has an inverse which is continuous, strictly increasing and surjective. We call this inverse the *natural logarithm* and we denote it:

$$\ln : \mathbb{R}_{>} \rightarrow \mathbb{R}, \quad \ln(x) = y, \text{ with } y \text{ such that } e^y = x.$$

Note that, because $e^{y_1} e^{y_2} = e^{y_1+y_2}$ we have $\ln(x_1 x_2) = \ln(x_1) + \ln(x_2)$. We are finally in the position to define irrational powers of real numbers. For all $\alpha \in \mathbb{R}$, $x \in \mathbb{R}_{>}$:

$$x^\alpha \equiv e^{\alpha \ln(x)}.$$

Exercise 7.5 (Convex and concave functions (Konvexe und konkave Funktionen)). The graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is the set of points $(x, f(x)) \in \mathbb{R}^2$. We say that $f : I \rightarrow \mathbb{R}$ is convex in the interval I if for any $[a, b] \subset I$ and $\forall t \in [0, 1]$:

$$f(ta + (1-t)b) \leq tf(a) + (1-t)f(b),$$

and we say it is concave if $f(ta + (1-t)b) \geq tf(a) + (1-t)f(b)$. This has a graphical interpretation: a function is convex if at an intermediate point $ta + (1-t)b$ it is below the straight line from $f(a)$ to $f(b)$.

Prove that:

- for any $n \in \mathbb{N}$, $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ with $f(x) = x^n$ is convex.
- e^x is convex on $\mathbb{R}$.
- $\ln(x)$ is concave on $\mathbb{R}_{>0}$.

$f(x) = x^n$ is convex. The cases $n = 0, 1$ are trivial. For $n \geq 2$ we proceed by induction:

$$[ta + (1-t)b]^{n+1} = [ta + (1-t)b]^n [ta + (1-t)b] \leq [ta^n + (1-t)b^n] [ta + (1-t)b] = t^2 a^{n+1} + (1-t)^2 b^{n+1} + t(1-t)[a^n b + ab^n],$$

where we used the induction hypothesis. A short manipulation shows that:

$$t^2 a^{n+1} + (1-t)^2 b^{n+1} + t(1-t)[a^n b + ab^n] = ta^{n+1} + (1-t)b^{n+1} - t(1-t)(a-b)^2 \left[\sum_{p=0}^{n-1} a^{n-1-p} b^p \right],$$

and convexity follows.

e^x is convex. Let us first prove it is convex on $\mathbb{R}_{>0}$. We have, using the previous point, that:

$$e^{ta+(1-t)b} = \sum_{n \geq 0} \frac{1}{n!} [ta + (1-t)b]^n \leq \sum_{n \geq 0} \frac{1}{n!} [ta^n + (1-t)b^n] = te^a + (1-t)e^b.$$

In general, we note that for any $x, y \in \mathbb{R}$, there exists a z such that $x+z > 0$ and $y+z > 0$ hence:

$$e^{tx+(1-t)y} = e^{-z} \left[e^{t(x+z)+(1-t)(y+z)} \right] \leq e^{-z} \left[te^{x+z} + (1-t)e^{y+z} \right] = te^x + (1-t)e^y.$$

$\ln(x)$ is concave. This follows from the previous point, combined with the observation that the inverse of a convex strictly increases function is concave. Indeed, assume $f : X \rightarrow Y$ is convex and strictly increasing. Then f^{-1} is also strictly increasing and

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2) \Rightarrow f^{-1}\left(f(tx_1 + (1-t)x_2)\right) \leq f^{-1}\left(tf(x_1) + (1-t)f(x_2)\right),$$

which, denoting $f(x_1) = y_1$ and $f(x_2) = y_2$ yields:

$$tf^{-1}(y_1) + (1-t)f^{-1}(y_2) \leq f^{-1}(ty_1 + (1-t)y_2).$$

7.3 The extreme value theorem

Continuous functions on compact sets attain their extrema. Note that the supremum c of any subset $S \subset \mathbb{R}$ can not be an interior point of S because for any ϵ , $B_\epsilon(c) = (c - \epsilon, c + \epsilon) \not\subset S$, as no element of S is larger than c . In fact the supremum:

- either is an isolated point of S (and necessarily belongs to S),
- or is a limit point of S (belonging or not to S).

Indeed, any point of S is either isolated or a limit point of S , hence if $c \in S$ we conclude. If $c \notin S$, then for any ϵ , $c - \epsilon$ is not the supremum of S hence $B_\epsilon(c) \cap S = B_\epsilon(c) \cap S \setminus \{c\} \neq \emptyset$, and c is a limit point of S .

Theorem 7.5 (**The extreme value theorem**). *Real valued continuous functions on compact domains attain their extrema, that is for any continuous function $f : K \rightarrow \mathbb{R}$ with compact domain K , there exists $x_m, x_M \in K$ such that $f(x_m) = \inf f(K)$ and $f(x_M) = \sup f(K)$.*

Proof. By definition, for any point $y \in f(K)$ there exists $x \in K$ such that $f(x) = y$. The continuous image $f(K)$ of the compact set K is compact and by Heine Borel a compact set in $\mathbb{R}$ is bounded and closed. As it is closed, $f(K)$ contains all the isolated points of $f(K)$ (if any) and all the limit points of $f(K)$, hence contains its supremum (and infimum). $\square$

We are now in the position to prove the fundamental theorem of algebra.

Theorem 7.6 (**The fundamental theorem of algebra (Fundamentalsatz der Algebra)**). *Any polynomial $P(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$ with complex coefficients $a_0, \dots, a_{n-1} \in \mathbb{C}$ admits a root in $\mathbb{C}$.*

Proof. We define $f(z) = |P(z)|$. It is easy to check that $f(z)$ is a continuous function on $\mathbb{C}$. The proof is divided into two steps:

$f(z)$ has a global minimum in $\mathbb{C}$. First we show that outside a big enough disk $B_R(0)$, the polynomial $P(z)$ has a large absolute value. Indeed, let R be a constant such that $R > \max\{1, 2 \sum_{i=0}^{n-1} |a_i|\}$. Then for $|z| > R$ we have:

$$|a_{n-1}z^{n-1} + \dots + a_0| \leq \sum_{i=0}^{n-1} |z|^i |a_i| \leq |z|^{n-1} \sum_{i=0}^{n-1} |a_i| < |z|^{n-1} \frac{R}{2} < \frac{|z|^n}{2}.$$

It follows that for $|z| > R$ we have:

$$f(z) = |P(z)| \geq |z|^n - |a_{n-1}z^{n-1} + \dots + a_0| > |z|^n - \frac{|z|^n}{2} = \frac{|z|^n}{2} > \frac{R^n}{2}.$$

Second we show that $f(z)$ has a minimum inside $\overline{B_R(0)}$. Indeed, the closed disk $\overline{B_R(0)}$ is closed and bounded, hence compact in $\mathbb{C}$. By the extreme value Theorem, the function $f(z)$ has a minimum in it, that is there exists $z_m \in \overline{B_R(0)}$ such that $f(z_m) \leq f(z)$ for all $z \in \overline{B_R(0)}$.

Third, we show that z_m is a global minimum in $\mathbb{C}$. Indeed, $f(z_m) \leq f(0) = |a_0| < \frac{R}{2} < \frac{R^n}{2}$, and outside the disk $f(z) > \frac{R^n}{2}$, therefore:

$$f(z_m) \leq f(z), \quad \forall z \in \mathbb{C}.$$

At its minimum, $f(z_m) = 0$. Observing that $z = z_m + (z - z_m)$, re expand the polynomial in terms of $z - z_m$:¹

$$P(z) = P(z_m) + c_k(z - z_m)^k + c_{k+1}(z - z_m)^{k+1} + \dots + c_n(z - z_m)^n,$$

¹Anticipating, this is just the pedestrian derivation of the fact that close to z_m the Taylor expansion at first non trivial order of $P(z)$ is $P(z_m + \delta z) \approx P(z_m) + \frac{1}{k!} P^{(k)}(z_m)(\delta z)^k$ for some $1 \leq k \leq n$ and $|P(z_m + \delta z)|$ decreases for $(\delta z)^k = -P(z_m) \frac{k!}{P^{(k)}(z_m)} t^k$.

where $c_n = 1$ and we denoted k the lowest degree such that $c_k \neq 0$. Note that k is at most n , as $c_n = 1$, or lower.

We assume that $P(z_m) \neq 0$. Then, as $c_k \neq 0$, there exists $a \neq 0$ one of the roots of order k of $-\frac{P(z_m)}{c_k}$, that is $a^k = -\frac{P(z_m)}{c_k}$. We now evaluate P at the point $z_m + ta$:

$$P(z_m + ta) = P(z_m) + c_k t^k \left(-\frac{P(z_m)}{c_k} \right) + \sum_{p=k+1}^n c_p (ta)^p = (1 - t^k)P(z_m) + t^{k+1} \sum_{p=k+1}^n c_p a^p t^{p-k-1}.$$

For $t < 1$ we have $|\sum_{p=k+1}^n c_p a^p t^{p-k-1}| \leq \sum_{p=k+1}^n |c_p a^p| < C$ for some constant C and

$$|P(z_m + ta)| \leq (1 - t^k)|P(z_m)| + t^{k+1}C = 1 - t^k|P(z_m)| \left(1 - t \frac{C}{|P(z_m)|} \right),$$

which is strictly smaller than $|P(z_m)|$ for $t < \frac{|P(z_m)|}{C}$ leading to a contradiction. We conclude that $P(z_m) = 0$.

□

Theorem 7.7 (In finite dimensions all norms are equivalent). *For any two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a finite dimensional Banach space V ($\mathbb{R}^N$ or $\mathbb{C}^N$), there exist two constants C_1 and C_2 such that:*

$$C_1 \|x\|_b \leq \|x\|_a \leq C_2 \|x\|_b.$$

Proof. We will show that all norms are equivalent to $\|x\|_1 = \sum_{i=1}^N |x^i|$.

Any norm $\|\cdot\|_a$ is continuous under $\|\cdot\|_1$. By the triangle inequality $|\|x_1\|_a - \|x_2\|_a| \leq \|x_1 - x_2\|_a$. On the other hand, choosing a basis e_i , we have:

$$\|x_1 - x_2\|_a \leq \sum_{i=1}^N |x_1^i - x_2^i| \|e_i\|_a \leq \|x_1 - x_2\|_1 \max_{i=1, \dots, N} \|e_i\|_a,$$

hence for any ϵ , choosing $\delta = \epsilon / \max_{i=1, \dots, N} \|e_i\|_a$ we have $\|x - x_0\|_1 < \delta \Rightarrow \|\|x_1\|_a - \|x_2\|_a\| < \|x_1 - x_2\|_a < \epsilon$.

The closed ball of radius 1 in $\|\cdot\|_1$, $B_1^{\|\cdot\|_1}(0) = \{x \in V \mid \|x\|_1 \leq 1\}$ is a bounded closed set in V , hence by the Heine-Borel theorem it is compact: it is here that we use the fact that V is finite dimensional. By the extreme value theorem, the continuous function $f(x) = \|x\|_a$ will attain a maximum and a minimum on this closed ball, which we denote $C_1 = \inf_{\|u\|_1=1} \|u\|_a$ and $C_2 = \sup_{\|u\|_1=1} \|u\|_a$ hence for any x :

$$C_1 \leq \left\| \frac{1}{\|x\|_1} x \right\|_a \leq C_2.$$

□

As a consequence we have the following result.

Theorem 7.8 (In finite dimensions all the linear operators are bounded). *If X is a finite dimensional normed vector space and Y is a normed vector space then any linear operator $T : X \rightarrow Y$ is bounded.*

Proof. Let us denote $e_1, \dots, e_n$ a Hamel basis in X . Then $\forall x \in X$, $x = \sum_{k=1}^n a_k e_k$:

$$\|T(x)\|_Y \leq \sum_{k=1}^n |a_k| \|T(e_k)\| \leq \left(\max_{k=1, \dots, n} \|T(e_k)\| \right) \sum_{k=1}^n |a_k| = \left(\max_{k=1, \dots, n} \|T(e_k)\| \right) \|x\|_1,$$

and we conclude as any norm on X is equivalent with $\|\cdot\|_1$.

□

7.4 Differentiation in one variable

The derivative of a function captures how the function behaves when one of the arguments is varied. For a function h continuous at x_0 we denote $\lim_{x \rightarrow x_0} h(x) = h(x_0)$.

Definition 7.2 (**Derivative**). Consider $f : \mathbb{R} \rightarrow V$ for $(V, \|\cdot\|)$ some Banach space. We say that f is differentiable at x_0 if the function: $\frac{f(x)-f(x_0)}{x-x_0}$ is a continuous function of x at $x = x_0$ ² and in this case we denote:

$$f'(x_0) \equiv \frac{df}{dx}(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}, \quad \left(\forall \epsilon, \exists \delta, \text{ such that } |x - x_0| < \delta \Rightarrow \left\| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right\| < \epsilon \right)$$

and we call $f'(x_0)$ the derivative of f at x_0 . f is called differentiable on an interval I if it is differentiable at all $x_0 \in I$ and it is called continuously differentiable if $f'(x_0)$ is a continuous function. We denote the vector space of continuously differentiable functions $\mathcal{C}^1(\mathbb{R}, V)$.

Note that $f(x) - f(x_0)$ is a vector in V and $(x - x_0)^{-1}$ a scalar, hence $f'(x)$ is a vector in V . Denoting e_k a basis in V , $f(x) = \sum_k f_k(x)e_k$ and $f'(x) = \sum_k f'_k(x)e_k$ ³. Remark that a priori $\frac{f(x)-f(x_0)}{x-x_0}$ is defined for $x \neq x_0$ and needs to be extended by continuity to x_0 . If f is differentiable at x_0 then f is continuous at x_0 , as for any ϵ , $\|f(x) - f(x_0)\| \leq (\epsilon + \|f'(x_0)\|)|x - x_0|$ if $|x - x_0|$ is small enough.

Theorem 7.9 (**Mean value inequality**). Let $f : [x_1, x_2] \rightarrow V$ a function continuous on $[x_1, x_2]$, differentiable on (x_1, x_2) . Then:

$$\|f(x_2) - f(x_1)\| \leq |x_2 - x_1| \sup_{x \in (x_1, x_2)} \|f'(x)\|.$$

Proof. Assume that there exists some $\epsilon > 0$ such that $\|f(x_2) - f(x_1)\| - (x_2 - x_1) \left(\sup_{x \in (x_1, x_2)} \|f'(x)\| + \epsilon \right) > 0$. Consider the function:

$$g_\epsilon(x) = \|f(x) - f(x_1)\| - (x - x_1) \left(\sup_{x \in (x_1, x_2)} \|f'(x)\| + \epsilon \right), \quad g_\epsilon(x_1) = 0, \quad g_\epsilon(x_2) > 0,$$

and denote $x_0 = \sup\{x \in [x_1, x_2], g_\epsilon(x) = 0\}$. As g_ϵ is continuous it is sequentially continuous and considering a sequence⁴ $x_n \in S$, $x_n \rightarrow x_0$ we conclude that $g_\epsilon(x_0) = 0$. Furthermore, as $g_\epsilon(x_2) > 0$, we have $x_0 < x_2$ and $g_\epsilon(x) > 0$ for $x_0 < x < x_2$ (were $g_\epsilon(x) < 0$ for some such x , by the intermediate value theorem there would exist x' with $x < x' < x_2$ with $g_\epsilon(x') = 0$ and x_0 would not be the supremum). For all x with $x_0 < x < x_2$ we have:

$$\begin{aligned} 0 &\leq g_\epsilon(x) - g_\epsilon(x_0) = \|f(x) - f(x_1)\| - \|f(x_0) - f(x_1)\| - (x - x_0) \left(\sup_{x \in (x_1, x_2)} \|f'(x)\| + \epsilon \right) \\ &\leq \|f(x) - f(x_0)\| - (x - x_0) \left(\sup_{x \in (x_1, x_2)} \|f'(x)\| + \epsilon \right), \end{aligned}$$

hence

$$\sup_{x \in (x_1, x_2)} \|f'(x)\| + \epsilon \leq \left\| \frac{f(x) - f(x_0)}{x - x_0} \right\|,$$

and sending $x \rightarrow x_0$ we obtain a contradiction. □

²Strictly speaking, $\frac{f(x)-f(x_0)}{x-x_0}$ can be continuously continued at x_0 .

³This holds for Hamel basis in finite dimensions, but also for Schauder basis in infinite dimensions. Indeed, it can be shown that in any Banach space with a Schauder basis $(e_n)_n$, the coordinate functionals $F_n : V \rightarrow \mathbb{R}$, $F_n(\sum_{p=1}^\infty \alpha_p e_p) = \alpha_n$ and the projections $P_m : V \rightarrow V$, $P_m(\sum_{p=1}^\infty \alpha_p e_p) = \sum_{p=1}^m \alpha_p e_p$ are continuous functions.

⁴If x_0 is the supremum of S there exists a sequence $x_n \in S$ with $x_n \rightarrow x_0$. If $x_0 \in S$ then the sequence $x_n = x_0$ belongs to S and converges to x_0 . If $x_0 \notin S$, as $x_0 - \epsilon$ is not the supremum of S there exist $x \in S$ with $x_0 - \epsilon < x < x_0$ hence $B_\epsilon(x_0) \cap S \setminus \{x_0\} \neq \emptyset$ and x_0 is a limit point of S hence the limit of some sequence in S from Theorem 4.1.

7.4.1 The case $f : \mathbb{R} \rightarrow \mathbb{R}$

An important particular class of functions are the real functions of one variable.

Theorem 7.10 (**Properties of derivatives**). Consider $f, g : \mathbb{R} \rightarrow \mathbb{R}$. We have:

- Leibnitz rule if f and g are differentiable at x_0 , then so is their product and $(f \cdot g)'(x_0) = f'(x_0) \cdot g(x_0) + f(x_0) \cdot g'(x_0)$.
- chain rule if g is differentiable at x_0 and f is differentiable at $g(x_0)$, then the composition $(f \circ g)(x) = f(g(x))$ is differentiable at x_0 and $(f \circ g)'(x_0) = (f' \circ g)(x_0) \cdot g'(x_0)$.
- power series if $f = \sum_{n \geq 0} c_n x^n$ is a power series with radius of convergence $r > 0$ then $\sum_{n \geq 1} n c_n x^{n-1}$ is a power series with radius of convergence r and its sum equals $f'(x)$.

Proof. We have:

$$(f \cdot g)'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{1}{x - x_0} \left(f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0) \right).$$

The composition is straightforward.

For the last part, it is easy to check that the radius of converge of $\sum_{n \geq 1} n c_n x^{n-1}$ is r . As a power series converges absolutely inside its domain of convergence we have:

$$h(x) = \frac{f(x) - f(x_0)}{x - x_0} = \sum_{n=1}^{\infty} c_n \sum_{p=0}^{n-1} x^p x_0^{n-p},$$

and the series converges absolutely for $|x|, |x_0| < \rho < r$. We have:

$$|h(x) - h(x_0)| \leq \sum_{n=2}^{\infty} |c_n| \sum_{p=0}^{n-1} |x^p - x_0^p| |x_0|^{n-p} \leq \sum_{n=2}^{\infty} \frac{1}{r^n} |x - x_0| \sum_{p=1}^{n-1} \sum_{q=1}^p |x|^q |x_0|^{p-q} |x_0|^{n-p} \leq |x - x_0| \sum_{n \geq 2} \frac{n(n-1)}{r^n} \rho^{n-2},$$

and the series converges absolutely as $\rho < r$. □

We have some additional properties.

Theorem 7.11 (**Extremum criterion, Rolle's theorem and mean value theorem**). Let $f : [x_1, x_2] \rightarrow \mathbb{R}$ continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) . Then:

- Extremum criterion if f has a local extremum (maximum or minimum) at $x_0 \in (x_1, x_2)$ then $f'(x_0) = 0$.
- Rolle's theorem if $f(x_1) = f(x_2)$ then there exists $x \in (x_1, x_2)$ such that $f'(x) = 0$.
- Mean value theorem $\exists x \in (x_1, x_2)$ such that $\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(x)$.

Proof. Assume that f attains a maximum at $x_0 \in (x_1, x_2)$. Then:

$$\lim_{x \nearrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \geq 0, \quad \lim_{x \searrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \leq 0,$$

and as f is differentiable at x_0 , the two limits must coincide which is possible only if $f'(x_0) = 0$.

For the second item, by the extreme value theorem, f must attain a maximum and a minimum in the interval $[x_1, x_2]$. If $f(x_1)$ is both minimum and maximum, then the function is constant and its derivative is identically zero; if not we invoke the first item.

For the last item, we apply the second one to:

$$g(x) = f(x) - \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1) .$$

□

A direct consequence of this is that, under certain circumstances, functions can be well approximated by series.

Theorem 7.12 (Taylor series). *Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$, continuous on the closed interval and $k+1$ times differentiable on the open interval between x and a . Then there exists ξ between a and x such that*

$$f(x) = f(a) + (x-a)f'(a) + \frac{1}{2!}(x-a)^2 f''(a) + \cdots + \frac{1}{k!}(x-a)^k f^{(k)}(a) + R_k(x) , \quad R_k(x) = \frac{1}{(k+1)!}(x-a)^{k+1} f^{(k+1)}(\xi) ,$$

where $f^{(k)}(a) = \frac{d^k f}{dx^k}(a)$ is the k 'th derivative of f at the point a .

Proof. Define $F(t) = f(t) + (x-t)f'(t) + \frac{1}{2!}(x-t)^2 f''(t) + \cdots + \frac{1}{k!}(x-t)^k f^{(k)}(t)$ for t in the interval comprised between x and a and observe that $F'(t) = \frac{1}{k!}(x-t)^k f^{(k+1)}(t)$, while $F(x) = f(x)$ and $F(a)$ are the explicit Taylor terms.

For any G , there exists ξ between x and a such that

$$\frac{F'(\xi)}{G'(\xi)} = \frac{F(x) - F(a)}{G(x) - G(a)} .$$

Indeed, the function:

$$H(u) = [F(x) - F(a)]G(u) - [G(x) - G(a)]F(u) ,$$

is such that $H(x) = H(a) = F(x)G(a) - G(x)F(a)$, hence by Rolle's theorem there exist $\xi \in (x, a)$ such that $H'(\xi) = 0$. Using $G(t) = (x-t)^{k+1}$ we have $F(x) - F(a) = R_k(x) = \frac{F'(\xi)}{G'(\xi)}(G(x) - G(a)) = \frac{1}{k!}(x-\xi)^k f^{(k+1)}(\xi) \frac{(x-a)^{k+1}}{(k+1)(x-\xi)^k}$. □

Monotonic and convex functions can be understood starting from their derivatives.

Theorem 7.13 (Convexity criterion). *Consider $f : [x_1, x_2] \rightarrow \mathbb{R}$ continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) . Then:*

- f monotonically increasing if and only if $f' \geq 0$.
- recall that f is called convex if $\forall t \in [0, 1]$, and $\forall y_1, y_2 \in (x_1, x_2)$, $f(ty_1 + (1-t)y_2) \leq tf(y_1) + (1-t)f(y_2)$. If f is twice continuously differentiable on (x_1, x_2) then f is convex if and only if $f''(x) \geq 0$ for all $x \in (x_1, x_2)$.

Proof. If f is monotonically increasing, then using the definition we have $f'(x) \geq 0$. Conversely, if $f' \geq 0$ then f is increasing on any interval by the mean value theorem.

Setting $t = \frac{x_2 - x}{x_2 - x_1}$, convexity is equivalent to:

$$\frac{f(x) - f(x_1)}{x - x_1} \leq \frac{f(x_2) - f(x)}{x_2 - x} , \quad x_1 < x < x_2 .$$

Assume that $f''(x) < 0$ for some $x \in (x_1, x_2)$. Then, as f'' is continuous, $f''(x) < 0$ for x in some interval, $x \in [y_1, y_2] \subset (x_1, x_2)$, and f' is strictly decreasing on this interval. But if f is convex on $[y_1, y_2]$, then:

$$\frac{f(x) - f(y_1)}{x - y_1} \leq \frac{f(y_2) - f(x)}{y_2 - x} , \quad y_1 < x < y_2 ,$$

and by the mean value theorem, there exist $z_1 \in (y_1, x)$ and $z_2 \in (x, y_2)$, hence $z_1 < z_2$ such that:

$$f'(z_1) = \frac{f(x) - f(y_1)}{x - y_1} \leq \frac{f(y_2) - f(x)}{y_2 - x} = f'(z_2),$$

contradicting the fact that f' is strictly decreasing on $[y_1, y_2]$. $\square$

Note that if $f'(x_0) = 0$ and $f''(x_0) > 0$ for some $x_0 \in (x_1, x_2)$, then f' must be monotonically increasing in some neighbourhood $x_0 \in (y_1, y_2) \subset (x_1, x_2)$ hence f' must change sign at x_0 that is $f' \leq 0$ on (y_1, x_0) and $f' \geq 0$ on (x_0, y_2) . It follows that f is decreasing on (y_1, x_0) and increasing on (x_0, y_2) , that is f has a local minimum in x_0 .

Definition 7.3 (**Critical points**). Let $f : (x_1, x_2) \rightarrow \mathbb{R}$ be a differentiable function in (x_1, x_2) . A point x_0 such that $f'(x_0) = 0$ is called a critical point of f . If f is twice continuously differentiable in (x_1, x_2) the critical point is a local minimum (respectively maximum) if $f''(x_0) > 0$ (respectively $f''(x_0) < 0$).

If $f''(x_0) = 0$ the point can be a minimum (e.g. $f(x) = x^4$ in $x = 0$), maximum (e.g. $f(x) = -x^4$ in $x = 0$) or inflection point (e.g. $f(x) = x^3$ in $x = 0$).

Undetermined limits can sometimes be computed using the following rule.

Theorem 7.14 (**L'Hôpital's rule**). Let c and L be real numbers or $\pm\infty$, and I an interval containing c if $c \in \mathbb{R}$ or open at c if $c = \pm\infty$. Let two functions $f, g : I \rightarrow \mathbb{R}$ differentiable on I (except possibly at c) such that $g'(x) \neq 0$ (except possibly at c). If $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = L$ and either $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$ or $\lim_{x \rightarrow c} |g(x)| = \infty$ then $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = L$.

Proof. We take the case $\lim_{x \nearrow c} f(x) = \lim_{x \nearrow c} g(x) = 0$ (the other cases can be deduced from this one by simple tricks). If needed, we continue f and g at c to $f(c) = g(c) = 0$. For any $a < c$, apply Rolle's theorem to the function $h : [a, c] \rightarrow \mathbb{R}$:

$$h(y) = (f(y) - f(a))(g(c) - g(a)) - (f(c) - f(a))(g(y) - g(a)) \Rightarrow \exists y \in (a, c), \quad \frac{f'(y)}{g'(y)} = \frac{f(c) - f(a)}{g(c) - g(a)} = \frac{f(a)}{g(a)}.$$

As $\lim_{y \nearrow c} \frac{f'(y)}{g'(y)} = L$, for any ϵ , $\exists \delta$ such that $c - y < \delta \Rightarrow |\frac{f'(y)}{g'(y)} - L| < \epsilon$. But then, for any a with $c - a < \delta$, $|\frac{f'(a)}{g'(a)} - L| = |\frac{f'(y)}{g'(y)} - L|$ for some $y \in (a, c)$ hence $|\frac{f'(a)}{g'(a)} - L| < \epsilon$. $\square$

Any function can be approximated by a series, with a controlled rest term.

Exercise 7.6 (Differentiable but not continuously differentiable). Prove that the function:

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases},$$

is differentiable but not continuously differentiable at $x = 0$. We have:

$$x \neq 0, \quad f'(x) = 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x}), \quad f'(0) = \lim_{x \rightarrow 0} \frac{x^2 \sin(\frac{1}{x})}{x} = 0,$$

hence $f'(x)$ exist for all $x \in \mathbb{R}$ but it is not continuous in 0.

Exercise 7.7 (Derivatives of simple functions). Let $f : I \rightarrow \mathbb{R}$ for $I \in \mathbb{R}$ some suitable interval. Prove that:

- $f(x) = c$, $f'(x) = 0$. Prove that if $f'(x) = 0$ on an interval, then $f(x) = c$ for some constant c .
- $f(x) = x$, $f'(x) = 1$.
- $f(x) = x^n$, $f'(x) = nx^{n-1}$.

- $f(x) = e^x$, $f'(x) = e^x$.
- $f(x) = \ln(x)$, $f'(x) = \frac{1}{x}$.
- recalling that $x < \sin x < \tan x$ prove that $\sin'(x) = \cos(x)$ and $\cos'(x) = -\sin(x)$.
- prove that $e^{-iz}[\cos(x) + i\sin(x)] = 1$. The complex sinus and cosinus functions are defined as

$$e^{\pm iz} = \cos(z) \pm i\sin(z) \Rightarrow \begin{aligned} \sin(z) &= \frac{e^{iz} - e^{-iz}}{2i} = \sum_{k \geq 0} \frac{1}{(2k+1)!} (-1)^k z^{2k+1}, \\ \cos(z) &= \frac{e^{iz} + e^{-iz}}{2} = \sum_{k \geq 0} \frac{1}{(2k)!} (-1)^k z^{2k}. \end{aligned}$$

Exercise 7.8 (Generalized binomial theorem). Apply Taylor's theorem to $\sin(x)$, $\cos(x)$, e^x , $\ln(1+x)$ and $(1+x)^\alpha$.

Part II

HöMa II

Chapter 8

Spectral theory in finite dimensions

In this chapter we focus mainly on square matrices, or endomorphisms of a vector space.

8.1 Projectors

Two subspaces $V_1, V_2 \subset V$ are called linearly independent if $v_1 + v_2 = 0$ with $v_1 \in V_1, v_2 \in V_2$ then $v_1 = v_2 = 0$. For instance, if $\{e_1, \dots, e_k, h_1, \dots, h_l\}$ is a set of linearly independent vectors then $\text{Span}\{e_1, \dots, e_k\}$ and $\text{Span}\{h_1, \dots, h_l\}$ are linearly independent subspaces. Two subspaces are linearly independent if and only if $V_1 \cap V_2 = \{0\}$.

Exercise 8.1. *Prove that two subspaces are linearly independent if and only if $V_1 \cap V_2 = \{0\}$. $v_1 \in V_1 \Rightarrow \alpha_1 v_1 \in V_1$ and $v_2 \in V_2 \Rightarrow \alpha_2 v_2 \in V_2$.*

It follows that $\alpha_1 v_1 + \alpha_2 v_2 = 0 \Leftrightarrow V_1 \ni \alpha_1 v_1 = -\alpha_2 v_2 \in V_2$, hence $V_1 \cap V_2 = \{0\}$ implies $\alpha_1 = \alpha_2 = 0$. Conversely, if V_1 and V_2 are linearly independent, assume $v \in V_1 \cap V_2$ with $v \neq 0$. Then $\frac{1}{2}v - \frac{1}{2}v = 0$ and seeing the first term as an element of V_1 and the second one as an element of V_2 , we have $\frac{1}{2}v = 0 = -\frac{1}{2}v$, a contradiction.

The *sum* of two vector subspaces $V_1, V_2 \subset V$ is the vector subspace $V_1 + V_2 = \{v_1 + v_2 \mid v_1 \in V_1, v_2 \in V_2\}$. The sum is *direct*, denoted $V_1 \oplus V_2$, if V_1 and V_2 are linearly independent, or equivalently $V_1 \cap V_2 = \{0\}$. If $V = V_1 \oplus V_2$, then any vector $v \in V$ decomposes uniquely¹ as $v = v_1 + v_2$ with $v_1 \in V_1$ and $v_2 \in V_2$ and consequently $\dim V = \dim V_1 + \dim V_2$.

Proposition 8.1 (Projectors). *An idempotent operator $P : V \rightarrow V$, $P = P^2$ is called a projection and:*

- $Px = x$, for all $x \in \text{Im } P$
- $x = Px + (I - P)x$ with $Px \in \text{Im } P$ and $(I - P)x \in \ker P$ and $V = \ker P \oplus \text{Im } P$.

Proof. First, for any $x \in \text{Im } P$, $x = Py = P^2y = P(Py) = Px$. Second, for any x , $P(I - P)x = 0$, hence $(I - P)x \in \ker P$. If $x \in \ker P \cap \text{Im } P$ then $x = Py$ and $Px = 0$ hence $P(Py) = 0$. As $P = P^2$ we conclude that $Py = 0$ and as $Py = x$ it follows that $x = 0$. $\square$

For any direct sum decomposition $V = V_1 \oplus V_2$, any $v \in V$ will write uniquely as $v = v_1 + v_2$ and we define $P_1(v) = v_1$ and $P_2(v) = v_2$. These operators are projectors and $I = P_1 + P_2$ and $P_1P_2 = P_2P_1 = 0$.

Note there is a subtlety: one needs the decomposition $V = V_1 \oplus V_2$ first in order to identify the two projectors P_1 and P_2 . Consider $V = \{v = (v^1, v^2)\}$ and V_1 the subspace $V_1 = \{\alpha(1, 0)\}$. For any $w \notin V_1$ we have a decomposition $V = V_1 \oplus \{\alpha w\}$ because any vector decomposes uniquely as $(v_1, v_2) = (v_1 - v_2 \frac{w_1}{w_2})(1, 0) + v_2 \frac{1}{w_2}(w_1, w_2)$ and the two projectors are $P_1(v_1, v_2) = (v_1 - v_2 \frac{w_1}{w_2})(1, 0)$ respectively $P_2(v_1, v_2) = v_2 \frac{1}{w_2}(w_1, w_2)$. We check directly that:

$$(P_1 + P_2)(v_1, v_2) = (v_1, v_2), \quad P_1P_2 = (v_2 \frac{w_1}{w_2} - v_2 \frac{w_1}{w_2})(1, 0) = 0 = P_2P_1.$$

¹Assume two decompositions exist $v = v_1 + v_2 = v'_1 + v'_2$. Then $V_1 \ni v_1 - v'_1 = v'_2 - v_2 \in V_2$ and we conclude.

The point is that there is no a priori notion of “projector on $V_1 \subsetneq V$ ”: in fact there are as many projectors on V_1 , as there are possible decompositions $V = V_1 \oplus V_2$. The “intuitively natural” projector $P_1(v_1, v_2) = v_1(1, 0)$ is just one possible choice, the one corresponding to $w = (0, 1)$, but all the other possible projectors on V_1 are as good as this one.

8.1.1 Projectors along a basis

Given a basis of vectors $f_1, \dots, f_N$, one can build the projector operators along the basis elements. For any $x \in \mathbb{R}^N$, the equation $x = \sum_i x^i e_i = \sum_j y^j f_j$ writes in matrix form:

$$\begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^n \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & \dots & f_N \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ \vdots \\ y^n \end{pmatrix},$$

where the matrix change basis $S = \begin{pmatrix} f_1 & f_2 & \dots & f_N \end{pmatrix}$ is the matrix of the new basis vectors f expressed as column vectors in the standard basis e . Then $S^{-1}x = y$, and the inverse matrix is a matrix of row vectors:

$$S^{-1} = \begin{pmatrix} L_1 \\ L_2 \\ \vdots \\ L_N \end{pmatrix}, \quad L_i x = y^i$$

As the projector along f_i is $P_i x = y^i f_i$, it follows that $P_i = f_i L_i$, that is the product between the column vector f_i and the row vector L_i corresponding to the i 'th the inverse matrix.

8.2 Eigenvalues and eigenvectors

Definition 8.1 (**Eigenvalues, eigenvectors**). Consider a linear operator $A : V \rightarrow V$, with V a finite dimensional vector space over $\mathbb{F}$. A pair (λ, v) with $\lambda \in \mathbb{F}$ and $v \in V, v \neq 0$ is called an eigenpair, λ an eigenvalue and v an eigenvector of A if $Av = \lambda v$.

If $z \in \mathbb{C}$ is not an eigenvalue of A , we say that z belongs to the resolvent set of A and in this case $(A - zI)$ has zero kernel and is injective. As $\dim(\text{Im}(A - zI)) = \dim V$ it follows that $A - zI$ is also surjective hence it is invertible. We call $R(z, A) = (A - zI)^{-1}$ the resolvent of A at z .

The eigenvalues are the values λ for which $\dim \ker(A - \lambda I) > 0$, that is the system of equations $(A - \lambda I)v = 0$ admits a non trivial solution (note that in order to write this system of equations we need to pick a basis). This system of linear equations with constant coefficients is compatible undetermined if and only if $\det(A - \lambda I) = 0$. The polynomial $\det(A - \lambda I) = (-\lambda)^{\dim V} + O(\lambda^{\dim V - 1})$ is called the *characteristic polynomial* of A and it is basis independent. If $\det(A - \lambda I) = 0$, performing Gaussian elimination on $A - \lambda I$ leads to a *RREF* of determinant zero, hence necessarily having at least a row of zeroes and consequently at most $N - 1$ pivots and at least one non dominant column, $\dim \ker(A - \lambda I) \geq 1$.

Let us consider the complex case $\mathbb{F} = \mathbb{C}$. By the fundamental theorem of algebra any polynomial factors over its complex roots² hence the characteristic polynomial of A writes as:

$$\det(A - \lambda I) = \prod_{i=1}^q (\lambda_i - \lambda)^{m_i},$$

²If p is a polynomial of degree n and $p(z_i) = 0$ then $p(z) = p(z) - p(z_i) = (z - z_i)q(z)$ where $q(z)$ has degree $n - 1$ and we iterate.

where λ_i , $i = 1, \dots, q$ with $q \leq \dim V$ are the distinct roots of the characteristic polynomial. It follows that $\det(A - \lambda I) = 0$ if and only if $\lambda = \lambda_i$ for some i , that is the eigenvalues of A are exactly the roots λ_i . We call m_i the *algebraic multiplicity* of the eigenvalue λ_i and we have $\sum_{i=1}^q m_i = \dim V$. The eigenvectors corresponding to the eigenvalue λ_i form the vector subspace $\ker(A - \lambda_i I) \neq \{0\}$. We call $\ker(A - \lambda_i I) \neq \{0\}$ the eigenspace corresponding to λ_i .

Both $\ker(A - \lambda_i I)$ and $\text{Im}(A - \lambda_i I)$ are A -invariant subspaces, that is:

$$\begin{aligned} v \in \ker(A - \lambda_i I) &\Rightarrow (A - \lambda_i I)Av = A(A - \lambda_i I)v = 0 \Rightarrow Av \in \ker(A - \lambda_i I), \\ v \in \text{Im}(A - \lambda_i I) &\Rightarrow Av = A(A - \lambda_i I)w = (A - \lambda_i I)Aw \Rightarrow Av \in \text{Im}(A - \lambda_i I). \end{aligned}$$

Theorem 8.1 (*The eigenspaces of an operator are linearly independent*). *The eigenspaces $\ker(A - \lambda_i I)$ corresponding to distinct eigenvalues $\lambda_i \neq \lambda_j$ of $A : V \rightarrow V$ are linearly independent. We call $\gamma_i = \dim \ker(A - \lambda_i I)$ the geometric multiplicity of λ_i and $1 \leq \gamma_i \leq m_i$, where m_i is the algebraic multiplicity of λ_i .*

Proof. If $v \in \ker(A - \lambda_i I) \cap \ker(A - \lambda_j I)$ then $Av = \lambda_i v = \lambda_j v$ and as $\lambda_i \neq \lambda_j$ we conclude that $v = 0$.³

For the second part, we have already proved that $1 \leq \gamma_i$.⁴ Choosing a basis in $\ker(A - \lambda_i I)$ and completing to a basis in V , the characteristic polynomial takes the form:

$$\begin{vmatrix} \lambda_i - \lambda & 0 \\ 0 & * \end{vmatrix} = (\lambda_i - \lambda)^{\gamma_i} R(\lambda),$$

with $R(\lambda)$ some polynomial. On the other hand, $\det(A - \lambda I) = (\lambda_i - \lambda)^{m_i} Q(\lambda)$ and λ_i is not a root of $Q(\lambda)$, and we conclude that $\gamma_i \leq m_i$. $\square$

In the real case $\mathbb{F} = \mathbb{R}$ the coefficients of the characteristic polynomial are real. This does not mean that the roots of the characteristic polynomial are real, but rather that they come in complex conjugated pairs.⁵ An example is given by the matrix:

$$\begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}, \quad \det(A - \lambda I) = (3 - \lambda)(-1 - \lambda) + 8 = 5 - 2\lambda + \lambda^2 = ((1 - 2i) - \lambda)((1 + 2i) - \lambda).$$

In this case $Av = \lambda v$ has a *real* solution $v \in \mathbb{R}^N$ if and only if the eigenvalue λ is real (for $\lambda \notin \mathbb{R}$, $Av = \lambda v$ will only admit solutions in $\mathbb{C}^N$). For real vector spaces we factor the characteristic polynomial as:

$$\det(A - \lambda I) = \prod_{i \geq 1} (\lambda_i - \lambda)^{m_i} \prod_{j \geq 1} (\lambda^2 - 2a_j \lambda + b_j)^{n_j}, \quad a_j^2 - b_j < 0,$$

and the terms $(\lambda^2 - 2a_j \lambda + b_j)^{n_j}$ correspond to complex eigenvalues $\lambda_j^\pm = a_j \pm i\sqrt{b_j - a_j^2}$ whose associated eigenvectors are complex. If $n_j = 0$, then the real case parallels the complex case. If not, one can always *complexify*, that is extend the vector space from V to $V_C = \{v = v_1 + iv_2, v_1, v_2 \in V\}$ endowed with the natural operations and search for eigenvalues of $A : V_C \rightarrow V_C$.

³Alternatively, we proceeded by induction. Consider $v_i \in \ker(A - \lambda_i I)$ for $i = 1, \dots, q$. For $q = 2$, if $\alpha_1 v_1 + \alpha_2 v_2 = 0$ then:

$$A(\alpha_1 v_1 + \alpha_2 v_2) - \lambda_2(\alpha_1 v_1 + \alpha_2 v_2) = (\lambda_1 - \lambda_2)\alpha_1 v_1 = 0,$$

hence $\alpha_1 = 0$ which in turn implies $\alpha_2 = 0$. Assuming the statement proven for q , if $\sum_{i=1}^{q+1} \alpha_i v_i = 0$ then:

$$0 = A\left(\sum_{i=1}^{q+1} \alpha_i v_i\right) - \lambda_{q+1}\left(\sum_{i=1}^{q+1} \alpha_i v_i\right) = \sum_{i=1}^q (\lambda_i - \lambda_{q+1})\alpha_i v_i = 0,$$

therefore $(\lambda_i - \lambda_{q+1})\alpha_i = 0$ for $i = 1, \dots, q$ and, as $\lambda_i - \lambda_{q+1} \neq 0$, by the induction hypothesis we conclude that $\alpha_i = 0$ for $i = 1, \dots, q$ and consequently $\alpha_{q+1} = 0$.

⁴If $\det(A - \lambda_i I) = 0$ then in the RREF of $A - \lambda_i I$ we must have at least a row of zeroes, hence at least a non dominant column and consequently $\ker(A - \lambda_i I)$ has dimension at least 1, as the dimension of the kernel is the number of non dominant columns.

⁵If $p(z) = 0$ then $\overline{p(z)} = p(\bar{z}) = 0$ for any polynomial with real coefficients $p(\cdot)$.

8.2.1 Diagonalizable matrices

Definition 8.2 (**Diagonalizable matrices**). A $N \times N$ matrix A is diagonalizable if it is similar to a diagonal matrix, $A = SDS^{-1}$, with D having non zero elements only on the diagonal.

Seen as an operator $A : V \rightarrow V$ with V a finite dimensional vector space, A is diagonalizable if V admits a basis $\{e_\nu\}_\nu$ such that every e_ν is an eigenvector of A , $Ae_\nu = \lambda_\nu e_\nu$ (the eigenvalues λ_ν do not need to be distinct). The basis e_ν is called an eigenbasis of A and in an eigenbasis the matrix elements of A are $a_{\mu\nu} = \theta^\mu(Ae_\nu) = \lambda_\mu \delta_{\mu\nu}$. Equivalently, A is diagonalizable if and only if V decomposes in A -invariant subspaces as:

$$V = \ker(A - \lambda_1 I) \oplus \cdots \oplus \ker(A - \lambda_q I),$$

with λ_i pairwise distinct. An eigenbasis of A is obtained by picking a basis in each of its eigenspaces and taking the union.

The transition matrix S such that $D = S^{-1}AS$ is the matrix of the eigenvectors $S = (e_1, \dots, e_N)$ expressed as column vectors in the original basis.

Not all the matrices are diagonalizable. Matrices that are not diagonalizable are called *defective*.

Theorem 8.2 (**Diagonalizability criterion**). An operator $A : V \rightarrow V$ over a complex vector space V is diagonalizable if and only if the algebraic and geometric multiplicities of all its eigenvalues coincide. For a real vector space, one needs to require furthermore that the algebraic multiplicities sum to $\dim V$ (this guarantees that the characteristic polynomial admits only real roots and factors over them).

In particular, if all the roots of the characteristic polynomial are distinct then the matrix is diagonalizable.

Proof. We have $m_i = \gamma_i$ for any i if and only if any $\ker(A - \lambda_i I)$ admits a basis of m_i linearly independent vectors. As $\sum_i m_i = \dim V$ and the subspaces $\ker(A - \lambda_i I)$ are linearly independent, the union of these bases is a basis in V consisting in eigenvectors of A . $\square$

If A is diagonalizable then, associated to the orthogonal decomposition $V = \bigoplus_i \ker(A - \lambda_i I)$ in eigenspaces of A , we have a family of projectors P_i on the eigenspaces $\ker(A - \lambda_i I)$: any $v \in V$ decomposes uniquely as $v = \sum_i v_i$ with $v_i \in \ker(A - \lambda_i I)$ and we define $P_i(v) = v_i$. We have:

- the P_i s are linear operators and $P_i P_j = P_i \delta_{ij}$ as $\ker(A - \lambda_i I)$ are linearly independent.
- $AP_i = P_i AP_i = P_i A$ as $\ker(A - \lambda_i I)$ are A invariant.⁶
- $I = \sum_{i=1}^q P_i$ as $V = \bigoplus_i \ker(A - \lambda_i I)$.
- when restricted to $\ker(A - \lambda_i I)$, A is the multiplication by λ_i , $AP_i = \lambda_i P_i$.
- $\text{Im}(A - \lambda_i I) = \bigoplus_{j=1, j \neq i}^q \ker(A - \lambda_j I)$, as $\forall w = (A - \lambda_i I)v$, we have $w = \sum_{j=1}^q P_j w$ and $P_j w = \sum_{j=1}^q P_j (A - \lambda_i I)v = \sum_{j=1, j \neq i}^q P_j (A - \lambda_j I)v$ and $\ker(A - \lambda_i I)$ are linearly independent.

Theorem 8.3 (**Simultaneous diagonalizability criterion**). Two operators $A, B : V \rightarrow V$ over a finite dimensional vector space are simultaneously diagonalizable, that is they admit a common eigenbasis $\{e_\nu\}_\nu$ with $Ae_\nu = \lambda_\nu e_\nu$ and $Be_\nu = \mu_\nu e_\nu$ if and only if they are diagonalizable and commute.

Proof. If A and B have a common eigenbasis then they commute on every element of the basis hence on any element in V .

Conversely, let us first prove that if B is diagonalizable on V and W is a B -invariant subspace $B(W) \subset W \subset V$ then B is diagonalizable on W . Denoting $\mu_1, \dots, \mu_k$ the distinct eigenvalues of B we have $V = \bigoplus_{i=1}^k \ker(B - \mu_i I)$ and any $w \in W$

⁶ As $v_i \in \ker(A - \lambda_i I) \Rightarrow Av_i \in \ker(A - \lambda_i I)$ we have $AP_i = P_i AP_i$. For the second equality, we note that $P_i A = P_i A \sum_j P_j = \sum_j P_i P_j A P_j = P_i AP_i$.

decomposes uniquely as $w = w_1 + \dots + w_k$ with $w_j \in \ker(B - \mu_j I)$. As W is B -invariant it follows that $B^m(w) \in W$ for any m , that is for any m we have $B^m(w) = \lambda_1^m w_1 + \dots + \lambda_k^m w_k \in W$. Selecting $m = 0, \dots, k-1$ we can solve for w_j as a linear combination⁷ $w_j = \sum_{l=0}^{k-1} c_{jl} B^l(w)$ hence $w_j \in W$ and $W = \bigoplus_{j=1}^k (\ker(B - \mu_j I) \cap W)$, that is $B|_W$ is diagonalizable.

Let us now assume A and B are diagonalizable on V and commute. For any λ_i eigenvalue of A , the eigenspace $V_i = \ker(A - \lambda_i I)$ is B invariant, $v \in \ker(A - \lambda_i I) \Rightarrow (A - \lambda_i I)(Bv) = B(A - \lambda_i I)v = 0$, hence $Bv \in \ker(A - \lambda_i I)$. By the previous paragraph $B|_{V_i}$ is also diagonalizable and:

$$V = \bigoplus_{i=1}^q \ker(A - \lambda_i I) = \bigoplus_{i=1}^q \bigoplus_{j=1}^k (\ker(B - \mu_j I) \cap \ker(A - \lambda_i I)).$$

A common eigenbasis for A and B is obtained by taking the union over i and j of bases in each $\ker(B - \mu_j I) \cap \ker(A - \lambda_i I)$. $\square$

8.2.2 Spectral decomposition of diagonalizable matrices

The *spectral decomposition* of the operator A consists in realising that A can be written as a sum of eigenvalues multiplying the projectors on its eigenspaces:

$$A = \sum_{i=1}^q A P_i = \sum_{i=1}^q \lambda_i P_i,$$

that is for any $v \in V$, $v = \sum_i v_i$ with $v_i \in \ker(A - \lambda_i I)$ and $Av = \sum_i Av_i = \sum_i \lambda_i v_i = \sum_i \lambda_i P_i v$. The main use of the spectral decomposition of a diagonalizable matrix A is that it allows us to evaluate arbitrary functions of A :

$$f(A) = \sum_i f(\lambda_i) P_i,$$

as this holds for polynomials in A and by extension for any function which can be expanded in Taylor series. In a basis in V which consists in the union of bases in the eigenspaces, both A and $f(A)$ are diagonal:

$$A = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_N \end{pmatrix}, \quad f(A) = \begin{pmatrix} f(\lambda_1) & & \\ & f(\lambda_2) & \\ & & \ddots \\ & & & f(\lambda_N) \end{pmatrix},$$

where in this equation some of the eigenvalues can be equal.

Exercise 8.2. Check directly that for any polynomial $P(A) = \sum_i f(\lambda_i) P_i$.

For instance the matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ has:

$$\begin{aligned} \det(A - \lambda I) &= (1 - \lambda)(2 - \lambda), & \lambda_1 &= 1, \lambda_2 = 2; \\ \ker(A - I) &= \ker \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \text{Span} \left\{ e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}, & \ker(A - 2I) &= \ker \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} = \text{Span} \left\{ e_1 + e_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}; \\ P_1 &= \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, & P_2 &= \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

where in order to identify the two projectors we note that $xe_1 + ye_2 = (x - y)e_1 + y(e_1 + e_2)$.

⁷The system is compatible determined as the eigenvalues are pairwise different. For instance $w_1 + w_2 = w$, $\lambda_1 w_1 + \lambda_2 w_2 = Bw$ is solved by $w_1 = \frac{Bw - \lambda_2 w}{\lambda_1 - \lambda_2}$ and $w_2 = \frac{\lambda_1 w - Bw}{\lambda_1 - \lambda_2}$.

For the matrix $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ seen as an operator over $\mathbb{C}^2$ we find:

$$\det(A - \lambda I) = (\cos \theta - \lambda)^2 + \sin^2 \theta = \lambda^2 - 2\lambda \cos \theta + 1, \quad \lambda_{\pm} = e^{\pm i\theta};$$

$$\ker(A - e^{i\theta}) = \ker \begin{pmatrix} -i \sin \theta & -\sin \theta \\ \sin \theta & -i \sin \theta \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ -i \end{pmatrix} \right\}, \quad \ker(A - e^{-i\theta}) = \ker \begin{pmatrix} i \sin \theta & -\sin \theta \\ \sin \theta & i \sin \theta \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ i \end{pmatrix} \right\};$$

$$P_1 = \begin{pmatrix} \frac{1}{2} & i\frac{1}{2} \\ -i\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad P_2 = \begin{pmatrix} \frac{1}{2} & -i\frac{1}{2} \\ i\frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

where in order to identify the two projector we note that $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{x+iy}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} + \frac{x-iy}{2} \begin{pmatrix} 1 \\ i \end{pmatrix}$.

Exercise 8.3. Consider the matrix:

$$A = \begin{pmatrix} 4 & 1 & 6 & -3 \\ -2 & 1 & 2 & -5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Compute the eigenvalues of A and their algebraic multiplicities.

We have the characteristic polynomial:

$$\begin{vmatrix} 4-\lambda & 1 & 6 & -3 \\ -2 & 1-\lambda & 2 & -5 \\ 0 & 0 & 1-\lambda & 1 \\ 0 & 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda) \begin{vmatrix} 4-\lambda & 1 \\ -2 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda) [(4-\lambda)(1-\lambda) + 2] = (2-\lambda)(1-\lambda) [6-5\lambda+\lambda^2] = (2-\lambda)(1-\lambda)(2-\lambda)(3-\lambda),$$

hence the eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 2$ and $\lambda_3 = 3$ with algebraic multiplicities $m_1 = 1$, $m_2 = 2$ and $m_3 = 1$.

Determine the geometric multiplicities of the eigenvalues and identify bases in the corresponding eigenspaces.

In order:

$$\begin{aligned} A - I &= \begin{pmatrix} 3 & 1 & 6 & -3 \\ -2 & 0 & 2 & -5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 3 & 1 & 6 & -3 \\ 0 & \frac{2}{3} & 2 + \frac{2}{3}6 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & \frac{1}{3} & 2 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \\ A - 2I &= \begin{pmatrix} 2 & 1 & 6 & -3 \\ -2 & -1 & 2 & -5 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & 6 & -3 \\ 0 & 0 & 8 & -8 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & \frac{1}{2} & 3 & -\frac{3}{2} \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & \frac{1}{2} & 0 & \frac{3}{2} \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \\ A - 3I &= \begin{pmatrix} 1 & 1 & 6 & -3 \\ -2 & -2 & 2 & -5 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \end{aligned}$$

hence we have $\ker(A - I) = \text{Span}\{f_1 = e_3 + e_1 - 9e_2\}$, $\ker(A - 2I) = \text{Span}\{f_2 = e_2 - \frac{1}{2}e_1, f_3 = e_4 - \frac{3}{2}e_1 + e_3\}$, $\ker(A - 3I) = \text{Span}\{f_4 = e_2 - e_1\}$ with geometric multiplicities $\gamma_1 = 1$, $\gamma_2 = 2$, $\gamma_3 = 1$ hence A is diagonalizable.

Identify the projectors on the eigenspaces, write the spectral decomposition of A and compute $\ln(A)$.

In order to find the projectors we need to expand an arbitrary vector on the basis of eigenvectors, $xe_1 + ye_2 + ze_3 + te_4 = \alpha f_1 + \beta f_2 + \gamma f_3 + \delta f_4$. This is a linear system:

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & -1 \\ -9 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}, \quad x = S\alpha,$$

which is solved by inverting the matrix $S^{-1}x = \alpha$. Regarding S^{-1} as a matrix of row vectors L_i , the projector along f_i is $P_i = f_i L_i$.

Instead of inverting the matrix mechanically, we can note that $t = \gamma$ and $z = \alpha + \gamma \Rightarrow \alpha = z - t$ and the remaining two equations write $x = (z - t) - \frac{1}{2}\beta - \frac{3}{2}t - \delta$ respectively $y = -9(z - t) + \beta + \delta$ or equivalently:

$$x - z + \frac{5}{2}t = -\frac{1}{2}\beta - \delta, \quad y + 9z - 9t = \beta + \delta \Rightarrow \beta = \frac{x - z + \frac{5}{2}t + y + 9z - 9t}{1/2} = \frac{x + y + 8z - \frac{13}{2}t}{1/2}, \quad \delta = \frac{y + 9z - 9t + 2x - 2z + 5t}{-1} = \frac{2x + y + 7z - 4t}{-1},$$

Leading to $S^{-1} = \begin{pmatrix} 0 & 0 & 1 & -1 \\ 2 & 2 & 16 & -13 \\ 0 & 0 & 0 & 1 \\ -2 & -1 & -7 & 4 \end{pmatrix}$, hence:

$$P_1 = \begin{pmatrix} 1 \\ -9 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & -1 \\ 0 & 0 & -9 & 9 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 2 & 2 & 16 & -13 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad P_3 = \begin{pmatrix} -3/2 \\ 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$P_4 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 2 & 1 & 7 & -4 \\ -2 & -1 & -7 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Finally, we get:

$$A = \begin{pmatrix} 4 & 1 & 6 & -3 \\ -2 & 1 & 2 & -5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & -1 \\ 0 & 0 & -9 & 9 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} + 2 \begin{pmatrix} -1 & -1 & -8 & 5 \\ 2 & 2 & 16 & -13 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} + 3 \begin{pmatrix} 2 & 1 & 7 & -4 \\ -2 & -1 & -7 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \ln(A) = \ln(2) \begin{pmatrix} -1 & -1 & -8 & 5 \\ 2 & 2 & 16 & -13 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \ln(3) \begin{pmatrix} 2 & 1 & 7 & -4 \\ -2 & -1 & -7 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

8.2.3 Decomposition in invariant subspaces for a generic matrix

We now consider the general case of a matrix which is not necessarily diagonalizable.

Theorem 8.4 (Decomposition in A -invariant subspaces). *Let V be a finite dimensional vector space. We denote the distinct eigenvalues of a linear operator $A : V \rightarrow V$ by $\lambda_1, \dots, \lambda_q$. Then there exist integers $n_1, \dots, n_q \geq 1$ such that⁸ V decomposes as:*

$$V = \ker(A - \lambda_1 I)^{n_1} \oplus \dots \oplus \ker(A - \lambda_q I)^{n_q},$$

where the subspaces $\ker(A - \lambda_i I)^{n_i}$ are A -invariant.

Proof. Consider λ an eigenvalue of A and denote $N_p = \ker(A - \lambda I)^p$ for $p \geq 1$ and we set $N_0 = \{0\}$. Clearly $N_0 \subset N_1 \subset N_2 \subset \dots \subset N_i \subset \dots$. If $N_i \neq N_{i+1}$, then $\dim N_{i+1} \geq \dim N_i + 1$, hence there exist n such that $N_{n-1} \neq N_n = N_{n+1}$. Furthermore:

For any $l \geq 1$, $N_{n+l} = N_{n+l+1}$. We proceed by induction. By the induction hypothesis, we have that $N_{n+l} = N_n$. Assume there exists v such that $v \in N_{n+l+1}$ but $v \notin N_{n+l}$, that is $(A - \lambda I)^{n+l+1}v = 0$ but $(A - \lambda I)^{n+l}v \neq 0$. Denoting $w = (A - \lambda I)v \neq 0$, we have $(A - \lambda I)^{n+l}w = 0$, but $(A - \lambda I)^{n+l-1}w \neq 0$, contradicting the induction hypothesis.

$\ker(A - \lambda I)^n \cap \text{Im}(A - \lambda I)^n = \{0\}$. Assume there exists $v \neq 0$ such that $(A - \lambda I)^n v = 0$ and $v = (A - \lambda I)^n w$ for some w . Then $(A - \lambda I)^{2n} w = (A - \lambda I)^n v = 0$, hence $w \in \ker(A - \lambda I)^{2n}$ and by the previous item $w \in \ker(A - \lambda I)^n$, hence $(A - \lambda I)^n w = 0$, contradicting the fact that $v \neq 0$.

$V = \ker(A - \lambda I)^n \oplus \text{Im}(A - \lambda I)^n$. The subspaces $\ker(A - \lambda I)^n$ and $\text{Im}(A - \lambda I)^n$ are linearly independent hence:

$$\dim \left(\ker(A - \lambda I)^n \oplus \text{Im}(A - \lambda I)^n \right) = \dim \ker(A - \lambda I)^n + \dim \text{Im}(A - \lambda I)^n = \dim V,$$

where we used the rank nullity theorem. It follows that $\ker(A - \lambda I)^n \oplus \text{Im}(A - \lambda I)^n$ is a subspace of V having the same dimension, hence must equal V .

Note that both $\ker(A - \lambda I)^n$ and $\text{Im}(A - \lambda I)^n$ are A -invariant subspaces. When restricted to $\ker(A - \lambda I)^n$, the operator A has only one eigenvalue, λ : if $Av = \mu v$ and $v \in \ker(A - \lambda I)^n$, then $(A - \lambda I)v = (\mu - \lambda)v \Rightarrow (A - \lambda I)^n v = (\mu - \lambda)^n v$, hence $\mu = \lambda$. When restricted to the subspace $\text{Im}(A - \lambda I)^n$, λ is not an eigenvalue of A : assuming the contrary, there exists $v \in \text{Im}(A - \lambda I)^n$ and $v \in \ker(A - \lambda I) \subset \ker(A - \lambda I)^n$ contradicting the fact that the $\text{Im}(A - \lambda I)^n$ and $\ker(A - \lambda I)^n$ are linearly independent.

⁸More precisely one should say minimal integers $n_1, \dots, n_q \geq 1$.

We proceed by induction: as $\dim \ker(A - \lambda I)^n \geq 1$, we use the decomposition $V = \ker(A - \lambda I)^n \oplus \operatorname{Im}(A - \lambda I)^n$ and proceed to the A -invariant subspace $\operatorname{Im}(A - \lambda I)^n$. $\square$

Let us consider the matrix:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \det(A - \lambda I) = (1 - \lambda)^2, \quad A - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

We have that the eigenvalue $\lambda_1 = 1$ has algebraic multiplicity 2, a basis in $\operatorname{Im}(A - I)$ is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and a basis in $\ker(A - I)$ is $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, hence $\lambda_1 = 1$ has geometric multiplicity 1.

Exercise 8.4. *Prove that for the three matrices below:*

$$\begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \quad \begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

λ is an eigenvalue of algebraic multiplicity 3, 3, 3 and 4 and geometric multiplicity 3, 2, 1 and 2 and find bases in the kernel respectively the image of $(A - \lambda I)$

8.2.4 Spectral decomposition for generic matrices

Associated to the orthogonal decomposition $V = \bigoplus_i \ker(A - \lambda_i I)^{n_i}$ in invariant subspaces of A , we have a family of projectors P_i on $\ker(A - \lambda_i I)^{n_i}$ and:

- $P_i P_j = P_i \delta_{ij}$ as $\ker(A - \lambda_i I)^{n_i}$ are linearly independent.
- $AP_i = P_i AP_i = P_i A$ as $\ker(A - \lambda_i I)^{n_i}$ are A invariant.
- $I = \sum_{i=1}^q P_i$ as $V = \bigoplus_i \ker(A - \lambda_i I)^{n_i}$.
- denoting $B_i = A - \lambda_i I$, we have $AP_i = (\lambda_i + B_i)P_i$ with B_i such that $B_i^{n_i} P_i = 0$.

We note that $BP_i = P_i BP_i = P_i B$ and $1 \leq n_i \leq m_i - \gamma_i + 1$, where m_i and γ_i are the algebraic and geometric multiplicities of λ_i , as $\gamma_i = \dim \ker B_i < \dim \ker B_i^2 < \dots < \dim \ker B_i^{n_i} = m_i$. Checking first for polynomials:

$$A^n = \sum_{i=1}^q (\lambda_i I + A - \lambda_i I)^n P_i = \sum_{i=1}^q \sum_{k=0}^n \binom{n}{k} \lambda_i^{n-k} (A - \lambda_i I)^k P_i = \sum_{i=1}^q \sum_{k=0}^n \frac{(x^n)^{(k)}}{k!} \Big|_{x=\lambda_i} (A - \lambda_i I)^k P_i,$$

we conclude that for any function of A we have:

$$f(A) = \sum_{i=1}^q \sum_{k \geq 0} \frac{f^{(k)}(\lambda_i)}{k!} (A - \lambda_i I)^k P_i,$$

where the sum over k is finite and $(A - \lambda_i I)^0 = I$. Note that for diagonalizable matrices the sum stops at $k = 0$, as $(A - \lambda_i I)P_i = 0$, and we recover the spectral decomposition of diagonalizable matrices. For instance, for $|z| > \|A\|_{\text{op}}$ we

have:⁹

$$\frac{1}{z-A} = \sum_{n \geq 0} \frac{1}{z^{n+1}} A^n = \sum_{i=1}^q \left(\sum_{n \geq 0} \frac{x^n}{z^{n+1}} \right)^{(k)} \Big|_{x=\lambda_i} (A - \lambda_i I)^k P_i = \sum_{i=1}^q \sum_{k \geq 0} \frac{1}{(z - \lambda_i)^{k+1}} (A - \lambda_i I)^k P_i.$$

As an example:

$$A = \begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}, \quad P_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \lambda_2 - \lambda_1 \end{pmatrix}, \quad B_1 P_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$B_2 = \begin{pmatrix} \lambda_1 - \lambda_2 & 1 & 0 \\ 0 & \lambda_1 - \lambda_2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B_2 P_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad f(\lambda) = \begin{pmatrix} f(\lambda_1) & f'(\lambda_1) & 0 \\ 0 & f(\lambda_1) & 0 \\ 0 & 0 & f(\lambda_2) \end{pmatrix}.$$

8.2.5 The Jordan normal form

When a matrix is not diagonalizable it can be brought by a similarity transformation to a simplified form, called the Jordan normal form, in which, beyond the diagonal entries, the matrix can have non zero entries only on the upper diagonal.

Theorem 8.5 (The Jordan normal form). *Let A be an arbitrary square matrix. Then there exists a complex similarity transformation (invertible matrix S) such that:*

$$A = SJS^{-1}, \quad J = \begin{pmatrix} J_{\lambda_1} & 0 & \dots & 0 \\ 0 & J_{\lambda_2} & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & J_{\lambda_r} \end{pmatrix}, \quad J_{\lambda_i} = \begin{pmatrix} \lambda_i & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda_i & 1 & \dots & 0 & 0 \\ \vdots & & & & & \\ \vdots & & & & & \\ 0 & 0 & 0 & \dots & \lambda_i & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda_i \end{pmatrix},$$

where the block matrices J_{λ_i} have diagonal entries the eigenvalues λ_i of A and upper diagonal entries 1. The blocks J_{λ_i} are called Jordan blocks. We emphasize that there can be several Jordan blocks corresponding to the same eigenvalue. Note that the case of A diagonalizable corresponds to all the Jordan blocks having size 1.

Proof. The decomposition of V in A -invariant subspaces writes as $V = \ker(A - \lambda_1 I)^{n_1} \oplus \dots \oplus \ker(A - \lambda_q I)^{n_q}$. Picking a basis in each invariant subspace, the matrix of A is block diagonal, hence we can restrict to one of the invariant subspaces $\ker(A - \lambda I)^n$.

Let us relabel $\ker(A - \lambda I)^n = V$ and $B = A - \lambda I$. Then $B : V \rightarrow V$ is such that $B^n = 0$ and following the proof of Theorem 8.4 we define $N_p = \ker B^p$ for $p \geq 1$ and we have $N_0 \subsetneq N_1 \subsetneq N_2 \subsetneq N_3 \subsetneq \dots N_n = N_{n+1} = \dots = V$.

As $N_{n-1} \subsetneq N_n$, there exist $e_1^{(n)} \in N_n \setminus N_{n-1}$ and in this case $\text{Span}\{e_1^{(n)}\} \cap N_{n-1} = 0$. If $\text{Span}\{e_1^{(n)}\} \oplus N_{n-1} \subsetneq N_n$, then iterating we find $e_1^{(n)}, \dots, e_{s_n}^{(n)}$ linearly independent vectors (which we call generalized eigenvectors of rank n) such that $N_n = \text{Span}\{e_1^{(n)}\} \oplus \dots \oplus \text{Span}\{e_{s_n}^{(n)}\} \oplus N_{n-1}$.

We pass to N_{n-1} . As $e_i^{(n)} \in N_n \setminus N_{n-1}$ we have $B(e_i^{(n)}) \in N_{n-1} \setminus N_{n-2}$ and the vectors $B(e_1^{(n)}), \dots, B(e_{s_n}^{(n)})$ are linearly independent and linearly independent from N_{n-2} . For any linear combination:

$$\beta_1 B(e_1^{(n)}) + \dots + \beta_{s_n} B(e_{s_n}^{(n)}) + v^{(n-2)} = 0, \quad v^{(n-2)} \in N_{n-2},$$

⁹As a glimpse into the future, this expression implies that the spectral projector can be written as a complex integral along a circle $C_\epsilon(\lambda_i)$ centered at λ_i and of radius ϵ , $P_i = \int_{C_\epsilon(\lambda_i)} \frac{dz}{2\pi i} \frac{1}{z-A}$ and that any holomorphic function can be represented as $f(A) = \int_\Gamma \frac{dz}{2\pi i} \frac{f(z)}{z-A}$ for Γ a union of simple curves such that the spectrum of A lies in the interior of the curves.

we apply B^{n-2} , recall that $e_1^{(n)}, \dots, e_{s_n}^{(n)}$ are linearly independent and conclude that $\beta_1 = \dots \beta_{s_n} = 0$ and consequently $v^{(n-2)} = 0$. If $\text{Span}\{B(e_1^{(n)})\} \oplus \dots \oplus \text{Span}\{B(e_{s_n}^{(n)})\} \oplus N_{n-2} \subsetneq N_{n-1}$, we identify iteratively $e_1^{(n-1)}, \dots, e_{s_{n-1}}^{(n-1)}$ additional vectors in $N_{n-1} \setminus N_{n-2}$ (generalized eigenvectors of rank $n-1$) linearly independent of $B(e_i^{(n)})$ such that:

$$N_{n-1} = \text{Span}\{B(e_1^{(n)})\} \oplus \dots \oplus \text{Span}\{B(e_{s_n}^{(n)})\} \oplus \text{Span}\{e_1^{(n-1)}\} \oplus \dots \oplus \text{Span}\{e_{s_{n-1}}^{(n-1)}\} \oplus N_{n-2}.$$

At next level, adding (if there exist) generalized eigenvectors of rank $n-2$ we get:

$$N_{n-2} = \text{Span}\{B^2(e_1^{(n)})\} \oplus \dots \oplus \text{Span}\{B^2(e_{s_n}^{(n)})\} \oplus \text{Span}\{B(e_1^{(n-1)})\} \oplus \dots \oplus \text{Span}\{B(e_{s_{n-1}}^{(n-1)})\} \oplus \text{Span}\{e_1^{(n-2)}\} \oplus \dots \oplus \text{Span}\{e_{s_{n-2}}^{(n-2)}\} \oplus N_{n-3},$$

and iterating we obtain at the end a basis in N_n in consisting in *Jordan* chains:

$$\begin{array}{ll} e_1^{(n)}, B(e_1^{(n)}), B^2(e_1^{(n)}), \dots, B^{n-1}(e_1^{(n)}) & \dots \quad e_{s_n}^{(n)}, B(e_{s_n}^{(n)}), B^2(e_{s_n}^{(n)}), \dots, B^{n-1}(e_{s_n}^{(n)}), \\ e_1^{(n-1)}, B(e_1^{(n-1)}), B^2(e_1^{(n-1)}), \dots, B^{n-2}(e_1^{(n-1)}) & \dots \quad e_{s_{n-1}}^{(n-1)}, B(e_{s_{n-1}}^{(n-1)}), B^2(e_{s_{n-1}}^{(n-1)}), \dots, B^{n-2}(e_{s_{n-1}}^{(n-1)}), \\ \dots & \\ e_1^{(2)}, B(e_1^{(2)}) & \dots \quad e_{s_2}^{(2)}, B(e_{s_2}^{(2)}), \\ e_1^{(1)} & \dots \quad e_{s_1}^{(1)}, \end{array}$$

where the last vector in each chain is a genuine eigenvector of A . The numbers s_i of Jordan chains of length i respect $s_n = \dim N_n - \dim N_{n-1}$, $s_n + s_{n-1} = \dim N_{n-1} - \dim N_{n-2}$ and so on up to $s_n + s_{n-1} + \dots + s_1 = \dim N_1$. On a Jordan chain of length l , denoting $x_q = B^{l-q}e_i^{(l)}$ and taking into account that $Bx_1 = 0$, we have $Ax_q = \lambda x_q + Bx_q = \lambda x_q + x_{q-1}$, and taking the x s as column vectors:

$$x_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots \quad x_l = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

A takes the form of a Jordan block.

The geometric multiplicity of λ is the numbers of independent Jordan chains $\gamma = \dim N_1 = \sum_{l \geq 1} s_l$, and its algebraic multiplicity is the dimension of the corresponding N_n , $m = \dim N_n = \sum_{l \geq 1} l s_l$. $\square$

Let us consider the matrix:

$$A = \begin{pmatrix} 3 & 4 & 1/4 & -3/4 \\ 0 & 2 & 0 & 0 \\ -1 & -1 & 1/4 & -3/4 \\ 1 & 1 & 7/4 & 11/4 \end{pmatrix}.$$

The characteristic polynomial of this matrix is:

$$\begin{aligned} \det(A - \lambda I) &= (2 - \lambda) \begin{vmatrix} 3 - \lambda & 1/4 & -3/4 \\ -1 & 1/4 - \lambda & -3/4 \\ 1 & 7/4 & 11/4 - \lambda \end{vmatrix} = (2 - \lambda) \begin{vmatrix} 3 - \lambda & 1/4 & -3/4 \\ 0 & 2 - \lambda & 2 - \lambda \\ 1 & 7/4 & 11/4 - \lambda \end{vmatrix} = (2 - \lambda) \begin{vmatrix} 3 - \lambda & 1/4 & -1 \\ 0 & 2 - \lambda & 0 \\ 1 & 7/4 & 1 - \lambda \end{vmatrix} \\ &= (2 - \lambda)^2 ((3 - \lambda)(1 - \lambda) + 1) = (2 - \lambda)^4, \end{aligned}$$

that is the matrix has eigenvalue 2 with algebraic multiplicity 4. Denoting $B = A - 2I$ we have:

$$B = \begin{pmatrix} 1 & 4 & 1/4 & -3/4 \\ 0 & 0 & 0 & 0 \\ -1 & -1 & -7/4 & -3/4 \\ 1 & 1 & 7/4 & 3/4 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 9/4 & 5/4 \\ 0 & 1 & -1/2 & -1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B^2 = \begin{pmatrix} 0 & 3 & -3/2 & -3/2 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & 3/2 & 3/2 \\ 0 & 3 & -3/2 & -3/2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 0 & 1 & -1/2 & -1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

and $B^3 = 0$. Thus 2 has geometric multiplicity 2 and we have that the invariant spaces are:

$$N_3 = \ker B^3 = \mathbb{R}^4, \quad N_2 = \ker B^2 = \text{Span}\left\{e_1, e_3 + \frac{1}{2}e_2, e_4 + \frac{1}{2}e_2\right\}, \quad N_1 = \ker B = \text{Span}\left\{e_3 + \frac{1}{2}e_2 - \frac{9}{4}e_1, e_4 + \frac{1}{2}e_2 - \frac{5}{4}e_1\right\}.$$

One can proceed in two ways: one can either search for generalized eigenvectors of rank 3, that is linearly independent vectors in $N_3 \setminus N_2$, build their Jordan chains, then search for linearly independent generalized eigenvectors of rank 2 in $N_2 \setminus N_1$, build their Jordan chains and so on. Alternatively, one can start from the eigenvectors of $A - 2I$ and backward generate the chains. We apply the first strategy and search for generalized eigenvectors of rank 3. As $\dim N_3 = 4$ and $\dim N_2 = 3$, we must chose f so that $N_3 = \text{Span}\{f\} \oplus N_2$ and we obtain a first Jordan chain. We take:¹⁰

$$f = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad B(f) = \begin{pmatrix} 4 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad B^2(f) = \begin{pmatrix} 3 \\ 0 \\ -3 \\ 3 \end{pmatrix}, \quad B^3(f) = 0.$$

As $\dim N_2 = 3$ and $\dim N_1 = 2$, it must be that $N_2 = \text{Span}\{B(f)\} \oplus N_1$ and there is no generalized eigenvector of rank 2. But as $\dim N_1 = 2$, there must exist a generalized eigenvector of rank 1, h so that $N_1 = \text{Span}\{h\} \oplus \text{Span}\{B^2(f)\}$. We take:

$$h = \begin{pmatrix} -\frac{7}{2} \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad S = (h, B^2(f), B(f), f) = \begin{pmatrix} -\frac{7}{2} & 3 & 4 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & -3 & -1 & 0 \\ 1 & 3 & 1 & 0 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} 0 & 0 & 1/2 & 1/2 \\ -1/9 & 0 & -5/12 & 1/36 \\ 1/3 & 0 & 3/4 & 5/12 \\ 0 & 1 & -1/2 & -1/2 \end{pmatrix},$$

and indeed:

$$S^{-1}AS = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Note that computing the dimensions of the invariant subspaces $N_i = \ker B^i$ is sufficient in order to identify the number of distinct Jordan chains and their length, hence fix the Jordan normal form of A .

Real matrices can be brought in the usual Jordan normal form by complex similarity transformations. If one restricts to real transformations, one obtains the real Jordan normal form.

Corollary 8.1 (Real Jordan normal form). *Let A be an arbitrary real square matrix. Then there exists a real similarity transformation (invertible matrix S) such that $A = SJS^{-1}$ where the Jordan normal form consists in usual Jordan blocks*

¹⁰ k non zero column vectors are linearly independent if the RREF of the matrix obtained by juxtaposing them consists in k dominant columns (every non dominant column yields a non trivial linear relation between the previous columns) and indeed the RREF of $(e_1, e_2, e_3 + \frac{1}{2}e_2, e_4 + \frac{1}{2}e_2)$ is so.

for the real eigenvalues of A , while for the couples of complex conjugated eigenvalues $\lambda_{\pm} = a \pm ib$ they take the form:

$$J_{\lambda_{\pm}} = \begin{pmatrix} C & I & \dots & 0 \\ 0 & C & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & C \end{pmatrix}, \quad C = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

We note that matrices which can be diagonalized over $\mathbb{C}$ also need non trivial blocks C if the eigenvalues are not real. If A is complex diagonalizable then its real Jordan form has either diagonal elements for the real eigenvalues or block diagonal blocks of C s for the complex ones.

Proof. This follows by observing that not only the eigenvalues come into complex conjugated pairs, but also the Jordan chains. Taking linear combinations $e + \tilde{e}$, $i(e - \tilde{e})$ of the complex conjugated chains we get to the real Jordan form. $\square$

Exercise 8.5. Diagonalize the rotation matrix $R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and find the projectors on its eigenspaces. We have $(\cos \theta - \lambda)^2 + \sin^2 \theta = 0 \Rightarrow \lambda^2 - 2\lambda \cos \theta + 1 = 0 \Rightarrow \lambda_{\pm} = \cos \theta \pm \sqrt{\cos^2 \theta - 1} \Rightarrow \lambda = e^{\pm i\theta}$. The eigenvectors are determined by Gaussian elimination:

$$R - e^{\pm i\theta} I = \begin{pmatrix} \mp i \sin \theta & -\sin \theta \\ \sin \theta & \mp i \sin \theta \end{pmatrix} \rightarrow \begin{pmatrix} \sin \theta & \mp i \sin \theta \\ \mp i \sin \theta & -\sin \theta \end{pmatrix} \rightarrow \begin{pmatrix} \sin \theta & \mp i \sin \theta \\ \mp i \sin \theta & -\sin \theta \end{pmatrix} \rightarrow \begin{pmatrix} \sin \theta & \mp i \sin \theta \\ 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \mp i \\ 0 & 0 \end{pmatrix},$$

leading to eigenvectors in the respective kernels $e_2 \pm i e_1 = \begin{pmatrix} \pm i \\ 1 \end{pmatrix}$, that is:

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} i \\ 1 \end{pmatrix} = e^{i\theta} \begin{pmatrix} i \\ 1 \end{pmatrix}, \quad \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} -i \\ 1 \end{pmatrix} = e^{-i\theta} \begin{pmatrix} -i \\ 1 \end{pmatrix}, \quad \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}.$$

Exercise 8.6. Diagonalize and find the real Jordan form, and the associated changes of basis for the matrix $A = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$.

We have $\det(A - \lambda I) = (3 - \lambda)(-1 - \lambda) + 8 = 5 - 2\lambda + \lambda^2 = (1 - 2i - \lambda)(1 + 2i - \lambda) \Rightarrow \lambda_{\pm} = 1 \pm 2i$. The complex eigenvectors are basis vectors in the respective eigenspaces:

$$A - \lambda_+ I = \begin{pmatrix} 2 - 2i & -2 \\ 4 & -2 - 2i \end{pmatrix} \rightarrow \begin{pmatrix} 4 & -2 - 2i \\ 2 - 2i & -2 \end{pmatrix} \rightarrow \begin{pmatrix} 4 & -2 - 2i \\ 0 & -2 + \frac{1}{2}(1 - 1)(-2 - 2i) \end{pmatrix} = \begin{pmatrix} 4 & -2 - 2i \\ 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -\frac{1+i}{2} \\ 0 & 0 \end{pmatrix}$$

with eigenvector $f_+ = e_2 + \frac{1+i}{2} e_1$, hence:

$$\begin{pmatrix} 1 + 2i & 0 \\ 0 & 1 - 2i \end{pmatrix} = \frac{1}{i} \begin{pmatrix} 1 & -\frac{1-i}{2} \\ -1 & \frac{1+i}{2} \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \end{pmatrix} = \begin{pmatrix} -i & \frac{1+i}{2} \\ -\frac{1-i}{2} & 1 \end{pmatrix} \begin{pmatrix} -\frac{1-i}{2} & \frac{1-i}{2} \\ 1 + 2i & 1 - 2i \end{pmatrix} = \begin{pmatrix} \frac{i+3+(1+i)(1+2i)}{2} & \frac{i-3+(1+i)(1-2i)}{2} \\ \frac{-i-3+(-i+1)(1+2i)}{2} & \frac{-i+3+(-i+1)(1-2i)}{2} \end{pmatrix},$$

and the Jordan form is obtained in the basis $h_1 = f_+ + f_- = 2e_2 + e_1$, $h_2 = \frac{f_+ - f_-}{i} = e_1$

$$\frac{1}{-2} \begin{pmatrix} 0 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$$

As an application of the Jordan normal form we have the following theorem.

Theorem 8.6 (**The Cayley-Hamilton theorem**). Let us denote the characteristic polynomial of the matrix A by $q(\lambda) = \det(A - \lambda I)$. Then $q(A) = 0$.

Note that this is non trivial – the reasoning “ $q(\lambda) = \det(A - \lambda I)$ hence $q(A) = \det(A - A) = 0$ ” is nonsensical. For 2×2 matrices for instance we have:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad q(\lambda) = \lambda^2 - \text{Tr}(A)\lambda + \det(A), \quad q(A) = A^2 - \text{Tr}(A)A + \det(A)I = 0.$$

Proof. The characteristic polynomial is $q(\lambda) = \prod_{i=1}^q (\lambda_i - \lambda)^{m_i}$ with λ_i the distinct eigenvalues of A , hence $q(A) = \prod_{i=1}^q (\lambda_i I - A)^{m_i}$. As $q(S^{-1}AS) = S^{-1}q(A)S$, it is enough to prove the theorem for a matrix brought in its Jordan form. Denoting the

$m_i \times m_i$ Jordan block corresponding λ_i by $\lambda_i I_{m_i} + \bar{B}_i$ with $\bar{B}_i^{m_i} = 0$, $q(J) = \prod_{i=1}^q (\lambda_i I - J)^{m_i}$ writes as:

$$\begin{pmatrix} -\bar{B}_1 & 0 & \dots & 0 \\ 0 & (\lambda_1 - \lambda_2)I_{m_2} - \bar{B}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & (\lambda_1 - \lambda_q)I_{m_q} - \bar{B}_q \end{pmatrix}^{m_1} \dots \begin{pmatrix} (\lambda_q - \lambda_1)I_{m_1} - \bar{B}_1 & 0 & \dots & 0 \\ 0 & (\lambda_q - \lambda_2)I_{m_2} - \bar{B}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -\bar{B}_q \end{pmatrix}^{m_q} = 0 ,$$

as in any diagonal block we have a term $\bar{B}_i^{m_i} = 0$. □

Exercise 8.7. Check directly that for J a $k \times k$ Jordan block we have:

$$f(J) = \begin{pmatrix} f(\lambda) & f'(\lambda) & \frac{f^{(2)}(\lambda)}{2!} & \frac{f^{(3)}(\lambda)}{3!} & \dots & \frac{f^{(k-1)}(\lambda)}{(k-1)!} \\ 0 & f(\lambda) & f'(\lambda) & \frac{f^{(2)}(\lambda)}{2!} & \dots & \frac{f^{(k-2)}(\lambda)}{(k-2)!} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & f(\lambda) \end{pmatrix} .$$

If $P(x)$ is the characteristic polynomial of a matrix, associated to a root λ of multiplicity k of P we have a Jordan block of size at most k , and when evaluating $P(J)$ we have $P(\lambda) = P'(\lambda) = \dots = P^{(k-1)}(\lambda) = 0$ and consequently $P(J) = 0$, proving again the Cayley-Hamilton theorem.

8.3 Bilinear, quadratic and sesquilinear forms

Definition 8.3 (**Bilinear and sesquilinear forms**). A bilinear form on a finite dimensional vector space is a function $q: V \times V \rightarrow \mathbb{F}$ which is linear in both arguments:

$$q(x, \alpha_1 y_1 + \alpha_2 y_2) = \alpha_1 q(x, y_1) + \alpha_2 q(x, y_2) , \quad q(\alpha_1 x_1 + \alpha_2 x_2, y) = \alpha_1 q(x_1, y) + \alpha_2 q(x_2, y) .$$

A bilinear form is called symmetric if $q(x, y) = q(y, x)$ for all x, y . A quadratic form is a bilinear form evaluated on the same argument, $q(x) = q(x, x)$.

A sesquilinear form on a finite dimensional complex vector space is a function $h: V \times V \rightarrow \mathbb{C}$ which is linear in the second argument and conjugated linear in the first:

$$h(x, \alpha_1 y_1 + \alpha_2 y_2) = \alpha_1 h(x, y_1) + \alpha_2 h(x, y_2) , \quad h(\alpha_1 x_1 + \alpha_2 x_2, y) = \bar{\alpha}_1 h(x_1, y) + \bar{\alpha}_2 h(x_2, y) .$$

A sesquilinear form is called Hermitian if $h(x, y) = \overline{h(y, x)}$ for all x, y .

We focus of the real case $q: V \times V \rightarrow \mathbb{R}$. Choosing a basis $\{e_i\}$ in V a bilinear form writes $q(x, y) = \sum_{i,j=1}^{\dim V} q(e_i, e_j) x^i y^j$, and we denote $q_{ij} = q(e_i, e_j)$ the components of q in the basis e_i . While $q(x, y)$ is basis independent, under a change of basis $\tilde{e}_i = \sum_j e_j S_{ji}$, the components of q change like

$$\tilde{q}_{kl} = q(\tilde{e}_k, \tilde{e}_l) = \sum_{ij} q_{ij} S_{ik} S_{jl} \Rightarrow \tilde{Q} = S^t Q S ,$$

where $(S^t)_{ki} = S_{ik}$ is the transposed of the matrix S .¹¹ Using column vector notation, a quadratic form writes $q(x) = x^t Q x$ and under a change of variables $x = Sy$ it becomes $\tilde{q}(y) = q(Sy) = y^t S^t Q S y$. This is to be contrasted with the behaviour of an operator under a change of basis $\tilde{A} = S^{-1} A S$.

Note that, while bilinear forms need not be symmetric, a quadratic form $q(x, x) = \sum_{i,j=1}^N q_{ij} x^i x^j$ can always be symmetrized, that is $q(x, x) = \sum_{i,j=1}^N \frac{q_{ij} + q_{ji}}{2} x^i x^j$. From now on we assume that the quadratic forms are symmetrized.

Theorem 8.7 (Lagrange reduction). *Any quadratic form $x^t Q x$ for $x \in \mathbb{R}^N$ can be reduced by a non singular change of variables $x = Sy$ (that is S is an invertible $N \times N$ matrix with real entries) to the standard form:*

$$(y^1)^2 + \dots + (y^p)^2 - (y^{p+1})^2 \dots - (y^r)^2,$$

and:

- if $p = N$ the form is called positive definite, or positive non degenerate, and in this case $\forall x \in \mathbb{R}^N \setminus \{0\}, x^t Q x > 0$.
- if no negative terms are present but $p < N$, in which case $x^t Q x \geq 0, \forall x \in \mathbb{R}^N$, the form is called positive semi definite.
- we have the corresponding negative definite and semi definite cases.
- the form is called indefinite if both positive and negative terms are present.

A quadratic form on a complex vector space can always be brought in the canonical form $(z^1)^2 + \dots + (z^r)^2$.

Proof. The proof proceeds by “completing the squares”. Assume there exist some $q_{ll} \neq 0$, then:

$$\sum_{i,j=1}^N q_{ij} x^i x^j = q_{ll} \left(x_{ll} + \sum_{j \neq l} \frac{q_{lj}}{q_{ll}} x^j \right)^2 + \sum_{j,k \neq l} \left(q_{jk} - \frac{q_{lj} q_{lk}}{q_{ll}} \right) x^j x^k,$$

and the last sum does not contain x_l . We define $y_l = \sqrt{|q_{ll}|} \left(x_{ll} + \sum_{j \neq l} \frac{q_{lj}}{q_{ll}} x^j \right)$ and iterate.

If $q_{ll} = 0$ for all l , but there exists $q_{ls} \neq 0$, we have:

$$\sum_{i,j=1}^N q_{ij} x^i x^j = \frac{q_{ls}}{2} \left[\left(x_l + x_s + \sum_{j \neq l,s} \frac{q_{lj} + q_{sj}}{q_{ls}} x^j \right)^2 - \left(x_l - x_s + \sum_{j \neq l,s} \frac{-q_{lj} + q_{sj}}{q_{ls}} x^j \right)^2 \right] + \sum_{i,j \neq l,s} \left(q_{ij} - \frac{q_{li} q_{sj} + q_{si} q_{lj}}{q_{ls}} \right) x^i x^j,$$

and the last sum does not contain x_l or x_s . We define $y_l = \sqrt{\frac{|q_{ls}|}{2}} \left(x_l + x_s + \sum_{j \neq l,s} \frac{q_{lj} + q_{sj}}{q_{ls}} x^j \right)$ and $y_s = \sqrt{\frac{|q_{ls}|}{2}} \left(-x_s + x_l + \sum_{j \neq l,s} \frac{-q_{lj} + q_{sj}}{q_{ls}} x^j \right)$ and iterate. $\square$

Exercise 8.8. Reduce the quadratic form $x_1^2 - 2x_1x_2 + x_2^2 + 2x_2x_3 + 2x_2x_4 - 2x_3x_4$. We have

$$(x_1 - x_2)^2 + 2x_2x_3 + 2x_2x_4 - 2x_3x_4 = (x_1 - x_2)^2 + \frac{1}{2}[(x_2 + x_3)^2 - (x_2 - x_3 - 2x_4)^2] + 2x_4^2.$$

Tensors. Let us reinstate for the moment the upper or lower positions of the indices. One can consider more general multi linear forms $T : V^p \times (V^*)^q \rightarrow \mathbb{F}$:

$$T(v_1 \dots v_p, \varphi^1, \dots, \varphi^q) = \sum_{i,j} v_1^{i_1} \dots v_p^{i_p} \varphi_{j_1}^1 \dots \varphi_{j_q}^q T_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q}, \quad T_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q} = T(e_{i_1} \dots e_{i_p}, \theta^{j_1} \dots \theta^{j_q}).$$

¹¹Sesquilinear forms on complex vector spaces change under a change of basis as $\tilde{Q} = S^\dagger Q S$ with $(S^\dagger)_{ki} = \overline{S_{ik}}$.

Under a change of basis $\tilde{e}_i = \sum_k e_k S^k_i$ and $\tilde{\theta}^j = \sum_l (S^{-1})^j_l \theta^l$ we have:

$$\tilde{T}_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q} = T(\tilde{e}_{i_1} \dots \tilde{e}_{i_p}, \tilde{\theta}^{j_1} \dots \tilde{\theta}^{j_q}) = \sum_{k,l} S^{k_1}_{i_1} \dots S^{k_p}_{i_p} (S^{-1})^{j_1}_{l_1} \dots (S^{-1})^{j_q}_{l_q} \tilde{T}_{k_1 \dots k_p}^{l_1 \dots l_q}.$$

This is a (p, q) tensor and the p lower indices are called covariant and the q upper ones are contravariant. As a multilinear form, $T \in (V^*)^{\otimes p} \otimes V^{\otimes q}$ and we have:

$$T = \sum_{i,j} T_{i_1 i_2 \dots i_p}^{j_1 j_2 \dots j_q} \theta^{i_1} \otimes \dots \otimes \theta^{i_p} \otimes e_{j_1} \otimes \dots \otimes e_{j_q}.$$

8.4 Scalar product

We have already introduced *norms* on vector spaces. A *norm* on the vector space $(V, \mathbb{F})$ is a function $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ such that:

- $\|v\| \geq 0$ and $\|v\| = 0$ if and only if $v = 0$.
- $\|\alpha x\| = |\alpha| \|x\|$, $\forall \alpha \in \mathbb{F}$ and $\forall x \in V$.
- $\|x + y\| \leq \|x\| + \|y\|$, $\forall x, y \in V$.

A Cauchy complete normed space is called a Banach space. For instance $\mathbb{R}^D$ and $\mathbb{C}^D$ are normed spaces with norm one of the p -norms $\|x\|_p = (\sum_{\mu=1}^D |x^\mu|^p)^{1/p}$. Remark that $\mathbb{Q}^D$ with $\|\cdot\|_p$ is a finite dimensional normed vector space over $\mathbb{Q}$ which is not Banach.

Definition 8.4 (**Scalar product**). An inner product (or scalar product) on a vector space V is a function: $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$

- in the real case $\mathbb{F} = \mathbb{R}$, a positive non degenerate symmetric bilinear form, that is
 - $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.
 - $\langle \alpha x + \beta y, u \rangle = \alpha \langle x, u \rangle + \beta \langle y, u \rangle$ and $\langle u, \alpha x + \beta y \rangle = \alpha \langle u, x \rangle + \beta \langle u, y \rangle$
 - $\langle x, y \rangle = \langle y, x \rangle$
- in the complex case $\mathbb{F} = \mathbb{C}$, a positive non degenerate Hermitian sesquilinear form, that is (denoting by bar the complex conjugation):
 - $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.
 - $\langle \alpha x + \beta y, u \rangle = \bar{\alpha} \langle x, u \rangle + \bar{\beta} \langle y, u \rangle$ and $\langle u, \alpha x + \beta y \rangle = \alpha \langle u, x \rangle + \beta \langle u, y \rangle$
 - $\overline{\langle x, y \rangle} = \langle y, x \rangle$

Any inner product induces a norm $\|x\| = \sqrt{\langle x, x \rangle}$. A Cauchy complete inner product space is called a Hilbert space.

For example, $\mathbb{R}^D$ and $\mathbb{C}^D$ are inner product spaces with inner product:

$$\langle x, y \rangle = \sum_{\mu=1}^D x^\mu y^\mu, \quad \langle z, w \rangle = \sum_{\mu=1}^D \bar{z}^\mu w^\mu,$$

and the corresponding norm is the norm $\|\cdot\|_2$ we introduced before. We routinely suppress the subscript 2 on the norm 2.

Theorem 8.8 (**The Cauchy-Schwarz inequality**). For all x, y in an inner product space V we have: $|\langle x, y \rangle| \leq \|x\| \|y\|$, with equality reached if and only if x and y are proportional. This holds also in infinite dimensions and for spaces that are not Cauchy complete.

Proof. We consider the complex case and we write:

$$0 \leq \|x - \lambda y\|^2 = \langle x - \lambda y, x - \lambda y \rangle = \|x\|^2 + |\lambda|^2 \|y\|^2 - \lambda \langle x, y \rangle - \bar{\lambda} \langle y, x \rangle,$$

and evaluate the last expression for $\lambda = \frac{\langle y, x \rangle}{\|y\|^2}$. □

Exercise 8.9. Note that the proof of the Cauchy-Schwarz inequality only requires linearity and positivity of the scalar product. Prove using the Cauchy-Schwarz inequality that the norm induced by the scalar product respects the triangle inequality. We have $\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \|y\|^2 + \langle x, y \rangle + \langle y, x \rangle \leq \|x\|^2 + \|y\|^2 + 2\|x\| \|y\|$.

Definition 8.5 (Orthonormal basis). Two vectors are called orthogonal if their inner product is zero. A basis $\{e_i\}_{i=1, \dim V}$ in a finite dimensional vector space V is called orthonormal with respect to the inner product $\langle \cdot, \cdot \rangle$ if $\langle e_i, e_j \rangle = \delta_{ij}$. Any vector decomposes into an orthonormal basis as:

$$v = \sum_{i=1}^{\dim V} \alpha_i e_i, \quad \langle e_i, v \rangle = \alpha_i,$$

and the forms in the dual basis $\theta^i(e_j) = \delta_{ij}$ write as $\theta^i(\cdot) = \langle e_i, \cdot \rangle$. In particular:

$$\|v\|^2 = \langle v, v \rangle = \langle v, \sum_{i=1}^{\dim V} \langle e_i, v \rangle e_i \rangle = \sum_{i=1}^{\dim V} \langle e_i, v \rangle \langle v, e_i \rangle = \sum_{i=1}^{\dim V} |\langle e_i, v \rangle|^2$$

The matrix elements of an operator in an orthonormal basis are $a_{ij} = \theta^i(Ae_j) = \langle e_i, Ae_j \rangle$.

Exercise 8.10. Prove that the standard basis in $\mathbb{R}^D$ and $\mathbb{C}^D$ is orthonormal.

Construction 8.1 (Gram-Schmidt process). Starting from any finite subset of nonzero vectors $x_1, \dots, x_n \in V$, one can construct an orthonormal system $e_1, \dots, e_p$ with the same span by the Gram-Schmidt process:

$$e_1 := \frac{1}{\|x_1\|} x_1, \text{ set } k = n = 1 \text{ and } \forall n \begin{cases} \text{if } x_n \notin \text{Span}\{e_1, \dots, e_k\} \text{ then } e_{k+1} := \frac{x_n - \sum_{j=1}^k \langle e_j, x_n \rangle e_j}{\|x_n - \sum_{j=1}^k \langle e_j, x_n \rangle e_j\|}, & k \rightarrow k+1, \quad n \rightarrow n+1 \\ \text{else } n \rightarrow n+1 \end{cases}.$$

It is easy to check that $\langle e_k, e_l \rangle = \delta_{kl}$. Applying the Gram-Schmidt process to any set spanning V one obtains an orthonormal basis in V .

Theorem 8.9 (Orthogonal complement, orthogonal decomposition). Let V be a finite dimensional Hilbert space and W a subspace. We define the orthogonal complement $W^\perp$ of W as the set of vectors orthogonal to W , that is $W^\perp = \{v \in V \mid \langle v, w \rangle = 0, \forall w \in W\}$. Then $W^\perp$ is a subspace of V and any $v \in V$ can be decomposed uniquely as a sum $v = y + z$ with $y \in W$ and $z \in W^\perp$, that is $V = W \oplus W^\perp$.

Proof. We pick a basis in W and use the Gram-Schmidt process to obtain an orthonormal basis in W . We complete this to a basis in V , and apply again the Gram-Schmidt process to obtain an orthonormal basis such that $e_1, \dots, e_{\dim W}$ are a basis in W and $e_{\dim W+1}, \dots, e_{\dim V}$ are orthogonal on W . Any $v \in V$ decomposes uniquely on this basis:

$$v = \sum_{i=1}^{\dim W} \alpha_i e_i + \sum_{i=\dim W+1}^{\dim V} \alpha_i e_i,$$

and all the statements follow. □

Definition 8.6 (**Adjoint of an operator**). The adjoint of an operator $A : V \rightarrow V$ is the operator $A^\dagger : V \rightarrow V$ such that $\langle x, Ay \rangle = \langle A^\dagger x, y \rangle$ for all $x, y \in V$. In an orthonormal basis, the adjoint has matrix elements (where complex conjugation is superfluous for real vector spaces):

$$(A^\dagger)_{ij} = \langle e_i, A^\dagger e_j \rangle = \overline{\langle A^\dagger e_j, e_i \rangle} = \overline{\langle e_j, A e_i \rangle} = \overline{a_{ji}},$$

that is the matrix $A^\dagger = \overline{A^t}$ is the complex conjugated of the transposed of A . An operator is called self adjoint if $A = A^\dagger$. A self adjoint operator is called symmetric in the real case.

Exercise 8.11. Prove that $(AB)^\dagger = B^\dagger A^\dagger$. Prove that if λ is an eigenvalue of A then $\bar{\lambda}$ is an eigenvalue of $A^\dagger$. We note that $\det(A^\dagger - \bar{\lambda}I) = \overline{\det(A^t - \lambda I)} = \overline{\det(A - \lambda I)}$.

Definition 8.7 (**Orthogonal projectors**). A projection operator $P : V \rightarrow V$, $P = P^2$ is called orthogonal if $P^\dagger = P$ and in this case the decomposition $V = \ker P \oplus \text{Im } P$ is an orthogonal decomposition as $\text{Im } P = (\ker P)^\perp$ (for all $x \in \ker P$ and $y \in \text{Im } P$, $y = Pz$ and $\langle x, y \rangle = \langle x, Pz \rangle = \langle Px, z \rangle = 0$).

For any subspace $W \subset V$, the orthogonal decomposition $V = W \oplus W^\perp$ is unique and the (unique) projector P corresponding to this decomposition is orthogonal. Indeed, for any $x, y \in V$, as $y = Py + (I - P)y$ with $Py \in W$ and $(I - P)y \in W^\perp$, we have $\langle Px, y \rangle = \langle Px, Py + (I - P)y \rangle = \langle Px, Py \rangle = \langle Px + (I - P)x, Py \rangle = \langle x, Py \rangle$. It follows that the scalar product selects a favored projector on a subspace W , out of the infinitely many corresponding to all the decompositions $V = W \oplus W'$. Conversely for any orthogonal projector there exists an associated orthogonal decomposition of V into its kernel and its image.¹² In short:

- direct sum decomposition exist in the absence of a scalar product and:
 - for every direct sum decomposition $V = W \oplus W'$ there exist an associated projector $P = P^2$ so that $x = Px + (I - P)x$ with $Px \in W$ and $(I - P)x \in W'$.
 - for any projector $P^2 = P$ we have $x = Px + (I - P)x$ and an associated direct sum decomposition $V = W \oplus W'$ where we denoted $W = \text{Im } P$ and $W' = \ker P = \text{Im}(I - P)$.
- orthogonal decompositions require a scalar product and:
 - for every $W \subset V$ there exists a unique orthogonal decomposition $V = W \oplus W^\perp$ and an associated orthogonal projector $P^\dagger = P = P^2$ so that $x = Px + (1 - P)x$ with $Px \in W$ and $(1 - P)x \in W^\perp$.
 - for any orthogonal projector $P^\dagger = P = P^2$ we have $x = Px + (1 - P)x$ and an associated orthogonal decomposition $V = W \oplus W^\perp$ where we denoted $W = \text{Im } P$ and $W^\perp = \ker P = \text{Im}(I - P) = (\text{Im } P)^\perp$.

Definition 8.8 (**Orthogonal and unitary matrices**). The matrix S of a change of basis $\tilde{e}_i = \sum_j e_j S_{ji}$ between two orthonormal bases in the vector space V is called:

- orthogonal if V is a real vector space.
- unitary if V is a complex vector space.

¹²Note that any projector gives a direct sum decomposition, but if the projector is not orthogonal the kernel and image, while linearly independent, are not necessarily orthogonal.

Observe that the matrix elements of S can be computed as $\langle e_i, \tilde{e}_j \rangle = S_{ij}$ and, taking into account that $v = \sum_i \langle e_i, v \rangle e_i$, we have (where the complex conjugation is superfluous for real vector spaces):

$$\begin{aligned} (SS^\dagger)_{ik} &= \sum_j S_{ij} \overline{S_{kj}} = \sum_j \langle e_i, \tilde{e}_j \rangle \overline{\langle e_k, \tilde{e}_j \rangle} = \sum_j \langle e_i, \tilde{e}_j \rangle \langle \tilde{e}_j, e_k \rangle = \langle e_i, \sum_j \langle \tilde{e}_j, e_k \rangle \tilde{e}_j \rangle = \langle e_i, e_k \rangle = \delta_{ik}, \\ (S^\dagger S)_{ki} &= \sum_j \overline{S_{jk}} S_{ji} = \sum_j \overline{\langle e_j, \tilde{e}_k \rangle} \langle e_j, \tilde{e}_i \rangle = \sum_j \langle \tilde{e}_k, e_j \rangle \langle e_j, \tilde{e}_i \rangle = \langle \tilde{e}_k, \sum_j \langle e_j, \tilde{e}_i \rangle e_j \rangle = \langle \tilde{e}_k, \tilde{e}_i \rangle = \delta_{ki}, \end{aligned}$$

that is $S^\dagger = S^{-1}$.

We denote orthogonal matrices by O and unitary matrices by U . We have $\det(O^t O) = (\det O)^2 = 1$ and $\det(U^\dagger U) = \det U^\dagger \det U = |\det U|^2 = 1$ hence $\det O = \pm 1$ and $\det U = e^{i\gamma}$. It is easy to show that the orthogonal (respectively unitary) matrices form a group, the orthogonal (resp. unitary) group.

8.4.1 The volume of a parallelepiped

The volume of a parallelepiped defined by N vectors is defined with respect to a reference basis.

Definition 8.9 (Volume). Let X be vector space with dimension N and fix a basis e_i . The signed volume of the parallelepiped defined by the N column vectors $h_i = \sum_j e_j H_{ji}$ is the determinant:

$$\begin{vmatrix} h_1 & \dots & h_N \end{vmatrix} = \det(H),$$

and the volume is the absolute value of the signed volume $\text{Vol}(h_1, \dots, h_N) = |\det(H)|$. Note that the volume of the parallelepiped defined by the basis vectors themselves is $\text{Vol}(e_1, \dots, e_N) = \det(I) = 1$.

If the vectors $h_1, \dots, h_N$ are linearly dependent then the volume is zero and taking into account that the dimension of the vector space is N we conclude that the volume is non zero if and only if $h_1, \dots, h_N$ form a basis.

We stress that the volume is defined in any finite dimensional vector space, even in the absence of an inner product, and it is dependent of the reference basis $\{e_i\}$.

In an inner product space, it is convenient to chose as reference basis an orthonormal basis, say $\{e_i\}$. The volume of the parallelepiped defined by the basis vectors of any other orthonormal basis $\tilde{e}_i = \sum_j e_j U_{ji}$ is $\text{Vol}(\tilde{e}_1, \dots, \tilde{e}_N) = |\det(U)| = 1$, and the volume of $h_1, \dots, h_N$ is the same in all the orthonormal bases. Indeed, under a change of orthonormal basis $\tilde{e}_i = \sum_j e_j U_{ji}$, the column vectors transform as $h_i = \sum_j e_j H_{ji} = \sum_l \tilde{e}_l \sum_i U_{li}^{-1} H_{ji}$ and $|\det(\tilde{H})| = |\det U^{-1}| |\det H| = |\det(H)|$.

Proposition 8.2 (Hadamard's inequality). The volume of a parallelepiped is smaller than the product of the edge lengths. Consider N column vectors h_i in some orthonormal basis. Then:

$$|\det(h_1 \ h_2 \ \dots \ h_N)| \leq \prod_{i=1}^N \|h_i\|,$$

with equality achieved if and only if the vectors h_i are all orthogonal.

Proof. Let us denote $\tilde{e}_1, \dots, \tilde{e}_n$ the orthonormal basis obtained by applying the Gram-Schmidt process to $h_1, \dots, h_N$. Then $h_j = \sum_{i \leq j} \tilde{e}_i r_{ij}$, or in components $h_{\alpha j} = \sum_{i \leq j} \tilde{e}_{\alpha i} r_{ij}$, which writes as:

$$\begin{pmatrix} h_1 & h_2 & \dots & h_N \end{pmatrix} = \begin{pmatrix} \tilde{e}_1 & \tilde{e}_2 & \dots & \tilde{e}_N \end{pmatrix} R, \quad R = \begin{pmatrix} r_{11} & r_{12} & \dots & r_{1N} \\ 0 & r_{22} & \dots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_{NN} \end{pmatrix},$$

where we remark that R is upper triangular. Now note that on the one hand $\|h_j\| = \|\sum_{i \leq j} \tilde{e}_i r_{ij}\| \geq |r_{jj}|$, while on the other $\det(H) = \det(\begin{pmatrix} \tilde{e}_1 & \tilde{e}_2 & \dots & \tilde{e}_N \end{pmatrix}) \det(R)$ and $|\det(R)| = \prod_{i=1}^N |r_{ii}|$, while $|\det(\begin{pmatrix} \tilde{e}_1 & \tilde{e}_2 & \dots & \tilde{e}_N \end{pmatrix})|$ as $\tilde{e}_i$ are an orthonormal basis. Finally, $\|h_j\| = |r_{jj}|$ only if $h_j = r_{jj}\tilde{e}_j$. $\square$

Observe that writing by the Gram–Schmidt process $h_1 = \tilde{e}_1$, $h_2 = r_{12}\tilde{e}_1 + \tilde{e}_2$, $h_3 = r_{13}\tilde{e}_1 + r_{23}\tilde{e}_2 + \tilde{e}_3$ and so on, where $h_2^\perp$ is orthogonal to h_1 , $h_3^\perp$ is orthogonal to $\text{Span}\{h_1, h_2\}$ and so on we have:

$$\text{Vol}(h_1, \dots, h_N) = |\det(h_1 \ h_2 \ \dots \ h_N)| = |\det(h_1 \ h_2^\perp \ \dots \ h_N^\perp)|.$$

8.5 Spectral decomposition for self adjoint, unitary and normal operators

The spectral representation is particularly simple for self adjoint operators.

Theorem 8.10 (**Spectral theorem for self adjoint operators**). *Let V be a finite dimensional vector space. The eigenvalues of a self adjoint operator $A : V \rightarrow V$ are real and the eigenvectors corresponding to distinct eigenvalues are orthogonal. Moreover, A is always diagonalizable that is V admits an orthonormal basis consisting in eigenvectors of A .*

Equivalently, a real symmetric matrix can be diagonalized by an orthogonal transformation $A = O^t D O$, and a complex Hermitian matrix can be diagonalized by a unitary transformation $A = U^\dagger D U$ and in both cases D is a diagonal matrix with real entries.

Proof. Let (λ, v) be an eigenpair of A ; then $\bar{\lambda} \langle v, v \rangle = \langle Av, v \rangle = \langle v, Av \rangle = \lambda \langle v, v \rangle$ and λ is real. Now, let $(\lambda_1, v_1), (\lambda_2, v_2)$ be two eigenpairs of A ; then $0 = \langle Av_1, v_2 \rangle - \langle v_1, Av_2 \rangle = (\bar{\lambda}_1 - \lambda_2) \langle v_1, v_2 \rangle$, hence if $\lambda_1 \neq \lambda_2$ then $\langle v_1, v_2 \rangle = 0$.

We proceed by induction. We consider the characteristic polynomial $\det(A - \lambda I) = 0$ and denote λ one of its roots. The linear system of equations $(A - \lambda)v = 0$ admits a nontrivial solution $v \neq 0$, that is there exists an eigenpair (λ, v) . We denote $v^\perp = \{w \in V \mid \langle w, v \rangle = 0\}$ the orthogonal complement of v . Then $\langle v, Av^\perp \rangle = \langle Av, v^\perp \rangle = \bar{\lambda} \langle v, v^\perp \rangle = 0$, hence $v^\perp$ is an invariant subspace of A and by the induction hypothesis there exists an orthonormal basis in $v^\perp$ consisting in eigenvectors of A . Supplementing this basis by $v/\|v\|$ we conclude. $\square$

In its eigenbasis a self adjoint operator writes as:

$$A(\cdot) = \sum_{\mu} \lambda_{\mu} \langle e_{\mu}, \cdot \rangle e_{\mu}, \quad f(A) = \sum_{\mu} f(\lambda_{\mu}) \langle e_{\mu}, \cdot \rangle e_{\mu},$$

where the λ_{μ} are the (not necessarily distinct) eigenvalues of A and the eigenvectors e_{μ} form an orthonormal basis.

Note that $P_{\mu} = \langle e_{\mu}, \cdot \rangle e_{\mu}$ is an orthogonal projector:

$$\langle x, P_{\mu} y \rangle = \langle x, \langle e_{\mu}, y \rangle e_{\mu} \rangle = \langle e_{\mu}, y \rangle \langle x, e_{\mu} \rangle = \overline{\langle e_{\mu}, x \rangle} \langle e_{\mu}, y \rangle = \langle \langle e_{\mu}, x \rangle e_{\mu}, y \rangle = \langle P_{\mu} x, y \rangle,$$

and $P_{\mu} P_{\nu} \sim \langle e_{\mu}, e_{\nu} \rangle = \delta_{\mu\nu}$, that is the projectors on the eigenspaces of a Hermitian matrix are mutually orthogonal orthogonal projectors:

$$A = \sum_i \lambda_i P_i, \quad P_i P_j = \delta_{ij} P_i, \quad P_i^\dagger = P_i, \quad \lambda_i \in \mathbb{R}.$$

Theorem 8.11 (**Spectral theorem for unitary operators**). *Let V be a finite dimensional vector space. The eigenvalues of a unitary operator $U : V \rightarrow V$ with $U^{-1} = U^\dagger$ are complex numbers with absolute value 1, the eigenvectors corresponding to distinct eigenvalues are orthogonal and U is diagonalizable in an orthonormal basis.*

Proof. Let (λ, v) and eigenpair of U . We have $\langle v, v \rangle = \langle v, U^\dagger U v \rangle = \langle U v, U v \rangle = |\lambda|^2 \langle v, v \rangle$. Now, let $(\lambda_1, v_1), (\lambda_2, v_2)$ be two eigenpairs of U with $\lambda_1 \neq \lambda_2$. Then $\langle v_1, v_2 \rangle = \langle U v_1, U v_2 \rangle = \overline{\lambda_1} \lambda_2 \langle v_1, v_2 \rangle$. As $\overline{\lambda_1} = \frac{1}{\lambda_1}$, it follows that $(1 - \frac{\lambda_2}{\lambda_1}) \langle v_1, v_2 \rangle = 0$ and we conclude.

Finally, take (λ, v) an eigenpair. The orthogonal complement $v^\perp$ is U invariant as $\langle Uw, v \rangle = \langle w, U^\dagger v \rangle = \bar{\lambda} \langle w, v \rangle$ and we proceed by induction. $\square$

Both the Hermitian and the unitary matrices are diagonalizable in orthonormal eigenbases.

Corollary 8.2 (**Unitary and Hermitian matrices**). *A matrix U is unitary if and only if $U = e^{iA}$ with A Hermitian.*

If B is real matrix, $A = iB$ is a Hermitian matrix if and only if B is antisymmetric $B^t = -B$. An matrix O is orthogonal if and only if $O = e^B$ for an antisymmetric matrix $B^t = -B$.

Proof. The spectral decomposition of a unitary matrix writes as $U = \sum_\mu e^{i\varphi_\mu} \langle e_\mu, \cdot \rangle e_\mu$ with $\varphi_\mu \in \mathbb{R}$ hence $A = \sum_\mu \varphi_\mu \langle e_\mu, \cdot \rangle e_\mu$ is Hermitian. $\square$

Theorem 8.12 (**Spectral theorem for normal operators**). *An operator $A : V \rightarrow V$ is called normal if it commutes with its adjoint, $[A, A^\dagger] = AA^\dagger - A^\dagger A = 0$. A can be diagonalized in a orthonormal basis if and only if A is normal.*

Proof. If A can be diagonalized in an orthonormal basis then $A = \sum_i \lambda_i P_i$ and P_i are orthogonal projectors $P_i^\dagger = P_i$, $P_i P_j = \delta_{ij} P_j$. Then $A^\dagger = \sum_i \bar{\lambda}_i P_i$ and $[A, A^\dagger] = 0$.

Assume A is normal, that is $[A, A^\dagger] = 0$. If (z, v) is an eigenpair of A then $(\bar{z}, v)$ is an eigenpair of $A^\dagger$. Indeed:

$$\|(A - zI)v\|^2 = \langle (A - zI)v, (A - zI)v \rangle = \langle v, (A^\dagger - \bar{z}I)(A - zI)v \rangle = \langle v, (A - zI)(A^\dagger - \bar{z}I)v \rangle = \langle (A^\dagger - \bar{z}I)v, (A^\dagger - \bar{z}I)v \rangle = \|(A^\dagger - \bar{z}I)v\|^2.$$

We proceed by induction and assume we have proven the statement for all dimension smaller than $\dim V$. The characteristic polynomial $\det(A - \lambda I)$ has a root z and there exists $v \neq 0$ so that $(A - z)v = 0$. Then $v^\perp$ is:

- A invariant because $\langle v, Aw \rangle = \langle A^\dagger v, w \rangle = \langle \bar{z}v, w \rangle = z \langle v, w \rangle$.
- $A^\dagger$ invariant because $\langle v, A^\dagger w \rangle = \langle Av, w \rangle = \langle zv, w \rangle = \bar{z} \langle v, w \rangle$.

It follows that $A_{v^\perp}$ and $A_{v^\perp}^\dagger$ exist, commute and are adjoint of each other. By the induction hypothesis $A_{v^\perp}$ is diagonalized in an orthonormal basis in $v^\perp$ and supplementing this with $v/\|v\|$ we obtain an orthonormal basis in V . $\square$

8.5.1 Singular values

One can define the adjoint of an operator between different Hilbert spaces.

Consider $M : X \rightarrow Y$, with both X and Y Hilbert spaces. The adjoint of M is the operator $M^\dagger : Y \rightarrow X$ such that $\langle y, Mx \rangle_Y = \langle M^\dagger y, x \rangle_X$ where the subscript denotes in which space the scalar product is taken. Note that the dimensions of X and Y do not need to be equal. The operator $M^\dagger M : X \rightarrow X$ is self adjoint:

$$\langle x, M^\dagger Mv \rangle_X = \langle Mx, Mv \rangle_Y = \langle M^\dagger Mx, v \rangle_X.$$

Definition 8.10 (**Positive operators**). *A self adjoint operator $A : X \rightarrow X$ is called positive if for all $x \in X$, $\langle x, Ax \rangle \geq 0$. An operator is positive if and only if it has positive eigenvalues.¹³ For any $M : X \rightarrow Y$ the operator $M^\dagger M$ is self adjoint and positive. Denoting s_μ^2 its positive eigenvalues and f_μ the corresponding eigenvectors, we have:*

$$M^\dagger M(\cdot) = \sum_i s_\mu^2 \langle f_\mu, \cdot \rangle f_\mu.$$

We call $s_\mu \geq 0$ the singular values of M .

¹³Denoting e_μ the eigenbasis of A , $\langle e_\mu, Ae_\mu \rangle = \lambda_\mu$ and $\langle v, Av \rangle = \sum_\mu \lambda_\mu |\langle e_\mu, v \rangle|^2$.

For any operator $M : X \rightarrow Y$ there exist two unitary transformations U acting on X and V acting on Y such that $M = VDU$, with $D_{\mu\mu} = s_\mu$ and all other elements of D being 0. Note that D is *not* a square matrix.

Theorem 8.13 (**Operator norm and singular values**). *The operator norm of an operator $A : X \rightarrow Y$ is:*

$$\|A\|_{\text{op.}} = \sup_{v \in V} \frac{\|Av\|}{\|v\|} = \sup_{v \in V, \|v\|=1} \|Av\|.$$

Recalling that in a Hilbert space $\|x\| = \sqrt{\langle x, x \rangle}$, the operator norm of A is the largest of its singular values. If $A : V \rightarrow V$ is self adjoint, its norm is the maximum of the absolute value of its eigenvalues.

Proof. We have $\|Av\|^2 = \langle Av, Av \rangle = \langle v, A^\dagger Av \rangle$ and denoting f_μ the eigenbasis of $A^\dagger A$ and s_μ its singular values, we have:

$$\langle v, A^\dagger Av \rangle = \sum_\mu s_\mu^2 \langle v, f_\mu \rangle \langle f_\mu, v \rangle \leq \max_\nu \{s_\nu^2\} \sum_\mu \langle v, f_\mu \rangle \langle f_\mu, v \rangle = \max_\nu \{s_\nu^2\} \|v\|^2,$$

and the bound is saturated for v the eigenvector corresponding to $s_\mu^2 = \max_\nu \{s_\nu^2\}$. The singular values of a self adjoint operator are the absolute values of its eigenvalues. $\square$

Exercise 8.12. *Let $M : X \rightarrow Y$ be a linear operator. Prove that M and $M^\dagger$ have the same singular values. Prove that $M = UDV$ with U and V unitary matrices and D a rectangular matrix whose only non zero entries lie on the main diagonal $d_{\mu\nu} = 0$ if $\mu \neq \nu$ and they equal the singular values $s_\mu \geq 0$ of M .*

8.6 Matrix norms

The space $B(X, Y)$ of linear operators $A : X \rightarrow Y$ with X and Y finite dimensional vector spaces is itself a finite dimensional vector space with the addition. A basis is given by the matrices $(E^{ij})_{kl} = \delta_k^i \delta_l^j$. Beyond the operator norm which we have already seen, $\|A\|_{\text{op.}} = \sup_{v \in V} \frac{\|Av\|}{\|v\|}$, one can endow $B(X, Y)$ with other norms. We have that $B(X, Y)$ is a Hilbert space with scalar product $\langle A, B \rangle = \text{Tr}(A^\dagger B)$ and the associated norm is called the Hilbert-Schmidt norm:

$$\|A\|_{\text{HS}}^2 = \text{Tr}(A^\dagger A) = \sum_{ij} \overline{a_{ij}} a_{ij}.$$

As $B(X, Y)$ is finite dimensional, from Theorem 7.7 we have that all these norms are equivalent.

Theorem 8.14 (**Diagonalizable matrices are dense**). *Diagonalizable matrices are dense in the space $\mathfrak{M}_{N \times N}(\mathbb{F})$ of $N \times N$ matrices in any norm.*

Proof. For convenience we will use the Hilbert-Schmidt norm. We aim to prove that for any matrix M , for any ϵ , there exists a diagonalizable matrix M^ϵ such that $\|M - M^\epsilon\|_{\text{HS}} \leq \epsilon$.

As any matrix can be brought in Jordan normal form, it suffices to prove the statement for a Jordan block. We consider M to be a Jordan block and perturb it to $M_{ij}^\epsilon = M_{ij} + j\epsilon \delta_{ij}$. The matrix M^ϵ has n distinct eigenvalues, hence it is diagonalizable (as all the eigenvalues have both algebraic and geometric multiplicity 1) and $\|M - M^\epsilon\|_{\text{HS}}^2 \leq \epsilon^2 \sum_{i=1}^N i^2$. $\square$

Corollary 8.3 (**$\det(A) = e^{\text{Tr} \ln A}$**). *For any matrix we have $\det(A) = e^{\text{Tr} \ln A}$.*

Proof. The statement trivially holds for diagonalizable matrices, hence by density it holds for all matrices. Alternatively, bringing A in Jordan normal form, both A and $\ln A$ are upper triangular, the determinant of A is the product of its eigenvalues (counted with multiplicity) and the trace of its logarithm is the sum of the logarithms of the eigenvalues, counted with multiplicity. $\square$

Exercise 8.13 (Pauli matrices). *Prove that the matrices:*

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & \imath \\ -\imath & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

form a basis in the space of 2×2 Hermitian matrices. Then:

- *compute the eigenvalues and eigenvectors of all these matrices.*
- *prove that $\sigma_i \sigma_j = \delta_{ij} + \imath \epsilon_{ijk} \sigma_k$.*
- *we denote $\vec{\sigma} = \sigma_i e_i$ with e_i the standard basis in $\mathbb{R}^3$. Prove that for any vector $\vec{x} = \sum_i x_i e_i$ in $\mathbb{R}^3$, $\vec{x} = \frac{1}{2} \text{tr}[(\vec{x} \cdot \vec{\sigma}) \vec{\sigma}]$.*
- *prove that $\det(\vec{x} \cdot \vec{\sigma}) = -\|\vec{x}\|^2$.*
- *prove that $[(\vec{x} \cdot \vec{\sigma}), (\vec{y} \cdot \vec{\sigma})] = 2\imath(\vec{x} \times \vec{y}) \cdot \vec{\sigma}$ and $\{(\vec{x} \cdot \vec{\sigma}), (\vec{y} \cdot \vec{\sigma})\} = 2(\vec{x} \cdot \vec{y})I$.*
- *compute $R_{\vec{n}}(\theta) = e^{-\imath \frac{\theta}{2}(\vec{n} \cdot \vec{\sigma})}$ if $\|\vec{n}\| = 1$. Compute $R_{\vec{n}}(\theta)R_{\vec{n}'}(\theta')$ and $R_{\vec{n}}(-\theta)(\vec{x} \cdot \vec{\sigma})R_{\vec{n}}(\theta)$.*

Chapter 9

Banach spaces

We recall from Chapter 5 that a vector space over the field of scalars $\mathbb{F} = \mathbb{R}, \mathbb{C}$ is a set endowed with two operations:

- the addition $+: V \times V \rightarrow V, (x, y) \rightarrow x + y$ such that $+$ is associative and commutative, 0 is the identity element for $+$ and every element $x \in V$ has an additive inverse, $-x$.
- multiplication by scalars $\cdot: \mathbb{F} \times V \rightarrow V, (\alpha, x) \rightarrow \alpha \cdot x$ such that:
 - $\cdot$ is compatible with the multiplication $\times$ in $\mathbb{F}$ that is $\alpha \cdot (\beta \cdot x) = (\alpha \times \beta) \cdot x$.
 - the identity of $\cdot$ is the multiplicative identity in $\mathbb{F}$, denoted 1 , that is $1 \cdot x = x$.
 - $\cdot$ is distributive with respect to both the addition in $\mathbb{F}$ and the addition in V , that is $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$ and $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$.

Note that $x = 1 \cdot x = (1 + 0) \cdot x = 1 \times x + 0 \cdot x$ hence $0 \cdot x = 0$. We suppress the $\cdot$ and $\times$ notation for the multiplication.

As we have seen, any vector space has a Hamel basis, that is a set of vectors $B \subset V$ such that:

- *Linearly independent.* For every finite subset $\{e_1, \dots, e_n\} \subset B$, $\alpha_1 e_1 + \dots + \alpha_n e_n = 0 \Rightarrow \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$.
- *Spanning.* Every vector in V writes as a finite linear combination of vectors in the basis, that is $\forall v \in V$ there exist $\beta_1, \beta_2, \dots, \beta_m \in \mathbb{F}$ and $\{e_1, \dots, e_m\} \subset B$ such that $v = \beta_1 e_1 + \dots + \beta_m e_m$, and the scalars $\beta_1, \dots, \beta_m$ are called the components of v in the basis B .

The drawback of Hamel basis is that they are useful only if they have a finite number of elements. If not, insisting that every vector writes as a finite linear combination makes Hamel bases huge: one often needs an uncountable infinity of basis vectors in order to finitely span any element in V .

Interesting examples of vector spaces of infinite dimension are the spaces of real (or complex) valued sequences respectively functions:

$$\left\{ x = (x_n)_{n \in \mathbb{N}}, x_n \in \mathbb{R} \mid (\alpha x + \beta y)_n = \alpha x_n + \beta y_n, \forall n \in \mathbb{N} \right\}, \quad \left\{ f : X \rightarrow \mathbb{R} \mid (\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x), \forall x \in X \right\}.$$

9.1 Banach spaces

Definition 9.1 (**Banach space**). A norm on the vector space $(V, \mathbb{F})$ is a function $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ such that:

- $\|v\| \geq 0$ and $\|v\| = 0$ if and only if $v = 0$.¹

¹It is in fact sufficient to require $\|v\| > 0$ if $v \neq 0$.

- $\|\alpha x\| = |\alpha| \|x\|$, $\forall \alpha \in \mathbb{F}$ and $\forall x \in V$.
- $\|x + y\| \leq \|x\| + \|y\|$, $\forall x, y \in V$.

Any normed vector space is a metric space with distance function $d(x, y) = \|x - y\|$. A Banach space is a Cauchy complete normed vector space.

For instance $\mathbb{R}^D$ and $\mathbb{C}^D$ are normed spaces (and in fact Banach spaces) with norm and of the p -norms $\|x\|_p = (\sum_{\mu=1}^D |x^\mu|^p)^{1/p}$. An less trivial example of a Banach space is given in the lemma below.

Proposition 9.1. *The set of bounded real (or complex)² valued functions on some set:*

$$\mathcal{B}(X, \mathbb{R}) = \left\{ f : X \rightarrow \mathbb{R} \left| \sup_{x \in X} |f(x)| = \|f\|_\infty < \infty \right. \right\},$$

$\mathcal{B}(X, \mathbb{R})$ with sup norm is a Banach space.

Proof. $\mathcal{B}(X, \mathbb{R})$ is a vector space with pointwise addition and multiplication. As $|f(x) + g(x)| \leq |f(x)| + |g(x)|$, $\forall x \in X$, $\|\cdot\|_\infty$ respects the triangle inequality and is a norm.

Let f_n be a Cauchy sequence in $\mathcal{B}(X, \mathbb{R})$, that is $\forall \epsilon, \exists N_\epsilon$ such that $k, l > N_\epsilon \Rightarrow \|f_k - f_l\|_\infty < \epsilon$:

- $\forall x \in X$, $|f_k(x) - f_l(x)| < \epsilon$ hence $f_k(x)$ is a Cauchy sequence in $\mathbb{R}$ and converges to some limit which we denote $\lim_n f_n(x) = f(x)$.
- as $\forall x \in X$, $|f_k(x) - f_l(x)| < \epsilon$ and any distance function is continuous, we can send $l \rightarrow \infty$ and conclude that $\forall x \in X$, $|f_k(x) - f(x)| < \epsilon$.
- it follows that $\forall x \in X$, $|f(x)| \leq |f_k(x)| + \epsilon$ hence $\|f\|_\infty \leq \|f_k\|_\infty + \epsilon$ and $f \in \mathcal{B}(X, \mathbb{R})$.
- it follows also that $\sup_{x \in X} |f_k(x) - f(x)| = \|f_k - f\|_\infty \leq \epsilon$.

□

Definition 9.2 (Separability). *A topological space X is called separable if it contains a countable dense subset, that is X contains a sequence (x_n) such that any open set contains at least one element of the sequence. If X is a metric space, it is separable in the standard topology if $\forall x \in X$, $\forall \epsilon > 0$, $\exists N$ such that $d(x_N, x) < \epsilon$.*

For example $\mathbb{R}$ is separable (in the standard topology) as $\mathbb{Q}$ is countable and dense in $\mathbb{R}$. Separability is a stringent requirement: one does sometimes encounter spaces that are not separable. If a topological space is *second countable*, that is its topology has a countable basis,³ then it is separable: one builds a countable dense subset by choosing one element per member of the basis. The converse does hold for metric spaces: denoting (x_n) a countable dense subset in X , the set $\{B_r(x_n) \mid r \in \mathbb{Q}_{>0}\}$ is countable (as it is the countable union of countable sets) and gives a basis for the topology of X . Indeed, any neighborhood U of $x \in X$ contains $B_{3\epsilon}(x)$ for some ϵ . As (x_n) is dense, there exist $x_p \in B_\epsilon(x)$. As $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists a rational number r such that $\epsilon < r < 2\epsilon$ hence $x \in B_r(x_p) \subset U$.

We define for $p \geq 1$ the spaces $\ell^p(\mathbb{F})$ of real or complex sequences as:

$$\ell^p(\mathbb{F}) = \left\{ x = (x_n) \in \mathbb{F} \left| \|x\|_p = \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p} < \infty \right. \right\}, \quad \ell^\infty(\mathbb{F}) = \left\{ x = (x_n) \in \mathbb{F} \left| \|x\|_\infty = \sup_{n \in \mathbb{N}} |x_n| < \infty \right. \right\}.$$

We have that if $1 \leq q \leq p$, $\|x\|_q \leq \|x\|_p$ hence $\ell^p \subset \ell^q$.

²Or any Banach space.

³That is there exist a countable set $(U_i)_{i=1}^\infty$ of open sets such that any open set can be written as the union of some subfamily of this set.

Proposition 9.2. *The spaces $(\ell^p(\mathbb{F}), \|\cdot\|_p)$ and $(\ell^\infty(\mathbb{F}), \|\cdot\|_\infty)$ are Banach spaces. $(\ell^p(\mathbb{F}), \|\cdot\|_p)$ for $1 \leq p < \infty$ are separable. $(\ell^\infty(\mathbb{F}), \|\cdot\|_\infty)$ is not separable.*

Proof. We consider real valued sequences. The fact that ℓ^p are normed vector spaces is straightforward.⁴ Let us prove that ℓ^p is complete (ℓ^∞ is a particular case of the previous proposition).

Let $x^{(n)}$ be a Cauchy sequence in ℓ^p , that is $\forall \epsilon, \exists N_\epsilon$ such that $k, l > N_\epsilon \Rightarrow \|x^{(k)} - x^{(l)}\|_p < \epsilon$:

- $\forall i, |x_i^{(k)} - x_i^{(l)}| < \epsilon$ hence $x_i^{(k)}$ is a Cauchy sequence in $\mathbb{R}$ and converges to some limit which we denote $\lim_n x_i^{(n)} = x_i$.
- we have $\|x^{(k)} - x^{(l)}\|_p < \epsilon$. Then for all I , $\sum_{i=0}^I |x_i^{(k)} - x_i^{(l)}|^p < \epsilon^p$ and sending $l \rightarrow \infty$ we have $\sum_{i=0}^I |x_i^{(k)} - x_i|^p \leq \epsilon^p$. As this holds for any I , taking the supremum over I we have $\|x^{(k)} - x\|_p \leq \epsilon$.
- using the triangle inequality $\|x\|_p \leq \|x^{(k)} - x\|_p + \|x^{(k)}\|_p \leq \|x^{(k)}\|_p + \epsilon$, hence $x \in \ell^p$.
- as $\|x^{(k)} - x\|_p \leq \epsilon$, $x^{(k)} \rightarrow x$ in ℓ^p .

The set ℓ^p has a countable dense subset. Consider the set of finite sequences of rational numbers and their union:

$$Q = \bigcup_{n=0}^{\infty} Q^{(n)}, \quad Q^{(n)} = \{(q_j) \subset \mathbb{Q} \mid q_j = 0, \quad j > n\}.$$

The sets $Q^{(n)}$ are countable as they are a Cartesian product of countable sets⁵ and so is their union Q . Also, $\forall n, Q^{(n)} \subset \ell^p \Rightarrow Q \subset \ell^p$. To show that this set is dense, consider $x \in \ell^p$. As $\sum_n |x_n|^p < \infty$ we have that $\forall \epsilon, \exists N_\epsilon$ such that $\sum_{n > N_\epsilon} |x_n|^p \leq \frac{\epsilon^p}{2^p}$. Chose $q \in Q^{(N_\epsilon)}$ such that for $0 \leq n \leq N_\epsilon$, $|x_n - q_n| < \frac{\epsilon}{2(N_\epsilon+1)}$. Then, using the triangle inequality:

$$\|x - q\|_p \leq \left(\sum_{n=0}^{N_\epsilon} |x_n - q_n|^p \right)^{1/p} + \left(\sum_{n > N_\epsilon} |x_n|^p \right)^{1/p} \leq \sum_{n=0}^{N_\epsilon} (|x_n - q_n|^p)^{1/p} + \left(\sum_{n > N_\epsilon} |x_n|^p \right)^{1/p} \leq \epsilon.$$

On the contrary, ℓ^∞ has no countable dense subset. For $I \subset \mathbb{N}$ we define the sequence $e_I \in \ell^\infty$:

$$(e_I)_n = \begin{cases} 1, & n \in I \\ 0, & n \notin I \end{cases},$$

hence $\|e_I - e_J\|_\infty = 1$ for $I \neq J$. By Cantor's Theorem 2.1, the set $\{B_{\frac{1}{2}}(e_I) \mid I \subset \mathbb{N}\}$ is uncountable hence it is an uncountably infinite collection of disjoint open balls in ℓ^∞ (due to the triangle inequality there can be no point at distance less than $1/2$ from two distinct e_I). Any dense subset of ℓ^∞ must contain an element in each of these balls hence it is uncountable. $\square$

A Schauder basis in a infinite dimensional space is similar to a Hamel basis, but with the difference that one is allowed to take countable infinite linear combinations. Such “bases” can be defined in generic topological vector spaces, but here we are only interested in the Banach space case.

Definition 9.3 (**Schauder basis**). *Let $(V, \mathbb{F}, \|\cdot\|)$ be a Banach space. A Schauder basis is a countable, ordered set $(e_n)_{n \in \mathbb{N}}$ such that $\forall x \in V$, there exists a unique sequence $(\alpha_n)_{n \in \mathbb{N}}$ such that:*

$$x = \lim_{N \rightarrow \infty} \sum_{n=0}^N \alpha_n e_n \equiv \sum_{n=0}^{\infty} \alpha_n e_n.$$

⁴Hint: use Hölder's inequality.

⁵A finite Cartesian product is an iterated pairwise product. As for the counting of $\mathbb{Q}$, the Cartesian product of two countable sets is counted by listing the two sets on the horizontal and vertical axes and stepping on the grid $s_{11} \rightarrow s_{21} \rightarrow s_{12} \rightarrow s_{13} \rightarrow s_{22} \rightarrow s_{31}$ and so on.

Note that the elements (e_n) must be ordered, as this series might not converge unconditionally. If $\{e_n\}$ is a Schauder basis of V then $V = \overline{\text{Span}\{e_n \mid n \in \mathbb{N}\}}$.⁶ The elements (e_n) in a Schauder basis are linearly independent (i.e. any finite subset of (e_n) is linearly independent).

The spaces ℓ^p have a Schauder basis given by the sequences e_n with elements $(e_n)_j = \delta_{nj}$. Every sequence $x = (x_n)$ writes as $x = \sum_{n \geq 0} x_n e_n$, and $\|x - \sum_{n=0}^N x_n e_n\|_p^p \leq \sum_{n > N} |x_n|^p \rightarrow_{N \rightarrow \infty} 0$. ℓ^∞ does not have a Schauder basis.

Theorem 9.1 (**A Banach space with a Schauder basis is separable**). *If a Banach space has a Schauder basis (e_n) then it is separable. The converse is not true: there exist separable Banach spaces with no Schauder basis.*

Proof. We define $Q^{(n)} = \{\sum_{j=0}^n q_j e_j \mid q_j \in \mathbb{Q} \text{ or } \mathbb{Q} + i\mathbb{Q}\}$, and we observe that $\bigcup_{n=0}^\infty Q^{(n)}$ is a countable union of countable sets hence a countable subset of V . To prove that it is dense, consider any $x \in V$. Then there exists a sequence (α_n) such that for any ϵ there exist N_ϵ such that $n \geq N_\epsilon$:

$$\|x - \sum_{k=0}^n \alpha_k e_k\| \leq \frac{\epsilon}{2}.$$

Choose for $k \leq N_\epsilon$, $q_k \in \mathbb{Q}$ (or $\mathbb{Q} + i\mathbb{Q}$) such that $|q_k - \alpha_k| \leq \frac{\epsilon}{2(N_\epsilon + 1)\|e_k\|}$ and denote $y = \sum_{k=0}^{N_\epsilon} q_k e_k \in Q$. Then:

$$\|x - y\| = \|x - \sum_{k=0}^{N_\epsilon} \alpha_k e_k + \sum_{k=0}^{N_\epsilon} (\alpha_k - q_k) e_k\| \leq \|x - \sum_{k=0}^{N_\epsilon} \alpha_k e_k\| + \sum_{k=0}^{N_\epsilon} |\alpha_k - q_k| \|e_k\| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

□

9.2 Linear operators

Theorem 9.2 (**A linear operator is continuous if and only if it is bounded**). *An linear operator $A : V \rightarrow Y$ from a normed space to a normed space $A(\alpha x + \beta y) = \alpha A(x) + \beta A(y)$ is a continuous map if and only if there exists some $c \in \mathbb{R}_{\geq 0}$ such that $\|Av\| \leq c\|v\|$ for all $v \in V$.*

Proof. We use here the $\epsilon - \delta$ definition of pointwise continuity. One implication is trivial: if $\|A(v) - A(w)\| = \|A(v - w)\| \leq c\|v - w\|$ then A is continuous at any v . Conversely, as $A(0) = A(x - x) = A(x) - A(x) = 0$, if A is continuous it is continuous at 0 and there exists δ such that $\|x\| < \delta \Rightarrow \|Ax\| \leq 1$. Then $\forall x$, $\|Ax\| = \|\frac{\|x\|}{\delta} A(\frac{x}{\|x\|} \delta)\| \leq \frac{1}{\delta} \|x\|$. □

Definition 9.4 (**Operator norm**). *The operator norm of A is:*

$$\|A\|_{\text{op.}} = \inf \left\{ c \geq 0 \mid \|Av\| \leq c\|v\|, \forall v \in V \right\} = \sup_{v \in V} \frac{\|Av\|}{\|v\|} = \sup_{v \in V, \|v\|=1} \|Av\|,$$

if it is finite.

Exercise 9.1. *Prove that $\|\cdot\|_{\text{op.}}$ is a norm. Prove it is submultiplicative $\|AB\|_{\text{op.}} \leq \|A\|_{\text{op.}} \|B\|_{\text{op.}}$.*

Exercise 9.2 (**The space of bounded linear operators between two Banach spaces is Banach**). *Let X and Y be Banach spaces. The space of bounded linear operators $(\mathcal{B}_{\text{op.}}(X, Y), \|\cdot\|_{\text{op.}})$ is a Banach space.*

Theorem 9.3 (**Bounded inverse theorem**). *Let $A : X \rightarrow Y$ a bijective bounded linear operator from one Banach space to another. Then A has a bounded inverse A^{-1} and A^{-1} is linear.*

⁶Note that the closure of the span of any set of vectors in a Banach space is always a vector space.

Proof. If it exists, the inverse of a linear operator is linear, as for any $y_1 = A(x_1)$ and $y_2 = A(x_2)$, $A^{-1}(\alpha_1 y_1 + \alpha_2 y_2) = A^{-1}(A(\alpha_1 x_1 + \alpha_2 x_2)) = \alpha_1 A^{-1}(y_1) + \alpha_2 A^{-1}(y_2)$.

In finite dimension the proof is simple: A is bounded hence continuous and maps $\overline{B_1(0)} \subset X$ to some subset of $\overline{B_{\|A\|}(0)} \subset Y$, hence it is a continuous mapping from a compact Hausdorff space (according to the Heine Borel theorem) to another one and from corollary 4.2 its inverse is continuous.

An alternative proof that the inverse of a continuous bijection between compact metric spaces is continuous goes as follows. Assume f^{-1} is not sequentially continuous, that is there $\exists(x_n), \exists\epsilon$ such that $\|f(x_n) - f(x)\| \rightarrow 0$ but $\|x_n - x\| > \epsilon$.⁷ As the domain is compact x_n has a convergent subsequence $x_{n_k} \rightarrow y$ and as f is continuous $f(x_{n_k}) \rightarrow f(y)$. As $f(x_n) \rightarrow f(x)$, we have $f(x) = f(y)$ hence $x = y$ by injectivity. But this is impossible as $|x_{n_k} - x| > \epsilon$.

The theorem holds in infinite dimensions, but its proof in this case is significantly more involved. Contrary to the finite dimensional case in which, due to the Heine Borel theorem, the unit ball is compact, the unit ball is *never* compact in infinite dimensions (which is known as Riesz's theorem). $\square$

We recall that a linear form on the vector space V over $\mathbb{F}$ is a linear function $\varphi : V \rightarrow \mathbb{F}$, $\varphi(\alpha x + \beta y) = \alpha\varphi(x) + \beta\varphi(y)$, for $x, y \in V$, $\alpha, \beta \in \mathbb{F}$. The dual space V^* is the vector space of linear functionals on V endowed with the vector space operations: $(\varphi_1 + \varphi_2)(x) = \varphi_1(x) + \varphi_2(x)$, $\varphi_1, \varphi_2 \in V^*$, $x \in V$ and $(\alpha \cdot \varphi)(x) = \alpha \cdot \varphi(x)$, $\varphi \in V^*$, $\alpha \in \mathbb{F}$, $x \in V$.

Definition 9.5 (**The continuous dual**). *In infinite dimensions, the continuous dual of a normed space $V, \|\cdot\|$ is the subset $V' = \{\varphi \in V^* \mid \|\varphi\|_{\text{op.}} < \infty\} \subset V^*$. The dual space V^* is almost useless in infinite dimensions: it is usually called algebraic dual and one usually calls dual the continuous dual. In infinite dimension there is no useful notion of dual of a Hamel basis: already the Hamel basis are quite useless.*

⁷We assume by contradiction that $\exists(x_n), \exists\epsilon$ such that $\forall\delta$ and $\forall N$ there exists $n > N$ with $\|f(x_n) - f(x)\| < \delta$ and $\|x_n - x\| > \epsilon$.

Chapter 10

Functions

An important role in physics and mathematics is played by the continuous functions.

10.1 Continuity

We give an alternative characterization of continuous functions in metric spaces.

Definition 10.1 (**The ϵ – δ criterion**). For X and Y metric spaces, a function $f : X \rightarrow Y$ is called continuous at x if and only if $\forall \epsilon > 0, \exists \delta_{\epsilon, x} > 0$ such that $d(x, x') < \delta_{\epsilon, x} \Rightarrow d(f(x), f(x')) < \epsilon$. Note that $\delta_{\epsilon, x}$ not only depends on ϵ , but also can depend on the point x . If f is continuous at x we denote $\lim_{x' \rightarrow x} f(x') = f(x)$.

A function between two metric spaces continuous at any $x \in X$ is called pointwise continuous.

As we have proven in Chapter 7, in metric spaces pointwise continuity, sequential continuity and continuity are equivalent. There exists several more refined notions of continuity in metric spaces.

Definition 10.2 (**Continuation, uniform and Lipschitz continuity**). For X and Y metric spaces, a function $f : X \rightarrow Y$:

- is pointwise continuous (already defined) if $\forall x, \forall \epsilon, \exists \delta_{\epsilon, x} : d(x, x') < \delta \Rightarrow d(f(x), f(x')) < \epsilon$.
- can be continued at a point x if there exists a function $\bar{f}$ whose domain includes x , is continuous at x and is such that $f = \bar{f}$ on the domain of f .
- is uniformly continuous on X if $\forall \epsilon, \exists \delta_{\epsilon}, \forall x, x' : d(x, x') < \delta \Rightarrow d(f(x), f(x')) < \epsilon$.
- is Lipschitz continuous with Lipschitz constant $L \geq 0$ on X if $\forall x, x' : d(f(x), f(x')) \leq L d(x, x')$.

Theorem 10.1 (**Heine–Cantor theorem (Satz von Heine)**). Let $f : K \rightarrow Y$ be a continuous functions on a compact metric space K valued in another metric space Y . If f is continuous then it is uniformly continuous.

Proof. As f is continuous, $\forall \epsilon, \forall x, \exists \delta_{\epsilon, x}$ such that $d(x, y) < \delta_{\epsilon, x} \Rightarrow d(f(x), f(y)) < \epsilon/2$.

Consider the open cover $(B_{\delta_{\epsilon, x}/2}(x))_{x \in K}$ of K made by the open balls of radius $\delta_{\epsilon, x}/2$ and select a finite subcover. We denote the centers of the balls in the subcover $x_1, \dots, x_n$ and we define $\delta_{\epsilon} = \min_{i=1, \dots, n} \{\delta_{\epsilon, x_i}/2\}$. For any $y, z \in K$ with $d(y, z) < \delta_{\epsilon}$, y must belong to one of the balls in the finite cover, say $y \in B_{\delta_{\epsilon, x_i}/2}(x_i)$. But then $d(z, x_i) \leq d(z, y) + d(y, x_i) < \delta_{\epsilon} + \delta_{\epsilon, x_i}/2 \leq \delta_{\epsilon, x_i}$. That is:

$$d(y, z) < \delta_{\epsilon} \Rightarrow \exists x_i, \quad d(y, x_i) \leq \delta_{\epsilon, x_i}, \quad d(z, x_i) \leq \delta_{\epsilon, x_i}.$$

But then $d(f(z), f(y)) \leq d(f(z), f(x_i)) + d(f(x_i), f(y)) < \epsilon$. □

Exercise 10.1. Prove that the absolute value $f(z) = |z|$ is Lipschitz continuous with constant 1.

10.2 The uniform limit theorem

We denote $\mathcal{C}(X, Y)$ the set of continuous functions with domain X and co domain Y . We are often interested in sequences of functions $f_n(x)$ and their limits $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. For instance one can consider sequences in $\mathcal{C}(X, Y)$, that is sequences of continuous functions. However, continuity does not survive taking naively such limits. For instance::

$$f_n : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}, \quad f_n(x) = x^n, \quad \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & x \in [0, 1) \\ 1, & x = 1 \\ \infty, & x > 1 \end{cases},$$

hence, while $f_n \in \mathcal{C}(\mathbb{R}_{\geq 0}, \mathbb{R}_{\geq 0})$, $f \notin \mathcal{C}(\mathbb{R}_{\geq 0}, \mathbb{R}_{\geq 0})$.

Definition 10.3 (**Pointwise and uniform convergence**). For X and Y metric spaces, a sequence of functions $f_n : X \rightarrow Y$ is said to converges pointwise to $f : X \rightarrow Y$ if $\forall x \in X, \lim_{n \rightarrow \infty} f_n(x) = f(x)$, that is

$$\forall x \in X, \forall \epsilon, \exists N_{\epsilon, x} : n > N_{\epsilon, x} \Rightarrow d(f_n(x), f(x)) < \epsilon.$$

This notion of convergence is called pointwise because $N_{\epsilon, x}$ can depend on x .

There exists stronger notions of convergence of sequences of functions. The sequence (f_n) is said to converge uniformly to f if:

$$\forall \epsilon, \exists N_\epsilon : \forall x \in X, n > N_\epsilon \Rightarrow d(f_n(x), f(x)) < \epsilon,$$

that is N_ϵ does not depend on the point x . The point of uniform convergence is that $f_n(x)$ approaches $f(x)$ everywhere at the same rate. If Y is a Banach space, uniform convergence is converge in $\|\cdot\|_\infty$.

Theorem 10.2 (**Uniform limit theorem**). For X and Y metric spaces, if the sequence $f_n \in \mathcal{C}(X, Y)$ converges uniformly to f then $f \in \mathcal{C}(X, Y)$.

Proof. For any ϵ , let N_ϵ such that for $n \geq N_\epsilon$, $d(f_n(x), f(x)) < \epsilon/3$ for all $x \in X$. Now, $\forall x, \exists \delta_{\epsilon, x}$ such that $d(x, y) < \delta_{\epsilon, x} \Rightarrow d(f_{N_\epsilon}(x), f_{N_\epsilon}(y)) < \epsilon/3$, hence $d(f(x), f(y)) \leq d(f(x), f_{N_\epsilon}(x)) + d(f_{N_\epsilon}(x), f_{N_\epsilon}(y)) + d(f_{N_\epsilon}(y), f(y)) < \epsilon$. $\square$

Corollary 10.1. Consider a power series centered at 0 and with radius of convergence r . For any $\rho < r$, the power series converges uniformly in $\overline{B_\rho(0)}$ and consequently defines a continuous function.

Proof. Consider any z with $|z| \leq \rho < r$. Pick δ such that $\rho < (r^{-1} + \delta)^{-1}$ (this always exists as $\rho < r$). As $r^{-1} = \limsup |a_n|^{1/n}$, there exists N such that for any $n > N$, $|a_n| < (r^{-1} + \delta)^n$ hence for any $n > N$, $|\sum_{k \geq n} a_k z^k| \leq \frac{[\rho(r^{-1} + \delta)]^n}{1 - \rho(r^{-1} + \delta)}$, which is smaller than any ϵ for $n > N_\epsilon$ for some large enough N_ϵ . $\square$

Corollary 10.2 (**$\mathcal{C}_b(X, Y)$ is a Banach space in $\|\cdot\|_\infty$**). If Y is a Banach space and X a metric space, the space $\mathcal{C}_b(X, Y)$ of bounded continuous functions on X is a Banach space with $\|\cdot\|_\infty$. If X is compact, any continuous function on X is bounded, that is $\mathcal{C}_b(X, Y) = \mathcal{C}(X, Y)$.

Proof. This follows from observing that the space of bounded functions on X is a Banach space, and the uniform limit theorem, which implies that the limit in $\|\cdot\|_\infty$ of a sequence of continuous functions is continuous.

For the second assertion, note that the image of a compact set via a continuous function is compact, hence bounded and close (see the proof of Heine Borel). $\square$

10.3 Differentiation in Banach spaces

For functions $F : X \rightarrow Y$ with X and Y general Banach spaces, the notion of derivative is replaced by the notion of differential.

Definition 10.4 (**Directional (Gateaux) differential**). *Let X and Y be Banach spaces, $U \subset X$ open and $F : U \rightarrow Y$. The Gateaux differential of F at $x \in U$ in the direction $v \in X$ is $DF(\cdot, \cdot) : U \times X \rightarrow Y$:*

$$DF(x; v) = \lim_{t \rightarrow 0} \frac{F(x + tv) - F(x)}{t} = \left. \frac{d}{dt} F(x + tv) \right|_{t=0}.$$

If the limit exists for all $v \in X$ we say that F is Gateaux differentiable at x . If X is a real vector space then the limit is taken for $t \in \mathbb{R}$; if X is a complex vector space, t is taken to zero in the complex plane.

At each point $x \in U$, the Gateaux differential defines a homogeneous function $DF(x; \cdot) : X \rightarrow Y$, $DF(x; \alpha v) = \alpha DF(x; v)$. However, this function is *not necessarily* linear in v , that is it is not necessarily additive, $DF(x; v+w) \neq DF(x; v) + DF(x; w)$. In infinite dimensions it might happen that $DF(x; v)$ is linear but discontinuous in v .

A second notion of differential is the following.

Definition 10.5 (**Fréchet differential**). *Let X and Y be Banach spaces and $U \subset X$ open. A function $F : U \rightarrow Y$ is called Fréchet differentiable at $x \in U$ if there exist a bounded linear operator $A : X \rightarrow Y$ such that:*

$$\lim_{\|h\|_X \rightarrow 0} \frac{\|F(x+h) - F(x) - Ah\|_Y}{\|h\|_X} = 0.$$

If A exists it is unique and we denote $A = \mathfrak{d}F(x) \equiv \mathfrak{d}F|_x$. F is said to be Fréchet continuously differentiable in U , or $\mathcal{C}^1$, if the map $x \mapsto \mathfrak{d}F(x) = \mathfrak{d}F|_x$ from U to the bounded linear operators from X to Y , $\mathcal{B}_{\text{op.}}(X, Y)$ endowed with the operator norm, is continuous:

$$\forall x \in U, \forall \epsilon, \exists \delta_{\epsilon, x}, \quad \|y - x\|_X \leq \delta_{\epsilon, x} \Rightarrow \|\mathfrak{d}F|_y - \mathfrak{d}F|_x\|_{\text{op.}} < \epsilon.$$

Observe that if F is a bounded linear operator then $\mathfrak{d}F = F$.

If the domain of F is $\mathbb{R}$, that is $F : \mathbb{R} \rightarrow Y$, then the Fréchet differential reduces to the usual derivative, that is $\mathfrak{d}F|_x : \mathbb{R} \rightarrow Y$ is the (linear) multiplication operator $v \mapsto \mathfrak{d}F|_x v = F'(x)v$ which is a vector in Y as $v \in \mathbb{R}$ and $F'(x) \in Y$.

The Fréchet derivative is often encountered in physics. Anticipating a bit, the space of trajectories of a mechanical system with one degree of freedom is the space of twice continuously differentiable real functions, $\mathcal{C}^2([a, b]) = \{q : [a, b] \rightarrow \mathbb{R} \mid \sup_{x \in (a, b)} \|q(t)\| + \sup_{x \in (a, b)} \|\dot{q}(t)\| + \sup_{x \in (a, b)} \|\ddot{q}(t)\| < \infty\}$ where we denoted time derivative by dots. We will see later that this is a Banach space. The Lagrangian is a function $L : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, $(q, \dot{q}) \rightarrow L(q, \dot{q})$ and the action is a functional $S : \mathcal{C}^2([a, b]) \rightarrow \mathbb{R}$, $S[q(t)] = \int_a^b dt L(q(t), \dot{q}(t))$. The Fréchet differential of the action is:

$$\mathfrak{d}S|_{q(t)}[h(t)] = \int_a^b dt \left[\frac{\partial L}{\partial q}(q(t), \dot{q}(t)) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}}(q(t), \dot{q}(t)) \right) \right] h(t) + \frac{\partial L}{\partial \dot{q}}(q(b), \dot{q}(b)) h(b) - \frac{\partial L}{\partial \dot{q}}(q(a), \dot{q}(a)) h(a).$$

We will see below that if the action has a minimum then $\mathfrak{d}S|_{q(t)} = 0$. Supplementing this with the constraint that the trajectories have fixed end points, $h(a) = h(b) = 0$, leads to the Euler Lagrange equations.¹

Theorem 10.3 (**Chain rule**). *Let X, Y, Z Banach spaces, $U \subset X$ open, $F : U \rightarrow Y$ a Fréchet differentiable function at $x_0 \in U$, V with $F(x_0) \in V \subset Y$ an open set and $G : V \subset Y \rightarrow Z$ Fréchet differentiable at $F(x_0)$. Then $G \circ F$ is Fréchet*

¹If $f(t)$ is continuous and $\int_a^b dt f(t)h(t) = 0$ for all continuous functions h such that $h(a) = h(b) = 0$, then $f(t) = 0$, $\forall t \in [a, b]$. Indeed, assume $f(t') > 0$ for some $t' \in (a, b)$ then by continuity $f(t) > 0$ on some interval $[t - \epsilon, t + \epsilon] \subset [a, b]$ and one can chose $h(t) = e^{-\frac{1}{\epsilon^2 - |t' - t|^2}}$ for $|t - t'| < \epsilon$ and $h(t) = 0$ otherwise.

differentiable at x_0 and:

$$\mathrm{d}(G \circ F)|_{x_0} = \mathrm{d}G|_{F(x_0)} \circ \mathrm{d}F|_{x_0} \in \mathcal{B}_{\mathrm{op.}}(X, Z) .$$

Proof. As G is Fréchet differentiable at $y_0 = F(x_0)$, for every $\zeta > 0$, there exists ϵ such that:

$$\forall y \in B_\epsilon(y_0) , \quad \|G(y) - G(y_0) - \mathrm{d}G|_{y_0}(y - y_0)\|_Z \leq \zeta \|y - y_0\|_Y .$$

As F is continuous $F^{-1}(B_\epsilon(y_0))$ is open and contains $x_0 \in F^{-1}(y_0)$ hence there exists δ such that, at the same time, $B_\delta(x_0) \subset F^{-1}(B_\epsilon(y_0))$ and:

$$\forall x \in B_\delta(x_0) , \quad \|F(x) - F(x_0) - \mathrm{d}F|_{x_0}(x - x_0)\|_Y \leq \zeta \|x - x_0\|_X .$$

We then have for $x \in B_\delta(x_0)$ that, denoting $F(x) = y$ and $F(x_0) = y_0$ and observing that $y \in B_\epsilon(y_0)$:

$$\begin{aligned} & \|G \circ F(x) - G \circ F(x_0) - (\mathrm{d}G|_{F(x_0)} \circ \mathrm{d}F|_{x_0})(x - x_0)\|_Z \\ & \leq \|G(y) - G(y_0) - \mathrm{d}G|_{y_0}(y - y_0)\|_Z + \left\| \mathrm{d}G|_{y_0} \left(F(x) - F(x_0) - \mathrm{d}F|_{x_0}(x - x_0) \right) \right\|_Z \\ & \leq \zeta \|y - y_0\|_Y + \|\mathrm{d}G|_{y_0}\|_{\mathrm{op.}} \zeta \|x - x_0\|_X \leq \zeta \left(\|\mathrm{d}F|_{x_0}\|_{\mathrm{op.}} + \zeta \right) \|x - x_0\|_X + \|\mathrm{d}G|_{y_0}\|_{\mathrm{op.}} \zeta \|x - x_0\|_X \end{aligned}$$

□

The relation between the Fréchet and the Gateaux differentials depends on the continuity properties of the differentials.

Theorem 10.4 (**Gateaux versus Fréchet**). *Let X, Y be Banach spaces and let $F : X \rightarrow Y$. Then*

- *if F is Fréchet differentiable at x_0 , then it is Gateaux differentiable at x_0 and the Gateaux and Fréchet differentials agree $DF(x_0; v) = \mathrm{d}F|_{x_0}(v)$. The converse is not true: for instance the Gateaux differential might fail to be a linear operator.*

If F is $\mathcal{C}^1$ then $\mathrm{d}F|_x$, which by the previous assertion equals $DF(x; \cdot)$, is a bounded linear operator for x in some neighborhood of x_0 and the map from X to $\mathcal{B}_{\mathrm{op.}}(X, Y)$, $x \mapsto \mathrm{d}F|_x = DF(x; \cdot)$ is continuous at x_0 .

- *if F is Gateaux differentiable at a point x_0 and the Gateaux differential $DF(x; \cdot)$ is a bounded linear operator in its second argument for x in some neighborhood of x_0 and furthermore the map from X to $\mathcal{B}_{\mathrm{op.}}(X, Y)$, $x \mapsto DF(x; \cdot)$ is continuous at x_0 then F is continuously Fréchet differentiable at x_0 and $\mathrm{d}F|_{x_0} = DF(x_0; \cdot)$.*

Proof. The linear function $G : \mathbb{R} \rightarrow X$, $G(t) = x_0 + tv$ is Fréchet differentiable in t at $t = 0$ and $\mathrm{d}G|_{t=0} = v$. If F is Fréchet differentiable at $x = G(0)$, then by the chain rule $F \circ G$ is Fréchet differentiable at $t = 0$ and we have:

$$DF(x_0; v) = \frac{d}{dt} F(x_0 + tv) \Big|_{t=0} = \frac{d}{dt} (F \circ G)(t) \Big|_{t=0} = \mathrm{d}(F \circ G)|_{t=0} = \mathrm{d}F|_{x_0} \circ \mathrm{d}G|_0 = \mathrm{d}F|_{x_0}(v) .$$

The remaining statements in this item are straightforward.

For the second item, consider the function $G(x_0 + th) = F(x_0 + th) - DF(x_0; th) = F(x_0 + th) - t DF(x_0; h)$. Its Gateaux differential in the direction h is $DG(x_0 + th; h) = \frac{d}{d\tau} G(x_0 + (t + \tau)h) \Big|_{\tau=0} = \frac{d}{dt} G(x_0 + th)$ which, substituting G in terms of F , is $DG(x_0 + th; h) = DF(x_0 + th; h) - DF(x_0; h)$. By the mean value inequality we have:

$$\|G(x_0 + h) - G(x_0)\| \leq \sup_{t \in (0,1)} \left\| \frac{d}{dt} G(x_0 + th) \right\| = \sup_{t \in (0,1)} \|DG(x_0 + th; h)\| ,$$

which, substituting G in terms of F is:

$$\|F(x_0 + h) - F(x_0) - DF(x_0; h)\| \leq \sup_{t \in (0,1)} \|DF(x_0 + th; h) - DF(x_0; h)\| \leq \sup_{t \in (0,1)} \|(DF(x_0 + th; \cdot) - DF(x_0; \cdot))h\| ,$$

where we used the linearity of $DF(x_0 + th; \cdot)$ and $DF(x_0; \cdot)$. It follows that:

$$\frac{\|F(x_0 + h) - F(x_0) - DF(x_0; h)\|}{\|h\|} \leq \sup_{t \in (0,1)} \|DF(x_0 + th; \cdot) - DF(x_0; \cdot)\|_{\text{op}} ,$$

and as $x \mapsto DF(x; \cdot)$ is continuous, $\forall \epsilon, \exists \delta$ such that $\|x_0 - x\| < \delta \Rightarrow \|DF(x; \cdot) - DF(x_0; \cdot)\|_{\text{op}} < \epsilon$, concluding the proof. $\square$

Thus for a continuously differentiable function the Fréchet and Gateaux differentials agree.

Theorem 10.5 (Generalized mean value inequality). *A subset set K of some Banach space (or more generally vector space) called convex if $\forall x_1, x_2 \in K$ and $t \in [0, 1]$, $(1-t)x_1 + tx_2 \in K$.*

If $F : U \subset X \rightarrow Y$ is continuously differentiable in U , then for any two $x_1, x_2 \in K \subset U$ with K some convex set:

$$\|F(x_2) - F(x_1)\| \leq \left(\sup_{x \in K} \|dF|_x\|_{\text{op}} \right) \|x_1 - x_2\| .$$

Proof. Applying the mean value inequality to $f(t) = F((1-t)x_1 + tx_2)$ we get:

$$\begin{aligned} \|F(x_2) - F(x_1)\| &\leq \sup_{t \in [0,1]} \left\| \frac{d}{dt} F((1-t)x_1 + tx_2) \right\| = \sup_{t \in [0,1]} \|DF((1-t)x_1 + tx_2; x_2 - x_1)\| \\ &= \sup_{t \in [0,1]} \|dF|_{(1-t)x_1 + tx_2}(x_2 - x_1)\| \leq \left(\sup_{t \in [0,1]} \|dF|_{(1-t)x_1 + tx_2}\|_{\text{op}} \right) \|x_2 - x_1\| \leq \left(\sup_{x \in K} \|dF|_x\|_{\text{op}} \right) \|x_1 - x_2\| . \end{aligned}$$

$\square$

Exercise 10.2. *Consider the Banach space $X = \mathcal{C}([0, 1], \mathbb{R})$ with norm $\|\cdot\|_\infty$ and the function $F : X \rightarrow \mathbb{R}$, $F(f) = \int_0^1 dt f^3(t)$. Prove that its Fréchet differential is $dF|_f : X \rightarrow \mathbb{R}$, $dF|_f(g) = \int_0^1 dt 3f^2(t)g(t)$.*

Exercise 10.3. *Compute the formal Fréchet differential of a formal power series $P(x) = \sum_{n \geq 0} a_n x^n$. We linearize the power series in $x - x_0$:*

$$P(x) = \sum_{n \geq 0} a_n (x_0 + x - x_0)^n = \sum_{n \geq 1} a_n \sum_{p=0}^{n-1} x_0^p (x - x_0) x_0^{n-p-1} + O((x - x_0)^2) \Rightarrow dP|_{x_0}(x - x_0) = \sum_{n \geq 1} a_n \sum_{p=0}^{n-1} x_0^p (x - x_0) x_0^{n-p-1} .$$

For the case of $\exp(x) = \sum_{k!} x^k$ we have $d \exp|_0 = 1$.

Exercise 10.4. *Denote $\mathcal{M}_{N \times N}(\mathbb{C})$ the space of $N \times N$ with complex entries. As all the norms are equivalent in finite dimensions, we endow the space with the norm $\|A\|_\infty = \max_{i,j} |a_{ij}|$. Compute the Fréchet differential of $f : \mathcal{M}_{N \times N}(\mathbb{C}) \rightarrow \mathcal{M}_{N \times N}(\mathbb{C})$, $f(A) = A^n$. From the previous exercise, $d f|_A(B) = \sum_{k=0}^{n-1} A^k B A^{n-1-k}$ and $\|(A+B)^n - A^n - d f|_A(B)\|_\infty \leq (n-1) \sum_{p=2}^{n-1} \|B\|_\infty^p \|A\|_\infty^{n-p}$.*

Exercise 10.5. *Compute the Fréchet differential of $F(x) = \det(x)$ which is a real valued function of a matrix x with real entries. We have for invertible x :*

$$\det(x+h) = \det(x) \det(1+x^{-1}h) = \det(x) e^{\text{tr} \ln(1+x^{-1}h)} = \det(x) e^{\text{tr}(x^{-1}h) + O(h^2)} = \det(x) \left(1 + \text{tr}(x^{-1}h) + O(h^2) \right) \Rightarrow d(\det)|_x(v) = \det(x) \text{tr}(x^{-1}v) .$$

The $O(h^2)$ terms are a finite sum of monomials of degree at least 2 in the entries of the matrix h , hence bounded by $\|h\|_\infty^2$. For x non invertible, denoting $x_{\hat{i}\hat{j}}$ the matrix x with the row i and column j deleted we have:

$$\det(x+h) = \det x + \sum_{i,j=1}^n h_{ij} (-1)^{i+j} \det x_{\hat{i}\hat{j}} = \text{Tr}(\text{adj}(x) h) ,$$

with $[\text{adj}(x)]_{ji} = (-1)^{i+j} \det x_{\hat{i}\hat{j}}$, and indeed when x is invertible $x^{-1} = \frac{1}{\det(x)} \text{adj}(x)$.

Exercise 10.6. Prove that the function $F(x, y) = \frac{x^2 y}{x^2 + y^2}$, $F(0, 0) = 0$ is Gateaux differentiable but not Fréchet differentiable at $(0, 0)$ in $\mathbb{R}^2$. Which of the conditions of the Theorem above is broken? For any non zero vector with components (v_x, v_y) , $DF((0, 0); (v_x, v_y)) = \frac{d}{dt} F(tv_x, tv_y) = F(v_x, v_y)$, and from the definition $DF((0, 0); (0, 0)) = 0$. This is not a linear function in v_x, v_y , hence F is not Fréchet differentiable: $\lim_{\|v\| \rightarrow 0} \frac{F(v_x, v_y) - Av}{\|v\|} \neq 0$ for any linear operator A .

Exercise 10.7. Prove that $f(x, y) = (x^2 + y^2) \sin \frac{1}{\sqrt{x^2 + y^2}}$, $f(0, 0) = 0$ is Fréchet differentiable in 0 but does not have continuous Gateaux differentials. Which of the conditions of the Theorem is broken? $\lim_{\|h\| \rightarrow 0} \frac{\|h\|^2 \sin \frac{1}{\|h\|}}{\|h\|} = 0$ hence the Fréchet differential of the function in 0 is 0. However,

$$\partial_x f = 2x \sin \frac{1}{\sqrt{x^2 + y^2}} + (x^2 + y^2) \frac{(-\frac{1}{2} 2x)}{(x^2 + y^2)^{3/2}} \cos \frac{1}{\sqrt{x^2 + y^2}} = 2x \sin \frac{1}{\sqrt{x^2 + y^2}} - \frac{x}{\sqrt{x^2 + y^2}} \cos \frac{1}{\sqrt{x^2 + y^2}} = 2r \sin \varphi \sin \frac{1}{r} - \sin \varphi \cos \frac{1}{r},$$

which is discontinuous for $r \rightarrow 0$. The function is Fréchet differentiable but not continuously Fréchet differentiable at $(0, 0)$.

10.4 The inverse function theorem

If a function between two Banach spaces is continuously differentiable and the differential is non degenerate, then it has a continuously differentiable inverse.

Theorem 10.6 (**The inverse function theorem**). Let X and Y be Banach spaces, $U \subset X$ an open subset and let $F : U \rightarrow Y$ be continuously differentiable on U . For any $x_0 \in U$ such that $dF|_{x_0}$ is invertible, there exists an open set $V_0 \in Y$ such that $F(x_0) \in V_0$ and another open set $U_0 \subset X$ such that $x_0 \in U_0$ and a continuously differentiable function $G : V_0 \rightarrow U_0$ such that $F \circ G = \text{id}|_{V_0}$, $G \circ F = \text{id}|_{U_0}$ and for $x \in U_0$, $dG|_{F(x)} = [dF|_x]^{-1}$.

Proof. Let us denote $D_0 = dF|_{x_0}$ which is a linear map from X to Y . As D_0 is bounded and invertible, by the Bounded invers theorem it has a bounded inverse, D_0^{-1} , which is a linear operator from Y to X .

Building G . As F is differentiable, there exist δ such that

$$\|F(x) - F(x_0) - D_0(x - x_0)\| \leq \frac{1}{2\|D_0^{-1}\|_{\text{op}}} \|x - x_0\|, \quad \forall x \in B_\delta(x_0),$$

and as $x \mapsto dF|_x$ is continuous, we can chose δ such that at the same time $\|dF|_x - D_0\|_{\text{op}} \leq \frac{1}{2\|D_0^{-1}\|_{\text{op}}}$, $\forall x \in B_\delta(x_0)$.²

Let us denote $y_0 = F(x_0)$, $s = \frac{\delta}{2\|D_0\|_{\text{op}}}$ and $V_0 = B_s(y_0)$. For $y \in V_0$ we define:

$$T^{(y)} : \overline{B_\delta(x_0)} \rightarrow X, \quad T^{(y)}(x) = x - D_0^{-1}(F(x) - y),$$

and remark that the fixed points of this map $T^{(y)}(x) = x$ are such that $F(x) = y$. We have:

- $T^{(y)}$ is a map from the complete metric space $\overline{B_\delta(x_0)}$ to itself. Indeed:

$$\begin{aligned} \|T^{(y)}(x) - x_0\| &= \|x - x_0 - D_0^{-1}(F(x) - F(x_0)) - D_0^{-1}(y_0 - y)\| = \|-D_0^{-1}(F(x) - F(x_0) - D_0(x - x_0)) - D_0^{-1}(y_0 - y)\| \\ &\leq \|D_0^{-1}\|_{\text{op}} \frac{1}{2\|D_0^{-1}\|_{\text{op}}} \|x - x_0\| + \|D_0^{-1}\|_{\text{op}} \|y - y_0\| < \delta. \end{aligned}$$

In fact $T(\overline{B_\delta(x_0)}) \subset B_\delta(x_0)$.

²In particular the series:

$$[dF|_x]^{-1} = \frac{1}{D_0 + (dF|_x - D_0)} = D_0^{-1} \sum_{k=0}^{\infty} (- (dF|_x - D_0) D_0^{-1})^k,$$

converges in operator norm for $x \in B_\delta(x_0)$, hence $dF|_x$ is invertible for all $x \in B_\delta(x_0)$.

- $T^{(y)}$ is a contradiction on this space. As $\overline{B_\delta(x_0)}$ is convex we have:

$$\|T^{(y)}(x_1) - T^{(y)}(x_2)\| \leq \sup_{x \in \overline{B_\delta(x_0)}} \|\mathrm{d}T^{(y)}|_x\|_{\mathrm{op}} \|x_1 - x_2\| = \sup_{x \in \overline{B_\delta(x_0)}} \|\mathrm{id} - D_0^{-1} \mathrm{d}F|_x\|_{\mathrm{op}} \|x_1 - x_2\| \leq \frac{1}{2} \|x_1 - x_2\| ,$$

where we recall that $\|\mathrm{d}F|_x - D_0\|_{\mathrm{op}} \leq \frac{1}{2\|D_0^{-1}\|_{\mathrm{op}}}$.

By the Banach fixed point theorem, $T^{(y)}$ has a fixed point $T^{(y)}(x) = x$ in $\overline{B_\delta(x_0)}$, and in fact in $B_\delta(x_0)$. We denote this fixed point $x = G(y)$, and G is a function $G : V_0 \rightarrow U_0 = F^{-1}(V_0) \cap B_\delta(x_0)$, which is an open set as $F^{-1}(V_0)$ is open by the continuity of F .

G is continuously differentiable. It is enough to check G is differentiable at y_0 (as any other $y \in V_0$ writes as $y = F(x)$ for some $x \in B_\delta(x_0)$ and one can rely on continuity arguments). Using the contracting property of $T^{(y)}$ for $x_1 = x_0$ and $x_2 = x = G(y)$, as $x_0 = T^{(y)}(x_0) + D_0^{-1}(F(x_0) - y)$ we have:

$$\|x - x_0\| \leq \|T^{(y)}(x) - T^{(y)}(x_0) - D_0^{-1}(F(x_0) - y)\| \leq \frac{1}{2} \|x - x_0\| + \|D_0^{-1}\|_{\mathrm{op}} \|y - y_0\| \Rightarrow \|x - x_0\| \leq 2\|D_0^{-1}\|_{\mathrm{op}} \|y - y_0\| ,$$

that is $\|G(y) - G(y_0)\| \leq 2\|D_0^{-1}\|_{\mathrm{op}} \|y - y_0\|$ and G is continuous at y_0 .

As F is differentiable at x_0 and $D_0 = \mathrm{d}F|_{x_0}$, $\forall \epsilon, \exists \zeta$ such that:

$$\begin{aligned} x \in B_\zeta(x_0) &\Rightarrow \|F(x) - F(x_0) - D_0(x - x_0)\| \leq \epsilon \|x - x_0\| \Leftrightarrow_{x=G(y), x_0=G(y_0)} \\ G(y) \in B_\zeta(x_0) &\Rightarrow \|D_0(G(y) - G(y_0)) - (y - y_0)\| \leq \epsilon \|G(y) - G(y_0)\| . \end{aligned}$$

As $\|G(y) - G(y_0)\| \leq 2\|D_0^{-1}\|_{\mathrm{op}} \|y - y_0\|$, choosing $r < \frac{\zeta}{2\|D_0^{-1}\|_{\mathrm{op}}}$, ensures that $G(B_r(y_0)) \subset B_\zeta(x_0)$, hence $\forall y \in B_r(y_0)$:

$$\begin{aligned} \|G(y) - G(y_0) - D_0^{-1}(y - y_0)\| &\leq \|D_0^{-1}\|_{\mathrm{op}} \|D_0(G(y) - G(y_0)) - (y - y_0)\| \\ &\leq \|D_0^{-1}\|_{\mathrm{op}} \epsilon \|G(y) - G(y_0)\| \leq 2\|D_0^{-1}\|_{\mathrm{op}}^2 \|y - y_0\| \epsilon , \end{aligned}$$

and G is differentiable at y_0 . Continuity of $y \mapsto \mathrm{d}G|_y = [\mathrm{d}F|_{G(y)}]^{-1}$ follows from the fact that it is a composition of continuous functions.

□

Let us consider for instance (see below for the Fréchet differential in finite dimensions) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$:

$$F^1(x^1, x^2) = x^1 , \quad F^2(x^1, x^2) = (x^1)^2 + (x^2)^2 , \quad \mathrm{d}F = \begin{pmatrix} 1 & 0 \\ 2x^1 & 2x^2 \end{pmatrix} , \quad \det(\mathrm{d}F|_x) = 2x^2 .$$

We have for $x_0 = (0, 1)$, $F(x_0) = (0, 1)$, and $\det(\mathrm{d}F|_{x_0}) = 2$, hence in some neighborhoods of $x_0 = (0, 1)$ and $y_0 = (0, 1)$, function F is invertible, and we have its inverse:

$$G^1(y^1, y^2) = y^1 , \quad G^2(y^1, y^2) = \sqrt{y^2 - (y^1)^2} , \quad \mathrm{d}G|_{y=F(x)} = \begin{pmatrix} 1 & 0 \\ \frac{-y^1}{\sqrt{y^2 - (y^1)^2}} & \frac{\frac{1}{2}}{\sqrt{y^2 - (y^1)^2}} \end{pmatrix}_{y=F(x)} = \begin{pmatrix} 1 & 0 \\ \frac{-x^1}{x^2} & \frac{1}{2x^2} \end{pmatrix} .$$

The function F is not locally invertible around the point $x_0 = (1, 0)$, $y_0 = (1, 1)$ as $\det(\mathrm{d}F|_{x_0}) = 0$.

10.5 The case $F : \mathbb{R}^N \rightarrow \mathbb{R}^M$

Recall that we denoted the dual of V by V^* , that is the vector space of linear functionals $\varphi : V \rightarrow \mathbb{F}$, $\varphi(\alpha x + \beta y) = \alpha\varphi(x) + \beta\varphi(y)$ endowed with the vector space operations $(\varphi_1 + \varphi_2)(x) = \varphi_1(x) + \varphi_2(x)$ and $(\alpha \cdot \varphi)(x) = \alpha \cdot \varphi(x)$. If the dimension of V is finite, given a basis $\{e_1, \dots, e_N\}$ in V , the dual basis in V^* consists in the linear functionals λ^j defined by $\lambda^j(e_i) = \delta_i^j$, that is $\lambda^j(\sum_{i=1}^N v^i e_i) = v^j$.

We denote the elements of $\mathbb{R}^M$ as $y = \sum_{j=1}^M y^j f_j$, in some basis $\{f_1, \dots, f_M\}$ and we denote the dual basis of linear forms $\mu^j : \mathbb{R}^M \rightarrow \mathbb{R}$, $\mu^j(f_i) = \delta_i^j$, that is $\mu^j(\sum_{i=1}^M y^i f_i) = \sum_{i=1}^M y^i \mu^j(f_i) = \sum_{i=1}^M y^i \delta_i^j = y^j$. As it is a linear operator, the Fréchet differential of μ^j is $d\mu^j|_{y_0} = \mu^j$. For any function $F : \mathbb{R}^N \rightarrow \mathbb{R}^M$, choosing some basis $\{e_1, \dots, e_N\}$ in $\mathbb{R}^N$, we write:

$$F\left(\sum_{i=1}^N x^i e_i\right) = \sum_{j=1}^M F^j(x^1, \dots, x^N) f_j,$$

where F^j are the components of F .

The function F is Fréchet differentiable at x_0 if and only if all $F^j : \mathbb{R}^N \rightarrow \mathbb{R}$ are Fréchet differentiable at x_0 : indeed, the sum of Fréchet differentiable functions is Fréchet differentiable; conversely, if F is Fréchet differentiable, as μ^j is Fréchet differentiable we have that $F^j = \mu^j \circ F$ is Fréchet differentiable and $dF^j|_{x_0} = d(\mu^j \circ F)|_{x_0} = \mu^j \circ dF|_{x_0}$.

We denote $x = \sum_i x^i e_i \in \mathbb{R}^N$ and we define the partial derivatives:

$$\frac{\partial F^j}{\partial x^i}(x_0) \equiv DF^j(x_0; e_i) = \frac{d}{dt} F^j(x_0 + te_i) \Big|_{t=0},$$

which are the Gateaux differentials along the basis elements.

If F is Fréchet differentiable then $\frac{\partial F^j}{\partial x^i}(x_0) = dF^j|_{x_0}(e_i)$ and the Fréchet differential is the linear operator $dF|_{x_0} : \mathbb{R}^N \rightarrow \mathbb{R}^M$ whose matrix elements are (denoting λ^i the basis dual to e_i):³

$$dF|_{x_0} = \sum_{j=1}^M \sum_{i=1}^N f_j (dF|_{x_0})^j_i \lambda^i, \quad (dF|_{x_0})^j_i = \mu^j(dF|_{x_0}(e_i)) = dF^j|_{x_0}(e_i) = \frac{\partial F^j}{\partial x^i}(x_0),$$

that is the matrix of the Fréchet differential $dF|_{x_0} v = J_F(x_0) v$ is the $M \times N$ Jacobian matrix:

$$J_F(x_0) = \begin{pmatrix} \frac{\partial F^1}{\partial x^1}(x_0) & \dots & \frac{\partial F^1}{\partial x^N}(x_0) \\ \vdots & & \\ \frac{\partial F^M}{\partial x^1}(x_0) & \dots & \frac{\partial F^M}{\partial x^N}(x_0) \end{pmatrix}.$$

As an example consider the mapping $F : (0, \pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3$:

$$F(\theta e_\theta + \varphi e_\varphi) = \sin \theta \cos \varphi f_1 + \sin \theta \sin \varphi f_2 + \cos \theta f_3, \quad dF|_{\theta_0, \varphi_0}(V) = \begin{pmatrix} \cos \theta_0 \cos \varphi_0 & -\sin \theta_0 \sin \varphi_0 \\ \cos \theta_0 \sin \varphi_0 & \sin \theta_0 \cos \varphi_0 \\ -\sin \theta_0 & 0 \end{pmatrix} \begin{pmatrix} V^\theta \\ V^\varphi \end{pmatrix},$$

and $dF|_{\theta_0, \varphi_0}$ maps $\mathbb{R}^2$, the two dimensional space tangent to $(0, \pi) \times (0, 2\pi)$ at the point (θ_0, φ_0) , to the two dimensional subspace of $\mathbb{R}^3$ tangent to the sphere at the point $F(\theta_0, \varphi_0)$.

Theorem 10.7 (In finite dimensions, continuous Gateaux differentials imply Fréchet differentiability). *Let $F : U \subset \mathbb{R}^N \rightarrow \mathbb{R}^M$. If F^j are continuously partially differentiable in U (i.e. differentiable in all the directions in $\mathbb{R}^N$ at all the points in U) then F is continuously Fréchet differentiable in U .*

³Any linear map $A : \mathbb{R}^N \rightarrow \mathbb{R}^M$, $Av = \sum_j f_j \mu^j(Av) = \sum_{j,i} f_j [\mu^j(Ae_i)] v^i \Rightarrow A = \sum_{i,j} f_j [\mu^j(Ae_i)] \lambda^i$.

Proof. It is enough to consider one component $F : U \subset \mathbb{R}^N \rightarrow \mathbb{R}$. The partial derivatives are instances of the Gateaux differentials, hence continuous. The Gateaux differential along an arbitrary direction writes, using the mean value theorem (where empty sums are zero):

$$\begin{aligned} DF(x; v) &= \lim_{t \rightarrow 0} \frac{F(x + t \sum_{i=1}^N v^i e_i) - F(x)}{t} = \lim_{t \rightarrow 0} \frac{\sum_{i=1}^N F(x + t \sum_{j=1}^i v^j e_j) - F(x + t \sum_{j=1}^{i-1} v^j e_j)}{t} \\ &= \lim_{t \rightarrow 0} \frac{\sum_{i=1}^N \frac{\partial F}{\partial x^i}(x + \xi_i t v^i e_i + t \sum_{j=1}^{i-1} v^j e_j) t v^i}{t}, \end{aligned}$$

for some $\xi_i \in (0, 1)$. Taking $t \rightarrow 0$ we conclude that $DF(x; v) = \sum_{i=1}^N \partial_i F(x) v^i$ hence is a linear operator in v . Using the Cauchy-Schwarz inequality (that is Hölder inequality for $p = q = 2$) this operator is bounded in norm by $\sqrt{\sum_i |\partial_i F(x)|^2}$ and continuous in x as $\|DF(x; \cdot) - DF(x_0; \cdot)\|_{\text{op}} \leq \sqrt{\sum_i |\partial_i F(x) - \partial_i F(x_0)|^2}$ $\square$

10.6 The implicit function theorem

The theorem below holds for general Banach spaces, but we will only present here the finite dimensional version.

Theorem 10.8 (**The implicit function theorem**). *Let $F : U \subset \mathbb{R}^{N+M} \rightarrow \mathbb{R}^M$ a continuously differentiable function and let $z_0 \in U$ be a solution of the equation $F(z_0) = 0$ such that $\mathbb{d}F|_{z_0}$ has maximal rank $\dim(\text{Im}(\mathbb{d}F|_{z_0})) = M$. We can parametrize $\mathbb{R}^{N+M} = \{(x, y) \mid x \in \mathbb{R}^N, y \in \mathbb{R}^M\}$, $z_0 = (x_0, y_0)$ and:*

$$\mathbb{d}F|_{(x_0, y_0)} = \begin{bmatrix} \frac{\partial F^1}{\partial x^1}(x_0, y_0) & \cdots & \frac{\partial F^1}{\partial x^N}(x_0, y_0) & \left| \frac{\partial F^1}{\partial y^1}(x_0, y_0) & \cdots & \frac{\partial F^1}{\partial y^M}(x_0, y_0) \right. \\ \vdots & \ddots & \vdots & \left| \vdots & \ddots & \vdots \right. \\ \frac{\partial F^M}{\partial x^1}(x_0, y_0) & \cdots & \frac{\partial F^M}{\partial x^N}(x_0, y_0) & \left| \frac{\partial F^M}{\partial y^1}(x_0, y_0) & \cdots & \frac{\partial F^M}{\partial y^M}(x_0, y_0) \right. \end{bmatrix} = \begin{bmatrix} \mathbb{d}_{\parallel} F|_{(x_0, y_0)} & \left| \mathbb{d}_{\perp} F|_{(x_0, y_0)} \right. \end{bmatrix},$$

with $\mathbb{d}_{\perp} F|_{(x_0, y_0)}$ invertible.

Then there exist open sets X_0, Y_0 such that $x_0 \in X_0 \subset \mathbb{R}^N$, $y_0 \in Y_0 \subset \mathbb{R}^M$, $X_0 \times Y_0 \subset U$ and a continuously differentiable map $f : X_0 \rightarrow Y_0$ such that

$$\forall (x, y) \in X_0 \times Y_0, \quad F(x, y) = 0 \Leftrightarrow f(x) = y,$$

that is $F(0)^{-1} \cap (X_0 \times Y_0) = \{(x, f(x)), x \in X_0\}$. Furthermore, as $F(x, f(x)) = 0$ we have:

$$\mathbb{d}_{\parallel} F|_{(x, f(x))} + \mathbb{d}_{\perp} F|_{(x, f(x))} \mathbb{d}f|_x = 0.$$

In words, in the neighborhood of a solution (x_0, y_0) , a system of M independent equations over $N + M$ unknowns is solved by the implicit functions $y = f(x)$ (and of course $y_0 = f(x_0)$).

Proof. We define $\Phi : U \rightarrow \mathbb{R}^{N+M}$, $\Phi(x, y) = (x, F(x, y))$, and Φ is continuously differentiable and its differential (denoting $\mathbb{I}$ the identity matrix):

$$\mathbb{d}\Phi = \begin{pmatrix} \mathbb{I} & 0 \\ \mathbb{d}_{\parallel} F & \mathbb{d}_{\perp} F \end{pmatrix},$$

is invertible at (x_0, y_0) , as $\mathbb{d}_{\perp} F$ is invertible. In fact, in view of Footnote 2, $\mathbb{d}_{\perp} F$ is non degenerate on some neighborhood of (x_0, y_0) . By the inverse function theorem, there exists an open neighborhood W_0 of $\Phi(x_0, y_0) = (x_0, 0)$, an open neighborhood U_0 of (x_0, y_0) and a function $\Psi : W_0 \rightarrow U_0$ which is the inverse of $\Phi|_{U_0}$. Note that $U_0 \supset U_0^{\parallel} \times U_0^{\perp}$ such that

$U_0^\parallel$ is an open neighborhood of x_0 in $\mathbb{R}^N$ and $U_0^\perp$ is an open neighborhood of y_0 in $\mathbb{R}^M$ and similarly for W_0 .⁴ We denote $Y_0 = U_0^\perp$ and $X_0 = U_0^\parallel \cap W_0^\parallel$.

We denote $\Psi = (\Psi_\parallel, \Psi_\perp)$ and we observe that:

$$(x, y) = \Phi(\Psi_\parallel(x, y), \Psi_\perp(x, y)) = \left(\Psi_\parallel(x, y), F(\Psi_\parallel(x, y), \Psi_\perp(x, y)) \right) \Rightarrow x = \Psi_\parallel(x, y), \quad y = F(x, \Psi_\perp(x, y)).$$

It follows that the function $f : \mathbb{R}^M \rightarrow \mathbb{R}^M$, $f(x) = \Psi_\perp(x, 0)$ is such that:

$$\begin{aligned} f(x) = y &\Rightarrow F(x, y) = F(x, f(x)) = F(x, \Psi_\perp(x, 0)) = 0, \\ F(x, y) = 0 &\Rightarrow (x, 0) = \Phi(x, y) \Rightarrow \Psi(x, 0) = (x, y) \Rightarrow \Psi_\perp(x, 0) = y \Rightarrow f(x) = y. \end{aligned}$$

Furthermore, f is continuously partially differentiable, as $\Psi_\perp$ is, hence f is continuously Fréchet differentiable. $\square$

Take for instance the function $F(x, y, z) = x^2 + y^2 + z^2 - 1$ and consider the solution $(x_0, y_0, z_0) = (0, 0, 1)$ of the equation $F(x, y, z) = 0$. We have:

$$\mathrm{d}F|_{(0,0,1)} = \begin{bmatrix} 2x & 2y & 2z \end{bmatrix}_{(0,0,1)} = \begin{bmatrix} 0 & 0 & 2 \end{bmatrix},$$

hence $\mathrm{d}_\perp F|_{(0,0,1)} = 2 \neq 0$ and in a neighborhood of $(x_0, y_0) = (0, 0)$ a solution of $F(x, y, z) = 0$ is the function $z(x, y) = \sqrt{1 - x^2 - y^2}$ which is continuously differentiable and such that:

$$\begin{bmatrix} 2x & 2y \end{bmatrix}_{x,y,z(x,y)} + \begin{bmatrix} 2z \end{bmatrix}_{x,y,z(x,y)} \begin{bmatrix} \frac{-x}{\sqrt{1-x^2-y^2}} & \frac{-y}{\sqrt{1-x^2-y^2}} \end{bmatrix}_{x,y} = 0.$$

The naive reader will think that despite the fact that $\partial_x F(0, 0, 1) = 0$, one can solve for $x = \sqrt{1 - y^2 - z^2}$ for (y, z) in some neighborhood of $(y_0, z_0) = (0, 1)$. However this is wrong as the supposed “solution” $\sqrt{1 - y^2 - z^2}$ is not *defined* for $z > 1$, hence is not defined (much less continuously differentiable) in any neighborhood of $(0, 1)$.

10.7 The case $F : \mathbb{R}^N \rightarrow \mathbb{R}$

We now restrict to real valued functions $F : \mathbb{R}^N \rightarrow \mathbb{R}$. We denote the partial derivatives $DF(x_0; e_i) = \frac{\partial F}{\partial x^i}(x_0) = \partial_i F(x_0)$.

Theorem 10.9 (**Commutation of partial derivatives**). *Assume that $F : U \subset \mathbb{R}^N \rightarrow \mathbb{R}$ is such that $\partial_i F, \partial_j F$ and $\partial_j \partial_i F$ exist and $\partial_j \partial_i F$ is continuous in x_0 . Then $\partial_i \partial_j F$ exists and $\partial_i \partial_j F = \partial_j \partial_i F$.*

Proof. It is enough to prove this for $n = 2$. Let us consider the finite difference quotient mimicking the mixed derivative:

$$\Xi(x^1, x^2) = \frac{F(x^1, x^2) - F(x_0^1, x^2) - F(x^1, x_0^2) + F(x_0^1, x_0^2)}{(x^1 - x_0^1)(x^2 - x_0^2)}.$$

As $\partial_1 F$ exists and is continuous $\lim_{x^1 \rightarrow x_0^1} \Xi(x^1, x^2) = (\partial_1 F(x_0^1, x^2) - \partial_1 F(x_0^1, x_0^2)) / (x^2 - x_0^2)$ and $\lim_{x^2 \rightarrow x_0^2} (\lim_{x^1 \rightarrow x_0^1} \Xi(x^1, x^2)) = \partial_2 \partial_1 F(x_0^1, x_0^2)$ exist by hypothesis.

We assume without loss of generality that $x^1 \geq x_0^1$ and $x^2 \geq x_0^2$. By the mean value theorem applied to $(x^1 - x_0^1)\Xi(x^1, x^2)$, $\exists \xi^1 \in [x_0^1, x^1]$ such that $(x^1 - x_0^1)\Xi(x^1, x^2) = (x^1 - x_0^1) \frac{\partial(x^1 - x_0^1)\Xi}{\partial x^1}[\xi^1, x^2]$, that is:

$$\Xi(x_1, x_2) = \frac{\partial_1 F(\xi^1, x^2) - \partial_1 F(\xi^1, x_0^2)}{(x^2 - x_0^2)},$$

⁴For any norm, any ball $B_r(x)$ in $\mathbb{R}^N$ contains some hypercube $\times_{i=1}^N (x^i - a, x^i + a)$. For instance, with $\|\cdot\|_2$ it is enough to ensure $Na^2 < r^2$. The hypercube is the Cartesian product of hypercubes. Finally, in any dimension, a hypercube contains a ball (with $\|\cdot\|_2$ the ball of radius a).

and applying again the mean value theorem to $(x^2 - x_0^2)\Xi[x_1, x_2]$ with $\Xi[x_1, x_2]$ expressed by the last equation we get that there exist $\xi^2 \in [x_0^2, x^2]$ such that $\Xi(x^1, x^2) = \partial_2 \partial_1 F(\xi^1, \xi^2)$.

Note that $\|\xi - x_0\| = \sqrt{|\xi^1 - x_0^1|^2 + |\xi^2 - x_0^2|^2} \leq \|x - x_0\|$. As $\partial_2 \partial_1 F$ is continuous, $\forall \epsilon, \exists \delta$ such that:

$$\|x - x_0\| < \delta \Rightarrow |\Xi(x_1, x_2) - \partial_2 \partial_1 F(x_0^1, x_0^2)| = \|\partial_2 \partial_1 F(\xi^1, \xi^2) - \partial_2 \partial_1 F(x_0^1, x_0^2)\| \leq \epsilon.$$

Taking $x^2 \rightarrow x_0^2$ in the original expression, we conclude that $\forall \epsilon, \exists \delta$ such that:

$$|x^1 - x_0^1| < \delta \Rightarrow \left| \lim_{x^2 \rightarrow x_0^2} \Xi(x_1, x_2) - \partial_2 \partial_1 F(x_0^1, x_0^2) \right| = \left| \frac{\partial_2 F(x^1, x_0^2) - \partial_2 F(x_0^1, x_0^2)}{x^1 - x_0^1} - \partial_2 \partial_1 F(x_0^1, x_0^2) \right| \leq \epsilon.$$

□

10.7.1 Critical points

Theorem 10.10 (Extremum criterion). *For any Banach space X , if $F : U \subset X \rightarrow \mathbb{R}$ is Fréchet differentiable in U and $x_0 \in U$ is a local extremum (maximum or minimum) then $dF|_{x_0} = 0$.*

Proof. If F Fréchet differentiable at x_0 it is Gateaux differentiable at x_0 and $dF|_{x_0}(v) = DF(x_0; v)$. If x_0 is a local maximum of F , then for any $v \in X$, $F(x_0 + tv)$ has a local maximum at $t = 0$ and by the extremum criterion in one variable $DF(x_0; v) = 0$, $\forall v$. □

A function $F : U \subset \mathbb{R}^N \rightarrow \mathbb{R}$ such that all its partial derivatives up to order k exist and are continuous is denoted $F \in \mathcal{C}^k(U, \mathbb{R})$. Such a function is Fréchet differentiable and we have:

$$dF|_{x_0}(v) = \sum_{i=1}^N v^i dF|_{x_0}(e_i) = \sum_{i=1}^N v^i \frac{\partial F}{\partial x^i}(x_0) = DF(x_0; v).$$

Observe that the $x \mapsto dF|_x$ is a map from X to the Banach space of bounded linear operators $\mathcal{B}_{\text{op.}}(X = \mathbb{R}^n, Y = \mathbb{R})$. If this map is Fréchet differentiable in x , its Fréchet differential $d^2 F|_x$ is a bounded linear map from X to $\mathcal{B}_{\text{op.}}(X, \mathbb{R})$, $v \mapsto d^2 F|_x(v)(\cdot)$, or equivalently a bilinear map from X to $\mathbb{R}$:

$$d^2 F|_{x_0}(v, u) \equiv d(dF|_{x_0}(u))|_{x_0}(v) = \sum_{i,j=1}^N u^i v^j d(dF|_{x_0}(e_i))|_{x_0}(e_j) = \sum_{i,j=1}^N u^i v^j \partial_i \partial_j F(x),$$

where $\partial_i \partial_j F(x)$, the real symmetric matrix of partial derivatives of second order of F , is called the *Hessian* matrix of F .

Theorem 10.11 (Convexity criterion). *Let U a convex open set. A function $F \in \mathcal{C}^2(U, \mathbb{R})$ is convex in U , $F((1-t)x_1 + tx_2) \leq (1-t)F(x_1) + tF(x_2)$ if and only if its Hessian is a positive semi defined matrix for all $x \in U$.*

Proof. We recall that a $N \times N$ matrix is called positive semi defined if $\forall v \in \mathbb{R}^N$, $\sum_{i,j=1}^N H_{ij} v^i v^j \geq 0$. As the Hessian is real and symmetric this is equivalent to having non negative eigenvalues. We note that F is convex if and only if, for any x_1 and x_2 the function $\tilde{F}(t) = F((1-t)x_1 + tx_2)$ is convex in $t \in [0, 1]$, that is $\tilde{F}((1-t)0 + t1) \leq (1-t)\tilde{F}(0) + t\tilde{F}(1)$. By the convexity criterion in one variable, this means that $\frac{d^2}{dt^2} \tilde{F} \geq 0$ for $t \in [0, 1]$ and:

$$\begin{aligned} \frac{d^2}{dt^2} F((1-t)x_1 + tx_2) &= D\left(\frac{d}{dt} F((1-t)x_1 + tx_2)\right)((1-t)x_1 + tx_2; x_2 - x_1) \\ &= D\left(DF((1-t)x_1 + tx_2; x_2 - x_1)\right)((1-t)x_1 + tx_2; x_2 - x_1) = \sum_{i,j=1}^N \left(\partial_i \partial_j F((1-t)x_1 + tx_2)\right)(x_2 - x_1)^i (x_2 - x_1)^j. \end{aligned}$$

□

Definition 10.6 (**Critical point**). Let $F : U \subset \mathbb{R}^N \rightarrow \mathbb{R}$ be a Fréchet differentiable function in U . A point x_0 such that $\mathfrak{d}F|_{x_0} = 0$, that is $\partial_i F(x_0) = 0$ for all i , is called a critical point of F . If $F \in \mathcal{C}^2(U, \mathbb{R})$ then the critical point is a local minimum (resp. maximum) if the Hessian $\partial_i \partial_j F(x_0)$ is positive (resp. negative) defined. If the Hessian is neither but is non degenerate⁵ then the critical point is called a saddle point. Note that as the Hessian is real symmetric, its eigenvectors are mutually orthogonal.

Critical points of functions $F : \mathbb{R}^N \rightarrow \mathbb{R}^M$ are defined as the points in which $\mathfrak{d}F$ is not surjective (the Jacobi matrix is not full rank). There is no notion of local extremum and no equivalent of the Hessian for $M > 1$.

The variation of F around the critical point x_0 along the direction v is:

$$F(x_0 + v) \approx F(x_0) + \frac{d}{dt} F(x_0 + tv)|_{t=0} + \frac{1}{2} \frac{d^2}{dt^2} F(x_0 + tv)|_{t=0} \approx F(x_0) + \frac{1}{2} \sum_{i,j=1}^N H_{ij}(x_0) v^i v^j,$$

and in the eigenbasis e_a of H , we write $v = \sum_a v^a e_a$ and $F(x_0 + v) = F(x_0) + \frac{1}{2} \sum_{a=1}^N \lambda_a (v^a)^2$. The variation of F in the eigendirection e_a is governed by the sign of the corresponding eigenvalue λ_i .

Consider for instance the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, $F(x, y) = \frac{x^2 + y^2}{2}$. We have $\mathfrak{d}F : \mathbb{R}^2 \rightarrow \mathbb{R}$, $\mathfrak{d}F = (\partial_x F, \partial_y F)$ and at a critical point $\mathfrak{d}F = 0 \Rightarrow \partial_x F = \partial_y F = 0$, hence at a critical point of $\mathfrak{d}F$ maps $\mathbb{R}^2$ to 0. The Hessian of F is the identity matrix, hence the critical point is a local minimum. It is convenient to consider the graph of F , that is:

$$\tilde{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad \tilde{F} = (x, y, F(x, y)), \quad \mathfrak{d}\tilde{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \partial_x F & \partial_y F \end{pmatrix}.$$

At the critical point $(x_0, y_0) = (0, 0)$ we have $\mathfrak{d}\tilde{F}|_{(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ and it maps $\mathbb{R}^2$ to the 2 dimensional space tangent at $(x_0, y_0, F(x_0, y_0)) = (0, 0, 0)$.

⁵That is all its eigenvalues are non zero and some are positive and some are negative eigenvalues

Chapter 11

The Riemann integral

11.1 Continuous functions on intervals and primitives

We will now define the *Riemann integral* of a continuous function, an important object in calculus. This requires some prerequisites.

Let $[a, b] \subset \mathbb{R}$ be some closed interval. We denote $\mathcal{C}([a, b], V)$ the spaces of continuous (hence bounded) functions $f : [a, b] \rightarrow V$, with V a Banach space.¹ A partition of the interval $[a, b]$ into sub intervals is:

$$[a, b] = \bigcup_{i=0}^{r-1} \Lambda_i, \quad \Lambda_i = [s_i, s_{i+1}) \in [a, b],$$

where $a \equiv s_0 \leq s_1 \leq \dots \leq s_r \equiv b$. We denote $\Pi = \{\Lambda_i \mid \Lambda_i \cap \Lambda_j = \emptyset, \bigcup_{i=0}^{r-1} \Lambda_i = [a, b]\}$ the set of parts in the partition. The refinement order on the set of partitions is $\Pi' \leq \Pi$ if $\forall \Lambda' \in \Pi', \exists \Lambda \in \Pi$ such that $\Lambda' \subset \Lambda$, and the meet of two partitions (i.e. the coarsest partition finer than both Π' and Π) is denoted $\Pi' \wedge \Pi$ and² it is computed as $\Pi' \wedge \Pi = \bigcup_{\Lambda' \in \Pi', \Lambda \in \Pi} (\Lambda' \cap \Lambda)$.

The mesh (sometimes inaccurately called norm) of Π is the largest length of an interval in Π , $\delta(\Pi) = \max_{i=0, \dots, r-1} |s_{i+1} - s_i|$. A set of *tags* consistent with the partition Π is a set of points constraint to belong to the intervals $\xi_i \in [s_i, s_{i+1})$, but otherwise free $\xi_\Pi = \{\xi_i \in \Lambda_i, i = 0, \dots, r-1\}$. The left sided Riemann sum of the function $f \in \mathcal{C}([a, b], V)$ in the partition Π and the general Riemann sum with arbitrary tags are:

$$S(f, \Pi) = \sum_{i=0}^{r-1} (s_{i+1} - s_i) f(s_i), \quad S(f, \Pi, \xi_\Pi) = \sum_{i=0}^{r-1} (s_{i+1} - s_i) f(\xi_i).$$

As we will see below, the exact position of the tags is not consequential: the notation $S(f, \Pi, \xi_\Pi)$ assumes some tags compatible with Π have been chosen.

Lemma 11.1. *For any continuous function $f \in \mathcal{C}([a, b], V)$ we have:*

- $\forall \epsilon, \exists \delta$ such that $\forall \Pi, \delta(\Pi) < \delta \Rightarrow \|S(f, \Pi, \xi_\Pi) - S(f, \Pi)\| < \epsilon$ for any choice of tags ξ_Π .
- $\forall \epsilon, \exists \delta$ such that $\forall \Pi, \delta(\Pi) < \delta, \tilde{\Pi} \leq \Pi \Rightarrow \|S(f, \tilde{\Pi}) - S(f, \Pi)\| < \epsilon$.
- $\forall \epsilon, \exists \delta$ such that $\forall \Pi_1, \forall \Pi_2, \delta(\Pi_1), \delta(\Pi_2) < \delta \Rightarrow \|S(f, \Pi_1) - S(f, \Pi_2)\| < \epsilon$.

¹The space $(\mathcal{C}([a, b], V), \|\cdot\|_{\infty, [a, b]})$ is itself a Banach space from Corollary 10.2. The same holds for $\mathcal{C}^k([a, b], V)$, the spaces of k -times continuously differentiable functions $f : [a, b] \rightarrow V$ with bounded derivative endowed with the norm $\|f\|_{\mathcal{C}^k, [a, b]} = \sum_{i=0}^k \|f^{(i)}\|_{\infty, [a, b]}$, see Proposition 17.2.

² $\wedge$ is the standard notation both for the meet of partitions and for the exterior product, although the two have nothing to do with each other.

Proof. The function f is continuous on the compact interval $[a, b]$ hence, from the Heine-Cantor theorem, f is uniformly continuous. Therefore, for any ϵ there exists a δ such that:

$$\forall t_1, t_2 \in [a, b], \quad |t_1 - t_2| < \delta \Rightarrow \|f(t_1) - f(t_2)\| < \frac{\epsilon}{b-a}.$$

If Π is a partition of mesh $\delta(\Pi) < \delta$. Then:

$$\|S(f, \Pi, \xi_\Pi) - S(f, \Pi)\| = \left\| \sum_{i=0}^{r-1} [f(\xi_i) - f(s_i)](s_{i+1} - s_i) \right\| \leq \sum_{i=0}^{r-1} \frac{\epsilon}{b-a} (s_{i+1} - s_i) < \epsilon.$$

If $\tilde{\Pi}$ is a refinement of Π , then for any $\tilde{s}_j$ extremity of an interval in $\tilde{\Pi}$, there exists an $i(j)$ such that $s_{i(j)} \leq \tilde{s}_j \leq s_{i(j)+1}$. Then $|\tilde{s}_j - s_{i(j)}| < \delta$ and $s_{i+1} - s_i = \sum_{j, i(j)=i} \tilde{s}_{j+1} - \tilde{s}_j$, therefore:

$$\|S(f, \tilde{\Pi}) - S(f, \Pi)\| = \left\| \sum_{j=0}^{\tilde{r}-1} (\tilde{s}_{j+1} - \tilde{s}_j) f(\tilde{s}_j) - \sum_{i=0}^{r-1} (s_{i+1} - s_i) f(s_i) \right\| = \left\| \sum_{j=0}^{\tilde{r}-1} (\tilde{s}_{j+1} - \tilde{s}_j) [f(\tilde{s}_j) - f(s_{i(j)})] \right\| \leq \frac{\epsilon}{b-a} \sum_{j=0}^{\tilde{r}-1} (\tilde{s}_{j+1} - \tilde{s}_j) = \epsilon.$$

The last assertion follows by choosing δ such that $\|S(f, \Pi_1 \wedge \Pi_2) - S(f, \Pi_1)\|, \|S(f, \Pi_1 \wedge \Pi_2) - S(f, \Pi_2)\| < \epsilon/2$. $\square$

Theorem 11.1 (**Existence of the Riemann integral**). *Let $I \subset \mathbb{R}$ be an interval, V a Banach space and $f : I \rightarrow V$ be a continuous function. Then for any $a, b \in I$ the sequence of left sided Riemann sums³:*

$$S_n(f; a, b) \equiv \sum_{k=0}^{n-1} \left(\frac{b-a}{n} \right) f\left(a + k \frac{b-a}{n}\right),$$

converges. We call this limit the Riemann integral from a to b of f and we denote it:

$$\int_a^b dx f(x) = \lim_{n \rightarrow \infty} S_n(f; a, b).$$

Proof. The limit is 0 if $a = b$. For $a \neq b$ we note that:

$$S_n(f; a, b) \equiv \sum_{k=0}^{n-1} \left(\frac{b-a}{n} \right) f\left(a + k \frac{b-a}{n}\right) = \sum_{p=1}^n \left(-\frac{a-b}{n} \right) f\left(\frac{p}{n}a + \frac{n-p}{n}b\right) = -S_n(f; b, a) + \frac{b-a}{n} (f(a) - f(b)),$$

hence we can assume that $a < b$. Using the last item of Lemma 11.1, we conclude that the sequence $S_n(f; a, b)$ is a Cauchy sequence hence converges in V . $\square$

We have decided to define the Riemann integral as the limit of the left sided Riemann sum in a certain sequence of partitions of increasingly smaller norm. In fact the precise sequence of partitions and the positions of the tags are irrelevant: from lemma 11.1 it follows that if f is continuous on $[a, b]$, then $\forall \epsilon, \exists \delta$ such that $\forall \Pi$ and $\forall \xi_\Pi, \delta(\Pi) < \delta \Rightarrow \left\| \int_a^b dx f(x) - S(f, \Pi, \xi_\Pi) \right\| < \epsilon$. The Riemann integral has a numbers of useful properties.

Lemma 11.2 (**Properties of the Riemann integral**). *Let I and interval, $a, b \in I$ with $a < b$, V a Banach space and $f \in \mathcal{C}(I, V)$ a continuous function. Then:*

- Orientation flip: $\int_a^b dx f(x) = -\int_b^a dx f(x)$.
- Linearity: $\int_a^b dx (\lambda_1 f_1 + \lambda_2 f_2) = \lambda_1 \int_a^b dx f_1(x) + \lambda_2 \int_a^b dx f_2(x); \int_a^b dx v = v(b-a)$.
- Standard bound: $\left\| \int_a^b dx f(x) \right\| \leq \int_a^b dx \|f(x)\| \leq \|f\|_{\infty, [a, b]} (b-a)$.

³Corresponding to equidistant partitions of mesh $1/n$.

- Additivity of intervals: $\int_a^b dx f(x) + \int_b^c dx f(x) = \int_a^c dx f(x)$.

Proof. The first assertions are direct consequences of the definition. For the last assertion, assume $a < b < c$. For any ϵ , let n_1 be such that $\left\| \int_a^b dx f(x) - S_n(f; a, b) \right\|, \left\| \int_b^c dx f(x) - S_n(f; b, c) \right\|, \left\| \int_a^c dx f(x) - S_n(f; a, c) \right\| < \epsilon/4$ for $n > n_1$. Also, $\forall \epsilon, \exists \delta$ such that $\delta(\Pi_1), \delta(\Pi_2) < \delta \Rightarrow \|S(f, \Pi_1) - S(f, \Pi_2)\| < \epsilon/4$, hence denoting $n_2 = \max\{\frac{|a-b|}{\delta}, \frac{|b-c|}{\delta}, \frac{|c-a|}{\delta}\}$ we have:

$$\begin{aligned} & \left\| \int_a^b dx f(x) + \int_b^c dx f(x) - \int_a^c dx f(x) \right\| \\ & \leq \left\| \int_a^b dx f(x) - S_n(f; a, b) \right\| + \left\| \int_a^c dx f(x) - S_n(f; a, c) \right\| + \left\| \int_b^c dx f(x) - S_n(f; b, c) \right\| \\ & \quad + \|S_n(f; a, b) + S_n(f; b, c) - S_n(f; a, c)\| < \epsilon, \end{aligned}$$

for $n > \max\{n_1, n_2\}$. □

Definition 11.1 (Primitives). Let $I^\circ = (a, b) \subset \mathbb{R}$ an open interval and V a Banach space. A function $F : I^\circ \rightarrow V$ is called a primitive of $f : I^\circ \rightarrow V$ if F is differentiable on I° and $F'(x) = f(x)$, $\forall x \in I^\circ$. If F is primitive of f then, for any constant v_0 , $F + v_0$ is a primitive of f .

Taking the primitive is a linear operation: if F_1 is a primitive of f_1 and F_2 a primitive of F_2 then $\lambda_1 F_1 + \lambda_2 F_2$ is a primitive of $\lambda_1 f_1 + \lambda_2 f_2$.

Theorem 11.2 (Fundamental theorem of calculus). Consider $I \subset \mathbb{R}$ some interval, V a Banach space and fix $a \in I$. For any continuous function $f \in \mathcal{C}(I, V)$ the Riemann integral:

$$F(x) = \int_a^x ds f(s),$$

is a primitive of f on I° . For any F a primitive of the continuous function f on $I^\circ \supset (a, b)$ we have:

$$\int_a^b ds f(s) = F(b) - F(a).$$

Proof. For any ϵ , let δ such that $|s - x_0| < \delta \Rightarrow \|f(s) - f(x_0)\| < \epsilon$. Then for $|x - x_0| < \delta$ we have:

$$\|F(x) - F(x_0) - (x - x_0)f(x_0)\| = \left\| \int_{x_0}^x ds (f(s) - f(x_0)) \right\| < \epsilon|x - x_0|.$$

The second assertion follows by noting that for any primitive F of f and for any partition of $[a, b]$, applying the mean value theorem to each interval and obtaining a tag per interval:

$$F(b) - F(a) = \sum_{i=0}^{r-1} [F(s_{i+1}) - F(s_i)] = \sum_{i=0}^{r-1} f(\xi_i)(s_{i+1} - s_i), \quad \xi_i \in (s_i, s_{i+1}),$$

and we conclude using Lemma 11.1. □

If $F \in C^{k+1}([0, 1], V)$, applying this theorem k times we obtain the Taylor formula with integral rest:

$$F(1) - F(0) = \sum_{q=0}^k \frac{1}{q!} F^{(q)}(0) + \frac{1}{k!} \int_0^1 du (1-u)^k F^{(k+1)}(u).$$

Lemma 11.3 (**The mean value theorem for integrals**). Let $a, b \in \mathbb{R}$, $a \leq b$ and $f : [a, b] \rightarrow \mathbb{R}$ a continuous function. Then there exist $\xi \in [a, b]$ such that:

$$\int_a^b dx f(x) = f(\xi) (b - a) ,$$

that is the primitive F of f is Lipschitz continuous on any compact $K \subset I$ with Lipschitz constant $\|f\|_{\infty, K}$.

Proof. This follows from the usual mean value theorem applied to $F(x) = \int_a^x ds f(s)$. □

11.1.1 Indefinite integrals, integration by parts and change of variables

In some cases one needs to evaluate indefinite Riemann integrals: for $I = (\alpha, \beta)$, $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$, we chose $x_0 \in (\alpha, \beta)$ and we define:

$$\int_{\alpha}^{\beta} ds f(s) = \lim_{a \rightarrow \alpha} \int_a^{x_0} ds f(s) + \lim_{b \rightarrow \beta} \int_{x_0}^b ds f(s) ,$$

if the limits exist. Note that, if it exists, $\int_{\alpha}^{\beta} ds f(s)$ is independent on the intermediate point x_0 .

The Leibnitz rule and the chain rule for differentials imply:

Change of variables. If $F \in \mathcal{C}^1(I, V)$ is a primitive of $f \in \mathcal{C}(I, V)$ and $\tau : J \rightarrow I$ is continuously differentiable then $F \circ \tau$ is a primitive of $(f \circ \tau) \tau'$:

$$\int_a^x ds f(s) = \int_{\alpha, \tau(\alpha)=a}^{\beta, \tau(\beta)=x} du f(\tau(u)) \tau'(u) .$$

Note that τ does not need to be a bijection (this is a non trivial point).

Integration by parts. A Banach space V is a Banach algebra if it is endowed with an associative multiplication $\cdot$ such that the unit element for the multiplication I has norm $\|I\| = 1$ and the norm is submultiplicative $\|A \cdot B\| \leq \|A\| \|B\|$. If V is a Banach algebra, $F \in \mathcal{C}^1(I, V)$ a primitive of $f \in \mathcal{C}(I, V)$ and G some function in $\mathcal{C}^1(I, V)$ then:

$$F(x) \cdot G(x) - \int_a^x ds F(s) \cdot G'(s) \quad \text{is a primitive of} \quad (f \cdot G)(x) = f(x) \cdot G(x) ,$$

or equivalently:

$$\int_a^x ds F'(s) G(s) = F(x) G(x) - F(a) G(a) - \int_a^x ds F(s) G'(s) .$$

11.1.2 Integrals depending on a parameter

The Cartesian product of topological spaces $\times_{i \in I} X_i$ can be endowed with either the *box topology* ($\times_{i \in I} U_i$, with U_i open in X_i) or with the *product topology* ($\times_{i \in I} U_i$, with U_i open in X_i and $U_j \neq X_j$ only for $j \in J \subset I$ with J a finite set). If I is a finite set then the two topologies coincide and a basis in the product topology is $\times_{i \in I} B_i$ with B_i running over a basis in the topology in X_i .

For a finite number of metric spaces $(X_1, d_1), \dots, (X_n, d_n)$ the product topology is generated by the metric $d(x, y) = \max_{i \in \{1, \dots, n\}} \{d_i(x_i, y_i)\}$.⁴ Indeed, in this metric an open ball is:

$$B_r(x_1, \dots, x_n) = \{(y_1, \dots, y_n) \mid d_i(x_i, y_i) < r, i = 1, \dots, n\} = \bigcap_{i=1}^n \{y_i \mid d(x_i, y_i) < r\} = \bigcap_{i=1}^n B_r^{(d_i)}(x_i) ,$$

where we denoted $B_r^{(d_i)}(x_i)$ the ball of radius r in X_i centered at x_i with respect to the distance function d_i . An open ball centered at $x = (x_1, \dots, x_n) \in U$ in the product topology is $\times_{i=1}^n B_{r_i}^{(d_i)}(x_i)$ and contains $B_r(x)$ with $r < \min_{i \in \{1, \dots, n\}} \{r_i\}$. Conversely $B_r(x)$ contains any $\times_{i=1}^n B_{r_i}(x_i)$ with $r_i < r$, $\forall i$.

⁴If n is finite this distance is equivalent to any other version $d(x, y) = (\sum_{i=1}^n [d_i(x_i, y_i)]^p)^{1/p}$.

Theorem 11.3 (**Differentiating under the integral sign**). Let $U \subset X$ some open set in a Banach space and $F : [a, b] \times U \rightarrow Y$ a continuous function with Y some Banach space. Then the function:

$$G : U \rightarrow Y, \quad G(x) = \int_a^b ds F(s, x),$$

is continuous. If furthermore Y is finite dimensional, the map $x \mapsto F(t, x)$ is Gateaux differentiable in the direction $v \in X$ and the map $(t, x) \mapsto DF(t, x; v)$ is continuous then G is Gateaux differentiable in the direction v and:

$$DG(x; v) = \int_a^b ds DF(s, x; v).$$

Similar statements hold for Fréchet differentiability.

Proof. As $F(t, x)$ is jointly continuous in the product topology in both arguments, for fixed $x \in U$, $t \mapsto F(t, x)$ is continuous and the integral exists. Chose some $x_0 \in U$, then $\forall \epsilon > 0$, as $(t, x) \rightarrow F(t, x)$ is continuous, $\forall t \in [a, b]$, $\exists \delta(t)$ such that:

$$|s - t| < \delta(t), \|x - x_0\| < \delta(t) \Rightarrow \|F(s, x) - F(t, x_0)\| < \frac{\epsilon}{2(b-a)}.$$

But then $\{B_{\delta(t)}(t) \mid t \in [a, b]\}$ is an open cover of $[a, b]$ and, as this is a compact set by the Heine-Borel theorem, there exists a finite subcover $\{B_{\delta(t_i)}(t_i) \mid i \in I\}$. We denote $\delta = \min_{i \in I} \{\delta(t_i)\}$.

We now take $\|x - x_0\| < \delta < \delta(t_i)$ and let $t \in [a, b]$ be arbitrary. Then $t \in B_{\delta(t_i)}(t_i)$ for some i therefore $\|t - t_i\| < \delta(t_i)$ and we have:

$$\|F(t, x_0) - F(t_i, x_0)\| < \frac{\epsilon}{2(b-a)}, \quad \|F(t, x) - F(t_i, x_0)\| < \frac{\epsilon}{2(b-a)} \quad \Rightarrow \quad \|F(t, x) - F(t, x_0)\| < \frac{\epsilon}{(b-a)},$$

hence for $\|x - x_0\| < \delta$ we obtain that:

$$\|G(x) - G(x_0)\| = \left\| \int_a^b ds (F(s, x) - F(s, x_0)) \right\| \leq (b-a) \|F(\cdot, x) - F(\cdot, x_0)\|_{\infty, [a, b]} < \epsilon,$$

hence G is continuous in x_0 .

The mean value theorem applied to the component i of F states that there exists $\xi_i(t) < \tau$ depending on t such that:

$$F^i(t, x_0 + \tau v) - F^i(t, x_0) - \tau DF^i(t, x_0; v) = \tau (DF^i(t, x_0 + \xi_i(t)v; v) - DF^i(t, x_0; v)),$$

As $\|x_0 + \xi(t)v - x_0\| = |\xi(t)| \|v\| < \tau$ (we take $\|v\| = 1$), from the continuity of the derivative we conclude that $\forall \epsilon$, there exists some $\delta(t)$ such that:

$$\|x_0 + \xi(t)v - x_0\| < |\tau| < \delta(t) \Rightarrow |F^i(t, x_0 + \tau v) - F^i(t, x_0) - \tau DF^i(t, x_0; v)| \leq |\tau| \frac{\epsilon}{(b-a)}.$$

Using again the same trick of extracting a finite cover of $[a, b]$ from $\{B_{\delta(t)} \mid t \in [a, b]\}$ and collecting all the components we conclude that (using the product topology on the finite dimensional space Y):

$$\forall \epsilon, \exists \delta \text{ such that } |\tau| < \delta, \forall t \in [a, b] \Rightarrow \|F(t, x_0 + \tau v) - F(t, x_0) - \tau DF(t, x_0; v)\| \leq |\tau| \frac{\epsilon}{b-a},$$

hence:

$$\|G(x_0 + \tau v) - G(x_0) - \int_a^b ds DF(s, x_0; v)\| \leq \int_a^b ds \|F(t, x_0 + \tau v) - F(t, x_0) - \tau DF(t, x_0; v)\| \leq \epsilon |\tau|,$$

and the theorem follows. $\square$

Combining this with the fundamental theorem of calculus, it follows that under the right regularity conditions:

$$\frac{\partial}{\partial x^i} \int_{a(x)}^{b(x)} dt f(t, x) = f(b(x), x) \frac{\partial b}{\partial x^i}(x) - f(a(x), x) \frac{\partial a}{\partial x^i}(x) + \int_{a(x)}^{b(x)} dt \frac{\partial f}{\partial x^i}(t, x) .$$

Exercise 11.1 (Primitives of simple functions of one real variable). *Find the primitives of:*

$$x^n, 1/x, e^x, \ln(x), \sin(x), \cos(x), \frac{1}{1+x^2}, \frac{1}{1-x^2}, \frac{1}{\sqrt{1-x^2}}, \frac{1}{\sqrt{1+x^2}} .$$

Exercise 11.2. *Assuming $\int_{-\infty}^{\infty} dx e^{-\frac{x^2}{2}} = \sqrt{2\pi}$, compute the moments of the normalized Gaußian integral $\int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} e^{-\frac{x^2}{2\alpha}} x^n$.*

11.1.3 Limitations of the Riemann integral

While perfectly suited for continuous functions on compact intervals, the Riemann integral does not easily generalize beyond this class. One says that a function is Riemann integrable if the partial Riemann sums converge (when sending the mesh of the partition to zero) to some limit which is independent of the tags. One can formulate necessary and sufficient conditions for a function to be Riemann integrable, but the latter are not particularly simple. Furthermore, in general, functions obtained as limits of sequences of Riemann integrable functions are not Riemann integrable, and so on.

For instance the characteristic function of the rational numbers: $\mathbf{1}_{\mathbb{Q}}(x) = 1$ for $x \in \mathbb{Q}$ and $\mathbf{1}_{\mathbb{Q}}(x) = 0$ otherwise is not integrable on any interval: the Riemann sums with only rational tags yield the length of the interval, while the ones with irrational tags yield zero.

These flaws are resolved when passing to the Lebesgue integral, which we will do later. In preparation for this generalization, we discuss now the Riemann integral in several variables from a slightly more general point of view.

11.2 The Riemann integral in $\mathbb{R}^n$

Below we will deal extensively with boundaries of subsets of $\mathbb{R}^n$. As we will see in the next chapter, if one embeds a n -dimensional manifold with boundary in $\mathbb{R}^n$, the manifold boundary of M and its topological boundary seen as a subset of $\mathbb{R}^n$ coincide. Below by boundary we mean manifold boundary, but we stress that this is also the topological boundary when embedding in the space with appropriate dimension.

11.2.1 The Jordan measure

We will now discuss the concept of Jordan measure. We will see later that this notion has limitations, in particular it is not a “measure” in the sense of measure theory⁵ and is superseded by the Lebesgue measure, which is. However, for an introduction to the theory of integration, the Jordan measure is very instructive.

We denote e_i the standard basis in $\mathbb{R}^n$, λ^i its dual basis and $x^i = \lambda^i(\sum_j x^j e_j)$.

Definition 11.2 (Boxes and elementary sets). *A box in $\mathbb{R}^n$ is the Cartesian product of n intervals:*

$$B = \bigtimes_{\nu=1}^n [a^\nu, b^\nu] = [a^1, b^1] \times \cdots \times [a^n, b^n] = \{x \in \mathbb{R}^n, a^i \leq x^i \leq b^i\} .$$

⁵Some authors prefer to talk about the Jordan content rather than the Jordan measure.

The measure of a box is: $\mu(B) = \prod_{i=1}^n |a^i - b^i| \geq 0$. The boundary (both manifold and topological as a subset of $\mathbb{R}^n$) of a box is:

$$\partial B = \bigcup_{i=1}^n \left([a^1, b^1] \times \cdots \times (\{a^i\} \cup \{b^i\}) \times \cdots \times [a^n, b^n] \right).$$

An elementary set is a finite union of boxes $E = \bigcup_{i=1}^k B_i$. Any elementary set can be written as the finite union of almost disjoint (i.e. $B_i^\circ \cap B_j^\circ = \emptyset$) boxes (as the union of two boxes can be written as a union of a finite number of almost disjoint boxes). The measure of an elementary set is:

$$\mu(E) = \sum_{i=1}^k \mu(B_i), \quad E = \bigcup_i B_i, \quad B_i^\circ \cap B_j^\circ = \emptyset,$$

and is independent on the decomposition of E into almost disjoint boxes.

Proposition 11.1. Under a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $x \rightarrow L(x)$ hence with differential $dL|_x = L$, the box B is pushed forward by L to a parallelepiped $L(B)$ and:

$$\mu(L(B)) = |\det L| \mu(B),$$

in particular the measure of a box is invariant under rotations and translations.

Proof. The measure of a box is the volume of the cube with sides $v_i = |a^i - b^i|e_i$ (no sum over i). Under a linear map $x \rightarrow L(x)$ the box B is pushed forward by L to the parallelepiped $L(B)$ with sides $f_i = L(v_i)$ and:⁶

$$\mu(L(B)) = |\det(f_1, f_2, f_3 \dots)| = |\det[L(v_1, v_2, v_3 \dots)]| = |\det L| |\det(v_1, v_2, v_3 \dots)|.$$

□

The measure of a box can be “measured” by looking at the intersection of the box with an increasingly refined grid.

Proposition 11.2. We denote $|M|$ is the cardinal of the set M . For any box we have:

$$\mu(B) = \lim_{k \rightarrow \infty} \frac{1}{k^n} \left| B \cap \left(\frac{1}{k} \mathbb{Z} \right)^n \right|, \quad \mu(\partial B) = 0,$$

where $(\frac{1}{k} \mathbb{Z})^n = \{(\frac{p^1}{k}, \dots, \frac{p^n}{k}) \mid p^i \in \mathbb{Z}\}$ and similarly for elementary sets.

Proof. Denoting $[\cdot]$ the integer part of a real number. The number of rational numbers of the form $\frac{p}{k}$ between a and b is comprised between $[k(b-a)]$ and $[k(b-a)] + 1$ hence:

$$\prod_{\mu=1}^n [k(b^\mu - a^\mu)] \leq \left| B \cap \left(\frac{1}{k} \mathbb{Z} \right)^n \right| \leq \prod_{\mu=1}^n ([k(b^\mu - a^\mu)] + 1).$$

Concerning the boundary, we observe that it splits into a union of boxes with one side of zero length, hence it is zero. Taking any decomposition of an elementary set into finitely many almost disjoint boxes one finds $\mu(E) = \lim_{k \rightarrow \infty} \frac{1}{k^n} |E \cap (\frac{1}{k} \mathbb{Z})^n|$. □

Definition 11.3 (The Jordan measure). For any subset $X \in \mathbb{R}^n$ we define:

⁶In order to compute the volume of the parallelepiped we need first to transform f_i into a set of orthogonal vectors by the Gram–Schmidt process $f_1 = f_1$, $f_2 = c_{12}f_1 + f_2^\perp$, $f_3 = c_{13}f_1 + c_{23}f_2 + f_3^\perp$ and so on, where $f_2^\perp$ is orthogonal to f_1 , $f_3^\perp$ is orthogonal to $\text{Span}\{f_1, f_2\}$ and so on. Then, arranging the vectors as column vectors:

$$\mu(L(B)) = |\det(f_1, f_2^\perp, f_3^\perp \dots)| = |\det(f_1, f_2, f_3 \dots)|$$

- the inner Jordan measure of X is $\mu_*(X) = \sup \{ \mu(E) \mid E \subseteq X, E \text{ elementary} \}$.
- the outer Jordan measure of X is $\mu^*(X) = \inf \{ \mu(E) \mid X \subseteq E, E \text{ elementary} \}$.

Note that we trivially have:

- $\forall X, \mu_*(X) \leq \mu^*(X)$ and if $X_1 \subset X_2$, $\mu_*(X_1) \leq \mu_*(X_2)$ and $\mu^*(X_1) \leq \mu^*(X_2)$.
- $\forall X_1, X_2, \mu^*(X_1 \cup X_2) \leq \mu^*(X_1) + \mu^*(X_2)$ and if $X_1 \cap X_2 = \emptyset$, $\mu_*(X_1 \cup X_2) \geq \mu_*(X_1) + \mu_*(X_2)$.

The inner and outer Jordan measure exist (but might be infinite) for any set. However, they do not necessarily agree. We define:

- the set X is called Jordan measurable if $\mu_*(X) = \mu^*(X) < \infty$. In this case the common value of the inner and outer Jordan measures is called the Jordan measure $\mu(X)$ of X . A set is called null if $\mu^*(X) = 0$, and a null set is measurable.

A useful way to compute the inner and outer Jordan measures of a set is to intersect it with an increasingly refined grid of boxes associated to a refining partition of $\mathbb{R}^n$: the subset $(\frac{1}{2^k}\mathbb{Z})^n$ of the grids $(\frac{1}{k}\mathbb{Z})^n$ which are constraint to be refinements of each other.

Proposition 11.3. Consider the set of boxes:

$$\mathcal{B}_k^n = \left\{ \left[\frac{a^1}{2^k}, \frac{a^1+1}{2^k} \right] \times \cdots \times \left[\frac{a^n}{2^k}, \frac{a^n+1}{2^k} \right] \mid a^1, \dots, a^n \in \mathbb{Z} \right\},$$

which partitions $\mathbb{R}^n$ into almost disjoint boxes of side length 2^{-k} (hence of measure 2^{-nk}). Any $q \in \mathcal{B}_k^n$ is the union of 2^n boxes in $\mathcal{B}_{k+1}^n$. For any bounded X we have:

$$\mu_*(X) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{q \in \mathcal{B}_k^n, q \subset X} q \right), \quad \mu^*(X) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{q \in \mathcal{B}_k^n, q \cap X \neq \emptyset} q \right).$$

Proof. Let us denote $q_k(X) = \bigcup_{q \in \mathcal{B}_k^n, q \cap X \neq \emptyset} q$. For any bounded X , this is a finite union of almost disjoint boxes, hence an elementary set; consequently $\mu^*(X) \leq \mu(q_k(X))$. Any $q \in \mathcal{B}_k^n$ is the union of 2^n boxes in $\mathcal{B}_{k+1}^n$, and if $q \cap X \neq \emptyset$ each of these sub boxes can intersect or not X . It follows that $\mu(q_k(X))$ is monotonically decreasing with k , hence convergent and $\mu^*(X) \leq \lim_{k \rightarrow \infty} \mu(q_k(X))$.

On the other hand, any elementary set E is the almost disjoint union of a finite number of boxes and we have from Proposition 11.2:

$$\mu(E) = \lim_{k \rightarrow \infty} \frac{1}{2^{nk}} |E \cap \frac{1}{2^k} \mathbb{Z}^n| = \lim_{k \rightarrow \infty} \mu(q_k(E)).$$

Now, as $\mu^*(X)$ is an infimum, $\forall N, \exists E_N$ an elementary set such that $E_N \supset X$ and $\mu(E_N) - \mu^*(X) < \frac{1}{2^N}$. We choose k_N large enough such that $\mu(q_{k_N}(E_N)) - \mu(E_N) < \frac{1}{2^N}$ then:

$$q_{k_N}(E_N) \supset E_N \supset X, \quad \mu(q_{k_N}(X)) - \mu^*(X) < \mu(q_{k_N}(E_N)) - \mu^*(X) = \mu(q_{k_N}(E_N)) - \mu(E_N) + \mu(E_N) - \mu^*(X) < \frac{1}{2^N},$$

hence the subsequence $\mu(q_{k_N}(X))$ converges to $\mu^*(X)$ and we conclude. $\mu_*(X)$ is dealt with similarly. $\square$

There exist many non Jordan measurable sets, for instance $\mu_*(\mathbb{Q}^n \cap [0, 1]^n) = 0$ and $\mu^*(\mathbb{Q}^n \cap [0, 1]^n) = 1$. In particular this shows that the outer measure of a countable union of null sets (each individual rational number) can be non zero: this is very unpleasant!

Theorem 11.4 (**Jordan measurability is equivalent to null boundary**). Let X be a bounded set $X \subset \mathbb{R}^n$. Then $\mu^*(\partial X) = \mu^*(X) - \mu_*(X)$, therefore:

$$\mu^*(\partial X) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset; q \cap X^c \neq \emptyset} q \right),$$

and the set X is Jordan measurable if and only if its boundary is a null set.

Proof. Consider . We assume X has no isolated points (isolated points have measure 0). As $X^\circ \subset X \subset \bar{X}$ we have $\mu^*(X^\circ) \leq \mu^*(X) \leq \mu^*(\bar{X})$ and $\mu_*(X^\circ) \leq \mu_*(X) \leq \mu_*(\bar{X})$.

As elementary sets are closed, if $X \in E$ then $\bar{X} \in E$ hence $\mu^*(\bar{X}) = \mu^*(X)$. Also, $\mu_*(X^\circ) = \mu_*(X)$. This is trivial if $\mu_*(X) = 0$. If $\mu_*(X) > 0$ then $E \subset X \Rightarrow E^\circ \subset X^\circ$ as any open set $U \subset X$ consists of interior points⁷ of X . By definition of the supremum, $\forall \epsilon, \exists E$ elementary set such that $\mu(E) > \mu_*(X) - \epsilon/2$. Denoting such a set:

$$E = \bigcup_{i=1}^p B_i, \quad B_i^\circ \cap B_j^\circ = \emptyset, \quad B_i = \bigtimes_{\mu=1}^n [a_i^\mu, b_i^\mu].$$

We define $E^{-\epsilon/2} = \bigcup_i B_i^{-\epsilon/2}$, where $B_i^{-\epsilon/2} = \bigtimes_{\mu=1}^n [a_i^\mu + \frac{\epsilon}{2}, b_i^\mu - \frac{\epsilon}{2}]$. Clearly $E^{-\epsilon/2} \subset E^\circ \subset X^\circ$ hence $\mu_*(X^\circ) > \mu(E^{-\epsilon/2})$. Chossing α small enough⁸ we can alwas render $\mu(E^{-\epsilon/2}) > \mu(E) - \epsilon/2$ hence:

$$\forall \epsilon, \mu_*(X^\circ) > \mu_*(X) - \epsilon \Rightarrow \mu_*(X^\circ) = \mu_*(X).$$

Consider a subset $Y \subset X$. The boxes in the grid $\mathcal{B}_k^n$ are partitioned as:

$$q \in \mathcal{B}_k^n, q \cap X \neq \emptyset : \begin{cases} \text{either} & q \subset Y \subset X \\ \text{or} & q \cap (X \setminus Y) \neq \emptyset \end{cases} \Rightarrow \mu_*(Y) + \mu^*(X \setminus Y) = \mu^*(X),$$

where in the last step we used Prop. 11.3. As $\partial X = \bar{X} \setminus X^\circ$ we have $\mu^*(\partial X) = \mu^*(\bar{X}) - \mu_*(X^\circ) = \mu^*(X) - \mu_*(X)$ and the theorem follows. $\square$

Being Jordan measurable is a restrictive condition: while many “nice” sets are Jordan measurable, many sets one would like to integrate over are not. This problem is solved by considering a generalization of the outer measure by allowing a countable box cover of X . This apparently small modification dramatically changes the properties of the outer measure.

Theorem 11.5 (**The graph of a continuous function on a compact domain is a Jordan null set**). Let $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}^r$ be a continuous function and A a compact set. Then the set $S = \{(x, f(x)) \mid x \in A\} \subset \mathbb{R}^{n+r}$, that is the graph of f , is a null set.

Proof. Let $\omega(\delta) = \sup\{\|f(x) - f(y)\| \mid x, y \in A, \|x - y\| < \delta\}$. As f is continuous on the compact set A , by the Heine Cantor theorem f is uniformly continuous, hence $\lim_{\delta \rightarrow 0} \omega(\delta) = 0$

Consider the grid of boxes $\mathcal{B}_k^{n+r}$ in $\mathbb{R}^{n+r}$. This is the Cartesian product of boxes $\mathcal{B}_k^n$ in $\mathbb{R}^n$ and $\mathcal{B}_k^r$ in $\mathbb{R}^r$: any box $q \in \mathcal{B}_k^{n+r}$ is the Cartesian product $q_\parallel \times q_\perp$ with $q_\parallel \in \mathcal{B}_k^n \subset \mathbb{R}^n$ and $q_\perp \in \mathcal{B}_k^r \subset \mathbb{R}^r$. Then:

$$\mu^*(S) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{q \in \mathcal{B}_k^{n+r}, q \cap S \neq \emptyset} q \right), \quad \mu \left(\bigcup_{q \in \mathcal{B}_k^{n+r}, q \cap S \neq \emptyset} q \right) = \sum_{q_\parallel \in \mathcal{B}_k^n, q_\parallel \cap A \neq \emptyset} \sum_{q_\perp \in \mathcal{B}_k^r, q_\parallel \times q_\perp \cap S \neq \emptyset} \mu(q_\parallel) \mu(q_\perp),$$

For any $x, y \in q_\parallel$, $\|x - y\| < 2^{-k}$ hence at $q_\parallel$ fixed $\sum_{q_\perp \in \mathcal{B}_k^r, q_\parallel \times q_\perp \cap S \neq \emptyset} \mu(q_\perp) \leq [\omega(2^{-k})]^r$ and $\mu^*(S) \leq \mu^*(A)[\omega(2^{-k})]^r \xrightarrow{k \rightarrow \infty} 0$. $\square$

This theorem is very useful: if a bounded submanifold of $\mathbb{R}^n$ is defined by a set of continuous constraints, then it is Jordan measurable. This is the case in many practical applications.

⁷ $\forall x \in U$ there exists a ball centred at x entirely comprised in U and consequently in X

⁸ We have $\mu(B_i) - \mu(B_i^{-\epsilon/2}) \leq \alpha \epsilon \sum_{\mu} \prod_{\nu \neq \mu} |a_i^\nu - b_i^\nu| = \alpha \epsilon \sum_{\mu} \frac{\mu(B_i)}{|b_i^\mu - a_i^\mu|}$

Exercise 11.3. *Prove that the set difference of two elementary sets is an elementary set.*

Exercise 11.4. *Prove that if X_1 and X_2 are Jordan measurable sets in $\mathbb{R}^n$ then:*

- $X_1 \cup X_2$ and $X_1 \cap X_2$ are Jordan measurable.
- if X_1 and X_2 are almost disjoint then $\mu(X_1 \cup X_2) = \mu(X_1) + \mu(X_2)$.
- if $X_1 \subset X_2$ then $X_2 \setminus X_1$ is Jordan measurable and $\mu(X_2 \setminus X_1) = \mu(X_2) - \mu(X_1)$.
- $\bar{X}_1$ is Jordan measurable and $\mu(\bar{X}_1) = \mu(X_1)$.

11.2.2 The multidimensional Riemann integral

Let X be a bounded Jordan measurable set in $\mathbb{R}^n$ and some function $f : \bar{X} \rightarrow V$, with V a Banach space. We intersect X with the increasingly refined grids $\mathcal{B}_k^n$, chose one tag per box $\xi = \{\xi_q\}_{q \in \mathcal{B}_k^n}$ and build the outer and inner Riemann sums:

$$S_k^*(f, \xi) = \sum_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset} \mu(q) f(\xi_q), \quad S_k^*(f, \xi) = \sum_{q \in \mathcal{B}_k^n; q \subset X} \mu(q) f(\xi_q),$$

where all the boxes have equal measure $\mu(q) = 2^{-nk}$.

Definition 11.4 (**Riemann integral**). *Consider a function $f : \bar{X} \rightarrow V$ with $X \subset \mathbb{R}^n$ a bounded Jordan measurable set and V a Banach space. The function f is said to be Riemann integrable on X if:*

$$\lim_{k \rightarrow \infty} S_k^*(f, \xi) = \lim_{k \rightarrow \infty} S_k^*(f, \xi) = I,$$

irrespective of the positions of the tags ξ . In this case we denote $I \equiv \int_X d^n x f(x) \equiv \int \dots \int_X dx^1 \dots dx^n f(x)$ and call I the Riemann integral of f on X .

One could rightly wonder under which conditions a function is Riemann integrable on a set. It turns out that continuity is a *sufficient* condition for the Riemann integral to exist. However, we stress that continuity is not *necessary* and some discontinuous functions can be Riemann integrated.

Lemma 11.4. *For a continuous function $f \in \mathcal{C}(\bar{X}, V)$ on some bounded Jordan measurable set $X \subset \mathbb{R}^n$ with values in a Banach space V we have:*

- $\forall \epsilon, \exists k_\epsilon$ such that for all $k > k_\epsilon$, $\|S_k^*(f, \xi) - S_k^*(f, \xi')\| < \epsilon$ for any tags ξ, ξ' .
- $\forall \epsilon, \exists k_\epsilon$ such that for all $k_1, k_2 > k_\epsilon$, $\|S_{k_1}^*(f, \xi^1) - S_{k_2}^*(f, \xi^2)\| < \epsilon$ for any tags ξ^1 and ξ^2 .

Similar statements hold for the inner Riemann sum $S_k^*(f, \xi)$.

Proof. As $\bar{X}$ is bounded and closed, it is compact. The function f is continuous on $\bar{X}$ hence, from the Heine-Cantor theorem, it is uniformly continuous. Then (using the product topology) for any ϵ there exists a k_ϵ such that for $k > k_\epsilon$:

$$\forall x_1, x_2 \in \bar{X}, \max_{\mu} |x_1^\mu - x_2^\mu| < 2^{-k} \Rightarrow \|f(x_1) - f(x_2)\| < \frac{\epsilon}{\mu(X)}.$$

As any two tags in the same box of $\mathcal{B}_k^n$ are closer to each other than 2^{-k} we have:

$$\|S_k^*(f, \xi) - S_k^*(f, \xi')\| = \left\| \sum_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset} \mu(q) [f(\xi_q) - f(\xi'_q)] \right\| \leq \sum_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset} \mu(q) \|f(\xi_q) - f(\xi'_q)\| \leq \frac{\epsilon}{\mu(X)} \mu \left(\bigcup_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset} q \right),$$

and we conclude by recalling that $\lim_{k \rightarrow \infty} \mu\left(\bigcup_{q \in \mathcal{B}_k^n, q \cap X \neq \emptyset} q\right) = \mu(X)$.

For the second part, we assume $k_2 > k_1$ and write:

$$\|S_{k_1}^*(f, \xi^1) - S_{k_2}^*(f, \xi^2)\| = \left\| \sum_{q \in \mathcal{B}_{k_1}^n; q \cap X \neq \emptyset} \mu(q) f(\xi_q^1) - \sum_{q' \in \mathcal{B}_{k_2}^n; q' \cap X \neq \emptyset} \mu(q') f(\xi_{q'}^2) \right\|.$$

Observe that for any box $q' \in \mathcal{B}_{k_2}^n$, $\exists q \in \mathcal{B}_{k_1}^n$ with $q' \subset q$. Let us denote $q'(q) = \{q' \in \mathcal{B}_{k_2}^n \mid q' \subset q \in \mathcal{B}_{k_1}^n\}$ and note that $\mu(q) = \sum_{q' \in q'(q)} \mu(q')$. The sums split as:

$$\|S_{k_1}^*(f, \xi^1) - S_{k_2}^*(f, \xi^2)\| \leq \left\| \sum_{q \in \mathcal{B}_{k_1}^n; q \cap X \neq \emptyset} \sum_{q' \in q'(q); q' \cap X \neq \emptyset} \mu(q') [f(\xi_q^1) - f(\xi_{q'}^2)] \right\| + \left\| \sum_{q \in \mathcal{B}_{k_1}^n; q \cap X \neq \emptyset} \sum_{q' \in q'(q); q' \cap X = \emptyset} \mu(q') f(\xi_{q'}^1) \right\|,$$

where we took into account that some of the boxes in $q'(q)$ might not intersect X (the refinement of the partition $\mathcal{B}_{k_1}^n$ will leave some refined boxes outside). The first term is bounded as before observing that $\|\xi_q^1 - \xi_{q'}^2\|_{q' \in q'(q)} < 2^{-k}$ (where this is the product topology norm). The second term is bounded by

$$\|f\|_{\infty, \bar{X}} \sum_{q \in \mathcal{B}_{k_1}^n; q \cap X \neq \emptyset; q \setminus X \neq \emptyset} \mu(q) \xrightarrow{k_1 \rightarrow \infty} \|f\|_{\infty, \bar{X}} \mu^*(\partial X) = 0,$$

as X is Jordan measurable. □

Theorem 11.6 (**Existence of the multiple Riemann integral for continuous functions**). *Any continuous function on a bounded Jordan measurable set $\bar{X} \subset \mathbb{R}^n$ with values in a Banach space, $f \in \mathcal{C}(\bar{X}, V)$ is Riemann integrable.*

Proof. From Lemma 11.4 we conclude that both $S_k^*(f, \xi)$ and $S_*^k(f, \xi)$ are Cauchy sequences hence convergent. As

$$\|S_k^*(f, \xi) - S_*^k(f, \xi)\| = \left\| \sum_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset, q \setminus X \neq \emptyset} \mu(q) f(\xi_q) \right\| \leq \|f\|_{\infty, \bar{X}} \sum_{q \in \mathcal{B}_k^n; q \cap X \neq \emptyset, q \setminus X \neq \emptyset} \mu(q) \xrightarrow{k \rightarrow \infty} \|f\|_{\infty, \bar{X}} \mu^*(\partial X) = 0,$$

we conclude the two limits are equal. □

As for the one dimensional case, the mutlidimensional Riemann integral is linear, additive on subdomains and bounded by:

$$\left\| \int_X d^n x f(x) \right\| \leq \mu(X) \|f\|_{\infty, \bar{X}}.$$

Once one knows that the Riemann integral exists, one can chose ones favorite sequence of increasingly refined partitions to compute it, not only $\mathcal{B}_k^n$. For continuous functions on measurable sets X , the contribution of the boundary ∂X will go to zero as f is bounded on ∂X and $\mu^*(\partial X) = 0$.

Definition 11.5 (**Normal domain**). *A domain $X \subset \mathbb{R}^n$ is normal with respect to the x^n axis if there exists a bounded Jordan measurable set $X' \subset \mathbb{R}^{n-1}$ and two continuous functions $\varphi_{n-1}, \psi_{n-1} : X' \rightarrow \mathbb{R}$ such that:*

$$X = \{(x^1, \dots, x^n) \mid (x^1, \dots, x^{n-1}) \in X', \varphi_{n-1}(x^1, \dots, x^{n-1}) \leq x^n \leq \psi_{n-1}(x^1, \dots, x^{n-1})\}.$$

Theorem 11.7 (**Fubini's theorem for continuous function on normal domains**). *If X is a normal domain then it is Jordan measurable and for any continuous function $f \in \mathcal{C}(\bar{X}, V)$:*

$$\int_X d^n x f(x) = \int_{X'} d^{n-1} x \int_{\varphi_{n-1}(x^1, \dots, x^{n-1})}^{\psi_{n-1}(x^1, \dots, x^{n-1})} d x^n f(x^1, \dots, x^n).$$

Proof. As φ_{n-1} and ψ_{n-1} are continuous and $\bar{X}'$ is compact, we conclude that X is measurable by Theorem 11.5.⁹ As f is continuous, all the integrals exist:

$$\int_X d^d x f(x) = \lim_{k \rightarrow \infty} S_*^k(f, \xi), \quad S_*^k(f, \xi) = \sum_{q \in \mathcal{B}_k^n, q \subset X} \mu(q) f(\xi_q),$$

where we decided to use the approximation of the integral by inner Riemann sums. The grid of boxes $\mathcal{B}_k^n$ is the Cartesian product of boxes $\mathcal{B}_k^{n-1}$ in $\mathbb{R}^{n-1}$ and $\mathcal{B}_k$ in $\mathbb{R}$: any box $q \in \mathcal{B}_k^n$ is the Cartesian product $q_{\parallel} \times q_{\perp}$ with $q_{\parallel} \in \mathcal{B}_k^{n-1}$ and $q_{\perp} \in \mathcal{B}_k$:

$$S_*^k(f, \xi) = \sum_{q_{\parallel} \in \mathcal{B}_k^{n-1}, q_{\parallel} \subset X'} \sum_{q_{\perp} \in \mathcal{B}_k, q_{\parallel} \times q_{\perp} \subset X} \mu(q_{\parallel}) \mu(q_{\perp}) f(\xi_{q_{\parallel}}^{\parallel}, \xi_{q_{\parallel}, q_{\perp}}^{\perp}),$$

where $\xi_{q_{\parallel}}^{\parallel} \in q_{\parallel}$ and $\xi_{q_{\parallel}, q_{\perp}}^{\perp} \in q_{\perp}$ are the components of the tags ξ_q along the parallel resp. orthogonal directions. Note that here we chose tags of the form $\xi_{q_{\parallel} \times q_{\perp}} = (\xi_{q_{\parallel}}^{\parallel}, \xi_{q_{\parallel}, q_{\perp}}^{\perp})$, that is we require that the projections on $\mathbb{R}^{n-1}$ of all the tags corresponding to boxes with the same $q_{\parallel}$ are equal (i.e. all the tags in boxes with equal $q_{\parallel}$ belong to the same one dimensional line orthogonal to $\mathbb{R}^{n-1}$).

We conclude by noting that for $q_{\parallel} \subset X'$ and some fixed $\xi_{q_{\parallel}}^{\parallel} \in q_{\parallel}$, $\{\xi_{q_{\parallel}, q_{\perp}}^{\perp}\} \times q_{\perp} \subset X$ if and only if $\varphi_{n-1}(\xi_{q_{\parallel}, q_{\perp}}^{\perp}) < q_{\perp} < \psi_{n-1}(\xi_{q_{\parallel}, q_{\perp}}^{\perp})$. $\square$

Exercise 11.5. Compute the volume of one eight of the 3-sphere, that is the integral of the function $f(x, y, z) = 1$ on the domain $D = x^2 + y^2 + z^2 \leq R^2$, $x, y, z > 0$. We have from Fubini:

$$\begin{aligned} \int_D dx dy dz &= R^3 \int_0^1 dx \int_0^{\sqrt{1-x^2}} dy \int_0^{\sqrt{1-x^2-y^2}} dz = R^3 \int_0^1 dx \int_0^{\sqrt{1-x^2}} dy \sqrt{1-x^2-y^2} \stackrel{y=u\sqrt{1-x^2}}{=} R^3 \int_0^1 dx (1-x^2) \int_0^1 du \sqrt{1-u^2} \\ &= R^3 \frac{2}{3} \int_0^{\pi/2} d\theta \cos^2 \theta = R^3 \frac{2}{3} \int_0^{\pi/2} d\theta \left(\frac{1+\cos 2\theta}{2} \right) = \frac{\pi}{6} R^3 \end{aligned}$$

11.2.3 Indefinite integrals, change of variables

Integrals over unbounded domains can sometimes be computed. For X an unbounded set, the Riemann integral over X is defined as the limit of Riemann integrals over a sequence of Jordan measurable sets X_k with $\bar{X}_k \subset X_{k+1}$ and $\bigcup_k X_k = X$.

Taking into account that the measure of a box is the volume form evaluated on the displacement vectors corresponding to the sides of the box, hence under a linear transformation $\mu(L(B)) = |\det L| \mu(B)$ the following theorem is straightforward.

Theorem 11.8 (Change of variable). Let $X \in \mathbb{R}^n$ be a Jordan measurable set and $\varphi : X \rightarrow \mathbb{R}^n$ a continuously differentiable bijection such that its Jacobian is non zero on $\varphi(X)$. Then:

$$\int_X d^n x f(x) = \int_{\varphi(X)} d^n u \det(d\varphi^{-1}|_u) f(\varphi^{-1}(u)) = \int_{\varphi(X)} d^n u \frac{\partial(x^1 \dots x^n)}{\partial(u^1, \dots, u^n)} f(x(u)),$$

where we denoted $x(u) \equiv \varphi^{-1}(u)$.

Exercise 11.6. Prove that $\int_{-\infty}^{\infty} dx e^{-\frac{x^2}{2}} = \sqrt{2\pi}$.

⁹The piece of the boundary $\{(x, x^n) | x \in \partial X', \phi_{n-1}(x) \leq x^n \leq \psi_{n-1}(x)\}$ is also of measure zero as $\partial X'$ is measure zero.

Chapter 12

Elements of differential geometry

Differential geometry is crucial for the study of classical mechanics, electromagnetism, gauge theory and general relativity.

12.1 Manifolds

Let M be a topological space. A *chart* (U, φ) on M consists in an open set $U \subset M$ and a homeomorphism (bicontinuous bijection) $\varphi : U \rightarrow \varphi(U) \subset \mathbb{R}^n$, which we abusively denote $\varphi : U \rightarrow \mathbb{R}^n$. A trivial example is $M = \mathbb{R}^n$ and $\varphi(x) = x$, but other examples are possible. A $\mathcal{C}^k$ *atlas* on M is a collection of charts $(U_\alpha, \varphi_\alpha)_{\alpha \in A}$ such that $M = \bigcup_{\alpha \in A} U_\alpha$ and $\forall \alpha, \beta$, the *transition map* $\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^n \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$ is a $\mathcal{C}^k$ map (i.e. k times differentiable with continuous partial derivatives of order k). The atlas is smooth if the transition maps are indefinitely differentiable.

Definition 12.1 (**Differentiable manifold**). A differentiable n -manifold (or manifold of dimension n) is a second countable, Hausdorff topological space M endowed with a $\mathcal{C}^k$ differentiable atlas $(U_\alpha, \varphi_\alpha)_{\alpha \in A}$, $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$. To simplify our life we consider from now on the atlas to be $\mathcal{C}^\infty$, in which case the manifold is called smooth.

Loosely speaking, a differentiable n -manifold M is a space that locally looks like $\mathbb{R}^n$, hence on which we can define derivatives. The subtlety of the atlas is that the mapping to $\mathbb{R}^n$ works only locally, in patches, and one needs to sew together the patches to get M . When sewing the patches one needs to make sure that the mappings of different patches to $\mathbb{R}^n$ are consistent with each other on the intersections. A chart $p \rightarrow \varphi(p) = (x^1(p), \dots, x^n(p))$ provides the local coordinates $x^1(p), \dots, x^n(p)$ of the point p in the patch U and the transition functions are changes of local coordinates.

Definition 12.2 (**Differentiable manifold with boundary**). We denote $\mathbb{R}_{\leq 0}^n = \mathbb{R}^{n-1} \times \mathbb{R}_{\leq 0} = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n \leq 0\}$. A differentiable n -manifold with boundary is a Hausdorff, second countable topological space such that for every point $p \in M$ there exists a neighborhood U_α , $p \in U_\alpha$ and either a homeomorphism $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$, $\varphi_\alpha(p) = (0, \dots, 0)$ (a regular chart) or a homeomorphism $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}_{\leq 0}^n$, $\varphi_\alpha(p) = (0, \dots, 0)$ (a boundary chart), with $\mathcal{C}^k$ transition maps $\varphi_\alpha \circ \varphi_\beta^{-1}$. We endow $\mathbb{R}_{\leq 0}^n$ with the induced subspace topology, i.e. the open sets in $\mathbb{R}_{\leq 0}^n$ consist in sets $W \cap \mathbb{R}_{\leq 0}^n$ with W open in $\mathbb{R}^n$.

A point $p \in M$ is called a regular point if $p \in U_\alpha$ for some regular chart $(U_\alpha, \varphi_\alpha)$, or a boundary point if not, and in this case $p \in U_\alpha$ for some boundary chart $(U_\alpha, \varphi_\alpha)$. The boundary ∂M of M is the set of boundary points of M and if M has dimension n , ∂M is a manifold of dimension $n - 1$ and $M \setminus \partial M$ is a manifold of dimension n without boundary. Note that $\partial \partial M = \emptyset$.

In words, M is manifold with boundary if for every point p there exist local coordinates φ around p , hence with $\varphi(p) = (0, \dots, 0)$, such that a neighborhood of p is parametrized:

- either by coordinates $(x^1, \dots, x^n) \in \mathbb{R}^n$, in which case the point is regular.

- or by coordinates $(x^1, \dots, x^n \leq 0) \in \mathbb{R}_{\leq 0}^n$, in which case the point is a boundary point.

Note that the points with local coordinates $(x^1, \dots, x^{n-1}, x^n = 0)$ in the neighborhood of a boundary point are themselves boundary points.¹

Although confusingly denoted similarly, the manifold boundary of ∂M of M is not the topological boundary of ∂M of M . For instance any open set in M has empty manifold boundary, but non empty topological boundary. The main point is that the manifold boundary is an intrinsic notion: it is defined without embedding the manifold into a bigger space. In opposition, the topological boundary is a relative notion: in order to identify the topological boundary of $Y \subset X$, one needs X .

An k -manifold N is immersed in an n -manifold M with $n \geq k$ if there exists a map $\phi : N \rightarrow M$, which is an immersion, that is ϕ is continuously differentiable² and its differential $d\phi$ is injective (it has zero kernel). In general ϕ is not required to be injective, that is $\phi(N) = \Sigma \subset M$ can have self intersections. N is embedded in M if ϕ is a diffeomorphism (a continuously differentiable bijection with continuously differentiable inverse) on Σ , where $\Sigma \subset M$ is endowed with the subspace topology, that is open sets in Σ are $W \cap \Sigma$, with W open in M . In this case ϕ is called an embedding. Equivalently, we have the following definition (we note that proving the equivalence of the two definitions is not at all trivial).

Definition 12.3 (Embedded submanifold). Let M be an n -dimensional manifold, and let k be an integer such that $0 \leq k \leq n$. A k -dimensional embedded submanifold (with or without boundary) of M is a subset $\Sigma \subset M$ such that every point of Σ is:

- either a regular point $p \in \Sigma$, in which case there exists a chart $(U \subset M, \varphi : U \rightarrow \mathbb{R}^n)$ around p , $\varphi(p) = (0, \dots, 0)$, such that $\varphi(\Sigma \cap U)$ is the intersection of a k -dimensional subspace $\mathbb{R}^k$ of $\mathbb{R}^n$ with $\varphi(U)$.
- or a boundary point $p \in \Sigma$, in which case there exists a chart $(U \subset M, \varphi : U \rightarrow \mathbb{R}^n)$ around p , $\varphi(p) = (0, \dots, 0)$, such that $\varphi(\Sigma \cap U)$ is the intersection of a half subspace $\mathbb{R}_{\leq 0}^k$ of $\mathbb{R}^n$ with $\varphi(U)$.

The pairs $(\Sigma \cap U, \varphi|_{\Sigma \cap U})$ form an atlas for the differential structure on Σ .

In local coordinates, Σ is embedded if there exists local coordinates φ on M around p , hence with $\varphi(p) = (0, \dots, 0)$, such that if p is a regular point of Σ then a neighborhood is parametrized by $(x^1, \dots, x^k, 0, 0 \dots 0)$, while if p is a boundary point of Σ a neighborhood is parametrized by $(x^1, \dots, x^k \leq 0, 0, 0 \dots 0)$. The points with coordinates $(x^1, \dots, x^k = 0, 0, 0 \dots 0)$ in a local chart around a boundary point are themselves boundary points.

If $k = n$ and Σ has no boundary then Σ is an open subset of M .

If M is a manifold with boundary and $p \in \partial M$, denoting $\varphi : U \subset M \rightarrow \mathbb{R}_{\leq 0}^n$ the boundary chart around p , $\varphi(U \cap \partial M) \subset \{(x^1, \dots, x^{n-1}, 0)\} = \mathbb{R}^{n-1}$ hence ∂M is embedded in M .³ We often consider manifolds embedded in $\mathbb{R}^n$. If M is a compact (as a topological space) n dimensional manifold with boundary embedded in $\mathbb{R}^n$, the manifold boundary of M and its topological boundary as a subset of $\mathbb{R}^n$ coincide. We often consider compact manifolds.

Any embedded submanifold $\Sigma \subset M$ as defined in Definition 12.3 has a natural embedding, the identity map $\phi : \Sigma \rightarrow M$, $\phi(p \in \Sigma) = p \in M$. This embedding can have a very non trivial form once one fixes coordinate charts on Σ and M . We stress that one can embed manifolds only in manifolds of higher (or equal) dimension. In particular, the differential of an embedding map $d\phi$ is full rank if and only if it is injective. Although not quite proper, when talking about embedded manifolds we sometimes ignore the distinction between N and its image $\phi(N) = \Sigma \subset M$ and loosely denote $N = \Sigma$.

¹For any $\tilde{p} = (x^1, \dots, x^n = 0)$ in the boundary chart (U, φ) centred at p , $\varphi(p) = 0$, one can build the translated chart $(U, \tilde{\varphi})$, $\tilde{\varphi}(r) = \varphi(r) - \varphi(\tilde{p})$, which is a boundary chart centred at $\tilde{p}$, $\tilde{\varphi}(\tilde{p}) = 0$. Note that we can not translate the boundary point to a regular point in U , as such a translation does not map open neighborhoods (in the subspace topology) of the boundary point into open neighborhoods of the regular point hence it is not an open map.

²To be precise one fixed coordinates in M and N and checks differentiability of $\varphi_M \circ \phi \circ \varphi_N^{-1} : U_N \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$

³Consequently, if Σ is a submanifold of M of the same dimension and $\partial \Sigma \neq \emptyset$, then $\partial \Sigma$ is a $n - 1$ dimensional submanifold embedded in M .

Examples of manifolds with no boundary are the sphere, the torus or the Klein bottle. Examples of manifolds with boundary are the cylinder and the Möbius strip.

Lemma 12.1 (**Graph embedding theorem**). *for $n > k$, let $f : \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ be a smooth function. Then its graph $(x, f(x)) \in \mathbb{R}^n$ is a smooth embedded k dimensional submanifold of $\mathbb{R}^n$. Every smooth k -dimensional submanifold embedded in $\mathbb{R}^n$ is locally the graph of a smooth function.*

From the implicit function theorem, the solution set of $n - k$ independent equations, $\Sigma = \{x \in \mathbb{R}^n | F^{k+1}(x), \dots, F^n(x) = 0\}$ with $\mathrm{d}F|_{F^{-1}(0)}$ of rank $n - k$ (maximal rank) is locally the graph of a function (the implicit function) and is therefore a k -dimensional submanifold embedded in $\mathbb{R}^n$. Combining the constraints into a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$, if $\mathrm{d}F|_{F^{-1}(0)}$ is surjective, then $F^{-1}(0)$ is an embedded submanifold.

Proof. The proof is tautological. Let us denote $x^1, \dots, x^n$ coordinates on $\mathbb{R}^n$. The graph of a smooth function $f : \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ is the subset $(x^1, \dots, x^k, x^{k+1} = f^1(x^1, \dots, x^k), \dots, x^n = f^{n-k}(x^1, \dots, x^k))$ and under the change of coordinates $\tilde{x}^1 = x^1$ up to $\tilde{x}^k = x^k$ and $\tilde{x}^{k+1} = x^{k+1} - f^1(x^1, \dots, x^k)$ up to $\tilde{x}^n = x^n - f^{n-k}(x^1, \dots, x^k)$ (whose Jacobian is 1 everywhere), it writes as $(\tilde{x}^1, \dots, \tilde{x}^k, 0, \dots, 0)$ hence the graph is an embedded submanifold.

If Σ is an embedded submanifold of $\mathbb{R}^n$ then in the chart φ of Definition 12.3 it is the graph of the function $f^i(x^1, \dots, x^k) = 0$ with $i = 1, \dots, n - k$. Changing to another coordinate system y , the coordinates in the chart φ are explicit functions of y , $x^1(y), x^2(y), \dots, x^n(y)$. Relabelling the functions $x^p(y) = F^p(y)$ for $k + 1 \leq p \leq n$, the submanifold Σ is defined in the coordinates y by the $n - k$ linearly independent constraints $F^{k+1}(y) = \dots = F^n(y) = 0$. As the F s are only x coordinates expressed in terms of y , the forms dF^p span a $n - k$ dimensional subspace of T_p^*M . $\square$

Theorem 12.1 (**Whitney embedding theorem**). *Any smooth real m -dimensional manifold without boundary can be smoothly embedded in $\mathbb{R}^{2m}$; any smooth real m -dimensional manifold with boundary can be smoothly embedded in $\mathbb{R}^{2m+1}$.*

Proof. The reader should search the literature available on the subject. $\square$

Exercise 12.1. *Recalling the stereographic projection, give an atlas consisting of two charts for $\mathbb{S}^1 = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 1\}$.*

We have $\varphi_N : \mathbb{S}^1 \setminus \{(0, 1)\} \rightarrow \mathbb{R}$, $\varphi_N(x, y) = \frac{x}{1-y}$, and $\varphi_S : \mathbb{S}^1 \setminus \{(0, -1)\} \rightarrow \mathbb{R}$, $\varphi_S(x, y) = \frac{x}{1+y}$. Denoting $s = \frac{x}{1+y}$ and $t = \frac{x}{1-y}$ we have $st = \frac{x^2}{1-y^2} = 1$ for $(x, y) \in \mathbb{S}^1 \setminus \{(0, 1), (0, -1)\}$. furthermore, $\varphi_S(\mathbb{S}^1 \setminus \{(0, 1), (0, -1)\}) = \{s \in \mathbb{R}, s \neq 0\}$ and $\varphi_N(\mathbb{S}^1 \setminus \{(0, 1), (0, -1)\}) = \{t \in \mathbb{R}, t \neq 0\}$ hence the transition map $\varphi_N \circ \varphi_S^{-1} : \{s \in \mathbb{R}, s \neq 0\} \rightarrow \{t \in \mathbb{R}, t \neq 0\}$ is $t(s) = \varphi_N \circ \varphi_S^{-1}(s) = \frac{1}{s}$, which is a smooth map for $\{s \in \mathbb{R}, s \neq 0\}$.

Exercise 12.2. *Using the implicit function theorem, find an atlas consisting in four charts for the set $\mathbb{S}^1 = F^{-1}(0)$, where $F(x, y) = x^2 + y^2 - 1 = 0$ and check that the transition functions are smooth. We have $\mathrm{d}F = (2x \quad 2y)$. We start from the point $(x, y) = (0, 1)$ separate $\mathrm{d}F = (2x | 2y)$ and solve for $y = f_N(x) = \sqrt{1 - x^2}$. This goes up to $x = \pm 1$. In the neighborhood of $(x, y) = (1, 0)$ we separate $\mathrm{d}F = (2y | 2x)$ and solve for $x = f_E(y) = \sqrt{1 - y^2}$ and so on. This covers $\mathbb{S}^1$ by four open sets:*

$$U_N = \{(x, \sqrt{1-x^2}), x \in (-1, 1)\}, \quad U_E = \{(\sqrt{1-y^2}, y), y \in (-1, 1)\}, \quad U_S = \{(x, -\sqrt{1-x^2}), x \in (-1, 1)\}, \quad U_W = \{(-\sqrt{1-y^2}, y), y \in (-1, 1)\},$$

and we have four charts:

$$\varphi_N : U_N \rightarrow \mathbb{R}, \quad \varphi_N((x, y) \in U_N) = x, \quad \varphi_E : U_E \rightarrow \mathbb{R}, \quad \varphi_E((x, y) \in U_E) = y, \quad \varphi_S : U_S \rightarrow \mathbb{R}, \quad \varphi_S((x, y) \in U_S) = x, \quad \varphi_W : U_W \rightarrow \mathbb{R}^2, \quad \varphi_W((x, y) \in U_E) = y.$$

We have $U_N \cap U_E = \{(x, \sqrt{1-x^2}), x \in (0, 1)\}$ and $\varphi_N(U_N \cap U_E) = (0, 1)$, $\varphi_E(U_N \cap U_E) = (0, 1)$ and transition map is $\varphi_E \circ \varphi_N^{-1} : (0, 1) \rightarrow (0, 1)$, $\varphi_E \circ \varphi_N^{-1}(x) = \sqrt{1-x^2}$ repeating the construction we get $U_S \cap U_E = \{(x, -\sqrt{1-x^2}), x \in (0, 1)\}$ and $\varphi_S \circ \varphi_E^{-1} : (-1, 0) \rightarrow (0, 1)$, $\varphi_S \circ \varphi_E^{-1}(y) = \sqrt{1-y^2}$, while $\varphi_E \circ \varphi_S^{-1}(x) = -\sqrt{1-x^2}$.

The embedding of $\mathbb{S}^1$ into $\mathbb{R}^2$ is the identity $\phi : \mathbb{S}^1 \rightarrow \mathbb{R}^2$, $\phi(p) = p$. However, in coordinate charts $\varphi_{\mathbb{S}^1}$ and $\varphi_{\mathbb{R}^2}$ this writes $\phi = \varphi_{\mathbb{R}^2} \circ \varphi_{\mathbb{S}^1}^{-1}$ and it might look quite complicated. For instance, using Cartesian coordinates in $\mathbb{R}^2$ and the north stereographic coordinates in $\mathbb{S}^1 \setminus (0, 1)$ we have:

$$\phi(t) = \left(x(t) = \frac{2t}{1+t^2}, y(t) = \frac{-1+t^2}{1+t^2} \right), \quad \mathrm{d}\phi = \begin{pmatrix} 2 \frac{1-t^2}{(1+t^2)^2} \\ 2 \frac{t}{(1+t^2)^2} \end{pmatrix},$$

which is injective, $\ker(d\phi) = 0$, for any $t \in \mathbb{R}$. Note that, as this is only valid in a coordinate system, it does not cover the entire $\mathbb{S}^1$: this expression does not map the north pole, where the north stereographic projection is not well defined.

Exercise 12.3. Write the embedding by the identity map of the paraboloid defined by the equation $F(x, y, z) = x^2 + y^2 - z = 0$ in a coordinate chart. $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ with $\phi(s, t) = (x(s, t), y(s, t), z(s, t)) = (s, s^2 + t^2)$ with $d\phi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2s & 2t \end{pmatrix}$.

While everything we present can be generalized to arbitrary manifolds, in order to simplify the understanding the reader can always consider that we discuss k dimensional manifolds embedded in $\mathbb{R}^n$. Such manifolds are specified by their embedding map which in coordinate systems writes as $\phi(s) = (x^1(s^1, \dots, s^k), \dots, x^n(s^1, \dots, s^k))$, with $d\phi$ maximal rank (that is k) or as the solution of $n - k$ equations $F^i(x) = 0$ for $i = k + 1, \dots, n$ with dF of maximal rank (that is $n - k$). The solution set of $F^k(x) \leq 0$, $F^{k+1}(x) = F^{k+2}(x) = \dots = F^n(x) = 0$ and with $\frac{\partial(F^k, \dots, F^n)}{\partial(x^1, \dots, x^n)}$ and $\frac{\partial(F^{k+1}, \dots, F^n)}{\partial(x^1, \dots, x^n)}$ of maximal rank (that is $n - k$ and $n - k - 1$ respectively) is an embedded $k + 1$ dimensional manifold with boundary.

12.2 The tangent and cotangent spaces

A function $f : M \rightarrow \mathbb{R}$ with M a n -dimensional manifold is said to belong to $\mathcal{C}^\infty(M)$ if $f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is infinitely differentiable for any chart (U, φ) .⁴ A *derivation* at $p \in M$ is a linear map $\mathcal{D}|_p : \mathcal{C}^\infty(M) \rightarrow \mathbb{R}$ that satisfies the Leibnitz rule:

$$\mathcal{D}|_p(fg) = \mathcal{D}|_p(f) \cdot g(p) + f(p) \cdot \mathcal{D}|_p(g).$$

In particular, as $\mathcal{D}|_p$ is linear, $\mathcal{D}|_p(f) = 0$ for constant f .

Definition 12.4 (**Tangent space**). The tangent space of M at p , denoted $T_p M$ is the vector space of all the derivations at p with vector addition and multiplication by scalars:

- $(\mathcal{D}_1 + \mathcal{D}_2)|_p(f) = \mathcal{D}_1|_p(f) + \mathcal{D}_2|_p(f),$
- $(a \cdot \mathcal{D})|_p(f) = a \cdot \mathcal{D}|_p(f).$

We commonly denote the elements in $T_p M$ by capital letters $V|_p, W|_p$ and so on.

Definition 12.5 (**Cotangent space**). The cotangent space of M at p , denoted $T_p^* M$ is the dual of $T_p M$, that is the vector space of all the linear forms $\alpha|_p : T_p M \rightarrow \mathbb{R}$, $\alpha|_p(a_1 V_1|_p + a_2 V_2|_p) = a_1 \alpha|_p(V_1|_p) + a_2 \alpha|_p(V_2|_p)$, with addition and multiplication by scalars $(a\alpha + b\beta)|_p(V|_p) = a\alpha|_p(V|_p) + b\beta|_p(V|_p)$. We commonly denote the elements in $T_p^* M$ by Greek letters.

Definition 12.6 (**Exterior differential of $f \in \mathcal{C}^\infty(M)$**). The exterior differential of a function $f \in \mathcal{C}^\infty(M)$ is the linear form $df|_p \in T_p^* M$ defined by $df|_p(V|_p) = V|_p(f)$, $\forall V|_p \in T_p M$.

Collecting all the tangent (resp. cotangent) space we obtain the tangent (resp. cotangent) bundle:

$$TM = \bigcup_{p \in M} \{p\} \times T_p M = \{(p, V|_p) | p \in M, V|_p \in T_p M\}, \quad T^*M = \bigcup_{p \in M} \{p\} \times T_p^* M = \{(p, \alpha|_p) | p \in M, \alpha|_p \in T_p^* M\}.$$

The elements of the tangent bundle are called *vector fields* and consist in a vector $V|_p$ in $T_p M$ for every point $p \in M$ and we are interested in the behaviour of $V|_p$ as the base point p changes. The elements in the cotangent bundle are called *form fields*.

⁴ $\mathcal{C}^\infty(M)$ is an associative algebra with respect to the pointwise sum and product and multiplication by scalars.

12.2.1 Local coordinates and bases

The tangent and cotangent spaces are intrinsic: their definition does not require the use of a chart. It is however instructive to see what information we can get about them if we specify one.

Consider (U, φ) a chart at $p \in U \subset M$, $\varphi : U \rightarrow \varphi(U) \subset \mathbb{R}^n$. We denote $\{e_1, \dots, e_n\}$ some fixed basis in $\mathbb{R}^n$, $x = \sum_i x^i e_i$ and $\{\lambda^1, \dots, \lambda^n\}$ its dual basis $\lambda^i(e_j) = \delta_j^i$. The x^i 's are the coordinates of x in the chosen basis and provide local coordinates on U .

For any $f : M \rightarrow \mathbb{R}$, we denote $f^\varphi = f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$. As f^φ is a function from a subset of $\mathbb{R}^n$ to $\mathbb{R}$, one can consider its Gateaux derivative at $\varphi(p) = x = (x^1, \dots, x^n) \in \mathbb{R}^n$ along some direction $V = \sum_{i=1}^n V^i e_i \in \mathbb{R}^n$:

$$\begin{aligned} \frac{d}{dt}(f \circ \varphi^{-1})(\varphi(p) + tV) \Big|_{t=0} &= \frac{d}{dt}(f^\varphi)(x + tV) \Big|_{x=\varphi(p)} \Big|_{t=0} = Df^\varphi(x; V) \Big|_{x=\varphi(p)} = \mathbb{d}f^\varphi \Big|_{x=\varphi(p)}(V) \\ &= \sum_{i=1}^n V^i \mathbb{d}f^\varphi \Big|_{x=\varphi(p)}(e_i) = \sum_{i=1}^n V^i \frac{\partial f^\varphi}{\partial x^i} \Big|_{x=\varphi(p)}, \end{aligned}$$

where $\mathbb{d}f^\varphi \Big|_{x=\varphi(p)}$ is the usual Fréchet differential of $f^\varphi : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$. It is easy to check that $\sum_{i=1}^n V^i \frac{\partial f^\varphi}{\partial x^i} \Big|_{x=\varphi(p)}$ is linear and respects the Leibnitz rule, hence it is a derivation at p . In fact, every derivation at p must be of this form.⁵

The transition functions between charts can be written as $x = \varphi_\alpha \varphi_\beta^{-1}(y)$ and are diffeomorphism from $\mathbb{R}^n$ to $\mathbb{R}^n$. The change of chart is just the composition with such a diffeomorphism. For instance $f^{\varphi_\alpha}(x) = f^{\varphi_\alpha}(\varphi_\alpha \varphi_\beta^{-1}(y)) = f^{\varphi_\beta}(y)$, and the partial derivatives of f^φ transform as dictated by the chain rule (see next section). It is then justified to change notation and render the chart φ implicit, that is we denote $f^\varphi(x)$ simply as $f(x)$. In order to simplify notation, unless absolutely necessary, we will suppress the base point $\Big|_p$, but keep in mind that the tangent and cotangent spaces are defined at a fixed base point.

With this notation, the tangent space in coordinates x is the space of first order differential operators:

$$T_p M = \left\{ V = \sum_{i=1}^n V^i \frac{\partial}{\partial x^i} \mid V^i \in \mathbb{R} \right\}, \quad V(f) = \sum_{i=1}^n V^i \frac{\partial f}{\partial x^i} = \mathbb{d}f(V),$$

that is $T_p M$ is nothing but $\mathbb{R}^n$ where the basis elements are identified as differential operators $e_i \leftrightarrow \frac{\partial}{\partial x^i}$. This identification comes from the fact that the Gateaux derivative in the direction e_i is a partial derivative, $\mathbb{d}f(e_i) = \frac{\partial f}{\partial x^i}$. We conclude that choosing a chart of local coordinates comes to choosing a basis in the tangent space.

The exterior differential of functions $f : M \rightarrow \mathbb{R}$ becomes, when written in coordinates, just the usual F chet differential:

$$df(V) = V(f) = \sum_{i=1}^n V^i \frac{\partial f}{\partial x^i}, \quad df\left(\frac{\partial}{\partial x^i}\right) = \frac{\partial f}{\partial x^i},$$

and yields the speed of variation of f in the direction V . An interesting set of functions are the projections along the directions e_i that is $x^j = \lambda^j \circ \varphi : M \rightarrow \mathbb{R}$ or, suppressing the chart, $x^j = \lambda^j$, where we see x^j as a function $x^j(x^1, \dots, x^n) = x^j$. The differential of the function x^j is then $dx^j\left(\frac{\partial}{\partial x^i}\right) = \frac{\partial x^j}{\partial x^i} = \delta_i^j$, that is:

$$dx^j\left(\sum_i V^i \frac{\partial}{\partial x^i}\right) = V^j.$$

⁵Any $\mathcal{D}|_p$ is mapped one to one to a derivation $\mathcal{D}|_{\varphi(p)}$ on $\mathcal{C}^\infty(\mathbb{R}^n)$ defined by $\mathcal{D}|_p(F) = \mathcal{D}|_{x=\varphi(p)}(F^\varphi)$. Denote $\mathcal{D}|_{x_0}$ a derivation on $\mathcal{C}^\infty(\mathbb{R}^n)$ at x_0 . Using Taylor's formula we write any $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as $f(x) = f(x_0) + \sum_{i=1}^n \frac{\partial f}{\partial x^i} \Big|_{x=x_0} (x^i - x_0^i) + R(x)$, where $R(x) = \sum_{i,j} (x^i - x_0^i)(x^j - x_0^j) h_{ij}(x)$. It follows from Leibnitz's rule that $\mathcal{D}|_{x_0}(R) \propto (x^i - x_0^i) \Big|_{x=x_0} Q(x_0) = 0$. Using the linearity of $\mathcal{D}|_{x_0}$ and the fact that it sends constants to 0, we conclude that $\mathcal{D}|_{x_0}(f) = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \Big|_{x_0} \cdot \mathcal{D}|_{x_0}(x^i)$.

It follows that the forms $\{dx^1, \dots, dx^n\}$ are dual to the basis of vectors $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$, $\lambda^i \leftrightarrow dx^i$, and span the cotangent space T_p^*M :

$$\alpha(V) = \alpha\left(\sum_i V^i \frac{\partial}{\partial x^i}\right) = \sum_i V^i \alpha\left(\frac{\partial}{\partial x^i}\right) = \sum_i \alpha\left(\frac{\partial}{\partial x^i}\right) dx^i(V), \quad T_p^*M = \left\{ \alpha = \sum_{i=1}^n \alpha_i dx^i \mid \alpha_i = \alpha\left(\frac{\partial}{\partial x^i}\right) \in \mathbb{R} \right\}.$$

In particular, as $df(\frac{\partial}{\partial x^i}) = \frac{\partial f}{\partial x^i}$ the exterior differential of any function decomposes on this basis as:

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i.$$

12.2.2 Curves

Definition 12.7 (Curve). A curve in a n -dimensional manifold M is a continuous map $\gamma: [a, b] \rightarrow M$. We often denote the image of the curve also by γ . A curve is called smooth ($\mathcal{C}^1$, etc.) if $\varphi \circ \gamma$ is smooth ($\mathcal{C}^1$, etc.) for any chart φ and it is called simple if γ is injective on (a, b) , that is γ has no self intersections except possibly at the end points. A curve is naturally oriented from $\gamma(a)$ to $\gamma(b)$.

Definition 12.8 (Integral curves). Given a $\mathcal{C}^1$ curve $\gamma: [a, b] \rightarrow M$, the tangent vector to γ is the vector $V|_{\gamma(t)}(f) = \frac{d}{dt}(f \circ \gamma)(t)$. Conversely, the integral curve of a vector field $V \in TM$ is the curve $\gamma: [a, b] \rightarrow M$ such that for all $f \in C^\infty(M)$, $\frac{d}{dt}(f \circ \gamma)(t) = V|_{\gamma(t)}(f)$. Equivalently, seen as a map from $[a, b] \rightarrow \mathbb{R}^n$ in some chart φ we have $\varphi \circ \gamma(t) = (x^1(t), \dots, x^n(t))$, the integral curve of the vector field $V = \sum_i V^i(x) \frac{\partial}{\partial x^i}$ respects $\frac{dx^i}{dt}(t) = V^i(x(t))$.

We will see later that, under mild assumptions (e.g. if $V^i(x)$ are Lipschitz continuous), any vector field V has locally integral curves, and through a point of M there passes only one integral curve of V . For example, the integral curve of the vector field $\frac{\partial}{\partial x^i}$ is $(\varphi \circ \gamma)(t) = (x^1, x^2, \dots, x^i + t, \dots, x^n)$. These curves are called coordinate curves and can be brought back to M by $\varphi^{-1}(x^1, x^2, \dots, x^i + t, \dots, x^n)$.

A curve writes in some chart as $\gamma^\varphi = \varphi \circ \gamma: [a, b] \rightarrow \mathbb{R}^n$ and we denote $\dot{\gamma}^\varphi(0) = \frac{d\gamma^\varphi}{dt}|_{t=0}$. One can build the tangent space at p using the curves with $\gamma(0) = p$ as the space of equivalence classes $[\dot{\gamma}]$ where $\gamma_1 \sim \gamma_2$ if $\frac{d\gamma_1}{dt}|_{t=0} = \frac{d\gamma_2}{dt}|_{t=0}$. For $f: M \rightarrow \mathbb{R}$, we have $f \circ \gamma: \mathbb{R} \rightarrow \mathbb{R}$ and denoting e_t the basis vector in $T_0\mathbb{R}$ and $\{f_1, \dots, f_n\}$, $\{\mu^1, \dots, \mu^n\}$ a basis and its dual in $T_{\gamma(0)}\mathbb{R}^n$ we have:

$$\begin{aligned} \frac{d}{dt}\gamma^\varphi|_{t=0} &= D\gamma^\varphi(0, e_t) = \mathbb{d}\gamma^\varphi|_0(e_t) = \sum_i f_i \mu^i [\mathbb{d}\gamma^\varphi|_0](e_t) = \sum_i f_i \mathbb{d}(\gamma^\varphi)^i|_0(e_t) = \sum_i f_i \frac{d(\gamma^\varphi)^i}{dt}|_{t=0}, \\ \frac{d}{dt}(f \circ \gamma)|_{t=0} &= D[f \circ \gamma](0; e_t) = d[f \circ \gamma]|_0(e_t) = df^\varphi|_{\gamma(0)} \cdot \mathbb{d}\gamma^\varphi|_0(e_t) = \sum_i df^\varphi|_{\gamma(0)}(f_i) \frac{d(\gamma^\varphi)^i}{dt}|_{t=0} = \sum_i \frac{d(\gamma^\varphi)^i}{dt}|_{t=0} \frac{\partial f^\varphi}{\partial x^i}|_{\gamma(0)}. \end{aligned}$$

Exercise 12.4. Let M a manifold with boundary and $p \in \partial M$. Compare $T_p\partial M$ and T_pM . In the boundary chart $\{(x^1, \dots, x^{n-1}, x^n \leq 0)\}$ we have $\partial M = \{(x^1, \dots, x^{n-1}, x^n = 0)\}$ hence $T_p\partial M = \{V = \sum_{i=1}^{n-1} V^i \partial_i\}$ and $T_pM = \{V = \sum_{i=1}^n V^i \partial_i \mid V^n \leq 0\}$. That is $T_p\partial M$ is missing one direction, and T_pM is only a half space.

Let $h: \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function and assume that the set $\Sigma = \{x \in \mathbb{R}^n \mid h(x) = 0\}$ is such that $dh|_\Sigma \neq 0$. Prove that $M = \{x \in \mathbb{R}^n \mid h(x) \leq 0\}$ is a manifold with boundary embedded in $\mathbb{R}^n$ and that $\Sigma = \partial M$. As $dh \neq 0$, for any $p \in \Sigma$ there exists i such that $\partial_i h \neq 0$. We introduce local coordinates $y = (x_1, \dots, x_{i-1}, h(x_1, \dots, x_n), x_{i+1}, \dots, x_n)$ with non zero Jacobian $\det(\frac{\partial y^j}{\partial x^i}) = \frac{\partial h}{\partial x^i}$ and we relabel the coordinates maintaining the orientation to put $y_n = h(x_1, \dots, x_n)$.

12.2.3 Coordinate transformations

Let us consider some coordinates x on a manifold, and a coordinate change $x = X(y)$, that is a change of chart. We denote the inverse transformation $Y(x)$ that is $X(Y(x)) = x$ and $Y(X(y)) = y$. Note that (recall the inverse function theorem):

$$\frac{\partial}{\partial x^i} [X^j(Y(x))] = \frac{\partial x^j}{\partial x^i} = \delta_i^j = \sum_k \frac{\partial X^j}{\partial y^k} \bigg|_{y=Y(x)} \frac{\partial Y^k}{\partial x^i} \bigg|_x = \delta_j^i, \quad \frac{\partial}{\partial y^j} Y^i(X(y)) = \sum_k \frac{\partial Y^i}{\partial x^k} \bigg|_{x=X(y)} \frac{\partial X^k}{\partial y^j} \bigg|_y = \delta_j^i.$$

It is standard to drop the capital letter notation for the functions in the change of variables (as the position of a symbol automatically tells us if we deal with a function or an argument) and to drop the arguments, to write in a somewhat sloppy notation:

$$\sum_k \frac{\partial x^i}{\partial y^k} \frac{\partial y^k}{\partial x^j} = \delta_j^i, \quad \sum_k \frac{\partial y^i}{\partial x^k} \frac{\partial x^k}{\partial y^j} = \delta_j^i,$$

and $\frac{\partial x^i}{\partial y^j}$ the Jacobian matrix of the change of coordinates $x(y)$ and $\frac{\partial y^j}{\partial x^i}$ its inverse. Of course, $\frac{\partial y^j}{\partial x^i}$ is also the Jacobian matrix of the inverse change of variables $y = y(x)$.

Upper indices are called contravariant and lower ones are called covariant. The behaviour of tangent vectors and cotangent forms under a change of coordinates $x = x(y)$ is dictated by the chain rule, that is:

$$dx^i = \sum_{j=1}^n \frac{\partial x^i}{\partial y^j} dy^j, \quad \frac{\partial}{\partial x^i} = \sum_{j=1}^n \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j}.$$

The first equation follows as in the new coordinate $df = \sum_j \frac{\partial f}{\partial y^j} dy^j$ for all f , including $f(x) = x^i$ which becomes in the new coordinates $x^i(y)$ while the second is just the chain rule:

$$\frac{\partial f}{\partial y^i} = \sum_{j=1}^n \frac{\partial x^j}{\partial y^i} \frac{\partial f}{\partial x^j},$$

multiplied to the left by $\frac{\partial y^i}{\partial x^k}$. For instance in spherical coordinates (r, θ, φ) on $\mathbb{R}^3$ we have $(x^1, x^2, x^3) = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$ and the Jacobian matrix is:

$$\left(\frac{\partial x^i}{\partial y^j} \right)_{i,j} = \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix},$$

which leads to the following transformation law for the basis vectors and forms:

$$\begin{pmatrix} \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x^1} & \frac{\partial}{\partial x^2} & \frac{\partial}{\partial x^3} \end{pmatrix} \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix},$$

$$\begin{pmatrix} dx^1 \\ dx^2 \\ dx^3 \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix} \begin{pmatrix} dr \\ d\theta \\ d\varphi \end{pmatrix}.$$

All contravariant, respectively covariant indices transform in the same way. A mixed tensor with covariant and contravariant indices is a multilinear form on a tensor product of tangent and cotangent spaces, $T : (T_p M)^q \times (T_p^* M)^p \rightarrow \mathbb{R}$, $T = \sum T(x)_{j_1 \dots j_q}^{i_1 \dots i_p} \partial_{x^{i_1}} \dots \partial_{x^{i_p}} dx^{j_1} \dots dx^{j_q}$, where $T(x)_{j_1 \dots j_q}^{i_1 \dots i_p}$ denotes the components of the tensor in the coordinate system x .

The tensor is invariant under coordinate changes, hence its components change like:

$$T(x)_{j_1 \dots j_q}^{i_1 \dots i_p} = \sum_{a,b} T(y)_{b_1 \dots b_q}^{a_1 \dots a_p} \frac{\partial x^{i_1}}{\partial y^{a_1}} \cdots \frac{\partial x^{i_p}}{\partial y^{a_p}} \frac{\partial y^{b_1}}{\partial x^{j_1}} \cdots \frac{\partial y^{b_q}}{\partial x^{j_q}} .$$

Quantities in which covariant and contravariant indices are contracted pairwise are invariant under change of coordinates. Only pairs of covariant and contravariant indices can be contracted. Indices can be contracted only between quantities (components, basis vectors, coordinates, etc.) which are expressed in the same basis.

From now on we will always work in some coordinate patch, and all the notions we discuss are extended to global notions by sewing if necessary such local patches.

12.3 (Pseudo-)Riemannian manifolds

The configuration space of a classical mechanical system is a manifold M , parametrized by coordinates q^μ , the generalized coordinates. The tangent bundle TM is the space of generalized coordinates and velocities (q, V) with $V = \sum V^\mu \frac{\partial}{\partial q^\mu}$. In physics textbooks the velocity V is often denoted $\dot{q}$. We denote dq^μ the dual forms $dq^\mu(V) = V^\mu$ and parametrize the cotangent bundle T^*M by (q, p) , with $p = \sum_\mu p_\mu dq^\mu$ generalized momenta. The cotangent bundle is also called the phase space.

Let us consider a pendulum. The position of the pendulum is specified by the generalized coordinate θ , the angle between the pendulum string and the vertical $x = r \sin \theta$, $y = -r \cos \theta$. The kinetic energy of the system is (in physics notation):

$$L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) = \frac{m}{2}((r \cos \theta \dot{\theta})^2 + (r \sin \theta \dot{\theta})^2) = \frac{m}{2}r^2 \dot{\theta}^2 .$$

As a second example, consider a particle constrained to move on a paraboloid in $\mathbb{R}^3$. We parametrize the paraboloid by $(s^1, s^2) \rightarrow \phi(s) = (x^1(s), x^2(s), x^3(s)) = (s^1, s^2, \frac{(s^1)^2 + (s^2)^2}{2})$. We use s^1 and s^2 as generalized coordinates (recall that the generalized coordinates must be independent) and a trajectory of the particle is specified by specifying these two coordinates as a function of time $(s^1(t), s^2(t))$. The velocity along a trajectory in generalized coordinates is $\frac{ds^1}{dt} \partial_{s^1} + \frac{ds^2}{dt} \partial_{s^2} = V^1 \partial_{s^1} + V^2 \partial_{s^2}$. The trajectory in Euclidean space is $x(s(t))$ and the velocity of our particle in $\mathbb{R}^3$ is:

$$\sum_j \left(\frac{d}{dt} x^j(s^1(t), s^2(t)) \right) \partial_{x^j} = \sum_{j,\mu} \frac{\partial x^j}{\partial s^\mu} \frac{ds^\mu}{dt} \partial_{x^j} = \sum_{j,\mu} \partial_{x^j} \frac{\partial x^j}{\partial s^\mu} V^\mu .$$

One recognizes the action of the Fréchet differential $d\phi$ of the map ϕ , which maps $T_s M$ to a two dimensional subspace of $T_{x(s)} \mathbb{R}^3$:

$$d\phi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ s^1 & s^2 \end{pmatrix}, \quad d\phi(V^1 \partial_{s^1} + V^2 \partial_{s^2}) = V^1 \partial_{x^1} + V^2 \partial_{x^2} + (s^1 V^1 + s^2 V^2) \partial_{x^3} .$$

The kinetic energy of the particle is:

$$\begin{aligned} L &= \frac{m}{2} \left[\left(\frac{dx^1}{dt} \right)^2 + \left(\frac{dx^2}{dt} \right)^2 + \left(\frac{dx^3}{dt} \right)^2 \right] = \frac{m}{2} \left[(V^1)^2 + (V^2)^2 + (s^1 V^1 + s^2 V^2)^2 \right] \\ &= \frac{m}{2} \sum_{\mu\nu} g_{\mu\nu} V^\mu V^\nu = \frac{m}{2} g(V, V), \quad g_{\mu\nu} = \sum_{i,j} \delta_{ij} \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\nu} . \end{aligned}$$

The non trivial bilinear form (i.e. linear in each argument) $g : T_s M \times T_s M \rightarrow \mathbb{R}$ is a *metric* on the tangent space. Any time independent holonomic constraints can be translated into a metric, that is the most general form of the kinetic energy of a mechanical system is $L = \frac{m}{2} \sum_{\mu\nu} g_{\mu\nu} V^\mu V^\nu$ for some $g_{\mu\nu}$ and V the speed in generalized coordinates.

The Lagrangian is a real valued function on TM . Choosing coordinates q^μ gives a basis $e_\mu = \partial_{q^\mu}$ in the tangent space $V = \sum_\mu V^\mu e_\mu$, and the Lagrangian writes $L(q^1, \dots, q^n, V^1, \dots, V^n)$. We denote L_q the restriction of the Lagrangian to $T_q M$, that is $L_q : T_q M \rightarrow \mathbb{R}$ with $L_q(V) = L(q, V)$. Furthermore, we denote $\mathbb{d}L_q|_V : T_q M \rightarrow \mathbb{R}$ the Fréchet differential of this restriction:

$$\mathbb{d}L_q|_V(W) = \sum_\mu W^\mu \mathbb{d}L_q|_V(e_\mu) = \sum_\mu W^\mu \frac{\partial L_q}{\partial V^\mu}(V) = \sum_\mu W^\mu \frac{\partial L}{\partial V^\mu}(q, V).$$

In particular the differential $\mathbb{d}L_q|_V$ is a one form on $T_q M$ as it maps vectors to the real numbers, that is $\mathbb{d}L_q|_V \in T_q^* M$. The energy function is a real valued function on TM defined as:

$$E(q, V) = \mathbb{d}L_q|_V(V) - L(q, V) = \sum_\mu V^\mu \frac{\partial L}{\partial V^\mu}(q, V) - L(q, V).$$

We stress that the energy function, like the Lagrangian, is a function of positions and velocities, that is it is a function on the tangent space. The fiber derivative of L is the mapping:

$$\mathcal{F}L : TM \rightarrow T^*M, \quad \mathcal{F}L(q, V) = (q, \mathbb{d}L_q|_V) = (q^1, \dots, \frac{\partial L}{\partial V^1}(q, V), \dots), \quad (q, \sum_\mu V^\mu \partial_{q^\mu}) \mapsto (q, \sum_\mu \frac{\partial L}{\partial V^\mu}(q, V) dq^\mu),$$

and takes the tangent space to the cotangent space.

The Hamiltonian (the Legendre transform of the Lagrangian) is the real valued function on the cotangent space defined as the composition of the energy function with the inverse of the fiber derivative:

$$H : T^*M \rightarrow \mathbb{R}, \quad H(q, p) = E \circ (\mathcal{F}L)^{-1}(q, p) = \left(\sum_\mu V^\mu \frac{\partial L}{\partial V^\mu}(q, V) - L(q, V) \right)_{V=V(q,p), p_\mu = \frac{\partial L}{\partial V^\mu}(q, V(q,p))}.$$

In words:

- we start from the coordinates and velocities (q, V) which are all independent variables.
- given a Lagrangian, we write the equations $p_\mu = \frac{\partial L}{\partial V^\mu}(q, V)$ which we solve (when is this possible?) and obtain $V^\mu(q, p)$, that is the velocities in terms of coordinates and momenta. We stress that the coordinates and momenta are independent variables.
- the Hamiltonian is obtained by substituting $V^\mu(q, p)$ in the energy function.

For a particle constrained to move on a surface, denoting $g_{\mu\nu}$ the metric and $g^{\mu\nu}$ its inverse, $g_{\mu\nu} g^{\nu\rho} = \delta_\mu^\rho$ we get (note that the metric is symmetric):

$$\begin{aligned} L(q, V) &= \frac{m}{2} \sum_{\mu, \nu} g_{\mu\nu} V^\mu V^\nu, \quad p_\mu = \frac{\partial L}{\partial V^\mu} = m \sum_\nu g_{\mu\nu} V^\nu \Rightarrow V^\nu = \frac{1}{m} \sum_\mu g^{\nu\mu} p_\mu, \\ E(q, V) &= m \sum_\mu V^\mu \left(\sum_\nu g_{\mu\nu} V^\nu \right) - L(q, V) = \frac{m}{2} \sum_{\mu, \nu} g_{\mu\nu} V^\mu V^\nu, \\ H(q, p) &= \frac{m}{2} \sum_{\mu, \nu} g_{\mu\nu} \left(\frac{1}{m} \sum_\rho g^{\mu\rho} p_\rho \right) \left(\frac{1}{m} \sum_\sigma g^{\nu\sigma} p_\sigma \right) = \frac{1}{2m} \sum_{\rho, \sigma} g^{\rho\sigma} p_\rho p_\sigma. \end{aligned}$$

Definition 12.9 ((Pseudo-)Riemannian manifolds). A bilinear form $g : T_p M \times T_p M \rightarrow \mathbb{R}$ that is:

- (non degenerate) $g(V, W) = 0, \forall W \Rightarrow V = 0$,

- (symmetric) $g(V, W) = g(W, V)$,
- (positively defined) $g(V, V) > 0$ for $V \neq 0$,

is called a metric or a scalar product. A manifold endowed with a metric field (i.e. the metric can depend on the basis point) is called Riemannian. Pseudo-Riemannian manifolds accommodate metrics that are only non degenerate and symmetric but not necessarily positively defined.

Note that if the metric is positively defined, then the non degenerate condition is equivalently stated as $g(V, V) = 0 \Leftrightarrow V = 0$.

A metric gives an isomorphism from $T_p M$ to $T_p^* M$. To simplify the notation, we suppress as much as possible the base point p , but we keep in mind that everything depends on $p \in M$ and implicitly on the coordinates $x^i = \lambda^i(\varphi(p))$ of its image in $\mathbb{R}^n$. We denote $g_{ij} = g(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j})$, hence $g(U, V) = \sum_{i,j=1}^n g_{ij} U^i V^j$, that is $g = \sum_{i,j=1}^n g_{ij} dx^i \otimes dx^j$. We stress that in general g_{ij} is a function (hence would be more appropriately denoted $g_{ij}(x)$) of $x = \varphi(p)$.

As g_{ij} is real symmetric, it is diagonalizable by an orthogonal transformation and all its eigenvalues are non zero (as g is non degenerate). It follows that the matrix g_{ij} is invertible and we denote its inverse g^{ij} , such that $\sum_j g^{ij} g_{jk} = \delta_k^i$. With the help of g we define an isomorphism between $T_p M$ and $T_p^* M$, called the *musical isomorphism* $V \in T_p M \rightarrow V^\flat \in T_p^* M$ and $\alpha \in T_p^* M \rightarrow \alpha^\sharp \in T_p M$:

$$V^\flat(W) = g(V, W) \Rightarrow V_j^\flat = \sum_{i=1}^n V^i g_{ij}, \quad g(\alpha^\sharp, W) = \alpha(W) \Rightarrow (\alpha^\sharp)^j = \sum_{i=1}^n g^{ij} \alpha_i,$$

The metric g also endows $T_p^* M$ with a scalar product:

$$\langle \alpha, \beta \rangle = g(\alpha^\sharp, \beta^\sharp) = \sum_{i,j,i',j'} g_{i'j'} g^{ii'} \alpha_i g^{jj'} \beta_j = \sum_{i,j} \beta_j g^{ji} \alpha_i \Rightarrow g^{ji} = \langle dx^i, dx^j \rangle$$

Exercise 12.5 (The gradient). For any function $f: M \rightarrow \mathbb{R}$, df is a form, that is an element of $T_p^* M$. We call the gradient of f , $\text{grad } f(x) \equiv \nabla f(x)$, the vector in $T_p M$ which maps to df via the musical isomorphism:

$$\nabla f = df^\sharp \Rightarrow \nabla f = \sum_{i=1}^n \frac{\partial f}{\partial x^i} g^{ij} \frac{\partial}{\partial x_j}.$$

Show that for Riemannian manifolds the gradient is the vector pointing into the direction of steepest ascent of f . Denoting $\|V\|^2 = g(V, V)$ we have, the variation of f in the direction V is the Gateaux derivative $\frac{d}{dt} f(x+tV)|_{t=0} = df(V) = \sum_i \partial_i f V^i = \sum_{j,k} g_{jk} (\sum_i \partial_i f g^{ij}) V^k = g(\nabla f, V)$. By the Cauchy-Schwarz inequality (which requires g to be positively defined) we have $|g(\nabla f, V)| \leq \|\nabla f\| \|V\|$. If g is nondegenerate and we impose $\|V\| = 1$, the inequality is satisfied only if $V = \frac{\nabla f}{\|\nabla f\|}$ (V decomposes in a component parallel to ∇f and an orthogonal component, and the inequality is satisfied only if the orthogonal component is 0). This can be summarized as: $df\left(\frac{\nabla f}{\|\nabla f\|}\right) = \|\nabla f\| = \sup_{V \in T_p M, \|V\|=1} \{|df(V)|\} = \|df\|_{\text{op}}$. In words, among all the curves at p , f increases the fastest along the ones with velocity parallel to ∇f .

12.4 Differential forms

Why differential forms? Differential forms are the objects that can be integrated over manifolds. They take a bit of getting used to.

We denote $S(k)$ the group of permutations of k elements. An elementary transposition $\tau = (ij)$ is a permutation which exchanges i and j and leaves all the other elements unchanged. Any permutation can be written in (many ways) as a product of elementary transpositions $\sigma = \prod_l (i_l j_l)$. We denote $\text{sgn}(\sigma)$ the signature of the permutations σ , that is 1 if the above product has an even number of elementary transpositions and -1 if the number is odd.

Definition 12.10 (**Linear and differential forms**). A k -linear form on $T_p M$ is a k -linear map $\eta : \times_{i=1}^k T_p M \rightarrow \mathbb{R}$, that is $(V_1, \dots, V_k) \mapsto \eta(V_1, \dots, V_k)$ is linear in each argument V_r . Denoting $V_r = \sum_{i_r} V_r^{i_r} \frac{\partial}{\partial x^{i_r}}$, hence $V_r^{i_r} = dx^{i_r}(V_r)$, for $r = 1, \dots, k$, the tensor product of k basis forms $dx^{i_1} \otimes \dots \otimes dx^{i_k}(V_1, \dots, V_k) := V_1^{i_1} \dots V_k^{i_k}$ is a k -linear form. A metric is a symmetric bilinear form. Any form writes in the standard basis as:

$$\eta(V_1, \dots, V_k) = \sum_i V_1^{i_1} \dots V_k^{i_k} \eta(\partial_{x^{i_1}}, \dots, \partial_{x^{i_k}}) = \left(\sum_i \eta_{i_1 \dots i_k} dx^{i_1} \otimes \dots \otimes dx^{i_k} \right) (V_1, \dots, V_k),$$

and $\eta(\partial_{x^{i_1}}, \dots, \partial_{x^{i_k}}) = \eta_{i_1 \dots i_k}$ are the components of the form.

A differential k -form, or a differential form of degree k , is an antisymmetric k -linear form, that is:

$$\alpha : \times_{i=1}^k T_p M \rightarrow \mathbb{R}, \quad \alpha(V_1, \dots, V_k) = \text{sgn}(\sigma) \alpha(V_{\sigma(1)}, \dots, V_{\sigma(k)}), \quad \forall \sigma \in S(k).$$

Examples of differential forms are the exterior products (see below) of 1-forms dx^i .⁶

$$dx^{i_1} \wedge \dots \wedge dx^{i_k} := \sum_{\sigma \in S(k)} \text{sgn}(\sigma) dx^{i_{\sigma(1)}} \otimes \dots \otimes dx^{i_{\sigma(k)}},$$

$$dx^{i_1} \wedge \dots \wedge dx^{i_k}(V_1 \dots V_k) = \sum_{\sigma \in S(k)} \text{sgn}(\sigma) V_1^{i_{\sigma(1)}} \dots V_k^{i_{\sigma(k)}} = \begin{vmatrix} dx^{i_1}(V_1) & \dots & dx^{i_k}(V_1) \\ \vdots & & \vdots \\ dx^{i_1}(V_k) & \dots & dx^{i_k}(V_k) \end{vmatrix}.$$

Unless otherwise specified, from now on “ k -form” will refer to a differential k -form. Due to multi linearity and antisymmetry any differential k -form writes uniquely as a linear combination:

$$\alpha = \sum_{i_1 \dots i_k} \alpha_{i_1 \dots i_k} dx^{i_1} \otimes \dots \otimes dx^{i_k} = \sum_{i_1 < i_2 < \dots < i_k} \alpha_{i_1, \dots, i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}, \quad \alpha_{i_1, \dots, i_k} = \alpha\left(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}}\right).$$

We denote I the ordered set $I = \{i_1 < i_2 < \dots < i_k\}$ and $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k}$, thus $\alpha = \sum_I \alpha_I dx^I$. The space of k -forms at p over the n -dimensional manifold M is denoted $\wedge^k T_p^* M$ and has dimension $\binom{n}{k}$. The maximal degree of a differential form is n : forms of degree higher than n are 0 while forms of degree n are proportional to $dx^1 \wedge \dots \wedge dx^n$ and are called top forms.

12.4.1 Operations on differential forms

Several operations on forms are relevant.

Definition 12.11 (**Exterior product**). Out of a k -form α and a p -form β one can build a $k+p$ form by taking the exterior product:

$$\alpha \wedge \beta(V_1, \dots, V_{k+p}) = \sum_{\alpha \in \text{Sh}(k, p)} \text{sgn}(\sigma) \alpha(V_{\sigma(1)}, \dots, V_{\sigma(k)}) \beta(V_{\sigma(k+1)}, \dots, V_{\sigma(k+p)}),$$

where $\text{Sh}(k, p) \in S(k+p)$ denotes the (k, p) shuffles, that is the permutations σ such that $\sigma(1) < \sigma(2) < \dots < \sigma(k)$ and $\sigma(k+1) < \dots < \sigma(k+p)$. The exterior product is associative, distributive with respect to the addition and compatible with the multiplication with scalars. In components, the exterior product writes $(\sum_I \alpha_I dx^I) \wedge (\sum_J \beta_J dx^J) = \sum_{I, J} \alpha_I \beta_J dx^I \wedge dx^J$. It is also graded commutative $\alpha \wedge \beta = (-1)^{kp} \beta \wedge \alpha$.

Exercise 12.6. Prove starting from the definition of the exterior product that $dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_{\sigma \in S(k)} \text{sgn}(\sigma) dx^{i_{\sigma(1)}} \otimes \dots \otimes dx^{i_{\sigma(k)}}$. Prove directly that the exterior product of dx^I and dx^J is the form we denoted $dx^I \wedge dx^J$. Prove that

⁶Some authors divide the right hand side of these equations by a $k!$, which changes some of the factors below.

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$\alpha \wedge \beta = (-1)^{kp} \beta \wedge \alpha$. As the exterior product is associative we proceed by induction:

$$(dx^{i_1} \wedge \dots \wedge dx^{i_k}) \wedge dx^{i_{k+1}}(V_1, \dots, V_{k+1}) = \sum_{\sigma \in \text{Sh}(k,1)} \text{sgn}(\sigma) \left(\sum_{\tau \in S(k)} \text{sgn}(\tau) V_{\sigma(1)}^{i_{\tau(1)}} \dots V_{\sigma(k)}^{i_{\tau(k)}} \right) V_{\sigma(k+1)}^{i_{k+1}} = \sum_{\alpha \in S(k+1)} \text{sgn}(\sigma) V_{\alpha(1)}^{i_1} \dots V_{\alpha(k)}^{i_k} V_{\alpha(k+1)}^{i_{k+1}},$$

where we used the fact that there are $k+1$ shuffles $(k,1)$ over $k+1$ elements, indexed by the choice of the element $k+1$ which we denote i_{k+1} and taking for each such shuffle all the permutations τ over the k ordered elements reconstructs all $(k+1)!$ permutations over $k+1$ elements $\alpha = \sigma \circ [\tau(i_{k+1})]$, where $\tau(i_{k+1})$ is the permutation over $k+1$ elements consisting in a cycle of length 1 for the element i_{k+1} and the permutation τ over the first k elements.

For any matrix $(k+p) \times (k+p)$ matrix B let S be the set of subsets of $k+p$ with exactly k elements. Then for every H in S :

$$\det(B) = \sum_{L \in S} \epsilon^{H,L} b_{H,L} c_{H,L},$$

with $b_{H,L}$ the minor of B obtained by deleting the rows with indices in H (always the same) and columns with indices in L (which is summed over), $c_{H,L}$ is the complementary minor obtained by keeping these lines and columns and $\epsilon^{H,L} = (-1)^{\sum_{h \in H} h + \sum_{l \in L} l}$ hence $(dx^{i_1} \wedge \dots \wedge dx^{i_k}) \wedge (dx^{i_{k+1}} \wedge \dots \wedge dx^{i_{k+p}}) = dx^{i_1} \wedge \dots \wedge dx^{i_k} \wedge dx^{i_{k+1}} \wedge \dots \wedge dx^{i_{k+p}}$.

Exercise 12.7. Prove that the exterior product is associative $(\alpha \wedge \beta) \wedge \eta = \alpha \wedge (\beta \wedge \eta)$. We have $(dx^I \wedge dx^J) \wedge dx^K = dx^I \wedge dx^J \wedge dx^K = dx^I \wedge (dx^J \wedge dx^K)$. One can also check directly, for instance:

$$\begin{aligned} dx^1 \wedge (dx^2 \wedge dx^3)(V_1, V_2, V_3) &= dx^1(V_1)(dx^2 \wedge dx^3)(V_2, V_3) - dx^1(V_2)(dx^2 \wedge dx^3)(V_1, V_3) + dx^1(V_3)(dx^2 \wedge dx^3)(V_1, V_2) \\ &= V_1^1(V_2^2 V_3^3 - V_2^3 V_3^2) - V_2^1(V_1^2 V_3^3 - V_1^3 V_3^2) + V_3^1(V_1^2 V_2^3 - V_1^3 V_2^2) = \begin{vmatrix} V_1^1 & V_1^2 & V_1^3 \\ V_2^1 & V_2^2 & V_2^3 \\ V_3^1 & V_3^2 & V_3^3 \end{vmatrix} \end{aligned}$$

and one can similarly compute $(dx^1 \wedge dx^2) \wedge dx^3$

Definition 12.12 (**Interior product**). The interior product of a k -form with a vector $V \in T_p M$ consists in setting the first argument of the form to V , that is $i_V(\alpha)(W_1, \dots, W_{k-1}) = \alpha(V, W_1, \dots, W_{k-1})$. This decreases the degree of the form by 1. One can show i_V is a graded derivation, that is it obeys a graded Leibnitz rule:

$$i_V(\alpha \wedge \beta) = (i_V \alpha) \wedge \beta + (-1)^{\deg(\alpha)} \alpha \wedge i_V \beta.$$

Exercise 12.8. Prove that $i_V(dx^{i_1} \wedge \dots \wedge dx^{i_k}) = \sum_{l=1}^k (-1)^{l-1} V^{i_l} dx^{i_1} \wedge \widehat{dx^{i_l}} \wedge \dots \wedge dx^{i_k}$, where a hat over an element in a list signifies that the element is absent from the list.

Exercise 12.9. Prove that $i_V(\alpha \wedge \beta) = (i_V \alpha) \wedge \beta + (-1)^{\deg(\alpha)} \alpha \wedge i_V \beta$.

Definition 12.13 (**The exterior derivative**). The exterior derivative $d : \wedge^k T_p^* M \rightarrow \wedge^{k+1} T_p^* M$ is the linear operator with the following additional properties:

- the functions $f : M \rightarrow \mathbb{R}$ are 0-forms and on zero forms df is the differential $df(V) = V(f)$,
- $d(df) = 0$ for a 0-form f ,
- $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg(\alpha)} \alpha \wedge d\beta$.

A form α is called closed if $d\alpha = 0$ and exact if $\alpha = d\beta$. Exact forms are closed.

Exercise 12.10. Prove that $d(\sum_I \alpha_I dx^I) = \sum_{i,I} \frac{\partial \alpha_I}{\partial x^i} dx^i \wedge dx^I$. Prove that $d^2 = 0$ for any form. We have $df = \sum_i \frac{\partial f}{\partial x^i} dx^i$ and we apply the graded Leibnitz rule treating α_I as a 0-form multiplying the k form dx^I .

Exercise 12.11. Prove that for any p -form:

$$d\alpha(V_0, \dots, V_p) = \sum_{i=0}^p (-1)^i V_i \left(\alpha(V_0, \dots, \widehat{V}_i, \dots, V_p) \right) + \sum_{i < j} (-1)^{i+j} \alpha([V_i, V_j], V_0, \dots, \widehat{V}_i, \dots, \widehat{V}_j, \dots, V_p),$$

where Lie bracket of the vector fields V and W is the vector field $[V, W](f) = V(W(f)) - W(V(f))$.

While the exterior derivative d becomes in a coordinate patch the Fréchet derivative $\mathfrak{d}$ for 0-forms f , this does not hold for higher order forms:

$$df = \mathfrak{d}f = \sum_i \frac{\partial f}{\partial x^i} dx^i, \quad d^2 f = \sum_{j,i} \frac{\partial^2 f}{\partial x^j \partial x^i} dx^j \wedge dx^i = 0, \quad \mathfrak{d}^2 f = \sum_{j,i} \frac{\partial^2 f}{\partial x^j \partial x^i} dx^j \otimes dx^i \neq 0.$$

We keep the symbol d for exterior derivatives and use $\mathfrak{d}$ for any other Fréchet derivative we encounter.

We have seen that a metric induces a mapping between $T_p M$ and $T_p^* M$. A metric can further be used to define a mapping between $\wedge^k T_p^* M$ and $\wedge^{n-k} T_p^* M$.

Definition 12.14 (The Hodge star). *The Hodge star $\star : \wedge^k T_p^* M \rightarrow \wedge^{n-k} T_p^* M$ is the linear operator which acts on basis forms as:*

$$\star(dx^{i_1} \wedge \dots \wedge dx^{i_k}) = \frac{\sqrt{|\det g|}}{(n-k)!} \sum_{j_1 \dots j_n} g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_n} dx^{j_{k+1}} \wedge \dots \wedge dx^{j_n},$$

where the Levi-Civita symbol $\epsilon_{j_1 \dots j_n}$ is zero unless $j_1, \dots, j_n$ is a permutation of $1, \dots, n$ and in that case it is the signature of that permutation.

Using ordered sets of indices, $\star dx^I = \sqrt{|\det g|} \epsilon^I_J dx^J$, where the indices are raised with g^{ij} . The action of the Hodge star on the components of a form is simply $\alpha = \sum_I \alpha_I dx^I \Rightarrow (\star \alpha)_J = \sqrt{|\det g|} \alpha^I \epsilon_{IJ}$. The Hodge star is used to define a preferred volume form in a pseudo-Riemannian manifold, the so called *unit volume*:

$$\star 1 = \frac{\sqrt{|\det g|}}{n!} \sum_{\sigma \in S(n)} \text{sgn}(\sigma) dx^{\sigma(1)} \wedge \dots \wedge dx^{\sigma(n)} = \sqrt{|\det g|} dx^1 \wedge \dots \wedge dx^n \equiv \omega.$$

On n vectors $V_1 \dots V_n$ with components $V_j = \sum_{i=1}^n (V_j)^i \frac{\partial}{\partial x^i}$, hence $g^{ik} g(\partial_k, V_j) = \sum_{k,i'} g^{ik} (V_j)^{i'} g_{ki'} = (V_j)^i = dx^i(V_j)$, the volume form evaluates as (recall that $\det(g^{ij}) = [\det g]^{-1}$):

$$\omega(V_1 \dots V_n) = \sqrt{\det g} \det(dx^i(V_j)) = \sqrt{\det g} \det\left(g^{ik} g\left(\frac{\partial}{\partial x^k}, V_j\right)\right) = \frac{1}{\sqrt{\det g}} \det(g(\partial_k, V_j)).$$

If V_i is an orthonormal basis $g(V_i, V_j) = \delta_{ij}$, then $\partial_i = \sum_k g(\partial_i, V_k) V_k$ and $g_{ij} = g(\partial_i, \partial_j) = \sum_k g(\partial_i, V_k) g(V_k, \partial_j)$ and consequently $\det g = \det[(g(\partial_i, V_j))]^2$. It follows that on an orthonormal basis the unit volume form evaluates to ± 1 (as a function of the orientation).

Several other operators on forms are often encountered: the co differential on a k form is defined as $\delta = s(-1)^{n(k-1)+1} \star d \star$ with $s = 1$ for Riemannian manifolds and $s = -1$ for Lorentzian ones (s is the sign of the determinant of the metric); the Laplace–deRham operator is $\Delta = \delta d + d\delta$.

Exercise 12.12. *Note we have a natural the scalar product between k -forms defined by the Gram determinant:*

$$\langle dx^{i_1} \wedge \dots \wedge dx^{i_k}, dx^{j_1} \wedge \dots \wedge dx^{j_k} \rangle = \begin{vmatrix} \langle dx^{i_1}, dx^{j_1} \rangle & \dots & \langle dx^{i_1}, dx^{j_k} \rangle \\ \vdots & & \vdots \\ \langle dx^{i_n}, dx^{j_1} \rangle & \dots & \langle dx^{i_n}, dx^{j_n} \rangle \end{vmatrix} = \det(g^{ij})_{i \in \{i_1, \dots, i_n\}, j \in \{j_1, \dots, j_n\}}.$$

Prove that $\alpha \wedge (\star \beta) = \langle \alpha, \beta \rangle \omega$ for any pair of forms of equal degree. We have

$$\langle \alpha, \beta \rangle = \sum_I \alpha_I \beta^I, \quad \alpha \wedge (\star \beta) = \sum_{I, I'} \alpha_I \beta_{I'} dx^I \wedge (\star dx^{I'}) = \sum_{I, I', J} \alpha_I \beta_{I'} dx^I \wedge (\sqrt{|\det g|} \epsilon^{I'}_J dx^J) = \sum_{I, I', J} \alpha_I \beta^{I'} dx^I \wedge (\sqrt{|\det g|} \epsilon_{I'J}) dx^J = \langle \alpha, \beta \rangle \omega,$$

where we note that $\epsilon_{I'J} dx^I \wedge dx^J$ is non zero only if $I = I'$ and in this case it equals $dx^1 \wedge \dots \wedge dx^n$.

Prove that $\star V^b = i_V \omega$. How about $\star(V_1^b \wedge \cdots \wedge V_k^b)$?

$$\sum_i V^i g_{ij} (\star dx^j) = \sum_i V^i g_{ij} \sqrt{|\det g|} \epsilon^j_{I} dx^I = \sum_i V^i \sqrt{|\det g|} \epsilon_{iJ} dx^J = \sum_i \sqrt{|\det g|} \epsilon_{iJ} dx^i (V) \wedge dx^J .$$

Where we use the fact that $i_V dx^I = \sum_i V^i \epsilon^I_{i, I \setminus i} dx^{I \setminus i}$ where we denote $\epsilon^I_{i, I \setminus i}$ the signature of the permutation bringing i to be the first element in the set, but not changing the order of the others.

$$\text{Similarly } \star(V_1^b \wedge \cdots \wedge V_k^b) = \sum_{i'_1, \dots, i'_k} V_1^{i'_1} g_{i'_1 i_1} \cdots V_k^{i'_k} g_{i'_k i_k} \sqrt{|\det g|} \epsilon^I_{i'_1 \dots i'_k} dx^I = \sum_i V_1^{i_1} \cdots V_k^{i_k} \sqrt{|\det g|} \epsilon_{i_1 \dots i_k} dx^I = i_{V_k} \cdots i_{V_1} \omega .$$

Exercise 12.13 (The divergence of a vector $\text{div} X$). Given a volume form ω , the divergence of a vector is defines as $(\text{div} X)\omega = d(i_X \omega)$. Give a formula for the divergence for the unit volume form. *Straightforwardly $\text{div} X = \frac{1}{\sqrt{|\det g|}} \partial_i (\sqrt{|\det g|} X^i)$*

Exercise 12.14. Compute Δf for functions and $\Delta \alpha$ for 1 forms.

12.4.2 Maxwell's equations

Maxwell's equations of electromagnetism are geometrical in nature. The space-time is a 4-dimensional Lorentzian manifold, that is the metric η has signature (signs of the eigenvalues) $(-, +, +, +)$.⁷ The gauge field is a 1-form $A = \sum_\mu A_\mu dx^\mu$. This form combines the familiar electric potential and magnetic vector potential. The field strength is $F = dA = \sum_{\mu, \nu} \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu$, and its components are the electric and magnetic fields. The sources (charges and currents) are a one form $J = \sum_\mu J_\mu dx^\mu$. Then Maxwell's equations write:⁸

$$dF = 0, \quad d(\star F) = \star J,$$

where $\star$ is the four dimensional Hodge star in the Lorentzian manifold and d the four dimensional exterior derivative. These equations are clearly invariant under a gauge transformation consisting in adding to A a total differential $A \rightarrow A + df$, as the field strength $F = dA$ is invariant. The Lorentz gauge condition writes $d(\star A) = 0$, thus in the vacuum $J = 0$ in the Lorentz gauge we have $d \star dA = 0 \Rightarrow \Delta A = 0$, that is the gauge potential obeys the wave equation. The four velocity V of a massif point particle is the tangent vector to its trajectory $\gamma(\tau)$, $V(f) = \frac{d}{d\tau}(f \circ \gamma)$, parametrized such that $\eta(V, V) = -1$ (due to our choice of signature). The Lorentz force acting on the particle is $q i_V F$ (or $\frac{q i_V F}{\sqrt{-\eta(V, V)}}$ in an arbitrary parametrization).

In order to pass to the three dimensional vector formulation of the Maxwell equations we consider a four dimensional metric $\sum_{\mu, \nu=1}^4 \eta_{\mu\nu} dx^\mu dx^\nu = -dt^2 + \sum_{i,j=1}^3 g_{ij} dx^i dx^j$, where g is a time independent space Riemannian metric and we separate the time and space components of all the quantities. From now on the indices i are summed from 1 to 3. The four volume form is $\sqrt{|\det g|} dt \wedge dx^1 \wedge dx^2 \wedge dx^3 = dt \wedge \omega$, with $\omega = \sqrt{|\det g|} dx^1 \wedge dx^2 \wedge dx^3$ the three dimensional volume form and denoting $\star_\eta$ the four dimensional Hodge star and $\star$ the three dimensional one, using $\star_\eta dx^I = \sqrt{|\det \eta|} \epsilon^I_J dx^J$ we obtain the four dimensional Hodge duals:

$$\begin{aligned} \star_\eta (dt \wedge dx^i) &= \sqrt{|\det g|} g^{0i} g_{j < k} dx^j \wedge dx^k = -(\star dx^i), & \star_\eta (dx^i \wedge dx^j)_{i < j} &= \sqrt{|\det g|} \epsilon^{ij}_{0k} dt \wedge dx^k = dt \wedge (\star(dx^i \wedge dx^j)), \\ \star_\eta dt &= \sqrt{|\det g|} \epsilon^0_{123} dx^1 \wedge dx^2 \wedge dx^3 = -\omega, & \star_\eta dx^i &= \sqrt{|\det g|} \epsilon^i_{0j < k} dt \wedge dx^j \wedge dx^k = -dt \wedge (\star dx^i). \end{aligned}$$

From now on we denote the four dimensional exterior differential as $d^{(4)}$ and we reserve d for the three dimensional exterior differential which acts on functions as $df = \sum_i \frac{\partial f}{\partial x_i} dx^i$. Finally, $\sharp, \flat$ denote the three dimensional musical isomorphism. We denote by an arrow the three dimensional vectors and we introduce the three dimensional differential operators $\nabla \cdot \vec{X}$

⁷Some authors prefer $(+, -, -, -)$.

⁸These are the Euler Lagrange equations for the electromagnetic action $S = -\frac{1}{4} \int F \wedge \star F$.

and $\nabla \times \vec{X}$ acting on three dimensional vectors $\vec{X}$:⁹

$$\begin{aligned} (\nabla \cdot \vec{X})\omega &= d(i_{\vec{X}}\omega), & \nabla \cdot \vec{X} &= \text{div} \vec{X} = \frac{1}{\sqrt{\det g}} \sum_i \partial_i (\sqrt{\det g} X^i), \\ d\vec{X}^\flat &= i_{\nabla \times \vec{X}}\omega, & \nabla \times \vec{X} &= \text{rot} \vec{X} = \frac{1}{\sqrt{\det g}} \sum_{ijk} \epsilon^{ijk} (\partial_j \sum_l g_{kl} X^l) \partial_i, \end{aligned}$$

Observe that $i_{\vec{X}}\omega = \star \vec{X}^\flat$ and we recall that for any function f we have defined $\text{grad} f = \nabla f = (df)^\sharp = \sum_{i,j} g^{ij} \partial_i f \partial_j$. Finally, the square of the Hodge star acting on k forms in n dimensions is $(-1)^{k(n-k)}$, hence it squares to the identity in odd dimensions.

The electric and magnetic potential φ and $\vec{A} = \sum_i A^i \partial_i$ are the four vector $\varphi \partial_t + \vec{A}$ associated to the gauge field, that is the gauge form field is $A = -\varphi dt + \vec{A}^\flat = -\varphi dt + \sum_i A_i dx^i = -\varphi dt + \sum_{i,j} g_{ij} A^i dx^j$ and the electromagnetic tensor writes:

$$F = d^{(4)}A = -d\varphi \wedge dt + \sum_i \partial_t A_i dt \wedge dx^i + d\vec{A}^\flat = (-d\varphi - \sum_i \partial_t A_i dx^i) \wedge dt + d\vec{A}^\flat = E \wedge dt + B,$$

and we identify the electric one form $E = -d\varphi - \partial_t \vec{A}^\flat$ and magnetic two form $B = d\vec{A}^\flat = \sum_{i,j} \frac{1}{2} (\partial_i A_j - \partial_j A_i) dx^i \wedge dx^j$. The electric and magnetic vector fields are defined respectively as $\vec{E} = E^\sharp$ and $\vec{B} = (\star B)^\sharp$ and write in terms of the potentials as:

$$\vec{E} = -\nabla \varphi - \partial_t \vec{A}, \quad \vec{B} = (\star d\vec{A}^\flat)^\sharp = (\star i_{\nabla \times \vec{A}} \omega)^\sharp = (\star \star (\nabla \times \vec{A})^\flat)^\sharp = \nabla \times \vec{A}.$$

The dual electromagnetic tensor is $\star_\eta F = \star_\eta (E \wedge dt) + \star_\eta B = \star E + dt \wedge \star B$. Denoting $\partial_t B$ the two form whose components are the partial derivatives with respect to the time t of the components of B we have:

$$\begin{aligned} d^{(4)}F &= dE \wedge dt + \partial_t B \wedge dt + dB = (d\vec{E}^\flat + \star \partial_t \vec{B}^\flat) \wedge dt + d(\star \vec{B}^\flat) \\ &= \star (\nabla \times \vec{E} + \partial_t \vec{B})^\flat \wedge dt + d(i_{\vec{B}} \omega) = \star (\nabla \times \vec{E} + \partial_t \vec{B})^\flat \wedge dt + (\nabla \cdot \vec{B}) \omega, \\ d^{(4)}(\star_\eta F) &= d(\star E) + \partial_t (\star E) \wedge dt - dt \wedge d(\star B) = (\nabla \cdot \vec{E}) \omega + (\star \partial_t \vec{E}^\flat - d\vec{B}^\flat) \wedge dt \\ &= (\nabla \cdot \vec{E}) \omega + \star (\partial_t \vec{E} - \nabla \times \vec{B})^\flat \wedge dt. \end{aligned}$$

The charge density ρ and current vector $\vec{j} = \sum j^i \partial_i$ form the sources four vector $\rho \partial_t + \vec{j}$ associated to the sources one form $J = -\rho dt + \vec{j}^\flat$ and $\star_\eta J = \rho \omega - \star \vec{j}^\flat \wedge dt$ therefore the Maxwell equations for three dimensional vectors write:

$$\nabla \cdot \vec{E} = \rho, \quad \nabla \times \vec{E} + \partial_t \vec{B} = 0, \quad \nabla \cdot \vec{B} = 0, \quad \partial_t \vec{E} - \nabla \times \vec{B} = -\vec{j}.$$

The sources respect the continuity equation:

$$0 = d^{(4)}(\star_\eta J) = \partial_t \rho dt \wedge \omega - d(\star \vec{j}^\flat) \wedge dt = \partial_t \rho dt \wedge \omega - d(i_{\vec{j}} \omega) \wedge dt = (\partial_t \rho + \nabla \cdot \vec{j}) dt \wedge \omega.$$

Note that the covariant Maxwell equations not only allow one to consider electromagnetism in any dimension (the rotational is defined only in 3 dimensions), but also do not assume a time independent ADM split of the metric into a foliation with constant laps function and zero shift vector.

⁹We have $i_H \omega = \frac{\sqrt{\det g}}{2} \sum_{ijk} \epsilon_{ijk} H^i dx^j \wedge dx^k$ and $\sum_i \epsilon_{ijk} \epsilon^{ilm} = \delta_j^l \delta_k^m - \delta_j^m \delta_k^l$.

12.5 Pushforward, pullback and the volume form

Consider two manifolds Σ and M and a function $\phi : \Sigma \rightarrow M$. We consider Σ to be k -dimensional, we denote $\varphi_\Sigma : \Sigma \rightarrow \mathbb{R}^k$ a chart around $p \in \Sigma$ and $s^\mu, \mu = 1, \dots, k$ local coordinates around $\varphi_\Sigma(p)$. We furthermore consider M to be n -dimensional, we denote $\varphi_M : M \rightarrow \mathbb{R}^n$ a chart around $\phi(p) \in M$ and $x^i, i = 1, \dots, n$ local coordinate around $\varphi_M(\phi(p))$. The function ϕ is called smooth if for any choice of coordinates $\varphi_M \circ \phi \circ \varphi_\Sigma^{-1} : \phi_\Sigma(U) \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$ is smooth. We denote somewhat abusively $\phi = (x^1(s), \dots, x^n(s))$ the map ϕ in the coordinates x on M and s on Σ .

The functions $f \in C^\infty(M)$ are *pulled back* via ϕ to functions in $\phi^*f \in C^\infty(\Sigma)$ by :

$$\phi^*f = f \circ \phi, \quad (f \circ \phi)(s) = f(x(s)).$$

We denote ϕ_* the *pushforward* that takes $V \in T_p\Sigma$ into the vector $\phi_*V \in T_pM$ which acts on functions $f \in C^\infty(M)$ as:

$$\begin{aligned} \phi_*V(f) &= V(\phi^*f) = V(f \circ \phi), \\ V &= \sum_\mu V^\mu(s) \frac{\partial}{\partial s^\mu} \mapsto \phi_*(V) = \sum_{i,\mu} V^\mu(s) \frac{\partial x^i}{\partial s^\mu} \bigg|_s \frac{\partial}{\partial x^i} \bigg|_{x(s)} \Rightarrow \phi_* \big|_{\phi(p)} = \sum_{i,\mu} \left(\frac{\partial}{\partial x^i} \right)_{\phi(p)} \left(\frac{\partial x^i}{\partial s^\mu} \right)_p (ds^\mu)_p, \end{aligned}$$

that is, in coordinates the pushforward is the Fréchet differential of $\varphi_M \circ \phi \circ \varphi_\Sigma^{-1} : \mathbb{R}^k \rightarrow \mathbb{R}^n$. In particular, $\phi_*(\partial_{s^\mu}) = \sum_i \frac{\partial x^i}{\partial s^\mu} \partial_{x^i}$.

While vector fields are pushed forward by ϕ_* , forms (be it differential or not), like functions, are pulled back by the *pullback* map $\phi^* : \wedge^q T_{\phi(p)}^*M \rightarrow \wedge^q T_p^*\Sigma$:

$$\phi^*\alpha \big|_p(V_1, \dots, V_q) = \alpha \big|_{\phi(p)}(\phi_*(V_1), \dots, \phi_*(V_q)), \quad \phi^*(\alpha \wedge \beta) = \phi^*\alpha \wedge \phi^*\beta.$$

In words, the pullback $\phi^*\alpha$ is the form on $T_p\Sigma$ which acting on any vectors in $T_p\Sigma$ reproduces the action of α on their image through the pushforward. The pullback is particularly simple on differentials of functions:

$$\phi^*(df)(V) = df(\phi_*V) = \phi_*V(f) = V(\phi^*f) = d(\phi^*f)(V).$$

Exercise 12.15. Prove that for any differential form α the pullback is compatible with the exterior derivative $\phi^*(d\alpha) = d(\phi^*\alpha)$ and with the exterior product $\phi^*(\alpha \wedge \beta) = (\phi^*\alpha) \wedge (\phi^*\beta)$.

In coordinates, as $\phi_*(\partial_{s^\mu}) = \sum_i \frac{\partial x^i}{\partial s^\mu} \partial_{x^i}$ we conclude that $\phi^*(dx^i) = \sum_\mu \frac{\partial x^i}{\partial s^\mu} ds^\mu$ and:

$$\alpha = \sum_{i_1 < \dots < i_l} \alpha_{i_1 \dots i_l}(x) dx^{i_1} \wedge \dots \wedge dx^{i_l} \mapsto \phi^*\alpha = \sum_{i_1 < \dots < i_l} \sum_{\mu_1 < \dots < \mu_l} \alpha_{i_1 \dots i_l}(x(s)) \begin{vmatrix} \frac{\partial x^{i_1}}{\partial s^{\mu_1}} & \dots & \frac{\partial x^{i_1}}{\partial s^{\mu_l}} \\ \vdots & & \vdots \\ \frac{\partial x^{i_l}}{\partial s^{\mu_1}} & \dots & \frac{\partial x^{i_l}}{\partial s^{\mu_l}} \end{vmatrix} ds^{\mu_1} \wedge \dots \wedge ds^{\mu_l}.$$

Consider a Riemannian manifold and its unit volume form $\star 1 = \sqrt{\det g(x)} dx^1 \wedge \dots \wedge dx^n \equiv \omega$. As we have mentioned, any embedded manifold $\Sigma \subset M$ has a natural embedding $\phi : \Sigma \rightarrow M$, $\phi(p \in \Sigma) = p \in M$, the identity. Contrary to appearances, this embedding is not trivial: the map $\varphi_M \circ \varphi_\Sigma^{-1} : \mathbb{R}^k \rightarrow \mathbb{R}^n$ can be quite involved, for instance $\phi(s) = (x^1(s) = s^1, x^2(s) = s^2, x^3(s) = \frac{(s^1)^2 + (s^2)^2}{2})$ is the embedding of a paraboloid in $\mathbb{R}^3$. The pushforward map of the natural embedding is $\phi_* = \sum_{i,\mu} \frac{\partial x^i}{\partial s^\mu} \partial_{x^i}$ and for the embedded paraboloid for instance this becomes:

$$\phi_* = \begin{pmatrix} \frac{\partial}{\partial x^1} & \frac{\partial}{\partial x^2} & \frac{\partial}{\partial x^3} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ s^1 & s^2 \end{pmatrix} \begin{pmatrix} ds^1 \\ ds^2 \end{pmatrix} \Rightarrow \phi_* = \frac{\partial}{\partial x^1} ds^1 + \frac{\partial}{\partial x^2} ds^2 + \frac{\partial}{\partial x^3} (s^1 ds^1 + s^2 ds^2),$$

where ∂_{x^1} , ∂_{x^2} and ∂_{x^3} are located at the base point $x^1 = s^1, x^2 = s^2, x^3 = \frac{(s^1)^2 + (s^2)^2}{2}$ in $\mathbb{R}^3$.

If M is a Riemannian manifold then the metric, which is a linear 2-form, can be pulledback to $G : T_p\Sigma \times T_p\Sigma \rightarrow \mathbb{R}$:

$$\begin{aligned} G(U, V) &= (\phi^*g)(U, V) = g(\phi_*(U), \phi_*(V)) = \sum_{i,j,\mu,\nu} g_{ij} \left(\frac{\partial x^i}{\partial s^\mu} \right) \left(\frac{\partial x^j}{\partial s^\nu} \right) U^\mu V^\nu \\ &= \left(\sum_{\mu,\nu} G_{\mu\nu} ds^\mu \otimes ds^\nu \right) (U, V), \quad G_{\mu\nu} = \sum_{i,j} g_{ij} \left(\frac{\partial x^i}{\partial s^\mu} \right) \left(\frac{\partial x^j}{\partial s^\nu} \right), \end{aligned}$$

and G is the metric on Σ which is consistent with the metric g on M (that is they agree on vectors). For the paraboloid the standard metric in $\mathbb{R}^3$ pulls back to $(ds^1)^2 + (ds^2)^2 + (s^1 ds^1 + s^2 ds^2)^2 = [1 + (s^1)^2](ds^1)^2 + [1 + (s^2)^2](ds^2)^2 + 2s^1 s^2 ds^1 ds^2$. The volume form¹⁰ on Σ is the volume form associated to the pulled back metric, $\omega_\Sigma = \sqrt{\det G} ds^1 \wedge \cdots \wedge ds^k$. For the paraboloid for instance we get: $\omega_\Sigma = \sqrt{1 + (s^1)^2 + (s^2)^2} ds^1 \wedge ds^2$.

Exercise 12.16. *How does the volume form change under a change of variables?* A change of variables is an embedding of a manifold in itself with the identity map. Under a change of coordinates $x = x(y)$ we have:

$$\frac{\partial}{\partial x^i} = \sum_{i'} \frac{\partial y^{i'}}{\partial x^i} \frac{\partial}{\partial y^{i'}}, \quad g(x)_{ij} = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) = \sum_{i',j'} \frac{\partial y^{i'}}{\partial x^i} \frac{\partial y^{j'}}{\partial x^j} g\left(\frac{\partial}{\partial y^{i'}}, \frac{\partial}{\partial y^{j'}}\right) = \sum_{i',j'} \frac{\partial y^{i'}}{\partial x^i} \frac{\partial y^{j'}}{\partial x^j} g(y)_{i'j'},$$

where $\frac{\partial y^{i'}}{\partial x^i}$ is the Jacobian of the change of variables. Expressed in the new coordinates the volume form is (we assume that the change preserves the orientation):

$$\sqrt{\det g(x)} dx^1 \wedge \cdots \wedge dx^n = \sqrt{\det g(y)} \det\left(\frac{\partial y}{\partial x}\right) \sum_{j_1 \dots j_n} \frac{\partial x^1}{\partial y^{j_1}} \cdots \frac{\partial x^n}{\partial y^{j_n}} dy^{j_1} \wedge \cdots \wedge dy^{j_n} = \sqrt{\det g(y)} \det\left(\frac{\partial y}{\partial x}\right) \det\left(\frac{\partial x}{\partial y}\right) dy^1 \wedge \cdots \wedge dy^n = \sqrt{\det g(y)} dy^1 \wedge \cdots \wedge dy^n,$$

as $\sum_i \frac{\partial y^{i'}}{\partial x^i} \frac{\partial x^i}{\partial y^{j'}} = \delta_{j'}^{i'}$.

Exercise 12.17. *In spherical coordinates (r, θ, φ) a point in $\mathbb{R}^3$ with metric $g_{ij} = \delta_{ij}$ writes $(x^1, x^2, x^3) = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$. Write the metric and the volume form in spherical coordinates. The sphere S^2 is defined as the embedded manifold with constant r . What is the volume form to the embedded sphere S^2 ?* The Jacobian is:

$$\frac{\partial(x^1, x^2, x^3)}{\partial(r, \theta, \varphi)} = \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix} \Rightarrow G = \begin{pmatrix} \sin \theta \cos \varphi & \sin \theta \sin \varphi & \cos \theta \\ r \cos \theta \cos \varphi & r \cos \theta \sin \varphi & -r \sin \theta \\ -r \sin \theta \sin \varphi & r \sin \theta \cos \varphi & 0 \end{pmatrix} \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

The pullback metric to the embedded sphere $(\theta, \varphi) \mapsto (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$ is $G_{\mu\nu} ds^\mu ds^\nu = r^2[(d\theta)^2 + \sin^2 \theta (d\varphi)^2]$.

12.6 The normal(s) to a subamifold

Lemma 12.2. *Let V be a vector space of dimension n and let $\alpha_1, \dots, \alpha_k \in V^*$ be linearly independent linear forms on V and β a linear form on V . Then, $\dim(\cap_{i=1}^k \ker \alpha_i) = n - k$ and $\cap_{i=1}^k \ker \alpha_i \subset \ker \beta$ if and only if $\beta = \sum_{i=1}^k a_i \alpha_i$.*

Proof. If $\beta = \sum_{i=1}^k a_i \alpha_i$ the implication is trivial.

For the converse, for any linear form, $\ker \alpha$ is a $n-1$ dimensional vector space by the rank nullity theorem. We proceeded by induction on k . Take $(\alpha_i)_{i=1 \dots k}$ linearly independent forms and select for $i = 1, \dots, k$ a $v_i \notin \ker \alpha_i$ (hence $\alpha_i(v_i) \neq 0$) and note by induction that, as the α s are independent $\cap_{j \neq i} \ker \alpha_j \not\subset \ker \alpha_i$ hence v_i can be chosen such that $\alpha_j(v_i) = 0$ for $j \neq i$.

Choose a basis $(e_b)_b$ in $K = \cap_{i=1}^k \ker \alpha_i$. Then $\{(v_i)_i(e_b)_b\}$ is a basis in V . Indeed, constructing $\cap_{i=1}^j \ker \alpha_i$ step by step, at each step $\cap_{i=1}^j \ker \alpha_i \not\subset \ker \alpha_{j+1}$ (by linear independence) hence at each step the dimension of $\cap_{i=1}^j \ker \alpha_i$ must strictly decrease. As this dimension can decrease by at most 1 at each step,¹¹ in the end $\dim K = n - k$. Choosing a basis $(e_b)_b$ in K , $\{v_1, (e_b)\}$ are $n - k + 1$ linearly independent vectors in $\cap_{i=2}^k \ker \alpha_i$ hence form a basis $\cap_{i=2}^k \ker \alpha_i$. Proceeding by induction we conclude that $\{(v_i)_i(e_b)_b\}$ is a basis in V .

¹⁰If Σ has dimension 2 we sometimes call this volume form the area form, while if Σ has dimension 1 we sometimes call it the length form.

¹¹Recall that if Y_1 and Y_2 are subspaces of X , then $\dim X + \dim(Y_1 \cap Y_2) \geq \dim(Y_1) + \dim(Y_2)$ and at each step the $\dim \ker \alpha_j = n - 1$.

Then for any β such that $\bigcap_{i=1}^k \ker \alpha_i \subset \ker \beta$, the linear form $\left[\prod_i \alpha_i(v_i) \right] \beta - \sum_i \beta(v_i) [\prod_{j \neq i} \alpha_j(v_j)] \alpha$ is identically zero as it is zero on the chosen basis. $\square$

Let $M = \mathbb{R}^n$ and let Σ be a smooth embedded submanifold of M of dimension $n - k$. As we have seen Σ can be specified by the natural embedding $\phi : \Sigma \rightarrow M$ with pushforward map ϕ_* . Alternatively, Σ can be specified as the solution set $\Sigma = \{x \in M \mid F^c(x) = 0, \quad c = 1, \dots, k\}$ of k independent constraints $F^c : M \rightarrow \mathbb{R}$ for which 0 is a regular value, that is $dF^c|_{\Sigma}$ are linearly independent. In other words, there exists $F : M \rightarrow \mathbb{R}^k$ such that dF is maximal rank on $F^{-1}(0) = \Sigma \subset M$. Note that for any $V \in T_p \Sigma$:

$$(\phi^* dF^c)(V) = dF^c(\phi_*(V)) = \phi_*(V)(F^c) = V(\phi^* F^c) = 0,$$

as $\phi^* F^c = F^c \circ \phi = F^c|_{\Sigma} = 0$, that is the forms $dF^c|_{\Sigma}$ are the *normal* to the submanifold Σ and $\phi_*(T_p \Sigma) \subset \ker(dF|_{\phi(p)}) = \bigcap_{c=1}^k \ker dF^c|_{\phi(p)}$. In fact, as dF^c are k linearly independent linear forms, from the previous lemma, $\dim(\bigcap_{c=1}^k \ker dF^c|_{\phi(p)}) = n - k$. On the other hand $\dim(\phi_*(T_p \Sigma)) = n - k$ and as both are vector spaces we conclude that:

$$\phi_*(T_p \Sigma) = \ker(dF|_{\phi(p)}) = \bigcap_{c=1}^k \ker dF^c|_{\phi(p)}.$$

In coordinate charts, using the embedding $\phi(s) = (x^1(s), \dots, x^n(s))$ we have:

$$\phi^*(dF^c) = \sum_{i,\mu} \partial_i F^c \frac{\partial x^i}{\partial s^\mu} ds^\mu = 0 \Leftrightarrow \sum_i \partial_i F^c(x(s)) \frac{\partial x^i}{\partial s^\mu} = 0,$$

hence, given the embedding, one can solve for the normal forms $dF^c|_{\phi(s)}$. We stress that the equation for the normal form can only be solved on the subset $\phi(\Sigma) \subset M$.

Consider for instance the embedded paraboloid with natural embedding $\phi(s) = (s^1, s^2, \frac{(s^1)^2 + (s^2)^2}{2})$. The paraboloid is alternatively defined by the constraint:

$$F(x) = x^3 - \frac{(x^1)^2 + (x^2)^2}{2} = 0, \quad dF = dx^3 - x^1 dx^1 - x^2 dx^2,$$

and 0 is a regular value for F (that is dF is surjective on $F^{-1}(0)$). For example, the point $(1, 1, 1)$ belongs to the paraboloid and $dF|_{(1,1,1)} = dx^3 - dx^1 - dx^2$. At a base point on the paraboloid $x^1 = s^1, x^2 = s^2, x^3 = \frac{(s^1)^2 + (s^2)^2}{2}$, the pullback of dF to Σ via the natural embedding is:

$$\phi^*(dF) = \phi^*(dx^3 - x^1 dx^1 - x^2 dx^2) = (s^1 ds^1 + s^2 ds^2) - s^1 ds^1 - s^2 ds^2 = 0.$$

Now, $\ker dF$ consists in the vectors $W = W^1 \partial_{x^1} + W^2 \partial_{x^2} + W^3 \partial_{x^3}$ such that $dF(W) = W^3 - x^1 W^1 - x^2 W^2 = 0$, that is the bidimensional space generated by the vectors $(1, 0, x^1)$ and $(0, 1, x^2)$. The pushforward map is the Fréchet differential:

$$\phi_* = \partial_{x^1} ds^1 + \partial_{x^2} ds^2 + \partial_{x^3} (s^1 ds^1 + s^2 ds^2),$$

and denoting $V = V^1 \partial_{s^1} + V^2 \partial_{s^2}$, the subspace $\phi_*(T_p \Sigma)$ is $\partial_{x^1} V^1 + \partial_{x^2} V^2 + \partial_{x^3} (s^1 V^1 + s^2 V^2)$ at the base point $x^1 = s^1, x^2 = s^2, x^3 = \frac{(s^1)^2 + (s^2)^2}{2}$, that is $\phi_*(T_p \Sigma) = \ker dF|_{\phi(p)}$.

If one is given only the embedding $\phi(s) = (s^1, s^2, \frac{(s^1)^2 + (s^2)^2}{2})$, the normal form is the form $\lambda = \sum_i \lambda_i dx^i$ such that $\phi^* \lambda = 0$, that is:

$$\lambda_1 ds^1 + \lambda_2 ds^2 + \lambda_3 (s^1 ds^1 + s^2 ds^2) = 0, \quad \lambda|_{\phi(s)} = \lambda_3 (-s^1 dx^1|_{\phi(s)} - s^2 dx^2|_{\phi(s)} + dx^3|_{\phi(s)}) = \lambda_3 dF|_{\phi(s)}.$$

12.6.1 Normal vectors

In general submanifolds have normal forms. If we endow $M = \mathbb{R}^n$ with a metric g (and consequently with a standard volume form ω), we can trade the normal forms for normal vectors by the musical isomorphism.

Normal vectors to Σ are the non trivial vectors $\nu_c = \sum_i \nu_c^i \partial_{x^i}$ in $T_{\phi(p)}M$ for $p \in \Sigma$ which are orthogonal on all the vectors in the push forward $\phi_*(T_p\Sigma) \subset T_{\phi(p)}M$, that is:

$$g(\nu, \phi_*(V)) = 0, \quad \forall V \in T_p\Sigma.$$

This writes in terms of the 1-forms $\nu_b = \sum_{ij} g_{ij} \nu^i dx^j \in T_{\phi(p)}^*M$ as $\nu_b(\phi_*(V)) = \phi_*\nu_b(V) = 0$. If Σ is specified by $n - k$ constraints, as $\phi_*(T_p\Sigma) = \bigcap_{c=1}^k \ker dF^c|_{\phi(p)}$ we obtain a basis of normal vectors $\nu_c = (dF^c)^\sharp$. If we are only given the embedding map, normalizing by $g(\nu, \nu) = 1$, in coordinates the unit normal vectors respect:

$$\phi^*\nu_b = 0 \Rightarrow \sum_{i,j} \nu_c^i g_{ij} \frac{\partial x^j}{\partial s^\mu} = 0, \quad \mu = 1 \dots n; \quad \sum_{ij} g_{ij} \nu_c^i \nu_c^j = 1,$$

And $n - k$ linearly independent unit normal vectors always exists as $\phi_* = d\phi$ has maximal rank (otherwise ϕ is not an embedding) and are determined only up to overall signs.

For the paraboloid embedded in $\mathbb{R}^3$ with standard metric, the normal vector is determined by:

$$\begin{pmatrix} \nu^1 & \nu^2 & \nu^3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ s^1 & s^2 \end{pmatrix} = 0 \Rightarrow \nu^1 + s^1 \nu^3 = \nu^2 + s^2 \nu^3 = 0, \quad (\nu^1)^2 + (\nu^2)^2 + (\nu^3)^2 = 1,$$

that is $\nu = \frac{-s^1}{\sqrt{1+(s^1)^2+(s^2)^2}} \partial_{x^1} + \frac{-s^2}{\sqrt{1+(s^1)^2+(s^2)^2}} \partial_{x^2} + \frac{1}{\sqrt{1+(s^1)^2+(s^2)^2}} \partial_{x^3}$, where ∂_{x^i} are located at the base point $x^1 = s^1, x^2 = s^2, x^3 = \frac{(s^1)^2 + (s^2)^2}{2}$ in $\mathbb{R}^3$.

The unit normals are particularly useful because the volume form on Σ , $\omega_\Sigma = \sqrt{\det G} ds^1 \wedge \dots \wedge ds^n$ with $G_{\mu\nu} = \sum_{i,j} g_{ij} \left(\frac{\partial x^i}{\partial s^\mu} \right) \left(\frac{\partial x^j}{\partial s^\nu} \right)$ the pullback of the metric on M to Σ is also the pullback of the volume form $\omega = \sqrt{\det g} dx^0 \wedge \dots \wedge dx^n$ on M evaluated on the unit normal:

$$\omega_\Sigma = \phi^*(i_\nu \omega), \quad \omega_\Sigma = \sqrt{\det g} \sum_i (-1)^i \nu^i \frac{\partial(x^0, \dots, \widehat{x^i}, \dots, x^n)}{\partial(s^1, \dots, s^n)} ds^1 \dots \wedge ds^n = \sqrt{\det G} ds^1 \dots \wedge ds^n.$$

This follows straightforwardly by observing that:

$$\begin{aligned} \sum_i (-1)^i \nu^i \frac{\partial(x^0, \dots, \widehat{x^i}, \dots, x^n)}{\partial(s^1, \dots, s^n)} &= \left| \nu^i \right|_{\frac{\partial x^i}{\partial s^\nu}} = \begin{vmatrix} \nu^0 & \dots & \nu^n \\ \frac{\partial x^0}{\partial s^1} & \dots & \frac{\partial x^n}{\partial s^1} \\ \vdots & & \vdots \\ \frac{\partial x^0}{\partial s^n} & \dots & \frac{\partial x^n}{\partial s^n} \end{vmatrix}, \\ \left(\nu^i \right)_{\frac{\partial x^i}{\partial s^\nu}} g_{ij} \left(\nu^j \right)_{\frac{\partial x^j}{\partial s^\nu}} &= \begin{pmatrix} \sum_{ij} g_{ij} \nu^i \nu^j & \sum_{ij} \nu^i g_{ij} \frac{\partial x^j}{\partial s^\nu} \\ \sum_{ij} \frac{\partial x^i}{\partial s^\nu} g_{ij} \nu^j & \sum_{ij} g_{ij} \frac{\partial x^i}{\partial s^\nu} \frac{\partial x^j}{\partial s^\mu} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & G \end{pmatrix}. \end{aligned}$$

For the paraboloid for instance:

$$i_\nu \omega = \frac{-s^1}{\sqrt{1+(s^1)^2+(s^2)^2}} dx^2 \wedge dx^3 - \frac{-s^2}{\sqrt{1+(s^1)^2+(s^2)^2}} dx^1 \wedge dx^3 + \frac{1}{\sqrt{1+(s^1)^2+(s^2)^2}} dx^1 \wedge dx^2 ,$$

$$\phi^*(i_\nu \omega) = \frac{-s^1 ds^2 \wedge (s^1 ds^1 + s^2 ds^2)}{\sqrt{1+(s^1)^2+(s^2)^2}} + \frac{s^2 ds^1 \wedge (s^1 ds^1 + s^2 ds^2)}{\sqrt{1+(s^1)^2+(s^2)^2}} + \frac{(ds^1 \wedge ds^2)}{\sqrt{1+(s^1)^2+(s^2)^2}} = \sqrt{1+(s^1)^2+(s^2)^2} ds^1 \wedge ds^2 .$$

reproducing the volume form associated to the pullback of the metric $(ds^1)^2 + (ds^2)^2 + (s^1 ds^1 + s^2 ds^2)^2$.

Exercise 12.18 (The normal bundle). *Let us consider n -dimensional manifold M and a k -dimensional submanifold Σ embedded in M . For any $p \in \Sigma$, the pushforward of $T_p \Sigma$ under the natural embedding (which we denote abusively also by $T_p \Sigma$) is a linear subspace of $T_p M$. We define the normal subspace to Σ at p as:*

$$N_p \Sigma = (T_p \Sigma)^\perp = \{N \in T_p M \mid g(N, X) = 0, \forall X \in T_p \Sigma\} .$$

As $\dim T_p \Sigma = k$ and $\dim T_p M = n$ we conclude that $\dim N_p \Sigma = n - k$, which we call the codimension of Σ . Fix an orthonormal basis $N_1, \dots, N_{n-k}$ in the normal space and prove that the volume form on Σ is (up to orientation) $\omega_\Sigma = \phi^*(i_{N_{n-k}} \dots i_{N_1} \omega)$.

Exercise 12.19. *The toroidal coordinate system (v, ξ, φ) with $v \in [0, \infty)$, $\xi \in [0, 2\pi)$, and $\varphi \in [0, 2\pi)$ in $\mathbb{R}^3$ is defined by:*

$$x = a \frac{\sinh(v)}{\cosh(v) - \cos(\xi)} \cos(\varphi) , \quad y = a \frac{\sinh(v)}{\cosh(v) - \cos(\xi)} \sin(\varphi) , \quad z = a \frac{\sin(\xi)}{\cosh(v) - \cos(\xi)} .$$

We denote the natural embedding of $\mathbb{R}^3$ in toroidal coordinates into $\mathbb{R}^3$ in Cartesian coordinates by $\phi(u, v, \varphi) = (x(u, v, \varphi), y(u, v, \varphi), z(u, v, \varphi))$ and $\eta = dx^2 + dy^2 + dz^2$ the Euclidean metric in Cartesian coordinates.

Compute the Fréchet differential $d\phi \equiv \phi_*$, the pullback of the Euclidean metric $g = \phi^*(\eta)$, the volume element ω in toroidal coordinates and the push forward of the basis vectors $\phi_*(\partial_\phi)$, $\phi_*(\partial_\varphi)$ and $\phi_*(\partial_v)$. Note that these vectors are mutually orthogonal. We note that $[1 - \cosh(v) \cos(\xi)]^2 + \sinh^2(v) \sin^2(\xi) = [\cosh(v) - \cos(\xi)]^2$. We have:

$$d\phi \equiv \phi_* = \frac{\partial}{\partial x^i} \frac{\partial x^i}{\partial s^\mu} ds^\mu = \begin{pmatrix} \partial_x & \partial_y & \partial_z \end{pmatrix} \begin{pmatrix} a \frac{1 - \cosh(v) \cos(\xi)}{[\cosh(v) - \cos(\xi)]^2} \cos(\varphi) & -a \frac{\sinh(v) \sin(\xi)}{[\cosh(v) - \cos(\xi)]^2} \cos(\varphi) & -a \frac{\sinh(v)}{\cosh(v) - \cos(\xi)} \sin(\varphi) \\ a \frac{1 - \cosh(v) \cos(\xi)}{[\cosh(v) - \cos(\xi)]^2} \sin(\varphi) & -a \frac{\sinh(v) \sin(\xi)}{[\cosh(v) - \cos(\xi)]^2} \sin(\varphi) & a \frac{\sinh(v)}{\cosh(v) - \cos(\xi)} \cos(\varphi) \\ -a \frac{\sin(\xi) \sinh(v)}{[\cosh(v) - \cos(\xi)]^2} & a \frac{\cosh(v) \cos(\xi) - 1}{[\cosh(v) - \cos(\xi)]^2} & 0 \end{pmatrix} \begin{pmatrix} dv \\ d\xi \\ d\varphi \end{pmatrix} .$$

We simplify the notation as follows $C_v = \cosh(v)$, $S_v = \sinh(v)$, $C_\xi = \cos(\xi)$, $S_\xi = \sin(\xi)$ and $C_\varphi = \cos(\varphi)$, $S_\varphi = \sin(\varphi)$. We have:

$$dx = a \frac{1 - C_v C_\xi}{(C_v - C_\xi)^2} C_\varphi dv - a \frac{S_v S_\xi}{(C_v - C_\xi)^2} C_\varphi d\xi - a \frac{S_v}{C_v - C_\xi} S_\varphi d\varphi , \quad dy = a \frac{1 - C_v C_\xi}{(C_v - C_\xi)^2} S_\varphi dv - a \frac{S_v S_\xi}{(C_v - C_\xi)^2} S_\varphi d\xi + a \frac{S_v}{C_v - C_\xi} C_\varphi d\varphi , \quad dz = -a \frac{S_v S_\xi}{(C_v - C_\xi)^2} dv - a \frac{1 - C_v C_\xi}{(C_v - C_\xi)^2} d\xi ,$$

hence:

$$\phi^*(\eta) = a^2 \frac{(1 - C_v C_\xi)^2 + S_v^2 S_\xi^2}{(C_v - C_\xi)^4} (dv^2 + d\xi^2) + a^2 \frac{S_v^2}{(C_v - C_\xi)^2} d\varphi^2 = \frac{a^2}{(C_v - C_\xi)^2} [dv^2 + d\xi^2 + S_v^2 d\varphi^2] = \phi^*(\eta) , \quad \omega = \frac{a^3 S_v}{(C_v - C_\xi)^3} dv \wedge d\varphi \wedge d\xi ,$$

and:

$$\phi_*(\partial_v) = \frac{a}{(C_v - C_\xi)^2} [(1 - C_v C_\xi) C_\varphi \partial_x + (1 - C_v C_\xi) S_\varphi \partial_y - S_v S_\xi \partial_z] , \quad \phi_*(\partial_\xi) = \frac{a}{(C_v - C_\xi)^2} [-S_v S_\xi C_\varphi \partial_x - S_v S_\xi S_\varphi \partial_y - (1 - C_v C_\xi) \partial_z] , \quad \phi_*(\partial_\varphi) = \frac{a S_v}{C_v - C_\xi} [-S_\varphi \partial_x + C_\varphi \partial_y] .$$

Prove that $x^2 + y^2 + z^2 + a^2 = 2a\sqrt{x^2 + y^2} \coth(v)$. It follows that the surface Σ with constant $v = v_0$ is defined by the constrained equation $F(x, y, z) = x^2 + y^2 + z^2 + a^2 - 2a\sqrt{x^2 + y^2} \coth(v_0) = 0$. Compute dF , $dF|_\Sigma$ and evaluate $dF|_\Sigma(\phi_*(\partial_\xi))$ and $dF|_\Sigma(\phi_*(\partial_\varphi))$. Prove that $\phi^*(dF)$ at Σ is normal to Σ . We have: $(x^2 + y^2)(C_v - C_\xi)^2 = a^2 S_v^2$, $z^2(C_v - C_\xi)^2 = a^2 S_\xi^2$ hence:

$$x^2 + y^2 + z^2 + a^2 - 2a\sqrt{x^2 + y^2} \coth(v) = a^2 \frac{S_v^2 + S_\xi^2}{(C_v - C_\xi)^2} + a^2 - 2a \frac{a S_v}{C_v - C_\xi} \frac{C_v}{S_v} = \frac{a^2}{(C_v - C_\xi)^2} \left[S_v^2 + S_\xi^2 + (C_v - C_\xi)^2 - 2C_v(C_v - C_\xi) \right] .$$

$$S_v^2 + S_\xi^2 + C_v^2 + C_\xi^2 - 2C_v C_\xi - 2C_v^2 + 2C_v C_\xi = S_\xi^2 + C_\xi^2 + S_v^2 - C_v^2 = 1 - 1 = 0$$

We have $dF = 2xdx + 2ydy + 2zdz - 2a \frac{xdx+ydy}{\sqrt{x^2+y^2}} \coth(v_0) = 2zdz + (xdx + ydy) \left(2 - 2a \frac{C_{v_0}/S_{v_0}}{\sqrt{x^2+y^2}} \right)$ and, identifying the points on Σ by their toroidal coordinates, we have:

$$\begin{aligned} dF|_{\Sigma} &= 2a \frac{S_{\xi}}{C_{v_0} - C_{\xi}} dz + \left(2 - 2a \frac{C_{v_0}/S_{v_0}}{aS_{v_0}/(C_{v_0} - C_{\xi})} \right) \left(a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} C_{\varphi} dx + a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} S_{\varphi} dy \right) = 2a \frac{S_{\xi}}{C_{v_0} - C_{\xi}} dz - 2 \frac{1 - C_{v_0} C_{\xi}}{S_{v_0}^2} a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} (C_{\varphi} dx + S_{\varphi} dy) \\ dF|_{\Sigma} &= -2a \frac{1 - C_{v_0} C_{\xi}}{S_{v_0} (C_{v_0} - C_{\xi})} (C_{\varphi} dx + S_{\varphi} dy) + 2a \frac{S_{\xi}}{C_{v_0} - C_{\xi}} dz, \quad dF|_{\Sigma}(\phi_*(\partial_{\xi})) = dF|_{\Sigma}(\phi_*(\partial_{\varphi})) = 0. \end{aligned}$$

As $\phi^*(dF) = d(F \circ \phi)$ we have $\phi^*(dF) = d\left(2a^2 \frac{S_{v_0}}{C_{v_0} - C_{\xi}} [\coth(v) - \coth(v_0)]\right) = \alpha_v dv + \beta_{\xi} [\coth(v) - \coth(v_0)] d\xi$, hence at Σ , where $v = v_0$, $\phi^*(dF)|_{\Sigma} \sim dv$, and it is normal to Σ .

Propose an embedding ψ of the surfaces Σ in $\mathbb{R}^3$, compute the differential $d\psi = \psi_*$, the pullback metric $\psi^*(\eta)$, the volume form for this pullback metric and compare it with $\psi^*(i_{\phi_*}(\hat{\partial}_v) dx \wedge dy \wedge dz)$, where $\hat{\partial}_v$ is a unit vector parallel to ∂_v . Evaluate $\psi^*(dF)$.

We propose $\psi(\xi, \varphi) = \phi(v_0, \xi, \varphi)$, then $d\psi \equiv \psi_* = \frac{\partial}{\partial \xi} \frac{\partial x^i}{\partial s^{\mu}} ds^{\mu} = \begin{pmatrix} \partial_x & \partial_y & \partial_z \end{pmatrix} \begin{pmatrix} -a \frac{S_{v_0} S_{\xi}}{[C_{v_0} - C_{\xi}]^2} C_{\varphi} & -a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} S_{\varphi} \\ -a \frac{S_{v_0} S_{\xi}}{[C_{v_0} - C_{\xi}]^2} S_{\varphi} & a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} C_{\varphi} \\ -a \frac{1 - C_{v_0} C_{\xi}}{[C_{v_0} - C_{\xi}]^2} & 0 \end{pmatrix} \begin{pmatrix} d\xi \\ d\varphi \end{pmatrix}$. We have:

$$dx = -a \frac{S_{v_0} S_{\xi}}{(C_{v_0} - C_{\xi})^2} C_{\varphi} d\xi - a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} S_{\varphi} d\varphi, \quad dy = -a \frac{S_{v_0} S_{\xi}}{(C_{v_0} - C_{\xi})^2} S_{\varphi} d\xi + a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} C_{\varphi} d\varphi, \quad dz = -a \frac{1 - C_{v_0} C_{\xi}}{(C_{v_0} - C_{\xi})^2} d\xi,$$

hence

$$\psi^*(\eta) = a^2 \frac{(1 - C_{v_0} C_{\xi})^2 + S_{v_0}^2 S_{\xi}^2}{(C_{v_0} - C_{\xi})^4} d\xi^2 + a^2 \frac{S_{v_0}^2}{(C_{v_0} - C_{\xi})^2} d\varphi^2 = \frac{a^2}{(C_{v_0} - C_{\xi})^2} [d\xi^2 + S_{v_0}^2 d\varphi^2] = \psi^*(\eta), \quad \omega_{\Sigma} = \frac{a^2 S_{v_0}}{(C_{v_0} - C_{\xi})^2} d\varphi \wedge d\xi,$$

On the other hand:

$$\begin{aligned} \phi^*(\hat{\partial}_v) &= \frac{1}{(C_v - C_{\xi})} \left[(1 - C_v C_{\xi}) C_{\varphi} \partial_x + (1 - C_v C_{\xi}) S_{\varphi} \partial_y - S_v S_{\xi} \partial_z \right], \quad i_{\phi_*}(\hat{\partial}_v) dx \wedge dy \wedge dz = \frac{(1 - C_v C_{\xi}) C_{\varphi}}{(C_v - C_{\xi})} dy \wedge dz - \frac{(1 - C_v C_{\xi}) S_{\varphi}}{(C_v - C_{\xi})} dx \wedge dz + \frac{(-S_v S_{\xi})}{(C_v - C_{\xi})} dx \wedge dy \\ \psi^*(i_{\phi_*}(\hat{\partial}_v) dx \wedge dy \wedge dz) &= \frac{(1 - C_{v_0} C_{\xi}) C_{\varphi}}{(C_{v_0} - C_{\xi})} \left(a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} C_{\varphi} d\varphi \right) \wedge \left(-a \frac{1 - C_{v_0} C_{\xi}}{(C_{v_0} - C_{\xi})^2} d\xi \right) - \frac{(1 - C_{v_0} C_{\xi}) S_{\varphi}}{(C_{v_0} - C_{\xi})} \left(-a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} S_{\varphi} d\varphi \right) \wedge \left(-a \frac{1 - C_{v_0} C_{\xi}}{(C_{v_0} - C_{\xi})^2} d\xi \right) \\ &\quad + \frac{(-S_{v_0} S_{\xi})}{(C_{v_0} - C_{\xi})} \left(-a \frac{S_{v_0} S_{\xi}}{(C_{v_0} - C_{\xi})^2} \right) \left(a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} \right) d\xi \wedge d\varphi = d\xi \wedge d\varphi \frac{a^2 S_{v_0}}{(C_{v_0} - C_{\xi})^4} [(1 - C_{v_0} C_{\xi})^2 + S_{v_0}^2 S_{\xi}^2] = -\omega_{\Sigma}. \end{aligned}$$

Finally $dF = 2xdx + 2ydy + 2zdz - 2a \frac{xdx+ydy}{\sqrt{x^2+y^2}} \coth(v_0)$ and:

$$\begin{aligned} \psi^*(dF) &= \left[a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} C_{\varphi} \left(-a \frac{S_{v_0} S_{\xi}}{(C_{v_0} - C_{\xi})^2} C_{\varphi} d\xi - a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} S_{\varphi} d\varphi \right) + a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} S_{\varphi} \left(-a \frac{S_{v_0} S_{\xi}}{(C_{v_0} - C_{\xi})^2} S_{\varphi} d\xi + a \frac{S_{v_0}}{C_{v_0} - C_{\xi}} C_{\varphi} d\varphi \right) \right] \left(2 - 2a \frac{C_{v_0}/S_{v_0}}{aS_{v_0}/(C_{v_0} - C_{\xi})} \right) \\ &\quad + 2a \frac{S_{\xi}}{C_{v_0} - C_{\xi}} \left(-a \frac{1 - C_{v_0} C_{\xi}}{(C_{v_0} - C_{\xi})^2} d\xi \right) \\ &= \left[-a^2 \frac{S_{v_0}^2 S_{\xi}}{(C_{v_0} - C_{\xi})^3} d\xi \right] \left[2 - 2 \frac{C_{v_0} (C_{v_0} - C_{\xi})}{S_{v_0}^2} \right] + 2a \frac{S_{\xi}}{C_{v_0} - C_{\xi}} \left(-a \frac{1 - C_{v_0} C_{\xi}}{(C_{v_0} - C_{\xi})^2} d\xi \right) = \left[-a^2 \frac{S_{v_0}^2 S_{\xi}}{(C_{v_0} - C_{\xi})^3} d\xi \right] 2(-1) \frac{1 - C_{v_0} C_{\xi}}{S_{v_0}^2} + 2a \frac{S_{\xi}}{C_{v_0} - C_{\xi}} \left(-a \frac{1 - C_{v_0} C_{\xi}}{(C_{v_0} - C_{\xi})^2} d\xi \right) = 0. \end{aligned}$$

Exercise 12.20. We consider the sphere $\mathbb{S}^2$ defined as the solution of the equation $F(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$. Determine the normal form, exterior normal and volume form on the sphere in Cartesian coordinates. We have $\frac{1}{2} dF|_{(x,y,z) \in F^{-1}(0)} = xdx + ydy + zdz$, hence $\nu|_{(x,y,z) \in F^{-1}(0)} = x\partial_x + y\partial_y + z\partial_z$ and $i_{\nu}(dx \wedge dy \wedge dz)|_{(x,y,z) \in F^{-1}(0)} = xdy \wedge dz - ydx \wedge dz + zdx \wedge dy$. Note that on the surface of the sphere $dz = -\frac{x}{z} dx - \frac{y}{z} dy$ hence we also have:

$$i_{\nu}(dx \wedge dy \wedge dz)|_{(x,y,z) \in F^{-1}(0)} = xdy \wedge \left(-\frac{x}{z} dx - \frac{y}{z} dy \right) - ydx \wedge \left(-\frac{x}{z} dx - \frac{y}{z} dy \right) + zdx \wedge dy = \left(\frac{x^2}{z} + \frac{y^2}{z} + z \right) dx \wedge dy = \frac{1}{z} dx \wedge dy.$$

As a crosscheck, in spherical coordinates $x = \sin \theta \cos \varphi$, $y = \sin \theta \sin \varphi$, $z = \cos \theta$ this form is $\frac{1}{\cos \theta} (\cos \theta \cos \varphi d\theta - \sin \theta \sin \varphi d\varphi) \wedge (\cos \theta \sin \varphi d\theta + \sin \theta \cos \varphi d\varphi) = \sin \theta d\theta \wedge d\varphi$, as expected.

The stereographic projection from the north pole of the sphere is:

$$\varphi: \mathbb{S}^2 \rightarrow \mathbb{R}^2, \quad \varphi(x, y, z) = (s, t), \quad s = \frac{x}{1-z}, \quad t = \frac{y}{1-z}.$$

The natural embedding of $\mathbb{S}^2$ in $\mathbb{R}^3$ writes using stereographic coordinates on $\mathbb{S}^2$ and Cartesian coordinates in $\mathbb{R}^3$ as:

$$\phi(s, t) = (x(s, t), y(s, t), z(s, t)) = \left(\frac{2s}{1+s^2+t^2}, \frac{2t}{1+s^2+t^2}, \frac{-1+s^2+t^2}{1+s^2+t^2} \right),$$

and note that this is valid only in the open subset $\mathbb{S}^2 \setminus (0, 0, 1)$, and one needs to consider the second stereographic projection in order to cover the full sphere. Compute the metric of the sphere in stereographic coordinates and the associated volume form. We have:

$$\begin{aligned} dx &= \frac{2(1+s^2+t^2)-4s^2}{(1+s^2+t^2)^2} ds - \frac{4st}{(1+s^2+t^2)^2} dt = \frac{2(1-s^2+t^2)}{(1+s^2+t^2)^2} ds - \frac{4st}{(1+s^2+t^2)^2} dt, & dy &= -\frac{4st}{(1+s^2+t^2)^2} ds + \frac{2(1+s^2-t^2)}{(1+s^2+t^2)^2} dt, \\ dz &= \frac{2s(1+s^2+t^2)-(-1+s^2+t^2)2s}{(1+s^2+t^2)^2} ds + \frac{2t(1+s^2+t^2)-(-1+s^2+t^2)2t}{(1+s^2+t^2)^2} dt = \frac{4s}{(1+s^2+t^2)^2} ds + \frac{4t}{(1+s^2+t^2)^2} dt \end{aligned}$$

the metric (line element) is:

$$\begin{aligned} dt^2 &= \frac{1}{(1+s^2+t^2)^4} [(4(1-s^2+t^2)^2 + 16s^2t^2 + 16s^2)ds^2 + (4(1+s^2-t^2)^2 + 16s^2t^2 + 16t^2)dt^2 + dsdt(-32st + 32st)] \\ &= \frac{4}{(1+s^2+t^2)^4} [(1+s^4+t^4-2s^2+2t^2-2s^2t^2+4s^2t^2+4s^2)ds^2 + \dots] = \frac{4}{(1+s^2+t^2)^2} (ds^2 + dt^2) \end{aligned}$$

and $dV = \frac{4}{(1+s^2+t^2)^2} ds \wedge dt$. This is to be compared with:

$$\phi^* \left(\frac{1}{z} dx \wedge dy \right) = \frac{1+s^2+t^2}{-1+s^2+t^2} \left(\frac{4(1-s^2+t^2)(1+s^2-t^2)-16s^2t^2}{(1+s^2+t^2)^4} \right) ds \wedge dt = 4 \frac{1-(s^2-t^2)^2-4s^2t^2}{(-1+s^2+t^2)(1+s^2+t^2)^3} ds \wedge dt = 4 \frac{1-(s^2+t^2)^2}{(-1+s^2+t^2)(1+s^2+t^2)^3} ds \wedge dt = \frac{-4}{(1+s^2+t^2)^2} ds \wedge dt.$$

12.6.2 Lagrange multipliers

We are interested in identifying the critical points of a function $f : M = \mathbb{R}^n \rightarrow \mathbb{R}$ subjected to the constraints $F^c(x) = 0$, $c = 1 \dots k$. To this end we consider the submanifold Σ of dimension $n - k$ defined by the constraints $F^c(x) = 0$, $c = 1 \dots k$, and we denote its embedding ϕ . We thus need to find the critical points of the restriction of f to Σ , namely $f \circ \phi$. Such critical points are the solutions of $d(f \circ \phi)|_p(V) = 0$, $\forall V \in T_p\Sigma$, that is:

$$d(f \circ \phi)(V) = d(\phi^* f)(V) = V(\phi^* f) = \phi_* V(f) = df(\phi_* V),$$

hence the critical points are such that $\bigcap_{c=1}^k \ker dF^c|_{\phi(p)} = \phi_*|_{\phi(p)}(T_p\Sigma) \subset \ker df|_{\phi(p)}$, which from Lemma 12.2 implies that $df|_{\phi(p)} = \sum_{c=1}^k \lambda_c dF^c|_{\phi(p)}$. In conclusion the n coordinates x^q , $q = 1, \dots, n$ (which are coordinates in M) of the critical points of $f \circ \phi$ and the k Lagrange multipliers λ_i , $i = 1, \dots, k$ are the solutions of the $n + k$ equations:

$$\partial_i f(x) = \sum_{c=1}^k \lambda_c \partial_i F^c(x), \quad i = 1, \dots, n, \quad F^c(x) = 0, \quad c = 1 \dots k,$$

that is the critical points of the Lagrange function $\mathcal{L}(x, \lambda_1, \dots, \lambda_k) = f(x) - \sum_{c=1}^k \lambda_c F^c(x)$ in all its variables.

Additional intuition can be gained from the case $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. The level sets of f are the solutions of the equation $f(x, y) - c = 0$, which exist (locally) by the implicit function theorem. The form df is normal to the tangent space to its level set $df(V) = 0$ (as df is a form on $T\mathbb{R}^2$, one needs to bring V from the tangent space to the level set into $T\mathbb{R}^2$ via a push forward). The constraint $F(x, y) = 0$ is solved by its own level set, and dF is normal to the tangent space to its level set. The function $f(x, y)$ is maximized with the constraint $F(x, y) = 0$ if the two tangent spaces (the one tangent to its level set and the one tangent to the level set of the constraint) are identical, once pushed forward to $T\mathbb{R}^2$, that is the forms df and dF are proportional. In this case $df(V) = 0$ for any vector tangent to $F(x, y) = 0$ hence f is stationary when moving along the constraint $F(x, y) = 0$. One can picture this as the point where a road through the mountains reaches the maximal height: the level sets of f are the curves of constant altitude (that is the contour lines, or “isolines” on a topographic map).

Exercise 12.21. Consider the submanifold of $\mathbb{R}^3$ determined by the constraints $F^1(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$ and $F^2(x, y, z) = x - y = 0$. Write the embedding map of the manifold in coordinates, determine dF^1 and dF^2 and find the minima and maxima of the function $f(x, y, z) = z + x$ on this manifold. We solve around the north pole $(0, 0, 1)$ and find a chart on the submanifold

$(s, s, \sqrt{1-2s^2})$ with $s \in (-1/\sqrt{2}, 1/\sqrt{2})$. We have $dF^1 = 2xdx + 2ydy + 2zdz$, $dF^2 = dx - dy$ and we need to solve:

$$\begin{aligned} dz + dx &= \lambda_1(2xdx + 2ydy + 2zdz) + \lambda_2(dx - dy) & \Rightarrow & \quad 1 = 2s\lambda_1, \quad 1 = 2x\lambda_1 + \lambda_2, \quad 0 = 2y\lambda_1 - \lambda_2 \\ x^2 + y^2 + z^2 - 1 &= 0, \quad x - y = 0, \end{aligned}$$

hence with solution $y = x, z = \pm\sqrt{1-2x^2}$, $\lambda_1 = \frac{1}{\pm 2\sqrt{1-2x^2}}$, $\lambda_2 = \frac{2x}{\pm 2\sqrt{1-2x^2}}$ hence $1 = \frac{2x+2x}{\pm 2\sqrt{1-2x^2}}$ that is $4x^2 = (1-2x^2)$ hence the four critical points are $x = y = \pm \frac{1}{\sqrt{6}}$, $z = \pm\sqrt{\frac{2}{3}}$, where the two signs are independent.

In infinite dimensions holonomic constraints $F_a(q, t) = 0$ are accounted for by introducing Lagrange multipliers as new degrees of freedom λ_a and considering the modified Lagrangian $\tilde{L}(q, \dot{q}, \lambda, t) = L(q, \dot{q}, \lambda, t) - \sum_a \lambda_a F_a(q, t)$ leading to the Euler-Lagrange equations for $\tilde{L}$:

$$-\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i}(q(t), \frac{dq}{dt}(t), \lambda(t), t) \right) + \frac{\partial L}{\partial q_i}(q(t), \frac{dq}{dt}(t), \lambda(t), t) = \sum_a \lambda_a(t) \frac{\partial F_a}{\partial q_i}(q(t), t), \quad F_a(q(t), t) = 0.$$

Take for instance a free particle in $\mathbb{R}^2$. In Cartesian coordinates (x, y) its Lagrangian writes $L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2)$. We aim to study the movement of the particle constraint to move on a circle of radius 1. A first approach is to identify appropriate independent generalized coordinates and write the Lagrangian in terms of them: in this case φ such that $x = \sin \varphi, y = \cos \varphi$ is a good generalized coordinate and the Lagrangian for this generalized coordinate writes $L = \frac{1}{2}\dot{\varphi}^2$, leading to the trajectories $\varphi = \varphi_0 + \omega t$, that is the particle moves at constant angular speed around the circle.

Assuming we were not able to identify the appropriate generalized coordinates, we use Lagrange multipliers. We can pass to polar coordinates $x = r \sin \varphi$, $\dot{x} = \dot{r} \sin \varphi + r\dot{\varphi} \cos \varphi$ and $y = \cos \varphi$, $\dot{y} = \dot{r} \cos \varphi - r\dot{\varphi} \sin \varphi$, $L(r, \dot{r}, \dot{\varphi}) = \frac{1}{2}(\dot{r}^2 + r^2\dot{\varphi}^2)$ and impose the constraint $r = 1$ by considering the modified Lagrangian $\tilde{L}(r, \dot{r}, \lambda, \dot{\varphi}) = \frac{1}{2}(\dot{r}^2 + r^2\dot{\varphi}^2) - \lambda(r - 1)$ leading to the Euler-Lagrange equations:

$$-\frac{d}{dt} \left(\frac{dr}{dt} \right) + r(t) \left(\frac{d\varphi}{dt} \right)^2 = \lambda(t), \quad -\frac{d}{dt} \left(\frac{d\varphi}{dt} \right) = 0 \quad r(t) - 1 = 0,$$

with solution $\varphi(t) = \varphi_0 + \omega t$, $r(t) = 1$ and $\lambda(t) = \omega^2$. If one does not have the idea to change to polar coordinates in the beginning, one gets $\tilde{L} = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) - \lambda(\sqrt{x^2 + y^2} - 1)$ with Euler Lagrange equations:

$$-\frac{d}{dt} \left(\frac{dx}{dt} \right) = \lambda(t) \frac{x(t)}{\sqrt{x(t)^2 + y(t)^2}}, \quad -\frac{d}{dt} \left(\frac{dy}{dt} \right) = \lambda(t) \frac{y(t)}{\sqrt{x(t)^2 + y(t)^2}}, \quad \sqrt{x(t)^2 + y(t)^2} = 1.$$

Observing that $x(t)^2 + y(t)^2$ is conserved, we can parametrize $x(t) = \sin \varphi(t)$, $y = \cos \varphi(t)$ and the equations become:

$$-\frac{d}{dt} \left(\frac{d\varphi}{dt} \cos \varphi \right) = \lambda \sin \varphi, \quad -\frac{d}{dt} \left(-\frac{d\varphi}{dt} \sin \varphi \right) = \lambda \cos \varphi \Leftrightarrow -\frac{d^2\varphi}{dt^2} \cos \varphi + \left(\frac{d\varphi}{dt} \right)^2 \sin \varphi = \lambda \sin \varphi, \quad \frac{d^2\varphi}{dt^2} \sin \varphi + \left(\frac{d\varphi}{dt} \right)^2 \cos \varphi = \lambda \cos \varphi.$$

A linear combination of the two equations yields $\frac{d^2\varphi}{dt^2}(\sin^2 \varphi + \cos^2 \varphi) = 0 \Rightarrow \varphi = \varphi_0 + \omega t$ while another linear combination yields $(\frac{d\varphi}{dt})^2(\sin^2 \varphi + \cos^2 \varphi) = \lambda(\sin^2 \varphi + \cos^2 \varphi)$, with solution $\lambda = \omega^2$.

12.7 The Lie derivative, the covariant derivative and geodesics

We use Einstein's convention that repeated indices are summed. A tensor with r contravariant and s covariant indices is a multilinear form on $T : (T_p^* M)^r \times (T_p M)^s \rightarrow \mathbb{R}$ and decomposes in the standard basis as $T_{j_1 \dots j_p}^{i_1 \dots i_r} \partial_{x^{i_1}} \dots \partial_{x^{i_r}} dx^{j_1} \dots dx^{j_p}$.

The Lie derivative. Let X be a vector field. The *flow* of X is the function $\Phi : \mathbb{R} \times M \rightarrow M$, $\Phi(t, x) \equiv \Phi^t(x)$ with $\Phi^0(x) = x$, solution of the differential equation $\frac{d}{dt} \Phi^t(x) = X(\Phi^t(x))$, that is $\Phi^t(x)$ at fixed x is the integral curve of X that

starts from x at $t = 0$. The flow $\Phi^t(x)$ takes the point x and evolves it along an integral curve of X for time t . The map $\Phi^t : M \rightarrow M$ pulls back forms in $T_{\Phi^t(x)}^* M$ to forms in $T_x^* M$ and pushes forward vectors in $T_x M$ to vectors in $T_{\Phi^t(x)} M$. It follows that Φ^{-t} “pushes forward” (but backward in time) vectors in $T_{\Phi^t(x)}^* M$ to vectors in $T_x^* M$.

Definition 12.15 (**The Lie derivative**). *The Lie derivative evaluates the change of a tensor field (including functions, vector fields and forms), along the flow defined by another vector field:*

- for functions $f : M \rightarrow \mathbb{R}$ the Lie derivative is the Gateaux derivative:

$$\mathcal{L}_X f = \frac{d}{dt}(f \circ \Phi^t) = df \circ \frac{d}{dt}\Phi^t = df(X) = X(f) .$$

- for vectors it is $\mathcal{L}_X Y = \frac{d}{dt}(\Phi_*^{-t} Y|_{\Phi^t})_{t=0}$ and for forms $\mathcal{L}_X \alpha = \frac{d}{dt}[(\Phi^t)^* \alpha|_{\Phi^t}]_{t=0}$.
- for arbitrary tensors it can be proven that the Lie derivative $\mathcal{L}_X T = \frac{d}{dt}[(\Phi^t)^* T|_{\Phi^t}]_{t=0}$ with $(\Phi^t)^* T|_{\Phi^t}$ the pullback of $T|_{\Phi^t}$ to $T_x M$ respects:

$$\mathcal{L}_X(S \otimes T) = \mathcal{L}_X S \otimes T + S \otimes \mathcal{L}_X T , \quad \mathcal{L}_X(T(A^1, \dots, A^n)) = \mathcal{L}_X T(A^1, \dots, A^n) + \sum_i T(A^1, \dots, \mathcal{L}_X A^i, \dots, A^n) ,$$

where A^i are vectors or one forms. In fact these two properties together with $[\mathcal{L}_X, d](f) = 0$, $\mathcal{L}_X(f) = X(f)$ uniquely fix $\mathcal{L}$.

In some chart the flow at linear order is $\Phi^t(x) \approx x + tX = y$. The differential is then $[\Phi_*^t]_i^j \approx \delta_i^j + t\partial_i X^j$, but recalling that vectors are pulled back by Φ^{-t} we have $(\Phi_*^{-t} Y)^j(x) \approx (\delta_i^j - t\partial_i X^j)Y^i(y)$. Taking into account that $Y^i(\Phi^t(x)) \approx Y^i + tX^s \partial_s Y^i$ we have:

$$[\mathcal{L}_X Y]^j = \frac{d}{dt}[(\delta_i^j - t\partial_i X^j)(Y^i + tX^s \partial_s Y^i)]_{t=0} = X^s \partial_s Y^j - Y^s \partial_s X^j \Rightarrow \mathcal{L}_X Y = [X, Y] .$$

For forms $\alpha(x)dx^{i_1} \wedge \dots \wedge dx^{i_k}$ we get (reinstating the sums for one moment):

$$\begin{aligned} \mathcal{L}_X \alpha &= \frac{d}{dt} \left[(\alpha + t \sum_{i=1}^n X^i \partial_i \alpha) \sum_{j_1 < \dots < j_k} \det[\delta_{j_1}^{i_1} + t\partial_{j_1} X^{i_1}] dx^{j_1} \wedge \dots \wedge dx^{j_k} \right] \\ &= \sum_{i=1}^n X^i \partial_i \alpha dx^{i_1} \wedge \dots \wedge dx^{i_n} + \alpha \sum_{l=1}^k \sum_{j_l=1}^n \partial_{j_l} X^{i_l} (-1)^{l-1} dx^{j_l} \wedge dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_l}} \dots \wedge dx^{i_k} , \end{aligned}$$

and¹² we conclude $\mathcal{L}_X \alpha = i_X d\alpha + di_X \alpha$. For tensors we use $dy^i = \frac{\partial y^i}{\partial x^j} dx^j \approx dx^i + t\partial_j X^i dx^j$, $\partial_{y^i} = \frac{\partial x^j}{\partial y^i} \frac{\partial}{\partial x^j} \approx \partial_{x^i} - t\partial_i X^j \partial_{x^j}$ and $T(\Phi^t(x)) = T_{b_1 \dots b_s}^{a_1 \dots a_r}(y) \partial_{y^{a_1}} \dots \partial_{y^{a_r}} dy^{b_1} \dots dy^{b_s}$ to write:

$$\begin{aligned} (\Phi^t)^* T|_{\Phi^t} &\approx (T_{b_1 \dots b_s}^{a_1 \dots a_r} + tX^i \partial_i T_{b_1 \dots b_s}^{a_1 \dots a_r})(\delta_{a_1}^{i_1} - t\partial_{a_1} X^{i_1}) \dots (\delta_{a_r}^{i_r} - t\partial_{a_r} X^{i_r})(\delta_{j_1}^{b_1} + t\partial_{j_1} X^{b_1}) \dots (\delta_{j_s}^{b_s} + t\partial_{j_s} X^{b_s}) \partial_{x^{i_1}} \dots \partial_{x^{i_r}} dx^{j_1} \dots dx^{j_s} \\ \mathcal{L}_X T_{b_1 \dots b_s}^{a_1 \dots a_r} &= X^i \partial_i T_{b_1 \dots b_s}^{a_1 \dots a_r} - \partial_i X^{a_1} T_{b_1 \dots b_s}^{i a_2 \dots a_r} - \dots - \partial_i X^{a_r} T_{b_1 \dots b_s}^{a_1 a_2 \dots i} + \partial_{b_1} X^i T_{i b_2 \dots b_s}^{a_1 \dots a_r} + \dots + \partial_{b_s} X^i T_{b_1 b_2 \dots i}^{a_1 \dots a_r} . \end{aligned}$$

The covariant derivative. Yet another notion of derivative is the covariant derivative. This derivative generalizes the directional derivative but behaves covariantly under a change of basis. While this derivative can be defined with no reference to an embedding, we prefer here to use one as it facilitates the comprehension. Moreover, Nash’s embedding theorem states

¹²We use:

$$\begin{aligned} i_X d\alpha &= i_X \sum_{j=1}^n \partial_j \alpha dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_{j=1}^n X^j \partial_j \alpha dx^{i_1} \wedge \dots \wedge dx^{i_k} + \sum_{l=1}^k \sum_{j=1}^n \partial_j \alpha (-1)^l X^{i_l} dx^j \wedge dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_l}} \dots \wedge dx^{i_k} , \\ di_X \alpha &= d \left(\sum_{l=1}^k \alpha X^{i_l} (-1)^{l-1} dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_l}} \dots \wedge dx^{i_k} \right) , \end{aligned}$$

that any Riemannian manifold can be isometrically embedded in $\mathbb{R}^n$ with standard metric for n large enough, that is there exists an embedding $f : M \rightarrow \mathbb{R}^n$ such that the metric g on M respects $g(V, W) = \delta(\mathbf{d}f(V), \mathbf{d}f(W))$ or equivalently $g_{\mu\nu} = \delta_{ij} \partial_{s^\mu} f^i \partial_{s^\nu} f^j$.

Consider a k -dimensional manifold Σ embedded in $\mathbb{R}^n$, via $s \mapsto x(s)$ endowed with standard metric δ_{ij} . The pushforward map from $T\Sigma$ to TM writes:

$$\phi_* = \frac{\partial}{\partial x^i} \frac{\partial x^i}{\partial s^\mu} ds^\mu,$$

and the metric δ on TM is pulled back on $T\Sigma$ to $g_{\mu\nu} = \delta_{ij} \partial_{s^\mu} x^i \partial_{s^\nu} x^j$. The vector $V = V^\mu \partial_{s^\mu} \in T\Sigma$ is pushed forward to $V = V^\mu \frac{\partial x^i}{\partial s^\mu} \frac{\partial}{\partial x^i}$. We have:¹³

$$\frac{\partial V}{\partial s^\nu} = \frac{\partial}{\partial s^\nu} \left(V^\mu \frac{\partial x^i}{\partial s^\mu} \right) \frac{\partial}{\partial x^i} = \left(\frac{\partial V^\mu}{\partial s^\nu} \frac{\partial x^i}{\partial s^\mu} + V^\mu \frac{\partial^2 x^i}{\partial s^\nu \partial s^\mu} \right) \frac{\partial}{\partial x^i}.$$

The second term above does not belong to $\mathbf{d}\phi(T\Sigma)$, but it can always be projected in a parallel and a normal component:

$$\frac{\partial^2 x^i}{\partial s^\nu \partial s^\mu} \frac{\partial}{\partial x^i} = \Gamma_{\nu\mu}^\rho \frac{\partial x^i}{\partial s^\rho} \frac{\partial}{\partial x^i} + N_{\nu\mu},$$

with some coefficients $\Gamma_{\mu\nu}^\rho$. Taking the scalar product with $\frac{\partial x^j}{\partial s^\lambda} \frac{\partial}{\partial x^j}$ and taking into account that $g^{\mu\nu} = \partial_{x^i} s^\mu \partial_{x^j} s^\nu$ we have:

$$\delta_{ij} \frac{\partial^2 x^i}{\partial s^\nu \partial s^\mu} \frac{\partial x^j}{\partial s^\lambda} = \Gamma_{\nu\mu}^\rho \delta_{ij} \frac{\partial x^i}{\partial s^\rho} \frac{\partial x^j}{\partial s^\lambda} \Rightarrow \Gamma_{\nu\mu}^\rho = \frac{\partial^2 x^i}{\partial s^\nu \partial s^\mu} \frac{\partial s^\rho}{\partial x^i}.$$

While this formula is very useful, it is not intrinsic as it depends on embedding the manifold into $\mathbb{R}^n$. We will derive below a more intrinsic formula for the coefficients Γ , which are called Christoffel symbols. The *covariant derivative* (in the Levi-Civita connection) of V along ∂_{s^ν} is the projection of $\frac{\partial V}{\partial s^\nu}$ onto the tangent space $\mathbf{d}(T\Sigma)$:

$$\nabla_\nu V = \nabla_{\frac{\partial}{\partial s^\nu}} V = \left(\frac{\partial V^\mu}{\partial s^\nu} + V^\rho \Gamma_{\nu\rho}^\mu \right) \frac{\partial x^i}{\partial s^\mu} \frac{\partial}{\partial x^i}.$$

Definition 12.16 (Covariant derivative). The covariant derivative ∇_X is:

- for functions $\nabla_X(f) = X(f)$,
- for vectors $\nabla_X V$ is linear in X , additive in V and respects $\nabla_X(fV) = (Xf)V + f\nabla_X V$. Denoting $e_\mu = \partial_\mu$ and $\nabla_\mu = \nabla_{e_\mu}$, we have:

$$\nabla_{X^\nu e_\nu} (V^\mu e_\mu) = X^\nu \nabla_\nu (V^\mu e_\mu) = X^\nu (\partial_\nu V^\mu e_\mu + V^\mu \nabla_\nu e_\mu) = X^\nu (\partial_\nu V^\mu e_\mu + V^\mu \Gamma_{\nu\rho}^\mu e_\rho) = X^\nu (\nabla_\nu V^\mu) e_\mu,$$

where we denoted $\nabla_\nu e_\mu = \Gamma_{\nu\mu}^\rho e_\rho$.

- for tensors $(\nabla_X T)(A^1, \dots, A^n) = \nabla_X(T(A^1, \dots, A^n)) - \sum_i T(A^1, \dots, \nabla_X A^i, \dots, A^n)$.

For instance on one forms we have:

$$(\nabla_X \alpha)(V) = X(\alpha(V)) - \alpha(\nabla_X V) = X^\nu (\partial_\nu \alpha_\mu) V^\mu + X^\nu \alpha_\mu \partial_\nu V^\mu - X^\nu \alpha_\mu [\partial_\nu V^\mu + V^\rho \Gamma_{\nu\rho}^\mu] \Rightarrow \nabla_\nu \alpha_\mu = \partial_\nu \alpha_\mu - \alpha_\rho \Gamma_{\nu\mu}^\rho$$

Exercise 12.22. Prove that under a change of basis $s = s(y)$, the covariant derivative transforms covariantly $\nabla_{\partial_{y^\nu}}(\cdot) = \frac{\partial s^\mu}{\partial y^\nu}(\cdot) \nabla_{\partial_{s^\mu}}(\cdot)$.

¹³Remark that ∂_{x^i} acts on functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ expressed in coordinates $f(x^i)$, hence $\partial_{s^\mu} \partial_{x^i} f = 0$. In other words, the basis vectors in $\mathbb{R}^n$ are independent of s^μ , the coordinates on the embedded manifold.

For the Levi-Civita connection, the coefficients $\Gamma_{\nu\mu}^\rho$ are the Christoffel symbols and can be computed (with no reference to any embedding) in terms of the metric g observing that:

$$\partial_{s^\rho} g_{\mu\nu} = \partial_{s^\rho} \left(\delta_{ij} \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\nu} \right) = \delta_{ij} \Gamma_{\rho\mu}^\sigma \frac{\partial x^i}{\partial s^\sigma} \frac{\partial x^j}{\partial s^\nu} + \delta_{ij} \Gamma_{\rho\nu}^\sigma \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\sigma} = \Gamma_{\rho\mu}^\sigma g_{\sigma\nu} + \Gamma_{\rho\nu}^\sigma g_{\mu\sigma} \Rightarrow \Gamma_{\mu\nu}^\rho = \frac{1}{2} g^{\rho\rho'} (\partial_\mu g_{\nu\rho'} + \partial_\nu g_{\rho'\mu} - \partial_{\rho'} g_{\mu\nu}) ,$$

and using $(\nabla_X g)(V_1, V_2) = d[g(V_1, V_2)](X) - g(\nabla_X V_1, V_2) - g(V_1, \nabla_X V_2)$ a short computation shows that $\nabla_X g = 0$ for all X , that is the Levi-Civita connection is metric compatible and torsion free (i.e. symmetric in the lower indices).¹⁴ Note that the connection coefficients Γ *do not transform covariantly* with the change of basis: their transformation law includes a non covariant piece which is designed to take out the non covariant piece of the partial derivative.

Geodesics and parallel transport. A curve γ is called a geodesic of the metric g if $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$, that is:

$$0 = \frac{d\gamma^\mu}{dt} \nabla_\mu \dot{\gamma} = \frac{d\gamma^\mu}{dt} \left(\frac{\partial}{\partial s^\mu} \dot{\gamma}^\rho + \Gamma_{\mu\nu}^\rho \dot{\gamma}^\nu \right) e_\rho \Rightarrow \ddot{\gamma}^\rho + \Gamma_{\mu\nu}^\rho \dot{\gamma}^\mu \dot{\gamma}^\nu = 0 ,$$

which is a second order ordinary differential equation (in the time variable) for the functions $\gamma^\mu(t)$. A geodesic is uniquely determined by specifying the position and speed at a point, or by specifying two points along it. The norm of the tangent vector is conserved along the geodesic, $d(g(\dot{\gamma}, \dot{\gamma}))(\dot{\gamma}) = \nabla_{\dot{\gamma}} g(\dot{\gamma}, \dot{\gamma}) + 2g(\nabla_{\dot{\gamma}} \dot{\gamma}, \dot{\gamma}) = 0$. It follows that the geodesic equation is not reparametrization invariant, and the time parameter along a geodesic is called the proper time. Along a geodesic the acceleration vector $\ddot{\gamma}$ is perpendicular to the tangent space, that is the acceleration is entirely due to the “bending” of the manifold.

The parallel transport of the vector $X|_p$ along the curve γ with $\gamma(0) = p$ is the solution of the differential equation with initial condition $\nabla_{\dot{\gamma}} X = (\dot{\gamma}^\mu \partial_\mu X^\rho + \Gamma_{\mu\nu}^\rho \dot{\gamma}^\mu X^\nu) e_\rho = 0$, $X|_{\gamma(0)} = X|_p$, or in components $\dot{X}^\rho + \Gamma_{\mu\nu}^\rho \dot{\gamma}^\mu X^\nu = 0$, which is a first order ordinary differential equation (in the time variable) for the functions $X^\rho(t)$. Note that the parallel transport equation is invariant under time reparametrization $t \rightarrow t(\tau)$, with $\frac{dt}{d\tau}(\tau) \neq 0$. A vector is parallel transported along a curve in an embedded manifold if the variation of the vector when moving along the curve, $\frac{d}{dt} X(\gamma(t))$, is orthogonal to the tangent plane to the manifold. The angle between vectors as well as their norm are conserved under parallel transport along any curve as $d[g(X, Y)](\dot{\gamma}) = 0$. We conclude that parallel transport along a close loop is a linear transformation from $T_p M$ to $T_p M$ which preserves the metric, hence it is a rotation. We note that the tangent vector $\dot{\gamma}$ to a geodesic is parallel transported by the geodesic.

Consider a free particle constraint to move on an embedded submanifold of $\mathbb{R}^n$. We denote $x^i(s)$ the embedding map and $g_{\mu\nu} = \delta_{ij} \partial_{s^\mu} x^i \partial_{s^\nu} x^j$ the induced metric. The Lagrangian of the particle is, in physics notation, $L = \frac{1}{2} g_{\mu\nu} \dot{s}^\mu \dot{s}^\nu$ and the Euler Lagrange equations write:

$$0 = \frac{d}{dt} \frac{\partial L}{\partial \dot{s}^\alpha} - \frac{\partial L}{\partial s^\alpha} = \frac{d}{dt} (g_{\alpha\nu} \dot{s}^\nu) - \frac{1}{2} (\partial_\alpha g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu = g_{\alpha\nu} \ddot{s}^\nu + \partial_\rho g_{\alpha\nu} \dot{s}^\rho \dot{s}^\nu - \frac{1}{2} (\partial_\alpha g_{\mu\nu}) \dot{s}^\mu \dot{s}^\nu = \ddot{s}^\beta + \Gamma_{\mu\nu}^\beta \dot{s}^\mu \dot{s}^\nu = 0 ,$$

that is the classical trajectories of the particle are the geodesics on the manifold. This gives an effective way to compute Christoffel symbols. For a particle on a sphere of radius 1 we have $g_{\mu\nu} ds^\mu ds^\nu = d\theta^2 + \sin^2 \theta d\varphi^2$, $L = \frac{1}{2} (\sin^2 \theta \dot{\varphi}^2 + \dot{\theta}^2)$ and:

$$\ddot{\theta} - \sin \theta \cos \theta \dot{\varphi}^2 = 0 , \quad \sin^2 \theta \ddot{\varphi} + 2 \sin \theta \cos \theta \dot{\theta} \dot{\varphi} = 0 ,$$

therefore the only non trivial Christoffel symbols are $\Gamma_{\varphi\varphi}^\theta = -\sin \theta \cos \theta$ and $\Gamma_{\theta\varphi}^\varphi = \Gamma_{\varphi\theta}^\varphi = \frac{\cos \theta}{\sin \theta}$. We note that the equator $\theta(t) = \frac{\pi}{2}$, $\varphi(t) = t$ is a geodesic on the sphere, as well as any great circle $\varphi(t) = \varphi_0$, $\theta(t) = t$. If we parallel transport the vector ∂_θ starting from the north pole to the equator along a great circle, then along the equator for an arc length $\Delta\varphi$ then back to the north pole along a great circle, taking into account that all the curves are geodesics and the parallel transport

¹⁴The Levi-Civita connection can be defined as the unique metric compatible torsion free connection.

along a geodesic preserves the angle between any vector and the tangent vector to the geodesic (as the latter is parallel transported by the geodesic), it is easy to see that ∂_θ comes back to the north pole rotated by $\Delta\varphi$.

The parallel transport of a vector $V^\theta \partial_\theta + V^\varphi \partial_\varphi$ along a general curve γ on the sphere is the solution of the equation:

$$\dot{V}^\theta + \Gamma_{\varphi\varphi}^\theta \dot{\gamma}^\varphi V^\varphi = \dot{V}^\theta - \sin\theta \cos\theta \dot{\gamma}^\varphi V^\varphi = 0, \quad \dot{V}^\varphi + \Gamma_{\theta\theta}^\varphi \dot{\gamma}^\theta V^\theta + \Gamma_{\varphi\theta}^\varphi \dot{\gamma}^\varphi V^\theta = \dot{V}^\varphi + \cot\theta \dot{\gamma}^\theta V^\varphi + \cot\theta \dot{\gamma}^\varphi V^\theta = 0,$$

with appropriate initial conditions. Along the coordinate curves on the sphere, this becomes:

Meridians. For $\theta(t) = \theta_0 \pm t, \varphi(t) = \varphi_0$ we have $\dot{V}^\theta = 0$ and $\dot{V}^\varphi + \cot\theta \dot{\theta} V^\varphi = 0$. Dividing the second equation by V^φ it rewrites as $\frac{d}{dt}[\ln(V^\varphi \sin\theta)] = 0$ therefore:

$$\begin{pmatrix} V^\theta \\ V^\varphi \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{\sin\theta_0}{\sin\theta} \end{pmatrix} \begin{pmatrix} V_0^\theta \\ V_0^\varphi \end{pmatrix}.$$

Paralells. For $\theta(t) = \theta_0, \varphi(t) = \varphi_0 \pm t$ we have $\dot{V}^\theta - \sin\theta_0 \cos\theta_0 \dot{\varphi} V^\varphi = 0$ and $\dot{V}^\varphi + \cot\theta_0 \dot{\varphi} V^\varphi = 0$ which writes:

$$\begin{pmatrix} \dot{V}^\theta \\ \dot{V}^\varphi \end{pmatrix} = \begin{pmatrix} 0 & -\sin\theta_0 \cos\theta_0 \dot{\varphi} \\ \cot\theta_0 \dot{\varphi} & 0 \end{pmatrix} \begin{pmatrix} V^\theta \\ V^\varphi \end{pmatrix} \Rightarrow \begin{pmatrix} V^\theta \\ V^\varphi \end{pmatrix} = e^{\begin{pmatrix} 0 & -\sin\theta_0 \cos\theta_0 \dot{\varphi} \\ \cot\theta_0 \dot{\varphi} & 0 \end{pmatrix} t} \begin{pmatrix} V_0^\theta \\ V_0^\varphi \end{pmatrix},$$

where we used the fact that $\dot{\varphi}$ is constant. The exponenetial is computed by going to an eigenbasis:

$$\begin{pmatrix} 0 & -\sin\theta_0 \cos\theta_0 \dot{\varphi} \\ \cot\theta_0 \dot{\varphi} & 0 \end{pmatrix} = \begin{pmatrix} \frac{\sin\theta_0}{\sqrt{1+\sin^2\theta_0}} & \frac{\sin\theta_0}{\sqrt{1+\sin^2\theta_0}} \\ \frac{-\imath}{\sqrt{1+\sin^2\theta_0}} & \frac{\imath}{\sqrt{1+\sin^2\theta_0}} \end{pmatrix} \begin{pmatrix} \imath \cos\theta_0 \dot{\varphi} & 0 \\ 0 & -\imath \cos\theta_0 \dot{\varphi} \end{pmatrix} \frac{1+\sin^2\theta_0}{2\imath \sin\theta_0} \begin{pmatrix} \frac{\imath}{\sqrt{1+\sin^2\theta_0}} & \frac{-\sin\theta_0}{\sqrt{1+\sin^2\theta_0}} \\ \frac{\imath}{\sqrt{1+\sin^2\theta_0}} & \frac{\sin\theta_0}{\sqrt{1+\sin^2\theta_0}} \end{pmatrix},$$

therefore the solution is:

$$\begin{pmatrix} V^\theta \\ V^\varphi \end{pmatrix} = \begin{pmatrix} \cos(\cos\theta_0 \dot{\varphi} t) & -\sin\theta_0 \sin(\cos\theta_0 \dot{\varphi} t) \\ \frac{\sin(\cos\theta_0 \dot{\varphi} t)}{\sin\theta_0} & \cos(\cos\theta_0 \dot{\varphi} t) \end{pmatrix} \begin{pmatrix} V_0^\theta \\ V_0^\varphi \end{pmatrix}.$$

12.8 The Poincarè lemma

Recall that forms such that $d\alpha = 0$ are called closed, and forms such that $\alpha = d\beta$ are called exact. If a form is exact then it is closed $d(d\beta) = 0$.

Theorem 12.2 (The Poincarè lemma). *Consider a continuously differentiable differential k -form α on a open convex domain $U \subset \mathbb{R}^n$. If the form α is closed then it is exact $d\alpha = 0 \Rightarrow \alpha = d\beta$ for some $k-1$ form β .*

Proof. Let $x_0 \in U$ and consider a k form $\omega = \sum_I \omega_I dx^I$ with I strictly increasing k indices. We define the $k-1$ form:

$$P(\omega) = \sum_{i_1 < \dots < i_k} \sum_{\alpha=1}^k (-1)^{\alpha+1} \left[\int_0^1 dt t^{k-1} \omega_{i_1 \dots i_k}(x_0 + t(x - x_0)) \right] (x^{i_\alpha} - x_0^{i_\alpha}) dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_\alpha}} \wedge \dots \wedge dx^{i_k}.$$

We will prove that $\omega = P(d\omega) + d(P(\omega))$. By linearity it is enough to prove this for forms of the form $f(x)dx^I$. In this case:

$$\begin{aligned} d(P(\omega)) &= \sum_{\alpha=1}^k (-1)^{\alpha+1} \int_0^1 dt t^{k-1} \sum_i \partial_i [f(x_0 + t(x - x_0))(x^{i_\alpha} - x_0^{i_\alpha})] dx^i \wedge dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_\alpha}} \wedge \dots \wedge dx^{i_k} \\ &= \sum_{\alpha=1}^k (-1)^{\alpha+1} \sum_i \int_0^1 dt t^{k-1} [t \partial_i f(x_0 + t(x - x_0))(x^{i_\alpha} - x_0^{i_\alpha}) + f(x_0 + t(x - x_0)) \delta_i^{i_\alpha}] dx^i \wedge dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_\alpha}} \wedge \dots \wedge dx^{i_k}, \end{aligned}$$

and, as $d\omega = \sum_i \partial_i f dx^i \wedge dx^I$ is a $k+1$ form we get:

$$\begin{aligned} P(d\omega) &= \sum_i \left(\int_0^1 dt t^k \partial_i f(x_0 + t(x - x_0)) \right) (x^i - x_0^i) dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ &\quad + \sum_i \sum_{\alpha=1}^k (-1)^{\alpha+2} \left(\int_0^1 dt t^k \partial_i f(x_0 + t(x - x_0)) \right) (x^{i_\alpha} - x_0^{i_\alpha}) dx^i \wedge dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_\alpha}} \wedge \cdots \wedge dx^{i_k}. \end{aligned}$$

Gathering the two results we get:

$$\begin{aligned} P(d\omega) + d(P(\omega)) &= \sum_i \left(\int_0^1 dt t^k \partial_i f(x_0 + t(x - x_0)) \right) (x^i - x_0^i) dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ &\quad + \sum_{\alpha=1}^k \int_0^1 dt t^{k-1} [f(x_0 + t(x - x_0))] dx^{i_1} \wedge \cdots \wedge dx^{i_\alpha} \wedge \cdots \wedge dx^{i_k} = \int_0^1 dt \left[t^k \frac{d}{dt} f + k t^{k-1} f \right] dx^{i_1} \wedge \cdots \wedge dx^{i_k}. \end{aligned}$$

□

The lemma is immediately generalizable to star shaped domains (convex for all the paths with one end point fixed).

In general, one defines the de Rham cohomology group as the group of the closed forms modulo the exact ones, $H_{\text{dR}}^k(M) = \ker(d_k)/\text{Im}(d_{k-1})$, where d_k designates the exterior derivative acting on k -forms. The Poincarè lemma states that the cohomology groups of star shape domains in $\mathbb{R}^n$ are trivial.

Exercise 12.23. *Prove that, in Euclidean space $\mathbb{R}^3$ with metric δ_{ij} , as $\nabla \cdot B = \partial_i B^i = 0$, any magnetic field is generated by a vector potential $B = \nabla \times A$, $B_i = \epsilon_{ijk} \partial_j A_k$.*

Chapter 13

The Stokes theorem

One can integrate differential two forms $\omega \in \wedge^2 T^* \mathbb{R}^n$ over surfaces embedded in $\mathbb{R}^n$. Let the surface Σ be parametrized by $(s^1, s^2) \in [0, 1] \times [0, 1]$ and embedded in $\mathbb{R}^n$ by the embedding map $\phi(s^1, s^2) = (x^1(s), \dots, x^n(s))$. We divide the square $[0, 1] \times [0, 1]$ into boxes of side length l . The box with vertices (s^1, s^2) , $(s^1 + l, s^2)$, $(s^1 + l, s^2 + l)$ and $(s^1, s^2 + l)$ is the box based at (s^1, s^2) and with sides the vectors $l\partial_{s^1}$ and $l\partial_{s^2}$ in $T_{(s^1, s^2)}\Sigma$. These two vector push forward to $\mathbb{R}^n$ and define a parallelogram at $\phi(s^1, s^2)$ with sides $\phi_*(l\partial_{s^1}) = l \sum_i \frac{\partial x^i}{\partial s^1} \partial_{x^i}$ and $\phi_*(l\partial_{s^2}) = l \sum_i \frac{\partial x^i}{\partial s^2} \partial_{x^i}$. A two form $\omega \in \wedge^2 T^* \mathbb{R}^n$ evaluates on these two vectors to a real number $\omega(\phi_*(l\partial_{s^1}), \phi_*(l\partial_{s^2})) = l^2 \omega(\phi_*(\partial_{s^1}), \phi_*(\partial_{s^2}))$ and one can construct the Riemann sum $\sum_{\text{boxes}} l^2 \omega(\phi_*(\partial_{s^1}), \phi_*(\partial_{s^2}))$. Taking the limit of infinitesimal boxes, one obtains the integral of the two form over the surface $\int_{\phi(\Sigma)} \omega$. Note that for this to work, ω needs to be a differential form: when the parallelogram degenerates the summand in the Riemann sum must be zero, hence $\omega(V, V) = 0$ and, together with linearity, $\omega(V_1 + V_2, V_1 + V_2) = 0$ implies $\omega(V_1, V_2) + \omega(V_2, V_1) = 0$.

In general, differential n -forms can be integrated on n -dimensional manifolds. We proceed in steps. Assume we have a n form on $\mathbb{R}^n$ written in the standard basis as $\omega = \omega(x) dx^1 \wedge \dots \wedge dx^n$. We define the integral of ω on some open (Jordan measurable) set V as the Riemann integral in n dimensions:

$$\int_V \omega = \int_V \omega(x) dx^1 \wedge \dots \wedge dx^n \stackrel{\text{def.}}{=} \int_V dx^1 \dots dx^n \omega(x) = \int_V d^n x \omega(x).$$

Note that here we implicitly endowed $\mathbb{R}^n$ with an orientation. If we have a chart (U, φ) in a n dimensional manifold and ω an n -form which writes in the chart φ as $\omega = \omega(x) dx^1 \wedge \dots \wedge dx^n$, then:

$$\int_U \omega \stackrel{\text{def.}}{=} \int_{\varphi(U)} \omega(x) dx^1 \wedge \dots \wedge dx^n = \int_{\varphi(U)} dx^1 \dots dx^n \omega(x) = \int_{\varphi(U)} d^n x \omega(x).$$

In order to simplify the notation, we suppress from now on the chart. Clearly the integral of a form is invariant under a coordinate change.

Finally, assume that the n dimensional manifold Σ is mapped into the l dimensional manifold M by the pushforward map $\phi : \Sigma \rightarrow M$ which we denote $\phi(u) = (x^1(u), \dots, x^l(u))$ with x some local coordinates on M and u local coordinates on Σ . For a n -form $\omega = \sum_{i_1 < \dots < i_n} \omega_{i_1 \dots i_n}(x) dx^{i_1} \wedge \dots \wedge dx^{i_n}$ on M , the integral of ω over $\phi(\Sigma)$ is the integral of the pullback $\phi^* \omega$ over Σ :

$$\int_{\phi(\Sigma)} \omega \stackrel{\text{def.}}{=} \int_{\Sigma} \phi^* \omega = \int_{\Sigma} \sum_{i_1 < \dots < i_n} \omega_{i_1 \dots i_n}(x(u)) \frac{\partial(x^{i_1}, \dots, x^{i_n})}{\partial(u^1, \dots, u^n)} du^1 \wedge \dots \wedge du^n.$$

13.1 Stokes's theorem for a box domain

We denote in the remainder of this chapter the manifold boundary of the manifold M by ∂M and we recall that a hat over an element in a list signifies that the element is omitted in the list.

For any function $f : [a, b] \rightarrow \mathbb{R}$, the one form df can be integrated over the closed interval (which is a one dimensional manifold):

$$\int_{[a,b]} df = \int_{[a,b]} \frac{df}{dx} dx = \int_a^b \frac{df}{dx} dx = f(b) - f(a) .$$

The attentive reader will note the minus sign between the two terms.

A one form can always be integrated along a curve. Take a curve (with no self intersections, to keep things simple) in $\mathbb{R}^2$, $\gamma : [a, b] \rightarrow \mathbb{R}^2$, $\gamma(t) = (x^1(t), x^2(t))$ and the one form $f_1 dx^1 + f_2 dx^2$. Then:

$$\begin{aligned} \int_{\gamma} (f_1 dx^1 + f_2 dx^2) &= \int_{[a,b]} \gamma^* (f_1 dx^1 + f_2 dx^2) = \int_a^b \left(f_1 \frac{dx^1}{dt} + f_2 \frac{dx^2}{dt} \right) dt , \\ \int_{\gamma = \gamma_1 \cup \gamma_2} (f_1 dx^1 + f_2 dx^2) &= \int_{\gamma_1} (f_1 dx^1 + f_2 dx^2) + \int_{\gamma_2} (f_1 dx^1 + f_2 dx^2) , \end{aligned}$$

for any split of the curve γ into two curves γ_1 and γ_2 . Of course the integral is invariant under reparametrizations of the curve $t \rightarrow t(\tau)$. Similarly, one integrates one forms over curves in any manifold.

Let us consider the box $B = [0, 1] \times [0, 1]$ in dimension 2. Its boundary ∂B consists in the segments $[0, 1] \times \{0\}$, $\{1\} \times [0, 1]$, $[0, 1] \times \{1\}$ and $\{0\} \times [0, 1]$, which we parametrize as the curves $\gamma_1(t) = (t, 0)$, $\gamma_2(t) = (1, t)$, $\gamma_3(t) = (1 - t, 1)$ and $\gamma_4(t) = (0, 1 - t)$, with $t \in [0, 1]$. As all the curves are oriented from $\gamma_i(0)$ to $\gamma_i(1)$, we see that these parameterizations correspond to *turning counterclockwise* around the box B . Relabelling some dummy variables and using the fundamental theorem of calculus, we have that for any two functions $f_1, f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$:

$$\begin{aligned} \int_{\partial B} (f_1 dx^1 + f_2 dx^2) &= \int_{\gamma_1 \cup \gamma_2 \cup \gamma_3 \cup \gamma_4} (f_1 dx^1 + f_2 dx^2) \\ &= \int_0^1 f_1(t, 0) dt + \int_0^1 f_2(1, t) dt + \int_0^1 f_1(1 - t, 1) (-1) dt + \int_0^1 f_2(0, 1 - t) (-1) dt \\ &= \int_0^1 f_1(x^1, 0) dx^1 + \int_0^1 f_2(1, x_2) dx^2 - \int_0^1 f_1(x^1, 1) dx^1 - \int_0^1 f_2(0, x_2) dx^2 = \int_0^1 \int_0^1 \left(\frac{\partial f_2}{\partial x_1} - \frac{\partial f_1}{\partial x_2} \right) dx^1 dx^2 . \end{aligned}$$

This generalized fundamental theorem of calculus can be written naturally using forms: denoting $\alpha = f_1 dx^1 + f_2 dx^2$, the right hand side is nothing but the integral of the form $d\alpha = (\partial_{x_1} f_2 - \partial_{x_2} f_1) dx^1 \wedge dx^2$ inside the box B , that is:

$$\int_{\partial B} \alpha = \int_B d\alpha .$$

This is a particular case of a much more general formula. The main difficulty in higher dimensions is to carefully track the orientation of the boundary pieces of the box. The counterclockwise orientation can also be tracked by noticing that the outward pointing normal along the pieces of the boundary is respectively $-\partial_{x^2}$, ∂_{x^1} , ∂_{x^2} , $-\partial_{x^1}$ and the volume form $dx^1 \wedge dx^2$ evaluated on the normal is respectively:

$$i_{\nu}(dx^1 \wedge dx^2) = \quad dx^1 \text{ on } [0, 1] \times \{0\} , \quad dx^2 \text{ on } \{1\} \times [0, 1] , \quad -dx^1 \text{ on } [0, 1] \times \{1\} , \quad -dx^2 \text{ on } \{0\} \times [0, 1] ,$$

hence the orientation with which the pieces of the boundary need to be counted is the orientation induced by the outward pointing normal. The attentive reader will note that the same happens in one dimension.

In higher dimension, we consider the box domain $B^k = [a^1, b^1] \times \dots \times [a^k, b^k]$ in $\mathbb{R}^k$ endowed with standard metric, and with volume form:

$$\omega_{B^k} = dx^1 \wedge \dots \wedge dx^k, \quad \int_{B^k} \rho(x) \omega_{B^k} = \int_{[a^1, b^1] \times \dots \times [a^k, b^k]} \rho(x) dx^1 \wedge \dots \wedge dx^k = \int_{a^1}^{b^1} \dots \int_{a^k}^{b^k} \rho(x) dx^1 \dots dx^k.$$

The volume form ω_{B^k} induces an orientation on the boundary of the box. The outward pointing normals are $\nu_{i, b^i} = \partial_{x^i}$, for the boundary component $[a^1, b^1] \times \dots \times \{b^i\} \times \dots \times [a^k, b^k]$, respectively $\nu_{i, a^i} = -\partial_{x^i}$, for $[a^1, b^1] \times \dots \times \{a^i\} \times \dots \times [a^k, b^k]$. We embedded the boundary back in B^k , that is $[a^1, b^1] \times \dots \times \{a^i\} \times \dots \times [a^k, b^k] \subset \partial B^k$ is parametrized as:

$$\phi_{i, a^i} : B_i^{k-1} \rightarrow B^k, \quad (u^1, \dots, \widehat{u^i}, \dots, u^k) \mapsto (x^1 = u^1, \dots, x^i = a^i, \dots, x^k = u^k),$$

and so on. The volume form on ∂B^k , $i_\nu \omega_{B^k}$, pulls back to the intrinsic volume form on B_i^{k-1} with a different sign on each component of the boundary:

$$\begin{aligned} \phi_{i, a^i}^* (i_{\nu_{i, a^i}} \omega_{B^k}) &= \phi_{i, a^i}^* \left(dx^1 \wedge \dots \wedge dx^k (-\partial_{x^i}) \right) = (-1)^i \phi_{i, a^i}^* \left(dx^1 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^k \right) = (-1)^i du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k, \\ \phi_{i, b^i}^* (i_{\nu_{i, b^i}} \omega_{B^k}) &= \phi_{i, b^i}^* \left(dx^1 \wedge \dots \wedge dx^k (\partial_{x^i}) \right) = (-1)^{i-1} \phi_{i, b^i}^* \left(dx^1 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^k \right) = (-1)^{i-1} du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k, \end{aligned}$$

and the intrinsic oriented volume form on B_i^{k-1} is $\omega_{B_i^{k-1}} = du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k$.¹ The oriented boundary of the box is the signed sum of the (one dimension lower) boundary boxes, that is for any $k-1$ form Ω on B^k we have:²

$$\partial B^k \equiv \sum_{i=1}^k (-1)^{i-1} [\phi_{i, b^i}(B_i^{k-1}) - \phi_{i, a^i}(B_i^{k-1})], \quad \int_{\partial B^k} \Omega = \sum_{i=1}^k (-1)^{i-1} \left[\int_{\phi_{i, b^i}(B_i^{k-1})} \Omega - \int_{\phi_{i, a^i}(B_i^{k-1})} \Omega \right],$$

and $\partial \partial B^k = 0$. Indeed, $\partial[0, 1] = \{1\} - \{0\}$ and:

$$\begin{aligned} \partial \partial[0, 1] \times [0, 1] &= \partial(\{1\} \times [0, 1] - \{0\} \times [0, 1] - [0, 1] \times \{1\} + [0, 1] \times \{0\}) \\ &= \{1\} \times \{1\} - \{1\} \times \{0\} - \{0\} \times \{1\} + \{0\} \times \{0\} - \{1\} \times \{1\} + \{0\} \times \{1\} + \{1\} \times \{0\} - \{0\} \times \{0\} = 0. \end{aligned}$$

Note that $\int_{[a, b]} f(x) dx = \int_a^b f(x) dx$ with $a \leq b$, that is the integral over a one dimensional box always goes from the lower limit to the upper limit of the interval.

Theorem 13.1 (Stokes' theorem on boxes). *For any smooth $k-1$ form Ω on the box $B^k \subset \mathbb{R}^k$ we have:*

$$\int_{B^k} d\Omega = \int_{\partial B^k} \Omega.$$

Proof. Consider a $k-1$ form on $\mathbb{R}^k$, $\Omega = \sum_{j=1}^k \Omega_j(x) dx^1 \wedge \dots \wedge \widehat{dx^j} \wedge \dots \wedge dx^k$. We have:

$$\phi_{i, a^i}^* \Omega = \sum_{j=1}^k \Omega_j(x) \frac{\partial(x^1, \dots, \widehat{x^j}, \dots, x^k)}{\partial(u^1, \dots, \widehat{u^i}, \dots, u^k)} \Big|_{x^i=a^i, x^j=u^j} du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k = \Omega_i(u^1, \dots, a^i, \dots, u^k) du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k,$$

¹We have $\int_{B_i^{k-1}} \rho(u) \omega_{B_i^{k-1}} = \int_{[a^1, b^1] \times \dots \times [\widehat{a^i, b^i}] \times \dots \times [a^k, b^k]} \rho(u) du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k = \int_{a^1}^{b^1} \dots \int_{a^i}^{b^i} \dots \int_{a^k}^{b^k} \rho(u) du^1 \wedge \dots \wedge \widehat{du^i} \wedge \dots \wedge du^k$.

²For instance the total area of the box (i.e. the volume of the boundary of the box) is:

$$\begin{aligned} \int_{\partial B^k} i_\nu \omega_{B^k} &= \sum_{i=1}^k (-1)^{i-1} \left[\int_{\phi_{i, b^i}(B_i^{k-1})} i_{\nu_{i, b^i}} \omega_{B^k} - \int_{\phi_{i, a^i}(B_i^{k-1})} i_{\nu_{i, a^i}} \omega_{B^k} \right] = \sum_{i=1}^k (-1)^{i-1} \left[\int_{B_i^{k-1}} \phi_{i, b^i}^* (i_{\nu_{i, b^i}} \omega_{B^k}) - \int_{B_i^{k-1}} \phi_{i, a^i}^* (i_{\nu_{i, a^i}} \omega_{B^k}) \right] \\ &= \sum_{i=1}^k 2 \int_{B_i^{k-1}} \omega_{B_i^{k-1}} = 2 \sum_{i=1}^k \prod_{j \neq i} |b^j - a^j|. \end{aligned}$$

and so on. In words, on the side $x^i = a^i$ of the box, the component $dx^1 \wedge \dots \widehat{dx^i} \wedge \dots dx^k$ pulls back trivially and all the other components pull back to zero. Collecting all the pieces of the boundary we have:

$$\begin{aligned} \int_{\partial B^k} \Omega &= \sum_{i=1}^k (-1)^{i-1} \left(\int_{\phi_{i,b^i}(B_i^{k-1})} \Omega - \int_{\phi_{i,a^i}(B_i^{k-1})} \Omega \right) = \sum_{i=1}^k (-1)^{i-1} \left(\int_{B_i^{k-1}} \phi_{i,b^i}^* \Omega - \int_{B_i^{k-1}} \phi_{i,a^i}^* \Omega \right) \\ &= \sum_{i=1}^k (-1)^{i-1} \left(\int_{B_i^{k-1}} \Omega_i(u^1, \dots, b^i, \dots, u^k) du^1 \wedge \dots \widehat{du^i} \wedge \dots du^k - \int_{B_i^{k-1}} \Omega_i(u^1, \dots, a^i, \dots, u^k) du^1 \wedge \dots \widehat{du^i} \wedge \dots du^k \right) \\ &= \sum_{i=1}^k (-1)^{i-1} \left(\int_{B_i^{k-1}} \Omega_i(u^1, \dots, b^i, \dots, u^k) du^1 \dots \widehat{du^i} \dots du^k - \int_{B_i^{k-1}} \Omega_i(u^1, \dots, a^i, \dots, u^k) du^1 \dots \widehat{du^i} \dots du^k \right). \end{aligned}$$

Using the fundamental theorem of calculus we have:

$$\Omega_i(u^1, \dots, b^i, \dots, u^k) - \Omega_i(u^1, \dots, a^i, \dots, u^k) = \int_{a^i}^{b^i} du^i \frac{\partial \Omega_i}{\partial x^i}(u^1, \dots, u^i, \dots, u^k),$$

and we conclude that:

$$\int_{\partial B^k} \Omega = \int_{B^k} \left(\sum_{i=1}^k (-1)^{i-1} \frac{\partial \Omega_i}{\partial x^i}(x) \right) dx^1 \dots dx^i \dots dx^k.$$

On the other hand $d\Omega = \sum_{j=1}^k \frac{\partial \Omega_j}{\partial x^j}(y) dx^j \wedge dx^1 \wedge \dots \widehat{dx^j} \wedge \dots dx^k = \sum_{j=1}^k (-1)^{j-1} \frac{\partial \Omega_j}{\partial x^j}(y) dx^1 \wedge \dots dx^j \wedge \dots dx^k$ and the theorem follows. $\square$

Consider for instance the form $x^1 dx^2$ on B^2 . Then on the one hand:

$$\int_{B^2} d(x^1 dx^2) = \int_{B^2} dx^1 \wedge dx^2 = \int_{[a^1, b^1] \times [a^2, b^2]} dx^1 \wedge dx^2 = \int_{a_1}^{b_1} dx_1 \int_{a_2}^{b_2} dx^2 = |b^1 - a^1| |b^2 - a^2|,$$

and on the other:

$$\phi_{1,b^1}^*(x^1 dx^2) = b^1 ds^2, \quad \phi_{1,a^1}^*(x^1 dx^2) = a^1 ds^2, \quad \phi_{2,\dots}(x^1 dx^2) = 0,$$

therefore:

$$\begin{aligned} \int_{\partial B^2} x^1 dx^2 &= \sum_{i=1}^2 (-1)^{i-1} \left[\int_{\phi_{i,b^i}(B^1)} x^1 dx^2 - \int_{\phi_{i,a^i}(B^1)} x^1 dx^2 \right] = \int_{[a^2, b^2]} b^1 ds^2 - \int_{[a^2, b^2]} a^1 ds^2 \\ &= \int_{a_2}^{b_2} b^1 ds^2 - \int_{a_2}^{b_2} a^1 ds^2 = |b^1 - a^1| |b^2 - a^2|. \end{aligned}$$

The Stokes theorem on boxes is a particular case of a much more general theorem on orientable manifolds with boundary. The problem in the general case is to identify properly the exterior normals to the manifold, or equivalently the correct notion of oriented boundary. The Stokes theorem is intrinsic, that is it holds for any k dimensional manifold. For instance one can consider k dimensional manifolds embedded in n dimensional ones. This follows from the remark that the pullback commutes with the exterior differential and with the signed boundary:

$$\int_D d\alpha = \int_{\phi(B^k)} d\alpha = \int_{B^k} \phi^*(d\alpha) = \int_{B^k} d(\phi^*\alpha) = \int_{\partial B^k} \phi^*\alpha = \int_{\phi(\partial B^k)} \alpha = \int_{\partial D} \alpha.$$

Exercise 13.1. Prove that for any form α , $\phi^*(d\alpha) = d(\phi^*\alpha)$. For one forms we have $\phi^*(d\alpha)(V) = d\alpha(\phi^*(V)) = \phi^*(V)(\alpha) = V(\alpha \circ \phi) = d(\phi^*\alpha)(V)$.

Exercise 13.2. Prove that if ϕ is a bicontinuous bijection $\partial\phi(B^k) = \phi(\partial B^k)$. As ϕ is continuous with continuous inverse, $\phi(B^k)$ is compact and ϕ

maps open sets to open sets, hence interior points to interior points.

Consider for instance $\mathbb{R}^3$ with flat metric and we label its coordinates x, y, z the one form $\alpha = z^2 dx + y^2 dy + x dz$ and the contour $\gamma = \partial\Sigma$ bounding the square Σ with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(1, 0, 1)$, $(0, 0, 1)$. The square Σ is two dimensional, but it is embedded in three dimensions.

We can test Stokes theorem directly for the square embedded in $\mathbb{R}^3$. On the one hand we have $d\alpha = (1 - 2z)dx \wedge dz$ and the embedding of B^2 in $\mathbb{R}^3$ is $\phi(B^2) = (s^1, 0, s^2) = (x(s), y(s), z(s))$, therefore:

$$\int_{\Sigma} d\alpha = \int_{\phi(B^2)} d\alpha = \int_{B^2} \phi^*(d\alpha) = \int_{B^2} (1 - 2z(s)) \frac{\partial(x, z)}{\partial(s^1, s^2)} ds^1 \wedge ds^2 = \int_0^1 ds^1 \int_0^1 ds^2 (1 - 2s^2) = 1 - 2 \int_0^1 ds^2 s^2 = 0.$$

The boundary is embedded in $\mathbb{R}^3$ as $\phi_{x,1}(t) = (1, 0, t) = (x(t), y(t), z(t))$, $\phi_{x,0}(t) = (t, 0, 1) = (x(t), y(t), z(t))$ and so on, hence:

$$\gamma = \partial\Sigma = \phi_{x,1}(B^1) - \phi_{x,0}(B^1) - \phi_{z,1}(B^1) + \phi_{z,0}(B^1), \quad \int_{\gamma} \alpha = \int_{\phi_{x,1}(B^1)} \alpha - \int_{\phi_{x,0}(B^1)} \alpha - \int_{\phi_{z,1}(B^1)} \alpha + \int_{\phi_{z,0}(B^1)} \alpha,$$

and for each piece we get:

$$\begin{aligned} \int_{\phi_{x,1}(B^1)} \alpha &= \int_{B^1} \phi_{x,1}^*(\alpha) = \int_{B^1} (z(t)^2 \frac{dx}{dt} + y(t)^2 \frac{dy}{dt} + x(t) \frac{dz}{dt}) dt = \int_0^1 dt = 1, \\ \int_{\phi_{x,0}(B^1)} \alpha &= 0, \quad \int_{\phi_{z,1}(B^1)} \alpha = 1, \quad \int_{\phi_{z,0}(B^1)} \alpha = 0, \end{aligned}$$

which add up to $\int_{\gamma} \alpha = 0$, as expected.

Alternatively we can first pullback to $\mathbb{R}^2$, $\phi^* \alpha = (s^2)^2 ds^1 + s^1 ds^2$, $d(\phi^* \alpha) = (1 - 2s^2) ds^1 \wedge ds^2$ and check the Stokes theorem on B^2 :

$$\int_{B^2} (1 - 2s^2) ds^1 \wedge ds^2 = \int_{\partial B^2} ((s^2)^2 ds^1 + s^1 ds^2) = 0.$$

13.1.1 Orientability

A n -dimensional manifold is orientable if it admits an atlas such that all the transition maps have positive Jacobian. Equivalently, a n -manifold is orientable if it admits a non-degenerate (that is nowhere zero) volume n -form.

Orientation of surfaces embedded in $\mathbb{R}^3$. A surface embedded in $\mathbb{R}^3$ is orientable if it admits a consistently defined normal vector field, that is a normal vector field which varies continuously when moving along the surface. This makes the volume form on the surface well defined and nowhere zero. Not all the two dimensional surfaces embedded in $\mathbb{R}^3$ are orientable: for instance the Möbius strip is not as a normal vector ν will continuously return to $-\nu$ when moving along the surface. A normal vector ν orients a surface Σ ,³ that is the positive volume form on Σ is the pullback of $i_{\nu} \omega$, with ω the volume form on $\mathbb{R}^3$. Once ν is chosen, the orientation of Σ induces an orientation on $\partial\Sigma$. It turns out that the induced orientation on $\partial\Sigma$ corresponds to the orientation of the right hand rule with respect to ν . This comes about as, in Cartesian coordinates the induced volume form on the plane $\text{Span}(\partial_1, \partial_2)$ is $i_{\partial_3}(dx^1 \wedge dx^2 \wedge dx^3) = dx^1 \wedge dx^2$ and going around the elementary square $\{[0, 1] \times \{0\}\} \cup \{\{1\} \times [0, 1]\} \cup \{[0, 1] \times \{1\}\} \cup \{\{0\} \times [0, 1]\}$ in this plane, the exterior normal ν is in order $-\partial_2, \partial_1, \partial_2, -\partial_1$ hence $i_{\nu}(dx^1 \wedge dx^2)$ is $dx^1, dx^2, -dx^1, -dx^2$, corresponding to a counterclockwise turn around ∂_3 .

Let us consider again the one form $\alpha = z^2 dx + y^2 dy + x dz$ with $d\alpha = (1 - 2z)dx \wedge dz$ and the contour γ consisting in the triangle Δ with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, encountered in this order. Our aim is to compute $\int_{\gamma} \alpha$. We do this in three ways:

³If the surface bounds some three dimensional manifold with boundary, then it is customary to choose the exterior normal, but if not one chooses one of the two possible normals

- *Direct computation.* We parametrize the three segments of γ as $\gamma_1 = (x = 1 - u, y = u, z = 0)$, $\gamma_2 = (x = 0, y = 1 - u, z = u)$ and $\gamma_3 = (x = u, y = 0, z = 1 - u)$ and compute:

$$\int_{\gamma} \alpha = \int du \left(\alpha_x \frac{dx}{du} + \alpha_y \frac{dy}{du} + \alpha_z \frac{dz}{du} \right) = \int_0^1 du u^2 + \int_0^1 du (1 - u)^2 (-1) + \int_0^1 du [(1 - u)^2 + u(-1)] = -\frac{1}{6}.$$

- *One triangle.* We denote T the triangle, that is the two dimensional manifold with boundary $T = \{(s, t) \mid 0 \leq s \leq 1, 0 \leq t \leq 1 - s\}$. The boundary of T has three components $\partial T_1 = \psi_1(u) = (s = 1 - u, t = 0)$, $\partial T_2 = \psi_2(u) = (s = 0, t = u)$ and $\partial T_3 = \psi_3(u) = (s = u, t = 1 - u)$, with normals $-\partial_t$, $-\partial_s$ and $\partial_s + \partial_t$. The triangle Δ is embedded in $\mathbb{R}^3$ by the embedding map $\Delta = \phi(T)$:

$$\phi : T \rightarrow \Delta \subset \mathbb{R}^3, \quad \phi(s, t) = (x = s, y = 1 - s - t, z = t), \quad \phi^*(d\alpha) = (1 - 2t)ds \wedge dt,$$

and $\phi(\partial T_1) = \gamma_1$, $\phi(\partial T_2) = \gamma_2$ respectively $\phi(\partial T_3) = \gamma_3$, with the parametrizations given previously.

The triangle Δ is alternatively defined by the constraint $F(x, y, z) = x + y + z - 1 = 0$ with normal form $dF = dx + dy + dz$. The normal vector to Δ which is consistent with the orientation of γ (see below) is $\nu = \partial_x + \partial_y + \partial_z$ with corresponding induced volume form on T :

$$\phi^*(i_{\partial_x + \partial_y + \partial_z}(dx \wedge dy \wedge dz)) = \phi^*(dy \wedge dz - dx \wedge dz + dx \wedge dy) = (-ds - dt) \wedge dt - ds \wedge dt + ds \wedge (-ds - dt) \sim -ds \wedge dt,$$

that is the positively oriented volume form on T corresponding to ν is $dt \wedge ds$. With this volume form, the induced positive volume forms on the boundary components are $\phi(\partial T_1) : \psi_1^*(i_{-\partial_t} dt \wedge ds) = du$, $\phi(\partial T_2) : \psi_2^*(i_{-\partial_s} dt \wedge ds) = du$ and $\phi(\partial T_3) : \psi_3^*(i_{\partial_s + \partial_t} dt \wedge ds) = 2du$, confirming that the induced orientation on ∂T is given by the right hand rule with respect to the normal ν .

It follows that, denoting ∂T the boundary of T oriented by the volume form $dt \wedge ds$ we have $\phi(\partial T) = \gamma$ and:

$$\begin{aligned} \int_{\gamma} \alpha &= \int_{\phi(\partial T)} \alpha = \int_{\partial T} \phi^* \alpha = \int_T d(\phi^* \alpha) = \int_T \phi^*(d\alpha) \stackrel{\int_{\Delta} d\alpha}{=} \int_T (1 - 2t) ds \wedge dt \\ &= \int_T (2t - 1) dt \wedge ds \stackrel{\text{because } dt \wedge ds \text{ is the positive volume form}}{=} \int_T (2t - 1) dt ds \\ &= \int_0^1 ds \int_0^{1-s} dt (2t - 1) = \int_0^1 ds [(1 - s)^2 - (1 - s)] = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}. \end{aligned}$$

We stress that the positive volume form on Σ plays a role when erasing the $\wedge$ product in order to go to the usual Riemann integral: one needs first to order the volume form on T as dictated by ν and then erase the $\wedge$.

- *Tetrahedral surface.* The curve γ bounds also (among others) the tetrahedral surface consisting in the three triangles Δ_{xy} with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, Δ_{yz} with vertices $(0, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ and Δ_{zx} with vertices $(0, 0, 0)$, $(0, 0, 1)$, $(1, 0, 0)$, each of them is embedded via the embeddings:

$$\begin{aligned} \phi_{xy} : T &\rightarrow \Delta_{xy} \subset \mathbb{R}^3, & \phi_{xy}(s, t) &= (x = s, y = t, z = 0), & \phi_{xy}^*(d\alpha) &= 0, \\ \phi_{yz} : T &\rightarrow \Delta_{yz} \subset \mathbb{R}^3, & \phi_{yz}(s, t) &= (x = 0, y = s, z = t), & \phi_{yz}^*(d\alpha) &= 0, \\ \phi_{zx} : T &\rightarrow \Delta_{zx} \subset \mathbb{R}^3, & \phi_{zx}(s, t) &= (x = s, y = 0, z = t), & \phi_{zx}^*(d\alpha) &= (1 - 2t)ds \wedge dt. \end{aligned}$$

As $d\alpha$ pulls back non trivially only on Δ_{zx} , we now focus on it. The triangle Δ_{zx} is defined by the constraint $F(x, y, z) = y = 0$ with normal form $dF = dy$. The normal vector to Δ which is consistent with the orientation of γ is ∂_y and the induced volume form on T is $\phi_{zx}^*(i_{\partial_y} dx \wedge dy \wedge dz) = -ds \wedge dt$. As before, denoting ∂T the boundary of T

oriented by $dt \wedge ds$, $\phi_{zx}(\partial T) \cup \phi_{xy}(\partial T) \cup \phi_{yz}(\partial T) = \gamma$ and:

$$\int_{\gamma} \alpha = \int_{\Delta_{zx}} d\alpha = \int_T \phi_{zx}^*(d\alpha) = \int_T (1-2t)ds \wedge dt = \int_T (2t-1)dt \wedge ds = \int_T (2t-1) dt ds = -\frac{1}{6}.$$

13.1.2 Euclidean $\mathbb{R}^n$

An orientable n -dimensional manifold with boundary is obtained by gluing n dimensional boxes along $n-1$ boundary boxes such that every boundary box is assigned opposite orientations by the two n -dimensional boxes to which it belongs,⁴ that is the exterior normals to the $n-1$ box assigned by the two n boxes point in opposite directions. The boundary components possess a consistent induced orientation (as they are gluings of oriented $n-1$ dimensional boxes). An example of an orientable surface with two boundary components is a cylinder, while a non orientable surface with one boundary component is the Möbius strip. We will discuss the proof of the general Stokes in the next section: for now we assume that for an arbitrary compact orientable n dimensional manifold M and an arbitrary smooth $n-1$ form Ω in M with compact support we have:

$$\int_M d\Omega = \int_{\partial M} \Omega,$$

where the boundary ∂M is endowed with the induced orientation. The boundary can have several connected components, in which case the right hand side splits into an integral per connected component.

This formula needs more precision. The boundary $\partial M \subset M$ is an embedded submanifold of M and denoting N the corresponding abstract manifold and $\phi : N \rightarrow \partial M \subset M$ the embedding map, the right hand side of the Stokes theorem is $\int_{\partial M} \Omega \equiv \epsilon \int_{\phi(N)} \Omega = \int_N \epsilon \phi^* \Omega$, where we included a sign ϵ in order to account for a possible discrepancy between the intrinsic and the induced orientation of N . In words, the form Ω , which is a form in $\wedge^{n-1} T_p M$, needs to be pulled back to $\wedge^{n-1} T_p N$ and integrated over N with the induced orientation. The sign ϵ is determined by requiring that the induced volume form on N is positively oriented, that is for $\Omega = i_\nu \omega$ with ω the volume form on M and ν the exterior (outward pointing) unit normal at ∂M , the integrand $\epsilon \phi^*(i_\nu \omega) = \rho(p) \omega_N$ with ω_N the positive volume form on N and $\rho(p) > 0$.

13.1.3 The many names of the Stokes theorem

One uses the Stokes theorem extensively and under many names in electromagnetism. Let us consider a n dimensional manifold with metric $\sum_{ij} g_{ij} dx^i dx^j$ and volume form (in physics notation) $dV = \omega = \sqrt{\det g(x)} dx^1 \wedge \dots \wedge dx^n$.

The divergence theorem. Let us consider a n dimensional orientable compact embedded submanifold $D \subset \mathbb{R}^n$ with boundary $\partial D = \Sigma$ and a vector field X . For the vector field $X = \sum_i X^i \partial_i$ the Stokes theorem yields:

$$\int_{\Sigma} i_X \omega = \int_D d(i_X \omega) = \int_D \operatorname{div} X \omega, \quad \operatorname{div} X = \sum_i \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i).$$

For simplicity we set $n = 3$ we assume Σ is defined by an embedding map $\phi : S \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $\phi(s^1, s^2) = (x^1(s), x^2(s), x^3(s))$. The exterior normal to the surface and the area element dA (in physics notation) are:

$$\sum_{ij} \nu^i g_{ij} \frac{\partial x^j}{\partial s^\mu} = 0, \quad \sum_{ij} g_{ij} \nu^i \nu^j = 1, \quad dA = \phi^*(i_\nu \omega) = \sqrt{\det g} \begin{vmatrix} \nu^1 & \nu^2 & \nu^3 \\ \frac{\partial x^1}{\partial s^1} & \frac{\partial x^2}{\partial s^1} & \frac{\partial x^3}{\partial s^1} \\ \frac{\partial x^1}{\partial s^2} & \frac{\partial x^2}{\partial s^2} & \frac{\partial x^3}{\partial s^2} \end{vmatrix} ds^1 \wedge ds^2.$$

Then $\int_{\Sigma} i_X \omega = \int_S \phi^*(i_X \omega)$ and $\phi^*(i_X \omega)(V_1, V_2) = \omega(X, \phi_*(V_1), \phi_*(V_2))$ for $V_1, V_2 \in T\Sigma$. The exterior normal is such that $g(\nu, \phi_*(V)) = 0$ for any $V \in T\Sigma$. The space orthogonal to ν has dimension $3-1=2$. Decomposing the vector X

⁴Note that we can deform the boxes by orientation preserving mappings, that is ϕ such that $d\phi$ has positive determinant.

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in a component parallel to the normal and an orthogonal one $X = g(X, \nu)\nu + X^\perp$, $\omega(X^\perp, \phi_*(V_1), \phi_*(V_2))$ is a three form evaluated on three linearly dependent vectors (as they are three vectors in a 2 dimensional vector space), hence is zero by anti symmetry, therefore $\phi^*(i_X \omega) = g(X, \nu)\phi^*(i_\nu \omega)$. Putting everything together we get (in physics notation)

$$\int_D \underbrace{dV}_{\sqrt{\det g} dx^1 \wedge dx^2 \wedge dx^3} \underbrace{\operatorname{div} X}_{\sum_i \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i)} = \int_S \underbrace{dA}_{\sqrt{\det g} \begin{vmatrix} \nu^1 & \nu^2 & \nu^3 \\ \frac{\partial x^1}{\partial s^1} & \frac{\partial x^2}{\partial s^1} & \frac{\partial x^3}{\partial s^1} \\ \frac{\partial x^1}{\partial s^2} & \frac{\partial x^2}{\partial s^2} & \frac{\partial x^3}{\partial s^2} \end{vmatrix} ds^1 \wedge ds^2} \underbrace{X \cdot \nu}_{\sum_{i,j} g_{ij} \nu^i X^j}.$$

Exercise 13.3. Write the Stokes theorem for a sphere of radius R bounding a ball of radius R in spherical coordinates. In spherical coordinates, the ball is parametrized as $x^1 = r \sin \theta \cos \varphi$, $x^2 = r \sin \theta \sin \varphi$ and $x^3 = r \cos \theta$ with metric and volume forms:

$$\sum_{i,j} \delta_{ij} \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\nu} ds^\mu ds^\nu = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\varphi^2, \quad dV = r^2 \sin \theta dr \wedge d\theta \wedge d\varphi.$$

In spherical coordinates the sphere is embedded as $(\Theta, \Phi) \rightarrow (r = R, \theta = \Theta, \varphi = \Phi)$, and the exterior unit normal to the sphere satisfies $\sum_{i,j} \nu^i g_{ij} \frac{\partial x^i}{\partial s^\mu} = \nu^\theta = \nu^\varphi = 0$ and consequently $\nu = \partial_r$. The unit volume form (area form) on the sphere can be computed in two ways. On the one hand $\phi^*(i_\nu \omega) = \phi^*(r^2 \sin \theta d\theta \wedge d\varphi) = R^2 \sin \Theta d\Theta \wedge d\Phi$, on the other the pullback of the metric $dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\varphi^2$ on the sphere and consequently the volume form is:

$$\sum_{i,j} g_{ij} \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\nu} ds^\mu ds^\nu = R^2 d\Theta^2 + R^2 \sin^2 \Theta d\Phi^2, \quad dA = R^2 \sin \Theta d\Theta \wedge d\Phi.$$

The Stokes theorem writes in this case as:

$$\int_D (r^2 \sin \theta dr \wedge d\theta \wedge d\varphi) \frac{1}{r^2 \sin \theta} \left[\partial_r (r^2 \sin \theta X^r) + \partial_\theta (r^2 \sin \theta X^\theta) + \partial_\varphi (r^2 \sin \theta X^\varphi) \right] = \int_S (R^2 \sin \Theta d\Theta \wedge d\Phi) X^r \Big|_{r=R, \theta=\Theta, \varphi=\Phi}.$$

Consider a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and denote $\partial_\nu f = \nu(f) = \sum_i \nu^i \partial_i f = g(\nu, \nabla f)$ the Gateaux derivative of f along ν . The Laplacian in arbitrary coordinates is the divergence of the gradient:

$$\Delta \varphi = \operatorname{div} \nabla \varphi = \sum_{i,j} \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} g^{ij} \partial_j \varphi).$$

Applying the Stokes theorem for the vector $X = \varphi \nabla \psi$ and taking into account that $\sum_{i,j} g_{ij} \nu^i X^j = \varphi \sum_i \nu^i \partial_i \psi = \varphi \partial_\nu \psi$, we get:

$$\int_S dA \varphi \partial_\nu \psi = \int_D dV \sum_{i,j} \frac{1}{\sqrt{\det g}} \partial_i [\sqrt{\det g} g^{ij} (\varphi \partial_j \psi)] = \int_D dV \varphi \Delta \psi + \int_D dV \sum_{i,j} g^{ij} \partial_i \varphi \partial_j \psi,$$

therefore $\int_S dA (\varphi \partial_\nu \psi - \psi \partial_\nu \varphi) = \int_D dV (\varphi \Delta \psi - \psi \Delta \varphi)$, which is also known as the Gauss-Ostrogradski theorem. For $n = 1$, the Stokes theorem $\int_{\partial D} (P dx + Q dy) = \int_D (\partial_x Q - \partial_y P) dx dy$, with ∂D oriented anticlockwise around D is also known as Green's theorem.

Kelvin-Stokes theorem. Another particular case is the Kelvin-Stokes theorem (or Stokes theorem, to increase the confusion). Let us consider an orientable two dimensional compact surface Σ bounded by a closed, smooth, simple curve $\partial \Sigma = \gamma$ embedded in $\mathbb{R}^3$ and a vector field B . We denote the embedding of the surface by $\phi : S \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $\phi(s^1, s^2) = (x^1(s), x^2(s), x^3(s))$, the one of the curve by $\varphi : I \subset \mathbb{R} \rightarrow \mathbb{R}^3$, $\varphi(t) = (x^1(t), x^2(t), x^3(t))$, the vector field $B = \sum_i B^i \partial_i$ and the associated one form $B^\flat = \sum_i B_i dx^i$, with $B_i = \sum_j g_{ij} B^j$. The Stokes theorem $\int_\gamma B^\flat = \int_\Sigma dB^\flat$ reads, denoting $T = \sum \frac{dx^i}{dt} \partial_i$ the tangent vector to the curve γ , as:

$$\int_\gamma B^\flat = \int_I \varphi^*(B^\flat) = \int_I \sum_i B_i \frac{dx^i}{dt} dt = \int_I dt B \cdot T,$$

where we used the physics notation for the scalar product $B \cdot T = g(B, T) = \sum_{i,j} g_{ij} B^i T^j$. On the other hand, we have:⁵

$$dB^b = \sum_{ij} \partial_i B_j dx^i \wedge dx^j = i_{\text{rot}B} \omega \quad \text{rot}B = \frac{1}{\sqrt{\det g}} \sum_{ijk} \epsilon^{ijk} (\partial_j \sum_l g_{kl} B^l) \partial_i ,$$

where $\text{rot}B$ is the curl, or rotational of B (also denoted $\nabla \times B$). Using $\int_{\Sigma} dB^b = \int_S \phi^*(dB^b) = \int_S \phi^*(i_{\text{rot}B} \omega)$ we conclude that:

$$\int_I dt \underbrace{B \cdot T}_{\sum_{i,j} g_{ij} B^i T^j} = \int_S \underbrace{dA}_{\sqrt{\det g} \begin{vmatrix} \nu^1 & \nu^2 & \nu^3 \\ \frac{\partial x^1}{\partial s^1} & \frac{\partial x^2}{\partial s^1} & \frac{\partial x^3}{\partial s^1} \\ \frac{\partial x^1}{\partial s^2} & \frac{\partial x^2}{\partial s^2} & \frac{\partial x^3}{\partial s^2} \end{vmatrix} ds^1 \wedge ds^2} \underbrace{\nu \cdot \left(\frac{1}{\sqrt{\det g}} \sum_{ijk} \epsilon^{ijk} (\partial_j \sum_l g_{kl} B^l) \partial_i \right)}_{\sum_{ij} g_{ij} \nu^i (\text{rot}B)^j} = \int_S dA \nu \cdot \text{rot}B ,$$

where ν is the normal oriented from the tangent vector field T by the right hand rule. Note that the equality holds for *any* surface bounded by γ .

For $\alpha = z^2 dx + y^2 dy + x dz$ with flat metric and Σ the square with vertices $(0,0,0)$, $(1,0,0)$, $(1,0,1)$, $(0,0,1)$ we get $\text{rot}\alpha^\sharp = e_y(2z-1)$, $dA = dx dz$ and $\nu = -e_y$ and $T = e_x, e_z, -e_x, -e_z$ along the segments of $\gamma = \partial\Sigma$.

Exercise 13.4. Rewrite Maxwell's equations for the electric and magnetic vector fields E and B , $\text{div}E = \rho$, $\text{div}B = 0$, $\text{rot}E = -\partial_t B$, $\text{rot}B = J + \partial_t E$ in integral form.

Exercise 13.5. Write the Stokes theorem for a disk domain centred at the origin and belonging to the x^1, x^2 plane in cylindrical coordinates, bounded by a circle of radius M . The cylindrical coordinates are $x^1 = r \cos \varphi$, $x^2 = r \sin \varphi$, $x^3 = z$. The metric in cylindrical coordinates is

$$\sum_{i,j} \delta_{ij} \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\nu} ds^\mu ds^\nu = dr^2 + r^2 d\varphi^2 + dz^2, \quad dV = r dr \wedge d\varphi \wedge dz .$$

The disk is embedded in cylindrical coordinates as $(R, \Phi) \rightarrow (r = R, \varphi = \Phi, z = 0)$ and the normal to the disk is ∂_z . The metric on the disk and, the area form are:

$$dR^2 + R^2 d\Phi^2, \quad dA = R \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} dR \wedge d\Phi = R dR \wedge d\Phi = \phi^*(i_{\partial_z}(r dr \wedge d\varphi \wedge dz)) .$$

The circle bounding the disk is embedded in cylindrical coordinates as $t \rightarrow (r = M, \varphi = \psi(t), z = 0)$, and the tangent vector to a path going around the circle is $T^\varphi = \frac{d\psi}{dt} \partial_\varphi$. For any vector field $B = B^r \partial_r + B^\varphi \partial_\varphi + B^z \partial_z$, we have:

$$\int_I dt M^2 B^\varphi \Big|_{r=M, \varphi=\psi(t), z=0} T^\varphi = \int_S (R dR \wedge d\Phi) \left(\frac{1}{r} [-\partial_\varphi(B^r) + \partial_r(r^2 B^\varphi)] \right) \Big|_{r=M, \varphi=\Phi, z=0} .$$

13.2 Triangulated spaces

The main point of this section is that one has a straightforward notion of boundary of simplices. One can then generalize Stokes's theorem to any orientable gluing of simplices and to any topological space which is diffeomorphic with such a gluing.

An abstract k -dimensional simplex σ is an oriented set of $k+1$ vertices $\sigma = [0, 1, \dots, k]$. We associate to the abstract simplex σ the geometrical standard k -simplex which is the convex set:

$$\left\{ (t^0, \dots, t^k) \in \mathbb{R}^{k+1} \mid \sum_{i=0}^k t^i = 1, \text{ and } t^i \geq 0, \forall i \right\},$$

whose geometrical vertices are located at $e_0, e_1, \dots, e_k$. For instance the geometric 0-simplex is a point in $\mathbb{R}$, the 1-simplex a segment in $\mathbb{R}^2$, the 2-simplex an equilateral triangle in $\mathbb{R}^3$, the 3-simplex an equilateral tetrahedron in $\mathbb{R}^4$ and so on. The coordinates $t^0, \dots, t^k$ are called the barycentric coordinates of the geometrical simplex and are not independent.

⁵We have $i_H \omega = \frac{\sqrt{\det g}}{2} \sum_{ijk} \epsilon_{ijk} H^i dx^j \wedge dx^k$ and $\sum_i \epsilon_{ijk} \epsilon^{ilm} = \delta_j^l \delta_k^m - \delta_j^m \delta_k^l$.

In order to identify independent coordinates we solve the constraint for the 0'th component and find the embedding diffeomorphism $\varphi : \Sigma^k \subset \mathbb{R}^k \rightarrow \mathbb{R}^{k+1}$ with:

$$\Sigma^k = \left\{ (u^1, \dots, u^k) \mid u^j \geq 0, \sum_{j=1}^k u^j \leq 1 \right\}, \quad (u^1, \dots, u^k) \mapsto (1 - \sum_{i=1}^k u^i, u^1, \dots, u^k).$$

The oriented volume form on Σ^k in independent coordinates is $du^1 \wedge \dots \wedge du^k$.

13.2.1 The boundary operator

We define the k -boundary operator on abstract simplices as the linear operator $\partial_k[0, 1, \dots, k] = \sum_{i=0}^k (-1)^i [0, \dots, \hat{i}, \dots, k]$ that is the alternated sum of the simplices having one less vertex. Note that:

$$\begin{aligned} \partial_{k-1} \partial_k[0, 1, \dots, k] &= \sum_{i=0}^k (-1)^i \partial_{k-1}[0, \dots, \hat{i}, \dots, k] \\ &= \sum_{i=0}^k \sum_{j < i} (-1)^i (-1)^j [0, \dots, \hat{j}, \dots, \hat{i}, \dots, k] + \sum_{i=0}^k \sum_{i < j} (-1)^i (-1)^{j-1} [0, \dots, \hat{i}, \dots, \hat{j}, \dots, k] = 0. \end{aligned}$$

For geometrical simplices the boundary of Σ^k (which is a set) is an alternated sum of sets. In barycentric coordinate this has a very simple form:

$$\partial_k \left\{ (t^0, \dots, t^k) \mid \sum t^i = 1 \right\} = \sum_{i=0}^k (-1)^i \left\{ (t^0, \dots, t^{i-1}, 0, t^{i+1}, \dots, t^k) \mid \sum t^i = 1 \right\},$$

as we simply identify on the right hand side the boundary simplices in barycentric coordinates. In independent coordinates the boundary operator takes a less symmetric form. We define the embeddings:

$$\epsilon_q : \Sigma^{k-1} \rightarrow \Sigma^k, \quad \epsilon_0(u^2, \dots, u^k) = (1 - \sum_{j=2}^k u^j, u^2, \dots, u^k), \quad \epsilon_q(u^1, \dots, \widehat{u^q}, \dots, u^k) = (u^1, \dots, u^{q-1}, 0, u^{q+1}, \dots, u^k),$$

which identify the boundaries of the simplex with lower dimensional simplices. The reader will note that ϵ_0 is special as it corresponds to the boundary simplex spanned by $(0, t^1, \dots, t^k)$ in barycentric coordinates. The boundary operator on geometrical simplices is then:

$$\partial_k \Sigma^k = \sum_{q=0}^k (-1)^q \epsilon_q(\Sigma^{k-1}).$$

Theorem 13.2 (Stokes' theorem on simplices). *For any smooth $k-1$ form Ω on the k -simplex Σ^k we have:*

$$\int_{\Sigma^k} d\Omega = \int_{\partial_k \Sigma^k} \Omega.$$

Proof. For $k=1$ this is the fundamental theorem of calculus. For $k \geq 2$ any $k-1$ form on Σ^k will write in independent coordinates as:

$$\Omega = \sum_{i=1}^k \omega_i(v) dv^1 \wedge \dots \wedge \widehat{dv^i} \wedge \dots \wedge dv^k.$$

Focusing on one component we get:

$$\begin{aligned}\int_{\partial_k \Sigma^k} \Omega &= \int_{\partial_k \Sigma^k} \omega(v) dv^1 \wedge \cdots \wedge \widehat{dv^i} \wedge \cdots \wedge dv^k = \sum_{q=0}^k (-1)^q \int_{\epsilon_q(\Sigma^{k-1})} \omega(v) dv^1 \wedge \cdots \wedge \widehat{dv^i} \wedge \cdots \wedge dv^k \\ &= \sum_{q=0}^k (-1)^q \int_{\Sigma^{k-1}} \epsilon_q^* \left(\omega(v) dv^1 \wedge \cdots \wedge \widehat{dv^i} \wedge \cdots \wedge dv^k \right).\end{aligned}$$

When computing the pullback, we observe that only the terms with $q = 0$ and $q = i$ contribute and they yield:

$$\begin{aligned}&\int_{\Sigma^{k-1}} \omega \left(1 - \sum_{j=2}^k u^j, u^2, \dots, u^i, \dots, u^k \right) \frac{\partial(1 - \sum_{j=2}^k u^j, u^2, \dots, \widehat{u^i}, \dots, u^k)}{\partial(u^2, \dots, u^k)} du^2 \wedge \cdots \wedge du^k \\ &\quad + (-1)^i \int_{\Sigma^{k-1}} \omega(u^1, \dots, u^i = 0, \dots, u^k) \frac{\partial(u^1 \dots \widehat{u^i} \dots u^k)}{\partial(u^1 \dots \widehat{u^i} \dots u^k)} du^1 \wedge \cdots \wedge \widehat{du^i} \wedge \cdots \wedge du^k \\ &= \int_{\Sigma^{k-1}} \omega \left(1 - \sum_{j=2}^k u^j, u^2, \dots, u^i, \dots, u^k \right) \frac{\partial(1 - \sum_{j=2}^k u^j, u^2, \dots, \widehat{u^i}, \dots, u^k)}{\partial(u^2, \dots, u^k)} du^2 \dots du^k \\ &\quad + (-1)^i \int_{\Sigma^{k-1}} \omega(u^1, \dots, u^i = 0, \dots, u^k) \frac{\partial(u^1 \dots \widehat{u^i} \dots u^k)}{\partial(u^1 \dots \widehat{u^i} \dots u^k)} du^1 \dots \widehat{du^i} \dots du^k,\end{aligned}$$

where we eliminated the wedge products recalling the oriented volume form on the $k-1$ simplex.

The first term is, computing the Jacobian, $(-1)^{i-1} \int_{\Sigma^{k-1}} \omega(1 - \sum_{j=2}^k u^j, u^2, \dots, u^i, \dots, u^k) du^2 \dots du^k$. As the integrals can be computed in any order, this is also:

$$(-1)^{i-1} \int_0^1 du^2 \dots \int \widehat{du^i} \dots \int_0^{1 - \sum_{j \neq i, k} u^j} du^k \int_0^{1 - \sum_{j \neq i} u^j} du^i \omega \left(1 - \sum_{j=2}^k u^j, u^2, \dots, u^i, \dots, u^k \right).$$

We now change variable on the last integral to $u^i = 1 - y^1 - \sum_{j \neq i} u^j$ and get:

$$(-1)^{i-1} \int_0^1 du^2 \dots \int \widehat{du^i} \dots \int_0^{1 - \sum_{j \neq i, k} u^j} du^k \int_{1 - \sum_{j \neq i} u^j}^0 (-dy^1) \omega(y^1, u^2, \dots, u^i = 1 - y^1 - \sum_{j \neq i} u^j, \dots, u^k).$$

Flipping the last integral, relabelling $y^1 = u^1$ and changing the integration order the two terms combine into:

$$\begin{aligned}&(-1)^{i-1} \int_{\Sigma^{k-1}} du^1 \dots \widehat{du^i} \dots du^k \left(\omega(u^1, \dots, u^i = 1 - \sum_{j \neq i} u^j, \dots, u^k) - \omega(u^1, \dots, u^i = 0, \dots, u^k) \right) \\ &= (-1)^{i-1} \int_0^1 du^1 \int_0^{1-u^1} du^2 \dots \int \widehat{du^i} \dots \int_0^{1 - \sum_{j \neq i, k} u^j} du^k \left(\omega(u^1, \dots, u^i = 1 - \sum_{j \neq i} u^j, \dots, u^k) - \omega(u^1, \dots, u^i = 0, \dots, u^k) \right).\end{aligned}$$

On the other hand:

$$d\Omega = (-1)^{i-1} \left(\frac{\partial}{\partial v^i} \omega(v) \right) dv^1 \wedge \cdots \wedge dv^k \Rightarrow \int_{\Sigma} d\Omega = \int_{\Sigma^k} dv^1 \dots dv^k \left((-1)^{i-1} \frac{\partial}{\partial v^i} \omega(v) \right),$$

which is also:

$$(-1)^{i-1} \int_0^1 dv^1 \int_0^{1-v^1} dv^2 \dots \int \widehat{dv^i} \dots \int_0^{1 - \sum_{j \neq i, k} v^j} dv^k \int_0^{1 - \sum_{j \neq i} v^j} dv^i \frac{\partial}{\partial v^i} \omega(v).$$

□

13.2.2 Simplicial (pseudo-)manifolds

Our aim is to generalize the Stokes theorem to more general spaces. Below we will generalize it to triangulated spaces which can be built by gluing simplices. This covers most of the manifolds you will encounter in your career and then some. However, there exist topological manifolds (i.e. such that the transition maps in the atlas are only continuous functions, not necessarily derivable) which do not admit triangulations.

Definition 13.1 (**Simplicial complex**). *A finite abstract simplicial complex is finite set A and a collection (multiset) of subsets of A such that if $\sigma \in \Delta$ and $\tau \subset \sigma$ then $\tau \in \Delta$. The dimension of a simplex $\sigma \in \Delta$ is $|\sigma| - 1$.*

An element $v \in A$ such that $\{v\} \in \Delta$ is called a *vertex* of Δ . Any element $\sigma \in \Delta$ is called a *simplex* of Δ . The proper subsets of σ , $\tau \subsetneq \sigma$ are called the *faces* or *subsimplices* of σ . For any $\tau \in \Delta$ we define:

- the *closed star* of τ is $\text{star}_\Delta(\tau) = \{\sigma \in \Delta, \tau \cup \sigma \in \Delta\}$.
- the *link* of τ is $\text{lk}_\Delta(\tau) = \{\sigma \in \Delta, \sigma \cap \tau = \emptyset, \tau \cup \sigma \in \Delta\}$.

The star is the topological cone over the link, $X \times [0, 1]_{/\sim}$ with $(x, t) \sim (x', t')$ if $x = x'$ and $t = t'$ or $t = t' = 1$.

An abstract simplicial complex can be lifted to a geometrical simplicial complex by promoting all the top simplices σ to geometric simplices $\Sigma^{|\sigma|-1}$. A topological space (in particular a differentiable manifold) is a triangulated space if it is homeomorphic with a geometric simplicial complex. From now on we will focus of abstract simplicial complexes, but the reader should keep in mind that they are homeomorphic with triangulated spaces: although we are doing algebra, we are discussing topological spaces.

Definition 13.2 (**Simplicial pseudomanifold**). *A closed (resp. with boundary) n -dimensional simplicial pseudomanifold is a finite abstract simplicial complex which is moreover:*

- non branching: *each $(n-1)$ simplex is the a face of precisely two (resp. one or two) n -simplices. For pseudomanifolds with boundary, the $n-1$ simplices that belong to only one n -simplex are called boundary simplices.*
- strongly connected: *any two n -simplices can be joined by a “strong chain” consisting in n -simplices in which each pair of neighbouring n -simplices have a common $(n-1)$ -simplex.*
- pure: *each simplex is the face of some n -simplex. We call the n -simplices the top simplices in the pseudomanifold.*

A simplicial pseudomanifold is normal if all its links are pseudomanifolds.

A simplicial pseudomanifold is a simplicial manifold if every link is isomorphic with a sphere (equivalently every star is isomorphic with a ball).

We focus on normal pseudomanifolds. The “connected sum” of two top simplices consists in gluing the two top n -simplices along their common $n-1$ simplex (or is the disjoint union of the two if they do not share a $n-1$ simplex). A normal simplicial pseudomanifold (closed or with boundary) is then identified with the sum of the top simplices constituting it $\Delta = \sum_i \sigma_i$.

Proposition 13.1 (**The boundary of a pseudomanifold is a pseudomanifold**). *The boundary of a n -dimensional pseudomanifold Δ is the simplicial complex consisting in the boundary simplices of Δ and their subsimplices and is the disjoint union of several closed $n-1$ dimensional pseudomanifolds.*

Proof. We assume the boundary has only one connected component. Starting from Δ we build the simplicial complex consisting in Δ , a new vertex v and all the simplices of the form $[v, \tau]$ with τ a boundary simplex of Δ . It can be shown that $\tilde{\Delta}$ is a pseudomanifold: the only slightly non trivial point is to prove that the new $n-1$ simplices added are non branching.⁶ The boundary of Δ is the link of v in $\tilde{\Delta}$ hence a pseudomanifold. $\square$

⁶By tracking the n simplices in Δ containing the $n-2$ subsimplices of the boundary $n-1$ simplices one concludes that any such $n-2$ simplex separates two boundary $n-1$ simplices.

Every simplicial pseudomanifold Δ can be subdivided via a barycentric subdivision into a *vertex colored* simplicial pseudomanifold $\tilde{\Delta}$. We proceeded as follows:

- for every simplex $\tau \in \Delta$ we introduce a vertex $\{v(\tau)\}$, the “barycenter of τ ”. We assign to this vertex the color $|\tau| - 1$, that is the dimension of the simplex τ .
- the simplices in $\tilde{\Delta}$ correspond to the strict inclusion sequences of simplices in Δ , that is:

$$\tilde{\Delta} = \left\{ [v(\tau_0), v(\tau_1), \dots, v(\tau_k)] \mid \tau_0 \subsetneq \tau_1 \subsetneq \dots \subsetneq \tau_k \in \Delta \right\}.$$

Exercise 13.6. *Prove that both the star and the link of a simplex in a simplicial complex are simplicial complexes.*

Exercise 13.7. *Prove that the links of a simplicial pseudomanifold are pure and non branching. They can however be disconnected, that is there exist pseudomanifolds which are not normal.*

One can wonder how common are manifolds that can be triangulated.

Theorem 13.3 (**Cairns–Whitehead theorem**). *Every smooth manifold admits an essentially unique compatible piecewise linear structure, that is it is diffeomorphic with a simplicial manifold.*

Proof. The interested reader can consult the literature on the topic. □

13.2.3 Stokes–Cartan theorem

The top simplices in a simplicial pseudomanifold can be assigned orientations which in turn induce via the ∂ boundary operators orientations of all of their subsimplices. The simplicial pseudomanifold is *orientable* if there exists an assignment of orientations of the top n -simplices such that any $n - 1$ simplex is assigned the two opposite orientations by the two top simplices to which it belongs, that is if σ_1 and σ_2 are two top simplices sharing a $n - 1$ simplex, then $\partial_n(\sigma_1 + \sigma_2)$ does not contain the common $n - 1$ simplex.

Proposition 13.2 (**Induced orientation**). *If a pseudomanifold with boundary is orientable then its boundary is orientable.*

Proof. Let $[0, 1 \dots n]$ and $-[0', 1, \dots, n]$ be top simplices sharing the $n - 1$ simplex $[1, \dots, n]$. We have

$$\partial_n[0, 1 \dots n] = [1, 2, \dots, n] - [0, 2 \dots n] + \dots, \quad \partial_n(-[0', 1, \dots, n]) = -[1, 2, \dots, n] + [0', 2 \dots n] + \dots.$$

that is the two n -simplices are oriented such that $[0, 1, \dots, n]$ drops out when taking the boundary ∂_n . The remaining $n - 1$ simplices are assigned orientations by the boundary operator ∂_n , e.g. $-[0, 2 \dots n]$ and $[0', 2 \dots n]$. Note that any $n - 2$ simplex, for instance $[2, \dots, n]$, is a subsimplex of two $n - 1$ simplices which are oriented such that:

$$\partial_{n-1}(-[0, 2 \dots n]) = -[2, \dots, n] + \dots, \quad \partial_{n-1}[0', 2 \dots n] = [2, \dots, n],$$

that is the orientations induced by the boundary operator ∂_n are such any $n - 2$ simplex is assigned opposite orientations by the ∂_{n-1} operator acting on the two $n - 1$ boundary simplices to which it belongs.

The result follows by gluing the top simplices one by one to obtain the pseudomanifold with boundary. □

As already mentioned, a topological space (in particular a differentiable manifold) is a triangulated space if it is homeomorphic with a geometric simplicial complex. We extend by linearity the integral of a k -form to sums of k -simplices in a geometric simplicial complex:

$$\int_{\sum_i \Sigma_i} \omega = \sum_i \int_{\Sigma_i} \omega.$$

Theorem 13.4 (**Stokes–Cartan theorem**). *For any smooth $k - 1$ form Ω on an orientable k -dimensional triangulated (pseudo-)manifold M we have:*

$$\int_M d\Omega = \int_{\partial M} \Omega .$$

Proof. We write $M = \sum \Sigma_i$ and apply Stokes theorem for each simplex. The interior $n - 1$ simplices are assigned opposite orientations by the top simplices to which they belong and drop from the sum. Only the boundary $n - 1$ simplices which triangulate ∂M survive. $\square$

Stokes' theorem holds for non orientable pseudomanifolds, provided one takes properly into account the contribution of the internal $n - 1$ simplices to the action of $\partial_n \sum_i \Sigma_i$.

Simplicial homology

The k -chain group is the Abelian group freely generated by the k -dimensional simplices in a simplicial complex:

$$C^k(\Delta, \mathbb{F}) = \left\{ \sum_{i=1}^N c_i \sigma_i \mid \sigma_i \in \Delta, |\sigma_i| = k + 1, \text{ and } c_i \in \mathbb{F} \right\},$$

for some field $\mathbb{F}$ (typically the integer numbers). The operator $\partial_k : C^k \rightarrow C^{k-1}$ is linear and is such that $\partial_{k-1} \partial_k = 0$.

We define the *cycles*, *boundaries* and *homology* subgroups:

$$Z^k = \ker(\partial_k), \quad B^k = \text{Im}(\partial_{k+1}), \quad H^k(\Delta, \mathbb{F}) = Z^k / B^k .$$

The k 'th *Betti number* of Δ is the rank (i.e. maximal number of linearly independent vectors) of H^k .

A cycle is a closed submanifold and a boundary is a cycle which is also the boundary of a submanifold. A homology class it is an equivalence class of cycles modulo boundaries. Each such a class is represented by a cycle which is not the boundary of any submanifold: the cycle represents a hole, namely a hypothetical manifold whose boundary would be that cycle, but which is "not there".

The computation of the homology groups uses the Smith normal form.

Proposition 13.3 (**The Smith normal form**). *Let A be a $m \times n$ matrix with integer entries. Then there exist invertible $m \times m$ and $n \times n$ matrices U and V with integer entries such that:*

$$A = U \begin{pmatrix} R & 0 \\ 0 & 0 \end{pmatrix} V, \quad R = \begin{pmatrix} a_1 & 0 & \dots & 0 \\ 0 & a_2 & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & a_r \end{pmatrix}, \quad a_i = \frac{d_i(A)}{d_{i-1}(A)} \in \mathbb{N},$$

where $d_i(A)$ is the greatest common divisor of the $i \times i$ minors of A and $d_0(A) = 1$.

Proof. The reader should search the literature on the topic. $\square$

Corollary 13.1. *Let B and A be $l \times m$ and $m \times n$ matrices with integer entries. Then:*

$$\ker B / \text{Im}(A) = \bigoplus_{i=1}^{\text{Rank}(A)} \left(\frac{\mathbb{Z}}{\alpha_i \mathbb{Z}} \right) \oplus \mathbb{Z}^{m - \text{Rank}(A) - \text{Rank}(B)} .$$

The Euler characteristic and the classification of two dimensional surfaces

We denote $f^p(\Delta)$ the number of simplices of dimension p in the simplicial complex, the Euler characteristic (or Euler character) of the simplicial complex is $\chi(\Delta) = \sum_{p \geq 0} (-1)^p f^p(\Delta)$.

Proposition 13.4 (Euler character and Betti numbers). *The Euler character of a pseudomanifold and its Betti numbers are related by:*

$$\chi(\Delta) = \sum_{i \geq 0} (-1)^i b_i(\Delta)$$

Proof. It is simpler to consider $\mathbb{F} = \mathbb{R}$. Then the rank of the cycle, boundary and homology groups is just the vector space dimension and ∂_k is a linear operator. We have on the one hand $\beta_k = \text{Rank } H^k = \text{Rank } Z^k - \text{Rank } B^k$. On the other, by the rank nullity theorem, $f^k(\Delta) = \text{Rank } Z^k + \text{Rank } B^{k-1}$ $\square$

Two dimensional surfaces with no boundary are classified by their orientability (hence a sign) and their Euler character $\chi = 2 - 2g$ in the orientable case respectively $2 - k$ in the non orientable one. The orientable genus g counts the number of holes. For instance, examples of triangulations of the torus $(+, g = 1)$, the $\mathbb{R}P^2$ plane $(-, k = 1)$ and the Klein Bottle $(-, k = 2)$ can be built starting from the patches with the identifications depicted in Fig. 13.1.

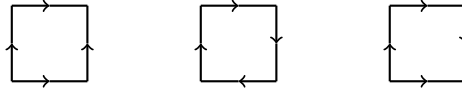
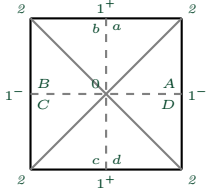


Figure 13.1: Identifications of the boundary of a quadrangle leading to different two dimensional manifolds.

Exercise 13.8. *Compute the simplicial homology groups of these surfaces. We first perform a barycentric subdivision of the square, which leads to a simplicial complex.*



For example, the $\mathbb{R}P^2$ gluing is the simplicial complex:

$$\Delta = \left\{ [01^+2]^a, [01^+2]^b, [01^+2]^c, [01^+2]^d, [01^-2]^A, [01^-2]^B, [01^-2]^C, [01^-2]^D, \right. \\ \left. [01^+]^{ab}, [01^+]^{cd}, [01^-]^{BC}, [01^-]^{AD}, [02]^aA, [02]^bB, [02]^cC, [02]^dD, [1^+2]^{ac}, [1^+2]^{bd}, [1^-2]^{BD}, [1^-2]^{AC}, [0], [1^+], [1^-], [2] \right\}$$

Let us compute $H^1(\Delta, \mathbb{Z}) = \ker(\partial_1) / \text{Im}(\partial_2)$. We have:

$$\partial_1(\sum N_{\alpha\beta}[\cdot, \cdot]^{\alpha\beta}) = 0 \Rightarrow \begin{cases} [0], & N_{ab} + N_{cd} + N_{BC} + N_{AD} + N_{aA} + N_{bB} + N_{cC} + N_{dD} = 0, \\ [1^+], & -N_{ab} - N_{cd} + N_{ac} + N_{bd} = 0 \\ [1^-], & -N_{BC} - N_{AD} + N_{BD} + N_{AC} = 0 \\ [2], & N_{aA} + N_{bB} + N_{cC} + N_{dD} + N_{ac} + N_{bd} + N_{BD} + N_{AC} = 0 \end{cases}$$

$$\partial_2(\sum N_{\alpha}[\cdot, \cdot, \cdot]^{\alpha}) = (N_a + N_b)[01^+]^{ab} + (N_c + N_d)[01^+]^{cd} + (N_B + N_C)[01^-]^{BC} + (N_A + N_D)[01^-]^{AD} \\ - (N_a + N_A)[02]^aA - (N_b + N_B)[02]^bB - (N_c + N_C)[02]^cC - (N_d + N_D)[02]^dD \\ + (N_a + N_c)[1^+2]^{ac} + (N_b + N_d)[1^+2]^{bd} + (N_B + N_D)[1^-2]^{BD} + (N_A + N_C)[1^-2]^{AC},$$

Part III

HöMa III

Chapter 14

Complex analysis

In this chapter we will study complex functions of one complex variable.

14.1 Differentiation in $\mathbb{C}$

Definition 14.1 (**Complex differentiability**). Let $U \subset \mathbb{C}$ be an open subset of $\mathbb{C}$. A function $f : U \rightarrow \mathbb{C}$ is called *complex differentiable* in $z_0 \in U$ if the function $\frac{f(z)-f(z_0)}{z-z_0}$ can be continuously continued in z_0 and in this case we denote:¹

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} .$$

A function is called *holomorphic* at z_0 if it is complex differentiable in a neighborhood of z_0 and holomorphic in U if it is complex differentiable in U .

For instance $z \rightarrow \bar{z}$ is not complex differentiable. Indeed, denoting $z = z_0 + re^{i\varphi}$ we have:

$$\frac{\bar{z} - \bar{z}_0}{z - z_0} = e^{-2i\varphi} ,$$

which does not have a well defined limit when sending $r \rightarrow 0$.

Denoting $z = x + iy$ and $f = u + iv$, the function f is mapped on a function $F : (\mathbb{R}^2, \|\cdot\|_2) \rightarrow (\mathbb{R}^2, \|\cdot\|_2)$, $F = (u, v)$. If f is complex differentiable at $z_0 = x_0 + iy_0$, then F it is Fréchet differentiable at (x_0, y_0) :

$$\begin{aligned} & \lim_{(x,y) \rightarrow (x_0,y_0)} \frac{\left\| \begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix} - \begin{pmatrix} u(x_0,y_0) \\ v(x_0,y_0) \end{pmatrix} - \begin{pmatrix} \operatorname{Re}(f'(z_0)) & -\operatorname{Im}(f'(z_0)) \\ \operatorname{Im}(f'(z_0)) & \operatorname{Re}(f'(z_0)) \end{pmatrix} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2}{\left\| \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2} \\ &= \lim_{z \rightarrow z_0} \frac{|f(z) - f(z_0) - f'(z_0)(z - z_0)|}{|z - z_0|} = 0 . \end{aligned}$$

¹In full detail, $\forall \epsilon, \exists \delta$ such that $|z - z_0| < \delta \Rightarrow \left| \frac{f(z)-f(z_0)}{z-z_0} - f'(z_0) \right| < \epsilon$.

Furthermore the partial derivatives of u and v are related:

$$\begin{aligned} f'(z_0) &= \lim_{t \rightarrow 0} \frac{u(x_0 + t, y_0) + iv(x_0 + t, y_0) - u(x_0, y_0) - iv(x_0, y_0)}{t} = \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0) \\ &= \lim_{t \rightarrow 0} \frac{u(x_0, y_0 + t) + iv(x_0, y_0 + t) - u(x_0, y_0) - iv(x_0, y_0)}{it} = -i \frac{\partial u}{\partial y}(x_0, y_0) + \frac{\partial v}{\partial y}(x_0, y_0) . \end{aligned}$$

Theorem 14.1 (**Cauchy-Riemann equations**). *A function $f : U \subset \mathbb{C} \rightarrow \mathbb{C}$, $f = u + iv$ is complex differentiable at $z_0 = x_0 + iy_0$ if and only if $F = (u, v)$ is Fréchet differentiable at (x_0, y_0) and the partial derivatives of u and v respect the Cauchy-Riemann equations:*

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0) , \quad \frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0) .$$

Observe that in this case $\partial_x^2 u = \partial_x \partial_y v = -\partial_y \partial_x u$ and similarly for v , that is:

$$\partial_x^2 u + \partial_y^2 u = \Delta u = 0 , \quad \partial_x^2 v + \partial_y^2 v = \Delta v = 0 .$$

The condition of Fréchet differentiability is a weak condition: from Theorem 10.7, if all the partial derivatives of u and v exist and are continuous, then $F = (u, v)$ is Fréchet differentiable.

Proof. We have already proven one implication. If $F = (u, v)$ is Fréchet differentiable then $dF = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}$ and using the Cauchy-Riemann equations we have:

$$\begin{aligned} \left| \frac{f(z) - f(z_0) - (\partial_x u + i \partial_x v)(z - z_0)}{z - z_0} \right| &= \frac{\left\| \begin{pmatrix} u(x, y) \\ v(x, y) \end{pmatrix} - \begin{pmatrix} u(x_0, y_0) \\ v(x_0, y_0) \end{pmatrix} - \begin{pmatrix} \partial_x u & -\partial_x v \\ \partial_x v & \partial_x u \end{pmatrix} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2}{\left\| \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2} \\ &= \frac{\left\| F(x, y) - F(x_0, y_0) - dF|_{(x_0, y_0)} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2}{\left\| \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2} \xrightarrow{(x, y) \rightarrow (x_0, y_0)} 0 . \end{aligned}$$

□

Consider the following examples:

- the function $f(z) = z^2 = x^2 - y^2 + i 2xy$:

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{z^2 - z_0^2}{z - z_0} = \lim_{z \rightarrow z_0} (z + z_0) = 2z_0 ; \quad u(x, y) = x^2 - y^2 , \quad v(x, y) = 2xy , \quad \partial_x u = 2x = \partial_y v , \quad \partial_y u = -2y = -\partial_x v ,$$

is holomorphic in $\mathbb{C}$.

- the function $f(z) = \bar{z}^2 = x^2 - y^2 - i 2xy$ with $\partial_x u = 2x \neq -2x = \partial_y v$ and $\partial_y u = -2y \neq 2y = -\partial_x v$ is not holomorphic at any point in $\mathbb{C}$.

- the function $f(z) = \frac{1}{z}$:

$$\begin{aligned} f'(z_0) &= -\frac{1}{z_0^2} ; & u &= \frac{x}{x^2 + y^2} , & v &= \frac{-y}{x^2 + y^2} , \\ \partial_x u &= \frac{-x^2 + y^2}{(x^2 + y^2)^2} , & \partial_y u &= \frac{-2xy}{(x^2 + y^2)^2} , & \partial_x v &= \frac{2xy}{(x^2 + y^2)^2} , & \partial_y v &= \frac{-x^2 + y^2}{(x^2 + y^2)^2} , \end{aligned}$$

is holomorphic in $\mathbb{C} \setminus \{0\}$ and in 0 fails to be holomorphic because the partial derivatives are not continuous.

For an arbitrary function of a complex variable and its complex conjugated, $f(z, \bar{z})$, we denote:

$$f(z, \bar{z}) = u(x, y) + \imath v(x, y) ; \quad \frac{\partial}{\partial z} = \frac{1}{2} (\partial_x - \imath \partial_y) , \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} (\partial_x + \imath \partial_y) ; \quad dz = dx + \imath dy , \quad d\bar{z} = dx - \imath dy ,$$

and the Cauchy-Riemann equations write $\partial_{\bar{z}} f(z, \bar{z}) = 0$.

Note that whenever F is Fréchet differentiable, taking into account that $\mathsf{d}F = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}$ we have:

$$\left\| F(x, y) - F(x_0, y_0) - \mathsf{d}F \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right\|_2 = |f(z) - f(z_0) - \partial_z f(z - z_0) - \partial_{\bar{z}} f(\bar{z} - \bar{z}_0)| ,$$

where we used:

$$\begin{aligned} \partial_z f(z - z_0) + \partial_{\bar{z}} f(\bar{z} - \bar{z}_0) &= (x - x_0)(\partial_z f + \partial_{\bar{z}} f) + \imath(y - y_0)(\partial_z f - \partial_{\bar{z}} f) \\ &= (x - x_0)\partial_x f + (y - y_0)\partial_y f = (x - x_0)(\partial_x u + \imath \partial_x v) + (y - y_0)(\partial_y u + \imath \partial_y v) , \end{aligned}$$

hence if f is complex differentiable (that is $\partial_{\bar{z}} f = 0$) then $f' = \partial_z f$.

Exercise 14.1 (Conformal mappings). *Consider $f : \mathbb{C} \rightarrow \mathbb{C}$ a complex differentiable function and $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ its associated real function. Prove that the linear map $\mathsf{d}F$ rescales all the vectors by a common factor Λ but preserves the angles. Such a mapping is called conformal. Let us denote e_x and e_y the standard unit vectors in the domain $T_{(x_0, y_0)} \mathbb{R}^2 \cong \mathbb{R}^2$ of $\mathsf{d}F$ and e_u and e_v the standard unit vectors in the codomain $T_{F(x_0, y_0)} \mathbb{R}^2 \cong \mathbb{R}^2$ of $\mathsf{d}F$. Acting on the basis vectors we get:*

$$\mathsf{d}F(e_x) = e_u du(e_x) + e_v dv(e_x) = e_u \partial_x u + e_v \partial_x v , \quad \mathsf{d}F(e_y) = e_u \partial_y u + e_v \partial_y v .$$

We endow $T_{F(x_0, y_0)} \mathbb{R}^2$ with the standard inner product $\langle e_u, e_u \rangle = \langle e_v, e_v \rangle = 1$ and $\langle e_u, e_v \rangle = 0$, and $T_{(x_0, y_0)} \mathbb{R}^2$ with $\langle e_x, e_x \rangle = \langle e_y, e_y \rangle = 1$ and $\langle e_x, e_y \rangle = 0$. We have:

$$\langle \mathsf{d}F(e_x), \mathsf{d}F(e_y) \rangle = \partial_x u \partial_y u + \partial_x v \partial_y v = 0 , \quad \|\mathsf{d}F(e_x)\|^2 = \|\mathsf{d}F(e_y)\|^2 = (\partial_x u)^2 + (\partial_x v)^2 = (\partial_y u)^2 + (\partial_y v)^2 \equiv \Lambda^2 > 0 ,$$

that is $\mathsf{d}F(e_x)$ and $\mathsf{d}F(e_y)$ are orthogonal vectors of equal norm. It follows that $\mathsf{d}F$ is a mapping from $T_{(x_0, y_0)} \mathbb{R}^2$ to $T_{F(x_0, y_0)} \mathbb{R}^2$ such that:

$$\|\mathsf{d}F(V)\| = \Lambda \|V\| , \quad \frac{\langle \mathsf{d}F(V), \mathsf{d}F(W) \rangle}{\|\mathsf{d}F(V)\| \|\mathsf{d}F(W)\|} = \frac{V^x W^x \|\mathsf{d}F(e_x)\|^2 + V^y W^y \|\mathsf{d}F(e_y)\|^2}{\Lambda^2 \|V\| \|W\|} = \frac{\langle V, W \rangle}{\|V\| \|W\|} ,$$

that is, as advertized, $\mathsf{d}F$ rescales all the vectors by a common factor Λ but preserves the angles.

In differential geometry terms, denoting g the standard metric on $\mathbb{R}^2$, we have the pushforward map $\mathsf{d}F : T_{(x_0, y_0)} \mathbb{R}^2 \rightarrow T_{F(x_0, y_0)} \mathbb{R}^2$ and under the associated pullback $F^* g = \Lambda g$, that is

$$F^* (du^2 + dv^2) = (\partial_x u dx + \partial_y u dy)^2 + (\partial_x v dx + \partial_y v dy)^2 = [(\partial_x u)^2 + (\partial_x v)^2] dx^2 + [(\partial_y u)^2 + (\partial_y v)^2] dy^2 + (2\partial_x u \partial_y u + 2\partial_x v \partial_y v) dx dy = \Lambda^2 (dx^2 + dy^2)$$

14.2 Analytic functions

Recall that power series converge absolutely inside their disk of convergence and that the terms in absolutely convergent series can be arbitrarily reshuffled without changing the sum.

Lemma 14.1. Any power series $P(z) = \sum_{k \geq 0} a_k z^k$ is holomorphic for all z_0 inside the convergence domain $|z_0| < r$, $r^{-1} = \limsup |a_n|^{1/n}$ and inside the convergence domain we have $P'(z) = \sum_{k \geq 1} k a_k z^{k-1}$.

Proof. It is easy to check that the radius of converge of $\sum_{n \geq 0} k a_k z^{k-1}$ is r . As a power series converges absolutely inside its domain of convergence we can rearrange the terms and write:

$$h(z) = \frac{P(z) - P(z_0)}{z - z_0} = \sum_{n \geq 1} a_n \sum_{p=0}^{n-1} z^p z_0^{n-1-p},$$

which converges absolutely for $|z|, |z_0| < \rho < r$. As $\forall \epsilon$, except at most for a finite number of terms, $|a_n| \leq (r^{-1} + \epsilon)^n$, we have:

$$\begin{aligned} |h(z) - h(z_0)| &\leq \sum_{n \geq 2} |a_n| \sum_{p=1}^{n-1} |z^p - z_0^p| |z_0|^{n-1-p} \leq \sum_{n \geq 2} (r^{-1} + \epsilon)^n |z - z_0| \sum_{p=1}^{n-1} \sum_{q=0}^{p-1} |z|^q |z_0|^{p-1-q} |z_0|^{n-1-p} \\ &\leq |z - z_0| \sum_{n \geq 2} (r^{-1} + \epsilon)^n n(n-1) \rho^{n-2} = 2(r^{-1} + \epsilon)^2 \frac{|z - z_0|}{(1 - \rho(r^{-1} + \epsilon))^3}, \end{aligned}$$

where we chose $\epsilon < \frac{1-\rho/r}{\rho}$, we conclude that $h(z)$ is continuous at z_0 . $\square$

Definition 14.2 (**Analytic functions**). A function $f : U \subset \mathbb{C} \rightarrow \mathbb{C}$ is called analytic in the open domain U if $\forall z_0 \in U$, there exists a sequence of coefficients $a_k(z_0)$ and a $\delta(z_0) > 0$ such that $\forall z \in B_{\delta(z_0)}(z_0) \subset U$, $f(z) = \sum_{k \geq 0} a_k(z_0) (z - z_0)^k$ (implicitly this implies that the series has non zero radius of convergence).

The following corollary follows from Lemma 14.1.

Corollary 14.1 (**Analytic functions are smooth**). An analytic function is indefinitely complex differentiable in its domain of definition U and $\forall z_0 \in U$, $f^{(k)}(z_0) = k! a_k(z_0)$.

Power series defined analytic functions.

Theorem 14.2 (**Power series define analytic functions inside their disk of converge**). The power series $P(z) = \sum_{k \geq 0} a_k z^k$ with radius of converge r defines an analytic function inside $B_r(0)$.

Proof. For any z_0 with $|z_0| < r$, and for any z such that $|z - z_0|$ is small enough we rewrite:

$$P(z) = \sum_{k \geq 0} a_k (z_0 + z - z_0)^k = \sum_{k \geq 0} a_k \sum_{p=0}^k \binom{k}{p} (z - z_0)^p z_0^{k-p} = \sum_{p \geq 0} (z - z_0)^p \sum_{k \geq p} \binom{k}{p} a_k z_0^{k-p},$$

where the last series is absolutely convergent for $|z - z_0|$ small enough as for any ϵ , for all but a finite number of k we have $|a_k| \leq (r^{-1} + \epsilon)^k$ hence:²

$$\left| \sum_{k \geq p} \binom{k}{p} a_k z_0^{k-p} \right| \leq \sum_{k \geq p} \binom{k}{p} (r^{-1} + \epsilon)^k |z_0|^{k-p} = \frac{(r^{-1} + \epsilon)^p}{[1 - (r^{-1} + \epsilon)|z_0|]^{p+1}}.$$

$\square$

Analytic functions are quite constrained. For instance their zeros are always isolated.

Lemma 14.2 (**A locally constant function on a connected topological space is globally constant**). Let X be a connected topological space and let $f : X \rightarrow Y$ be a locally constant function. Then f is constant in X .

²We have $\frac{x^p}{(1-x)^{p+1}} = \frac{x^p}{p!} \left(\frac{d}{dx}\right)^p \frac{1}{1-x} = \sum_{k \geq p} \binom{k}{p} x^k$. We use Fubini's theorem for series Theorem 4.14 to commute the sums.

Proof. By definition, X is connected if it can not be written as the disjoint union of two non empty open sets. In particular the complement of any non empty open set can not be open hence the only sets that are both open and closed at the same time are X and $\emptyset$.

We choose an arbitrary point $a \in X$ and we denote $c = f(a) \in Y$. We define $E = \{x \in X \mid f(x) = c\}$ so that $a \in E$ hence $E \neq \emptyset$. For every $x \in E$, by the definition of a locally constant function, there exists an open neighborhood $U(x)$ of x such that f is constant in $U(x)$, that is $f(U(x)) = c$ hence $U(x) \subset E$ and E is open.

We consider the complement of E in X , that is $F = \{x \in X \mid f(x) \neq c\}$. Any point $z \in F$ has an open neighborhood $V(z)$ where f is constant and since $z \in F$, this constant value cannot be c , so $f(y) \neq c$ for all $y \in V(z)$ thus $V(z) \subset F$, F is an open set and E is closed. $\square$

Lemma 14.3 (**The zeros of analytic functions are isolated**). *Let $f : U \rightarrow \mathbb{C}$ be an analytic function in the connected open domain U . Then either $f(U) = 0$ or the zeros of f are isolated, that is if $f(z_0) = 0$ for some $z_0 \in U$, then $\exists \delta > 0$ such that $f(z) \neq 0$ for $z \in B_\delta(z_0) \setminus \{z_0\} \subset U$.*

Proof. If f is analytic at z_0 , $f(z_0) = 0$, but f is not identically 0 in U , then $\exists \delta > 0$ such that for $z \in B_\delta(z_0) \subset U$, $f(z) = (z - z_0)^p \sum_{k \geq 0} a_k(z_0) (z - z_0)^k$, for some $p \geq 1$ and $a_0(z_0) \neq 0$. It follows that $g(z) = \sum_{k \geq 0} a_k(z_0) (z - z_0)^k$ is analytic at z_0 and $g(z_0) \neq 0$. As g is complex differentiable at z_0 , then $g(z) \neq 0$ in some neighborhood of z_0 and we conclude. $\square$

14.2.1 Analytic continuation

Definition 14.3 (**Analytic continuation**). *Consider the function $f : U \rightarrow \mathbb{C}$ analytic in U . If there exists a function $\tilde{f} : \tilde{U} \rightarrow \mathbb{C}$ analytic in the domain $\tilde{U}$ such that $\tilde{f}|_{\tilde{U} \cap U \neq \emptyset} = f|_{\tilde{U} \cap U \neq \emptyset}$ we say that $\tilde{f}$ is an analytic continuation of f to $\tilde{U}$.*

An example is the function $f(z) = \sum_{n \geq 0} z^n$ which is a priori analytic in the domain of convergence of the series, that is $B_1(0)$. For all $z \in B_1(0)$ the series can be resummed to $1/(1 - z)$. The function $\tilde{f} = \frac{1}{1-z}$ is analytic in $\mathbb{C} \setminus \{1\}$:

$$\tilde{f}(z) = \sum_{k \geq 0} \frac{1}{(1 - z_0)^{k+1}} (z - z_0)^k,$$

and the series converges for $|z - z_0| < |1 - z_0|$. As $\tilde{f}$ coincides with f in $B_1(0)$, $\tilde{f}$ is the analytic continuation of f to $\mathbb{C} \setminus \{1\}$.

Lemma 14.4 (**Unicity of analytic continuation on connected domains**). *If f and g are two analytic functions in a connected open domain U and $f = g$ on a subset $M \subset U$ which has an accumulation point (a limit point) in U then $f = g$ in the whole of U . Let $U \subset U'$ open domains in $\mathbb{C}$ with U' connected and let an analytic function $f : U \rightarrow \mathbb{C}$. Then f admits at most one analytic continuation to U' .*

Proof. The function $f - g$ is analytic in U and zero on M , hence the accumulation point of M is a non isolated zero of $f - g$, therefore $f - g$ is identically zero on U . Any two analytic continuations to U' coincide on U which has accumulation points hence are identical. $\square$

Despite this uniqueness theorem, the maximal analytic continuation of a functions is subtle. The fact that if two analytic continuations of a function exist on some connected domain then they coincide does not guarantee that there exist one unique continuation of a function to a connected domain.

Definition 14.4 (**Curve**). *A curve (path or contour) in $\mathbb{C}$ is a continuous map $\gamma : [0, 1] \rightarrow \mathbb{C}$, $t \mapsto \gamma(t) = \gamma^x(t) + i\gamma^y(t)$. We often denote the image of the curve by $\gamma([0, 1]) = C$. A curve is called smooth if $\gamma \in C^\infty([0, 1], \mathbb{C})$ and it is called simple if γ is injective on $(0, 1)$, that is C has no self intersections except possibly at the end points.*

Definition 14.5 (**Simply connected domain**). A domain $U \subset \mathbb{C}$ is called simply connected if any two points $z, w \in U$ can be connected by a continuous path $\gamma : [0, 1] \rightarrow \mathbb{C}$, $\gamma(0) = z, \gamma(1) = w$ and any two paths in U connecting the same end points can be continuously deformed into each other. Formally, for any $\gamma, \tilde{\gamma} : [0, 1] \rightarrow \mathbb{C}$ with $\gamma(0) = \tilde{\gamma}(0) = z$ and $\gamma(1) = \tilde{\gamma}(1) = w$, there exists a function $F : [0, 1] \times [0, 1] \rightarrow U$ continuous in the product topology (a homotopy) such that $F(t, 0) = \gamma(t)$ and $F(t, 1) = \tilde{\gamma}(t)$.

A function $f(z) = \sum_{k \geq 0} a_k (z - z_0)^k$ analytic in some open disk $B_r(z_0)$, admits a unique analytic continuation to some maximal open simply connected domain $V \supset B_r(z_0)$. Any point $v \in V$ can be reached starting from z_0 by gluing overlapping disks centered along a path from z_0 to v and, as V is simply connected, the analytic continuation at v is independent on the path. On the contrary, in principle there does not exist a unique analytic continuation of a function on arbitrary connected domains: connected domain can have holes (hence not be simply connected) such that the same point can be reached by non-equivalent paths. When one considers maximal continuations to simply connected domains, these analytic continuations do not need to coincide on the intersection of their domains.

Theorem 14.3 (**Riemann mapping theorem**). If U is a non-empty simply connected open subset of $\mathbb{C}$ which is not all of $\mathbb{C}$, then there exists a biholomorphic mapping f (i.e. a bijective holomorphic mapping whose inverse is also holomorphic) from U onto the open unit disk $\{z \in \mathbb{C}, |z| < 1\}$. Note that as a holomorphic function is a conformal mapping of the plane, the Riemann mapping preserves the angles, seen as a map from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Proof. The reader is invited to consult the literature on the topic. □

Exercise 14.2 (**Examples of complex functions**). Discuss the maximal analytic continuations of the complex functions z^n, e^z and $1/z$.

The functions $f(z) = z^n$ and $e^z = \sum_{n \geq 0} \frac{1}{n!} z^n$ are analytic in $\mathbb{C}$. The function $f(z) = 1/z$ is analytic in $\mathbb{C} \setminus \{0\}$, and is said to have a pole of order 1 at $z = 0$.

14.2.2 The complex logarithm

The complex logarithm aims to be the inverse of the complex exponential $f(z) = e^z$. However, as e^z is not injective on $\mathbb{C}$ ($e^{i2\pi k} = 1$ for $k \in \mathbb{Z}$), its “inverse” is *multivalued*: a logarithm of z is any complex number w such that $e^w = z$. Using polar coordinates $z = |z|e^{i\theta} \neq 0$, any $w_k = \ln|z| + i(\theta + 2k\pi)$ is a complex logarithm of z . Observe that $\ln(0)$ is ill defined as there exists no complex number w such that $e^w = 0$.

A *branch* of the complex logarithm is a continuous function $L(z)$ defined on a simply connected domain $U \subset \mathbb{C}$ such that $L(z)$ is a logarithm of z for all $z \in U$. A branch can be maximally analytically extended to a simply connected sub domain of $\mathbb{C} \setminus \{0\}$, that is a domain of the form $\mathbb{C} \setminus \text{Cut}$ where Cut is a branch cut in the complex plane, that is a simple curve (with no self intersections) connecting 0 to complex ∞ . The *principal branch* of the logarithm has $\text{Cut} = \mathbb{R}_{\leq 0}$, that is for $z = |z|e^{i\theta} \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, $\ln(z)$ is the complex number $w_k = \ln|z| + i(\theta + 2k\pi)$ such that $\text{Im}(w_k) \in (-\pi, \pi)$. We can extend this to a discontinuous function on $\mathbb{C}$ by allowing $\text{Im}(w_k) \in (-\pi, \pi]$.

Taking only rays in the complex plane as cuts, the branches of the logarithm are identified by the images of the imaginary part of the logarithm, $\text{Im } L(\mathbb{C} \setminus \text{Cut})$. The branch $(-\pi, \pi]$ and the branch $(0, 2\pi]$ agree on the upper half plane, but differ on the lower half plane. One can stitch all the branches of the logarithm together to obtain a *Riemann surface* which looks like an infinite corkscrew when embedded in $\mathbb{R}^3$.

Setting the cut on $\mathbb{R}_-$, we define the Riemann surface as the gluing of an infinity of copies of $\mathbb{C} \setminus \{0\}$ along $\mathbb{R}_-$, that is $R = \bigcup I_k$, with $I_k = \{e^z \mid (2k-1)\pi < \text{Im}(z) \leq (2k+1)\pi\} \cong \mathbb{C} \setminus \{0\}$. In words, each I_k is a copy of $\mathbb{C} \setminus \{0\}$ and is the image under the exponential function of the strip $\sigma_k = \{x + iy \mid x \in \mathbb{R}, y \in ((2k-1)\pi, (2k+1)\pi]\} \in \mathbb{C}$. Gluing together the strips σ_k to obtain $\mathbb{C}$, one glues together the sheets I_k and builds R . Note that one jumps from one sheet to the the next when turning continuously around the origin.

The exponential function $\exp : \mathbb{C} \rightarrow \mathbb{C}$ is single valued hence invertible to $\ln : \mathbb{C} \rightarrow \mathbb{C}$. In effect we trade the multivalued logarithm defined on $\mathbb{C}$ for a single valued logarithm on the Riemann surface R .

14.3 Contour integrals

As holomorphic functions are continuous, one can integrate them along smooth (in fact continuously differentiable suffices) simple curves. Denoting $\gamma : [0, 1] \rightarrow \mathbb{C}$ a smooth curve and C its image, we define:

$$\int_C dz f(z) \equiv \int_0^1 dt \dot{\gamma}(t) f(\gamma(t)),$$

where $\dot{\gamma}(t) \equiv \frac{d\gamma}{dt}$. We will assume from now on that $\dot{\gamma} \neq 0$, that is the parametrization of C is locally injective. One can in fact integrate complex valued functions that are only continuous, or even only measurable (we will see more details about this later).

The integral is invariant under reparametrization $\gamma \rightarrow \gamma \circ t$ for any function $t \in \mathcal{C}^1([a, b], [0, 1])$ with $t(a) = 0$, $t(b) = 1$ and $\frac{dt}{ds} > 0$ (in order to preserve local injectivity) we have:

$$\int_a^b ds \frac{d}{ds} \gamma(t(s)) f(\gamma(t(s))) = \int_a^b ds \left(\frac{d}{dt} \gamma \right)(t(s)) \frac{dt}{ds}(s) f(\gamma(t(s))) = \int_{t(a)=0}^{t(b)=1} dt \dot{\gamma}(t) f(\gamma(t)).$$

For instance, if we parameterize the unit circle as $z = e^{it}$ with $t \in [0, 2\pi]$. We have:

$$\int_{C_1(0)} dz z^n = \int_0^{2\pi} dt i e^{it} e^{int} = i \int_0^{2\pi} dt [\cos((n+1)t) + i \sin((n+1)t)] = \begin{cases} \frac{i}{n+1} [\sin((n+1)t) - i \cos((n+1)t)]_0^{2\pi} = 0, & n \neq -1 \\ 2\pi i, & n = -1 \end{cases},$$

while the integral over the upper semicircle only is:

$$i \int_0^\pi dt [\cos((n+1)t) + i \sin((n+1)t)] = \begin{cases} \frac{i}{n+1} [\sin((n+1)t) - i \cos((n+1)t)]_0^\pi = \frac{i}{n+1} [-i(-1)^{n+1} - 1] = \frac{(-1)^{n+1}-1}{n+1}, & n \neq -1 \\ \pi i, & n = -1 \end{cases}.$$

If we close the semicircle by a straight segment we need to add up to this $\int_{-1}^1 dt t^n = \frac{1-(-1)^{n+1}}{n+1}$, hence $\int dz z^n$ over the closed semicircle is 0 for $n \geq 0$.

The length of the curve C is (recall that the parametrization of C is locally injective):

$$l(C) = \int_0^1 dt |\dot{\gamma}(t)|, \quad \left| \int_C dz f(z) \right| \leq l(C) \sup_{z \in C} |f(z)|,$$

and the length is invariant under reparametrization. The length of the circle is $\int_0^{2\pi} dt |ie^{it}| = 2\pi$. Note that the integral switches sign if we reverse the orientation of the path:

$$\begin{aligned} \int_C dz f(z) &= \int_0^1 dt \dot{\gamma}(t) f(\gamma(t)) \stackrel{t=1-u}{=} \int_1^0 (-du) \left[-\frac{d}{du} \gamma(1-u) \right] f(\gamma(1-u)) \\ &\stackrel{\tilde{\gamma}(u)=\gamma(1-u)}{=} - \int_0^1 du \dot{\tilde{\gamma}}(u) f(\tilde{\gamma}(u)) = - \int_{C^{-1}} dz f(z). \end{aligned}$$

The complex plane $\mathbb{C}$ is canonically identified with the plane $\mathbb{R}^2$ by the mapping $z = x + iy \rightarrow (x, y)$, hence any complex curve $\gamma(t) = x(t) + iy(t)$ is canonically identified with the curve $(x(t), y(t)) \subset \mathbb{R}^2$. Denoting $f = v + iw$, we observe that the contour integral defined above is nothing but the integral along the one dimensional submanifold of $\mathbb{R}^2$ defined by the

embedding $\gamma = (x(t), y(t))$:

$$\begin{aligned} \int_C dz f(z) &= \int_0^1 dt \left(\frac{dx}{dt} + i \frac{dy}{dt} \right) (u + iv) = \int_0^1 dt \left[u \frac{dx}{dt} - v \frac{dy}{dt} + i \left(v \frac{dx}{dt} + u \frac{dy}{dt} \right) \right] \\ &= \int_{[0,1]} \gamma^*(udx - vdy) + i \int_{[0,1]} \gamma^*(vdx + udy) = \int_{C=\gamma([0,1])} (udx - vdy) + i \int_{C=\gamma([0,1])} (vdx + udy) = \int_C (dx + idy)(u + iv), \end{aligned}$$

with γ^* the pullback by γ and where we denoted somewhat abusively $C = \gamma([0, 1]) \subset \mathbb{R}^2$.

Simple closed curves separate an interior and an exterior components.

Theorem 14.4 (Jordan curve theorem). *A simple closed curve in $\mathbb{R}^2$ is the image of a continuous map $\gamma : [0, 1] \rightarrow \mathbb{R}^2$ such that $\gamma(0) = \gamma(1)$ and the restriction of γ to $[0, 1)$ is (globally) injective, that is $C = \gamma([0, 1])$ is a continuous loop with no self intersections. Let C be a closed simple curve in the plane $\mathbb{R}^2$. Then its complement, $\mathbb{R}^2 \setminus C$ consists of exactly two connected components. One of these components is bounded and simply connected (the interior) and the other one is unbounded (the exterior). The curve C is the manifold boundary of each component.*

Proof. Despite being intuitively obvious, the proof of this theorem is surprisingly subtle. □

A smooth closed simple curve γ is positively oriented if the interior of $C = \gamma([0, 1])$ is to the left of the tangent vector $\dot{\gamma}(t)$. Unless otherwise specified, integrals along closed simple curves are always considered to follow the positive orientation.

Theorem 14.5 (Cauchy's integral theorem). *Let f be a holomorphic function in some simply connected domain $U \subset \mathbb{C}$ and $C \subset U$ a smooth closed simple curve in U . Then:*

$$\int_C dz f(z) = 0.$$

Proof. We present three different proofs of this formula.

Using the Stokes theorem. As C is a closed simple curve, it separates an interior and an exterior component. The interior of C , which we denote D with $\partial D = C$, is fully contained in U : any two points of C divide C into two homotopically equivalent arcs. By Stokes theorem:

$$\int_{C=\partial D} (udx - vdy) = \int_D (-\partial_y u - \partial_x v) dx \wedge dy, \quad \int_{C=\partial D} (vdx + udy) = \int_D (-\partial_y v + \partial_x u) dx \wedge dy,$$

and both terms are zero by the Cauchy-Riemann equations $\partial_x u = \partial_y v$, $\partial_y u = -\partial_x v$.

Using the Stokes theorem with complex variables. One can use the Stokes theorem in complex variables. We have $dz = dx + idy$, $d\bar{z} = dx - idy$, $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$. The exterior differential on a function is then $dh = \partial_x h dx + \partial_y h dy = \partial_z h dz + \partial_{\bar{z}} h d\bar{z}$. The complex valued one form $f(z) dz$ with f holomorphic is closed as $d(f dz) = \partial_z f dz \wedge dz + \partial_{\bar{z}} f d\bar{z} \wedge dz = 0$ hence $0 = \int_D d(f dz) = \int_{C=\partial D} f dz$. Note that the integral of a complex 2 form on the domain $D \subset \mathbb{C}$ is the integral on D seen as a subset in $\mathbb{R}^2$: slightly abusing notation we have $\int_D d\bar{z} \wedge dz \omega(z, \bar{z}) = \int_D (dx - idy) \wedge (dx + idy) \omega(x + iy, x - iy) = 2i \int_D dx dy \omega(x + iy, x - iy)$.

Direct proof. We first observe that if X is a Cauchy complete metric space and $\{A_k\}_{k \in \mathbb{N}}$ is a sequence of nonempty nested closed sets (that is $A_{k+1} \subset A_k$ for all k) such that $\lim_{k \rightarrow \infty} \text{diam}(A_k) = 0$, where $\text{diam}(A) = \sup_{x, y \in A} \{d(x, y)\}$, then there exist a unique $x \in X$ such that $\bigcap_{k=1}^{\infty} A_k = \{x\}$. To prove this, we pick a sequence of elements $x_k \in A_k$. This is a Cauchy sequence as $x_l \in A_k$ for $l \geq k$ and $\lim_{k \rightarrow \infty} \text{diam}(A_k) = 0$. As X is Cauchy complete this sequence converges and let us denote $x = \lim_k x_k$. As any A_l is closed, $x \in A_l$ hence $x \in \bigcap_{k=1}^{\infty} A_k$. If there exist $y \in \bigcap_{k=1}^{\infty} A_k$, $y \neq x$ then x and y can not both belong to A_l when $\text{diam}(A_l) < d(x, y)$, hence x is unique.

In order to prove the theorem, assume the integration contour is a square of side length 1, and divide it in four equal squares of side length $\frac{1}{2}$. As the integrals over the sides cancel:

$$\int_{\square_1} dz f(z) = \sum_{i=1}^4 \int_{\square_{\frac{1}{2}}^{(i)}} dz f(z) \Rightarrow \left| \int_{\square_1} dz f(z) \right| \leq 4 \left| \int_{\square_{\frac{1}{2}}} dz f(z) \right| ,$$

where we picked the square in the subdivision with the largest absolute value of the integral. Iterating, we get a sequence of nested boxes and:

$$\left| \int_{\square_1} dz f(z) \right| \leq 4^n \left| \int_{\square_{\frac{1}{2^n}}} dz f(z) \right| .$$

We call z_0 the point contained in all the boxes $\square_{\frac{1}{2^n}}$ and we note, by a direct computation that $\int_{\square_{\frac{1}{2^n}}} dz = 0$ and $\int_{\square_{\frac{1}{2^n}}} dz (z - z_0) = 0$,³ hence:

$$\left| \int_{\square_{\frac{1}{2^n}}} dz f(z) \right| = \left| \int_{\square_{\frac{1}{2^n}}} dz [f(z) - f(z_0) - f'(z_0)(z - z_0)] \right| \leq 4 \frac{1}{2^n} \sup_{z \in \square_{\frac{1}{2^n}}} \left| f(z) - f(z_0) - f'(z_0)(z - z_0) \right| .$$

Now, as f is complex differentiable at z_0 , $\forall \epsilon, \exists N_\epsilon$ such that $n > N_\epsilon \Rightarrow \sup_{z \in \square_{\frac{1}{2^n}}} \left| f(z) - f(z_0) - f'(z_0)(z - z_0) \right| \leq \epsilon |z - z_0| \leq \epsilon \frac{1}{2^{n-1}}$ hence $\forall \epsilon, \left| \int_{\square_1} dz f(z) \right| \leq \epsilon 4^n \frac{8}{4^n}$ and we conclude. Arbitrary smooth curves can be approximated arbitrarily well by gluing squares. □

As an obvious corollary, the complex integral of a holomorphic function along paths in a simply connected domain depends only on the end points. One can then define complex primitives of holomorphic functions as follows. If $f : U \rightarrow \mathbb{C}$ is holomorphic in a simply connected domain U and a is some fixed point in U , then the function:

$$F(z) = \int_a^z dw f(w) , \quad z \in U ,$$

where $\int_a^z$ denotes any path from a to z fully contained in U is a complex primitive of f , $F'(z) = f(z)$. To prove this, we first note that, by concatenating paths that if $\gamma_1(1) = \gamma_2(0)$ then:

$$(\gamma_1 \sqcup \gamma_2)(t) = \begin{cases} \gamma_1(t) , & 0 \leq t < 1 \\ \gamma_2(t) , & 1 \leq t < 2 \end{cases} \Rightarrow \int_0^1 dt' \dot{\gamma}_1(t') f(\gamma_1(t')) + \int_1^2 dt' \dot{\gamma}_2(t') f(\gamma_2(t')) = \int_0^2 dt \frac{d(\gamma_1 \sqcup \gamma_2)}{dt}(t) f((\gamma_1 \sqcup \gamma_2)(t)) ,$$

that is $\int_a^b dz f(z) + \int_b^c dz f(z) = \int_a^c dz f(z)$. Then, for any $z \in U$, there exists $B_\epsilon(z) \subset U$ and for any $w \in B_\epsilon(z)$, denoting $\gamma(t) = w + t(z - w)$ we have:

$$\left| \frac{F(z) - F(w)}{z - w} - f(z) \right| = \left| \frac{1}{z - w} \int_w^z dv f(v) - f(z) \right| = \left| \frac{1}{z - w} \int_0^1 dt \dot{\gamma}(t) (f(\gamma(t)) - f(z)) \right| \leq \sup_{t \in [0,1]} |f(w + t(z - w)) - f(z)| ,$$

and by the continuity of f , $\lim_{z \rightarrow w} \sup_{t \in [0,1]} |f(w + t(z - w)) - f(z)| = 0$.

³For instance: $\int dz z = \int_0^1 dx (x + i0) + \int_0^1 i dy (1 + iy) + \int_1^0 dx (x + i) + \int_1^0 i dy (0 + iy) = \frac{1}{2} + i - \frac{1}{2} - i + \frac{1}{2} = 0$.

14.3.1 Cauchy's integral formula

Let us denote the circle of center z and radius R by $C_R(z) = \partial B_R(z)$ which can be parametrized as $\gamma(\varphi) = z + Re^{i\varphi}$, $\varphi \in [0, 2\pi]$. Then:

$$\frac{1}{2\pi i} \int_{C_R(z)} dw \frac{1}{w-z} = \frac{1}{2\pi i} \int_0^{2\pi} d\varphi \left(\frac{d\gamma}{d\varphi}(\varphi) \right) \frac{1}{\gamma(\varphi)-z} = \frac{1}{2\pi i} \int_0^{2\pi} d\varphi (iRe^{i\varphi}) \frac{1}{Re^{i\varphi}} = 1.$$

Theorem 14.6 (**Cauchy's integral formula**). *Let f be a holomorphic function in some simply connected domain U and let $z_0 \in U$. Then $B_R(z_0) \subset U$ for some R and $\forall z \in B_R(z_0)$:*

$$f(z) = \frac{1}{2\pi i} \int_{C_R(z_0)} dw \frac{f(w)}{w-z}.$$

Proof. We have that $\forall z \in B_R(z_0)$, $B_r(z) \subset B_R(z_0)$ for r small enough and the function $f(w)(w-z)^{-1}$ is holomorphic in the annulus domain bounded by $C_r(z)$ and $C_R(z_0)$. Joining $C_r(z)$ and $C_R(z_0)$ along the shortest possible segment C , we build the contour $C_r(z) \cup C \cup C_R(z_0) \cup C^{-1}$ which bounds a simply connected domain D in which $f(w)(w-z)^{-1}$ is holomorphic. By the Cauchy integral theorem we have $\int_{C_r(z) \cup C \cup C_R(z_0) \cup C^{-1}} dw f(w)(w-z)^{-1} = 0$ and $\int_C dz f(w)(w-z)^{-1} = -\int_{C^{-1}} dz f(w)(w-z)^{-1}$ hence:

$$\frac{1}{2\pi i} \int_{C_R(z_0)} dw \frac{f(w)}{w-z} = \frac{1}{2\pi i} \int_{C_r(z)} dw \frac{f(w)}{w-z},$$

where $r > 0$ can be chosen arbitrarily small.

Now, the function $g(w) = \frac{f(w)-f(z)}{w-z}$ is continuous in $B_r(z)$ hence $|g(w)|$ attains a maximum on the compact set $\overline{B_r(z)}$, $M = \sup_{w \in \overline{B_r(z)}} |g(w)| < \infty$ and:

$$\left| \frac{1}{2\pi i} \int_{C_R(z_0)} dw \frac{f(w)}{w-z} - f(z) \right| = \left| \frac{1}{2\pi i} \int_{C_r(z)} dw \frac{f(w)-f(z)}{w-z} \right| \leq \frac{1}{2\pi} M l(C_r(z)) = Mr \xrightarrow{r \rightarrow 0} 0.$$

□

Recalling that if $f(z)$ is holomorphic then $f'(z) = \partial_z f = \frac{1}{2} \partial_x f - \frac{i}{2} \partial_y f$ and using the properties of differentiation under the integral sign, Theorem 11.3, we conclude by induction that $f^{(n)}$ is complex differentiable in $B_r(z_0)$ and:

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{C_R(z_0)} dw \frac{f(w)}{(w-z)^{n+1}}.$$

We can finally state one of the fundamental results concerning holomorphic functions. We have already seen that analytic functions are holomorphic in their domain of convergence. The converse of this statement is also true.

Theorem 14.7 (**Holomorphic functions are analytic**). *Let $f : U \rightarrow \mathbb{C}$ be a function holomorphic in U (open, simply connected) and let $z_0 \in U$ and R such that $\overline{B_R(z_0)} \subset U$. Then $\forall z \in B_R(z_0)$, f admits the convergent power series expansion:*

$$f(z) = \sum_{n=0}^{\infty} a_n(z_0) (z-z_0)^n, \quad a_n(z_0) = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_{C_R(z_0)} dw \frac{f(w)}{(w-z_0)^{n+1}},$$

that is f is analytic in U .

Proof. If f is holomorphic in U then using the Cauchy integral formula:

$$\left| f(z) - \sum_{n=0}^N a_n(z_0) (z-z_0)^n \right| = \left| \frac{1}{2\pi i} \int_{C_R(z_0)} dw \left[\frac{f(w)}{w-z} - \sum_{n=0}^N \frac{f(w)}{(w-z_0)^{n+1}} (z-z_0)^n \right] \right| = \left| \frac{1}{2\pi i} \int_{C_R(z_0)} dw f(w) \sum_{n=N+1}^{\infty} \frac{(z-z_0)^n}{(w-z_0)^{n+1}} \right|,$$

where the series converges as $|z - z_0| = \rho < R = |w - z_0|$, and the above is bounded by:

$$\sup_{w \in C_R(z_0)} |f(w)| R \frac{\rho^{N+1}}{R^{N+2}} \frac{1}{1 - \frac{\rho}{R}} \xrightarrow{N \rightarrow \infty} 0.$$

□

14.3.2 The spectral theorem for matrices revisited

Recall that for any linear operator on a finite dimensional vector space $A : V \rightarrow V$ one has an orthogonal decomposition in A -invariant subspaces $V = \bigoplus_i \ker(A - \lambda_i I)^{n_i}$, that is there exists a family of projectors P_i on $\ker(A - \lambda_i I)^{n_i}$ and:

- $P_i P_j = P_i \delta_{ij}$ as $\ker(A - \lambda_i I)^{n_i}$ are linearly independent.
- $AP_i = P_i AP_i = P_i A$ as $\ker(A - \lambda_i I)^{n_i}$ are A invariant.
- $I = \sum_{i=1}^q P_i$ as $V = \bigoplus_i \ker(A - \lambda_i I)^{n_i}$.
- denoting $B_i = A - \lambda_i I$, we have $AP_i = (\lambda_i + B_i)P_i$ with B_i such that $B_i^{n_i} P_i = 0$.

We note that $BP_i = P_i BP_i = P_i B$ and $1 \leq n_i \leq m_i - \gamma_i + 1$, where m_i and γ_i are the algebraic and geometric multiplicities of λ_i , as $\gamma_i = \dim \ker B_i < \dim \ker B_i^2 < \dots < \dim \ker B_i^{n_i} = m_i$. Checking first for polynomials, we conclude that for any function of A we have:

$$f(A) = \sum_{i=1}^q \sum_{k=0}^{n_i} \frac{f^{(k)}(\lambda_i)}{k!} (A - \lambda_i I)^k P_i,$$

and the sum over k is finite and $(A - \lambda_i I)^0 = I$. For diagonalizable matrices the sum stops at $k = 0$, as $(A - \lambda_i I)P_i = 0$, and we recover the spectral decomposition of diagonalizable matrices. For instance we have for $|z| > \|A\|_{\text{op}}$:

$$\frac{1}{z - A} = \sum_{i=1}^q \sum_{k=0}^{\infty} \frac{1}{(z - \lambda_i)^{k+1}} (A - \lambda_i I)^k P_i.$$

In fact this equality holds for all $z \neq \lambda_i$, not only for $|z| > \|A\|_{\text{op}}$. Indeed:

$$\begin{aligned} (z - A) \sum_{i=1}^q \sum_{k=0}^{\infty} \frac{1}{(z - \lambda_i)^{k+1}} (A - \lambda_i I)^k P_i &= \sum_{i=1}^q \sum_{k=0}^{\infty} \frac{1}{(z - \lambda_i)^{k+1}} (A - \lambda_i I)^k (z - \lambda_i I - A + \lambda_i I) P_i \\ &= \sum_{i=1}^q \sum_{k=0}^{\infty} \frac{1}{(z - \lambda_i)^k} (A - \lambda_i I)^k P_i - \sum_{i=1}^q \sum_{k=0}^{\infty} \frac{1}{(z - \lambda_i)^{k+1}} (A - \lambda_i I)^{k+1} P_i = \sum_i P_i = I. \end{aligned}$$

Integrating over a small circle of radius ϵ around λ_j we have:⁴

$$\int_{C_\epsilon(\lambda_j)} \frac{dz}{2\pi i} \frac{1}{z - A} = \sum_{i=1}^q \sum_{k=0}^{\infty} \left(\int_{C_\epsilon(\lambda_j)} \frac{dz}{2\pi i} \frac{1}{(z - \lambda_i)^{k+1}} \right) (A - \lambda_i I)^k P_i = \sum_{i=1}^q \sum_{k=0}^{\infty} \delta_{ij} \delta_{k0} (A - \lambda_i I)^k P_i = P_j,$$

yielding an integral representation of the projector P_i in terms of a contour integral of the *resolvent* $(z - A)^{-1}$. Furthermore, for any holomorphic function $f : U \rightarrow \mathbb{C}$ and for $\Gamma \subset U$ a union of simple curves such that the spectrum of A lies in the

⁴Note that for any z dependent matrix $R(z)$ we have for matrix elements in some basis $[\int dz R(z)]_{ij} = \int dz [R(z)]_{ij}$.

interior of the curves we have:

$$\begin{aligned}
\int_{\Gamma} \frac{dz}{2\pi i} \frac{f(z)}{z-A} &= \sum_{j=1}^q \int_{C_{\epsilon}(\lambda_j)} \frac{dz}{2\pi i} \frac{f(z)}{z-A} = \sum_{j=1}^q \int_{C_{\epsilon}(\lambda_j)} \frac{dz}{2\pi i} \sum_{i=1}^q \sum_{k \geq 0} \frac{f(z)}{(z-\lambda_i)^{k+1}} (A-\lambda_i I)^k P_i \\
&= \sum_{j=1}^q \sum_{i=1}^q \delta_{ij} \int_{C_{\epsilon}(\lambda_j)} \frac{dz}{2\pi i} \sum_{k \geq 0} \frac{f(z)}{(z-\lambda_i)^{k+1}} (A-\lambda_i I)^k P_i = \sum_{i=1}^q \int_{C_{\epsilon}(\lambda_i)} \frac{dz}{2\pi i} \sum_{k \geq 0} \frac{f(z)}{(z-\lambda_i)^{k+1}} (A-\lambda_i I)^k P_i \\
&= \sum_{i=1}^q \sum_{k \geq 0} \frac{f^{(k)}(\lambda_i)}{k!} (A-\lambda_i I)^k P_i = f(A).
\end{aligned}$$

14.3.3 Entire functions

Definition 14.6 (**Entire functions**). A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is said to be an entire function if it is holomorphic (hence analytic) in the entire complex plane.

The Taylor series of an entire function around any point has infinite radius of converge. Consider for instance the expansion around zero $f(z) = \sum_{n \geq 0} a_n z^n$. Taking a circle of radius R in Theorem 14.7 we conclude that $|a_n| \leq M_R R^{-n}$ where $M_R = \sup_{|w| \leq R} \{|f(w)|\} < \infty$ (as we take the supremum of a continuous function over a compact set). We thus have $\limsup |a_n|^{1/n} \leq \lim_{n \rightarrow \infty} M_R^{1/n} R^{-1} = R^{-1}$ and as R is arbitrary the series has infinite radius of convergence.

Lemma 14.5 (**Little Picard theorem**). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an entire but non constant function. Then f takes all, except at most one, complex values. That is, either $f(\mathbb{C}) = \mathbb{C}$ or $f(\mathbb{C}) = \mathbb{C} \setminus \{z\}$. In particular any bounded entire function is constant.

Proof. The reader is invited to find the proof of the little Picard theorem in the literature. We focus on the weaker second statement, called Liouville's theorem. Assume $|f(z)| < M$ for any $z \in \mathbb{C}$. As f is entire, $f(z) = \sum_{k \geq 0} a_k z^k$ and:

$$a_n = \frac{1}{2\pi i} \int_{C_R} d\zeta \frac{f(\zeta)}{\zeta^{n+1}} \Rightarrow |a_n| < \frac{M}{R^n} \xrightarrow{R \rightarrow \infty} 0, \quad \forall n \geq 1.$$

□

14.4 Critical points of holomorphic functions and the steepest descent method

Theorem 14.8 (**The maximum modulus principle**). Let $f : U \rightarrow \mathbb{C}$ be a non constant holomorphic function in U (open, simply connected). Then $|f(z)|$ does not have any maxima in the interior of U , that is $\forall z \in U$ and $\forall \delta > 0$, $\exists \omega \in B_{\delta}(z) \cap U$ such that $|f(\omega)| > |f(z)|$.

In particular the supremum of $|f(z)|$ in $\bar{U}$ is attained on the boundary ∂U .

Proof. Pick r such that $B_r(z) \subset U$. From Cauchy's integral formula we have, parametrizing $w = z + re^{i\varphi}$:

$$f(z) = \frac{1}{2\pi i} \int_{C_r(z)} dw \frac{f(w)}{w-z} = \frac{1}{2\pi} \int_0^{2\pi} d\varphi f(z + re^{i\varphi}) \Rightarrow |f(z)| \leq \frac{1}{2\pi} \int_0^{2\pi} d\varphi |f(z + re^{i\varphi})| \leq \sup_{\varphi} |f(z + re^{i\varphi})|,$$

where the sup is attained as $[0, 2\pi]$ is a compact set and $|f(z + re^{i\varphi})|$ is continuous in φ .

Thus $|f(z)| \leq |f(\omega)|$ for some $\omega \in C_r(z)$, and equality is only achieved if $|f|$ is constant on $C_r(z)$ (as $|f(z)|$ is smaller than the average of $|f|$ on the circle, it is equal to the supremum of $|f|$ on the circle only if the average of and the supremum of $|f|$ on the circle are equal, that is $|f|$ is constant on the circle). As this holds for any r sufficiently small, this implies

that f is constant inside some small enough $B_r(z)$ hence (due to the unicity of the analytic continuation) it is constant in U . $\square$

While the absolute value of a holomorphic function can not have a maximum, the function can have critical points. Consider a function $f : \mathbb{C} \rightarrow \mathbb{C}$ analytic in some domain $U \subset \mathbb{C}$ and having a critical point z_c , $f'(z_c) = 0$.

Proposition 14.1 (Critical points of f are saddle points of u and v). *The gradients of the real and imaginary parts of a holomorphic function are orthogonal.*

Any critical point of a holomorphic function f is a saddle point for its real and imaginary parts.

Proof. Denoting $z = x + iy$ and $f = u + iv$, the Cauchy Riemann equations write $\partial_x u = \partial_y v$ and $\partial_y u = -\partial_x v$. Denoting g the standard metric on $\mathbb{R}^2$, we have $g(\nabla u, \nabla v) = \partial_x u \partial_x v + \partial_y u \partial_y v = 0$ by the Cauchy Riemann equations.

Let us denote z_c a critical point of f , $\partial_z f(z_c) = 0$ and as f is holomorphic $\partial_{\bar{z}} f(z_c) = 0$, hence $\partial_x u = \partial_y u = \partial_x v = \partial_y v = 0$. The Hessians of u and v at z_c are:

$$H_u = \begin{pmatrix} \partial_x^2 u & \partial_x \partial_y u \\ \partial_y \partial_x u & \partial_y^2 u \end{pmatrix} = \begin{pmatrix} \partial_x^2 u & -\partial_x^2 v \\ -\partial_x^2 v & -\partial_x^2 u \end{pmatrix}, \quad H_v = \begin{pmatrix} \partial_x^2 v & \partial_x \partial_y v \\ \partial_y \partial_x v & \partial_y^2 v \end{pmatrix} = \begin{pmatrix} \partial_x^2 v & \partial_x^2 u \\ \partial_x^2 u & -\partial_x^2 v \end{pmatrix},$$

which both have the two eigenvalues $\pm \sqrt{(\partial_x^2 u)^2 + (\partial_x^2 v)^2}$. Denoting partial derivatives as indices and $\lambda = \sqrt{u_{xx}^2 + v_{xx}^2}$ the eigenvectors of the Hessians are:

$$H_u : \begin{cases} \lambda, & \begin{pmatrix} \frac{v_{xx}}{\sqrt{2\lambda(\lambda - u_{xx})}} \\ -\sqrt{\frac{\lambda - u_{xx}}{2\lambda}} \end{pmatrix} \\ -\lambda, & \begin{pmatrix} \frac{v_{xx}}{\sqrt{2\lambda(\lambda + u_{xx})}} \\ \sqrt{\frac{\lambda + u_{xx}}{2\lambda}} \end{pmatrix} \end{cases}, \quad H_v : \begin{cases} \lambda, & \begin{pmatrix} \frac{u_{xx}}{\sqrt{2\lambda(\lambda - v_{xx})}} \\ \sqrt{\frac{\lambda - v_{xx}}{2\lambda}} \end{pmatrix} \\ -\lambda, & \begin{pmatrix} \frac{u_{xx}}{\sqrt{2\lambda(\lambda + v_{xx})}} \\ -\sqrt{\frac{\lambda + v_{xx}}{2\lambda}} \end{pmatrix} \end{cases}.$$

$\square$

The *gradient flow* of a function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is loosely speaking the set of trajectories $\gamma : \mathbb{R} \rightarrow \mathbb{R}^N$ solutions of the differential equation $\frac{d\gamma}{dt} = -\nabla f$, or in components $\frac{d\gamma^\mu}{dt} = -g^{\mu\nu} \partial_\nu f$, for all the possible initial conditions $\gamma(0) = x_0$ (see Chapter 19 for a formal definition of the flow of a vector field).

Let us consider f an holomorphic function. The gradient flow of the real part of f is the set of solutions of the differential equation:

$$\frac{dx}{dt} = -\partial_x u, \quad \frac{dy}{dt} = -\partial_y u, \quad \Rightarrow \quad \frac{d\bar{z}}{dt} = -\frac{\partial f}{\partial z},$$

where we used the notation $\partial_z f = \frac{1}{2}(\partial_x - i\partial_y)(u + iv) = \partial_x u - i\partial_y u$ in order to rewrite the differential equation as a complex flow equation. Along a trajectory of this gradient flow (i.e. any solution of the differential equation) the imaginary part of f is conserved:

$$\frac{dv}{dt} = \frac{1}{2i} \frac{d}{dt} (f - \bar{f}) = \frac{1}{2i} \left(\frac{dz}{dt} \frac{\partial f}{\partial z} - \frac{d\bar{z}}{dt} \frac{\partial \bar{f}}{\partial \bar{z}} \right) = \frac{1}{2i} \left(-\frac{\partial \bar{f}}{\partial \bar{z}} \frac{\partial f}{\partial z} + \frac{\partial f}{\partial z} \frac{\partial \bar{f}}{\partial \bar{z}} \right) = 0,$$

and the real part of f decreases:

$$\frac{du}{dt} = -\frac{1}{2} \left| \frac{\partial f}{\partial z} \right|^2.$$

At a critical point z_c of f , that is such that $\partial_z f(z_c) = 0$, we define the *upward* (denoted $\mathcal{K}$) and the *downward* (denoted $\mathcal{J}$) paths as the trajectories solving the gradient flow with initial conditions:

$$\lim_{t \rightarrow \infty} z(t) = z_c \text{ (upward)}, \quad \lim_{t \rightarrow -\infty} z(t) = z_c \text{ (downward)},$$

that is z_c is attractive along the upward path and repulsive along the downward one. The upward flow approaches z_c along the eigenvector of the Hessian H_u with positive eigenvalue, while the downward flow leaves the fixed point along the eigenvector with negative eigenvalue. Generically a flow line will not connect two different critical points of f as the imaginary part of f is typically different at different critical points.

We are interested in evaluating integrals like:

$$Z = \int_C dz e^{-f(z)},$$

where $f(z)$ is holomorphic and C is a “good” integrate path in $\mathbb{C}$ along which the integral is convergent. As an example one can consider $f(z) = \frac{1}{2}z^2 + \frac{g}{4!}z^4$ and $C = \mathbb{R}$.

The steepest descent method consists in decomposing such an integral into a linear combinations of integrals over paths passing through the critical points of $f(z)$. The downward flows of $-f(z)$ are “good” path of integration as the imaginary part of $-f$ is constant and the real part $-\text{Re}(f(z))$ is maximal at z_c and decreases at infinity:

$$\int_{\mathcal{J}} dz e^{-f(z)} = e^{-i\text{Im}(f(z_c))} \int_{\mathcal{J}} dw e^{-\text{Re}(f(z_c+w))},$$

where we parametrize the downward path $\mathcal{J}$ as $z_c + w$. Assuming we have several critical points Z_c^i , we have:

$$Z = \sum_i n_i \int_{\mathcal{J}_i} dw_i e^{-f(z_c^i+w_i)},$$

where $n_i = 0, \pm 1$ counts (with signs) the number of intersections between the original integration path C and the upward paths $\mathcal{K}_i$ of $-f$. Indeed, assume C intersects some upward path $\mathcal{K}$ corresponding to some z_c . As z_c is attractive along $\mathcal{K}$, when one relaxes the integration contour C to flow along the gradient flow, a neighborhood of the intersection point will be mapped onto the downward path $\mathcal{J}$ which passes through z_c .

The integral over a downward path is evaluated by a Laplace method. First we note that:

$$f(z_c + w) = f(z_c) + \frac{1}{2}f''(z_c)w^2 + \dots = f(z_c) + \frac{1}{2}|f''(z_c)|e^{i\varphi}w^2 + \dots,$$

with $f''(z_c) = |f''(z_c)|e^{i\varphi}$. It follows that $\mathcal{J}$ is locally parametrized by $w = \rho e^{-i\frac{\varphi}{2}}$, as this is the path so that $\text{Im}(f(z_c)) = \text{Im}(f(z))$, and $-\text{Re}(f(z)) = -\text{Re}(f(z_c)) - \frac{1}{2}|f''(z_c)|\rho^2$ is maximal for $\rho = 0$. We have:

$$\begin{aligned} \int_{\mathcal{J}} dz e^{-f(z)} &= e^{-f(z_c)} \int dw e^{-\frac{1}{2}|f''(z_c)|e^{i\varphi}w^2 + \dots} = e^{-f(z_c)} e^{-i\frac{\varphi}{2}} \int d\rho e^{-\frac{1}{2}|f''(z_c)|\rho^2 + \dots} \\ &= e^{-f(z_c)} \sqrt{\frac{2\pi}{e^{i\varphi}|f''(z_c)|}} (1 + \dots) = e^{-f(z_c)} \sqrt{\frac{2\pi}{f''(z_c)}} (1 + \dots), \end{aligned}$$

where the corrections are computed as normalized Gaussian integrals with covariance $|f''(x_c)|^{-1}$ by Taylor expanding in ρ^k , $k \geq 3$ and commuting the Taylor sums and the integral.

Exercise 14.3. *In order to evaluate the subleading terms in a Laplace approximation one needs to compute Gaussian integrals. Prove that:*

$$\int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} e^{-\frac{1}{2\alpha}x^2} x^{2q} = \alpha^q \frac{(2q)!}{2^q q!},$$

and construct a graphical representation of the result.

Exercise 14.4. *Derive the saddle point approximation of the Gamma function $\Gamma(z) = \int_0^{\infty} dt t^{z-1} e^{-t} dt$.*

14.5 Meromorphic functions and the residue theorem

Definition 14.7 (**Meromorphic functions and poles**). A function $f : U \rightarrow \mathbb{C}$ is meromorphic in U if every point of U has a neighbourhood such that either f or $1/f$ (or both) is (are) holomorphic in it.

Note that if f is holomorphic at z_0 and $f(z_0) \neq 0$, then $1/f$ is holomorphic at z_0 and $[1/f]'(z_0) = -f'(z_0)/f(z_0)^2$. If $f : U \rightarrow \mathbb{C}$ is meromorphic in U then $\forall z_0 \in U$:

- either f is holomorphic (hence analytic) at z_0 . Then in some neighborhood of z_0 we have $f(z) = (z - z_0)^p g(z)$ with $p \geq 0$ and $g(z)$ analytic and non zero at z_0 . If $p > 0$ then z_0 is a *zero of order p* of f .
- or f is not holomorphic at z_0 , but $1/f$ is holomorphic at z_0 . Then in some neighborhood of z_0 we have $\frac{1}{f(z)} = (z - z_0)^n g(z)$ with $n > 1$ and $g(z)$ analytic and non zero at z_0 . It follows that $g(z) \neq 0$ in some neighborhood of z_0 , hence $(z - z_0)^n f(z) = \frac{1}{g(z)}$ is complex differentiable in this neighborhood hence holomorphic at z_0 . In this case we say that z_0 is a *pole of order n* of f .

The poles and the zeros of meromorphic functions are isolated. As for any $z_0 \in U$ there exists an integer n such that $(z - z_0)^n f(z)$ is holomorphic and nonzero in some neighborhood of z_0 , labeling intelligently some coefficients we obtain that $f(z)$ always admits a *Laurent series* representation:

$$(z - z_0)^n f(z) = \sum_{k \geq 0} a_{-n+k} (z - z_0)^k \Rightarrow f(z) = \sum_{k \geq -n} a_k (z - z_0)^k = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n+1}} + \cdots + \frac{a_{-1}}{(z - z_0)} + a_0 + a_1(z - z_0) + \cdots ,$$

and the series converges in some punctured disk $B_\delta(z_0) \setminus \{z_0\}$ which belongs to the analyticity domain of $[(z - z_0)^n f(z)]^{-1}$ and is such that $[(z - z_0)^n f(z)]^{-1} \neq 0$, hence $a_{-n} \neq 0$. We call the part comprised of the (finitely many) strictly negative powers of the Laurent series *the principal part*; the principal is present only if z_0 is a pole of f .

Consider for instance the function $f(z) = \frac{1}{z-1} + \frac{1}{z-2}$. Its Laurent expansions around the poles $z_0 = 1$, $z_0 = 2$ and the regular point $z_0 = 3$ are:

$$\frac{1}{z-1} + \frac{1}{z-2} = \begin{cases} \frac{1}{z-1} + \frac{1}{-1+(z-1)} = \frac{1}{z-1} - \sum_{n \geq 0} (z-1)^n \\ \frac{1}{1+(z-2)} + \frac{1}{z-2} = \frac{1}{z-2} + \sum_{n \geq 0} (-1)^n (z-2)^n \\ \frac{1}{2+(z-3)} + \frac{1}{1+(z-3)} = \sum_{n \geq 0} (-1)^n \left(\frac{1}{2^{n+1}} + 1 \right) (z-3)^n \end{cases} .$$

As a second example we note that the function $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$ is entire (as the exponential is entire) and has simple zeros for $z = n\pi$ with $n \in \mathbb{Z}$, $\sin(n\pi + \epsilon) = \frac{(-1)^n}{2i} (2i\epsilon + O(\epsilon^3))$. It follows that the function $1/\sin z$ is meromorphic, has poles of order one at $n\pi$ and has no zeros in the complex plane.

Theorem 14.9 (**The residue theorem**). Let $f : U \rightarrow \mathbb{C}$ a meromorphic function in the open simply connect set $U \subset \mathbb{C}$ and let $z_0 \in U$ be a pole of order n of f . It follows that in a punctured disk around z_0 we have the Laurent series representation

$$f(z) = \sum_{k \geq -n} a_k (z - z_0)^k = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n+1}} + \cdots + \frac{a_{-1}}{(z - z_0)} + a_0 + a_1(z - z_0) + \cdots ,$$

convergent in some punctured disk around z_0 . We call a_{-1} , that is the first coefficient in the principal part of the Laurent series, the *residue of f at the pole z_0* and we denote it $\text{Res}(f, z_0)$. Then for R such that $B_R(z_0) \subset U$ does not contain any other pole of f we have:

$$\int_{C_R(z_0)} dz f(z) = 2\pi i \text{Res}(f, z_0) .$$

This generalizes trivially to arbitrary simple curves encircling a finit number of poles.

Proof. By applying the Cauchy's integral formula to the holomorphic function $(z - z_0)^n f(z) = \sum_{k \geq -n} a_k (z - z_0)^{n+k}$ we find that the coefficients a_k of the Laurent series of f are the coefficients of the term $(z - z_0)^{n+k}$ of the Taylor series of $(z - z_0)^n f(z)$ hence:

$$a_k = \frac{1}{2\pi i} \int_{C_R(z_0)} dw \frac{(w - z_0)^n f(w)}{(w - z_0)^{n+k+1}} = \frac{1}{2\pi i} \int_{C_R(z_0)} dw \frac{f(w)}{(w - z_0)^{k+1}} .$$

Alternatively, as for R small enough:

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n+1}} + \cdots + \frac{a_{-1}}{(z - z_0)} + \tilde{f}(z) ,$$

integrating around the circle $\int_{C_R(z_0)} dw \tilde{f}(w) = 0$ as $\tilde{f}$ is holomorphic in $B_R(z_0)$, while the other integrals are:

$$\int_{C_R(z_0)} dz \frac{1}{(z - z_0)^q} \stackrel{z=z_0+Re^{i\varphi}}{=} \int_0^{2\pi} d\varphi (iRe^{i\varphi}) \frac{1}{R^q e^{iq\varphi}} = 2\pi i \delta_{q1} .$$

□

Note that if z_0 is a pole of order n of f , then the residue of f at z_0 can be found by the formula:

$$\text{Res}(f, z_0) = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \frac{d^{n-1}}{dz^{n-1}} ((z - z_0)^n f(z)) .$$

Exercise 14.5 (The Euler Gamma function). *Let us define the Euler Gamma function*

$$\Gamma(z) = \int_0^\infty dt t^{z-1} e^{-t} , \quad \text{Re}(z) > 0$$

- Prove that the integral is absolutely convergent in the region $\text{Re}(z) > 0$.
- Prove that, in the region of convergence $\Gamma(z+1) = z\Gamma(z)$.
- Use the recursion relation to maximally analytically extended $\Gamma(z)$.
- Prove that $\int_0^\infty dt t^{z-1} (e^{-t} - 1)$ is a convergent representation of the Γ function in the region $0 > \text{Re}(z) > -1$.
- Identify the poles of $\Gamma(z)$ and compute the residues at the poles.
- Prove that $e^{-z} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) z^{-s} ds$, where $c > 0$ and z^{-s} is defined on using the principal branch of the logarithm.
- Prove that $\frac{1}{\Gamma(z)} = \frac{1}{2\pi i} \int_C dt (-t)^{-z} e^{-t}$ with C a Hankel contour turning counter clockwise around the origin, $C = [\infty - i, -i] \cup \{-i e^{-i\varphi} | \varphi = 0, \pi\} \cup [i, \infty + i]$

Exercise 14.6. Prove that the functions $f(z) = (e^{\beta z} \pm 1)^{-1}$ are meromorphic, identify their poles and compute the residues at these poles.

14.5.1 Isolated singularities of complex functions

Proposition 14.2. Consider a function f holomorphic in the annulus $\{z \in \mathbb{C} | r < |z - z_0| < R\}$. Then in the annulus f admits a unique representation as a Laurent series:

$$f(z) = \sum_{n=-\infty}^{\infty} (z - z_0)^n a_n , \quad a_n = \frac{1}{2\pi i} \int_{C_\rho(z_0)} dz \frac{f(z)}{(z - z_0)^{n+1}} , \quad r < \rho < R .$$

Proof. First note that the Laurent series converges as pushing ρ to $R_- < R$ for $n > 0$ and to $r_+ > r$ for $n < 0$ we have for $k \geq 0$ $|a_k| \leq R_-^{-k} \|f\|_{\infty, C_{R_-}(z_0)}$ $|a_{-k}| \leq r_+^k \|f\|_{\infty, C_{r_+}(z_0)}$, hence:

$$\left| \sum_{n=-\infty}^{\infty} (z - z_0)^n a_n \right| \leq a_0 + \|f\|_{\infty, C_{r_+}(z_0)} \sum_{k>0} r_+^k \frac{1}{|z - z_0|^k} + \|f\|_{\infty, C_{R_-}(z_0)} \sum_{k>0} \frac{1}{R_-^k} |z - z_0|^k,$$

which converges for $r_+ < |z - z_0| < R_-$.

To prove the existence of the Laurent series decomposition, for $r < r_+ < R_- < R$ by the Cauchy integral formula, for z in the annulus:

$$f(z) = \frac{1}{2\pi i} \int_{C_{R_-}(z_0)} dw \frac{f(w)}{(w - z)} - \frac{1}{2\pi i} \int_{C_{r_+}(z_0)} dw \frac{f(w)}{(w - z)}.$$

Using the binomial theorem $(w - z)^{-1} = \sum_{n \geq 0} \frac{(z - z_0)^n}{(w - z_0)^{n+1}}$ for the first integral and $(w - z)^{-1} = -\sum_{n \geq 0} \frac{(w - z_0)^n}{(z - z_0)^{n+1}}$ for the second, commuting sums and integrals (we will see later this is allowed by the dominated convergence theorem) and relabelling some terms we conclude. $\square$

Definition 14.8 (Isolated singularities). A point z_0 is an isolated singularity of a function f if there exists a disk $B_r(z_0)$ such that f is holomorphic in $B_r(z_0) \setminus \{z_0\}$. An isolated singularity of $f(z)$ is either:

- removable if $\lim_{z \rightarrow z_0} f(z)$ exists, that is the negative powers in the Laurent series of f at z_0 are absent. In this case f admits an analytic continuation to $B_r(z_0)$.
- a pole, if $\lim_{z \rightarrow z_0} f(z)$ does not exist but $\lim_{z \rightarrow z_0} \frac{1}{f(z)}$ exists, that is the Laurent series has finitely many terms with negative powers.
- an essential singularity if neither $\lim_{z \rightarrow z_0} f(z)$ nor $\lim_{z \rightarrow z_0} \frac{1}{f(z)}$ exist, that is the Laurent series has indefinitely many terms with negative powers.

We note that with minimal effort the residue theorem extends to isolated singularities (in practice this means it extends to essential singularities). The positions of the poles and the residues at those poles fix a meromorphic function.

Theorem 14.10 (Mittag-Leffler's theorem). Let $U \subset \mathbb{C}$ be an open set and $E \subset U$ be a subset whose limit points (if any) occur on the boundary of U . For each $a \in E$, let $p_a(z)$ be a polynomial in $1/(z - a)$ of the form:

$$p_a(z) = \sum_{n=1}^{N_a} \frac{c_{a,n}}{(z - a)^n},$$

that is with no constant coefficient.

Then there exists a meromorphic function $f : U \rightarrow \mathbb{C}$ whose poles are precisely the elements of E and such that for $a \in E$, the function $f(z) - p_a(z)$ has only a removable singularity at a (that is the principal part of the Laurent series of f at a is $p_a(z)$) and for any other meromorphic function $g : U \rightarrow \mathbb{C}$ with these properties $f(z) - g(z)$ is holomorphic on U .

Proof. The idea is to pick $f = \sum_{a \in E} p_a(z)$. Proving the convergence of the sum is non trivial if E is infinite. $\square$

A closely related result is the following.

Theorem 14.11 (Mittag-Leffler's expansion theorem). Let f be a meromorphic function in $\mathbb{C}$ having only simple poles (i.e. poles of order 1) and which is continuous (or has a removable singularity) at 0. Denote $X = \{z_1, z_2, \dots\}$ the set of poles of f . For $n \rightarrow \infty$, let D_n be the disk of radius R_n centered at the origin such that $\lim_{n \rightarrow \infty} R_n = \infty$ and its boundary ∂D_n does not contain any pole of f . Assume there exists $M \in \mathbb{R}_{>0}$ such that $\forall n, \sup_{z \in \partial D_n} |f(z)| < M$. Then:

$$f(z) = f(0) + \sum_{z_n \in X} \text{Res}(f, z_n) \left(\frac{1}{z - z_n} + \frac{1}{z_n} \right), \quad \text{Res}(f, z_n) = \lim_{z \rightarrow z_n} (z - z_n) f(z).$$

Proof. Let $w \in \mathbb{C} \setminus X$. Then $\frac{f(z)}{z-w}$ has simple poles at $X \cup \{w\}$. Take n large enough such that $w \in D_n$ and denote X_n the set of poles of f inside the disk D_n . By the residue theorem:

$$\frac{1}{2\pi i} \int_{\partial D_n} dz \frac{f(z)}{z-w} = \text{Res}\left(\frac{f(z)}{z-w}, w\right) + \sum_{z_k \in X_n} \text{Res}\left(\frac{f(z)}{z-w}, z_k\right) = f(w) + \sum_{z_k \in X_n} \frac{1}{z_k - w} \text{Res}(f, z_k).$$

subtracting the same expression with $w = 0$ we find:

$$\frac{1}{2\pi i} \int_{\partial D_n} dz f(z) \left(\frac{1}{z-w} - \frac{1}{z} \right) = f(w) - f(0) + \sum_{z_k \in X_n} \text{Res}(f, z_k) \left(\frac{1}{z_k - w} - \frac{1}{z_k} \right).$$

In order to conclude we need to prove that the integral vanishes in the $n \rightarrow \infty$ limit, but:

$$\frac{1}{2\pi} \left| \int_{\partial D_n} dz f(z) \left(\frac{1}{z-w} - \frac{1}{z} \right) \right| = \frac{1}{2\pi} \left| \int_{\partial D_n} dz f(z) \left(\frac{w}{z(z-w)} \right) \right| \leq M \frac{R_n |w|}{R_n(R_n - |w|)} \rightarrow_{n \rightarrow \infty} 0.$$

□

Exercise 14.7 (Pole expansion of $\coth(z)$). *Prove that:*

$$\coth(z) = \sum_{n=-\infty}^{\infty} \frac{z}{n^2 \pi^2 + z^2}.$$

We have $\coth(z) - \frac{1}{z} = \frac{e^{2z}+1}{e^{2z}-1} - \frac{1}{z}$ has poles at $z = in\pi$ with $n \in \mathbb{Z} \setminus \{0\}$ with residues 1 and respects the conditions of the theorem, hence:

$$\coth(z) - \frac{1}{z} = \sum_{n=1}^{\infty} \left(\frac{1}{z - in\pi} + \frac{1}{z + in\pi} \right) \Rightarrow \coth(z) = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 + n^2 \pi^2}.$$

Exercise 14.8. *Prove that $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$, with $\Gamma(z)$ the Euler Gamma function and deduce $\Gamma(1/2) = \sqrt{\pi}$.*

Lemma 14.6 (**Picard theorem**). *If a complex function f has an essential singularity at a point z_0 then on any punctured disk $B_r(z_0) \setminus \{z_0\}$, with $r < r_0$ for some r_0 , $f(z)$ it takes all possible complex values, with at most one exception.*

Proof. The reader should consult the literature on the subject.

□

14.6 Borel summation

We use K as a dustbin notation for irrelevant (real positive) multiplicative constants, and R and ρ for the important (real positive) constants.

Borel summable series. A formal power series $A(z) = \sum_{k=0}^{\infty} a_k z^k$ is called a *Borel summable series along the positive real axis* if the series

$$B(t) = \sum_{k=0}^{\infty} \frac{a_k}{k!} t^k,$$

is absolutely convergent in some disk $|t| < \rho$ and $B(t)$ admits an analytic continuation in a strip of width ρ along the positive real axis such that for t in this strip $|B(t)| < K e^{|t|/R}$ for some real positive R . The function $B(t)$ is called the *Borel transform* of $A(z)$ and the *Borel sum* of $A(z)$ is the Laplace transform of its Borel transform:

$$f(z) = \frac{1}{z} \int_0^{\infty} dt e^{-t/z} B(t).$$

It is easy to check that the function f is analytic in a disk of diameter R tangent to the imaginary axis at the origin, $\text{Disk}_R = \{z \in \mathbb{C} \mid \text{Re}(1/z) > 1/R\}$:

$$\text{Re}\left(\frac{2}{R(1+re^{i\varphi})}\right) > R^{-1} \Leftrightarrow \frac{2(1+r\cos\varphi)}{R(1+r^2+2r\cos\varphi)} > R^{-1} \Leftrightarrow 2 > 2+r^2 \Leftrightarrow r < 1.$$

Clearly, if it exists, the Borel sum of a series is unique. This raises the following question: given a function $h(z)$ whose asymptotic series at zero is the Borel summable series $A(z)$, does the Borel resummation of $A(z)$ reconstruct $h(z)$? That is, is $f(z) = h(z)$? The answer to this question is *no* in general: for instance the function $e^{-1/z}$ is asymptotic (along the positive real axis) at 0 to the Borel summable series $a_k = 0$. It turns out that one can formulate necessary and sufficient conditions for $h(z)$ which ensure that it is indeed the Borel sum of its asymptotic series.

Borel summable functions. A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is called a *Borel summable function along the positive real axis* if it is analytic in a disk Disk_R and has an asymptotic series at 0 (which can have zero radius of convergence):

$$f(z) = \sum_{k=0}^{q-1} a_k z^k + R_q(z),$$

such that the rest term of order q obeys the bound:

$$|R_q(z)| \leq K q! q^\beta \rho^{-q} |z|^q, \quad z \in \text{Disk}_R,$$

for some fixed $\beta \in \mathbb{R}_+$. The positive real axis is selected by the position of the center of Disk_R . We call $\text{Disk}_R = \{z \in \mathbb{C} \mid \text{Re}(1/z) > 1/R\}$ a *Sokal disk*.

These two notions are intimately related: the Borel sums of Borel summable series are Borel summable functions (this is straightforward to prove). Moreover, the asymptotic series of Borel summable functions are Borel summable series.

Theorem 14.12 (Nevanlinna-Sokal). *Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a Borel summable function, hence analytic in some Disk_R and obeying the bound $|R_q(z)| \leq K q! q^\beta \rho^{-q} |z|^q$ for $z \in \text{Disk}_R$ with some fixed β . Then:*

- the Borel transform of the asymptotic series of f ,

$$B(t) = \sum_{k=0}^{\infty} \frac{1}{k!} a_k t^k,$$

is convergent in a disk of radius ρ in t and it defines an analytic function in this domain.

- $B(t)$ can be analytically continued to the strip $\{t \in \mathbb{C} \mid \text{dist}(t, \mathbb{R}_+) < \rho\}$ and in this strip it obeys an exponential bound $|B(t)| < K e^{|t|/R}$.
- for all $z \in \text{Disk}_R$ we can reconstruct the function $f(z)$ by the absolutely convergent Laplace transform:

$$f(z) = \frac{1}{z} \int_0^\infty dt e^{-t/z} B(t).$$

Proof. We first observe that:

$$|a_q| = \lim_{z \rightarrow 0, z \in \text{Disk}_R} \left| \frac{a_q z^q + R_{q+1}(z)}{z^q} \right| = \lim_{z \rightarrow 0, z \in \text{Disk}_R} \frac{|R_q(z)|}{|z|^q} \leq K q! q^\beta \rho^{-q},$$

hence $B(t)$ is an integer power series which converges in the disk $|t| < \rho$ and defines an analytic function in this domain.

Let us recall Hankel's contour integral representation of the inverse of the Gamma function:

$$\frac{1}{2\pi i} \oint_{\text{Re}(z^{-1})=r^{-1}} dz e^{x/z} z^{k-1} = \frac{1}{2\pi i} \int_{r^{-1}-i\infty}^{r^{-1}+i\infty} dw e^{xw} w^{-k-1} = \frac{x^k}{\Gamma(k+1)},$$

which holds for any $r, x \in \mathbb{R}_+$ and $k \in \mathbb{C}$. We define for $x \in \mathbb{R}_{>0}$ the function:

$$b_0(x) = \frac{1}{2\pi i} \oint_{\text{Re}(z^{-1})=r^{-1}} dz e^{x/z} z^{-1} f(z),$$

which is an r independent function for $r < R$ as long as the integral converges because the integration contour is fully contain in Disk_R . Substituting the asymptotic expansion of $f(z)$ up to order q we obtain:

$$b_0(x) = \sum_{k=0}^{q-1} \frac{a_k}{k!} x^k + \frac{1}{2\pi i} \oint_{\text{Re}(z^{-1})=r^{-1}} dz e^{x/z} z^{-1} R_q(z),$$

and changing variables to $w = 1/z$ and using the bound on R_q , the reminder term above is bounded by:

$$K q! q^\beta \rho^{-q} e^{x/r} \int_{-\infty}^{\infty} dv \frac{1}{|r^{-1} + iv|^{q+1}} \leq K q! q^\beta \rho^{-q} e^{x/r} r^q \int_{-\infty}^{\infty} dv \frac{1}{(1+v^2)^{\frac{q+1}{2}}},$$

and the integral in the last line is always bounded by the case $q = 1$ in which case it is π and can be absorbed in K . Choosing $r = x/q$ (which is possible for q large enough $q > x/R$) and using the Stirling upper bound on the Gamma function, the reminder is finally bounded by $K q^{\beta+1/2} (x/\rho)^q$ hence goes to zero in the $q \rightarrow \infty$ limit as long as $x < \rho$. It follows that for $0 < x < \rho$, $b_0(x) = B(t)|_{t=x}$.

We now define $b_m(x) = \frac{d^m}{dx^m} b_0(x)$, which computes to:

$$b_m(x) = a_m + \frac{1}{2\pi i} \oint_{\text{Re}(z^{-1})=r^{-1}} dz e^{x/z} z^{-m-1} R_{m+1}(z),$$

and we have the bound $|b_m(x)| \leq K (m+1)! (m+1)^\beta \rho^{-m-1} e^{x/r}$ (note that this bound covers also the term $|a_m| < K m! m^\beta \rho^{-m}$).

The sum $B_x(t) = \sum_{m \geq 0} \frac{(t-x)^m}{m!} b_m(x)$ defines an analytic function in t as long as it converges. As:

$$|B_x(t)| \leq K e^{x/r} \sum_{m \geq 0} (m+1)^{\beta+1} (|t-x|/\rho)^m,$$

we conclude that $B_x(t)$ is analytic in a disk of radius ρ centered at x . It is immediate to check that $B_x(t) = B_{x'}(t)$ as long as they both converge, hence $B_x(t)$ is the analytic continuation of $B(t)$ to the strip $\{t \in \mathbb{C} \mid \text{dist}(t, \mathbb{R}_+) < \rho\}$ and it obeys the appropriate exponential bound. The last point follows by noting that for $z' \in \text{Disk}_r$:

$$\frac{1}{z'} \int_0^\infty dx e^{-x/z'} \frac{1}{2\pi i} \oint_{\text{Re}(z^{-1})=r^{-1}} dz e^{x/z} z^{-1} f(z) = \frac{1}{2\pi i} \oint_{\text{Re}(z^{-1})=r^{-1}} dz \frac{1}{z-z'} f(z) = f(z'),$$

by Cauchy's theorem. □

We emphasize that both for series and for functions, Borel summability is directional:

- for series, Borel summability along a direction requires the unimpeded analytic continuation of $B(t)$ in a thin strip centered on that direction.
- for functions, Borel summability along a direction requires analyticity and bound on the Taylor rest terms in a Sokal disk (with 0 on its boundary) centered on that direction.

Clearly, the singularities of the Borel transform $B(t)$ are associated to directions along which the function $f(z)$ ceases to be Borel summable.

The Stieltjes transform

Let $\rho : I \rightarrow \mathbb{R}$ be a continuations function on some interval $I \subset \mathbb{R}$ (or more generally a measure, see Chapter 16). The Stieltjes transform of ρ :

$$S(z) = \int_I dt \frac{1}{z-t} \rho(t) ,$$

is an analytic function on $\mathbb{C} \setminus I$. The Stieltjes transform is inverted as:

$$\rho(x) = \lim_{\epsilon \rightarrow 0^+} \frac{S(x - i\epsilon) - S(x + i\epsilon)}{2\pi i} .$$

Indeed, using Cauchy's integral formula we have for any $z \in \mathbb{C} \setminus I$ and for a small enough circle $C_\epsilon(z)$ around z :

$$S(z) = \int_{C_\epsilon(z)} \frac{dw}{2\pi i} \frac{S(w)}{w-z} .$$

Deforming the contour to go above and below I and taking into account the orientation of $C_\epsilon(z)$ we have:

$$S(z) = \int_I \frac{dt}{2\pi i} \frac{1}{t-z} \lim_{\epsilon \rightarrow 0^+} [S(x + i\epsilon) - S(x - i\epsilon)] .$$

Chapter 15

Hilbert spaces (Hilbert Raum)

We recall from Chapter 9 that a vector space over the field of scalars $\mathbb{F} = \mathbb{R}, \mathbb{C}$ is a set endowed with two operations:

- the addition $+: V \times V \rightarrow V$, $(x, y) \rightarrow x + y$ such that $+$ is associative and commutative, 0 is the identity element for $+$ and every element $x \in V$ has an additive inverse, $-x$.
- multiplication by scalars $\cdot: \mathbb{F} \times V \rightarrow V$, $(\alpha, x) \rightarrow \alpha \cdot x$ such that:
 - $\cdot$ is compatible with the multiplication $\times$ in $\mathbb{F}$ that is $\alpha \cdot (\beta \cdot x) = (\alpha \times \beta) \cdot x$.
 - the identity of $\cdot$ is the multiplicative identity in $\mathbb{F}$, denoted 1 , that is $1 \cdot x = x$.
 - $\cdot$ is distributive with respect to both the addition in $\mathbb{F}$ and the addition in V , that is $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$ and $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$.

In other words V is closed under finite linear combinations of vectors with scalar coefficients. Note that $x = 1 \cdot x = (1 + 0) \cdot x = 1 \cdot x + 0 \cdot x$ hence $0 \cdot x = 0$. We suppress the $\cdot$ and $\times$ notation for the multiplication.

A *norm* on the vector space $(V, \mathbb{F})$ is a function $\|\cdot\|: V \rightarrow \mathbb{R}_{\geq 0}$ such that:

- $\|v\| \geq 0$ and $\|v\| = 0$ if and only if $v = 0$.
- $\|\alpha x\| = |\alpha| \|x\|$, $\forall \alpha \in \mathbb{F}$ and $\forall x \in V$
- $\|x + y\| \leq \|x\| + \|y\|$

Any normed vector space is a metric space with distance function $d(x, y) = \|x - y\|$. A *Banach space* is a Cauchy complete normed vector space.

15.1 Properties of Hilbert spaces

Inner product spaces are topological vector spaces with additional structure.

Definition 15.1 (**Inner product, Hilbert space**). An inner product on a vector space V is a function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{F}$

- in the real case $\mathbb{F} = \mathbb{R}$, a positive non degenerate symmetric bilinear form, that is
 - $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.
 - $\langle \alpha x + \beta y, u \rangle = \alpha \langle x, u \rangle + \beta \langle y, u \rangle$ and $\langle u, \alpha x + \beta y \rangle = \alpha \langle u, x \rangle + \beta \langle u, y \rangle$
 - $\langle x, y \rangle = \langle y, x \rangle$

- in the complex case $\mathbb{F} = \mathbb{C}$, a positive non degenerate Hermitian sesquilinear form, that is (denoting by bar the complex conjugation):

- $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.
- $\langle \alpha x + \beta y, u \rangle = \bar{\alpha} \langle x, u \rangle + \bar{\beta} \langle y, u \rangle$ and $\langle u, \alpha x + \beta y \rangle = \alpha \langle u, x \rangle + \beta \langle u, y \rangle$
- $\overline{\langle x, y \rangle} = \langle y, x \rangle$

Any inner product induces a norm $\|x\| = \sqrt{\langle x, x \rangle}$.¹ A Hilbert space is an inner product space which is Cauchy complete (Banach) with respect to distance function induced by this norm.

For example, $\mathbb{R}^D$ and $\mathbb{C}^D$ and $\ell^2 = \{x(n) |_{n \in \mathbb{N}} | \sum_{n \geq 0} |x(n)|^2 < \infty\}$ are Hilbert spaces with inner product:

$$\langle x, y \rangle = \sum_{\mu=1}^D x^\mu y^\mu, \quad \langle z, w \rangle = \sum_{\mu=1}^D \bar{z}^\mu w^\mu, \quad \langle x, y \rangle = \sum_{n=0}^{\infty} \overline{x(n)} y(n),$$

as we already established (Proposition 9.2) that $\ell^2, \|x\|_2 = \sqrt{\langle x, x \rangle}$ is Cauchy complete.

Exercise 15.1. Consider H a real Hilbert space and the function $f : H \rightarrow \mathbb{R}, F(x) = \|x\|$. Prove that for $x \neq 0$, the Fréchet differential $dF|_x$ is the linear functional $\langle \frac{x}{\|x\|}, \cdot \rangle$. We have:

$$\begin{aligned} \frac{\left| \|x+h\| - \|x\| - \left\langle \frac{x}{\|x\|}, h \right\rangle \right|}{\|h\|} &= \frac{\left| \|x\| \|x+h\| - \langle x, x+h \rangle - \langle x, h \rangle \right|}{\|x\| \|h\|} = \frac{\left| \|x\|^2 \|x+h\|^2 - \langle x, x+h \rangle^2 \right|}{\|x\| \|h\| \left| \|x\| \|x+h\| + \langle x, x+h \rangle \right|} = \frac{\left| \langle x, x \rangle \langle x+h, x+h \rangle - \langle x, x+h \rangle^2 \right|}{\|x\| \|h\| \left| \|x\| \|x+h\| + \langle x, x+h \rangle \right|} \\ &= \frac{\left| \langle x, x \rangle^2 + \langle x, x \rangle \langle h, h \rangle + 2\langle x, x \rangle \langle x, h \rangle - \langle x, x \rangle^2 - \langle x, h \rangle^2 - 2\langle x, x \rangle \langle x, h \rangle \right|}{\|x\| \|h\| \left| \|x\| \|x+h\| + \langle x, x+h \rangle \right|} = \frac{\left| \langle x, x \rangle \langle h, h \rangle - \langle x, h \rangle^2 \right|}{\|x\| \|h\| \left| \|x\| \|x+h\| + \langle x, x+h \rangle \right|} \end{aligned}$$

For $\|h\| < \|x\|$, using Cauchy-Schwarz we have that $\langle x, x+h \rangle = \langle x, x \rangle + \langle x, h \rangle \geq \|x\|^2 - \|x\| \|h\|$ and $\|x+h\| > \|x\| - \|h\|$ hence $\|x\| \|x+h\| + \langle x, x+h \rangle \geq 2\|x\|^2 - 2\|x\| \|h\|$, hence:

$$\frac{\left| \|x+h\| - \|x\| - \left\langle \frac{x}{\|x\|}, h \right\rangle \right|}{\|h\|} \leq \frac{2\|x\|^2 \|h\|^2}{\|x\| \|h\| (2\|x\|^2 - 2\|x\| \|h\|)} \xrightarrow{h \rightarrow 0} 0$$

Theorem 15.1 (The Cauchy Schwarz inequality). For all x, y in an inner product space V we have: $|\langle x, y \rangle| \leq \|x\| \|y\|$, with equality reached if and only if x and y are proportional.

Proof. We consider the complex case and we write:

$$0 \leq \|x - \lambda y\|^2 = \langle x - \lambda y, x - \lambda y \rangle = \|x\|^2 + |\lambda|^2 \|y\|^2 - \lambda \langle x, y \rangle - \bar{\lambda} \langle y, x \rangle,$$

and evaluate the last expression for $\lambda = \frac{\langle y, x \rangle}{\|y\|^2}$. □

Theorem 15.2 (The parallelogram law). A norm $\|\cdot\|$ in a normed vector space derives from an inner product $\|v\|^2 = \langle v, v \rangle$ if and only if $\forall x, y \in V$ the Parallelogram law is satisfied:

$$2\|x\|^2 + 2\|y\|^2 = \|x+y\|^2 + \|x-y\|^2.$$

Proof. One direction is straightforward. For the other, we define:

$$\langle x, y \rangle = \frac{\|x+y\|^2 - \|x-y\|^2}{4}, \quad \langle x, y \rangle = \frac{\|x+y\|^2 - \|x-y\|^2}{4} + i \frac{\|ix-y\|^2 - \|ix+y\|^2}{4},$$

and we need to check that these are indeed inner products. We focus on the real case.

¹In order to prove that $\|\cdot\|$ respects the triangle inequality, one uses the Cauchy Schwarz inequality proven below.

$\langle \cdot, \cdot \rangle$ is symmetric non degenerate and positive. Trivially $\langle x, y \rangle = \langle y, x \rangle$ and $\langle x, x \rangle > 0$ for $x \neq 0$.

We have $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$. By the parallelogram law we get:

$$\begin{aligned} 2\|x + z\|^2 + 2\|y\|^2 &= \|x + z + y\|^2 + \|x + z - y\|^2 \Rightarrow \|x + z + y\|^2 = \begin{cases} 2\|x + z\|^2 + 2\|y\|^2 - \|x + z - y\|^2 \\ 2\|y + z\|^2 + 2\|x\|^2 - \|y + z - x\|^2 \end{cases} , \\ \|x + z + y\|^2 &= \|x + z\|^2 + \|y\|^2 - \frac{1}{2}\|x + z - y\|^2 + \|y + z\|^2 + \|x\|^2 - \frac{1}{2}\|y + z - x\|^2 \end{aligned}$$

Where we used the fact that the parallelogram law is symmetric in x and y and we took the average. Doing the same, this time for $-z$ instead of z we get:

$$\begin{aligned} \|x - z + y\|^2 &= \|x - z\|^2 + \|y\|^2 - \frac{1}{2}\|x - z - y\|^2 + \|y - z\|^2 + \|x\|^2 - \frac{1}{2}\|y - z - x\|^2 \\ &= \|x - z\|^2 + \|y\|^2 - \frac{1}{2}\|x - z + y\|^2 + \|y - z\|^2 + \|x\|^2 - \frac{1}{2}\|y - z + x\|^2 , \end{aligned}$$

and finally:

$$\langle x + y, z \rangle = \frac{\|x + y + z\|^2 - \|x + y - z\|^2}{4} = \frac{\|x + z\|^2 - \|x - z\|^2}{4} + \frac{\|y + z\|^2 - \|y - z\|^2}{4} = \langle x, z \rangle + \langle y, z \rangle .$$

$\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ for all $\lambda \in \mathbb{Q}$. For $\lambda = 1$ this is obviously true and using the previous step we prove by induction that:

$$\langle (n+1)x, y \rangle = \langle nx + x, y \rangle = \langle nx, y \rangle + \langle x, y \rangle = n\langle x, y \rangle + \langle x, y \rangle = (n+1)\langle x, y \rangle ,$$

and as $\langle -x, y \rangle = -\langle x, y \rangle$ this extends to $\lambda \in \mathbb{Z}$. For $\lambda = \frac{p}{q}$ with $q \in \mathbb{Z}_{\neq 0}$, with $x' = \frac{1}{q}x$ we get:

$$q\langle \lambda x, y \rangle = q\langle px', y \rangle = p\langle qx', y \rangle = p\langle x, y \rangle ,$$

and we conclude dividing by q that $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ for all $\lambda \in \mathbb{Q}$.

$\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ for all $\lambda \in \mathbb{R}$. Using the parallelogram law we have:

$$\langle x, y \rangle = \frac{\|x + y\|^2 - \|x\|^2 - \|y\|^2}{2} \Rightarrow \|x\|^2 + \|y\|^2 + 2\langle x, y \rangle = \|x + y\|^2$$

As $|\langle x, y \rangle| \leq \|x\| \|y\|$ we conclude that $-2\|x\| \|y\| \leq 2\langle x, y \rangle \leq 2\|x\| \|y\|$ hence $|\langle x, y \rangle| \leq \|x\| \|y\|$. For any $\lambda \in \mathbb{R}$, $r \in \mathbb{Q}$ we have:

$$|\lambda \langle x, y \rangle - \langle \lambda x, y \rangle| = |(\lambda - r)\langle x, y \rangle - \langle (\lambda - r)x, y \rangle| \leq |\lambda - r| |\langle x, y \rangle| + |\langle (\lambda - r)x, y \rangle| \leq |\lambda - r| \|x\| \|y\| + \|(\lambda - r)x\| \|y\| = |\lambda - r| \|x\| \|y\| ,$$

and we conclude using the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$.

□

Any finite dimensional subspace of a Hilbert space is closed.² For infinite dimensional subspaces this is not necessarily true (we will see later that for instance the closure in norm L^2 of the continuous L^2 functions is the entire space L^2).

²This holds for Banach spaces. Consider a sequence $(x_n) \in N$ that converges to some $x \in V$. Then for n large enough $\|x_n\| \leq \|x\| + \epsilon$ hence x_n is a bounded sequence in N . The closed ball of radius $\|x\| + \epsilon$ in N is compact (by Heine Borel, as N is finite dimensional) hence it is sequentially compact and x_n has a limit point inside this ball in N . We conclude by the unicity of the limit.

Theorem 15.3 (**Orthogonal complement**). *Let V a Hilbert space and W a closed subspace, which can be infinite dimensional. We define the orthogonal complement $W^\perp$ as the set of vectors orthogonal to W , that is $W^\perp = \{v \in V \mid \langle v, w \rangle = 0, \forall w \in W\}$. Then $W^\perp$ is a closed subspace of V and any $x \in V$ can be decomposed uniquely as a sum $x = y + z$ with $y \in W$ and $z \in W^\perp$. We denote this $V = W \oplus W^\perp$, and we note that $V^\perp = 0$.*

Proof. As it is closed under the addition of vectors and multiplication by scalars, $W^\perp$ is a subspace. Consider $(v_n)_n$ a sequence in $W^\perp$ converging to some limit $v_n \rightarrow v$. As $\langle v_n, w \rangle = 0$ for all $w \in W$ and the scalar product is continuous in either of its arguments (by Cauchy-Schwarz, $\|\langle v_n - v, w \rangle\| \leq \|v_n - v\| \|w\|$), $\langle v, w \rangle = 0$ for all $w \in W$ hence $W^\perp$ is closed. Observe that if $v \in W \cap W^\perp$, then $\langle v, v \rangle = 0$ and $v = 0$.

The idea is that y is the element of W which is closest to x . Let us define for $w \in W$, $f(w) = \|x - w\|^2$ and let us denote $\delta = \inf_{w \in W} f(w)$ (which exists as f is a positive function). Then there exists a sequence $y_n \in W$ such that $\lim_{n \rightarrow \infty} f(y_n) = \delta$ (as δ is either an element or an accumulation point of $f(W)$). As $\frac{y_n + y_m}{2} \in W$, the parallelogram identity implies:

$$\|(x - y_n) + (x - y_m)\|^2 + \|(x - y_n) - (x - y_m)\|^2 = 2\|x - y_n\|^2 + 2\|x - y_m\|^2 \Rightarrow 4f\left(\frac{y_n + y_m}{2}\right) + \|y_n - y_m\|^2 = 2f(y_n) + 2f(y_m),$$

therefore $\|y_n - y_m\|^2 \leq 2f(y_n) + 2f(y_m) - 4\delta$, and taking into account that $\lim_m f(y_m) = \lim_n f(y_n) = \delta$, y_n is a Cauchy sequence ($\forall \epsilon, \exists N_\epsilon$ such that $n, m > N_\epsilon$, $\|y_n - y_m\| < \epsilon$) hence converges in V to some y , and $y \in W$ as W is closed.

Now take any $w \in W$. The function:

$$g(t) = f(y + tw) = \langle x - (y + tw), x - (y + tw) \rangle = \|x - y\|^2 + t(\langle x - y, w \rangle + \langle w, x - y \rangle) + t^2\|w\|^2,$$

has a minimum at $t = 0$, hence $g'(0) = 0$ thus $\langle x - y, w \rangle + \langle w, x - y \rangle = 0$. The same holds for $\tilde{g}(t) = g(it)$, hence $\langle x - y, w \rangle - \langle w, x - y \rangle = 0$ and we conclude that $\langle x - y, w \rangle = 0$. As this holds for all $w \in W$, $x - y \in W^\perp$.

Finally, in order to prove that the decomposition is unique, assume two decompositions exist $x = y_1 + z_1 = y_2 + z_2$. Then $y_1 - y_2 = z_2 - z_1 \in W \cap W^\perp = \{0\}$, we conclude. $\square$

Construction 15.1 (**Gram-Schmidt process**). *We recall that the span of a set of vectors is the (vector space) of finite linear combinations of those vectors.*

Consider a countable subset $(x_n)_{n \in \mathbb{N}}$ in a Hilbert space, with $x_n \neq 0$. One can construct the orthonormal system $(e_k)_{k \in \mathbb{N}}$ with the same span as $(x_n)_{n \in \mathbb{N}}$ by taking the following process:

$$e_0 := \frac{1}{\|x_0\|} x_0, \text{ set } k = n = 0 \text{ and } \forall n \begin{cases} \text{if } x_n \notin \text{Span}\{e_0, \dots, e_k\} \text{ then } e_{k+1} := \frac{x_n - \sum_{j=0}^k \langle e_j, x_n \rangle e_j}{\|x_n - \sum_{j=0}^k \langle e_j, x_n \rangle e_j\|}, & k \rightarrow k+1, \quad n \rightarrow n+1 \\ \text{else } n \rightarrow n+1 \end{cases}.$$

It is easy to check that $\langle e_k, e_l \rangle = \delta_{kl}$ and clearly $(e_n)_{n \in \mathbb{N}}$ has at most countably many elements.

Theorem 15.4 (**A separable Hilbert space has a Hilbert basis**). *A Hilbert space is separable (i.e. it has a countable dense subset³) if and only if it admits a countable orthonormal Hilbert basis (e_n) , that is an orthonormal sequence $(e_n)_{n \in \mathbb{N}}$ such that $\forall x \in V$:*

$$x = \lim_{N \rightarrow \infty} \sum_{k=0}^N \langle e_k, x \rangle e_k \equiv \sum_{k=0}^{\infty} \langle e_k, x \rangle e_k,$$

where the equal sign in this equation means convergence in norm and the series is absolutely convergent (in particular the order of summation does not matter). It follows that every separable Hilbert space is isomorphic with ℓ^2 .

A countable orthonormal set $(e_n)_{n \in \mathbb{N}}$ in a Hilbert space is a Hilbert basis if and only if $\forall n, \langle e_n, v \rangle = 0 \Leftrightarrow v = 0$.

³Recall that a X set is dense in V if any open in V contains elements in X , or equivalently any element in $x \in V$ is the limit of some subsequence of x_n .

Proof. One implication follows from Theorem 9.1 as V is a Banach space with a Schauder basis.

For the reciprocal, assume V has a countable dense subset $(x_n)_{n \in \mathbb{N}}$. We build the orthonormal set $(e_n)_{n \in \mathbb{N}}$ using the Gram-Schmidt process. We have $x_n \in \text{Span}\{e_0, \dots, e_n\} \Rightarrow x_n = \sum_{k=0}^n \beta_k e_k$ and by orthonormality $\langle e_k, x_n \rangle = \sum_{k'=0}^n \beta_k \delta_{k'k}$, hence:

$$x_n = \sum_{k=0}^n \langle e_k, x_n \rangle e_k.$$

As $(e_n)_{n \in \mathbb{N}}$ is an orthonormal system, for any $x \in V$ and any finite $I \subset \mathbb{N}$, we have Bessel's inequality:

$$0 \leq \|x - \sum_{k \in I} \langle e_k, x \rangle e_k\|^2 = \|x\|^2 - 2 \sum_{k \in I} \langle e_k, x \rangle \langle x, e_k \rangle + \sum_{k \in I} |\langle e_k, x \rangle|^2 = \|x\|^2 - \sum_{k \in I} |\langle e_k, x \rangle|^2.$$

It follows that $s_N = \sum_{k=0}^N |\langle e_k, x \rangle|^2$ is monotonic and bounded in $\mathbb{R}$ hence convergent and $\lim_{N \rightarrow \infty} s_N \leq \|x\|^2$ and consequently I above can also be taken infinite. In particular, $y_N = \sum_{k=0}^N \langle e_k, x \rangle e_k$ is a Cauchy sequence in V :

$$\|y_M - y_N\|^2 = \left\| \sum_{k=N+1}^M \langle e_k, x \rangle e_k \right\|^2 = \sum_{k=N+1}^M |\langle e_k, x \rangle|^2 = s_M - s_N,$$

hence converges to some limit. As $\forall I \subset \mathbb{N}$ we have $\sum_{k \in I} |\langle e_k, x \rangle|^2 \leq \|x\|^2$, the series is absolutely convergent and the order of the summation does not matter.

In order to conclude, it is enough to prove that $y_N \rightarrow x$.

As $(x_n)_{n \in \mathbb{N}}$ is dense, $\forall x \in V$ there exists a subsequence $x_{n_i} \rightarrow x$, hence $\forall \epsilon$ and $\forall n$ there exists some $N' > n$ such that $x_{N'} \in B_{\epsilon/3}(x)$. On the other hand, as y_N is Cauchy we have that $\forall \epsilon$ there exists N_ϵ such that $\|y_n - y_l\| < \frac{\epsilon}{3}$ for $k, l > N_\epsilon$. Then for $n > N_\epsilon$:

$$\begin{aligned} \|x - y_n\| &= \|x - x_{N'} + x_{N'} - y_{N'} + y_{N'} - y_n\| = \|x - x_{N'} + \sum_{k=0}^{N'} \langle e_k, x_{N'} - x \rangle e_k + y_{N'} - y_n\| \\ &\leq \|x - x_{N'}\| + \left\| \sum_{k=0}^{N'} \langle e_k, x_{N'} - x \rangle e_k \right\| + \|y_{N'} - y_n\| \leq \|x - x_{N'}\| + \sqrt{\sum_{k=0}^{N'} |\langle e_k, x_{N'} - x \rangle|^2} + \|y_{N'} - y_n\| \\ &\leq 2\|x - x_{N'}\| + \|y_{N'} - y_n\| \leq \epsilon. \end{aligned}$$

where we used Bessel's inequality applied to $x_{N'} - x$.

For the last point, if there exist x such that $\langle e_n, x \rangle = 0$ for all n , then $x \in \overline{\text{Span}(e_0, \dots)}^\perp$ and $\overline{\text{Span}(e_0, \dots)} = V$ if and only if $e_0, \dots$ is a Hilbert basis. $\square$

Remark that by Cauchy-Schwarz the scalar product is continuous in either of its arguments hence:

$$\|x\|^2 = \langle x, \sum_{n \geq 0} \langle e_n, x \rangle e_n \rangle = \sum_{n \geq 0} |\langle e_n, x \rangle|^2.$$

Let us go back to the space of sequences $\ell^2 = \{x(n) |_{n \in \mathbb{N}_{>0}} \mid \sum_{n \geq 1} |x(n)|^2 < \infty\}$. A Hilbert basis in this space is given by the sequences e_n which are zero, except for the n 'th element which is 1, $e_1 = (1, 0, 0, \dots)$, $e_2 = (0, 1, 0, \dots)$ and so on, as $\langle e_n, e_m \rangle = \delta_{nm}$ and any x writes as $x = \sum_{n \geq 1} x(n) e_n$ with $x(n) = \langle e_n, x \rangle$.

Give two Hilbert spaces H_1 and H_2 , the algebraic tensor product of H_1 and H_2 is the linear span (that is space of finite linear combinations) of $\phi_1 \otimes \phi_2$ with $\phi_1 \in H_1$ and $\phi_2 \in H_2$. The Hilbert space tensor product $H_1 \otimes H_2$ is the closure of the algebraic tensor product $H_1 \otimes H_2 = \overline{\text{Span}\{\phi_1 \otimes \phi_2 \mid \phi_1 \in H_1, \phi_2 \in H_2\}}$ in the norm $\langle \phi_1 \otimes \phi_2, \psi_1 \otimes \psi_2 \rangle = \langle \phi_1, \psi_1 \rangle \langle \phi_2, \psi_2 \rangle$. We have that $H_1 \otimes H_2$ is a Hilbert space and given Hilbert bases $\{e_i\}$ in H_1 and $\{f_j\}$ in H_2 , $\{e_i \otimes f_j\}_{i,j}$ is a Hilbert basis in $H_1 \otimes H_2$.

15.2 Operators on Hilbert spaces

Recall that a linear operator $A : V \rightarrow Y$ from a normed space to a normed space $A(\alpha x + \beta y) = \alpha A(x) + \beta A(y)$ is a continuous map if and only if there exists some $c \in \mathbb{R}_{\geq 0}$ such that $\|Av\| \leq c\|v\|$ for all $v \in V$, that is the operator is *bounded*. Indeed, if $\|A(v) - A(w)\| = \|A(v - w)\| \leq c\|v - w\|$ the A is continuous at any v . Conversely, as $A(0) = A(x - x) = A(x) - A(x) = 0$, if A is continuous it is continuous at 0 and there exists δ such that $\|x\| < \delta \Rightarrow \|Ax\| \leq 1$. Then $\forall x, \|Ax\| = \left\| \frac{\|x\|}{\delta} A\left(\frac{x}{\|x\|}\delta\right) \right\| \leq \frac{1}{\delta}\|x\|$.

The *operator norm* of A is

$$\|A\|_{\text{op.}} = \inf \left\{ c \geq 0 \mid \|Av\| \leq c\|v\|, \forall v \in V \right\} = \sup_{v \in V} \frac{\|Av\|}{\|v\|} = \sup_{\|v\|=1} \|Av\|.$$

15.2.1 Linear forms and the dual space

Definition 15.2 (**The continuous dual**). A linear form or linear functional on the vector space V over $\mathbb{F}$ is a linear function $\varphi : V \rightarrow \mathbb{F}$, $\varphi(\alpha x + \beta y) = \alpha\varphi(x) + \beta\varphi(y)$, for $x, y \in V$, $\alpha, \beta \in \mathbb{F}$

The dual space V^* is the vector space of linear functionals on V endowed with the vector space operations:

- $(\varphi_1 + \varphi_2)(x) = \varphi_1(x) + \varphi_2(x)$, $\varphi_1, \varphi_2 \in V^*$, $x \in V$,
- $(\alpha \cdot \varphi)(x) = \alpha \cdot \varphi(x)$, $\varphi \in V^*$, $\alpha \in \mathbb{F}$, $x \in V$.

The elements of the dual are also called covectors.

In infinite dimensions, the continuous dual of a normed space $V, \|\cdot\|$ is the subset $V' = \{\varphi \in V^* \mid \|\varphi\|_{\text{op.}} < \infty\} \subset V^*$. The dual space V^* is almost useless in infinite dimensions: it is usually called algebraic dual and one usually calls dual the continuous dual.

There exists a natural homomorphic injection of V into $V^{**} = \{\Phi : V^* \rightarrow \mathbb{F} \mid \Phi \text{ linear}\}$ (which can be promoted to an isomorphic bijection in finite dimension) $v \rightarrow \Phi(v)$ defined by $(\Phi(v))(\varphi) = \varphi(v)$, $\forall v \in V$ and $\forall \varphi \in V^*$.

In infinite dimension there is no useful notion of dual basis: already the Hamel basis are quite useless. However, any bilinear (sesquilinear in the complex case) form on $g(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}, \mathbb{C}$ will give a mapping of V into its dual $v \mapsto g(v, \cdot)$. In particular, for an infinite dimensional Hilbert space, the inner product gives an isomorphism between V and its continuous dual V' , by the Riesz-Fréchet representation theorem.

Theorem 15.5 (**Riesz-Fréchet representation theorem**). Let V be a Hilbert space. For every bounded linear functional $\varphi \in V'$, there exists a unique element $v_\varphi \in V$ such that $\varphi(x) = \langle v_\varphi, x \rangle$ for all $x \in V$ and furthermore v_φ is the unique vector in $(\ker \varphi)^\perp$ with $\|\varphi\|_{\text{op.}} = \|v_\varphi\|$.

Proof. As φ is bounded (hence continuous), $\ker \varphi$ is a closed subspace of V and $V = \ker \varphi \oplus (\ker \varphi)^\perp$. If $(\ker \varphi)^\perp = \{0\}$, then $\varphi = 0$ and $v_\varphi = 0$. Else, there exists $v \in (\ker \varphi)^\perp$, $v \neq 0$, and we can normalize it to $\|v\| = 1$. Chose $v_\varphi = \overline{\varphi(v)} v \in (\ker \varphi)^\perp$. We have $\|v_\varphi\| = |\varphi(v)|$ and $\varphi(v_\varphi) = |\varphi(v)|^2$ and for all $x \in V$:

$$\langle v_\varphi, x \rangle = \langle v_\varphi, x - \frac{\varphi(x)}{\|v_\varphi\|^2} v_\varphi \rangle + \langle v_\varphi, \frac{\varphi(x)}{\|v_\varphi\|^2} v_\varphi \rangle = 0 + \frac{\varphi(x)}{\|v_\varphi\|^2} \|v_\varphi\|^2 = \varphi(x),$$

where the first term is zero because $\varphi(x - \frac{\varphi(x)}{\|v_\varphi\|^2} v_\varphi) = \varphi(x) - \frac{\varphi(x)}{\|v_\varphi\|^2} \varphi(v_\varphi) = 0$, that is $x - \frac{\varphi(x)}{\|v_\varphi\|^2} v_\varphi \in \ker(\varphi)$, while $v_\varphi \in (\ker \varphi)^\perp$.

In order to prove unicity, consider two such vectors v_φ and v'_φ . Then:

$$\langle v_\varphi - v'_\varphi, v_\varphi - v'_\varphi \rangle = \langle v_\varphi, v_\varphi - v'_\varphi \rangle - \langle v'_\varphi, v_\varphi - v'_\varphi \rangle = \varphi(v_\varphi - v'_\varphi) - \varphi(v_\varphi - v'_\varphi) = 0.$$

Finally, from Cauchy-Schwarz inequality we have $|\varphi(x)| = \langle v_\varphi, x \rangle \leq \|v_\varphi\| \|x\|$, and the inequality is saturated if x and v_φ are proportional. $\square$

An example of (continuous) dual spaces in infinite dimensions are ℓ^p and ℓ^q with $1/p + 1/q = 1$. Denoting $e_n \in \ell^p$, $(e_n)_j = \delta_{nj}$ the standard Schauder basis in ℓ^p , we associate to any $\varphi \in (\ell^p)'$ the sequence $\varphi(e_n) \in \ell^q$. Conversely, given $(a_n) \in \ell^q$, the corresponding linear functional on ℓ^p is $\varphi(x) = \sum_n a_n x_n$, and by Hölder inequality $\|\varphi(x)\| \leq \|a\|_q \|x\|_p$.

15.2.2 Operators from V to V

The space of bounded linear operators on a Hilbert space, $B(V) = \{A : V \rightarrow V, \|A\|_{\text{op.}} < \infty\}$ is a Banach space. For any Cauchy sequence of bounded linear operators A_n and for any v we have $\|A_n v - A_k v\| \leq \|A_n - A_k\|_{\text{op.}} \|v\|$ consequently the sequence $A_n v$ at fixed v is Cauchy hence convergent as V is Cauchy complete. We denote the pointwise limit $Av = \lim_{n \rightarrow \infty} A_n v$. As A_n is Cauchy in operator norm, for any ϵ we have for n and k large enough $\|A_n v - A_k v\| \leq \epsilon \|v\|$ for any v . As any norm is continuous, $\|A_n v - Av\| \leq \epsilon \|v\|$ for all v , hence on the one hand $\|Av\| \leq (\|A_n\|_{\text{op.}} + \epsilon) \|v\|$ and A is bounded, and on the other $\|A_n - A\|_{\text{op.}} \leq \epsilon$.

A simple class of bounded linear operators on Hilbert (Banach) spaces consists in the compact operators. Such operators are the closest one can come to matrices in infinite dimensions.

Definition 15.3 (**Compact operator**). *A bounded linear operator T on a Hilbert (Banach) space is called compact if for any bounded set $B \in V$, $\overline{T(B)}$ is a compact set. Equivalently (from theorem 4.4), a bounded linear operator $T : V \rightarrow V$ is compact if and only if any bounded sequence $(x_n)_{n \in \mathbb{N}} \in V$ has a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ such that $T x_{n_k}$ converges.*

A large class of operators in Hilbert (Banach) spaces are compact.

Proposition 15.1 (**Examples of compact operators**). *Let V be a Hilbert (Banach) space. Then:*

- *a bounded operator $P : V \rightarrow V$ is called finite rank if its image is a finite dimensional vector space. Finite rank operator are compact.*
- *if T_n is a sequence of compact operators and $T_n \xrightarrow{\|\cdot\|_{\text{op}}} T$ then T is a compact operator.*

Proof. For the first point, as P is bounded, for any bounded sequence x_n , $P(x_n)$ is a bounded sequence. As $\text{Im}(P)$ is a finite dimensional vector space, by the Bolzano Weierstrass theorem $P(x_n)$ admits a convergent subsequence.

For the second point, consider T_m compact operators with $\|T_m - T\|_{\text{op.}} \rightarrow 0$. For any sequence x_n with $\|x_n\| < 1$, on the one hand $T_m(x_n)$ (fixed m) has a Cauchy (convergent) subsequence. On the other hand, $\|T_m(x_n) - T(x_n)\| \leq \|T_m - T\|_{\text{op.}}$, hence $T_m(x_n)$ (fixed n) converges to $T(x_n)$ uniformly in n .

We will now select a Cauchy subsequence of $T(x_n)$. We will do this by a diagonal trick. Let $x_{n_{(1,i)}}$ be a subsequence of x_n such that $T_1(x_{n_{(1,i)}})$ is Cauchy. Iteratively, chose $x_{n_{(k+1,i)}}$ a subsequence of $x_{n_{(k,i)}}$ such that $T_{k+1}(x_{n_{(k+1,i)}})$ is Cauchy:

$$\begin{array}{cccc} T_1 : & x_{n_{(1,1)}} & x_{n_{(1,2)}} & x_{n_{(1,3)}} & \dots \\ T_2 : & x_{n_{(2,1)}} & x_{n_{(2,2)}} & x_{n_{(2,3)}} & \dots \\ T_3 : & x_{n_{(3,1)}} & x_{n_{(3,2)}} & x_{n_{(3,3)}} & \dots \\ & \vdots & & & \end{array}$$

Now take $x_{n_i} = x_{n_{(i,i)}}$: by construction $T_m(x_{n_i})$ is Cauchy for every m as for $i \geq m$, $x_{n_{(i,i)}}$ is a subsequence of $x_{n_{(m,i)}}$. Taking into account that:

$$\|T(x_{n_i}) - T(x_{n_j})\| \leq \|T(x_{n_i}) - T_m(x_{n_i}) + T_m(x_{n_i}) - T_m(x_{n_j}) + T_m(x_{n_j}) - T(x_{n_j})\| \leq 2\|T - T_m\|_{\text{op.}} + \|T_m(x_{n_i}) - T_m(x_{n_j})\|,$$

for every ϵ choosing m sufficiently large such that $\|T - T_m\|_{\text{op.}} \leq \epsilon/3$, there exists N_ϵ such that $i, j > N_\epsilon$, $\|T_m(x_{n_i}) - T_m(x_{n_j})\| < \epsilon/3$ hence $T(x_{n_i})$ is Cauchy. $\square$

An example of a finite rank operator is the *projector on v* . For $v \in V$, with $\|v\| = 1$, the projector on v is $P_v(x) = \langle v, x \rangle v$ for all $x \in V$.

The operator $T : \ell^2 \rightarrow \ell^2$ with $T(\sum_{n \geq 1} x(n)e_n) = \sum_{n \geq 1} \lambda_n x(n)e_n$ with $\lim_{n \rightarrow \infty} \lambda_n = 0$ is the limit in operator norm of its finite rank truncations $T_N(x) = \sum_{n=1}^N \lambda_n x(n)e_n$, as $\|T_N(x) - T(x)\|^2 = \sum_{n > N} |\lambda_n|^2 |x(n)|^2 \leq \sup_{n' > N} |\lambda_{n'}|^2 \|x\|^2$, hence it is compact.

However, in infinite dimensions, there exist many interesting operators which are not compact. For instance, the identity operator is not compact. To see this, consider the sequence $\{e_n\}$ consisting in the basis vectors in a Hilbert basis. This sequence has no Cauchy subsequence: for any subsequence $(e_{n_i})_{i \in \mathbb{N}}$, if $i \neq j$ then $\|e_{n_i} - e_{n_j}\| = \sqrt{2}$. In particular the identity is not the limit in operator norm of any sequence of finite rank operators. When studying operators on Hilbert spaces one sometimes considers weaker notion of convergence than convergence in operator norm. We say that a sequence of linear operators $A_n : V \rightarrow V$ converges to A strongly respectively weakly if:⁴

$$\forall v \in V, \quad \lim_{n \rightarrow \infty} A_n v = A v, \quad \forall v_1, v_2 \in V, \quad \lim_{n \rightarrow \infty} \langle v_1 | A_n v_2 \rangle \rightarrow \langle v_1 | A v_2 \rangle.$$

For instance, considering a Hilbert basis $\{e_n\}$, the identity is the strong limit of the sequence of finite rank operators $I_N(v) = \sum_{n=1}^N \langle e_n, v \rangle e_n$: for any $v \in V$, by the definition of a Hilbert basis, $\lim_{N \rightarrow \infty} I_N(v) = v$.

A crucial role in physics is played by the self-adjoint operators.

Definition 15.4 (*Self-adjoint operator, the bounded case*). Let $A : V \rightarrow V$ be a bounded linear operator on a Hilbert space. The adjoint of A is the operator $A^\dagger$ such that $\langle y, Ax \rangle = \langle A^\dagger y, x \rangle$ for all $x, y \in V$.

$A^\dagger$ always exists. The functional $\langle y, Ax \rangle$ at fixed y is a bounded linear functional in $x \in V$ as $|\langle y, Ax \rangle| \leq \|y\| \|A\|_{\text{op}} \|x\|$. From Riesz-Fréchet, there exists z_y such that $\langle y, Ax \rangle = \langle z_y, x \rangle$ and we define $A^\dagger(y) = z_y$. As $\|A^\dagger(y)\| = \|z_y\| = \|\langle y, A \cdot \rangle\|_{\text{op}} = \sup_{x, \|x\|=1} |\langle y, Ax \rangle| \leq \|A\| \|y\|$, we have $\|A^\dagger\|_{\text{op}} \leq \|A\|_{\text{op}}$. A bounded operator is called self-adjoint if $A = A^\dagger$.

An orthogonal projector is a bounded self-adjoint idempotent operator $P^\dagger = P$, $P^2 = P$. As $(1 - P)P = 0$, it follows that $\text{Im}(P) \subset \ker(1 - P)$ and as $v \in \ker(1 - P) \Rightarrow v = Pv \in \text{Im}(P)$ we have $\text{Im}(P) = \ker(1 - P)$, which is a closed subspace as $1 - P$ is bounded. For $x \in \text{Im}(P)$ and $y \in \ker(P)$ we have $\langle x, y \rangle = \langle Pv, y \rangle = \langle v, Py \rangle = 0$ hence $\text{Im}(P)^\perp = \ker P$ and $V = \text{Im}(P) \oplus \text{Im}(P)^\perp = \text{Im}(P) \oplus \ker P = \text{Im}(P) \oplus \text{Im}(1 - P)$.

On top of the bounded linear operators, one needs to deal also with unbounded ones. An *unbounded* linear operator $A : V \rightarrow V$ with V a Hilbert space is a pair $(A, D(A))$ with $D(A)$ a linear subspace of X and A a linear operator from $D(A)$ to V . We call $D(A)$ the *domain* of A and if $D(A)$ is dense in V then $\langle w, v \rangle = 0, \forall w \in D(A) \Rightarrow v = 0$.⁵ An unbounded operator is such that there exists a sequence $x_n \in D(A)$ with $\|x_n\| = 1$ such that $\|Ax_n\| \rightarrow \infty$.

Definition 15.5 (*Self-adjoint operator, the unbounded case*). Consider A an unbounded operator with dense domain on a Hilbert space V , that is $A : D(A) \rightarrow V$ with $\overline{D(A)} = V$. We denote $D(A^\dagger)$ the set of all $y \in V$ such that $\langle y, Ax \rangle$ at fixed y is a bounded linear functional in $x \in D(A)$. This set is non empty as $0 \in D(A^\dagger)$.

The bounded linear functional $\langle y, Ax \rangle$ at fixed y , originally defined for $x \in D(A)$, can be extended to a bounded linear functional in $x \in V$.⁶ Then from Riesz-Fréchet there exists z_y such that $\langle y, Ax \rangle = \langle z_y, x \rangle$, and z_y is unique as $D(A)$ is dense. We define the adjoint $A^\dagger y = z_y$, and the domain of the adjoint is $D(A^\dagger)$.

An operator is called symmetric if $\langle y, Ax \rangle = \langle Ay, x \rangle$ for all $x, y \in D(A)$. If A is symmetric, then $D(A) \subseteq D(A^\dagger)$. Indeed, for any $y \in D(A)$, $|\langle y, Ax \rangle| = |\langle Ay, x \rangle| \leq \|x\| \|Ay\|$ that is the linear functional $\langle y, Ax \rangle$ at fixed y is bounded on $D(A)$ (by $\|Ay\|$) hence $y \in D(A^\dagger)$.

An unbounded operator is called self-adjoint if A is symmetric and $D(A) = D(A^\dagger)$. A is called essentially self-adjoint if there exist a unique self-adjoint operator which coincides with A on $D(A)$.

⁴These two notions of convergence correspond to non metric topologies on the space of operators.

⁵The function $\langle w, v \rangle$ is continuous in w and zero on a dense subset hence it is zero on V .

⁶As $D(A)$ is dense, this follows from Proposition 17.3. More generally, one can extend a bounded linear functional even if its domain is not dense by the Hahn-Banach continuous extension theorem.

Definition 15.6 (Closed operator). An operator A is called closed if for any $x_n \in D(A)$, $x_n \rightarrow x$ and $Ax_n \rightarrow y$ then $x \in D(A)$ and $Ax = y$. Equivalently, A is closed if $D(A)$ is Cauchy complete with respect to the norm $\sqrt{\|x\|^2 + \|Ax\|^2}$. The closure of an operator is the smallest closed linear extension of the operator. A symmetric operator with dense domain always admits a closure. An operator is closed if $A - zI$ admits a bounded inverse for at least one value $z \in \mathbb{C}$. Most operators one encounters are closable.⁷

For instance, consider the operator $A : \ell^2 \rightarrow \ell^2$, $Ax = \sum_{n \geq 1} n \langle e_n, x \rangle e_n = \sum_{n \geq 1} n x(n) e_n$. This operator is unbounded as for any $c > 0$, if $n > c$, $\|Ae_n\| \geq c\|e_n\|$. Its (maximal) domain consists in the sequences x that go to zero sufficiently fast $D(A) = \{\sum_n n^2 |x(n)|^2 < \infty\}$. $D(A)$ is dense in ℓ^2 as the finite truncations $x_N = \sum_{n=1}^N x(n) e_n$ of any $x \in \ell^2$ are in $D(A)$ and $\|x_N - x\|_{N \rightarrow \infty} \rightarrow 0$. Any basis vector e_n belongs to $D(A^\dagger)$, as $|\langle e_n, Ax \rangle| = n |\langle e_n, x \rangle| \leq n \|x\|$. A is symmetric as by Cauchy-Schwarz, for $x, y \in D(A)$ we have $|\langle y, Ax \rangle| \leq \|y\| \|Ax\|$ hence $\sum_n n \bar{y}(n) x(n)$ converges absolutely and $\langle y, Ax \rangle = \langle Ay, x \rangle$, that is A is symmetric. In fact we will see later that A is self-adjoint, as it is symmetric with real spectrum.

There are two issues at play here: the unboundedness versus the boundedness of the operators and the domain of definition. Bounded densely defined operators can be extended to bounded globally defined operators, hence the unbounded operators are unbounded when restricted to their domain of definition.

15.3 Spectral theory

In physics notation, the scalar product is a bracket consisting on a bra (covector) acting on a ket (a vector) $\langle x, y \rangle \equiv \langle x|y \rangle = \langle x| |y \rangle$. A matrix element is $\langle x|A|y \rangle = \langle x, Ay \rangle = \langle A^\dagger x, y \rangle$. The object $|x\rangle \langle y|$ acts on (vectors) kets and produces kets: $(|x\rangle \langle y|) |z\rangle = |x\rangle \langle y|z\rangle$, hence is an operator. Given a Hilbert basis, $|v\rangle = \sum_n |e_n\rangle \langle e_n|v\rangle$ hence the identity operator writes $1 = \sum_n |e_n\rangle \langle e_n|$ and any operator is $A = \sum_{m,n} |e_n\rangle \langle e_n|A|e_m\rangle \langle e_m|$.

A pair (λ, v) with $\lambda \in \mathbb{C}$ and $v \in V$, $\|v\| = 1$ is an eigenpair of an operator if $Av = \lambda v$. We call λ an eigenvalue of A and v an eigenvector.⁸ The eigenvalues of a self-adjoint operator are real and the eigenvectors corresponding to distinct eigenvalues are orthogonal:

- let (λ, v) an eigenpair, then $\bar{\lambda} \langle v, v \rangle = \langle Av, v \rangle = \langle v, Av \rangle = \lambda \langle v, v \rangle$.
- let $(\lambda_1, v_1), (\lambda_2, v_2)$ two eigenpairs, then $0 = \langle Av_1, v_2 \rangle - \langle v_1, Av_2 \rangle = (\bar{\lambda}_1 - \lambda_2) \langle v_1, v_2 \rangle$.

Proposition 15.2. Let A be a self-adjoint operator A on a finite dimensional Hilbert space V . Then V admits an orthonormal basis consisting in eigenvectors of A .

Proof. Consider the characteristic polynomial $\det(A - \lambda I) = 0$, and denote λ one of its roots. Then the linear system of equations $(A - \lambda)v = 0$ admits a nontrivial solution $v \neq 0$, $Av = \lambda v$, that is there exists an eigenpair (λ, v) . We denote $v^\perp = \{w \in V \mid \langle w, v \rangle = 0\}$ the orthogonal complement of v . Then $\langle v, Av^\perp \rangle = \langle Av, v^\perp \rangle = \bar{\lambda} \langle v, v^\perp \rangle = 0$ and we proceed by induction. $\square$

In infinite dimensions the situation is more delicate. A first class of operators with a relatively simple structure are the compact self-adjoint operators. Such operators play an important role in the theory of differential equations as integral operators are usually compact and self-adjoint.

Theorem 15.6 (Spectral theorem for compact self-adjoint operators). For every compact self-adjoint operator T on a real or complex separable Hilbert space V , there exists a Hilbert basis f_n of V consisting of eigenvectors of T , $Tf_n = \lambda_n f_n$

⁷Using some notions from subsequent chapters, $C[0, 1]$ is dense in $L^2[0, 1]$ in norm L^2 , $T : C[0, 1] \rightarrow \mathbb{R}$, $T(f) = f(0)$ is a linear operator. The sequence of functions $f_n(t) = 1 - nt$ for $t \in [0, 1/n]$ and zero otherwise is such that $f_n \rightarrow 0$ in norm L^2 and $T(f_n) = 1 \rightarrow 1$ in $\mathbb{R}$ but $T(0) = 0$ hence T is not closed.

⁸Note that eigenvectors are defined up to a global phase.

whose corresponding eigenvalues $\lambda_n \in \mathbb{R}$ are ordered $|\lambda_1| > |\lambda_2| > \dots$ such that $\lambda_n \rightarrow 0$. In physics notation:

$$1 = \sum_{n \in \mathbb{N}} |f_n\rangle \langle f_n|, \quad T|f_n\rangle = \lambda_n |f_n\rangle, \quad \lambda_n \rightarrow 0.$$

The eigenspaces corresponding to non zero eigenvalues are finite dimensional.

The set of compact self-adjoint operators is the closure in norm of the finite rank self-adjoint operators.

Proof. As T is compact (hence bounded) and self-adjoint we have that the eigenvalues λ of T are real, the eigenspace corresponding to the eigenvalue λ , $N_\lambda = \ker(T - \lambda I)$, is a closed subspace of V (by continuity of T) and $N_{\lambda_1} \perp N_{\lambda_2}$ if $\lambda_1 \neq \lambda_2$. If $v \in N_\lambda^\perp$ and $w \in N_\lambda$, then $\langle T(v), w \rangle = \langle v, Tw \rangle = \lambda \langle v, w \rangle = 0$, that is $T(N_\lambda^\perp) \subset N_\lambda^\perp$.

We have that the restriction $T|_{N_\lambda}$ is compact. Indeed, any sequence $x_n \in N_\lambda$ with $\|x_n\| \leq 1$ is a bounded sequence in V hence it has a subsequence x_{n_i} such that $T(x_{n_i})$ converges. In particular, as the identity in infinite dimensions is not a compact operator, and $T|_{N_\lambda} = \lambda I|_{N_\lambda}$, then we conclude that for $\lambda \neq 0$, N_λ must be finite dimensional. The eigenspace corresponding to the zero eigenvalue, $N_0 = \ker T$ can be infinite dimensional.

For any bounded self adjoint operator $\|T\| = \sup_{\|x\|=1} |\langle Tx, x \rangle|$. Indeed, denoting $M = \sup_{\|x\|=1} |\langle Tx, x \rangle|$ we have $M \leq \|T\|$ and on the other hand $\|v\| = \sup_{\|y\|=1} |\operatorname{Re}\langle v, y \rangle|$ (from Cauchy Schwarz we have the inequality, and setting $y = v/\|v\|$ we conclude), therefore (note that we use explicitly $\langle Ty, x \rangle = \langle y, Tx \rangle = \overline{\langle Tx, y \rangle}$):

$$\|T\| = \sup_{\|x\|=1} \|Tx\| = \sup_{\|x\|=1, \|y\|=1} |\operatorname{Re}\langle Tx, y \rangle| = \sup_{\|x\|=1, \|y\|=1} \left| \frac{\langle T(x+y), x+y \rangle - \langle T(x-y), x-y \rangle}{4} \right| \leq \sup_{\|x\|=1, \|y\|=1} \frac{1}{4} M(\|x+y\|^2 + \|x-y\|^2),$$

and using the parallelogram law we get $\|T\| \leq \frac{1}{2} M(\|x\|^2 + \|y\|^2) = M$.

Now, if T is compact and self-adjoint then either $\|T\|$ or $-\|T\|$ is an eigenvalue of T . Indeed, pick a sequence x_n with $\|x_n\| = 1$ such that $|\langle Tx_n, x_n \rangle| \rightarrow \|T\|$ (such a sequence exists, as the supremum of a set is an accumulation point hence there must exist a sequence of elements in the set with limit the supremum), that is either $\langle Tx_n, x_n \rangle \rightarrow \|T\|$ or $\langle Tx_n, x_n \rangle \rightarrow -\|T\|$ (note that $\langle Tx, x \rangle$ is real as $\langle Tx, x \rangle = \langle x, Tx \rangle = \overline{\langle Tx, x \rangle}$). Denote the limit $M = \pm\|T\|$. Then :

$$0 \leq \|Tx_n - Mx_n\|^2 = \|Tx_n\|^2 + M^2\|x_n\|^2 - 2M\langle Tx_n, x_n \rangle \leq 2M^2 - 2M\langle Tx_n, x_n \rangle \xrightarrow{n \rightarrow \infty} 0.$$

As T is compact there exists a subsequence (x_{n_j}) such that $T(x_{n_j}) \xrightarrow{j \rightarrow \infty} y$ for some y . As $\|Tx_n - Mx_n\|^2 \rightarrow 0$, we have $Mx_{n_j} \rightarrow y$, $y \neq 0$ as $M \neq 0$ hence $x_{n_j} \rightarrow \frac{1}{M}y$ and $Ty = My$.

Let $\lambda_1 = \|T\|$. Then either λ_1 or $-\lambda_1$ is an eigenvalue of T and we use the orthogonal decomposition:

$$V = N_{\lambda_1} \oplus N_{-\lambda_1} \oplus V_2,$$

where one of the two eigenspace might be trivial and then we ignore it. The restriction $T|_{V_2}$ has norm strictly less than $\lambda_1 = \|T\|$, otherwise T would have an eigenvector with eigenvalue $\pm\lambda_1$ belonging to V_2 which is impossible as $V_2 \perp N_{\pm\lambda_1}$. Iterating, we obtain:

$$V = N_{\lambda_1} \oplus N_{-\lambda_1} \oplus \dots \oplus N_{\lambda_n} \oplus N_{-\lambda_n} \oplus V_{n+1},$$

with $0 < \lambda_n < \lambda_{n-1} < \dots < \lambda_1$. If at some point $\|T|_{V_{n+1}}\| = 0$ then $V_{n+1} = N_0$, which can be infinity dimensional, the process stops and T is finite rank. Otherwise we obtain a countable sequence of eigenvalues with $\lambda_n \rightarrow 0$ (assuming λ_n converges to some non zero constant, we can again conclude that, similarly to the identity, the operator can not be compact).

A basis can be chosen by choosing progressively bases in the finite dimensional N_{λ_i} s and completing with a basis in N_0 (which can be infinite dimensional). If x is orthogonal on all the $N_{\pm\lambda_i}$ for $i < j$ then $x \in V_j$ hence $\|Tx\| \leq \lambda_j\|x\|$ and requiring this to hold for all n , we obtain $\|Tx\| = 0$. Thus either $x = 0$ if the kernel of T is trivial, or $x \in N_0$ and can be decomposed on a Hilbert basis in N_0 . $\square$

The eigenvectors of a non compact (bounded or unbounded) self-adjoint operator might or might not form an orthogonal basis. The eigenvalues are generalized to the notion of spectrum.

Definition 15.7 (Spectrum of an operator). Let $A : D(A) \rightarrow V$ an operator on a Hilbert (or Banach) space. A complex number z belongs to the resolvent set of A if the operator $A - zI : D(A) \rightarrow V$ has a bounded (hence everywhere defined) inverse $R(z; A) = (A - zI)^{-1} : V \rightarrow D(A)$ with $\|(A - zI)^{-1}\|_{\text{op.}} \leq \infty$ which we call the resolvent.

A point not in the resolvent set is in the spectrum, $\lambda \in \sigma(A)$. Furthermore:

- (the resolvent does not exist) if $A - \lambda I$ is not injective, that is there exists $v \in D(A)$ such that $(A - \lambda I)v = 0$ then λ is an eigenvalue of A .

The discrete spectrum of A , $\sigma_d(A)$ is the set of eigenvalues that are isolated points of $\sigma(A)$ and have finite multiplicity (that is the dimension of the λ -eigenspace is finite). Note that eigenvalues, even with finite multiplicity do not necessarily lie in the discrete spectrum: this happens if such eigenvalues are not isolate points in the spectrum.

- (the resolvent is a densely defined unbounded operator) if $A - \lambda I$ is injective and has dense range but it is not surjective, then λ belongs to the continuous spectrum of A , $\lambda \in \sigma_c(A)$. In words, any x can be arbitrarily well approximated by $(A - \lambda I)(y)$ with $y \in D(A)$, but some x 's can never be reached.
- (the resolvent is not densely defined) if $A - \lambda I$ is injective and does not have dense range then λ belongs to the residual spectrum, $\lambda \in \sigma_{\text{res}}(A)$. This kind of spectrum is absent for densely defined self-adjoint operators.⁹

While the discrete spectrum is a union of isolated points, the continuous and residual spectra can contain intervals. It can be shown that the spectrum of an operator is always a closed set.

Proposition 15.3 (Spectral radius). If T is a bounded operator with $\|T\|_{\text{op.}} < 1$ then $1 - T$ is invertible and $(1 - T)^{-1} = \sum_{n \geq 0} T^n$. It follows that if A is bounded, then for $|z| > \|A\|_{\text{op.}}$, $\|(A - zI)^{-1}\|_{\text{op.}} = \left\| -\sum_{n \geq 0} \frac{A^n}{z^{n+1}} \right\|_{\text{op.}} \leq \frac{1}{|z| - \|A\|_{\text{op.}}}$, that is the spectrum of A is included in a disk of radius $\|A\|_{\text{op.}}$, which we consequently call the spectral radius of A .

Proof. Recall that the norm $\|\cdot\|_{\text{op.}}$ is sub-multiplicative, that is $\|T^2\|_{\text{op.}} \leq \|T\|_{\text{op.}}^2$. As $\|T\|_{\text{op.}} < 1$, there exist ρ such that $\|T\|_{\text{op.}} < \rho < 1$. Then $R_N = \sum_{n=1}^N T^n$ is a Cauchy sequence in operator norm (if $M > N$ $\|R_N - R_M\|_{\text{op.}} \leq \rho^{N+1} \frac{1}{1-\rho}$) hence convergent, as the space $B(V)$ of bounded linear operators on V is a Banach space. Furthermore $(1 - T)R_N = 1 - T^{N+1}$ and, as the product of operators is continuous in $B(V)$, $\|A(X - Y)\|_{\text{op.}} \leq \|A\|_{\text{op.}} \|X - Y\|_{\text{op.}}$, and $\|T^{N+1}\|_{\text{op.}} \leq \rho^{N+1} \xrightarrow{n \rightarrow \infty} 0$ we conclude. $\square$

Proposition 15.4 (Symmetric operators with real spectrum). An unbounded densely defined operator is symmetric if and only if the quadratic form $\langle Ax, x \rangle$ for $x \in D(A)$ is real. Recall that symmetric densely defined operators can be closed.

The spectrum of a closed self-adjoint operator is a subset of the real numbers. A closed symmetric operator A is self-adjoint if and only if it has real spectrum, $\sigma(A) \subset \mathbb{R}$.

Proof. For the first part, we note that if the form is real then $\langle Ax, x \rangle = \overline{\langle Ax, x \rangle} = \langle x, Ax \rangle$ and the converse is similar. For the rest, the reader should search the relevant literature. $\square$

Self-adjoint operators only have eigenvalues and continuous spectrum. One can check whether a number belongs to the spectrum directly.

⁹If λ is in the residual spectrum then $A - \lambda$ is injective but there exists $y \in \overline{(A - \lambda)(D(A))}^\perp$, $y \neq 0$. Then $\langle Ax, y \rangle = \bar{\lambda} \langle x, y \rangle$ for all $x \in D(A)$, hence $\langle Ax, y \rangle$ is a bounded linear functional in $x \in D(A)$ therefore $y \in D(A^\dagger) = D(A)$. Then $\langle x, (A - \bar{\lambda})y \rangle = 0$ for all $x \in D(A)$ and choosing $x_n \rightarrow (A - \bar{\lambda})y$ (which is possible as $D(A)$ is dense) we have $\|(A - \bar{\lambda})y\| = 0$ hence $Ay = \bar{\lambda}y$. But then $\bar{\lambda} \langle y, y \rangle = \langle y, Ay \rangle = \langle Ay, y \rangle = \langle \bar{\lambda}y, y \rangle = \bar{\lambda} \langle y, y \rangle$ hence λ is real and $(A - \lambda)y = 0$, that is $A - \lambda$ is not injective.

Theorem 15.7 (Weyl's criterion). *A point λ belongs to the spectrum $\sigma(A)$ of a (closed) densely defined self-adjoint operator A if and only if there exists a sequence $(v_n)_n \in D(A)$ with $\|v_n\| = 1$ such that $\lim_{n \rightarrow \infty} \|(A - \lambda)v_n\| = 0$.*

Furthermore, λ belongs to the essential spectrum $\lambda \in \sigma_{ess}(A) = \sigma(A) \setminus \sigma_d(A)$ (this is the continuous spectrum, the eigenvalues with infinite multiplicity and the eigenvalues with finite multiplicity that are not isolated) if and only if there exists such a sequence $(v_n)_n$ which is such that it contains no convergent subsequence. For instance the sequence of basis vectors $(e_n)_n$ in a Hilbert basis with $\|e_n\| = 1$ does not have any convergent subsequence as $\|e_i - e_j\| = \sqrt{2}$ if $i \neq j$.

Proof. If $\lambda \in \sigma(A)$ and A is self-adjoint, then λ is either an eigenvalue or it belongs to the continuous spectrum. If λ is an eigenvalue, choosing $v_n = v$ the corresponding eigenvector yields a sequence with the required properties. If $\lambda \in \sigma_c(A)$ then the resolvent $R(\lambda; A)$ is a densely defined unbounded operator and there exists a sequence x_n in the domain of $R(\lambda; A)$ with $\|x_n\| = 1$ such that $\|R(\lambda; A)x_n\| \rightarrow \infty$ hence $y_n = \frac{1}{\|R(\lambda; A)x_n\|} R(\lambda; A)x_n$ is such that $\|y_n\| = 1$ and $(A - \lambda I)y_n \rightarrow 0$. Note that if y_n converges, as A is closed then λ is an eigenvalue and $A - \lambda I$ is not injective

If an eigenvalue has infinite multiplicity or is an accumulation point for eigenvalues, then one can pick sequences with no convergent subsequences. $\square$

For a compact self-adjoint operator T :

- if the dimension of the kernel of T is infinite dimensional, the eigenvalue 0 of T has infinite multiplicity hence 0 belongs to the essential spectrum of T .
- if the kernel is non trivial but has finite dimensions then the non zero eigenvalues are such that $\lambda_n \rightarrow 0$. Thus 0 is an eigenvalue with finite multiplicity, but it is not isolated and belongs to the essential spectrum of T .
- if the kernel is trivial, we have $\lambda_n \rightarrow 0$ and picking one element v_n with $\|v_n\| = 1$ per N_{λ_n} we obtain a sequence with no convergent subsequence such that $\|Tv_n\| \rightarrow 0$, hence 0 belongs to the essential spectrum of T . Then 0 must belong to the continuous spectrum, as 0 is not an eigenvalue. A variation of the example below shows that T has dense image but is not surjective.

Consider the Hilbert space ℓ^2 . We have:

- the operator $T(\sum_{n \geq 1} v(n)e_n) = \sum_{n \geq 1} \frac{1}{n} v(n)e_n$ is bounded, $\|T\| = 1$, compact ($1/n \rightarrow 0$), with zero kernel. It follows that 0 is in its continuous spectrum. The resolvent at zero $(T - 0I)^{-1}$ is $T^{-1}(\sum_{n \geq 1} v(n)e_n) = \sum_{n \geq 1} n v(n)e_n$, and as we have seen is an unbounded but densely defined symmetric operator $D(T^{-1}) = \{x \mid \sum_{n \geq 1} n^2 |x(n)|^2 < \infty\}$. Clearly this is ℓ^2 (as the finite truncations belong to $D(T^{-1})$ and are dense), but $\sum_{n \geq 1} \frac{1}{n} e_n \notin D(T^{-1})$.
- The operator $A(\sum_{n \geq 1} v(n)e_n) = \sum_{n \geq 1} n v(n)e_n$ is unbounded, densely defined and symmetric. For any $z \notin \mathbb{R}$, the resolvent of this operator is:

$$\frac{1}{A - zI}(v) = R(z; A)(v) = \sum_{n \geq 1} \frac{1}{n - z} \langle e_n, v \rangle e_n, \quad \|R(z; A)(v)\|^2 = \sum_{n \geq 1} \frac{1}{|n - z|^2} |\langle e_n, v \rangle|^2 \leq \frac{1}{|\operatorname{Im}(z)|^2} \|v\|^2,$$

hence it is a bounded operator $\|R(z; A)\|_{\text{op}}^2 \leq |\operatorname{Im}(z)|^{-2}$. It follows that the spectrum of A is a subset of the real line, hence A is self-adjoint.

- The three kinds of essential spectrum are exemplified by $T(\sum_{n \geq 1} v(n)e_n) = \lambda_1 v(1)e_1 + \lambda_2 v(2)e_2 + \lambda \sum_{n \geq 3} v(n)e_n$, $T(\sum_{n \geq 1} v(n)e_n) = \lambda v(1)e_1 + \sum_{n \geq 2} (\lambda + \frac{1}{n}) v(n)e_n$ and $T(\sum_{n \geq 1} v(n)e_n) = \sum_{n \geq 1} (\lambda + \frac{1}{n}) v(n)e_n$. The first operator has an eigenvalue, λ , with infinite multiplicity; the second one has an eigenvalue λ which is not an isolated point; for the third operator λ is in the continuous spectrum as $T - \lambda I$ has dense image but is not surjective.

Exercise 15.2. Let the operator $B : \ell^2 \rightarrow \ell^2$, $B(e_1) = 0$, $B(e_n) = e_{n-1}$. Prove that $B - I$ has dense range but is not surjective. The range contains $-e_1, e_1 - e_2, e_2 - e_3$ and so on. As it is a subspace it contains e_1, e_2 etc. hence it is dense. However, $(1/n)$ does not belong to the range. Assume $(B - I)(a) = (1/n)$ that is $a_{n-1} - a_n = 1/n$ hence $a_n = a_1 + 1 + \frac{1}{2} + \dots + \frac{1}{n}$, which is a contradiction as $a_n \rightarrow 0$ if $(a_n) \in \ell^2$.

15.4 Quantum mechanics

The spectral theorem for non compact operators is significantly more involved. For non compact self-adjoint operators for instance, bounded or not, it involves a sum over the eigenvalues and an integral over the continuous spectrum. Roughly (for physicists) the spectral theorem states that for any self-adjoint operator A we can represent the identity operator as:

$$1 = \sum_i |a_i\rangle \langle a_i| + \int_{\sigma_{ess}(A)} da |a\rangle \langle a| ,$$

where $|a_i\rangle$ are eigenvectors $A|a_i\rangle = a_i|a_i\rangle$ and $|a\rangle$ are “generalized eigenvectors” for which $A|a\rangle = a|a\rangle$ but $|a\rangle$ does not belong to the Hilbert space: the point is that the sequence v_n in the Weyl criterion might converge in some larger space in which V is embedded. Typically the Hilbert space is a space of square integrable functions and the generalized eigenvectors are functions which respect the eigenvalue equation but are not square integrable. For mathematicians, the integral over the continuous spectrum is an integral over a projection valued measure. Note that A itself writes then as:

$$A = \sum_i |a_i\rangle a_i \langle a_i| + \int_{\sigma_{ess}(A)} da |a\rangle a \langle a| .$$

We will discuss more rigorously the continuous spectrum later. We are now in the position to state the three postulates of quantum mechanics.

Postulate 1: States of isolated quantum systems. The state of an isolated quantum system is an equivalence class of kets $|\psi\rangle$ in a (usually separable) complex Hilbert space H of norm 1, $\langle\psi|\psi\rangle = 1$, where two kets are equivalent if they differ by a phase. By definition such a state is also called *pure* and the *density matrix* associated to a pure state is the projector $\rho = |\psi\rangle \langle\psi|$ which is a class function for equivalent kets.

Postulate 1a: States of non-isolated quantum systems. The Hilbert space of a composite quantum system consisting in two parts 1 and 2 with Hilbert spaces H_1 and H_2 is the tensor product $H = H_1 \otimes H_2$. The state of a non-isolated quantum system is a *density matrix* ρ , that is a positive semi definite Hermitian operator of trace 1 on the Hilbert space H , $\rho = \rho^\dagger$, $\rho \geq 0$, $\text{Tr} \rho = 1$.

The states of non isolated system are further classified into:

- *Pure states.* A pure state of the composite system is a rank one projector, that is a density matrix $\rho = |\psi\rangle \langle\psi|$ with $|\psi\rangle$ a ket in $H = H_1 \otimes H_2$. A pure state is called a *product state*, or *separable* if $|\psi\rangle$ can be written as a tensor product $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, hence $\rho = \rho_1 \otimes \rho_2$, where both $\rho_\eta = |\psi_\eta\rangle \langle\psi_\eta|$ are rank one projectors. A pure state which is not separable is called *entangled*.¹⁰

¹⁰In finite dimensions, denoting $|a_i\rangle$ a basis in H_1 and $|b_j\rangle$ a basis in H_2 we have $|\psi\rangle = \sum_{i,j} c_{ij} |a_i\rangle \otimes |b_j\rangle = \sum_{i,j} c_{ij} |a_i b_j\rangle$ and using the singular values decomposition of c_{ij} , $|\psi\rangle = \sum_k \sqrt{p_k} |u_k\rangle \otimes |v_k\rangle$ with $|u_k\rangle$ and $|v_k\rangle$ bases in H_1 respectively H_2 and $p_k > 0$, $\sum_k p_k = 1$. A pure state is separable if the singular value decomposition leads to only one term and entangled if not, hence for pure states, entanglement is easy to detect.

- *Mixed states.* States of non isolated quantum systems that are not pure are called mixed.¹¹ A mixed state is called *separable* if it is a convex combination of tensor product states:

$$\rho = \sum_k p_k \rho_1^k \otimes \rho_2^k \quad p_k \geq 0, \quad \sum_k p_k = 1,$$

where ρ_η^k are mixed states of the subsystems, and *entangled* if not. For generic mixed states, entanglement is very hard to detect.

Postulate 2: Observables and measurements. The observables of a quantum system are (bounded or unbounded densely defined) self-adjoint operators on the Hilbert space. Many common observables, like momentum, angular momentum, position for non compact systems, etc. are unbounded.

Postulate 2a: Expectations of observables. The expectation of the operator A in a pure state $|\psi\rangle$ is $\bar{A} = \langle\psi|A|\psi\rangle$.¹² For mixed states, the expectation is $\bar{A} = \text{Tr}(A\rho)$.

Postulate 2b: Outcomes of measurements. The expectation of A writes in terms of the spectral representation as:

$$\begin{aligned} \langle\psi|A|\psi\rangle &= \langle\psi|A|\left(\sum_i |a_i\rangle\langle a_i| + \int da_{\sigma_{ess}(A)} |a\rangle\langle a|\right)|\psi\rangle = \sum_i \langle\psi|a_i\rangle a_i \langle a_i|\psi\rangle + \int_{\sigma_{ess}(A)} da \langle\psi|a\rangle a \langle a|\psi\rangle, \\ \text{Tr}(A\rho) &= \sum_i a_i \langle a_i|\rho|a_i\rangle + \int_{\sigma_{ess}(A)} da a \langle a|\rho|a\rangle. \end{aligned}$$

The *only possible outcomes* of a measurement of A are $a \in \sigma_d(A) \cup \sigma_{ess}(A)$. For a in the discrete spectrum, $\sum_{i, a_i=a} |\langle a_i|\psi\rangle|^2$, respectively $\sum_{i, a_i=a} \langle a_i|\rho|a_i\rangle$ (where we account for the fact that the eigenvalue a can have a finite multiplicity) is the probability of obtaining the result a . For the continuous spectrum, the probability of obtaining a result comprised between a and $a + da$ is $dP(a) = da |\langle a|\psi\rangle|^2 = da \langle a|\rho|a\rangle$, that is $|\langle a|\psi\rangle|^2$, respectively $\langle a|\rho|a\rangle$, is the *probability density* of a . The normalization of the state ensures that the probabilities add up to 1:

$$1 = \langle\psi|\psi\rangle = \sum_i |\langle a_i|\psi\rangle|^2 + \int_{\sigma_{ess}(A)} da |\langle a|\psi\rangle|^2, \quad 1 = \text{Tr} \rho = \sum_i \langle a_i|\rho|a_i\rangle + \int_{\sigma_{ess}(A)} da \langle a|\rho|a\rangle.$$

Postulate 2c: Projective measurements. Assume that we have prepared a system in the state $|\psi\rangle$ (or ρ) and a measurement of A yields the value a . If one immediately measures A again, one always obtains again the same result a , that is the measurement is projective.

In detail, if a belongs to the discrete spectrum, we denote the projector on the corresponding eigenspace $\mathcal{P} = \sum_{i, a_i=a} |a_i\rangle\langle a_i|$ and if a belongs to the essential spectrum and the precision of the measurement is δa we denote $\mathcal{P} = \int_a^{a+\delta a} da |a\rangle\langle a|$ the corresponding projector. The measurement projects the state on the corresponding eigenspace, that is after the measurement the state of the system is:

$$|\psi'\rangle = \frac{\mathcal{P}|\psi\rangle}{\sqrt{\langle\psi|\mathcal{P}|\psi\rangle}}, \quad \rho' = \frac{\mathcal{P}\rho\mathcal{P}}{\text{Tr}(\mathcal{P}\rho\mathcal{P})}.$$

¹¹A mixed state of the subsystem 1 can be obtained for instance as a *partial state* of a pure state of the whole system: $\rho = \text{Tr}_2(|\psi\rangle\langle\psi|) = \sum_k p_k |u_k\rangle\langle u_k|$, but non isolated quantum system admit more general mixed states, like generalized partial states for multipartite decompositions of isolated quantum systems.

¹²The variance of A is $(\Delta A)^2 = \overline{A^2} - \bar{A}^2 = \langle\psi|(A - \bar{A})(A - \bar{A})|\psi\rangle$. The Cauchy-Schwarz inequality implies:

$$(\Delta A)^2(\Delta B)^2 \geq |\langle\psi|(A - \bar{A})(B - \bar{B})|\psi\rangle|^2 \geq \left| \frac{\langle\psi|(A - \bar{A})(B - \bar{B})|\psi\rangle - \langle\psi|(B - \bar{B})(A - \bar{A})|\psi\rangle}{2i} \right|^2 = \frac{1}{4} |\langle\psi|[A, B]|\psi\rangle|^2,$$

and for mixed states we get $(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4} |\text{Tr}(\rho[A, B])|^2$. For A and B the position and momentum operators X and P with $[X, P] = i\hbar$, this yields Heisenberg's uncertainty principle.

Postulate 3: Time evolution. The (maybe time dependent) Hamiltonian of a system is a self adjoint (bounded or unbounded densely defined) operator $\mathcal{H}(t)$. The time evolution is the strongly continuous one parameter family of unitary operators $U_{t,t_0} : H \rightarrow H$ with $t, t_0 \in \mathbb{R}$, that is $U_{t,t_0} U_{t_0,t_0}^\dagger = U_{t,t_0}^\dagger U_{t,t_0} = U_{t,t} = 1$, respecting $U_{t,t_1} U_{t_1,t_0} = U_{t,t_0}$ and obeying the Schrödinger equation:

$$i \lim_{\epsilon \rightarrow 0} \frac{U_{t+\epsilon,t} |\psi\rangle - |\psi\rangle}{\epsilon} = \mathcal{H}(t) |\psi\rangle ,$$

for all $|\psi\rangle$ in the domain of $\mathcal{H}(t)$. We note that $\lim_{\epsilon \rightarrow 0} U_{t+\epsilon,t} |\psi\rangle = |\psi\rangle$, that is $\lim_{\epsilon \rightarrow 0} U_{t+\epsilon,t} = 1$ in the strong sense. One has $\lim_{\epsilon \rightarrow 0} U_{t+\epsilon,t} = 1$ in norm if and only if the Hamiltonian is bounded. The evolution operator can be written, at least for short enough times, as the time ordered exponential (or Dyson series):

$$U_{t,t_0} = \sum_{n \geq 0} (-i)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \dots \int_{t_0}^{t_2} dt_1 \mathcal{H}(t_n) \mathcal{H}(t_{n-1}) \dots \mathcal{H}(t_1) .$$

In the Schrödinger picture we consider that the state evolves in time: if the system is in the pure state $|\psi_{t_0}\rangle$ or has a density matrix ρ_{t_0} at time t_0 , then at time t the system is in the state $|\psi_t\rangle = U_{t,t_0} |\psi_{t_0}\rangle$ or $\rho_t = U_{t,t_0} \rho_{t_0} U_{t,t_0}^\dagger$, and the Schrödinger equation becomes:

$$i \frac{d}{dt} |\psi_t\rangle = \mathcal{H}(t) |\psi_t\rangle , \quad i \frac{d}{dt} \rho_t = [\mathcal{H}(t), \rho_t] .$$

In the Heisenberg picture, we consider that the states are fixed but the observables evolve in time $A_H(t) = U_{t,t_0}^\dagger A_S(t) U_{t,t_0}$, where A_S denotes the (possibly time dependent) observable in the Schrödinger picture, hence:

$$i \frac{d}{dt} A_H(t) = [A_H(t), \mathcal{H}(t)] + i \left(\frac{\partial A_S}{\partial t} \right)_H .$$

The two pictures clearly lead to the same expectations.

If the Hamiltonian $\mathcal{H}$ is time independent, then $U_{t,t_0} = e^{-i(t-t_0)\mathcal{H}}$ globally and the evolution operator is a strongly continuous unitary group of operators. This holds in particular for isolated quantum systems.

Note that there is slight tension between the continuous unitary evolution postulate and the projective measurement one, as a measurement should be a kind of quantum evolution, but the instantaneous projection is not a continuous unitary evolution. In the spirit of the Copenhagen interpretation we ignore such issues.

In order to predict the outcome of a measurement of an observable A in quantum mechanics one needs to:

- identify the Hilbert space. This is usually a space of “wave functions” $\psi(x)$ with scalar product $\langle \psi_1 | \psi_2 \rangle = \int d^d x \bar{\psi}_1(x) \psi_2(x)$. The wave functions must have good boundedness properties and asymptotic behaviour to ensure $\langle \psi | \psi \rangle < \infty$. As the Hilbert space is Cauchy complete, these functions lack regularity (not continuous or derivable).
- identify an appropriate dense domain $D(A)$ such that A is self-adjoint. Usually A is some differential operator, hence its domain only contains functions that are regular enough (sufficiently derivable) and maybe respect additional conditions in order to ensure that A is symmetric and $D(A) = D(A^\dagger)$. Of course, they also need to have good boundedness properties and asymptotic behaviour, in order to belong to the Hilbert space.
- identify the eigenvectors and the generalized eigenvectors by solving the eigenvalue equation $A|a\rangle = a|a\rangle$. First one determines the genuine eigenvectors, that is the vectors $|a_i\rangle$ in the domain $D(A)$ which respect the eigenvalue equation, hence are regular enough and $\langle a_i | a_i \rangle < \infty$. The generalized eigenvectors are solutions of the eigenvalue equation (hence are regular enough) but are not appropriately bounded, that is $\langle a | a \rangle = \infty$. Such generalized eigenvectors do not lie in the Hilbert space and we will see later that they can be understood in the sense of distributions.

Chapter 16

The Lebesgue integral, first contact

The Riemann integral has several limitations:

- one can only integrate functions with a countable number of discontinuities.
- many sets, e.g. $\mathbb{Q}$, are not Jordan measurable hence one can not Riemann integrate over them.
- one can not naively commute the Riemann integral with limits of sequences or series of functions. This is a very serious limitation.

In this chapter we discuss a more general notion of integral which overcomes these limitations. As we have seen, the existence of the Riemann integral of a function depends on the properties of its domain, most importantly on the existence of a notion of “volume” or measure. A first attempt is to use the Jordan measure. To simplify our life we focus on the one dimensional case $f : \mathbb{R} \rightarrow V$:

- a box is a closed interval $B = [a, b]$, its boundary is $\partial B = \{b\} \cup \{a\}$ and its measure $\mu(B) = |b - a|$. We have $\mu(B) = \lim_{k \rightarrow \infty} \frac{1}{k} |B \cap (\mathbb{Z}/k)|$ and $\mu(\partial B) = 0$.
- an elementary set is a finite union of boxes and can be written as a finite union of almost disjoint intervals $E = \bigcup_{i=1}^k B_i$ with $B_i^\circ \cap B_j^\circ = \emptyset$ and $\mu(E) = \sum_{i=1}^k \mu(B_i)$.
- the *inner Jordan measure* of $X \subset \mathbb{R}$ is $\mu_*(X) = \sup \{ \mu(E) \mid E \subseteq X, E \text{ elementary} \}$ and the *outer Jordan measure* of X is $\mu^*(X) = \inf \{ \mu(E) \mid X \subseteq E, E \text{ elementary} \}$; X is called *Jordan measurable* if $\mu_*(X) = \mu^*(X) < \infty$ and the Jordan measure of $\mu(X)$ of X is the common value of its inner and outer measures; X is called a null set if $\mu^*(X) = 0$.
- the set $\mathcal{B}_k = \{ [\frac{a}{2^k}, \frac{a+1}{2^k}] \mid a \in \mathbb{Z} \}$ partitions $\mathbb{R}$ into almost disjoint boxes of side length (measure) 2^{-k} . Any $q \in \mathcal{B}_k$ is the union of 2 boxes in $\mathcal{B}_{k+1}$ and $\mu_*(X) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{q \in \mathcal{B}_k, q \subset X} q \right)$, $\mu^*(X) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{q \in \mathcal{B}_k, q \cap X \neq \emptyset} q \right)$.
- a set $X \subset \mathbb{R}$ is Jordan measurable if and only if its boundary is a null set $\mu(\partial X) = 0$.

The last condition is quite restrictive. In the standard topology the set $\mathbb{Q} \cap [0, 1]$ has no interior points $(\mathbb{Q} \cap [0, 1])^\circ = \emptyset$, but its closure is the full interval $\overline{(\mathbb{Q} \cap [0, 1])} = [0, 1]$. Thus $\partial(\mathbb{Q} \cap [0, 1]) = [0, 1]$ which is of Jordan measure 1.

16.1 The Lebesgue measure

We will now discuss a more general notion of measure, the Lebesgue measure, which will allow us to integrate a very large class of functions over a very large class of domains. Many of the results presented here are very useful but technical to prove: some proofs are omitted below.

In this chapter we will deal with quantities which can be infinite.

Definition 16.1 (**The extend real numbers**). The extend real numbers $\mathbb{R} \cup \{-\infty\} \cup \{\infty\}$ are the real numbers with two additional elements ∞ and $-\infty$ which are larger respectively smaller than every real number.

In particular any monotonic sequence converges in the extended real numbers: if it is bounded it converges to a finite limit, and if it is unbounded it converges to one of the two infinities. In measure and integration theory $0 \cdot \infty = 0$ (one takes 0 as 0, but considers ∞ as the limit of a sequence).

Exercise 16.1. Prove that any open set $U \subset \mathbb{R}$ can be written uniquely as the countable union of disjoint open intervals. For $x \in U$, let $I_x = (a_x, b_x)$ the largest open interval such that $x \in I_x \subset U$. This is constructed by taking $a_x = \inf_a \{a \mid (a, x) \subset U\}$ and similarly for b_x . Then $U = \bigcup_{x \in U} I_x$. Suppose that two intervals I_y and I_x intersect. Then $I_y \cup I_x$ is an open interval in U and contains x (resp. y). As I_x (resp. I_y) is maximal, it follows that $I_x \cup I_y \subset I_x$ (resp. $I_x \cup I_y \subset I_y$) hence $I_x \cup I_y = I_x = I_y$. Therefore any two distinct intervals in the collection $\{I_x\}_{x \in U}$ must be disjoint. Finally, any I_x must contain a rational number and if I_x and I_y are distinct they must contain different rationals, hence $U = \sum_{q \in U \cap \mathbb{Q}} I_q$ and the union is countable.

Definition 16.2 (**The Lebesgue outer measure**). For any $X \subset \mathbb{R}$, we define the Lebesgue outer measure of X as:

$$m^*(X) = \inf_{\{I_i = (a_i, b_i)\}_{i \in \mathbb{N}}, X \subset \bigcup_{i=1}^{\infty} I_i} \sum_{i=1}^{\infty} |a_i - b_i|,$$

where the infimum is take over all countable coverings $\{I_i\}_{i \in \mathbb{N}}$ of X , $X \subset \bigcup_{i=1}^{\infty} I_i$ by open intervals $I_i = (a_i, b_i)$. This infimum can be finite or ∞ .

An open box in $\mathbb{R}^d$ is a Cartesian product of open intervals $B = (a^1, b^1) \times \dots \times (a^d, b^d)$ and its volume is $\text{Vol}(B) = \prod_{c=1}^d |a^c - b^c|$. For $X \subset \mathbb{R}^d$ we define:

$$m^*(X) = \inf_{\{B_i\}_{i \in \mathbb{N}} \text{ open boxes }, X \subset \bigcup_{i=1}^{\infty} B_i} \sum_{i=1}^{\infty} \text{Vol}(B_i).$$

Exercise 16.2. Prove that we can drop both “countable” and “open” from the definition. Prove that we can restrict to coverings by disjoint intervals.

Like the Jordan outer measure, the Lebesgue outer measure is defined for every set and $m^*(X) \leq \mu^*(X)$. The only difference between the Jordan outer measure and the Lebesgue outer measure is that the latter allows countable coverings of X .

Exercise 16.3. Prove that $m^*([a, b]) = b - a$. As $(a - \epsilon, b + \epsilon)$ is an open interval covering $[a, b]$ we have $m^*([a, b]) \leq b - a + 2\epsilon$ hence $m^*([a, b]) \leq b - a$.

As $[a, b]$ is compact, any open cover admits a finite subcover and the total length of the subcover is not greater than the one the original cover, it suffices to consider finite covers $\{I_i\}_{1 \leq i \leq n}$. Since $a \in [a, b]$ it belongs to one of the intervals in the cover, say $a \in I_1 = (a_1, b_1)$. If $b_1 < b$, then $b_1 \in (a, b)$ and $b_1 \notin (a_1, b_1)$ hence b_1 must belong to an interval in the cover $b_1 \in I_2 = (a_2, b_2)$. Iterating we obtain a sequence of intervals (a_1, b_1) , (a_2, b_2) etc. such that $a_i < b_{i-1} < b_i$ and the process must terminate at some (a_k, b_k) . But the process only terminates if $a_k < b < b_k$. Hence $\sum l(I_n) \geq \sum l(a_i, b_i) = b_k - a_k + b_{k-1} - a_{k-1} + \dots + b_1 - a_1 \geq b_k - a_1 \geq b - a$.

Exercise 16.4. Prove that for $I_i = (a_i, b_i)$, with $i \in \mathbb{N}$ countably many disjoint finite open intervals, the exterior measure of their union $I = \bigcup_{i=1}^{\infty} I_i$ is $m^*(I) = \sum_{i=1}^{\infty} (b_i - a_i)$ (may be infinite). It is trivial that $m^*(I) \leq \sum_{i=1}^{\infty} (b_i - a_i)$. Let $I \subset \bigcup_{i=1}^{\infty} (c_i, d_i)$ and define the compact set $A_n = \bigcup_{i=1}^n [a_i + \frac{\epsilon}{2^{n+1}}, b_i - \frac{\epsilon}{2^{n+1}}]$ which admits a finite subcover $A_n \subset \bigcup_{i=1}^n (c_i, d_i)$ and $\sum_{i=1}^n (b_i - a_i) - \epsilon \leq \sum_{i=1}^n (d_i - c_i) \leq \sum_{i=1}^{\infty} (d_i - c_i)$. As n and ϵ are arbitrary we have $\sum_{i=1}^{\infty} (b_i - a_i) \leq \sum_{i=1}^{\infty} (d_i - c_i)$.

Exercise 16.5. Prove that the Lebesgue outer measure of any countable set is zero. List the elements in the countable set S as $S = \{s_1, s_2, \dots\}$ and for any ϵ pick the open cover $S \subset \bigcup_{i=1}^{\infty} (s_i - \frac{\epsilon}{2^i}, s_i + \frac{\epsilon}{2^i})$. Then $m^*(S) \leq \epsilon \sum_{i=1}^{\infty} \frac{1}{2^i} = \epsilon$.

Exercise 16.6 (The Cantor set). *The Cantor set is defined as $\cap_{n=0}^{\infty} C_n$ where $C_0 = [0, 1]$ and C_n is obtained by “taking out the open middle third intervals”:*

$$C_1 = [0, 1] \setminus \left(\frac{1}{2}, \frac{2}{3}\right) = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right], \quad C_2 = C_1 \setminus (1/9, 2/9) \setminus (7/9, 8/9), \dots$$

that is $C = \{x = \sum_{i \geq 1} \frac{1}{3^i} a_i, \quad a_i \in \{0, 2\}\}$. Prove C has outer measure zero and is uncountable. We have $m^(C) \leq m^*(C_n) = (2/3)^n$ hence C has outer measure 0. In order to prove that C is uncountable we map:*

$$\sum_{i \geq 1} \frac{1}{3^i} a_i \rightarrow \sum_{i \geq 1} \frac{1}{2^i} b(a_i), \quad b(0) = 0, \quad b(2) = 1,$$

and we note that the image of this map is all the numbers between $[0, 1]$, written in decimal representation in basis 2.

Recall that the Jordan outer measure of $\mathbb{Q} \cap [0, 1]$, which is a countable set, is 1. This can be seen by intersecting this set with the increasingly refined grid of boxes, $\mathcal{B}_k^n$, as discussed in Chapter 11. As the Jordan outer measure requires to cover the set $\mathbb{Q} \cap [0, 1]$ with a finite number of boxes, the “smallest box” in the cover has finite size. In particular, any finite cover of $\mathbb{Q} \cap [0, 1]$ by boxes contains in fact all the elements in $[0, 1]$.¹ The Lebesgue outer measure on the contrary allows for countable covers and one can use increasingly smaller “boxes” to cover $\mathbb{Q} \cap [0, 1]$, with the size of the “smallest box” going to zero.

Lemma 16.1. *The Lebesgue outer measure respects:*

- $m^*(\emptyset) = 0$.
- (monotone) $Y \subset X \Rightarrow m^*(Y) \leq m^*(X)$.
- (countably subadditive) $m^*\left(\bigcup_{i=1}^{\infty} X_i\right) \leq \sum_{i=1}^{\infty} m^*(X_i)$.

If X_i are compact and disjoint then $m^\left(\bigcup_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} m^*(X_i)$.*

Proof. We focus on the one dimensional case. The first two statements are trivial.

For the third one, as $m^*(X_i)$ is an infimum, for any ϵ there exist coverings $X_i \subset \bigcup_{k=1}^{\infty} (a_{i,k}, b_{i,k})$ such that

$$\sum_{k=1}^{\infty} |a_{i,k} - b_{i,k}| \leq m^*(X_i) + \frac{\epsilon}{2^i}.$$

As countable unions of countable sets are countable, $\{(a_{i,k}, b_{i,k})\}_{1 \leq i, k \leq \infty}$ cover X and

$$m^*(X) \leq \sum_{i,k \geq 1} |a_{i,k} - b_{i,k}| \leq \sum_{i=1}^{\infty} m^*(X_i) + \epsilon.$$

For the last point, we note that every open cover of $\bigcup_{i=1}^{\infty} X_i$ covers $\bigcup_{i=1}^N X_i$ which is compact if the X_i s are compact. One can extract a finite subcover of $\bigcup_{i=1}^N X_i$ out of which one can build appropriate open covers for each X_i with $i \leq N$. $\square$

Lemma 16.2 (**Outer Lebesgue measure of compact sets**). *For any compact set $X \in \mathbb{R}^d$, the outer Jordan and Lebesgue measures agree, $\mu^*(X) = m^*(X)$.*

Proof. It suffices to prove that $\mu^*(X) \leq m^*(X)$. For every ϵ , taking a countable covering of X such that $\sum_i \text{Vol}(B_i) < m^*(X) + \epsilon$ and extracting a finite subcovering we find $\mu^*(X) < m^*(X) + \epsilon$. $\square$

Definition 16.3 (**Zero Lebesgue measure sets**). *A set $X \subset \mathbb{R}^d$ is said to have zero Lebesgue measure if $m^*(X) = 0$. Any countable set has Lebesgue measure zero. A property $P(x)$, $x \in \mathbb{R}^d$ is said to hold almost everywhere if it fails at most on a set of measure zero.*

¹Consider a finite cover $\mathbb{Q} \cap [0, 1] \subset \bigcup_{i=1}^n [a_i, b_i]$ and assume there exist $x \notin \bigcup_{i=1}^n [a_i, b_i]$. Then for all i either $x < a_i$ or $x > b_i$. Chose $b = \max_i \{b_i, x > b_i\}$ and $a = \min_i \{a_i, x < a_i\}$. Then the nonempty real interval (b, a) does not belong to $\bigcup_{i=1}^n [a_i, b_i]$. As $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $q \in \mathbb{Q} \cap (b, a)$ contradicting the hypothesis that $\bigcup_{i=1}^n [a_i, b_i]$ covers $\mathbb{Q} \cap [0, 1]$.

Theorem 16.1 (**Lebesgue–Vitali theorem on Riemann integration**). *A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable on $[a, b]$ if and only if it is continuous almost everywhere.*

Proof. The reader should search the literature on the topic. □

Note that the indicator function of the rational numbers in the interval $[0, 1]$, $\mathbf{1}_{\mathbb{Q} \cap [0, 1]}(x)$ which is 1 for all the rational numbers in $[0, 1]$ and 0 for the irrational ones is discontinuous at any $x \in [0, 1]$, as any x can be approached by sequences of only rational respectively only irrational numbers, hence it is not Riemann integrable. We will see below that this function is Lebesgue integrable and its Lebesgue integral is zero.

The Lebesgue outer measure is a particular case of an outer measure.

Definition 16.4 (**Outer measure**). *Let V be some set. A function $\nu^* : \mathcal{P}(V) \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$ from the power set of V to the positive extended real numbers such that $\nu^*(\emptyset) = 0$ and ν^* is monotone and countably subadditive is called an outer measure on V .*

Observe that the Jordan “outer measure” is not an outer measure in the sense of this definition as it is not countably subadditive. Indeed, for any rational number $q \in [0, 1]$, $\mu^*(\{q\}) = 0$, but $\mu^*(\mathbb{Q} \cap [0, 1]) = 1$.

16.1.1 Lebesgue measurable sets

Definition 16.5 (**Lebesgue measurable sets; Carathéodory’s criterion**). *The set $X \subset \mathbb{R}^d$ is called Lebesgue measurable if it respects Carathéodory’s criterion, that is for any $Y \subset \mathbb{R}^d$ we have:²*

$$m^*(Y) = m^*(Y \cap X) + m^*(Y \cap X^c), \quad X^c = \mathbb{R}^d \setminus X.$$

Note that Y itself could be measurable or not. For X Lebesgue measurable, we denote $m(X) = m^(X)$ the Lebesgue measure of X . Measurable sets are defined identically for an arbitrary outer measure ν^* on a generic set V , and the measure of a measurable set X is $\nu(X) = \nu^*(X)$.*

We denote $\mathfrak{M} \subset \mathcal{P}(\mathbb{R}^d)$ the set of Lebesgue measurable sets and $\mathfrak{M}(\nu) \subset \mathcal{P}(V)$ the set of measurable sets with respect to the outer measure ν^ .*

Exercise 16.7. *Prove that if $\nu^*(X) = 0$ then X is measurable and (by definition) $\nu(X) = 0$. For any Y , as $Y \cap X^c \subset Y$, $\nu^*(Y \cap X^c) \leq \nu^*(Y)$.*

On the other hand, from subadditivity of ν^ , we have that $\nu^*(Y) \leq \nu^*(Y \cap X) + \nu^*(Y \cap X^c)$ but as $Y \cap X \subset X$, $\nu^*(Y \cap X) \leq \nu^*(X) = 0$, hence $\nu^*(Y) \leq \nu^*(Y \cap X^c)$. Hence $\nu^*(Y) = \nu^*(Y \cap X^c) = \nu^*(Y \cap X^c) + \nu^*(Y \cap X)$ and X respects the Carathéodory criterion.*

Exercise 16.8. *Let $X \subset [0, 1]$. We define the Lebesgue inner measure of X as:*

$$m_*(X) = 1 - m^*(X^c), \quad X^c = [0, 1] \setminus X.$$

Prove that $m_(X) \leq m^*(X)$. Recall that $m^*(X) + m^*(X^c) \geq m(X \cup X^c)$. Prove that a set is Lebesgue measurable if and only if $m_*(X) = m^*(X)$. Prove that the inner Jordan measure of X is smaller than its inner Lebesgue measure $\mu_*(X) \leq m_*(X)$. Prove that if X is Jordan measurable then it is Lebesgue measurable and in this case $\mu(X) = m(X)$.*

Exercise 16.9. *Prove that open, closed and half open half closed intervals are Lebesgue measurable.*

Proposition 16.1 (**Properties of Lebesgue measurable sets: practically any set you will ever encounter is Lebesgue measurable**). *We consider the outer measure ν^* on the set V . We have the following properties:*

²Note that $Y \cap X^c = Y \setminus X$.

- X is measurable if and only if X^c is measurable.
- if X_1, X_2 are measurable then $X_1 \cup X_2$, $X_1 \setminus X_2$ and $X_1 \cap X_2$ are also measurable.

Proof. By subadditivity, $\nu^*(Y) \leq \nu^*(Y \cap (X_1 \cup X_2)) + \nu^*(Y \setminus (X_1 \cup X_2))$. In order to conclude that $X_1 \cup X_2$ is measurable we need to prove the $\geq$ inequality. As X_1, X_2 are measurable we have for any Y that:

$$\nu^*(Y) = \nu^*(Y \cap X_1) + \nu^*(Y \setminus X_1), \quad \nu^*(Y \setminus X_1) = \nu^*((Y \setminus X_1) \cap X_2) + \nu^*((Y \setminus X_1) \setminus X_2),$$

Taking into account that $(Y \cap X_1) \cup ((Y \setminus X_1) \cap X_2) = Y \cap (X_1 \cup X_2)$ and $(Y \setminus X_1) \setminus X_2 = Y \setminus (X_1 \cup X_2)$ we have:

$$\nu^*(Y) = \nu^*(Y \cap X_1) + \nu^*((Y \setminus X_1) \cap X_2) + \nu^*((Y \setminus X_1) \setminus X_2) \geq \nu^*(Y \cap (X_1 \cup X_2)) + \nu^*(Y \setminus (X_1 \cup X_2)).$$

The rest follows by noting that $X_1 \setminus X_2 = X_1 \cap X_2^c = (X_1^c \cup X_2)^c$. □

- if $\{X_i\}_{i \in \mathbb{N}}$ are measurable (not necessarily disjoint) then $\bigcup_{i=1}^{\infty} X_i$ and $\bigcap_{i=1}^{\infty} X_i$ are measurable.

Proof. We can assume that the X_i s are disjoint (if not set $X'_i = X_{i+1} \setminus (\bigcup_{j=1}^{i-1} X_j)$, which are measurable and disjoint). By countable subadditivity, for any Y and any X_i we have $\nu^*(Y) \leq \nu^*(Y \cap (\bigcup_{i=1}^{\infty} X_i)) + \nu^*(Y \setminus (\bigcup_{i=1}^{\infty} X_i))$. It remains to prove $\geq$. By the previous point, for any finite N we have:

$$\nu^*(Y) = \nu^*(Y \cap (\bigcup_{i=1}^N X_i)) + \nu^*(Y \setminus (\bigcup_{i=1}^N X_i)) \geq \nu^*(Y \cap (\bigcup_{i=1}^N X_i)) + \nu^*(Y \setminus (\bigcup_{i=1}^{\infty} X_i)).$$

Assume inductively that $\nu^*(Y \cap (\bigcup_{i=1}^{N-1} X_i)) = \sum_{i=1}^{N-1} \nu^*(Y \cap X_i)$. Denote $A_N = \bigcup_{i=1}^N X_i$; as X_N is measurable and disjoint from $X_1, \dots, X_{N-1}$ we have:

$$\nu^*(Y \cap A_N) = \nu^*((Y \cap A_N) \cap X_N) + \nu^*((Y \cap A_N) \setminus X_N) = \nu^*(Y \cap X_N) + \nu^*(Y \cap A_{N-1}).$$

Then $\nu^*(Y) \geq \sum_{i=1}^N \nu^*(Y \cap X_i) + \nu^*(Y \setminus (\bigcup_{i=1}^{\infty} X_i))$ for any N and sending N to infinity we have:

$$\nu^*(Y) \geq \sum_{i=1}^{\infty} \nu^*(Y \cap X_i) + \nu^*(Y \setminus (\bigcup_{i=1}^{\infty} X_i)) \geq \nu^*(\bigcup_{i=1}^{\infty} (Y \cap X_i)) + \nu^*(Y \setminus (\bigcup_{i=1}^{\infty} X_i)) = \nu^*(Y \cap (\bigcup_{i=1}^{\infty} X_i)) + \nu^*(Y \setminus (\bigcup_{i=1}^{\infty} X_i)),$$

where we used the subadditivity. □

- if $\{X_i\}_{i \in \mathbb{N}}$ are measurable and disjoint then $\bigcup_{i=1}^{\infty} X_i$ is measurable and $\nu(\bigcup_{i=1}^{\infty} X_i) = \sum_{i=1}^{\infty} \nu(X_i)$.

Proof. By countable subadditivity, for any sets X_i we have $\nu^*(\bigcup_{i=1}^{\infty} X_i) \leq \sum_{i=1}^{\infty} \nu^*(X_i)$. It remains to prove $\geq$. By monotonicity of ν^* we have $\nu^*(\bigcup_{i=1}^{\infty} X_i) \geq \nu^*(\bigcup_{i=1}^N X_i)$.

Assume inductively that $\nu^*(\bigcup_{i=1}^{N-1} X_i) = \sum_{i=1}^{N-1} \nu^*(X_i)$. Denote $A = \bigcup_{i=1}^N X_i$; as X_N is measurable and disjoint from $X_1, \dots, X_{N-1}$ we have:

$$\nu^*(A) = \nu^*(A \cap X_N) + \nu^*(A \setminus X_N) = \nu^*(X_N) + \nu^*(\bigcup_{i=1}^{N-1} X_i).$$

□

Some properties specific to the Lebesgue measure on $\mathbb{R}^d$ are listed below:

- all open sets and all closed sets are Lebesgue measurable.

Proof. Take $d = 1$. Open intervals are Lebesgue measurable; any open set is a countable union of open intervals; any closed set is the complement of an open set.

Note that as $Y = (Y \cap X) \cup (Y \cap X^c)$ and m^* is subadditive, $m^*(Y) \leq m^*(Y \cap X) + m^*(Y \cap X^c)$ for any X and Y . It remains to prove the converse inequality if X is an interval. Take $I = (a, b)$, recall that $m^*(I) = |a - b|$ and take any set Y . For any ϵ , there exists a cover of Y by open intervals $Y \subset \bigcup_n I_n$ such that $\sum_n m^*(I_n) \leq m^*(Y) + \epsilon$. Set $J_n = I \cap I_n$ and $J'_n = I^c \cap I_n$. Then $m^*(I_n) = m^*(J_n) + m^*(J'_n)$, $Y \cap I = \bigcup_n J_n$, $Y \cap I^c = \bigcup_n J'_n$. Thus $m^*(Y \cap I) + m^*(Y \cap I^c) \leq m^*(Y) + \epsilon$. $\square$

- X is Lebesgue measurable if and only if $\forall \epsilon$ there exists a closed set F and an open set G with $F \subset X \subset G$ and $m^*(G \setminus X) < \epsilon$ respectively $m^*(X \setminus F) < \epsilon$.

Proof. For any ϵ and for any set X , there exists a open set (a countable union of intervals) such that $X \subset G$ and $m^*(G) \leq m^*(X) + \epsilon$. If X is measurable then $m^*(G) = m^*(G \cap X) + m^*(G \setminus X)$ and $G \cap X = X$ we conclude. For $F \subset X$, $X \setminus F = F^c \setminus X^c$ hence there exist a closed $F \subset X$ such that $m^*(X \setminus F) < \epsilon$.

Conversely, by subadditivity, for every Y we have $m^*(Y) \leq m^*(Y \cap X) + m^*(Y \setminus X)$. For any ϵ , take $G \supset X$ with $m^*(G \setminus X) < \epsilon$. As G is open it is measurable, hence $m^*(Y) = m^*(Y \cap G) + m^*(Y \setminus G) \geq m^*(Y \cap X) + m^*(Y \setminus G)$. As $G \supset X$ we have $(Y \setminus G) \cup (G \setminus X) \supset Y \setminus X$ and by subadditivity and monotonicity, $m^*(Y \setminus G) + m^*(G \setminus X) \geq m^*(Y \setminus X)$, hence $m^*(Y) \geq m^*(Y \cap X) + m^*(Y \setminus X) - \epsilon$ and we conclude. $\square$

- the set of Borel sets $\mathcal{X}$ is the smallest set of sets closed under countable union, countable intersection and complement which contains all the open and all the closed sets; all the Borel sets are Lebesgue measurable. There exist Lebesgue measurable but non Borel sets.
- if $X \in \mathbb{R}^d$ is Lebesgue measurable and L is a linear map, $L(X)$ is Lebesgue measurable and $m(L(X)) = |\det L| m(X)$.
- there exist non Lebesgue measurable sets, but they are hard to construct explicitly (requires the axiom of choice).

Proof. Consider the set $\mathbb{R}/\mathbb{Q} = \{[r] | r \sim r' \text{ if } r - r' \in \mathbb{Q}\}$, and build the Vitali set $V \subset [0, 1]$ by picking one representative in every class $[r] \in \mathbb{R}/\mathbb{Q}$. Then any two elements $u, v \in V$, $u - v \notin \mathbb{Q}$. Taking a counting q_i of the rational numbers in between $[-1, 1]$ and defining $V_k = q + q_k$ we have $[0, 1] \subset \bigcup_{k \in \mathbb{N}} V_k \subset [-1, 2]$, where the first inclusion follows by noting that for any $r \in [0, 1]$, denoting v the representative of $[r]$ which belongs to V , then $r - v = q_k \in \mathbb{Q} \cap [-1, 1]$ for some q_k hence $r \in V_k$. Denoting λ the Lebesgue measure and taking into account it is invariant by translation, if V is Lebesgue measurable we have $1 = \lambda([0, 1]) \leq \sum_{k \in \mathbb{N}} \lambda(V_k) = \sum_{k \in \mathbb{N}} \lambda(V) \leq \lambda([-1, 2]) = 3$. But this is impossible as $\sum_{k \in \mathbb{N}} \lambda(V)$ can only be 0 (if $\lambda(V) = 0$) or ∞ (if not). $\square$

16.1.2 Measure spaces

Definition 16.6 (**Measure spaces**). A set V is called a measure space if:

- there exists a σ -ring $\mathfrak{M}$ on V , that is a set of subsets of V such that if $A_i \in \mathfrak{M}$ for $i \in \mathbb{N}$ then $\bigcup_{i \in \mathbb{N}} A_i \in \mathfrak{M}$ and if $A_1, A_2 \in \mathfrak{M}$ then $A_1 \setminus A_2 \in \mathfrak{M}$. In particular, if $A_i \in \mathfrak{M}$ for $i \in \mathbb{N}$ then $\bigcap_{i \in \mathbb{N}} A_i = A_1 \setminus \bigcup_{i \in \mathbb{N}} (A_1 \setminus A_i) \in \mathfrak{M}$. In words, $\mathfrak{M}$ is closed under countable unions, set differences and consequently countable intersections. We will call $\mathfrak{M}$ the set of measurable sets. If $V \in \mathfrak{M}$, then $\mathfrak{M}$ is called a σ -algebra.
- there exist a countably additive function $\nu : \mathfrak{M} \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$, i.e. if $A_i \cap A_j = \emptyset$ for $i \neq j$ then $\nu(\bigcup_{i \in \mathbb{N}} A_i) = \sum_{i \in \mathbb{N}} \nu(A_i)$. The function ν is called a measure on V .

If $V \in \mathfrak{M}$ (that is $\mathfrak{M}$ is a σ -algebra) then V is called measurable. We will only consider measurable measure spaces, that is $(V, \mathfrak{M}, \nu)$ with $\mathfrak{M}$ a σ -algebra of measurable sets.

Exercise 16.10. Prove that if ν is a countably additive function then:

$$\nu(\emptyset) = 0, \quad \nu(A_1 \cup A_2) + \nu(A_1 \cap A_2) = \nu(A_1) + \nu(A_2), \quad \forall A_1, A_2 \in \mathfrak{M}.$$

Exercise 16.11. Prove that for any measure space if $A_i \in \mathfrak{M}$ such that $A_i \subset A_{i+1}$ then $\nu(\bigcup_{i \in \mathbb{N}} A_i) = \lim_{n \rightarrow \infty} \nu(A_n)$. Set $B_1 = A_1$ and $B_n = A_n - A_{n-1}$, hence B_n are mutually disjoint. By additivity, as $A_n = \bigcup_{i=1}^n B_i$ we have $\nu(A_n) = \sum_{i=1}^n \nu(B_i)$. As $\bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i$, we have by countable additivity $\nu(\bigcup_{i \in \mathbb{N}} A_i) = \sum_{i=1}^{\infty} \nu(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \nu(B_i)$ and we conclude.

Exercise 16.12. Prove that for any measure space if $A_i \in \mathfrak{M}$ such that $A_{i+1} \subset A_i$ and $\nu(A_k) < \infty$ for some k then $\nu(\bigcap_{i \in \mathbb{N}} A_i) = \lim_{n \rightarrow \infty} \nu(A_n)$. Assume $\nu(A_1) < \infty$ and set $B_n = A_n - A_{n+1}$. Then $A_1 - A_n = \bigcup_{i=k}^{n-1} B_i$. By the previous exercise, $\lim_{n \rightarrow \infty} \nu(A_1 - A_n) = \nu(\bigcup_{i=1}^{\infty} B_i) = \nu(A_1 - \bigcap_{n \in \mathbb{N}} A_n)$. By additivity, $\nu(A_1 \setminus A_n) = \nu(A_1) - \nu(A_n)$ resp. $\nu(A_1 \setminus \bigcap_{n \in \mathbb{N}} A_n) = \nu(A_1) - \nu(\bigcap_{n \in \mathbb{N}} A_n)$.

Exercise 16.13. Prove that the counting measure $\lambda(X)$ which assigns to any finite subset of $\mathbb{R}$ its cardinal $\lambda(X) = |X|$, and ∞ to any infinite set, is a measure.

Exercise 16.14. Prove that the Dirac measure $\delta_x(A) = \mathbf{1}_A(x)$ which is 0 if $x \notin A$ and 1 if $x \in A$ is a measure.

Exercise 16.15 (Algebra generating a σ -algebra). Let $(X, \mathfrak{M}, \mu)$ be a measurable measure space of finite measure, hence $\mathfrak{M}$ is a σ -algebra (closed under countable union, complements, countable intersections, $\emptyset, X \in \mathfrak{M}$ and $\mu(X) < \infty$). Let $\mathfrak{A}$ be an algebra of measurable sets, that is $\emptyset, X \in \mathfrak{A}$ and $\mathfrak{A}$ is closed under finite unions and complements (hence finite intersections). We denote the symmetric difference of two sets $A \Delta B = (A \cup B) \setminus (A \cap B)$. Assume $\mathfrak{A}$ generates $\mathfrak{M}$, that is $\mathfrak{M}$ is the smallest σ -algebra containing $\mathfrak{A}$. Then for all $B \in \mathfrak{M}$ and for all $\epsilon > 0$ there exists $A \in \mathfrak{A}$ such that $\mu(A \Delta B) = \mu(A \cup B) - \mu(A \cap B) < \epsilon$. We denote $\mathfrak{S} = \{B \in \mathfrak{M} \mid \forall \epsilon, \exists A, \mu(A \cup B) - \mu(A \cap B) < \epsilon\}$. It is enough to prove $\mathfrak{S}$ is a sigma algebra. Clearly $\emptyset, X \in \mathfrak{A} \subset \mathfrak{S}$. As $(A \cup B) \setminus (A \cap B) = (A^c \cup B^c) \setminus (A^c \cap B^c)$, $B \in \mathfrak{S} \implies B^c \in \mathfrak{S}$. In order to prove countable additivity we first prove $\mathfrak{S}$ is stable by finite unions: for $B_1, B_2 \in \mathfrak{S}$, chose $A_1, A_2 \in \mathfrak{A}$ such that $\mu(B_1 \Delta A_1) < \epsilon/2$. As $(B_1 \cup B_2) \Delta (A_1 \cup A_2) \subset (B_1 \Delta A_1) \cup (B_2 \Delta A_2)$ and $A_1 \cup A_2 \in \mathfrak{A}$, it follows from monotonicity that $B_1 \cup B_2 \in \mathfrak{S}$. For countable unions, let $\{B_k\}_{k \in \mathbb{N}} \subset \mathfrak{S}$ mutually disjoint and let $\{A_k\}_{k \in \mathbb{N}} \subset \mathfrak{A}$ such that $\mu(B_k \Delta A_k) < \epsilon 2^{-k}$. Let N such that $\mu(\bigcup_{j \geq N+1} B_j) < \epsilon/2$ (this is possible because $\mu(\bigcup_{j=1}^N B_j) = \sum_{j=1}^N \mu(B_j)$ is a monotonically increasing sequence bounded by $\mu(X)$, hence convergent). Let $A = \bigcup_{j=1}^N A_j$, taking into account that $(\bigcup_{j \geq 1} B_j) \Delta A \subset \bigcup_{j=1}^N (B_j \Delta A_j) \cup \bigcup_{j \geq N+1} B_j$ we conclude by sub-additivity.

Exercise 16.16 (Tensor product of σ -algebras). Let $\mathfrak{M}$ and $\mathfrak{N}$ to σ -algebras. We define $\mathfrak{E}_0 = \{A \times B \mid A \in \mathfrak{M}, B \in \mathfrak{N}\}$ the tensor product sets and $\mathfrak{E} = \{\bigcup_{i=1}^n C_i \mid C_i \in \mathfrak{E}_0, n \in \mathbb{N}\}$ the finite unions of tensor product sets. Prove that $\mathfrak{E}$ is an algebra. Prove that that sets in $\mathfrak{E}$ can be written as finite disjoint unions of sets in $\mathfrak{E}_0$. We denote $\mathfrak{M} \otimes \mathfrak{N}$ the σ -algebra generated by $\mathfrak{E}$.

An immediate consequence of Proposition 16.1 is the following.

Theorem 16.2 ($\mathbb{R}^d$ with Lebesgue measure is a measure space). For any set V endowed with an outer measure ν^* , the set of measurable sets with respect to ν^* , $\mathfrak{M}(\nu)$ (that is the sets which respect Carathéodory's criterion) is a σ -ring and ν^* restricted to $\mathfrak{M}(\nu)$ is countably additive, hence V is a measure space.

The set $\mathbb{R}^d$ with $\mathfrak{M}$ the Lebesgue measurable sets and the Lebesgue measure is a measure space.

From now on, “measurable” means measurable with respect to some measure.

16.2 The Lebesgue integral

Now that we have refined the notion of measure of a set, we will refine the notion of integral. As this is zero cost, we will consider integrals with respect to an arbitrary measure ν on a measure space V , of which the Lebesgue measure is an example.

Definition 16.7 (**Measurable simple functions**). A simple function is a function that takes only finitely many values.

A non-negative measurable simple function is a finite linear combination of indicator functions of disjoint measurable sets with non negative coefficients. In detail, let $\{A_i\}_{1 \leq i \leq n} \subset V$ with $\bigcup_{i=1}^n A_i = V$ be n mutually disjoint measurable sets and denote $\mathbf{1}_{A_i} : V \rightarrow \{0, 1\}$, $\mathbf{1}_{A_i}(x) = 1$ if $x \in A_i$ and 0 if not the indicator functions of these sets. Let $\{a_i\}_{1 \leq i \leq n}$ with $a_i \in \mathbb{R}_{\geq 0}$ be n non-negative real numbers. A measurable non-negative simple function $s : V \rightarrow \mathbb{R}_{\geq 0}$ is a function of the form:

$$s(x) = \sum_{i=1}^n a_i \mathbf{1}_{A_i}(x) .$$

A step function is a simple function associated to the partition of an interval $[a, b]$ into a finite number of disjoint intervals. The function $\mathbf{1}_{\mathbb{Q}}$ is a simple function which is not a step function.

Theorem 16.3 (**Measurable functions**). A function $f : V \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$ with V a measure space is called measurable if $\forall a \in \mathbb{R}$ the set $\{x | f(x) > a\}$ is measurable.³ If f is measurable then the set $\{x | f(x) \geq a\} = \bigcap_{n=1}^{\infty} \{x | f(x) > a - 1/n\}$ is measurable.

Any non-negative measurable function $f : V \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$ is the pointwise limit of a monotonic increasing sequence of measurable non-negative simple functions taking only rational values.

Proof. Let f be a non-negative measurable function. For each $n \in \mathbb{N}$, subdivide $\mathbb{R}_{\geq 0}$ into $2^{2n} + 1$ disjoint intervals:

$$I_{n,k} = \left[\frac{k}{2^n}, \frac{k+1}{2^n} \right) , \quad k = 0, 1, \dots, 2^{2n} - 1 , \quad I_{n,2^{2n}} = [2^n, \infty) .$$

As f is measurable, the sets $A_{n,k} = f^{-1}(I_{n,k})$ for $k = 0, 1, \dots, 2^{2n}$ are measurable and f is the pointwise limit of the increasing sequence of measurable simple functions:

$$s_n = \sum_{k=0}^{2^{2n}} \frac{k}{2^n} \mathbf{1}_{A_{n,k}} ,$$

as whenever $f(x) < \infty$, $|f(x) - s_n(x)| < 2^{-n}$, and $s_n(x) = 2^n$ if $f(x) = \infty$. □

Exercise 16.17. Prove that any continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is measurable with the Lebesgue measure. (c, ∞) is open hence $f^{-1}(c, \infty)$ is open hence Borel.

Exercise 16.18. If f_n is a sequence of measurable functions, then the functions $g(x) = \sup_n f_n(x)$ and $h(x) = \limsup f_n(x)$ are measurable. $\{x | g(x) > a\} = \bigcup_{n=1}^{\infty} \{x | f_n(x) > a\}$. $h(x) = \inf g_m(x)$, $g_m(x) = \sup_{n \geq m} f_n(x)$.

Exercise 16.19. Prove that the sum of two measurable functions is measurable. We have that $f(x) + g(x) > t$ if and only if $f(x) > t - g(x)$, which holds if and only if there exists $r \in \mathbb{Q}$ such that $f(x) > r > t - g(x)$ we have:

$$\{x | f(x) + g(x) > t\} = \bigcup_{r \in \mathbb{Q}} \left(\{x | f(x) > r\} \cap \{x | g(x) > t - r\} \right) .$$

To simplify notation we will omit the word “measurable” when referring to measurable simple functions, and call such function simple functions.

Definition 16.8 (**Lebesgue integral**). Let $X \subset \mathbb{R}^d$ be a measurable set. The Lebesgue integral of a non-negative simple function $s : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ with respect to the Lebesgue measure is:

$$\int_X d^d x \left(\sum_{i=1}^n a_i \mathbf{1}_{A_i}(x) \right) = \sum_{i=1}^n a_i m(X \cap A_i) ,$$

³More generally, a function between two measure spaces is called measurable if the preimage of any measurable set is measurable.

where the contribution is zero if $a_i = 0$ (even when $m(X \cap A_i) = \infty$). Observe that for $d = 1$ this reproduces the Riemann integral if the simple function is a step function, but covers a much larger class of functions. The Lebesgue integral of a non-negative measurable function $f: \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$ with respect to the Lebesgue measure is:

$$\int_X d^d x f(x) = \sup_{\text{simple functions } 0 \leq s(x) \leq f(x)} \left\{ \int_X d^d x s(x) \right\},$$

which can be ∞ .

This is immediately generalized to an arbitrary measure ν on a measure space V , $X \subset V$ measurable set and $f: X \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$ measurable function:

$$\int_X d\nu f(x) = \sup_{\text{simple functions } 0 \leq s(x) \leq f(x)} \left\{ \int_X d\nu s(x) \right\}, \quad \int_X d\nu \left(\sum_{i=1}^n a_i \mathbf{1}_{A_i}(x) \right) = \sum_{i=1}^n a_i \nu(X \cap A_i).$$

A measurable function is called Lebesgue integrable if its Lebesgue integral is finite.

An explicit sequence of non-negative simple functions which approximates the Lebesgue integral of f increasingly accurately is the sequence s_n in Theorem 16.3, that is the Lebesgue integral of f can be approximated as:

$$\int_X d\nu f(x) = \lim_{n \rightarrow \infty} \int_X d\nu s_n, \quad s_n = \sum_{k=0}^{2^n-1} \frac{k}{2^n} \nu(f^{-1}([\frac{k}{2^n}, \frac{k+1}{2^n})) \cap X) + 2^n \nu(f^{-1}([2^n, \infty)) \cap X),$$

which writes simply as:

$$\sum_{k=1}^{2^{2n}-1} \frac{1}{2^n} [k \nu(f^{-1}([\frac{k}{2^n}, \infty)) \cap X) - k \nu(f^{-1}([\frac{k+1}{2^n}, \infty)) \cap X)] + 2^n \nu(f^{-1}([2^n, \infty)) \cap X) = \sum_{k=1}^{2^{2n}-1} \frac{1}{2^n} \nu(f^{-1}([\frac{k}{2^n}, \infty)) \cap X).$$

The attentive reader will recognize that this is nothing but a Riemann sum approximating the one dimensional Riemann integral of the function $\tilde{f}(t) = \nu(\{x \in V \cap X | f(x) > t\})$,⁴ that is $\int_X d\nu f(x) = \int_0^\infty dt \tilde{f}(t)$ where the right hand side is just the normal (improper) Riemann integral of $\tilde{f}(t)$. In effect, in order to compute the Lebesgue integral of a function one partitions the range of f and sums the volume of the horizontal slabs of the preimage.

From now on “ $\int_X d^d x$ ” denotes the Lebesgue integral with respect to the Lebesgue measure and “ $\int_X d\nu$ ” denotes the Lebesgue integral with respect to the measure ν .

The Lebesgue integral is readily extended to real valued functions by splitting $f = f^+ - f^-$ with $f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = -\min\{f(x), 0\}$, and f is Lebesgue integrable if each one of f^+ and f^- is integrable.

Exercise 16.20. *Extend the Lebesgue integral to real and complex valued measurable functions.*

Exercise 16.21. *Prove that the Lebesgue integral reproduces the Riemann integral whenever the latter is defined.*

Exercise 16.22. *Show that for any measurable function $\rho: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, the function $\nu(A) = \int_A dx \rho(x)$ for A Lebesgue measurable is a measure. We call $\rho(x) = \frac{d\nu}{dx}$ the Radon–Nikodym derivative of ν with respect to the Lebesgue measure.*

Exercise 16.23. *Prove that for any function $f: X \rightarrow \mathbb{R}_{\geq 0}$, if $x_0 \in X$ then $\int_X d\delta_{x_0} f(x) = f(x_0)$ and if $x \notin X$ then $\int_X d\delta_{x_0} f(x) = 0$. We sometimes denote improperly as $\int_X d\delta_{x_0} f(x) = \int_X dx \delta(x - x_0) f(x)$. For a simple function $\int_X d\delta_{x_0} s(x) = \sum_{i=1}^n a_i \delta_{x_0}(X \cap A_i) = a_{i_0}$, with $x_0 \in X \cap A_{i_0}$, hence $\int_X d\delta_{x_0} s(x) = s(x_0)$ and taking an increasing sequence of simple functions that converges pointwise to f we conclude.*

⁴The strict inequality in the definition of $\tilde{f}$ comes from the fact that the Riemann sum starts at $k = 1$.

Exercise 16.24 (Tensor product measure). Let ν_1 be a measure on X_1 and ν_2 a measure on X_2 . A measure on $X_1 \times X_2$ satisfying $(\nu_1 \times \nu_2)(B_1 \times B_2) = \nu_1(B_1)\nu_2(B_2)$ for B_1 measurable with respect to ν_1 and B_2 with respect to ν_2 is called the product measure on $X_1 \times X_2$, where the product is zero if one of the factors is zero (even if the second one is infinite). The product measure is not unique in general, but it is unique if both X_1 and X_2 can be written as countable unions of measurable sets of finite measure⁵ (i.e. are σ -finite). In this case, any $E \in X_1 \times X_2$, $\nu_1 \times \nu_2(E) = \inf \sum_{n \in \mathbb{N}} \nu_1(A_n)\nu_2(B_n)$ for $\bigcup_{n \in \mathbb{N}} (A_n \times B_n) \supset E$ and A_n (resp. B_n) is measurable with respect to ν_1 (resp. ν_2), that is:

$$\nu_1 \times \nu_2(E) = \int_{X_2} \nu_1(E^y) d\nu_2(y) = \int_{X_1} \nu_2(E_x) d\nu_1(x),$$

where $E^y = \{x \in X_1 | (x, y) \in E\}$ and $E_x = \{y \in X_2 | (x, y) \in E\}$ and are both measurable sets.

Consider the square $[0, 1] \times [0, 1]$ and its diagonal $\Delta = \{(x, x) | x \in [0, 1]\}$. Prove that Δ :

- Δ is measurable and has zero measure in the product of Lebesgue measures $m \times m$. As $[0, 1]$ is σ -finite for the Lebesgue measure, the product measure is unique. We cover the diagonal by A_n , the union of squares stashed along the diagonal of side length $\frac{1}{2^n}$. Then $\Delta = \bigcap_{n \geq 1} A_n$, $A_{n+1} \subset A_n$ and $m \times m(A_n) = 2^n \cdot \frac{1}{4^n}$.
- as $[0, 1]$ is not σ -finite for the counting measure, the product measures between the Lebesgue and the counting measure $m \times \lambda$ is not unique. Two versions are:

$$(m \times \lambda)_1(E) = \int_{[0,1]} m(E^y) d\lambda(y), \quad (m \times \lambda)_2(E) = \int_{[0,1]} \lambda(E_x) dm(x).$$

Compute the measure of Δ in the two product measures.

We have $\Delta^y = \{y\}$ and $\Delta_x = \{x\}$ and $\int_{[0,1]} m(\Delta^y) d\lambda(y) = \int_{[0,1]} 0 d\lambda(y) = 0$ respectively $\int_{[0,1]} \lambda(\Delta_x) dm(x) = \int_{[0,1]} dm(x) = 1$.

Theorem 16.4 (Properties of the Lebesgue integral). Let X be a measurable set with respect to the measure ν and denote $L(X, \nu)$ the set of Lebesgue integrable functions on X . The Lebesgue integral has the following properties:

- (monotonicity) if f, g are Lebesgue integrable and $0 \leq f(x) \leq g(x)$ then $\int_X d\nu f(x) \leq \int_X d\nu g(x)$; if f is integrable $\int_X d\nu cf(x) = c \int_X d\nu f(x)$.

Proof. The first statement follows by noting that any simple function with $0 \leq s \leq f$ is such that $0 \leq s \leq g$. □

- (respects equivalence up to a set of measure zero) if $\nu(X) = 0$ and f is integrable then $\int_X d\nu f(x) = 0$; if f, g are measurable functions and $f = g$ almost everywhere then $\int_X d\nu f(x) = \int_X d\nu g(x)$.
- (the integral is countably additive) for $f : X \rightarrow \mathbb{R}_{\geq 0}$ a measurable function, the function:

$$\phi(A) = \int_A d\nu f(x), \quad A \in \mathfrak{M}(\nu),$$

is countably additive on $\mathfrak{M}(\nu)$.

Proof. Let $A = \bigcup_{n=1}^{\infty} A_n$ with $A_i \cap A_j = \emptyset$. We aim to prove that $\phi(A) = \sum_{n=1}^{\infty} \phi(A_n)$. If f is the indicator function of some measurable set $f = \mathbf{1}_E$ then this is equivalent to $\nu(A \cap E) = \sum_{i=1}^n \nu(A_i \cap E)$ which holds as ν is countably additive. If f is a simple function, the conclusion holds again.

In the general case, for every measurable simple function $0 \leq s \leq f$ we have (using the definition of the integral for the last implication):

$$\int_A d\nu s(x) = \sum_{n=1}^{\infty} \int_{A_n} d\nu s(x) \leq \sum_{n=1}^{\infty} \phi(A_n) \Rightarrow \phi(A) \leq \sum_{n=1}^{\infty} \phi(A_n).$$

⁵This is known as Carathéodory's extension theorem.

As $f \geq \mathbf{1}_{A_n} f$ we have $\phi(A) \geq \phi(A_n)$ and the conclusion holds if $\phi(A_n) = \infty$. We thus assume $\phi(A_n) < \infty$ for all n . Then for any ϵ we can choose a measurable simple function $0 \leq s \leq f$ such that:

$$\int_{A_1} d\nu s(x) \geq \int_{A_1} d\nu f(x) - \epsilon, \quad \int_{A_2} d\nu s(x) \geq \int_{A_2} d\nu f(x) - \epsilon \Rightarrow \phi(A_1 \cup A_2) \geq \int_{A_1 \cup A_2} d\nu s(x) \geq \phi(A_1) + \phi(A_2) - 2\epsilon,$$

hence for every n , $\phi(A_1 \cup \dots \cup A_n) \geq \phi(A_1) + \dots + \phi(A_n)$ and as $A \supset A_1 \cup \dots \cup A_n$ we have $\phi(A) \geq \phi(A_1) + \dots + \phi(A_n)$ for any n and we conclude. $\square$

- (standard bound) if f is integrable then $|f|$ is integrable and $|\int_X d\nu f(x)| \leq \int_X d\nu |f(x)|$; if g is integrable, f is measurable and $|f| < g$ then f is integrable.

Proof. Separate in $f = f^+ - f^-$. $\square$

- (Beppo Levi or monotone convergence theorem) if $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of non-negative measurable functions such that $f_n(x) \leq f_{n+1}(x)$ then the pointwise limit function $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ (where the limit is taken in the extended real numbers, including ∞) is measurable and:

$$\lim_{n \rightarrow \infty} \int_X d\nu f_n(x) = \int_X d\nu f(x),$$

which can be infinite.

Proof. We have that $f(x)$ is measurable as $\{x | f(x) > a\} = \bigcup_n \{x | f_n(x) > a\}$. As $\int_X d\nu f_n(x)$ is monotonic, $\lim_{n \rightarrow \infty} \int_X d\nu f_n(x) = \alpha \leq \int_X d\nu f(x)$.

Now, choose c with $0 < c < 1$, let s be a simple function with $0 \leq s \leq f$ and define $A_n = \{x | f_n(x) \geq cs(x)\}$, which is a measurable set. As f_n is monotonic, $A_1 \subset A_2 \subset \dots$ and as $f_n(x) \rightarrow f(x)$, we have $X = \bigcup_{n=1}^{\infty} A_n = \bigcup_{i=1}^{\infty} (A_{i+1} \setminus A_i)$.

For every n we have $\int_X d\nu f_n(x) \geq \int_{A_n} d\nu f_n(x) \geq c \int_{A_n} d\nu s(x)$ and as the integral is countably additive:

$$\alpha = \lim_{n \rightarrow \infty} \int_X d\nu f_n(x) \geq c \lim_{n \rightarrow \infty} \int_{A_n} d\nu s(x) = c \lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} \int_{A_{i+1} \setminus A_i} d\nu s(x) = c \int_{\bigcup_{i=1}^{\infty} (A_{i+1} \setminus A_i)} d\nu s(x),$$

and taking the supremum over s and sending $c \rightarrow 1$ we conclude that $\alpha \geq \int_X d\nu f(x)$. $\square$

- (linearity) if f_1, f_2 are Lebesgue integrable on X then $f_1 + f_2$ is Lebesgue integrable and:

$$\int_X d\nu [f_1(x) + f_2(x)] = \int_X d\nu f_1(x) + \int_X d\nu f_2(x).$$

In particular, in conjunction with monotone convergence theorem, we conclude that if $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of non-negative measurable functions and $f(x) = \sum_{n=1}^{\infty} f_n(x)$ then $\int_X d\nu f(x) = \sum_{n=1}^{\infty} \int_X d\nu f_n(x)$.

Proof. If f_1, f_2 are non-negative simple functions $s = \sum_{i=1}^n a_i \mathbf{1}_{A_i}(x)$ and $v = \sum_{j=1}^m b_j \mathbf{1}_{B_j}(x)$ we have:

$$a \int_X d\nu s(x) + b \int_X d\nu v(x) = a \sum_{i=1}^n a_i \nu(A_i) + b \sum_{j=1}^m b_j \nu(B_j) = \sum_{i=1}^n \sum_{j=1}^m (aa_i + bb_j) \nu(A_i \cap B_j),$$

where we used the fact that $\bigcup_{i=1}^n A_i = \bigcup_{j=1}^m B_j = X$. If not f_1, f_2 are non-negative and measurable, choose $s_n \rightarrow f_1$ and $v_n \rightarrow f_2$ simple, set $w_n = s_n + v_n$ and use the monotone convergence theorem to conclude. If f_1 and f_2 change sign, divide the integration range into subsets where they have fixed sign. $\square$

- (Fatou's lemma) if $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of non-negative measurable functions then:

$$\int_X d\nu \liminf_n f_n(x) \leq \liminf_n \int_X d\nu f_n(x) .$$

Proof. Set $g_n(x) = \inf_{l \geq n} f_l(x)$. Then g_n are measurable, $0 \leq g_1(x) \leq g_2(x) \leq \dots$, moreover $\lim_{n \rightarrow \infty} g_n(x) \rightarrow f(x) = \liminf_n f_n(x)$ and $g_n(x) \leq f_n(x)$. By the monotone convergence theorem:

$$\int_X d\nu f(x) = \lim_n \int_X d\nu g_n(x) = \liminf_n \int_X d\nu g_n(x) \leq \liminf_n \int_X d\nu f_n(x) ,$$

where we used the fact that $\int_X d\nu g_n(x)$ is monotonically increasing. $\square$

- (Lebesgue's dominated convergence theorem) if $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of measurable functions with pointwise limit $\lim_n f_n(x) = f(x)$ and there exists a Lebesgue integrable non-negative function $g : X \rightarrow \mathbb{R}_{\geq 0}$, $\int_X d\nu g(x) < \infty$ such that $\forall n, |f_n(x)| \leq g(x)$ almost everywhere then f is Lebesgue integrable and:

$$\lim_n \int_X d\nu f_n(x) = \int_X d\nu f(x) .$$

Proof. Without loss of generality we can assume that $\forall n, |f_n(x)| \leq g(x)$ holds everywhere. From the standard bound, f_n and f are integrable and, since $f_n + g > 0$ by Fatou's lemma:

$$\int_X d\nu [f(x) + g(x)] \leq \liminf_n \int_X d\nu [f_n(x) + g(x)] \Rightarrow \int_X d\nu f(x) \leq \liminf_n \int_X d\nu f_n(x) .$$

Since $g - f_n \geq 0$ we similarly obtain:

$$-\int_X d\nu f(x) \leq \liminf_n \left[-\int_X d\nu f_n(x) \right] \Rightarrow \int_X d\nu f(x) \geq \limsup_n \int_X d\nu f_n(x) ,$$

As $\limsup_n \int_X d\nu f_n(x) \geq \liminf_n \int_X d\nu f_n(x)$ we conclude that $\limsup_n \int_X d\nu f_n(x) = \liminf_n \int_X d\nu f_n(x)$ hence $\lim_n \int_X d\nu f_n(x)$ exists and equals the common value of $\limsup$ and $\liminf$. $\square$

Furthermore:

- (Fubini-Tonelli theorem) if ν_1 is a measure on X_1 , ν_2 a measure on X_2 , both X_1 and X_2 are σ -finite⁶ and $f : X_1 \times X_2$ is a measurable function with respect to the product measure $\nu_1 \times \nu_2$, then:

$$\int_{X_1} d\nu_1(x) \left(\int_{X_2} d\nu_2(y) |f(x, y)| \right) = \int_{X_2} d\nu_2(y) \left(\int_{X_1} d\nu_1(x) |f(x, y)| \right) = \int_{X_1 \times X_2} d(\nu_1 \times \nu_2) |f(x, y)| ,$$

and if any of these integrals is finite then they all are and:

$$\int_{X_1} d\nu_1 \left(\int_{X_2} d\nu_2 f \right) = \int_{X_2} d\nu_2 \left(\int_{X_1} d\nu_1 f \right) = \int d(\nu_1 \times \nu_2) f .$$

- (change of variables) if X_1 and X_2 are measure spaces, $T : X_1 \rightarrow X_2$ is a one to one measurable function, and $\nu(\cdot)$ is a measure on X_1 then $\nu \circ T^{-1}$ is a measure on X_2 , a measurable function $f : X_2 \rightarrow \mathbb{R}, \mathbb{C}$ is integrable with respect to $\nu \circ T^{-1}$ if and only if $f \circ T$ is integrable with respect to ν and:

$$\int_{T(A)} d(\nu \circ T^{-1}) f = \int_A d\nu f \circ T , \quad A \subset X_1 .$$

⁶That is each of them can be covered by a countable union of measurable sets of finite measure. This is needed in order for the product measure to be uniquely defined.

- (differentiation under the integral sign) Let X be an open subset of $\mathbb{R}$ and Ω a measure space. If $f : X \times \Omega \rightarrow \mathbb{R}$ and:
 - $f(x, \omega)$ is Lebesgue integrable in ω for all $x \in X$,
 - for almost all $\omega \in \Omega$ the partial derivative $\frac{\partial f}{\partial x}(x, \omega)$ exists,
 - there exist a Lebesgue integrable function $\theta(\omega)$ such that $|\partial_x f(x, \omega)| \leq \theta(\omega)$ for all $x \in X$ almost everywhere in Ω ,

then for all $x \in X$:

$$\frac{d}{dx} \int_{\Omega} d\omega f(x, \omega) = \int_{\Omega} d\omega \partial_x f(x, \omega) .$$

As an example of non commutation of integral and limit we can take $f_n(x) = n \mathbf{1}_{[0, 1/n]}(x)$, which is neither monotonic nor dominated.

Let f be a function $f : [0, 1] \times [0, 1]$ continuous everywhere except at $(0, 0)$. As f is continuous and (c, ∞) is open, $f^{-1}(c, \infty)$ is open hence Borel (and consequently Lebesgue measurable), that is f is a measurable function. Consider $f = \frac{x^2 - y^2}{(x^2 + y^2)^2}$ and compute:

$$\begin{aligned} \int_0^1 dx \int_0^1 dy \frac{x^2 - y^2}{(x^2 + y^2)^2} &= \int_0^1 dx \left[\frac{y}{x^2 + y^2} \right]_0^1 = \int_0^1 dx \frac{1}{1 + x^2} = \frac{\pi}{4} , \\ \int_0^1 dy \int_0^1 dx \frac{x^2 - y^2}{(x^2 + y^2)^2} &= \int_0^1 dy \left[\frac{-x}{x^2 + y^2} \right]_0^1 = \int_0^1 dx \frac{-1}{1 + x^2} = -\frac{\pi}{4} . \end{aligned}$$

In order to understand what went wrong, we compute:

$$\int_0^1 dx \int_0^1 dy \left| \frac{x^2 - y^2}{(x^2 + y^2)^2} \right| \geq \int_0^1 dx \int_0^x dy \frac{x^2 - y^2}{(x^2 + y^2)^2} = \int_0^1 dx \left[\frac{y}{x^2 + y^2} \right]_0^x = \int_0^1 dx \frac{1}{2x} = \infty .$$

16.2.1 The L^p spaces

Let X be a measure space with measure ν . We set $1 \leq p < \infty$ and we define the space $L^p(X, \mathbb{C})$ as the space of (equivalence classes up to equality almost everywhere) measurable functions $f : X \rightarrow \mathbb{C}$ such that:

$$\|f\|_p \equiv \left(\int_X d\nu |f(x)|^p \right)^{1/p} < \infty .$$

These are straightforward generalization of the ℓ^p spaces we have encountered previously.

Theorem 16.5 (*L^p spaces are Banach spaces, L^2 is a separable Hilbert space*). The spaces $L^p(X, \mathbb{C})$, $1 \leq p < \infty$, endowed with $\|\cdot\|_p$ are Banach spaces, separable for $p < \infty$. The space $L^2(X, \mathbb{C})$ endowed with inner product:

$$\langle f, g \rangle = \int_X d\nu \overline{f(x)} g(x) ,$$

is a separable Hilbert space.

Proof. The reader should consult the literature on the topic.

We first prove that the L^p norms respect the triangle inequality. As a prerequisite we prove Hölder's inequality: for $p, q \in [1, \infty)$ with $1/p + 1/q = 1$ we have $\|fg\|_1 \leq \|f\|_p \|g\|_q$. Without loss of generality we assume that the right hand side is neither zero nor infinity and we note that the claim is trivial if $p = 1$. We have⁷ $|u(s)v(s)| \leq \frac{|u(s)|^p}{p} + \frac{|v(s)|^q}{q}$, hence

⁷As the logarithm is concave, for $a, b > 0$ and $1/p + 1/q = 1$ we have pointwise $\ln(\frac{1}{p}a^p + \frac{1}{q}b^q) \geq \frac{1}{p} \ln a^p + \frac{1}{q} \ln b^q = \ln(ab)$.

$\|uv\|_1 \leq \frac{\|u\|_p^p}{p} + \frac{\|v\|_q^q}{q}$, and the claim follows by taking $u = f/\|f\|_p$ and $v = g/\|g\|_q$. Now, as the function $|\cdot|^p$ is convex, we have:

$$\left| \frac{f+g}{2} \right|^p \leq \frac{1}{2}|f|^p + \frac{1}{2}|g|^p \Rightarrow |f+g|^p \leq 2^{p-1}(|f|^p + |g|^p),$$

hence if $f, g \in L^p$ then $f+g \in L^p$. Then:

$$\begin{aligned} \|f+g\|_p^p &= \int d\nu |f+g| |f+g|^{p-1} \leq \int d\nu (|f|+|g|) |f+g|^{p-1} = \int d\nu |f| |f+g|^{p-1} + \int d\nu |g| |f+g|^{p-1} \\ &\leq (\|f\|_p + \|g\|_p) \left(\int d\nu |f+g|^{(p-1)\frac{p}{p-1}} \right)^{\frac{p-1}{p}}, \end{aligned}$$

and we conclude.

We will now prove that $L^2(X, \mathbb{R})$ is Cauchy complete. Consider a Cauchy sequence of functions f_n in L^2 . It is enough to prove that the subsequence $\|f_n - f_m\|_{L^2} \leq 2^{-\min(m,n)}$ converges in L^2 norm to some function $f \in L^2$. Let g_n :

$$g_1 = |f_1|, \quad g_n = |f_1| + |f_2 - f_1| + \cdots + |f_n - f_{n-1}|,$$

which is a sequence of nonnegative functions such that:

$$\|g_n\|_{L^2} = \left(\int_X d\nu |g_n(x)|^2 \right)^{1/2} \leq \|f_1\|_{L^2} + \sum_{n \geq 2} \|f_n - f_{n-1}\|_{L^2} \leq \|f_1\|_{L^2} + 1 < \infty.$$

The sequence g_n^2 is a monotone sequence of measurable functions⁸ that converges pointwise to some $g^2(x)$, an extended valued function (i.e. $g^2(x)$ can be $+\infty$). From the monotone convergence theorem $g^2(x)$ is measurable, and $\lim_n \|g_n\|_{L^2}^2 = \lim_n \int_X d\nu |g_n(x)|^2 = \int_X d\nu |g(x)|^2 < \infty$.

It follows that $|g(x)|^2$ is integrable, hence $g^2(x)$ and consequently $g(x)$ are finite almost everywhere. From $|f_n(x) - f_m(x)| \leq |g_n(x) - g_m(x)|$ it follows that for x outside a set of measure zero $f_n(x)$ is a Cauchy sequence and converges to some $f(x)$. As $|f_n| \leq g_n \leq g$ almost everywhere we conclude from the dominated convergence theorem that $f \in L^2$. As on the other hand $|f_n(x) - f(x)|^2 \leq 4g^2(x)$ we have (using again the dominated convergence theorem):

$$\lim_n \int_X d\nu |f_n(x) - f(x)|^2 = 0 \Rightarrow \lim_n \|f_n - f\|_{L^2} = 0.$$

□

Theorem 16.6 ($\mathcal{C}_c^\infty(\mathbb{R}, \mathbb{C})$ is dense in $L^p(\mathbb{R}, \mathbb{C})$). A function $f : X \rightarrow \mathbb{C}$ is said to have compact support if the closure of the support of f , $\overline{\{x \in X | f(x) \neq 0\}}$ is compact. The space $\mathcal{C}_c^\infty(\mathbb{R}, \mathbb{C})$ of smooth functions with compact support is dense in $L^p(\mathbb{R}, \mathbb{C})$ with Lebesgue measure $d^d x$, in the sense of norm L^p for $1 \leq p < \infty$. The same holds in $\mathbb{R}^d$.

Proof. Without loss of generality we restrict to non negative real valued functions $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$. The proof is divided in three steps:

The simple functions are dense in $L^p(\mathbb{R})$. As $f \in L^p$, $|f|^p$ is integrable hence measurable. Therefore $|f|^p$ is the pointwise limit of an increasing sequence of measurable non negative simple functions t_N and denoting $s_N = t_N^{1/p}$ we obtain an increasing sequence of measurable non negative simple functions s_N such that pointwise $s_N(x) \rightarrow f(x)$ (we write $f = f^+ - f^-$ is f takes negative values). As $|f(x) - s_N(x)|^p \leq |f(x)|^p$ everywhere, by the dominated convergence theorem $\|f - s_N\|_{L^p} \rightarrow 0$. Observe that s_N is non zero on a set of finite measure.

⁸If g_n is measurable then g_n^2 is measurable.

Measurable simple functions are finite linear combinations of indicator functions of measurable sets (taking only rational values) therefore it is enough to prove that any indicator function $\mathbf{1}_A$ of a measurable set A with finite measure can be arbitrarily well approximated in L^p norm by a continuous function with compact support.

Smooth approximations of indicator functions for compact sets. Let $K \subset U$ with K compact and U open. Then there exist a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that:

- g is smooth and $0 \leq g(x) \leq 1$, for all $x \in \mathbb{R}$,
- $g(x) = 1$ for $x \in K$ and $g(x) = 0$ for $x \notin U$.

As $K \subset U$, for any $x \in K$, there exists δ_x such that $(x - 2\delta_x, x + 2\delta_x) \in U$. Considering the open cover $K \subset \bigcup_{x \in K} (x - \delta_x, x + \delta_x)$ we extract the finite cover $K \subset \bigcup_{i=1}^{i_m} (x_i - \delta_{x_i}, x_i + \delta_{x_i})$ and $[x_i - \delta_{x_i}, x_i + \delta_{x_i}] \subset U$.

As U is an open subset of $\mathbb{R}$, it is the countable union of disjoint open intervals $U = \bigcup_{j \in \mathbb{N}} I_j$. Only finitely many of these intervals contain balls in the finite cover of K . Say I_1 is one of them and say it contains the closed balls $[x_i - \delta_{x_i}, x_i + \delta_{x_i}] \in I_1$ for $i = 1, \dots, i_{m_1}$. We denote $I_1 = (a, b)$ and $a' = \min_{i=1, \dots, i_{m_1}} \{x_i - \delta_{x_i}\}$ respectively $b' = \max_{i=1, \dots, i_{m_1}} \{x_i + \delta_{x_i}\}$. We define g to be a smooth interpolation between 0 and 1 on the interval (a, a') :

$$g(x)|_{x \in (a, a')} = \frac{1}{\eta(1)} \eta\left(\frac{2x - a - a'}{a' - a}\right), \quad \eta(x) = \int_{-1}^x dy e^{-\frac{1}{1-y^2}},$$

$g(x) = 1$ for $x \in [a', b']$ and $g(x)$ is a smooth interpolation between 1 and 0 on (b', b) . We repeat the procedure for all the intervals that contain balls in the finite cover of K and set $g(x) = 0$ outside U .

Smooth approximations of indicator functions for measurable sets. From Proposition 16.1, if A is measurable then $\forall \epsilon$ there exist F a closed set and G an open one such that $F \subset A \subset G$ and $m(G \setminus F) < \epsilon$. If A has finite measure, then both F and G are bounded and F is compact. By the previous item, there exist a smooth function g such that:

$$|g(x) - \mathbf{1}_A(x)| \leq 1, \quad x \in G \setminus F, \quad |g(x) - \mathbf{1}_A(x)| = 0, \quad x \notin G \setminus F,$$

therefore:

$$\int_{\mathbb{R}} |g(x) - \mathbf{1}_A(x)|^p dx \leq \int_{G \setminus F} dx = m(G \setminus F) < \epsilon,$$

hence smooth functions are dense in the set of indicator functions of measurable sets of finite measure. □

Exercise 16.25 (Tensor product of L^2 spaces). Denoting $L^2(X, \mu)$ the space of functions $f : X \rightarrow \mathbb{C}$ that are square integrable $\int d\mu |f|^2 < \infty$. Prove that $L^2(X, \mu) \otimes L^2(Y, \nu) = L^2(X \times Y, \mu \times \nu)$, where $\otimes$ denotes the Hilbert space tensor product (and we assume μ and ν are σ -finite so that the product measure exists). In particular a Hilbert basis in $L^2(X \times Y, \mu \times \nu)$ can be built from the tensor product of a Hilbert basis in $L^2(X, \mu)$ with one in $L^2(Y, \nu)$. We denote $\mathfrak{M}$ and $\mathfrak{N}$ the σ -algebras of measurable sets in X respectively Y . As indicator function are dense in $L^2(X \times Y, \mathbb{C}, \mu \times \nu)$, it is enough to prove that for every $C \in \mathfrak{M} \times \mathfrak{N}$ measurable in the product measure such that $\mu \times \nu(C) < \infty$, the indicator function $\mathbf{1}_C(x, y) \in \overline{\text{Span}\{\mathbf{1}_A(x)\mathbf{1}_B(y)\}}$ with $A \in \mathfrak{M}$ and $B \in \mathfrak{N}$. We first assume X and Y have finite measure (hence also $X \times Y$). We define $\mathfrak{E}_0 = \{A \times B \mid A \in \mathfrak{M}, B \in \mathfrak{N}\}$ the tensor product sets and $\mathfrak{E} = \{\bigcup_{i=1}^n C_i \mid C_i \in \mathfrak{E}_0, n \in \mathbb{N}\}$ the finite unions of tensor product sets which generate $\mathfrak{M} \otimes \mathfrak{N}$. The indicator functions of the sets in $\mathfrak{E}_0$ are tensor products. The sets in $\mathfrak{E}$ can be written as finite disjoint unions of sets in $\mathfrak{E}_0$, hence their indicator functions are finite linear combinations of tensor products of indicator functions. Let $C \in \mathfrak{M} \times \mathfrak{N}$: by the exercise 16.15 there exists $E \in \mathfrak{E}$ such that $\mu \otimes \nu(C \Delta E) < \epsilon$ hence $\|\mathbf{1}_C - \mathbf{1}_E\|^2 = \|\mathbf{1}_{C \Delta E}\|^2 = \mu(\mathbf{1}_{C \Delta E}) < \epsilon$.

We now consider the case when X and Y are σ -finite. There exist increasing sequences of sets of finite measure E_n and F_n such that $X = \bigcup_{n \in \mathbb{N}} E_n$ and $Y = \bigcup_{n \in \mathbb{N}} F_n$. Let $f \in L^2(X \times Y)$ and define $f_n = \mathbf{1}_{E_n \times F_n} f$. For any $(x, y) \in X, Y$ we have $\lim_n \mathbf{1}_{E_n \times F_n}(x, y) = \lim_n \mathbf{1}_{E_n}(x) \mathbf{1}_{F_n}(y) = 1$ hence $f_n \rightarrow f$ pointwise and, as $|f_n(x, y)| \leq |f(x, y)|$, by the dominated convergence theorem $f_n \rightarrow f$ in L^2 . Let $\epsilon > 0$ and k such that $\|f - f_k\| < \epsilon$, and note that f_k is zero outside $E_k \times F_k$. We note that $(E_k, \mathfrak{M}|_{E_k}, \mu)$ and $(F_k, \mathfrak{N}|_{F_k}, \nu)$ are finite measure and $\mathfrak{M}|_{E_k} \otimes \mathfrak{N}|_{F_k} = \mathfrak{M} \otimes \mathfrak{N}|_{E_k \times F_k}$. By the finite measure case, there exists $g \in L^2(E_k \times F_k) \subset L^2(X \times Y)$ with $\|g - f_k\|_{E_k \times F_k} < \epsilon$ hence $\|g - f\| \leq \|g - f_k\| + \|f_k - f\| < 2\epsilon$.

Exercise 16.26 (Hilbert-Schmidt operators). Consider $\Omega \in \mathbb{R}$ and the operator $K : L^2(\Omega) \rightarrow L^2(\Omega)$ defined by $(Kf)(x) = \int_{\Omega} dy K(x, y) f(y)$ with $K(\cdot, \cdot) : \Omega \times \Omega \rightarrow \mathbb{C}$ a kernel such that $\int_{\Omega \times \Omega} dx dy |K(x, y)|^2 = \|K\|_{L^2(\Omega \times \Omega)}^2 < \infty$ (hence $\int dy |K(x, y)|^2 <$

∞ almost everywhere). Prove that K is compact. We first prove that K is bounded. We have:

$$|Ku(x)| = \int dy K(x, y)u(y) \leq \left(\int dy |K(x, y)|^2 \right)^{1/2} \left(\int dy |u(y)|^2 \right)^{1/2} \Rightarrow \|Ku\|^2 = \int dx |(Ku)(x)|^2 \leq \|u\|^2 \int dx \int dy |K(x, y)|^2.$$

Let $\{e_k\}_{k \in \mathbb{N}}$ be a Hilbert basis in $L^2(\Omega)$. At fixed x , $K(x, t)$ is such that $\int dt |K(x, t)|^2 < \infty$ hence $\overline{K(x, \cdot)} \in L^2$ and expands on the basis e_k as $\overline{K(x, t)} = \sum_k e_k(t) \int du \overline{e_k(u)} \overline{K(x, u)}$. It follows that:

$$\sum_k \|Ke_k\|^2 = \sum_k \int dx |(Ke_k)(x)|^2 \stackrel{\text{Beppo-L}}{=} \int dx \sum_k |(Ke_k)(x)|^2 = \int dx \sum_k \int dt K(x, t)e_k(t) \int du \overline{K(x, u)} \overline{e_k(u)} \stackrel{K \text{ cont. op.}}{=} \int dx \int dt |K(x, t)|^2.$$

An operator such that $\sum_k \|Ke_k\| < \infty$ for $\{e_k\}$ a Hilbert basis is called a Hilbert-Schmidt operator. Any Hilbert-Schmidt operator is the limit in norm of a sequence of finite rank operators. For any f we have $f = \sum_k \langle e_k, f \rangle e_k$ hence $Kf = \sum \langle e_k, f \rangle Ke_k$. The operator $K_N f = \sum_{k=1}^N \langle e_k, f \rangle Ke_k$ is finite rank (as its image is spanned by $\{Ke_i\}_{i=1 \dots N}$) and:

$$\|(K - K_N)f\| = \left\| \sum_{k>N} \langle e_k, f \rangle Ke_k \right\| \leq \sum_{k>N} |\langle e_k, f \rangle| \|Ke_k\| \leq \left(\sum_{k>N} |\langle e_k, f \rangle|^2 \right)^{1/2} \left(\sum_{k>N} \|Ke_k\|^2 \right)^{1/2} \leq \|f\| \left(\sum_{k>N} \|Ke_k\|^2 \right)^{1/2}.$$

Chapter 17

The Fourier transform

17.1 Fourier series

The functional space $L^2([0, T], \mathbb{C})$ space with inner product $\langle f, g \rangle = \int_0^T dt \overline{f(t)} g(t)$ and induced norm $\|f\| = \sqrt{\langle f, f \rangle}$ is a Hilbert space.¹ We will now show it is separable by exhibiting a Hilbert basis, that is a countable orthonormal set $\{e_k\}_{k \in \mathbb{Z}} \in L^2([0, T], \mathbb{C})$ such that any $f \in L^2([0, T], \mathbb{C})$ writes uniquely as:

$$f(x) = \lim_{N \rightarrow \infty} \sum_{|k| \leq N} \langle e_k, f \rangle e_k(x) \equiv \sum_{k \in \mathbb{Z}} \langle e_k, f \rangle e_k(x), \quad \langle e_k, f \rangle = \int_0^T dt \overline{e_k(t)} f(t),$$

where the convergence is in norm $\lim_{N \rightarrow \infty} \|f - \sum_{|k| \leq N} \langle e_k, f \rangle e_k\|$. Once a Hilbert basis is found, a countable dense subset is obtained as the span $\sum_{k \in \mathbb{Z}} (\mathbb{Q} + i\mathbb{Q})e_k$.

Exercise 17.1 (The Lebesgue integral of a L^1 function is continuous). *Let $f \in L^1([0, T], \mathbb{C})$ and define:*

$$F(x) = \int_0^x dt f(t).$$

Prove that $F(x)$ is continuous on $[0, T]$ (i.e. continuous on the interior and the one sided limits exist at the end points). Prove that if $F(x) = 0$ for all $x \in [0, T]$ then $f(t)$ is zero almost everywhere on $[0, T]$. For any $x \in [0, T]$ consider a sequence $(x_n)_{n \in \mathbb{N}}$ with $\lim_n x_n = x$. Then:

$$\lim_{n \rightarrow \infty} \int_0^{x_n} dt f(t) = \lim_{n \rightarrow \infty} \int_0^T dt f(t) \mathbf{1}_{[0, x_n]}(t) = \int_0^T dt f(t) \lim_{n \rightarrow \infty} \mathbf{1}_{[0, x_n]}(t),$$

where we used the dominated convergence theorem and the fact that $f \in L^1$. As $x_n \rightarrow x$, for any $t \neq x$, $\exists N$ such that $n > N$, $\mathbf{1}_{[0, x_n]}(t) = \mathbf{1}_{[0, x]}(t)$, therefore the pointwise limit $\lim_{n \rightarrow \infty} \mathbf{1}_{[0, x_n]}(t)$ equals $\mathbf{1}_{[0, x]}(t)$ almost everywhere (with the possible exception of $t = x$, which is of measure 0).

For the second point, we note first that if $g \geq 0$ and $\int_X dt g(t) = 0$ on some measurable set X , then $g = 0$ almost everywhere on X : assume $m(\{x | g(x) > \epsilon\}) > 0$, then $\int_X dt g(t) \geq \int_X dt \epsilon \mathbf{1}_{\{x | g(x) > \epsilon\}} = \epsilon m(\{x | g(x) > \epsilon\}) > 0$, and as $\{x | g(x) > 0\} = \bigcup_n \{x | g(x) > 1/n\}$ we conclude.

Now, assume $F(x) = \int_0^x dt f(t) = 0$, $\forall x \in [0, T]$ for some function f . Then $\int_I dt f(t)$ is zero on any interval I , hence on any open set, hence on any closed set. Denote $S = \{x \in (0, T) | f(x) > 0\}$ and assume $m(S) > 0$. Take $F \subset S$ a closed set such that $m(S \setminus F) < m(S)/2$, and by additivity $m(F) \geq m(S)/2 > 0$. As $\int_F dt f(t) = 0$ and $f \geq 0$ on F , $f = 0$ almost everywhere on F . But this is impossible as $m(F) > 0$ and $f > 0$ on F . Proceeding similarly for $f(x) < 0$, we conclude.

Exercise 17.2. *Prove that $L^2([0, T], \mathbb{C}) \subset L^1([0, T], \mathbb{C})$. Separating a function $f \in L^2$ as $f = f|_{|f| \leq 1} + f|_{|f| \geq 1}$ we conclude that on intervals (or more generally on domains with finite measure) $\int_X dt |f| \leq m(X) + \int_X dt |f|^2$.*

Theorem 17.1 (Riesz–Fischer theorem, the Fourier basis). *The set of functions:*

$$e_k(x) = \frac{1}{\sqrt{T}} \exp \left\{ ik \frac{2\pi}{T} x \right\}, \quad k \in \mathbb{Z},$$

¹The only non trivial part is to prove it is Cauchy complete.

is a Hilbert basis in $L^2([0, T], \mathbb{C})$.

Proof. The set $\{e_k\}_{k \in \mathbb{Z}}$ is an orthonormal set:

$$\langle e_k, e_p \rangle = \frac{1}{T} \int_0^T dt e^{i \frac{2\pi}{T} (p-k)t} = \delta_{kp}.$$

In order to conclude, it is enough to prove that if $\langle e_n, f \rangle = 0$ for all n then f is zero almost everywhere.

For any $f \in L^2([0, T], \mathbb{C})$ we have $f(t), tf(t) \in L^1([0, T], \mathbb{C})$ and $g(x) = \int_0^x dt f(t) + \frac{1}{T} \int_0^T dt tf(t)$ is continuous on $[0, T]$. If the Fourier coefficients of f are 0, so are the Fourier coefficients of g :

$$\begin{aligned} \sqrt{T} \langle e_0, g \rangle &= \int_0^T dx g(x) = \int_0^T dx \int_0^x dt f(t) + T \frac{1}{T} \int_0^T dt tf(t) = \int_0^T dt f(t) \int_t^T dx + \int_0^T dt tf(t) \\ &= \int_0^T dt f(t)(T-t) + \int_0^T dt tf(t) = 0, \end{aligned}$$

as $\int_0^T dt f(t) = \sqrt{T} \langle e_0, T \rangle = 0$. For $k \neq 0$, denoting $\omega_k = \frac{2\pi}{T} k$ we have:

$$\sqrt{T} \langle e_k, g \rangle = \int_0^T dx e^{-i\omega_k x} \left[\int_0^x dt f(t) + \frac{1}{T} \int_0^T dt tf(t) \right] = \int_0^T dt f(t) \int_t^T dx e^{-i\omega_k x} = \int_0^T dt f(t) \frac{1 - e^{-i\omega_k T}}{-i\omega_k} = 0,$$

as the two terms reproduce $\langle e_0, T \rangle = \langle e_k, T \rangle = 0$.

From the fundamental theorem of calculus, $h(x) = \int_0^x dt g(t) + \frac{1}{T} \int_0^T dt tg(t)$ is continuous on $[0, T]$ and continuously differentiable on $(0, T)$, and if the Fourier coefficients of g are zero then the Fourier coefficients of h are zero. But then from Lemma 17.1 below we conclude that h is identically 0, hence $g = h'$ is identically 0 on² $[0, T]$. As $g(0) = \frac{1}{T} \int_0^T dt tf(t) = 0$, we conclude that $g(x) = \int_0^x dt f(t)$ and as $g(x) = 0$ on $[0, T]$, f is zero almost everywhere in $[0, T]$. $\square$

If the function f has additional regularity properties then the convergence of the Fourier series is improved.

Lemma 17.1 (**The Fourier series converges pointwise for $\mathcal{C}^1$ functions**). *The Fourier series of a complex valued function h continuous on $[0, T]$ and continuously differentiable on $(0, T)$ converges pointwise in $(0, T)$ to h :*

$$\sum_{k \in \mathbb{Z}} \langle e_k, h \rangle e_k(x) = h(x), \quad \forall x \in (0, T).$$

Proof. We denote $S_N[h]$ the partial sum $S_N[h](x) = \sum_{k=-N}^N \langle e_k, h \rangle e_k(x)$ and we observe that:

$$S_N[h](x) = \int_0^T dt h(t) D_N(x-t), \quad D_N(s) = \frac{1}{T} \sum_{k=-N}^N e^{i \frac{2\pi}{T} ks} = \frac{1}{T} \frac{e^{i \frac{2\pi}{T} (N+1)s} - e^{-i \frac{2\pi}{T} Ns}}{e^{i \frac{2\pi}{T} s} - 1}.$$

A direct computation yields:

$$\int_0^T dt D_N(x-t) = \frac{1}{T} \sum_{k=-N}^N \int_0^T dt e^{i \frac{2\pi}{T} k(x-t)} = \frac{1}{T} \sum_{k=-N}^N T \delta_{k0} = 1,$$

in particular for constant functions the partial Fourier sum converges. It follows that:

$$\begin{aligned} S_N[h](x) - h(x) &= \int_0^T dt [h(t) - h(x)] D_N(x-t) = \frac{1}{T} \int_0^T dt \frac{h(t) - h(x)}{e^{i \frac{2\pi}{T} (x-t)} - 1} \left[e^{i \frac{2\pi}{T} (N+1)(x-t)} - e^{-i \frac{2\pi}{T} N(x-t)} \right] \\ &= A_{N+1} \frac{1}{\sqrt{T}} e^{i \frac{2\pi}{T} (N+1)x} - A_{-N} \frac{1}{\sqrt{T}} e^{i \frac{2\pi}{T} (-N)x}, \end{aligned}$$

²A priori both h and g are zero on $(0, T)$, but as they are continuous on $[0, T]$, they are zero on $[0, T]$.

where:

$$A_k = \int_0^T dt \tilde{h}(t) \frac{1}{\sqrt{T}} e^{-i\frac{2\pi}{T}kt}, \quad \tilde{h}(t) = \frac{h(t) - h(x)}{e^{i\frac{2\pi}{T}(x-t)} - 1},$$

that is A_k is the k 'th Fourier coefficient of the function $\tilde{h}(t)$. As h is continuously differentiable on $(0, T)$ and $x \in (0, T)$, $\tilde{h}(t)$ is continuous at x , hence in the entire interval $[0, T]$ (note that for $t \in [0, T]$ and $x \in (0, T)$, $|x - t| < T$, hence the only zero of $v(t) = e^{i\frac{2\pi}{T}(x-t)} - 1$ for $t \in (0, T)$ lies at $t = x$). Thus $\tilde{h}(t)$ is bounded on $[0, T]$ and belongs to $L^2([0, T], \mathbb{C})$. From the Bessel inequality $\sum_{k \in \mathbb{Z}} |A_k|^2 \leq \|\tilde{h}\|^2 < \infty$ we conclude that $\lim_{N \rightarrow \infty} A_{\pm N} = 0$ hence $S_N[h](x)$ converges to $h(x)$.

The proof fails for $x = 0, T$ as continuity of h on $[0, T]$ and continuous differentiability on $(0, T)$ do not guarantee that $\lim_{t \rightarrow 0} \frac{h(t) - h(0)}{e^{i\frac{2\pi}{T}(0-t)} - 1}$ is finite, hence $\tilde{h}(t)$ can fail to be continuous at the end points of the interval and might diverge such that it does not belong to L^2 . $\square$

Proposition 17.1 (**Derivative and Fourier series for $\mathcal{C}^2$ functions**). *The convergence of the Fourier series is improved to uniform convergence in the interval $[0, T]$ if f is continuous on $[0, T]$, continuously differentiable on $(0, T)$ and $f(0) = f(T)$.*

If $f : [0, T] \rightarrow \mathbb{C}$ is $\mathcal{C}^2$ (twice continuously differentiable) on $(0, T)$, both f and f' are continuous on $[0, T]$ and $f(0) = f(T)$ then:

$$\frac{df}{dx}(x) = \sum_{k \in \mathbb{Z}} \langle e_k, f \rangle \frac{de_k}{dx}(x), \quad \forall x \in (0, T).$$

If furthermore $f'(0) = f'(T)$ then the series converges uniformly to f' for $x \in [0, T]$ (hence including at the end points).

Proof. The pointwise convergence in Lemma 17.1 is improved to uniform convergence on the interval $[0, T]$ for functions which extend to $\mathcal{C}^1$ functions by periodicity.

We have Weierstrass's criterion for absolute convergence. If $f_n : A \rightarrow \mathbb{R}$ with $n \geq 1$ such that $\sup_{x \in A} |f_n(x)| = M_n < \infty$ and $\sum_{n \geq 1} M_n$ converges then $S_N = \sum_{n=1}^N f_n(x)$ converges absolutely and uniformly on A . Indeed, as $|S_N(x) - S_M(x)| \leq \sum_{n=N+1}^M M_n$ then $S_N(x)$ converges pointwise to some $S(x)$. At the same time, $|S_N(x) - S_M(x)| \leq \sum_{n=N+1}^M M_n < \epsilon \Rightarrow \lim_{M \rightarrow \infty} |S_N(x) - S_M(x)| = |S_N(x) - S(x)| < \epsilon$ for any x .

Consider f to be $\mathcal{C}^1$ on $[0, T]$ with $f(0) = f(T)$. Let us denote $\hat{f}(k) = \langle e_k, f \rangle$. Integrating by parts we have $\hat{f}'(k) = i\frac{2\pi}{T}k\hat{f}(k)$, where we use $f(0) = f(T)$. By Bessel's inequality we have $\sum_{k \in \mathbb{Z}} k^2 |\hat{f}(k)|^2 = \sum_{k \in \mathbb{Z}} |\hat{f}'(k)|^2 \leq \|f'\|^2 < \infty$ and the Cauchy-Schwarz inequality writes:

$$\sum_{k \in \mathbb{Z}} |\hat{f}(k)| \leq \left(\sum_{k \in \mathbb{Z}} k^2 |\hat{f}(k)|^2 \right)^{1/2} \left(\sum_{k \geq 1} \frac{1}{k^2} \right)^{1/2}.$$

From Weierstrass's criterion we conclude that $\sum_{k \in \mathbb{Z}} \frac{1}{\sqrt{T}} \hat{f}(k) e^{i\frac{2\pi}{T}kx}$ converges absolutely and uniformly. As from Lemma 17.1 the Fourier series converges pointwise to f on $(0, T)$ and f is continuous, from the uniform limit theorem the series converges uniformly to f on $[0, T]$. The second statement follows from Lemma 17.2 below. $\square$

Theorem 17.2 (**Properties of the Fourier series**). *Consider a function $f \in L^2([0, T], \mathbb{C})$ and its (Fourier) expansion on the Hilbert basis $e_k(x) = \frac{e^{i\frac{2\pi}{T}kx}}{\sqrt{T}}$:*

$$f(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e_k(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) \frac{e^{i\frac{2\pi}{T}kx}}{\sqrt{T}}, \quad \hat{f}(k) = \langle e_k, f \rangle = \int_0^T dt \frac{e^{-i\frac{2\pi}{T}kt}}{\sqrt{T}} f(t).$$

Then:

- f as well as its Fourier series can be continued by periodicity to the entire real axis as $e_k(x)$ can.

- (Parseval's theorem) from Bessel's inequality it follows that $\sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2$ converges hence in particular $\lim_{k \rightarrow \infty} \hat{f}(k) = 0$. As the norm is a continuous function³ we have $\|f\|^2 = \|\lim_{N \rightarrow \infty} \sum_{k=-N}^N \langle e_k, f \rangle e_k\|^2 = \lim_{N \rightarrow \infty} \sum_{k=-N}^N |\langle e_k, f \rangle|^2$ hence:

$$\|f\|^2 = \int_0^T dt |f(t)|^2 = \sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2.$$

- we denote $\mathcal{F}(f)(n) = \hat{f}(n)$ and the map $\mathcal{F} : L^2([0, T], \mathbb{C}) \rightarrow \ell^2(\mathbb{C})$ from the t space to the “frequency” space n is such that:
 - $\mathcal{F}$ is one to one. The uniqueness of the Fourier series follows from the fact that e_k is a Hilbert basis. Conversely, for any ℓ^2 sequence $(a_k)_{k \in \mathbb{Z}}$, $S_N(x) = \sum_{|k| \leq N} a_k e_k(x)$ is a Cauchy sequence of functions in L^2 hence it is convergent.
 - $\mathcal{F}$ is linear $\mathcal{F}(af + bg) = a\mathcal{F}(f) + b\mathcal{F}(g)$.
 - $\overline{\mathcal{F}(f)(k)} = \mathcal{F}(\bar{f})(-k)$ and denoting $f^-(x) = f(-x)$, $\mathcal{F}(f^-)(k) = \mathcal{F}(f)(-k)$.
 - denoting $f_{x_0} = f(x - x_0)$, $\mathcal{F}(f_{x_0})(k) = e^{-i\frac{2\pi}{T}kx_0} \mathcal{F}(f)(k)$
 - $\mathcal{F}(e^{i\frac{2\pi}{T}k_0x} f)(k) = \mathcal{F}(f)(k - k_0)$.

Proof. The remaining points are straightforward. □

We usually denote the set of frequencies in the Fourier series as $\omega_k = \frac{2\pi}{T}k = \omega k$.

Exercise 17.3. Find the Fourier series of $f : [-\pi, \pi] \rightarrow \mathbb{R}$ for:

- $f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases}$
- $f(x) = x$.
- $f(x) = x^2$. Prove that $\sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{6}$.
- $f(x) = x(\pi - x)$. Prove that $\sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^3} = \frac{\pi^3}{32}$.
- $f(x) = x^p$.

17.1.1 Canonical quantization and the spectrum of the $-i\frac{d}{dx}$ differential operator

The canonical quantization prescription of a classical system tells us that classical observables should be promoted to quantum observables (that is operators on Hilbert space) such that the commutator of any two quantum observables is i times the Poisson bracket of the classical versions of the observables. In particular the classical momentum and position x, p (more general canonical coordinates) with $\{x, p\} = \partial_x x \partial_p p - \partial_p x \partial_x p$ become operators with commutator $[X, P] = i$. In the position basis $f(x) = \langle x | f \rangle$ with $|x\rangle$ generalised eigenvector of X , $X|x\rangle = x|x\rangle$ we have $\langle x | X | f \rangle = xf(x)$ that is the position operator is the multiplication operator $(Xf)(x) = xf(x)$ and the momentum operator is a derivative $(Pf)(x) = (-i\frac{d}{dx}f)(x)$.

We are interested in identifying the spectrum of the differential operator $A = -i\frac{d}{dx}$ on the Hilbert space $L^2([0, T], \mathbb{C})$. We need first to chose a domain $D(A)$ for this operator. Let's say we chose $\mathcal{C}^1([0, T], \mathbb{C})$ functions, which are dense in L^2 . The operator A is not symmetric, but it becomes symmetric if one imposes *boundary conditions*, let's say $f(0) = f(T) = 0$:

$$\left\langle -i\frac{d}{dx}f, g \right\rangle = \int_0^T dx \overline{(-i)f'(x)}g(x) = i[\bar{f}g]_0^T + \int_0^T dx \overline{f(x)}(-i)g'(x) = i[\bar{f}(T)g(T) - \bar{f}(0)g(0)] + \left\langle f, -i\frac{d}{dx}g \right\rangle.$$

³For any x_n and x we have $\|x_n\| = \|x_n + x - x\| \leq \|x_n\| + \|x_n - x\|$ and $\|x\| = \|x + x_n - x_n\| \leq \|x_n\| + \|x_n - x\|$ hence $|\|x_n\| - \|x\|| \leq \|x_n - x\|$.

Now A on $D(A) = \{f \in C^1, f(0) = f(T) = 0\}$ is symmetric but it is not self-adjoint as $D(A^\dagger)$ is too big: the functions in $D(A^\dagger)$ are not constraint at the end points. In particular A is a symmetric operator with non real spectrum as for any $z \in \mathbb{C}$, the operator $-i\frac{d}{dx} - zI$ on $D(A)$ does not have dense range because $e^{i\bar{z}x} \in L^2([0, T], \mathbb{C})$ is orthogonal to it:

$$\langle -i\frac{d}{dx}f - zf, e^{i\bar{z}x} \rangle = \int_0^T dt \overline{[-if'(t) - zf(t)]} e^{i\bar{z}t} = i[\bar{f}e^{i\bar{z}t}]_0^T + \int_0^T \overline{f(t)}(-i\frac{d}{dt} - \bar{z})e^{i\bar{z}t} = 0.$$

We have imposed “too many” boundary conditions. In order to obtain self-adjoint operators one needs to impose “just the right amount” of boundary conditions: in this case requiring periodicity $f(0) = f(T)$ suffices. With periodic boundary conditions, A is an unbounded essentially self-adjoint operator.

One can search for the eigenvalues of A using the Fourier basis:

$$f(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e_k(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) \frac{e^{i\frac{2\pi}{T}kx}}{\sqrt{T}}, \quad \hat{f}(k) = \langle e_k, f \rangle = \int_0^T dt \frac{e^{-i\frac{2\pi}{T}kt}}{\sqrt{T}} f(t).$$

The eigenvalue equation writes $Af - \alpha f = 0$ that is $0 = \sum_{k \in \mathbb{Z}} \langle e_k, Af - \alpha f \rangle e_k = \sum_{k \in \mathbb{Z}} \langle (A - \alpha)e_k, f \rangle e_k$, where we used the fact that e_k belongs to the domain of A and we can send A on the other term in the scalar product. This computes to $0 = \sum_{k \in \mathbb{Z}} \hat{f}(k) \frac{e^{i\frac{2\pi}{T}kx}}{\sqrt{T}} (\frac{2\pi}{T}k - \alpha)$ which, as e_k is a basis, is solved by:

$$\forall k, \quad \hat{f}(k) \left(\frac{2\pi}{T}k - \alpha \right) = 0 \Rightarrow \begin{cases} \text{either} & \hat{f}(k) = 0 \\ \text{or} & \alpha = \frac{2\pi}{T}k \end{cases},$$

that is the eigenvalues of the operator $-i\frac{d}{dx}$ with periodic boundary conditions are $\alpha_k = \frac{2\pi}{T}k$ and the eigenfunctions are the Fourier modes $e_k(x) \sim e^{i\frac{2\pi}{T}kx}$. As the Fourier basis is a basis, $A - zI$ is full range for $z \neq \frac{2\pi}{T}k$ and the operator does not have continuous spectrum.

Alternatively, one can search for eigenvectors and generalized eigenvectors by solving the differential equation $Af = -if' = \alpha f$ with $f(0) = f(T)$ directly. One finds solutions $f = e^{i\alpha x}$ with $\alpha T = 1$ leading to the unique family of eigenvalues $\alpha_k = \frac{2\pi}{T}k$ with eigenfunctions $e_k \in L^2([0, T], \mathbb{C})$. As all the solutions of the eigenvalue equation belong to the Hilbert space, A with periodic boundary conditions has no generalized eigenvectors and no essential spectrum. The Fourier basis is the eigenbasis of the momentum operator with periodic boundary conditions on the Hilbert space $L^2([0, T], \mathbb{C})$:

$$I = \sum_{k \in \mathbb{Z}} |e_k\rangle \langle e_k|, \quad P = \sum_{k \in \mathbb{Z}} |e_k\rangle \frac{2\pi}{T}k \langle e_k|, \\ \langle f|g \rangle = \int_0^T dt \overline{f(t)}g(t), \quad \langle x|e_k \rangle = \frac{e^{i\frac{2\pi}{T}kx}}{\sqrt{T}}, \quad \langle x|P|f \rangle = \sum_{k \in \mathbb{Z}} \left(\frac{2\pi}{T}k \right) \frac{e^{i\frac{2\pi}{T}kx}}{\sqrt{T}} \int_0^T dt \frac{e^{-i\frac{2\pi}{T}kt}}{\sqrt{T}} f(t).$$

We conclude that the boundary conditions are part of the definition of the operator and constrain its spectrum. Contrary to the boundary conditions, we will see below that an *initial condition* of the type $f(0) = f_0$ guarantees the existence and unicity of the solution of the differential equation $-if' - \alpha f = 0$ for any complex α , $f(x) = f_0 e^{i\alpha x}$. One would naively conclude that any complex α is a eigenvalue of $-id/dx$ with associated eigenfunction $e^{i\alpha x} \in L^2([0, T])$. However, for arbitrary α , $e^{i\alpha x}$ is not periodic with period T , hence it is not an eigenfunction of $-id/dx$ with periodic boundary conditions: the initial conditions fix the overall coefficient f_0 , while the boundary conditions restrict the values of α , that is the space of admissible solutions. The logic is the following: the boundary conditions are part of the definition of the operator and quantize its spectrum, that is the possible eigenvalues α_k ; an initial conditions serves to fix a vector in the eigenspace corresponding to a given eigenvalue. From the differential equation without boundary conditions one concludes that “whatever sensible boundary conditioned I pick (i.e. whatever self-adjoint extension of the operator) the eigenspaces of the operator will have dimension equal to the number of independent initial conditions”.

Exercise 17.4. Find the spectrum and eigenfunctions of the operator $-\frac{d^2}{dx^2}$ acting on $f \in \mathcal{C}^\infty([0, T], \mathbb{C})$ with periodic, anti periodic and Dirichlet boundary conditions that is $f(0) = f(T)$, $f'(0) = f'(T)$, and $f(0) = -f(T)$, $f'(0) = -f'(T)$ respectively $f(0) = f(T) = 0$. Note that as the operator is second order (two derivatives) we need two boundary conditions (see also Sturm Liouville problems below). We first check that each of the three gives a self adjoint extension, that is the boundary terms cancel when we use the same boundary conditions on f and g (and we need to use them on both f and g : $\int dx f''g = f'g|_0^T - \int dx f'g' = f'g|_0^T - f'g|_0^T + \int dx f'g''$). We then solve the eigenvalue equation $-u'' = \alpha u \Rightarrow u = Ae^{i\sqrt{\alpha}x} + Be^{-i\sqrt{\alpha}x}$. Imposing the boundary conditions we get:

- for periodic $A + B = Ae^{i\sqrt{\alpha}T} + Be^{-i\sqrt{\alpha}T}$ and $A - B = Ae^{i\sqrt{\alpha}T} - Be^{-i\sqrt{\alpha}T}$ and adding up we have $e^{i\sqrt{\alpha}T} = 1$ hence $\alpha_k = \left(\frac{2\pi}{T}\right)^2 k^2$, $k \geq 1$ with eigenfunctions $e^{i\frac{2\pi}{T}kx}$ and $e^{-i\frac{2\pi}{T}kx}$ and $\alpha_0 = 0$, $f_0 = 1$.
- for antiperiodic $e^{i\sqrt{\alpha}T} = -1$ hence $\alpha_k = \left(\frac{2\pi}{T}(k + \frac{1}{2})\right)^2$, $k \geq 0$ with eigenfunctions $e^{i\frac{2\pi}{T}(k+\frac{1}{2})x}$ and $e^{-i\frac{2\pi}{T}(k+\frac{1}{2})x}$.
- for Dirichlet, $u(0) = A + B = 0 \Rightarrow u = \sin(\sqrt{\alpha}x)$ and $u(T) = 0 \Rightarrow \alpha_k = \frac{\pi^2}{T^2} k^2$ for $k \geq 1$.

17.1.2 Summing over the Fourier frequencies, Matsubara weighting function

We use $\omega_k = \frac{2\pi}{T}k$. One is often interested in resumming Fourier series. That is, given the list of Fourier coefficients $b(i\omega_k)$ (say $b(i\omega_k) = \frac{1}{\sqrt{T}} \frac{e^{\alpha T} - 1}{\alpha - i\omega_k}$) to reconstruct explicitly the Fourier sum:

$$f(x) = \sum_{k \in \mathbb{Z}} b(i\omega_k) \frac{e^{i\omega_k x}}{\sqrt{T}}, \quad 0 < x < T.$$

This can be done as follows. Let us define the weighting function, which is a function of one complex variable z with first order poles located at $i\omega_k$ and with residues 1:

$$h(z) = \frac{T}{e^{Tz} - 1}, \quad f(x) = \sum_{k \in \mathbb{Z}} b(i\omega_k) \frac{e^{i\omega_k x}}{\sqrt{T}} = \left(\int_{-\epsilon - i\infty}^{-\epsilon + i\infty} + \int_{\epsilon - i\infty}^{\epsilon + i\infty} \right) \frac{dz}{2\pi i} h(z) b(z) \frac{e^{zx}}{\sqrt{T}}.$$

Note that

$$\frac{e^{zx}}{e^{Tz} - 1} \sim \begin{cases} e^{-(T-x)\text{Re}(z)}, & \text{Re}(z) \rightarrow +\infty \\ e^{-x\text{Re}(z)}, & \text{Re}(z) \rightarrow -\infty \end{cases},$$

hence, unless $b(z)$ grows exponentially, both contours can be pushed to infinity. In the process the contours cross the poles α_i of $b(z)$ and leave negatively oriented contours around such poles:

$$f(x) = - \sum_{\alpha_i \text{ pole of } b} \int_{C_\epsilon(\alpha)} \frac{dz}{2\pi i} b(z) \sqrt{T} \frac{e^{zx}}{e^{Tz} - 1} = - \sum_{\alpha_i \text{ pole of } b} \text{Res} \left(b(z) \sqrt{T} \frac{e^{zx}}{e^{Tz} - 1}, \alpha_i \right).$$

For instance

$$\sum_{k \in \mathbb{Z}} \left(\frac{1}{\sqrt{T}} \frac{e^{\alpha T} - 1}{\alpha - i\omega_k} \right) \frac{e^{i\omega_k x}}{\sqrt{T}} = -\text{Res} \left(\left(\frac{1}{\sqrt{T}} \frac{e^{\alpha T} - 1}{\alpha - z} \right) \sqrt{T} \frac{e^{zx}}{e^{Tz} - 1}, \alpha \right) = e^{z\alpha}.$$

Exercise 17.5. Compute $\frac{1}{T} \sum_{k \in \mathbb{Z}} \frac{1}{(\alpha - i\omega_k)^p} e^{i\omega_k x}$.

Functions with anti periodic boundary conditions $f(T) = -f(0)$ expand on the fermionic Matsubara frequencies $\omega_k^F = \frac{2\pi}{T}(k + \frac{1}{2})$ with corresponding weighting function $h_F(z) = \frac{-T}{e^{Tz} + 1}$:

$$f(x) = \sum_{k \in \mathbb{Z}} b(i\omega_k^F) \frac{e^{i\omega_k^F x}}{\sqrt{T}} = \left(\int_{-\epsilon - i\infty}^{-\epsilon + i\infty} + \int_{\epsilon - i\infty}^{\epsilon + i\infty} \right) \frac{dz}{2\pi i} h_F(z) b(z) \frac{e^{zx}}{\sqrt{T}}.$$

17.1.3 Other bases in $L^2([0, T], \mathbb{C})$

Theorem 17.3 (The polynomials form a basis in $L^2([0, T], \mathbb{C})$). The polynomials $1, x, x^2, \dots, x^n, \dots$ are such that $\text{Span}\{1, x, x^2, \dots, x^n, \dots\} = L^2([0, T], \mathbb{C})$ and one can select a Hilbert basis by applying the Gram-Schmidt process, hence polynomials are dense (in L^2 norm) in $L^2([0, T], \mathbb{C})$.

Proof. In order to prove that the polynomials are spanning, it is enough to prove that, for f in L^2 :

$$\int_0^T dx x^q f(x) = 0, \quad \forall q \in \mathbb{N} \quad \Rightarrow f = 0.$$

We express any function in L^2 in the Fourier basis $f(x) = \sum_{k \in \mathbb{Z}} \hat{f}(k) \frac{e^{i \frac{2\pi}{T} k x}}{\sqrt{T}}$ with:

$$\hat{f}(k) = \int_0^T dt \frac{e^{-i \frac{2\pi}{T} k t}}{\sqrt{T}} f(t) = \frac{1}{\sqrt{T}} \int_0^T dt \sum_{q \geq 0} \frac{(-i\omega_k)^q}{q!} t^q f(t),$$

and we observe that for any n we have:

$$\left| \sum_{q=0}^n \frac{(-i\omega_k)^q}{q!} t^q f(t) \right| \leq e^{|\omega_k|t} |f(t)|, \quad \int_0^T dt e^{|\omega_k|t} |f(t)| \leq \left(\int_0^T dt e^{2|\omega_k|t} \right)^{1/2} \left(\int_0^T dt |f(t)|^2 \right)^{1/2},$$

where we used the Cauchy Schwarz inequality. It follows from Lebesgue's dominated convergence theorem that we can commute the sum and the integral to get:

$$\hat{f}(k) = \sum_{q \geq 0} \frac{(-i\omega_k)^q}{q!} \frac{1}{\sqrt{T}} \int_0^T dt t^q f(t),$$

thus if f is orthogonal on $\{t^q, q \in \mathbb{N}\}$ then $\hat{f}(k) = 0$ for all k and f is zero almost everywhere. $\square$

In words, any L^2 function can be arbitrarily well approximated in L^2 norm by a polynomial. This can be improved to the supremum norm for continuous functions.

Theorem 17.4 (**Weierstrass approximation theorem**). *The polynomials are dense in $\mathcal{C}([a, b], \mathbb{C})$ in the supremum norm, that is for any continuous function $f : [a, b] \rightarrow \mathbb{C}$, and for any ϵ there exists a polynomial $P_\epsilon(x)$ such that $\sup_{x \in [a, b]} |f(x) - P_\epsilon(x)| < \epsilon$.*

Proof. We will provide a direct proof of this theorem. Let us define:

$$S_t f(x) = \int_{-\infty}^{\infty} du f(u) \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}}.$$

If f is continuous with compact support, by the Heine-Cantor theorem it is uniformly continuous, hence for any ϵ there exists δ such that $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \epsilon/2$. Let us denote $M = \sup_x |f(x)|$ (which is finite as f is continuous with compact support). It is easy to show that:⁴

$$\forall \epsilon, \delta, \exists t_0 \text{ s.t. } t < t_0 \Rightarrow \int_{|x-u|>\delta} du \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}} < \frac{\epsilon}{4M},$$

so that for $t < t_0$ we have:

$$|S_t f(x) - f(x)| \leq \int_{-\infty}^{\infty} du |f(u) - f(x)| \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}} \leq \int_{|x-u|<\delta} du |f(u) - f(x)| \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}} + \int_{|x-u|>\delta} du 2M \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}} < \epsilon,$$

irrespective of x , hence $\lim_{t \rightarrow 0} \|S_t(f) - f\|_\infty = 0$ that is $S_t(f)$ converges uniformly to f .

Any continuous function on f on $[a, b]$ can be extended to a continuous function with compact support by defining $f(x) = f(a)(x - a + 1)$ for $x \in [a - 1, a)$ and $f(x) = -f(b)(x - b - 1)$ for $x \in (b, b + 1]$ and $f(x) = 0$ for $x \notin [a - 1, b + 1]$. In

⁴Prove this equation.

particular there exists R with $f(u) = 0$ if $|u| > R$ and:

$$S_t f(x) = \int_{-R}^R du f(u) \frac{e^{-\frac{(x-u)^2}{2t}}}{\sqrt{2\pi t}} .$$

Let us chose t such that $\|S_t(f) - f\|_\infty < \epsilon/2$. There exists N such that:⁵

$$\left| \frac{e^{-\frac{(x-u)^2}{2t}}}{\sqrt{2\pi t}} - \frac{1}{\sqrt{2\pi t}} \sum_{k=0}^N \frac{1}{k!} \left(-\frac{(x-u)^2}{2t} \right)^k \right| < \frac{\epsilon}{4RM} ,$$

for all x, u with $|x|, |u| < R$. Defining the polynomial:

$$P(x) = \sum_{k=0}^N \frac{1}{k!} \int_{-R}^R du f(u) \frac{1}{\sqrt{2\pi t}} \left(-\frac{(x-u)^2}{2t} \right)^k ,$$

and gathering everything we have:

$$\|f - P\|_\infty \leq \|f - S_t(f)\|_\infty + \|S_t(f) - P\|_\infty < \frac{\epsilon}{2} + \frac{\epsilon}{2} .$$

□

In the course of this proof we encountered a *delta sequence*, that is a family of functions:

$$\delta_t(u - x) = \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}} , \quad \int du \delta_t(x - u) = 1 , \quad \lim_{t \rightarrow 0} \int du f(u) \delta_t(x - u) = f(x) ,$$

where the last equality holds uniformly in x (that is in sup norm) if f is uniformly continuous and bounded. We call:

$$S_t(f)(x) = \int du f(u) \frac{e^{-\frac{(u-x)^2}{2t}}}{\sqrt{2\pi t}} ,$$

the Weierstrass transform of f .

17.2 The Schwartz space

We search for metric spaces of regular functions. Ultimately we will be interested in dense linear subspaces of L^2 , which can serve as domains of differentail operators.

Lemma 17.2 (**Uniform convergence in $\mathcal{C}^1$**). *If $g_k \in \mathcal{C}^1(\mathbb{R}, Y)$ converges uniformly (in sup norm) to g and g'_k converges uniformly to h then $g \in \mathcal{C}^1(\mathbb{R}, Y)$, $h \in \mathcal{C}(\mathbb{R}, Y)$ and $g' = h$.*

Proof. Indeed, from the uniform limit theorem, if g_k and g'_k are continuous and converge uniformly then the limits g and h are continuous. By the fundamental theorem of calculus:

$$g_k(x) = g_k(a) + \int_a^x dt g'_k(t) .$$

⁵Prove such an N exists.

As the sequence $g'_k(t)$ converges uniformly to $h(t)$, for any ϵ , for some n large enough $|g'_k(t)| < |h(t)| + \epsilon$. Taking the limit $k \rightarrow \infty$ and using Lebesgue dominated convergence theorem we have:

$$g(x) = g(a) + \int_a^x dt h(t),$$

therefore $g(x)$ is continuously differentiable and $g' = h$. $\square$

Proposition 17.2 ($\mathcal{C}^k(I, \mathbb{C})$ is Banach in $\|\cdot\|_{\mathcal{C}^k}$). *Let $I \subset \mathbb{R}$ some closed interval. The vector space of k times continuously differentiable functions⁶ on I with continuous k derivative endowed with the norm $\|f\|_{\mathcal{C}^k} = \sum_{i=0}^k \sup_{t \in I} |f^{(i)}(t)|$ is a Banach space. A variation on the Weierstrass approximation theorem proves that the polynomials $1, t, t^2, \dots$ form a Schauder basis in this space.*

The space of bounded $\mathcal{C}^k$ functions on $\mathbb{R}$, $\{f \in \mathcal{C}^k(\mathbb{R}, \mathbb{C}) \mid \|f\|_{\mathcal{C}^k} < \infty\}$ is also a Banach space. One can not use the same trick to define a norm on $\mathcal{C}^\infty(\mathbb{R}, \mathbb{C})$, as the sum might not converge: the space of bounded smooth functions with bounded derivatives $\{f \in \mathcal{C}^\infty(\mathbb{R}, \mathbb{C}) \mid \|f^{(k)}\|_\infty < \infty, \forall k\}$ is a complete metric space with distance function $d(f, g) = \sum_{k \geq 0} \frac{1}{2^k} \frac{\|f^{(k)} - g^{(k)}\|_\infty}{1 + \|f^{(k)} - g^{(k)}\|_\infty}$.

Proof. Let us first consider the continuous functions $\mathcal{C}^0$. Any Cauchy sequence in $\mathcal{C}^0$ with sup norm is such that $\|f_n - f_m\|_\infty \rightarrow 0$ hence pointwise $f_n(x) \rightarrow f(x)$. As the absolute value is continuous $|f_l(x) - f_k(x)| < \epsilon, \forall x \Rightarrow |f(x) - f_k(x)| < \epsilon, \forall x$ hence $\sup_x |f(x) - f_k(x)| = \|f - f_k\|_\infty < \epsilon$. As f is the uniform limit of continuous function, by the uniform limit theorem 10.2 it is continuous.

In order to prove $\mathcal{C}^k$ is Cauchy complete we use Lemma 17.2: if $g_n \in \mathcal{C}^1$, $g_n \rightarrow g$ uniformly and $g'_n \rightarrow h$ uniformly, then $g \in \mathcal{C}^1$ and $g' = h$. Concerning $\mathcal{C}^\infty$, see below. $\square$

However, the $\mathcal{C}^k$ functions are not necessarily integrable at infinity, hence $\mathcal{C}^k$ is not a subset of L^2 . Two spaces of functions which are at the same time regular enough and appropriately bounded at infinity play a role below. First, the space of smooth functions with compact support $\mathcal{D} = \{f \in \mathcal{C}^\infty(\mathbb{R}^n, \mathbb{C}) \mid \{x \mid f(x) \neq 0\} \text{ is compact}\}$. Denoting $|\cdot|$ the 2-norm on $\mathbb{R}^n$, an example of a smooth function with compact support is given by the bump function:

$$f(x) = \begin{cases} e^{-\frac{1}{1-x^2}}, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}.$$

Second, the space of smooth functions with rapid decrease.

Definition 17.1 (**The Schwartz space**). *We denote multi indices $\alpha = (\alpha_1 \dots \alpha_n) \in \mathbb{N}^n$ and for $x = (x^1, \dots, x^n) \in \mathbb{R}^n$ we denote $x^\alpha = (x^1)^{\alpha_1} \dots (x^n)^{\alpha_n}$ respectively $\partial^\beta = \partial_{x_1}^{\beta_1} \dots \partial_{x_n}^{\beta_n}$.*

The Schwartz space is the vector space of smooth rapidly decreasing functions on $\mathbb{R}^n$, that is smooth functions which, together with all their partial derivatives, decrease at infinity faster than any polynomial:

$$\mathcal{S}(\mathbb{R}^n, \mathbb{C}) = \{f \in \mathcal{C}^\infty(\mathbb{R}^n, \mathbb{C}) \mid \forall \alpha, \beta \in \mathbb{N}^n, \|f\|_{\alpha, \beta} < \infty\}, \quad \|f\|_{\alpha, \beta} = \sup_{x \in \mathbb{R}^n} |x^\alpha (\partial^\beta f)(x)|.$$

Any smooth function with compact support is a Schwartz function. Any polynomial multiplying a Gaussian $P(x)e^{-\frac{x^2}{2}}$ is also a Schwartz function.

Note that each individual $\|\cdot\|_{\alpha, \beta}$ is not a norm, but a *seminorm*, that is $\|f\|_{\alpha, \beta} = 0$ at fixed α, β does not imply $f = 0$. Indeed, if $\|f\|_{\alpha, \beta} = 0$ then $\partial^\beta f(x) = 0$ (this holds also for $x = 0$ by continuity). This admits as non trivial solutions polynomials f of degree lower than β .

⁶Where we take the one sided derivatives for the end points of the interval.

Lemma 17.3 (**The Schwartz functions are uniformly continuous and bounded**). *A Schwartz function and all its derivatives are uniformly continuous.*

Proof. As $\|f\|_{\alpha,0} < \infty$ it follows that $\lim_{|x| \rightarrow \infty} f(x) = 0$ hence $\forall \epsilon, \exists R$ such that $|x| > R \Rightarrow |f(x)| < \epsilon$. As $\overline{B_{2R}(0)}$ is compact, f is uniformly continuous on this closed ball and $\exists \delta$ such that $|x|, |y| < 2R$ and $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$.

Then, $\forall \epsilon, \exists \delta$ such that $|x - y| < \delta$ implies:

- if $|x|, |y| \leq 2R$ then $|f(x) - f(y)| < \epsilon$.
- if $|y| > 2R$ then $|x| \geq |y| - |x - y| \geq 2R - \delta > R$ hence $|f(x) - f(y)| < |f(x)| + |f(y)| < 2\epsilon$.

□

An immediate consequence of the density of compactly supported smooth functions in L^p (see Theorem 16.6) is the following remark.

Theorem 17.5 (**The Schwartz functions are dense in L^2 with L^2 norm**). *The space of Schwartz functions $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$ is a dense linear subspace of $L^2(\mathbb{R}^n, \mathbb{C})$ in the sense of the L^2 norm.*

While $\mathcal{S}$ is dense in L^2 in the sense of the L^2 norm, one can wonder whether $\mathcal{S}$ is a Banach space with respect to some other norms. The answer to this question is *no*. One can define:

- norms on $\mathcal{S}$ in which $\mathcal{S}$ is *not* Cauchy complete.
- a distance function (not coming from a norm) in which $\mathcal{S}$ is Cauchy complete.

Theorem 17.6 (**The Schwartz space $\mathcal{S}$ is Cauchy complete with respect to the distance d**). *The functions $\|\cdot\|_{\alpha,\beta}$ are seminorms on $\mathcal{S}$. $\mathcal{S}$ is Cauchy complete with respect to the distance function:*

$$d(f, g) = \sum_{\alpha, \beta} 2^{-|\alpha| - |\beta|} \frac{\|f - g\|_{\alpha, \beta}}{1 + \|f - g\|_{\alpha, \beta}},$$

with $|\alpha| = \alpha_1 + \dots + \alpha_n$. The distance $d(f, g)$ is not a norm, as it does not behave properly under multiplication by scalars. We say that f_k converges to f in $\mathcal{S}$, $f_k \xrightarrow{\mathcal{S}} f$, if $\forall \alpha, \beta, \|f_k - f\|_{\alpha, \beta} \rightarrow 0 \Leftrightarrow d(f_k, f) \rightarrow 0$.

Proof. In order to prove that d is a distance one needs to check the triangle inequality. It suffices to check that for any norm:

$$\frac{\|u + v\|}{1 + \|u + v\|} \leq \frac{\|u\|}{1 + \|u\|} + \frac{\|v\|}{1 + \|v\|}.$$

In order to prove completeness, assumes f_k is Cauchy with respect to $d(\cdot, \cdot)$. Then it is Cauchy with respect to all $\|\cdot\|_{\alpha, \beta}$, that is $\lim_{k \rightarrow \infty} x^\alpha \partial^\beta f_k = f_{\alpha, \beta}$ uniformly (in sup norm) to some function $f_{\alpha, \beta}$. We conclude by induction and using Lemma 17.2.

Finally, we must prove that $d(f_n, f) = \sum_{k \geq 0} \frac{1}{2^k} \frac{\|f_n - f\|_k}{1 + \|f_n - f\|_k} \rightarrow 0$ if and only if $\forall k, \|f_n - f\|_k \rightarrow 0$. For any k fixed, if $\epsilon < \frac{1}{2^k}$, then $d(f_n, f) < \epsilon \Rightarrow \|f_n - f\|_k \leq \frac{2^k \epsilon}{1 - 2^k \epsilon}$ hence $d(f_n, f) \rightarrow 0 \Rightarrow \|f_n - f\|_k \rightarrow 0$. Conversely, assume $\forall k, \|f_n - f\|_k \rightarrow 0$. For any ϵ , chose k_ϵ such that $\frac{1}{2^{k_\epsilon}} < \frac{\epsilon}{2}$ and N_ϵ such that for $0 \leq k \leq k_\epsilon$, if $n > N_\epsilon$ then $\|f_n - f\|_k < \frac{\epsilon}{4}$. Then for $n > N_\epsilon$:

$$d(f_n, f) = \sum_{k \geq 0} \frac{1}{2^k} \frac{\|f_n - f\|_k}{1 + \|f_n - f\|_k} \leq \sum_{k=0}^{k_\epsilon} \frac{1}{2^k} \frac{\epsilon}{4} + \sum_{k=k_\epsilon+1}^{\infty} \frac{1}{2^k} \leq \frac{\epsilon}{4} \sum_{k=0}^{\infty} \frac{1}{2^k} + \frac{1}{2^{k_\epsilon}} \sum_{p=0}^{\infty} \frac{1}{2^p} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2}.$$

□

The Schwartz space is the prototype of topological vectors space that is Cauchy complete with respect to a distance function induced by seminorms $\|\cdot\|_{\alpha, \beta}$. Another example is $\mathcal{C}^\infty$ with seminorms $\|f\|_{0, \beta}$. Such spaces are called Fréchet spaces.

17.3 The Fourier transform

Definition 17.2 (**The Fourier transform**). We denote the scalar product $px = \sum_{\mu=1}^n p_\mu x^\mu$. The Fourier transform and the inverse Fourier transform of $f : \mathbb{R}^n \rightarrow \mathbb{C}$ respectively $g : \mathbb{R}^n \rightarrow \mathbb{C}$ are defined as the integrals:

$$\mathcal{F}(f)(p) \equiv \hat{f}(p) = \int d^n x f(x) e^{-ipx}, \quad \mathcal{F}^{-1}(g)(x) \equiv \check{g}(x) = \int \frac{d^n p}{(2\pi)^n} g(p) e^{ipx},$$

whenever they converge.

A sufficient condition for the Fourier transform to exist is for f to be an L^1 function. In this case $\|\hat{f}\|_\infty \leq \|f\|_1$, but $\hat{f}$ might not belong to any L^p space with finite p . The natural domain of definition of the Fourier transform is the Schwartz space.

For instance, we have the following Fourier transforms and inverse Fourier transforms:

- for $\gamma \in \mathbb{R}_{>0}$ we have:

$$\int_{-\infty}^{\infty} dx e^{-\gamma|x|} e^{-ipx} = \int_{-\infty}^0 dx e^{\gamma x} e^{-ipx} + \int_0^{\infty} dx e^{-\gamma x} e^{-ipx} = \frac{1}{\gamma - ip} + \frac{1}{\gamma + ip} = \frac{2\gamma}{\gamma^2 + p^2}.$$

- the inverse Fourier transform:

$$\int \frac{dp}{2\pi} \frac{2\gamma}{p^2 + \gamma^2} e^{ipx} = \int \frac{dp}{2\pi} \frac{2\gamma}{(p + i\gamma)(p - i\gamma)} e^{ipx} = \begin{cases} x > 0 & 2\pi i \operatorname{Res}(\frac{1}{2\pi} \frac{2\gamma}{(p+i\gamma)(p-i\gamma)} e^{ipx}, i\gamma) = e^{-|x|} \\ x < 0 & 2\pi i \operatorname{Res}(\frac{1}{2\pi} \frac{2\gamma}{(p+i\gamma)(p-i\gamma)} e^{ipx}, -i\gamma) = e^{-|x|} \end{cases}$$

- the Fourier transform of the indicator function on an interval:

$$\int_{-\infty}^{\infty} dx e^{-ipx} \mathbf{1}_{[-T, T]}(x) = \frac{e^{-ipT} - e^{ipT}}{-ip} = 2 \frac{\sin(pT)}{p}.$$

Theorem 17.7 (**Properties of the Fourier transform**). Consider a function $f \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$. Then:

- the Fourier transform $\hat{f}(p)$ is absolutely convergent for any p . Furthermore, we have $\mathcal{F}(-i\partial_{x^i} f)(p) = p_i \hat{f}(p)$ and $\mathcal{F}(x^i f)(p) = i\partial_{p_i} \hat{f}(p)$, hence $\hat{f}$ is also a Schwartz function hence $\mathcal{F}$ is continuous in the sense $f_n \xrightarrow{\mathcal{S}} f \Rightarrow \mathcal{F}(f_n) \xrightarrow{\mathcal{S}} \mathcal{F}(f)$.

Proof. We have:

$$\|p^\alpha \partial_p^\beta \hat{f}(p)\|_\infty = \sup_p \left| \int d^n x [\partial_x^\alpha (x^\beta f(x))] e^{-ipx} \right| \leq \|(1 + x^{2n}) \partial_x^\alpha (x^\beta f(x))\|_\infty \int d^n x \frac{1}{1 + x^{2n}},$$

hence $\|\hat{f}\|_{\alpha, \beta}$ is bounded by some constant times a finite linear combination of seminorms $\|\cdot\|_{\alpha', \beta'}$ of f . $\square$

- we have $\overline{\mathcal{F}(f)(p)} = \mathcal{F}(\bar{f})(-p)$; denoting $f_{x_0} = f(x - x_0)$ we have $\mathcal{F}(f_{x_0})(p) = e^{-ipx_0} \mathcal{F}(f)(p)$ and $\mathcal{F}(e^{ip_0 x} f) = \mathcal{F}(f)(p - p_0)$.
- (inverse) the inverse Fourier transform is the inverse of the Fourier transform.

Proof. As we only integrate Schwartz functions, we can always use Fubini's theorem to change the order of integration, derivative under the integral sign and so on.

In order to prove that the inverse transform is the inverse of the transform, consider a Schwartz function f and its Fourier transform:

$$\mathcal{F}(f)(p) = \int d^n x f(x) e^{-ipx}.$$

Let be a positive real number $t > 0$. By the dominated converge theorem, we have:

$$\int \frac{d^n p}{(2\pi)^n} e^{ipx} \mathcal{F}(f)(p) = \lim_{t \rightarrow 0} \int \frac{d^n p}{(2\pi)^n} e^{-\frac{t}{2} p^2 + ipx} \mathcal{F}(f)(p).$$

As all the integrals are absolutely convergent we have at fixed $t > 0$:

$$\int \frac{d^n p}{(2\pi)^n} e^{-\frac{t}{2} p^2 + ipx} \int d^n u f(u) e^{-ipu} = \int d^n u f(u) \frac{e^{-\frac{(x-u)^2}{2t}}}{(\sqrt{2\pi t})^n} = S_t f(x).$$

In order to conclude that for all x (i.e. pointwise) $\mathcal{F}^{-1} \mathcal{F}(f)(x) = f(x)$, it is enough to prove that $\lim_{t \rightarrow 0} S_t f(x) = f(x)$. In fact, the Weierstrass transform converges in sup norm to f as f is a Schwartz function, that is uniformly continuous and bounded.

Let us prove pointwise convergence by an alternative method. We have:

$$\int d^n u f(u) \frac{e^{-\frac{(x-u)^2}{2t}}}{(\sqrt{2\pi t})^n} = \int d^n v f(x + \sqrt{t}v) \frac{e^{-\frac{v^2}{2}}}{(\sqrt{2\pi})^n},$$

and:

$$\left| f(x + \sqrt{t}v) \frac{e^{-\frac{v^2}{2}}}{(\sqrt{2\pi})^n} \right| \leq \|f\|_{\infty} \frac{e^{-\frac{v^2}{2}}}{(\sqrt{2\pi})^n}, \quad \int d^n u \|f\|_{\infty} \frac{e^{-\frac{v^2}{2}}}{(\sqrt{2\pi})^n} = \|f\|_{\infty},$$

and using Lebesgue's dominated convergence theorem we conclude:

$$\lim_{t \rightarrow 0} \int d^n v f(x + \sqrt{t}v) \frac{e^{-\frac{v^2}{2}}}{(\sqrt{2\pi})^n} = \int d^n v \frac{e^{-\frac{v^2}{2}}}{(\sqrt{2\pi})^n} \lim_{t \rightarrow 0} f(x + \sqrt{t}v) = f(x).$$

□

- (Plancherel's theorem) We have for $f, g \in \mathcal{S}(\mathbb{R}^n, \mathbb{C})$:

$$\langle f, g \rangle = \int d^n x \overline{f(x)} g(x) = \int d^n x \int \frac{d^n p}{(2\pi)^n} \overline{f(x)} \hat{g}(p) e^{ipx} = \int \frac{d^n p}{(2\pi)^n} \overline{\hat{f}(p)} \hat{g}(p).$$

In particular this implies that for $f \in \mathcal{S}$ we have $\|\mathcal{F}(f)\|_{L^2} = \|f\|_{L^2}$ hence $\mathcal{F}$ is a bounded densely defined operators on L^2 with norm $\|\mathcal{F}\|_{\text{op.}} = 1$ in the sense of L^2 norm.

- (convolution) we define the convolution of two functions by $(f * g)(x) = \int d^n u f(u) g(x - u)$. If f and g are Schwartz functions, then so is $f * g$ and

$$\mathcal{F}(f * g)(p) = \int d^n x \int d^n u f(u) g(x - u) e^{-ipx} = \int d^n u \int d^n y f(u) g(y) e^{-ip(u+y)} = \hat{f}(p) \hat{g}(p)$$

As the Fourier transform is bounded and densely define on L^2 , it can be uniquely extended to L^2 .

Proposition 17.3 (Linear extension of bounded operators). Let X and Y be Banach spaces and let A be a bounded linear operator from a dense subspace $D(A) \subset X$ to Y , $A : D(A) \rightarrow Y$, $\|A\|_{D(A)} = \sup_{x \in D(A)} \frac{\|Ax\|}{\|x\|} < \infty$. Then there exists a unique bounded linear extension of A to the whole of X , denoted $\bar{A} : X \rightarrow Y$ and $\|\bar{A}\|_X = \|A\|_{D(A)}$.

Proof. For any $x \in X$, pick a sequence $(x_n)_n \in D(A)$ with $x_n \rightarrow x$. Such a sequence always exists as $D(A)$ is dense in X . Then the sequence $A(x_n)$ is Cauchy, as $\|Ax_n - Ax_m\| \leq \|A\|_{D(A)} \|x_n - x_m\|$ hence it is convergent in Y . We denote $\bar{A}x$ the limit of this sequence. The operator $\bar{A}$ thus defined:

- is well defined on X : if x_n and y_n are two sequences with the same limit $x_n, y_n \rightarrow x$ then $\|A(x_n - y_n)\| \leq \|A\|_{D(A)} \|x_n - y_n\| \xrightarrow{n \rightarrow \infty} 0$.
- coincides with A on $D(A)$, as one can chose the sequence $x_n = x \in D(A)$ and $\bar{A}x = \lim Ax_n = Ax$.
- is linear as $\bar{A}(\alpha x + \beta y) = \lim_{n,p} A(\alpha x_n + \beta y_p)$ with $x_n \rightarrow x$ and $y_p \rightarrow y$.
- has norm $\|\bar{A}\|_X = \sup_{x \in X, \|x\|=1} \|\bar{A}x\| \geq \sup_{x \in D(A), \|x\|=1} \|\bar{A}x\| = \sup_{x \in D(A), \|x\|=1} \|Ax\| = \|A\|_{D(A)}$. Conversely, for any ϵ and $x \in X$ with $\|x\| = 1$, take $x_n \in D(A)$ such that $x_n \rightarrow x$ and $\|x_n\| = 1$ (if $x_n \rightarrow x$ and $\|x\| = 1$, then $\frac{1}{\|x_n\|} x_n \rightarrow x$). Then there exists N_ϵ such that $\forall n > N_\epsilon$, $\|\bar{A}x\| \leq \|Ax_n\| + \epsilon \leq \|A\|_{D(A)} + \epsilon$ and we conclude.

□

It follows that the Fourier transform can be uniquely extended to a continuous mapping $\mathcal{F} : L^2 \rightarrow L^2$ with operator norm 1 (from Plancherel theorem; note that we use the measure $d^d p / (2\pi)^d$ for the scalar product in Fourier space). The properties of the Fourier transform extended to L^2 , in particular the Fourier transform preserves the scalar product $\langle \mathcal{F}(f), \mathcal{F}(g) \rangle = \langle f, g \rangle$ for any $f, g \in L^2$.

The Plancherel theorem can be used to compute some non trivial definite integrals. For instance, while $e^{-\gamma|x|}$ and $\mathbf{1}_{[-T,T]}(x)$ are not Schwartz functions, as they are not smooth, they are in L^2 and, after extending the Fourier transform to L^2 one can use the fact that the Fourier transform is a unitary operator on L^2 to conclude:

$$\int_{-\infty}^{\infty} \frac{dp}{2\pi} \frac{\sin(pT)}{p(p^2 + \gamma^2)} = \int_{-\infty}^{\infty} dx \left(\frac{1}{2} \mathbf{1}_{[-T,T]}(x) \right) \left(\frac{1}{2\gamma} e^{-\gamma|x|} \right) = \frac{1}{4\gamma} \int_{-T}^T dx e^{-\gamma|x|} = \frac{1 - e^{-\gamma T}}{2\gamma^2}.$$

Contrary to the Fourier series, which is an expansion in the Hilbert basis $e^{i\omega_k t}$ of $L^2([0, T])$, the Fourier transform is an expansion on plane waves e^{ipx} which do not belong to $L^2(\mathbb{R}^n)$.

The Laplace and Mellin transforms

Closely related to the Fourier transform, one can defined other transforms of functions which are relevant in practical applications. The Laplace and inverse Laplace transforms of a function are defined as:

$$\mathcal{L}(f)(s) \equiv \hat{f}(s) = \int_0^{\infty} dx e^{-sx} f(x), \quad \mathcal{L}^{-1}(\hat{f})(x) = \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{ds}{2\pi i} e^{sx} \hat{f}(s),$$

where γ is to the right of all the singularities of $\hat{f}(s)$ in the complex plane. The Mellin and inverse Mellin transforms are defined as:

$$\mathcal{M}(f)(s) \equiv \hat{f}(s) = \int_0^{\infty} dx x^{s-1} f(x), \quad \mathcal{M}^{-1}(\hat{f})(x) = \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{ds}{2\pi i} x^{-s} \hat{f}(s).$$

The inverse Mellin transform is the inverse of the Mellin transform if f is piecewise continuous and the Mellin transform converges in a vertical strip in the complex plane.

Exercise 17.6. Prove that the Fourier transform is a continuous operator on the Schwartz space.

Exercise 17.7. Compute the Fourier transform of the Gaussian $\frac{1}{\sqrt{2\pi\epsilon}} e^{-\frac{x^2}{2\epsilon}}$. Discuss the $\epsilon \rightarrow 0$ limit.

Exercise 17.8. Compute the Laplace transform of $x^{\alpha-1}$ and the inverse transform of the result.

Exercise 17.9. *Compute the Fourier and Laplace transforms of $f(t) = e^{-t}$, $t \geq 0$ and $f(t) = 0$ for $t < 0$ and compute the inverse transforms of the result. Do the same for $e^{-a|t|}$ with $a > 0$.*

Chapter 18

Distributions

Many of the results we derive in this chapter can be reproduced if one uses as test functions the smooth functions with compact support. We prefer to use Schwartz functions, as this is the natural domain of definition of the Fourier transform.

18.1 Tempered distributions

We have endowed $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$ with the topology induced by the distance function:

$$d(f, g) = \sum_{\alpha, \beta} 2^{-|\alpha| - |\beta|} \frac{\|f - g\|_{\alpha, \beta}}{1 + \|f - g\|_{\alpha, \beta}}, \quad \|f\|_{\alpha, \beta} = \sup_{x \in \mathbb{R}^n} |x^\alpha (\partial^\beta f)(x)|.$$

The continuous dual of L^2 is, by the Riesz-Fréchet theorem, is identified with L^2 . A linear subspace, like $\mathcal{S}$ will accommodate a large dual.

Definition 18.1 (**Tempered distributions**). *The continuous dual space $\mathcal{S}'(\mathbb{R}^n, \mathbb{C})$ of $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$ is the space of continuous linear functionals on $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$:*

$$\mathcal{S}'(\mathbb{R}^n, \mathbb{C}) = \left\{ T : \mathcal{S}(\mathbb{R}^n, \mathbb{C}) \rightarrow \mathbb{C} \mid T(\alpha f + \beta g) = \alpha T(f) + \beta T(g), \ f_k \xrightarrow{\mathcal{S}} f \Rightarrow T(f_k) \xrightarrow{\mathbb{C}} T(f) \right\}.$$

The elements of $\mathcal{S}'$ are called tempered distributions and the space $\mathcal{S}'$ is endowed with the weak $$ topology (“pointwise convergence” where the points are Schwartz functions):*

$$T_k \xrightarrow{\mathcal{S}'} T \text{ if } \forall f \in \mathcal{S}, \ T_k(f) \xrightarrow{\mathbb{C}} T(f).$$

We say that $T_1 = T_2$ in the sense of distributions, or in the weak sense, if for any $f \in \mathcal{S}$, $(T_1, f) = (T_2, f)$.

As the integral is linear, to any measurable function ϕ which grows at most polynomial at infinity one can associate a tempered distribution $T_\phi \in \mathcal{S}'(\mathbb{R}^n, \mathbb{C})$ defined by:

$$T_\phi(f) = (\phi, f) = \int d^n x \ \phi(x) f(x),$$

where we used a pairing notation.¹ In particular $\mathcal{S}(\mathbb{R}^n, \mathbb{C}) \subset \mathcal{S}'(\mathbb{R}^n, \mathbb{C})$.

¹If one uses test functions with compact support one can accommodate distributions which grow even faster at infinity, e.g. exponentially.

We define the product of a (sufficiently regular) function and a distribution, the derivative of a distribution and the Fourier transform of a distribution as:²

$$(hT, f) = (T, hf), \quad (\partial^\beta T, f) = (T, (-1)^{|\beta|} \partial^\beta f), \quad (\mathcal{F}(T), f) = (T, \mathcal{F}(f)).$$

Note that, by definition, any distribution is arbitrarily derivable. Moreover, the Fourier transform of any tempered distribution is a tempered distribution and the Fourier transform is a continuous operator from $\mathcal{S}'$ to $\mathcal{S}'$. Indeed, if $T_n \xrightarrow{\mathcal{S}'} T$ then for all f , $(\mathcal{F}(T_n), f) = (T_n, \mathcal{F}(f)) \xrightarrow{\mathbb{C}} (T, \mathcal{F}(f)) = (\mathcal{F}(T), f)$, that is $\mathcal{F}(T_n) \xrightarrow{\mathcal{S}'} \mathcal{F}(T)$.

Exercise 18.1. *Prove that the Fourier transform of a tempered distribution is a tempered distribution. The Fourier transform sends Schwartz functions to Schwartz functions.*

We will use the kernel notation $(\phi, f) = \int d^n x \phi(x) f(x)$ for any distribution T_ϕ , but we stress that the “kernel” $\phi(x)$ does not necessarily belong to $\mathcal{S}$. Indeed, $\phi(x)$ might not even be a function, as exemplified by the Dirac δ distribution below.

Theorem 18.1 (**The Schwartz kernel theorem**). *For any continuous linear operator $A : \mathcal{S}(\mathbb{R}^m) \rightarrow \mathcal{S}'(\mathbb{R}^n)$, there exists a kernel $K \in \mathcal{S}'(\mathbb{R}^{m+n})$ such that:*

$$(A(h), f) = (K, f \otimes h) \stackrel{\text{denoted}}{=} \int d^n x d^m y K(x, y) f(x) h(y).$$

The same holds for test functions $\mathcal{D} = C^\infty(\Omega_m \in \mathbb{R}^m)$ with Ω compact, semi norms $\|f\|_\alpha = \sup_{x \in \Omega} |\partial^\alpha f|$, and continuous operators from $\mathcal{D}$ to the dual of some $C^\infty(\Omega_n \in \mathbb{R}^n)$.

Proof. The reader should search the literature on the topic. □

Some examples of distributions are:

- the *Dirac delta* distribution:

$$(\delta_{x_0}, f) = f(x_0) = \int d\delta_{x_0}(x) f(x) \stackrel{\text{denoted}}{=} \int d^n x \delta(x - x_0) f(x),$$

that is the distribution which returns the value of the test function at the point x_0 . Rigorously speaking the Dirac distribution is an integral with respect to the Dirac measures $\delta_{x_0}(A) = \mathbf{1}_A(x_0)$, but in the literature this is often loosely denoted by $d^n x \delta(x - x_0)$. The derivatives of the Dirac delta are:

$$(\partial^\beta \delta_{x_0}, f) = (-1)^{|\beta|} \partial^\beta f(x_0) = \int d\delta_{x_0}(x) (-1)^{|\beta|} \partial^\beta f(x) \stackrel{\text{denoted}}{=} \int d^n x \partial_x^\beta \delta(x - x_0) f(x),$$

Note that there is a risk associated to using the notation $dx \delta(x - x_0)$. The (often encountered) expression $\int_a^d dx \delta(x - x_0) f(x)$ is ambiguous as it can denote any of the following:

$$\begin{aligned} \int dx \delta(x - x_0) \mathbf{1}_{(a,b)}(x) f(x), & \quad \int dx \delta(x - x_0) \mathbf{1}_{[a,b)}(x) f(x), \\ \int dx \delta(x - x_0) \mathbf{1}_{(a,b]}(x) f(x), & \quad \int dx \delta(x - x_0) \mathbf{1}_{[a,b]}(x) f(x). \end{aligned}$$

If these were integrals with respect to the Lebesgue measure dx , as the notation suggests, then all theses expressions would be equal. As long as $x_0 \in (a, b)$ or $x_0 \in \mathbb{R} \setminus [a, b]$, all of them agree (and yield $f(x_0)$ respectively 0). The

²For instance $\int d^n y (\int d^n x e^{-iyx} \phi(x)) f(y) = \int d^n x \phi(x) \int d^n y e^{-iyx} f(y)$.

problem arises when $x_0 = a$ or $x_0 = b$, in which case one needs to recall that the integral is in fact an integral with respect to the Dirac measure and that $\int_I d\delta_{x_0} f(x)$ is 0 if $x_0 \notin I$ and $f(x_0)$ if $x_0 \in I$.

Exercise 18.2. Let f a function and denote x_i the solutions of the equation $f(x) = 0$. Then $\delta(f(x)) = \sum_i \frac{1}{|f'(x_i)|} \delta(x - x_i)$. We split in intervals of injectivity containing the x_i . On one interval, $\int dx \delta(f(x)) h(x) = \int dx |f'(x)| \frac{1}{|f'(x)|} \delta(f(x)) h(x) = \int_{y=f(x)} dy \frac{1}{|f'(f^{-1}(y))|} \delta(y) h(f^{-1}(y)) = \frac{1}{f'(f^{-1}(0))} h(f^{-1}(0)) = \frac{1}{|f'(x_0)|} h(x_0)$.

- the Heaviside Θ function and its associated distribution are:

$$\Theta_{t_0}(t) = \Theta(t - t_0) = \begin{cases} 1, & t - t_0 \geq 0 \\ 0, & t - t_0 < 0 \end{cases}, \quad (\Theta_{t_0}, f) = \int dt \Theta(t - t_0) f(t) = \int_{t_0}^{\infty} dt f(t).$$

The derivative of the Heaviside distribution is the Dirac δ :

$$\left(\frac{d}{dt}\Theta_{t_0}, f\right) = (\Theta_{t_0}, -\frac{d}{dt}f) = \int_{t_0}^{\infty} dt \left(-\frac{d}{dt}f\right) = f(t_0) = (\delta_{t_0}, f).$$

- the distribution³ $(t - t_0)\Theta_{t_0}(t)$ is such that

$$\left(\frac{d}{dt}[(t - t_0)\Theta_{t_0}(t)], f\right) = \int_{t_0}^{\infty} dt (t - t_0)\left(-\frac{d}{dt}f\right) = \int_{t_0}^{\infty} dt f(t) = (\Theta_{t_0}, f).$$

Note that $\delta(x - y)$ can be seen as the Schwartz kernel of the operator $A : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n)$, $A(h) = h$. Indeed, for any test function f , $(A(h), f) = (h, f) = \int d^n x h(x) f(x)$ and the kernel K is the distribution defined as $(K, h \otimes f) = \int d^n x h(x) f(x)$, often denoted as a double integral $\int d^n x d^n y \delta(x - y) h(y) f(x)$. The point of the Schwartz kernel theorem is that any reasonable linear operator A on the Schwartz space is an integral operator with a kernel:

$$(Ah)(y) = \int dx K(x, y) f(x).$$

In particular this puts integral and differential operators on the same footing: differential operators also have distributional kernels. We note the kernel K is a distribution: if $K(x, y)$ is a measurable function with $\int dx dy |K(x, y)|^2 < \infty$ the A is necessarily a compact operator. For more general operators the kernel is more singular. Note that the kernel of a product of operators is the integral of the product of the kernels:

$$(PQ)(x, y) = \int dz P(x, z) Q(z, y).$$

Proposition 18.1 (Convolution of tempered distributions with Schwartz functions). We consider $\phi \in \mathcal{S}'$, $f \in \mathcal{S}$, and we denote $f^-(x) = f(-x)$. Then:

$$(\phi \star f, g) \equiv (\phi, f^- \star g),$$

is a tempered distribution and the convolution product $\star$ is continuous in both its arguments. We have:

$$(\phi \star f, g) = \int d^d x \phi(x) \int d^n u f(-x + u) g(u) = \int d^n u \left(\int d^n x \phi(x) f(u - x) \right) g(u).$$

³The derivative of $|t - t_0|$ in the sense of distributions is:

$$\left(\frac{d}{dt}|t - t_0|, f\right) = \int_{-\infty}^{\infty} dt |t - t_0| \left(-\frac{d}{dt}f\right) = \int_{-\infty}^{t_0} dt (t_0 - t) \left(-\frac{d}{dt}f\right) + \int_{t_0}^{\infty} dt (t - t_0) \left(-\frac{d}{dt}f\right) = \int_{-\infty}^{\infty} dt \left[\Theta(t - t_0) - \Theta(-t + t_0) \right] f(t) = (\Theta_{t_0} - \Theta_{-t_0}^-, f).$$

Proof. The fact that $\phi \star f$ is a tempered distribution follows from the continuity of the ordinary convolution product on $\mathcal{S}$. The continuity of the convolution product on $\mathcal{S} \times \mathcal{S}'$ is straightforward. $\square$

Tempered distributions can be convoluted with more general functions, but we stress that this has limits: tempered distributions can not be convoluted together. For instance $1(x) = 1$ is a tempered distribution (it is just the integral), but $1 \star 1$ does not exist.

Exercise 18.3. Prove that in the sense of distributions $\delta(x - x_0) = \int \frac{d^n p}{(2\pi)^n} e^{-ip(x-x_0)}$. This can also be written as $\delta_{x_0} = \int \frac{d^n p}{(2\pi)^n} e^{ipx_0} (e^{-ipx}, \cdot)$. The equality $\delta(x - x_0) = \int \frac{d^n p}{(2\pi)^n} e^{-ip(x-x_0)}$ holds in the sense of distributions as:

$$\int \frac{d^n p}{(2\pi)^n} (e^{-ip(x-x_0)}, f) = \int \frac{d^n p}{(2\pi)^n} \int d^d x e^{-ip(x-x_0)} f(x) = \int \frac{d^n p}{(2\pi)^n} e^{ipx_0} \mathcal{F}(f)(p) = \mathcal{F}^{-1}(\mathcal{F}(f))(x_0) = f(x_0) = (\delta_{x_0}, f).$$

The crucial point is that, in the sense of distributions, the right hand side takes the form $\int (2\pi)^{-n} d^n p e^{ipx_0} (e^{-ipx}, f)$, that is one first applies the distribution e^{-ipx} to f and then integrates over p , not the other way around. This is crucial because Fubini's theorem can not be applied as the double integral of the absolute value is not convergent (nothing depends on p in the integrand), hence the integrals over p and x can not be freely interchanged.

Exercise 18.4 (Nascent delta). A nascent delta sequence is a sequence of functions η_ϵ such that $\int dx \eta_\epsilon(x) = 1$ and $\lim_{\epsilon \rightarrow 0} \int dx \eta_\epsilon(x) f(x) = f(0)$ for any $f \in \mathcal{S}$. Let $\eta: \mathbb{R} \rightarrow \mathbb{R}_+$ a function with compact support such that $\int dx \eta(x) = 1$. Prove that $\eta_\epsilon(x) = \frac{1}{\epsilon} \eta(\frac{x}{\epsilon})$ is a nascent delta sequence. $\lim_{\epsilon \rightarrow 0} \int dx \frac{1}{\epsilon} \eta(\frac{x}{\epsilon}) f(x) = \lim_{\epsilon \rightarrow 0} \int dy \eta(y) f(\epsilon y) = f(0)$ by Lebesgue's dominated converge theorem.

Let $\phi_t(x) = \frac{1}{(2\pi t)^{n/2}} e^{-\frac{|x-x_0|^2}{2t}}$. Prove that in the sense of distributions $\lim_{t \rightarrow 0} \phi_t = \delta_{x_0}$. Compute $\mathcal{F}(\phi_t)$ and $\mathcal{F}(\delta_{x_0})$ and check directly that in the sense of distributions $\lim_{t \rightarrow 0} \mathcal{F}(\phi_t) = \mathcal{F}(\delta_{x_0})$. We have $(\phi_t, f) = \int \frac{du}{\sqrt{2\pi t}} e^{-\frac{(u-x_0)^2}{2t}} f(u) = \int \frac{dv}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} f(x_0 + \sqrt{t}v) \rightarrow f(x_0)$ where we used Lebesgue's dominated convergence theorem. It is straightforward that $\mathcal{F}(\phi_t)(p) = \int dx \phi_t(x) e^{-ipx} = e^{-\frac{1}{2} p^2 t - ipx_0}$. On the other hand $(\mathcal{F}(\delta_{x_0}), f) = (\delta_{x_0}, \mathcal{F}(f)) = \int dp f(p) e^{-ipx_0}$ hence $\mathcal{F}(\delta_{x_0})(p) = e^{-ipx_0}$. We have:

$$\lim_{t \rightarrow 0} (\mathcal{F}(\phi_t), f) = \lim_{t \rightarrow 0} \left(\int dp e^{-\frac{1}{2} p^2 t - ipx_0} f(p) \right) \stackrel{LDCT}{=} \int dp e^{-ipx_0} f(p) = (\mathcal{F}(\delta_{x_0}), f).$$

Remark that:

$$\begin{aligned} (\delta_{x_0}, f) &= \lim_{t \rightarrow 0} (\phi_t, f) = \lim_{t \rightarrow 0} \int dx \frac{1}{\sqrt{2\pi t}} e^{-\frac{|x-x_0|^2}{2t}} f(x) = \lim_{t \rightarrow 0} \int dx \int \frac{dp}{2\pi} e^{-\frac{1}{2} p^2 t - ipx_0} f(x) \\ &\stackrel{\text{Fubini at fixed } t}{=} \lim_{t \rightarrow 0} \int \frac{dp}{2\pi} e^{-\frac{1}{2} p^2 t} \underbrace{\int dx e^{-ip(x-x_0)} f(x)}_{e^{ipx_0} \hat{f}(p)} \stackrel{LDCT}{=} \int \frac{dp}{2\pi} \lim_{t \rightarrow 0} e^{-\frac{1}{2} p^2 t} \hat{f}(p) = \int \frac{dp}{2\pi} e^{ipx_0} (e^{-ipx}, f). \end{aligned}$$

Exercise 18.5. Compute $\mathcal{F}(\Theta_{x_0})$ and $\mathcal{F}(\delta_{x_0})$ in the sense of distributions. Note that the natural variable of the Fourier transform is usually called p . We have:

$$(\mathcal{F}(\Theta_{x_0}), f) = (\Theta_{x_0}, \mathcal{F}(f)) = \int_{x_0}^{\infty} dx \mathcal{F}(f)(x) = \lim_{\epsilon \searrow 0} \int_{x_0}^{\infty} dx e^{-\epsilon x} \mathcal{F}(f)(x) \stackrel{\text{Fubini at fixed } \epsilon}{=} \lim_{\epsilon \searrow 0} \int_{x_0}^{\infty} dx \int dp f(p) e^{-\epsilon x - ipx} = \lim_{\epsilon \searrow 0} \int dp f(p) \frac{e^{-(\epsilon+ip)x_0}}{\epsilon+ip} = \left(\lim_{\epsilon \searrow 0} \frac{-ie^{-(\epsilon+ip)x_0}}{p-i\epsilon}, f \right).$$

For the second one we have:

$$(\mathcal{F}(\delta_{x_0}), f) = (\delta_{x_0}, \mathcal{F}(f)) = \mathcal{F}(f)(x_0) = \int dp f(p) e^{-ipx_0} = (e^{-ipx_0}, f),$$

and we note that $\mathcal{F}(\delta_0) = 1$. Also, we note:

$$(\delta_{x_0}, f) = (\mathcal{F}^{-1} \mathcal{F}(\delta_{x_0}), f) = (\mathcal{F}(\delta_{x_0}), \mathcal{F}^{-1}(f)) = \int dp e^{-ipx_0} \int \frac{du}{2\pi} f(u) e^{-iup} = \int \frac{dp}{2\pi} e^{-ipx_0} (e^{ipx}, f)$$

Exercise 18.6. Recall that for functions $\mathcal{F}(\partial_x f)(p) = \int dx (\partial_x f) e^{-ipx} = ip \mathcal{F}(f)(p)$ and $\mathcal{F}(xf)(p) = i \partial_p \mathcal{F}(f)(p)$. Compute $\mathcal{F}(\partial_x \phi)$ and $\mathcal{F}(x\phi)$ in the sense of distributions. We have

$$(\mathcal{F}(\partial_x \phi), f) = (\partial_x \phi, \mathcal{F}(f)) = (\phi, -\partial_x \mathcal{F}(f)) = (\phi, \mathcal{F}(ipf)) = (ip \mathcal{F}(\phi), f), \quad (\mathcal{F}(x\phi), f) = (\phi, x \mathcal{F}(f)) = (\phi, \mathcal{F}(-i \partial_p f)) = (i \partial_p \mathcal{F}(\phi), f)$$

Exercise 18.7. Compute $x \delta'_{x_0}$. We have $(x \delta'_{x_0}, f) = (\delta'_{x_0}, xf) = (\delta_{x_0}, -f - xf') = -(\delta_{x_0}, f) - x_0 (\delta_{x_0}, f')$ hence $x \delta'_{x_0} = -\delta_{x_0} + x_0 \delta'_{x_0}$.

Exercise 18.8. Prove that in the sense of distributions $x \delta_{x_0} = x_0 \delta_{x_0}$ and $-i \frac{d}{dx} e^{ipx} = p e^{ipx}$.

Exercise 18.9. Prove that in the sense of distributions $\delta \star f = f$ for any Schwartz function f . We have $(\delta_0 \star f, g) = (\delta_0, f \star g) = (\delta_0, f \star g) =$

$$(\delta_0, \int d^n u f^-(x-u) g(u)) = (\delta_0, \int d^n u f(u-x) g(u)) = \int d^n u f(u) g(u) = (f, g).$$

18.1.1 The Sokhotski–Plemelj formula

We will prove a very useful identity for distributions in dimension one. The function $1/x$ is not integrable at zero, hence it does not belong to $\mathcal{S}'(\mathbb{R})$. However, there is a version of the distribution $1/x$ which is a tempered distribution, the so called *principal value* $1/x$:

$$(\text{P.V.} \left(\frac{1}{x} \right), f) = \lim_{\epsilon \rightarrow 0} \left[\int_{-\infty}^{-\epsilon} dx \frac{1}{x} f(x) + \int_{\epsilon}^{\infty} dx \frac{1}{x} f(x) \right] = \int_0^{\infty} dx \frac{f(x) - f(-x)}{x}.$$

Indeed, the integral converges absolutely both in 0 and at infinity.

Proposition 18.2 ([Sokhotski–Plemelj formula](#)). *In the sense of distributions we have:*

$$\lim_{\epsilon \rightarrow 0} \frac{1}{x \pm i\epsilon} = \mp i\pi \delta(x) + \text{P.V.} \left(\frac{1}{x} \right).$$

Proof. For any test function $f \in \mathcal{S}(\mathbb{R})$ we have:

$$\int_{-\infty}^{\infty} dx \frac{f(x)}{x \pm i\epsilon} = \mp i \int_{-\infty}^{\infty} dx \frac{\epsilon}{x^2 + \epsilon^2} f(x) + \int_{-\infty}^{\infty} dx \frac{x}{x^2 + \epsilon^2} f(x)$$

We can commute the limits and the integrals due to Lebesgue's dominated convergence theorem. The first integral is:

$$\int_{-\infty}^{\infty} dx \frac{\epsilon}{x^2 + \epsilon^2} f(x) = \int_{-\infty}^{\infty} dy \frac{1}{y^2 + 1} f(\epsilon y) \xrightarrow{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} dy \frac{1}{y^2 + 1} f(0) = \pi f(0),$$

while the second one is:

$$\int_{-\infty}^{\infty} dx \frac{x}{x^2 + \epsilon^2} f(x) = \int_0^{\infty} dx \frac{x}{x^2 + \epsilon^2} (f(x) - f(-x)) \xrightarrow{\epsilon \rightarrow 0} \int_0^{\infty} dx \frac{f(x) - f(-x)}{x}.$$

□

18.1.2 The Dirac comb and the Poisson summation formula

The Dirac comb is the infinite sum of δ pics $D(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$. Observe that $D(t)$ is periodic of period T , so that one can expand it in Fourier series. Choosing as principal period $(-T/2, T/2)$ we have:

$$D(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T} \left(\int_{-T/2}^{T/2} du e^{-in \frac{2\pi}{T} u} D(u) \right) e^{in \frac{2\pi}{T} t} = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{in \frac{2\pi}{T} t}.$$

On the other hand, one can compute the Fourier transform:

$$\hat{D}(\omega) = \int_{-\infty}^{\infty} dt D(t) e^{-i\omega t} = \frac{1}{T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} dt e^{in \frac{2\pi}{T} t} e^{-i\omega t} = \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\omega - n \frac{2\pi}{T}),$$

that is Fourier transform of the Dirac comb is the Dirac comb.

Proposition 18.3 ([The Poisson summation formula](#)). *For any function $f \in \mathcal{S}$ we have:*

$$\sum_{k=-\infty}^{\infty} f(kT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} \hat{f}(n \frac{2\pi}{T})$$

Proof. We have

$$(D, f) = (\hat{D}, \check{f}) = \int d\omega \left(\frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\omega - n \frac{2\pi}{T}) \right) \int \frac{du}{2\pi} f(u) e^{i\omega u} = \frac{1}{T} \sum_{n=-\infty}^{\infty} \int du f(u) e^{in \frac{2\pi}{T} u}.$$

□

More generally, one can prove:

$$\sum_{k=-\infty}^{\infty} f(x + kT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} \hat{f}(n \frac{2\pi}{T}) e^{in \frac{2\pi}{T} x},$$

by using a shifted comb $D_x(t) = \sum_{k \in \mathbb{Z}} \delta(t - x - kT)$.

18.1.3 The Riemann zeta function

Applying $\sum_{k=-\infty}^{\infty} \delta(t - kT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{in \frac{2\pi}{T} t}$ on the function e^{-t^2} we get:

$$\sum_{k=-\infty}^{\infty} e^{-k^2 T^2} = \frac{1}{T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} dt e^{in \frac{2\pi}{T} t} e^{-t^2} = \frac{\sqrt{\pi}}{T} \sum_{n=-\infty}^{\infty} e^{-(\frac{\pi}{T})^2 n^2} \xrightarrow{T=\sqrt{\pi x}} \sum_{k=-\infty}^{\infty} e^{-\pi k^2 x} = \frac{1}{\sqrt{x}} \sum_{n=-\infty}^{\infty} e^{-\frac{\pi}{x} n^2}.$$

Defining $\omega(x) = \sum_{k=1}^{\infty} e^{-\pi k^2 x}$ this becomes $1 + 2\omega(x) = x^{-1/2} [1 + 2\omega(1/x)]$, that is:

$$\omega\left(\frac{1}{x}\right) = -\frac{1}{2} + \frac{1}{2} \sqrt{x} + \sqrt{x} \omega(x).$$

We define the Riemann zeta function by $\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}$ for $\text{Re}(s) > 1$. We aim to maximally analytically continue this function. We note that in the original region of convergence we have:

$$\Gamma\left(\frac{s}{2}\right) = \pi^{s/2} n^s \int_0^{\infty} dx x^{\frac{s}{2}-1} e^{-\pi n^2 x} \Rightarrow \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \int_0^{\infty} dx x^{\frac{s}{2}-1} \omega(x) = \int_1^{\infty} dx \left[x^{-\frac{s}{2}-1} \omega\left(\frac{1}{x}\right) + x^{\frac{s}{2}-1} \omega(x) \right],$$

where we split the integration interval at 1 and changed variable $x \rightarrow 1/x$ in the first integral. We then get:

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = -\frac{1}{2} \int_1^{\infty} dx x^{-\frac{s}{2}-1} + \frac{1}{2} \int_1^{\infty} dx x^{-\frac{s-1}{2}-1} + \int_1^{\infty} dx (x^{-\frac{s+1}{2}} + x^{\frac{s}{2}-1}) \omega(x) = \frac{1}{s(s-1)} + \int_1^{\infty} dx (x^{-\frac{s+1}{2}} + x^{\frac{s}{2}-1}) \omega(x),$$

where the integral converges for any s . In particular, as the integral is invariant under $s \rightarrow 1-s$, the (analytic continuation of the) Riemann zeta function respects:

$$\zeta(s) = \frac{\pi^{s-\frac{1}{2}} \Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)} \zeta(1-s).$$

Recalling that the Euler Γ function has no zeros in the complex plane and it has simple poles at the negative integers and 0 we conclude that:

- rewriting the integral formula as:

$$\zeta(s) = \frac{\pi^{s/2}}{2 \Gamma(1 + \frac{s}{2})(s-1)} + \frac{\pi^{s/2}}{\Gamma(\frac{s}{2})} \int_1^{\infty} dx (x^{-\frac{s+1}{2}} + x^{\frac{s}{2}-1}) \omega(x),$$

we have that $\zeta(s)$ is meromorphic in $\mathbb{C}$, has a pole of order 1 in $s = 1$ with residue $\lim_{s \rightarrow 1} (s-1) \zeta(s) = \frac{\pi^{1/2}}{2 \Gamma(\frac{3}{2})} = 1$, and $\zeta(0) = -\frac{1}{2}$.

- for $n \in \mathbb{N}_{\geq 1}$ we have $\zeta(-2n) \sim \frac{1}{\Gamma(-n)}\zeta(1+2n) = 0$, hence the even negative integers are trivial zeros of the Riemann zeta function.
- $\zeta(-1) = \frac{\pi^{-\frac{3}{2}}\Gamma(1)}{\Gamma(-\frac{1}{2})}\zeta(2) = \frac{\pi^{-\frac{3}{2}}(-1/2)}{\sqrt{\pi}}\frac{\pi^2}{6} = -\frac{1}{12}$.
- the non trivial zeros of the zeta function should be located on the line $\operatorname{Re}(s) = \frac{1}{2}$.

18.2 Differential operators and fundamental solutions

Let $P(\partial)$ be a linear partial differential operator $P(\partial) = \sum_{\alpha} h_{\alpha}(x)\partial^{\alpha}$ and $P^{\dagger}(\partial) = \sum_{\alpha} (-1)^{|\alpha|}\partial^{\alpha}[h_{\alpha}(x) \cdot]$ its formal hermitian conjugate (adjoint). We are interested in solving the inhomogenous linear differential equation $P(\partial)F = f$, that is in determining the distribution F for arbitrary f , which in first instance we can assume to be Schwartz.

We first assume $P(\partial)$ has constant coefficients, in which case $P^{\dagger}(\partial) = P(-\partial)$. The Malgrange–Ehrenpreis theorem states that in this case a *fundamental solution* of P always exists, that is a distribution G such that:

$$P(\partial)G = \delta.$$

The fundamental solution is not unique: adding a solution of the homogeneous equation $P(\partial)G_0 = 0$ will lead to an equally valid fundamental solution. As the convolution product of Schwartz functions is commutative and for any two Schwartz functions and any linear differential operator with constant coefficients $(P(\partial)g) \star f = \int d^n u [P(\partial)g](x-u) f(u) = P(\partial)(g \star f)$, a solution of the inhomogenous equation is $F = G \star f$:

$$\left(P(\partial)(G \star f), g\right) = \left(G \star f, P^{\dagger}(\partial)g\right) = \left(G, (P^{\dagger}(\partial)g) \star f^{-}\right) = \left(G, P^{\dagger}(\partial)(g \star f^{-})\right) = (P(\partial)G, g \star f^{-}) = (\delta, g \star f^{-}) = (\delta \star f, g) = (f, g).$$

In order to find fundamental solutions for $P(\partial)$ with constant coefficients, it is often a good idea to Fourier transform. Recall that for Schwartz functions the Fourier transform is defined as:

$$\mathcal{F}(f)(p) = \hat{f}(p) = \int d^n x f(x) e^{-ipx},$$

while for a distribution ϕ we have $(\mathcal{F}(\phi), g) = (\phi, \mathcal{F}(g))$. As $\mathcal{F}(\delta) = 1$ in the sense of distributions, taking into account that the natural variable of the Fourier transform is usually denoted p , the fundamental solution respects:

$$(1, g)_p = (\mathcal{F}(P(\partial)G), g)_p = (P(\partial)G, \mathcal{F}(g))_x = (G, P(-\partial)\mathcal{F}(g))_x = (G, \mathcal{F}(P(ip)g))_x = (P(ip)\mathcal{F}(G), g)_p,$$

hence it suffices to solve $\mathcal{F}(P(\partial)G) = P(ip)\hat{G}(p) = 1$ and Fourier transform back. The solution in Fourier space is $\hat{G}(p) = "1/P(ip)"^4$ and the fundamental solution is obtained by taking the inverse Fourier transform of $\hat{G}(p)$ in the sense of distributions:

$$G(x) = \int \frac{d^n p}{(2\pi)^n} \hat{G}(p) e^{ipx} = \int \frac{d^n p}{(2\pi)^n} \frac{1}{P(ip)} e^{ipx},$$

where, if need be, one takes into account that the above should be interpreted as a distribution $\int \frac{d^n p}{(2\pi)^n} \frac{1}{P(ip)} (e^{ipx}, \cdot)_x$.

Examples of fundamental solutions for one dimensional differential operators are:

- for $P(\frac{d}{dt}) = (\frac{d}{dt})^{n+1}$ we have $G(t) = \frac{t^n}{n!}\Theta(t)$ as:

$$\left(\frac{d}{dt} \left(\frac{t^n}{n!}\Theta(t)\right), g\right) = \int_{-\infty}^{\infty} dt \frac{t^n}{n!}\Theta(t)(-1)^{n+1}g^{(n+1)}(t) = \int_0^{\infty} dt \frac{t^n}{n!}(-1)^{n+1}g^{(n+1)}(t) = \int_0^{\infty} dt (-1)g'(t) = g(0) = (\delta, g).$$

⁴The proof that this inverse always exists in the sens of distributions is the non trivial part of the proof of the Malgrange–Ehrenpreis theorem.

- for $P(\frac{d}{dt}) = (\frac{d}{dt} + \gamma)^2$ with $\gamma \geq 0$ we have $G(t) = te^{-\gamma t}\Theta(t)$ as:

$$\left((\frac{d}{dt} + \gamma)^2 (te^{-\gamma t}\Theta(t)), g\right) = \int_0^\infty dt te^{-\gamma t} (-\frac{d}{dt} + \gamma)^2 g(t) = \int_0^\infty dt e^{-\gamma t} (-\frac{d}{dt} + \gamma)g(t) = g(0) .$$

- for $P(\frac{d}{dt}) = \frac{d^2}{dt^2} + 2\gamma\frac{d}{dt} + \omega_0^2$ with $\gamma \geq 0$ we have $G(t) = \Theta(t)e^{-\gamma t}\frac{\sin(\omega t)}{\omega}$ with $\omega = \sqrt{\omega_0^2 - \gamma^2}$. To see this, we use $G(t) = \int \frac{dp}{2\pi} \hat{G}(p) e^{ipt}$ to get:

$$(-p^2 + 2ip\gamma + \omega_0^2)\hat{G}(p) = 1 \Rightarrow G(t) = \int_{-\infty}^\infty \frac{dp}{2\pi} \frac{(-1)e^{ipt}}{[p - (\gamma + \sqrt{\omega_0^2 - \gamma^2})][p - (\gamma - \sqrt{\omega_0^2 - \gamma^2})]} .$$

For $t < 0$ we close the contour of integration with a semicircle at infinity in the lower half plane of p . As the contour traps no poles, the integral is zero. For $t > 0$, we close it in the upper half plane and we trap both poles with positive contours. Using the residue theorem we have:

$$\begin{aligned} G(t) &= \Theta(t) i \left[\lim_{p \rightarrow \gamma + \sqrt{\omega_0^2 - \gamma^2}} \frac{(-1)e^{ipt}}{p - (\gamma - \sqrt{\omega_0^2 - \gamma^2})} + \lim_{p \rightarrow \gamma - \sqrt{\omega_0^2 - \gamma^2}} \frac{(-1)e^{ipt}}{p - (\gamma + \sqrt{\omega_0^2 - \gamma^2})} \right] = \\ &= \Theta(t) (-i) e^{-\gamma t} \left[\frac{e^{i\omega t}}{2\omega} + \frac{e^{-i\omega t}}{-2\omega} \right] = \Theta(t) e^{-\gamma t} \frac{\sin(\omega t)}{\omega} . \end{aligned}$$

Exercise 18.10. Find a solution of the forced harmonic oscillator equation:

$$\left(\frac{d^2}{dt^2} + 2\gamma\frac{d}{dt} + \omega_0^2\right)F = e^{i\alpha t} .$$

We have

$$F(u) = \int_{-\infty}^\infty dt \Theta(t) e^{-\gamma t} \frac{\sin(\omega t)}{\omega} e^{i\alpha(u-t)} = e^{i\alpha u} \int_0^\infty dt \frac{e^{-(\gamma - i\omega + i\alpha)t} - e^{-(\gamma + i\omega + i\alpha)t}}{2i\omega} = \frac{e^{i\alpha u}}{2i\omega} \left[\frac{1}{\gamma - i\omega + i\alpha} - \frac{1}{\gamma + i\omega + i\alpha} \right] = \frac{e^{i\alpha u}}{(\gamma + i\alpha)^2 + \omega^2} = \frac{e^{i\alpha u}}{\omega_0^2 - \alpha^2 + 2i\alpha\gamma}$$

18.2.1 The Laplacian

The Laplace operator in dimension d is $\Delta = \sum_{i=1}^d (\partial_i)^2$. We are interested in the Laplacian acting on functions $f: \mathbb{R}^d \rightarrow \mathbb{C}$. This operator has the following properties:

- The a priori domain of definition of Δ is the space of Schwartz functions, $\Delta: \mathcal{S}(\mathbb{R}^d) \rightarrow \mathcal{S}(\mathbb{R}^d)$.
- Δ is a linear and *continuous operator* on $\mathcal{S}(\mathbb{R}^d)$ with distance function d , because for all $\|\cdot\|_{\alpha,\beta}$ seminorms we have $\|\Delta f_n - \Delta f\|_{\alpha,\beta} \leq \sum_{i=1}^d \|f_n - f\|_{\alpha,\beta'_i}$, where $\beta'_i = (\beta_1, \dots, \beta_i + 2, \dots, \beta_d)$. As $\mathcal{S}$ is not a Banach space, the questions whether Δ is “bounded” does not make sense.
- In Fourier transform $\mathcal{F}(-\Delta g)(p) = p^2 \hat{g}(p)$. It follows that Δ does not have zero modes in $\mathcal{S}(\mathbb{R}^d)$ hence it *is injective*. But Δ is *not surjective* on $\mathcal{S}(\mathbb{R}^d)$ as $\text{Range}(\Delta) = \{h \in \mathcal{S}(\mathbb{R}^d) \mid \|\frac{1}{p^2} \hat{h}\|_{\alpha,\beta} < \infty\}$ is a strict subset of $\mathcal{S}(\mathbb{R}^d)$. In particular, $\hat{h} \in \text{Range}(\Delta)$ only if for $p \rightarrow 0$, $\hat{h}(p)$ goes to zero at least as fast as p^2 .
- The *fractional Laplacian* $(-\Delta)^\zeta$ is the operator such that $\mathcal{F}((-\Delta)^\zeta g)(p) = p^{2\zeta} \hat{g}(p)$.
- As $\Delta: \mathcal{S}(\mathbb{R}^d) \rightarrow \mathcal{S}(\mathbb{R}^d) \subset \mathcal{S}'(\mathbb{R}^d)$ is a continuous operator, it has an associated kernel:

$$((\Delta h), f) = \int d^d x d^d y \Delta(x, y) h(x) g(y) = \int d^d x d^d y \left[\sum (\partial_{x^i})^2 \delta(x - y) \right] h(x) g(y) .$$

- $-\Delta$ is a positive operator on $\mathcal{S}(\mathbb{R}^d)$:

$$\langle g, (-\Delta)g \rangle = \int d^d x \overline{g(x)} (-\Delta g)(x) = \int d^d x \sum_{i=1}^d \overline{\partial_i g(x)} \partial_i g(x) \geq 0.$$

- $-\Delta$ is densely defined on $L^2(\mathbb{R}^d)$, as $\mathcal{S}(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$ and is a symmetric operator:

$$\forall f, g \in \mathcal{S}(\mathbb{R}^d) \quad \langle f, -\Delta g \rangle = \int d^d x \sum_{i=1}^d (\partial_i \bar{f})(\partial_i g) = \langle -\Delta f, g \rangle,$$

where the boundary terms are zero. Taking the Fourier transform, the resolvent $(-\Delta - z)^{-1}$ acting on $f \in L^2(\mathbb{R}^d)$ is such that:

$$\|(-\Delta - z)^{-1} f\|^2 = \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 - \bar{z}} \overline{\hat{f}(p)} \frac{1}{p^2 - z} \hat{f}(p) \leq \frac{1}{\inf_{p \in \mathbb{R}^d} (|p^2 - \operatorname{Re}(z)|^2 + |\operatorname{Im}(z)|^2)} \|f\|^2,$$

hence $-\Delta$ has real spectrum. It follows that $-\Delta$ is self adjoint on $L^2(\mathbb{R}^d)$. In fact the spectrum of $-\Delta$ consists in all the positive real numbers and any $x \in \mathbb{R}_+$ is in the continuous spectrum as $(p^2 - x)\hat{f}(p) = 0$ implies $\hat{f}(p) = 0$ almost everywhere.

- $-\Delta$ is an *unbounded* operator on $L^2(\mathbb{R}^d)$, hence *not continuous in norm* L^2 .

As it is not surjective, $-\Delta$ can not be inverted on $\mathcal{S}(\mathbb{R}^d)$. However, one can determine one of its fundamental solution which plays the role of an inverse in the sense of distributions. To this end, we solve in the sense of distributions $(-\Delta)G = \delta$, which in momentum space becomes $p^2 \hat{G}(p) = 1$ hence $\hat{G}(p) = \frac{1}{p^2}$. All that remains to be done is to Fourier transform back:

$$(G, g) = (\hat{G}, \check{g}) = \int d^d p \frac{1}{p^2} \int \frac{d^d y}{(2\pi)^d} e^{ipy} g(y).$$

However, there is a subtlety. The equation $-\Delta G = \delta$ holds in the sense of distributions, that is $((-\Delta G), g) = (G, (-\Delta)g)$ and the image of $-\Delta$ consists in test functions that, in momentum space behave like p^2 for $p \rightarrow 0$. A priori the distribution G is supported only on the range of $-\Delta$ which does not cover the whole of $\mathcal{S}(\mathbb{R}^d)$: for instance a necessary condition for a function to belong to the support of G is $\hat{h}(0) = 0 = \int d^d x h(x)$, which is respected by only a subset of the Schwartz functions. It is not a priori clear that G can be extended to the whole of $\mathcal{S}$. Keeping this in mind, we invert Fourier transform:

$$(G, g) = \int d^d p \hat{G}(p) \check{g}(p) = \int d^d y \left(\int \frac{d^d p}{(2\pi)^d} \frac{e^{ipy}}{p^2} \right) g(y).$$

We will in fact compute the slightly more general inverse Fourier transform of $1/p^{2\zeta}$, but we are interested in the case $\zeta = 1$. Using the integral representation of the Γ function we have:

$$\begin{aligned} G(y) &= \int \frac{d^d p}{(2\pi)^d} e^{ipy} \frac{1}{\Gamma(\zeta)} \int_0^\infty d\alpha \alpha^{\zeta-1} e^{-\alpha p^2} = \frac{1}{(4\pi)^{d/2} \Gamma(\zeta)} \int_0^\infty d\alpha \alpha^{\zeta-\frac{d}{2}-1} e^{-\frac{|y|^2}{4\alpha}} \\ &= \frac{4^{\frac{d-2\zeta}{2}}}{(4\pi)^{d/2} \Gamma(\zeta)} \frac{1}{|y|^{d-2\zeta}} \int_0^\infty du u^{\zeta-\frac{d}{2}-1} e^{-\frac{u}{u}} = \frac{4^{-\zeta}}{\pi^{d/2} \Gamma(\zeta)} \frac{1}{|y|^{d-2\zeta}} \int_0^\infty du u^{\frac{d}{2}-\zeta-1} e^{-u} = \frac{4^{-\zeta}}{\pi^{d/2} \Gamma(\zeta)} \frac{1}{|y|^{d-2\zeta}} \Gamma\left(\frac{d}{2} - \zeta\right). \end{aligned}$$

For $d \geq 3$ we can just set $\zeta = 1$ and obtain the fundamental solution of the Laplacian:

$$G(y) = \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \frac{1}{|y|^{d-2}}.$$

For $d = 2$, $\zeta = 1$ the argument of the $\Gamma(d/2 - \zeta)$ becomes 0 and the Gamma function diverges. In order to find the fundamental solution in this case we need to set $d = 1$, $\zeta = 1 - \epsilon$ and send $\epsilon \rightarrow 0$:

$$G(y) = \frac{1}{4\pi} \lim_{\epsilon \rightarrow 0} \left(\frac{1}{|y|^{2\epsilon}} \Gamma(\epsilon) \right) = \frac{1}{4\pi} \lim_{\epsilon \rightarrow 0} \left(e^{-2\epsilon \ln |y|} \frac{\Gamma(1 + \epsilon)}{\epsilon} \right) = \frac{1}{4\pi} \lim_{\epsilon \rightarrow 0} \left(\frac{1}{\epsilon} - 2 \ln |y| \right) = -\frac{1}{2\pi} \ln |y| + \lim_{\epsilon \rightarrow 0} \frac{1}{4\pi\epsilon} .$$

The second term is divergent, which seems to indicate a problem. However, as the a priori domain of G contains only functions h such that $(1, h) = 0$, the second term is $\lim_{\epsilon \rightarrow 0} (\frac{1}{\epsilon}, h) = 0$ and the fundamental solution is:

$$G(y) = -\frac{1}{2\pi} \ln |y| .$$

The subtraction of the infinite constant term can be seen as shifting the fundamental solution by a solution of the homogeneous equation $(-\Delta)G_0 = 0$, $G_0 = \text{constant}$. Now G can be extended to the whole of $\mathcal{S}$.

In electromagnetism $\text{div} E = \rho$ and $E = -\text{grad} \varphi$ hence $(-\Delta)\varphi = \rho$. The electric potential generated by the charge density ρ is then:

$$\varphi = G \star \rho = \int d^d u G(x - u) f(u) \sim \begin{cases} \int d^d u \frac{\rho(u)}{|x - u|^{d-2}} \\ \int d^2 u \rho(u) \ln |x - u| \end{cases} .$$

18.2.2 The d'Alembert operator

The d'Alembert operator $\square = \partial_t^2 - \sum_{i=1}^d \partial_i \partial^i$ corresponds to the Laplacian, but for a Lorentzian manifold with metric $\eta = (-, +, +, +)$. Contrary to the Laplacian, it has zero modes, hence it is not injective. Our aim is to find a fundamental solution of $\square$, that is to compute:⁵

$$G(t, x) = \int \frac{d\omega d^d p}{(2\pi)^{d+1}} \frac{e^{-i\omega t + i p x}}{-\omega^2 + p^2} = \int \frac{d\omega d^d p}{(2\pi)^{d+1}} \frac{e^{-i\omega t + i p x}}{2|p|} \left(\frac{1}{|p| - \omega} + \frac{1}{|p| + \omega} \right) .$$

As $|e^{-i\omega t}| = e^{t \text{Im}(\omega)}$, for $t > 0$, the contour of integration in ω can be completed by a semicircle in the lower half plane of ω , while for $t < 0$ it can be completed by a semicircle in the upper half plane.

One immediately remarks that this integral is in fact not well defined, as the integration contour traverses a non integrable singularity at $\omega = \pm |p|$. One adopts a regularization prescription, consisting in shifting the poles by an ϵ amount. We can chose to either move both poles in the same direction, or to move them in distinct directions:⁶ the last case is called Feynman $i\epsilon$ prescription. Separating the two poles, in each case we get (beware of the orientation of the contour):

$$\begin{aligned} G^{\text{ret}}(t, x) &= \lim_{\epsilon \rightarrow 0} \int \frac{d\omega d^d p}{(2\pi)^{d+1}} \frac{e^{-i\omega t + i p x}}{2|p|} \left(\frac{1}{|p| - i\epsilon - \omega} + \frac{1}{|p| + i\epsilon + \omega} \right) = i\Theta(t) \int \frac{d^d p}{(2\pi)^d} \frac{e^{i p x}}{2|p|} (e^{-it|p|} - e^{it|p|}) , \\ G^{\text{adv}}(t, x) &= \lim_{\epsilon \rightarrow 0} \int \frac{d\omega d^d p}{(2\pi)^{d+1}} \frac{e^{-i\omega t + i p x}}{2|p|} \left(\frac{1}{|p| + i\epsilon - \omega} + \frac{1}{|p| - i\epsilon + \omega} \right) = i\Theta(-t) \int \frac{d^d p}{(2\pi)^d} \frac{e^{i p x}}{2|p|} (-e^{-it|p|} + e^{it|p|}) , \\ G^{\text{Feyn}}(t, x) &= \lim_{\epsilon \rightarrow 0} \int \frac{d\omega d^d p}{(2\pi)^{d+1}} \frac{e^{-i\omega t + i p x}}{2|p|} \left(\frac{1}{|p| - i\epsilon - \omega} + \frac{1}{|p| - i\epsilon + \omega} \right) = i \int \frac{d^d p}{(2\pi)^d} \frac{e^{i p x}}{2|p|} (\Theta(t)e^{-it|p|} + \Theta(-t)e^{it|p|}) . \end{aligned}$$

⁵The $-$ sign on the ω in the exponential is purely conventional and can be replaced by a $+$.

⁶For the advanced and retarded cases, this has no neat interpretation. On the contrary, the Feynman case is a consistent complex rotation of $\pi-/2$ to imaginary frequencies and times. Indeed, the Euclidean Laplacian $p^2 + \omega_E^2$ becomes under a complex rotation $\omega_E = e^{i\frac{\pi}{2}} \omega$ the operator $p^2 - \omega^2 + i0$.

Using the same method for the Euclidean Laplacian in $d + 1$ dimension, $\hat{G}^{\text{Eucl.}}(\omega_E, p) = p^2 + \omega_E^2$, taking into account that $|e^{i\omega_E t_E}| = e^{-t_E \text{Im}(\omega_E)}$, we get:

$$G^{\text{Eucl.}}(t_E, x) = \lim_{\epsilon \rightarrow 0} \int \frac{d\omega_E d^d p}{(2\pi)^{d+1}} \frac{e^{i\omega_E t_E + i p x}}{2|p|} \left(\frac{1}{|p| + i\omega_E} + \frac{1}{|p| - i\omega_E} \right) = \int \frac{d^d p}{(2\pi)^d} \frac{e^{i p x}}{2|p|} \left(\Theta(t_E) e^{-t_E |p|} + \Theta(-t_E) e^{t_E |p|} \right).$$

Let us focus on the case $d = 3$, hence $d^3 p = |p|^2 d|p| \sin \theta d\theta d\varphi$ and, taking into account that $\theta \in [0, \pi]$:

$$\int \frac{\sin \theta d\theta d\varphi}{(2\pi)^3} e^{i|p||x| \cos \theta} = \frac{-1}{(2\pi)^2} \int_1^{-1} du e^{i|p||x|u} = \frac{1}{(2\pi)^2} \frac{e^{i|p||x|} - e^{-i|p||x|}}{i|p||x|}.$$

Gathering everything we have:

$$\begin{aligned} G^{\text{ret.}}(t, x) &= \frac{1}{2(2\pi)^2} \Theta(t) \frac{1}{|x|} \left[\int_0^\infty d|p| \left(e^{-i|p||t|+i|p||x|} - e^{-i|p||t|-i|p||x|} - e^{i|p||t|+i|p||x|} + e^{i|p||t|-i|p||x|} \right) \right], \\ G^{\text{adv.}}(t, x) &= G^{\text{ret.}}(-t, x), \\ G^{\text{Feyn.}}(t, x) &= \frac{1}{2(2\pi)^2} \frac{1}{|x|} \left[\Theta(t) + \Theta(-t) \right] \int_0^\infty d|p| \left(e^{-i|p||t|+i|p||x|} - e^{-i|p||t|-i|p||x|} \right), \end{aligned}$$

while in the Euclidean case one gets:

$$G^{\text{Eucl.}}(t_E, x) = \frac{1}{i} \frac{1}{2(2\pi)^2} \frac{1}{|x|} \left[\Theta(t_E) + \Theta(-t_E) \right] \int_0^\infty d|p| \left(e^{-|p||t_E|+i|p||x|} - e^{-|p||t_E|-i|p||x|} \right).$$

The integrals are computed by regularizing with a $-\epsilon|p|$ factor. For the retarded case we have four contributions, and using the Sokhotski-Plemelj formula we have:

$$\frac{1}{\epsilon + i(t - |x|)} - \frac{1}{\epsilon + i(t + |x|)} - \frac{1}{\epsilon + i(-t - |x|)} + \frac{1}{\epsilon + i(-t + |x|)} = 2\pi\delta(t - |x|) - 2\pi\delta(t + |x|),$$

and the second term drops because $t, |x| > 0$, therefore:

$$G^{\text{ret.}}(t, x) = \frac{1}{4\pi} \Theta(t) \frac{\delta(t - |x|)}{|x|}, \quad \int dt' d^3 x' G^{\text{ret.}}(t - t', x - x') f(t', x') = \frac{1}{4\pi} \int d^3 x' \frac{1}{|x - x'|} f(t - |x - x'|, x').$$

The retarded solution propagates to source from the past to the future. The advanced and Feynman solutions include propagation from the future to the past.

For the Feynman prescription we get:

$$\frac{1}{\epsilon + i(|t| - |x|)} - \frac{1}{\epsilon + i(|t| + |x|)} = \pi\delta(|t| - |x|) + \pi\delta(|t| + |x|) + \frac{1}{i} \text{P.V.} \left(\frac{1}{|t| - |x|} - \frac{1}{|t| + |x|} \right).$$

and the second delta again drops. As $\Theta(t) + \Theta(-t) = 1$ we have:

$$G^{\text{Feyn.}}(t, x) = \frac{1}{2(2\pi)^2} \frac{1}{|x|} \left[\pi\delta(|t| - |x|) + \frac{1}{i} \text{P.V.} \left(\frac{1}{|t| - |x|} - \frac{1}{|t| + |x|} \right) \right].$$

Things become more transparent if one reinstates the ϵ :

$$G^{\text{Feyn.}}(t, x) = \frac{1}{2(2\pi)^2} \frac{1}{|x|} \left(\frac{1}{\epsilon + i(|t| - |x|)} - \frac{1}{\epsilon + i(|t| + |x|)} \right) = \frac{i}{4\pi^2} \frac{1}{(\epsilon + i|t|)^2 + |x|^2} = \frac{i}{4\pi^2} \frac{1}{x^2 - t^2 + i\epsilon},$$

which is to be compared to the Euclidean solution:

$$G^{\text{Eucl}}(t_E, x) = \frac{1}{i} \frac{1}{2(2\pi)^2} \frac{1}{|x|} \left(\frac{1}{|t_E| - i|x|} - \frac{1}{|t_E| + i|x|} \right) = \frac{1}{4\pi^2} \frac{1}{x^2 + t_E^2}.$$

The real time solution can be obtained from the Euclidean solution $(x - x')^2 + (t_E - t'_E)^2$, by turning to real time $t_E = e^{i\frac{\pi-}{2}} t$. The additional i comes from the fact that the fundamental solution acts on imaginary time:

$$\int dt'_E dx' G^{\text{Eucl}}(t_E - t'_E, x - x') f_E(t_E, x), \quad \int dt' dx' G^{\text{Feyn}}(t - t', x - x') f(t', x'),$$

18.2.3 Green's functions

If $A = P(\partial)$ does not have constant coefficients, in stead of using fundamental solutions in order to solve inhomogenous partial differential equations, one uses the closely related notion of Green's function.

Let $A = \sum_{\alpha} h_{\alpha}(x) \partial^{\alpha}$ be a linear partial differential operator that maps $\mathcal{S}(\mathbb{R}^d)$ to $\mathcal{S}'(\mathbb{R}^d)$ having a kernel:⁷

$$(Ag)(x) = (A_x g)(x) = \sum_{\alpha} h_{\alpha}(x) \partial^{\alpha} g(x) = \int dy A(x, y) g(y) \quad A(x, y) = \sum_{\alpha} h_{\alpha}(x) \partial_x^{\alpha} \delta(x - y),$$

and we recall that the identity operator has kernel $\delta(x - y)$ as $(If)(x) = f(x) = \int \delta(x - y) f(y)$ and the operator product writes in terms of kernel as $(PQ)(x, y) = \int dz P(x, z) Q(z, y)$.

A Green's function G is the kernel of the operator inverse of A in distributions, that is:

$$AG = I \Rightarrow (A_x G)(x, y) = A_x G(x, y) = \delta(x - y), \quad (AGf)(x) = \int dy A_x G(x, y) f(y) = f(x),$$

and $F = Gf$ is solutions of the inhomogenous equation $AF = f$. If A has constant coefficients and $\tilde{G}$ is a fundamental solution $A_x \tilde{G} = \delta$, then $G(x, z) = \tilde{G}(x - z)$ is a Green function $(A_x G)(x - z) = \delta(x - z)$. In particular, Green functions that correspond to fundamental solutions are translation invariant $G(x + a, y + a) = G(x, y)$. There exist Green function that are not translation invariant, hence do not correspond to fundamental solutions.

If A is a positive, closed, self adjoint operator, the Hille–Yosida theorem⁸ guarantees that there exists a heat kernel expansion for the Green's function:

$$G = \frac{1}{A} = \int_0^{\infty} dt e^{-tA}, \quad \text{where} \quad \frac{d}{dt} e^{-tA} = -A e^{-tA}, \quad e^{-tA}|_{t=0} = I,$$

where the equality should be interpreted in the sense of distributions, that is $G(x, y)$ is the operator whose kernel is:

$$G(x, y) = \int_0^{\infty} dt (e^{-tA})(x, y), \quad \frac{d}{dt} e^{-tA}(x, y) = -A_x e^{-tA}(x, y), \quad (e^{-tA})(x, y)|_{t=0} = \delta(x - y).$$

Indeed $AG = \int_0^{\infty} A e^{-tA} = \int_0^{\infty} dt (-\frac{d}{dt}) e^{-tA} = -e^{-tA}|_0^{\infty} = I$, as long as $e^{-\infty A} = 0$. As an example, let us consider the operator (related to the Hamiltonian of the quantum harmonic oscillator) $H_x = -\Delta + \Omega^2 x^2$. We define the *Mehler kernel*:

$$K_t(x, y) = \frac{\Omega^{d/2}}{[4\pi \sinh(\Omega t)]^{d/2}} e^{-\frac{\Omega \cosh(\Omega t)}{4 \sinh(\Omega t)} (x^2 + y^2) - \frac{\Omega}{2 \sinh(\Omega t)} xy},$$

⁷We have $(Ag, f) = \int dx dy f(x) A(x, y) g(y) = \int dx dy [\sum_{\alpha} h_{\alpha}(x) \partial_x^{\alpha} \delta(x - y)] f(x) g(y) = \int dx dy [\sum_{\alpha} (-1)^{|\alpha|} \partial_x^{\alpha} (h_{\alpha}(x) f(x))] \delta(x - y) g(y) = \int dx (Ag)(x) f(x)$.

⁸The Hille–Yosida theorem states that if A is a closed operator with dense domain such that every $\lambda > \omega$ belongs to the resolvent set of A and $\|(A - \lambda I)^{-n}\| \leq M(\lambda - \omega)^n$ then A generates a strongly continuous semi group $T(0) = I$, $T(t + s) = T(t)T(s)$, $Ax = \lim_{h \searrow 0} \frac{1}{h} (T(h)x - x)$ such that $\|T(t)\| \leq M e^{\omega t}$

and it is easy to check that:

$$\lim_{t \rightarrow 0} K_t(x, y) = \delta(x - y), \quad \partial_t K_t(x, y) = -H_x K_t(x, y) \quad \Rightarrow \quad K_t(x, y) = (e^{-tH_x})(x, y),$$

leading to the explicitly non translation invariant Green function:⁹

$$G(x, y) = \left(\frac{1}{-\Delta + \Omega^2 x^2} \right) (x, y) = \int_0^\infty dt K_t(x, y) = \int_0^\infty dt \frac{\Omega^{d/2}}{[4\pi \sinh(\Omega t)]^{d/2}} e^{-\frac{\Omega \cosh(\Omega t)}{4 \sinh(\Omega t)}(x^2 + y^2) - \frac{\Omega}{2 \sinh(\Omega t)} xy}.$$

18.2.4 The heat diffusion operator and the heat kernel

For particle diffusion, we have the particle flux is proportional with minus the gradient of the density $j = -\nabla u$, and the continuity equation $\partial_t u + \nabla \cdot j = 0$ becomes $\partial_t u - \Delta u = 0$. The diffusion operator $D = \partial_t - \sum_{i=1}^d (\partial_i^2)$ singles out one coordinate in $\mathbb{R}^{d+1}$, $x = (t, x^1, \dots, x^d)$, and the equation $DG = \delta$ becomes in Fourier space:

$$(\imath\omega + p^2)\hat{G}(\omega, p) = 1, \quad G(t, y) = \int \frac{d\omega d^d p}{(2\pi)^{d+1}} \frac{e^{\imath\omega t + \imath p y}}{\imath\omega + p^2},$$

leading to the fundamental solution:

$$G(t, y) = \int \frac{d\omega}{2\pi} e^{\imath\omega t} \int_0^\infty d\alpha \frac{1}{(4\pi\alpha)^{d/2}} e^{-\frac{|y|^2}{4\alpha} - \imath\omega\alpha} = \int_0^\infty d\alpha \delta(t - \alpha) \frac{1}{(4\pi\alpha)^{d/2}} e^{-\frac{|y|^2}{4\alpha}} = \frac{1}{(4\pi t)^{d/2}} e^{-\frac{|y|^2}{4t}} \Theta(t).$$

One can check that $(\partial_t - \Delta_x) \frac{1}{(4\pi t)^{d/2}} e^{-\frac{|x-y|^2}{4t}} = 0$ and $\lim_{t \rightarrow 0} \frac{1}{(4\pi t)^{d/2}} e^{-\frac{|x-y|^2}{4t}} = \delta(x - y)$ hence $\frac{1}{(4\pi t)^{d/2}} e^{-\frac{|x-y|^2}{4t}}$ is in fact the heat kernel $(e^{-t(-\Delta)})(x, y)$. One can write a translation invariant Green function associated to the fundamental solution we found $G_{(t,x)(s,y)} = G(t - s, x - y) = \Theta(t - s)(e^{-(t-s)(-\Delta)})(x, y)$.

In general, for any operator admitting a heat kernel e^{-tA} , the diffusion operator $\partial_t + A$ admits a Green function $G_{(t,x)(s,y)} = \Theta(t - s)(e^{-(t-s)A})_{xy}$, that is:

$$(\partial_t + A)G = \partial_t G_{(t,x)(s,y)} + A_x G_{(t,x)(s,y)} = \delta(t - s)(e^{-(t-s)A})_{xy} = \delta(t - s)\delta(x - y),$$

which in general is not translation invariant. The inhomogeneous diffusion equation with initial condition $(\partial_t + A)u(t) = F(t)$, $u(t_0) = u_0$ admits the unique solution in terms of the heat kernel:

$$u(t) = e^{-(t-t_0)A} u_0 + \int_{t_0}^\infty ds \theta(t - s) e^{-(t-s)A} F(s) = e^{-(t-t_0)A} u_0 + \int_{t_0}^t ds e^{-(t-s)A} F(s),$$

where the second term involves the Green function, and the first one only the heat kernel.

This applies also to operators in $L^2(\Omega)$ with Ω compact. Take for example the Laplacian in one dimension with Dirichlet boundary conditions $u(0) = u(L) = 0$ (perfectly absorbing walls). Its eigenfunctions are $-\Delta_D f_n = \lambda_n f_n \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2$, $f_n = \sin\left(\frac{n\pi}{L}x\right)$ with $n \geq 1$ and form an orthogonal basis (we will see later that this is a well posed Sturm Liouville problem). The heat kernel can be expanded in this basis:

$$e^{-t(-\Delta_D)}(x, y) = \sum_{n \geq 1} \frac{1}{\|f_n\|^2} f_n(x) e^{-t\left(\frac{n\pi}{L}\right)^2} \overline{f_n(y)},$$

⁹As: $H_x G(x, y) = \int_0^\infty dt H_x K_t(x, y) = \int_0^\infty dt (-\partial_t K_t(x, y)) = \lim_{t \rightarrow 0} K_t(x, y) = \delta(x - y)$.

hence the solution of the diffusion equation $[\partial_t + (-\Delta_D)]u(t, x) = F(t, x)$ with initial condition $u(0, x) = u_0(x)$ is:

$$u(t, x) = \sum_{n \geq 1} \frac{1}{\|f_n\|^2} f_n(x) \left[e^{-t(\frac{n\pi}{L})^2} \int_0^L dy \overline{f_n(y)} u_0(y) + \int_0^t ds e^{-(t-s)(\frac{n\pi}{L})^2} \int_0^L dy \overline{f_n(y)} F(s, y) \right].$$

In order to prove uniqueness, assume u_1, u_2 are two solutions, then $w = u_1 - u_2$ is a solution of the homogeneous Dirichlet problem with zero initial condition and $\partial_t \int_x w^2 = 2 \int_x w \partial_t w = 2 \int_x w \Delta_{\text{Dirichlet}} w = 2w w'|_0^L - 2 \int_x w' w' \leq 0$, and taking into account that $\int_x w^2 \geq 0$ and $\int_x w^2|_{t_0} = 0$ we conclude that $w^2(t, x) = 0$ for all t and x .

For diffusion of particles in a box, or heat diffusion with impenetrable walls, one uses Neumann boundary conditions $u'(0) = u'(L) = 0$ (the particles, heat, etc. can not leave the box). We obtain the eigenpairs $\lambda_n = (\frac{n\pi}{L})^2$, $f_n = \cos(\frac{n\pi}{L}x)$ with $n \geq 0$ which form an orthogonal basis (again this is a well posed Sturm Liouville problem) and:

$$e^{-t(-\Delta_N)}(x, y) = \frac{1}{L} + \sum_{n \geq 1} \frac{1}{\|f_n\|^2} f_n(x) e^{-t(\frac{n\pi}{L})^2} \overline{f_n(y)}.$$

Contrary to the Dirichlet problem, the diffusion equation with no source and Neumann boundary conditions preserves the total number of particles (total heat, etc.) $\int dx u(t, x)$ as $\partial_t \int dx u = \int dx \partial_t u = \int dx \Delta u = u'|_0^L$. This can also be verified directly as $\int dx f_n(x) = 0$ for $n \geq 1$ hence:

$$\int_0^L dx \int_0^L dy e^{-t(-\Delta_N)}(x, y) u_0(y) = \frac{1}{L} \int_0^L dx \int_0^L dy u_0(y) = \int_0^L dy u_0(y).$$

Particle conservation holds for the diffusion in the entire space $\mathbb{R}$.

18.3 Generalized eigenvectors

We present in this section a non rigorous discussion of continuous spectrum of an operator. The point is that $\mathcal{S} \subset L^2 = (L^2)' \subset \mathcal{S}'$ and usually singular Weyl sequences corresponding to λ in the essential spectrum $\sigma_{ess}(A)$ of an operator $((x_n)_n$ with $\|x_n\|_{L^2} = 1$, $\|(A - \lambda)x_n\|_{L^2} \rightarrow 0$ with no convergent subsequence $x_{n_i} \not\rightarrow^{L^2}$) actually converge in $\mathcal{S}'$, $x_n \xrightarrow{\mathcal{S}'} x$, that is in the sense of distributions.

Let us consider the position and momentum operators, X and P , acting on $L^2(\mathbb{R})$, $(Xf)(t) = tf(t)$ and $(Pf)(t) = -i f'(t)$. Both are unbounded densely defined (their domain includes $\mathcal{S}$) essentially self-adjoint operators on $L^2(\mathbb{R})$. Our aim is to understand the spectral decomposition of these operators.

The position operator. The eigenvalue equation $Xf = xf$ of the position operator $(Xf)(t) = tf(t)$ does not have any solution for $f \in L^2(\mathbb{R})$. Indeed, $(t-x)f(t) = 0 \Rightarrow f(t) = 0$ for $t \neq x$ hence $f = 0$ in L^2 . The operator $X - x$ is then injective and its image is dense (this is true for any densely defined self adjoint operator). However, it is never surjective: $\mathbf{1}_{[x-1, x+1]} \in L^2$ does not belong to the image of $X - x$.¹⁰ Hence any real x belongs to the continuous spectrum of the position operator X . By Weyl's criterion, for x in the continuous spectrum, the eigenvalue equation $(X - x)f = 0$ “almost has” a solution, in the sense that there exists a sequence of functions $f_n \in L^2$ with $\|f_n\| = 1$ such that $\lim_{n \rightarrow \infty} \|(X - x)f_n\| = 0$, but f_n has no convergent subsequence. We will now exhibit such a sequence.

The subspace of function with support in $[x - \frac{\epsilon}{2}, x + \frac{\epsilon}{2}]$ is invariant under the action of the position operator X , and for any function in this subspace $(Xf)(t) = tf(t) \approx xf(t)$. An “almost” solution can be built from a shrinking rescaled indicator function centered at x :

$$f_n(t) = 2^{\frac{n}{2}} \mathbf{1}_{[x - \frac{1}{2^{n+1}}, x + \frac{1}{2^{n+1}}]}(t), \quad \|f_n\|^2 = \int dt |f_n(t)|^2 = 1.$$

¹⁰The presumptive preimage $[(X - x)^{-1} \mathbf{1}_{[x-1, x+1]}](t) = (t - x)^{-1} \mathbf{1}_{[x-1, x+1]}(t)$ is not square integrable.

This sequence has no convergent subsequence because for $p > n$ we have:

$$\|f_n - f_p\|^2 = \int dt |f_n(t) - f_p(t)|^2 = 2^n \left(\frac{1}{2^{n+1}} - \frac{1}{2^{p+1}} \right) 2 + (2^{\frac{n}{2}} - 2^{\frac{p}{2}})^2 \frac{1}{2^p} = 2 - 2^{1-\frac{p-n}{2}} > 2 - \sqrt{2},$$

hence f_n does not have any Cauchy subsequence. On the other hand:

$$\|(X - x)f_n\|^2 = \int dt |(t - x)f_n(t)|^2 = \int_{x-\frac{1}{2^{n+1}}}^{x+\frac{1}{2^{n+1}}} dt (t - x)^2 2^n = 2^n \left[\frac{u^3}{3} \right]_{-\frac{1}{2^{n+1}}}^{\frac{1}{2^{n+1}}} = \frac{2^{n+1}}{3 \cdot 2^{3n+3}} \xrightarrow{n \rightarrow \infty} 0,$$

as advertised.

While f_n does not converge in L^2 , a rescaled version of it does converge to a non trivial limit in the sense of distributions. This rescaled version has divergent L^2 norm and is called in the physics literature a “non normalizable” state. We consider an indicator function with a different normalization:

$$\delta_x^\epsilon(t) = \frac{1}{\epsilon} \mathbf{1}_{[x-\frac{\epsilon}{2}, x+\frac{\epsilon}{2}]}(t).$$

For $\epsilon > 0$ this is an element in $L^2(\mathbb{R})$ and in physics notation $\delta_x^\epsilon = |x^\epsilon\rangle$. We still have $(X\delta_x^\epsilon)(t) = t\delta_x^\epsilon(t) \approx x\delta_x^\epsilon(t)$, that is $X\delta_x^\epsilon \approx x\delta_x^\epsilon$, in the sense that $\|(X - x)\delta_x^\epsilon\|^2 = \frac{1}{\epsilon^2} \left[\frac{u^3}{3} \right]_{-\frac{\epsilon}{2}}^{\frac{\epsilon}{2}} = \frac{\epsilon}{12}$. Let us now pick the grid of points $x_n = n\epsilon$ on the real axis. The family of vectors we introduced is orthogonal:

$$\langle \delta_{x_i}^\epsilon, \delta_{x_j}^\epsilon \rangle = \int dt \delta_{x_i}^\epsilon(t) \delta_{x_j}^\epsilon(t) = \frac{1}{\epsilon} \delta_{ij},$$

and is “almost” complete, in the sense that any vector can be approximated by:

$$f \approx \sum_i \frac{1}{\|\delta_{x_i}^\epsilon\|^2} \langle \delta_{x_i}^\epsilon, f \rangle \delta_{x_i}^\epsilon, \quad f(t) \approx \sum_i \epsilon \left(\int dt' \delta_{x_i}^\epsilon(t') f(t') \right) \delta_{x_i}^\epsilon(t) = \sum_i \left(\frac{1}{\epsilon} \int_{x_i-\frac{\epsilon}{2}}^{x_i+\frac{\epsilon}{2}} dt' f(t') \right) \mathbf{1}_{[x_i-\frac{\epsilon}{2}, x_i+\frac{\epsilon}{2}]}(t),$$

that is f is well approximated by a simple (step) function which assigns to each interval $[x_i - \frac{\epsilon}{2}, x_i + \frac{\epsilon}{2}]$ the average of f over that interval.

All the approximate equalities become exact in the $\epsilon \rightarrow 0$ limit. The only problem is that the “almost eigenvector” gets out of L^2 and the limit must be taken in the sense of distributions:

$$\lim_{\epsilon \rightarrow 0} \delta_x^\epsilon = \delta_x, \quad \lim_{\epsilon \rightarrow 0} \int dt \delta_x^\epsilon(t) f(t) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_{x-\frac{\epsilon}{2}}^{x+\frac{\epsilon}{2}} dt f(t) = f(x) = (\delta_x, f).$$

We call $\delta_x = |x\rangle$, the generalized eigenvector of the position corresponding to the generalized eigenvalue x . The kernel of the distribution δ_x is denoted $\delta_x(t) = \delta(t - x)$ and:¹¹

$$f(x) = \langle \delta_x, f \rangle = \int dt \delta(t - x) f(t), \quad \delta_x(t) = \langle \delta_t, \delta_x \rangle = \int du \delta_t(u) \delta_x(u) = \delta(t - x).$$

In physics notation the kernel is $\delta(t - x) = \langle \delta_t, \delta_x \rangle = \langle t|x \rangle$ hence the above becomes:

$$f(x) = \langle x|f \rangle = \int dt \langle x|t \rangle \langle t|f \rangle, \quad |f \rangle = \int dt |t \rangle \langle t|f \rangle = \int dt f(t) |t \rangle,$$

that is $f(t)$ is the component of the vector $|f \rangle$ in the basis $|t \rangle$. In particular $\langle t|x \rangle = \delta(t - x)$ is the component of the vector $|x \rangle$ in the basis $|t \rangle$ and $|x \rangle = \int dt \langle t|x \rangle |t \rangle = \int dt \delta(t - x) |t \rangle$. We have, in the sense of distributions:

¹¹We note that: $\int dx g(x) \delta_x = g$ both pointwise and in distributions. The scalar product $\langle \delta_t, \delta_x \rangle$ should be understood as a distribution, that is: $\int dx \langle \delta_t, \delta_x \rangle g(x) = \langle \delta_t, \int dx g(x) \delta_x \rangle = \langle \delta_t, g \rangle = g(t)$.

- “eigenvalue equation” $X|x\rangle = x|x\rangle$ in the sense that for any $|f\rangle$ we have:

$$\langle x|X|f\rangle = \langle x|Xf\rangle = \langle \delta_x, Xf\rangle = \int dt \delta(t-x) t f(t) = x f(x) = x \langle x|f\rangle .$$

- “generalized orthogonality” $\langle x|x'\rangle = \langle \delta_x, \delta_{x'}\rangle = \int dt \delta(t-x) \delta(t-x') = \delta(x-x')$.
- “completeness relation” $1 = \int dx |x\rangle \langle x|$, that is for any $|f\rangle$ and $|g\rangle$:

$$|f\rangle = \left(\int dx |x\rangle \langle x| \right) |f\rangle , \quad \langle f|g\rangle = \int dt \langle f|t\rangle \langle t|g\rangle = \int dt \overline{f(t)} g(t) .$$

The momentum operator. The momentum operator is similar. The eigenvalue equation $(Pf)(t) = -i \frac{df}{dt}(t) = pf(t)$ is solved by e^{ipt} , but this is a distribution, not an $L^2(\mathbb{R})$ function. We build “almost eigenvectors” of the momentum:

$$\psi_p^\epsilon(x) = \int \frac{dq}{2\pi} e^{iqx} \frac{1}{\epsilon} \mathbf{1}_{[p-\frac{\epsilon}{2}, p+\frac{\epsilon}{2}]}(q) , \quad P\psi_p^\epsilon \approx p\psi_p^\epsilon ,$$

which are states in L^2 with momentum very peaked on p^{12} but with norm $\|\psi_p^\epsilon\| \sim \epsilon^{-1}$. Taking a grid $p_n = n\epsilon$ we have, again using Plancherel:

$$\langle \psi_{p_k}^\epsilon, \psi_{p_l}^\epsilon \rangle = \int \frac{dq}{2\pi} \frac{1}{\epsilon} \mathbf{1}_{[p_k-\frac{\epsilon}{2}, p_k+\frac{\epsilon}{2}]}(q) \frac{1}{\epsilon} \mathbf{1}_{[p_l-\frac{\epsilon}{2}, p_l+\frac{\epsilon}{2}]}(q) = \frac{1}{2\pi\epsilon} \delta_{kl} ,$$

and:

$$\begin{aligned} f &\approx \sum_i \frac{1}{\|\psi_{p_i}^\epsilon\|^2} \langle \psi_{p_i}^\epsilon, f \rangle \psi_{p_i}^\epsilon , \\ f(x) &\approx \sum_i 2\pi\epsilon \left(\int \frac{dq'}{2\pi} \frac{1}{\epsilon} \mathbf{1}_{[p_i-\frac{\epsilon}{2}, p_i+\frac{\epsilon}{2}]}(q') \hat{f}(q') \right) \left(\int \frac{dq}{2\pi} e^{iqx} \frac{1}{\epsilon} \mathbf{1}_{[p_i-\frac{\epsilon}{2}, p_i+\frac{\epsilon}{2}]}(q) \right) \\ &= \int \frac{dq}{2\pi} e^{iqx} \sum_i \left(\frac{1}{\epsilon} \int dp' \mathbf{1}_{[p_i-\frac{\epsilon}{2}, p_i+\frac{\epsilon}{2}]}(p') \hat{f}(p') \right) \mathbf{1}_{[p_i-\frac{\epsilon}{2}, p_i+\frac{\epsilon}{2}]}(q) , \end{aligned}$$

that is f is well approximated by the inverse Fourier transform of a step function approximation of its Fourier transform. Again $\epsilon \rightarrow 0$ renders all the approximate equalities exact, but sends the eigenfunction outside the Hilbert space $\lim_{\epsilon \rightarrow 0} \psi_p^\epsilon(x) = \psi_p(x) = e^{ipx}$. We have, in the sense of distributions and using the physics notation $\psi_p = |p\rangle$:

- the “overlap” is $e^{ipx} = \psi_p(x) = \langle \delta_x, \psi_p \rangle = \langle x|p\rangle$.
- “eigenvalue equation” $P|p\rangle = p|p\rangle$ in the sense that for any $|f\rangle$:

$$\langle p|P|f\rangle = \langle p|Pf\rangle = \langle \psi_p, Pf\rangle = \int dt e^{-ipt} \left(-i \frac{df}{dt} \right)(t) = p \int dt e^{ipt} f(t) = p \langle \psi_p, f \rangle = p \langle p|f \rangle .$$

- relation with Fourier modes:

$$\langle p|f\rangle = \int dx \langle p|x\rangle \langle x|f\rangle = \int dx e^{-ipx} f(x) = \hat{f}(p) .$$

- “generalized orthogonality” $\langle p|p'\rangle = \langle \psi_p, \psi_{p'} \rangle = \int dx e^{-ipx} e^{ip'x} = 2\pi \delta(p-p')$.
- “completeness relation” $1 = \int \frac{dp}{2\pi} |p\rangle \langle p|$ in the sense that for any $|f\rangle$ and $|g\rangle$:

$$|f\rangle = \int \frac{dp}{2\pi} |p\rangle \langle p|f\rangle , \quad \langle x|f\rangle = \int \frac{dp}{2\pi} \langle x|p\rangle \langle p|f\rangle , \quad \langle f|g\rangle = \int \frac{dp}{2\pi} \langle f|p\rangle \langle p|g\rangle = \int \frac{dp}{2\pi} \overline{\hat{f}(p)} \hat{g}(p) = \int dx \overline{f(x)} g(x) .$$

¹²Using the Plancherel identity $\|(P-p)\psi_p^\epsilon\|^2 = \frac{1}{\epsilon^2} \int \frac{dq}{2\pi} |q-p|^2 \mathbf{1}_{[p-\frac{\epsilon}{2}, p+\frac{\epsilon}{2}]}(p) \sim \epsilon$.

In general. For any self adjoint operator A we have a discrete spectrum $\{a_i\}_{i \in \mathbb{N}}$ and an essential one, which is a union of (may be infinite) intervals $\sigma_{ess}(A) = \cup_j [p_j, q_j]$ and:

- the discrete spectrum corresponds to eigenvectors $A|a_i\rangle = a_i|a_i\rangle$. The “generalized eigenvectors” are such that $A|a\rangle = a|a\rangle$ but do not belong to the Hilbert space.
- generalized orthogonality, $\langle a_i|a_j\rangle = \delta_{ij}$, $\langle a_i|a\rangle = 0$ and $\langle a|a'\rangle = \delta(a - a')$.
- completeness relation $1 = \sum_i |a_i\rangle \langle a_i| + \int_{\sigma_{ess}(A)} da |a\rangle \langle a|$.

The physical idea is that, while $|a\rangle$ does not belong to the Hilbert space, one can always choose a wave packet picked on the generalized eigenvectors with eigenvalue close to a , which does belong to the Hilbert space. That is, one should always be able to approximate $|a\rangle$ by a state:

$$|\psi^\epsilon\rangle = \int da' \langle a'|\psi^\epsilon\rangle |a'\rangle, \quad \langle a'|\psi^\epsilon\rangle \sim \frac{1}{\epsilon} \mathbf{1}_{[a-\frac{\epsilon}{2}, a+\frac{\epsilon}{2}]}(a'),$$

where the normalization has been chose such that $\lim_{\epsilon \rightarrow 0} |\psi^\epsilon\rangle = |a\rangle$. However, while $|\psi^\epsilon\rangle$ belongs to the Hilbert space, $\langle \psi^\epsilon|\psi^\epsilon\rangle = \frac{1}{\epsilon}$ hence diverges when sending $\epsilon \rightarrow 0$. Thus $|a\rangle$ can be approximated by sequences of states in the Hilbert space, but only if the latter have divergent norm. Such states are sometimes called non normalizable and one makes sense of the $\epsilon \rightarrow 0$ limit in the sense of distributions.

There are various more rigorous ways to make sense of the continuous part of the spectrum, either via projection valued measures, or via direct integrals of Hilbert spaces.

The Schrödinger equation. As already mentioned, the state of an isolated quantum system is a ket $|\psi\rangle$ with $\langle \psi|\psi\rangle = 1$ in a Hilbert space. In the Schrödinger picture the state evolves in time to $|\psi(t)\rangle$, and the evolution is governed by the Schrödinger equation:

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad |\psi(0)\rangle = |\psi_0\rangle \quad \Rightarrow \quad |\psi(t)\rangle = e^{-itH} |\psi_0\rangle, \quad e^{-itH} = \sum_{n \geq 0} \frac{(-itH)^n}{n!},$$

where H is the time independent Hamiltonian. The Hamiltonian is a self-adjoint operator, most commonly unbounded: $H = P^2 + V$ where the kinetic energy P^2 can be arbitrarily large. The solution of this equation is written in terms of a complete set of eigenvectors of the Hamiltonian (which might include generalized eigenvectors) as:

$$|\psi(t)\rangle = \sum_{E_i} e^{-itE_i} \langle E_i|\psi_0\rangle |E_i\rangle + \int dE e^{-itE} \langle E|\psi_0\rangle |E\rangle,$$

thus in order to solve the Schrödinger equation one must:

- identify the appropriate Hilbert space, usually a space L^2 of wave functions $\psi(x)$ with scalar product $\langle \psi_1|\psi_2\rangle = \int d^d x \bar{\psi}_1(x) \psi_2(x)$. The wave functions $\psi(x) \in L^2$ have good boundedness properties and asymptotic behaviour to ensure $\langle \psi|\psi\rangle < \infty$. As the Hilbert space is Cauchy complete, these functions lack regularity (not continuous or derivable).
- identify an appropriate dense domain $D(H)$ such that the Hamiltonian is self-adjoint. As the Hamiltonian is some differential operator, usually second order, its domain only contains functions that are regular enough (at least $\mathcal{C}^1$, if the potential is discontinuous) and maybe respect additional conditions in order to ensure that H is symmetric and $D(H) = D(H^\dagger)$, e.g. boundary conditions if the x domain is compact. Of course, they also need to belong to L^2 , hence have good boundedness properties and asymptotic behaviour.

- identify the eigenvectors and the generalized eigenvectors by solving the eigenvalue equation $H|E\rangle = E|E\rangle$. The eigenvectors are vectors $|E_i\rangle$ in the domain of H which respect the eigenvalue equation, hence are regular enough and $\langle E_i|E_i\rangle < \infty$. The generalized eigenvectors are solutions of the eigenvalue equation (hence are regular enough) but are not appropriately bounded, that is $\langle E|E\rangle = \infty$.

Exercise 18.11. *Prove that if $|\psi(t)\rangle$ is a solution of the Schrödinger equation, then $\langle\psi(t)|\psi(t)\rangle = 1$.*

Exercise 18.12 (The one dimensional, one sided finite potential well.). *Discuss the energy spectrum of the one dimensional self-adjoint Hamiltonian operator $H = -\Delta + V(x)$ for a particle in a potential well $V(x) = \infty$ for $x \leq 0$, $V(x) = -V_0$ for $0 \leq x \leq L$ and $V(x) = 0$ from $x \geq L$.*

In order for the equation $-\Delta\psi + V(x)\psi = E\psi$ to make sense, the eigenfunctions ψ need to be twice differentiable, that is ψ must be continuous with continuous derivative. However, as $V(x)$ is discontinuous, the second derivative ψ'' must also be discontinuous. Furthermore, as $V(x)$ is infinite for $x < 0$, $\psi(x) = 0$ for $x < 0$ and as ψ must be continuous, $\psi(0) = 0$. Finally, we require that the eigenfunctions are tempered distributions, such that the projection of any $f \in L^2$ on such an eigenfunction is defined $\langle\psi, f\rangle = \int dx \psi(x)f(x)$ via an approximation by Schwartz functions.

We will see below that the regularity conditions for ψ (continuity, differentiability, no divergence at infinity) lead to both the quantization of the spectrum of energy for the bound states (i.e. the discrete spectrum) and to the generalized orthogonality of all the states.

The asymptotic value of the potential is $\lim_{x \rightarrow \infty} V(x) = 0$. The modes with negative energy correspond to bound states, while the ones with positive energy correspond to diffusion states and are not normalizable. As $\langle\psi|H + V_0|\psi\rangle = \psi\psi'|_0^\infty + \|\psi'\|^2 + \|\sqrt{V_0}\mathbf{1}_{[L,\infty)}\psi\|^2 > 0$ we conclude that the eigenvalues of H must be $E > -V_0$, hence we aim to solve $H\psi = E\psi$ with $E \in (-V_0, \infty)$. Taking into account that $\psi(0) = 0$ and separating the two regions we get:

$$0 \leq x \leq L : -\Delta\psi = (E + V_0)\psi \Rightarrow \psi^I(x) = A(e^{\imath\sqrt{E+V_0}x} - e^{-\imath\sqrt{E+V_0}x}), \quad x \geq L : -\Delta\psi = E\psi \Rightarrow \psi^{II}(x) = C e^{\imath\sqrt{E}x} + D e^{-\imath\sqrt{E}x}.$$

Bound states. For negative energy states, $E = -|E|$ but $V_0 - |E| > 0$. As the eigenfunction can not diverge exponentially at infinity, we have:

$$\psi(x) = 2\imath A \sin(\sqrt{V_0 - |E|x}) \mathbf{1}_{[0,L]}(x) + C e^{-\sqrt{|E|x}} \mathbf{1}_{[L,\infty)}(x).$$

The continuity of ψ and its derivative at L impose $2\imath A \sin(\sqrt{V_0 - |E|}L) = C e^{-\sqrt{|E|}L}$ and $2\imath A \sqrt{V_0 - |E|} \cos(\sqrt{V_0 - |E|}L) = -\sqrt{|E|} C e^{-\sqrt{|E|}L}$, which on the one side fixes $C = 2\imath A \sin(\sqrt{V_0 - |E|}L) e^{\sqrt{|E|}L}$ and on the other imposes $\tan(\sqrt{V_0 - |E|}L) = -\sqrt{V_0 - |E|}/\sqrt{|E|}$.

The eigenvalue equation $\tan(\sqrt{V_0 - |E|}L) = -\sqrt{V_0 - |E|}/\sqrt{|E|}$ admits at most a finite number of solutions, corresponding to the intersections of the graphs of the function $\tan(x)$ and $-x(L^2 V_0 - x^2)^{-1/2}$ for $x > 0$. As the latter has a vertical asymptote at $x = L\sqrt{V_0}$, the number of solutions is the number of vertical asymptotes of the tangent comprised between $x = 0$ and $x = L\sqrt{V_0}$, that is $N \geq 0$ such that $\frac{\pi}{2} + (N-1)\pi \leq L\sqrt{V_0} < \frac{\pi}{2} + N\pi$. In particular there is no bound state if $L\sqrt{V_0} \leq \pi/2$. We denote the possible discrete values of bound states energy by E_k and the corresponding eigenfunctions $\psi_k(x)$ are orthogonal and normalizable. Indeed, denoting $s_k = \sqrt{V_0 - |E_k|}$ we have for $k \neq l$:

$$\begin{aligned} \frac{\langle\psi_k|\psi_l\rangle}{\tilde{A}_k \tilde{A}_l} &= \int_0^L dx (e^{-\imath s_k x} - e^{\imath s_k x})(e^{\imath s_l x} - e^{-\imath s_l x}) + (-2\imath \sin(s_k L) e^{\sqrt{|E_k|}L})(2\imath \sin(s_l L) e^{\sqrt{|E_l|}L}) \int_L^\infty dx e^{-(\sqrt{|E_k|} + \sqrt{|E_l|})x} \\ &= \frac{2 \sin[(s_k - s_l)L]}{s_k - s_l} - \frac{2 \sin[(s_k + s_l)L]}{s_k + s_l} + \frac{4 \sin(s_k L) \sin(s_l L)}{\sqrt{|E_k|} + \sqrt{|E_l|}} = \frac{2(s_k + s_l) \sin[(s_k - s_l)L] - 2(s_k - s_l) \sin[(s_k + s_l)L]}{s_k^2 - s_l^2} + \frac{4 \sin(s_k L) \sin(s_l L)}{\sqrt{|E_k|} + \sqrt{|E_l|}} \\ &= 2 \frac{(s_k + s_l)[\sin(s_k L) \cos(s_l L) - \cos(s_k L) \sin(s_l L)] - (s_k - s_l)[\sin(s_k L) \cos(s_l L) + \cos(s_k L) \sin(s_l L)]}{s_k^2 - s_l^2} + \frac{4 \sin(s_k L) \sin(s_l L)}{\sqrt{|E_k|} + \sqrt{|E_l|}} \\ &= 4 \frac{s_l \sin(s_k L) \cos(s_l L) - s_k \cos(s_k L) \sin(s_l L)}{s_k^2 - s_l^2} + \frac{4 \sin(s_k L) \sin(s_l L)}{\sqrt{|E_k|} + \sqrt{|E_l|}} \\ &= 4 \frac{-\sqrt{|E_l|} \sin(s_k L) \sin(s_l L) + \sqrt{|E_k|} \sin(s_k L) \sin(s_l L)}{s_k^2 - s_l^2} + 4 \frac{\sin(s_k L) \sin(s_l L)}{\sqrt{|E_k|} + \sqrt{|E_l|}} = 4 \sin(s_k L) \sin(s_l L) \left(\frac{\sqrt{|E_k|} - \sqrt{|E_l|}}{|E_l| - |E_k|} + \frac{1}{\sqrt{|E_l|} + \sqrt{|E_k|}} \right) = 0, \end{aligned}$$

where for the first integral we used:

$$\frac{e^{-\imath s L + \imath s' L} - 1}{-\imath s + \imath s'} - \frac{e^{\imath s L + \imath s' L} - 1}{\imath s + \imath s'} - \frac{e^{-\imath s L - \imath s' L} - 1}{-\imath s - \imath s'} + \frac{e^{\imath s L - \imath s' L} - 1}{\imath s - \imath s'} = \frac{e^{\imath s L - \imath s' L} - e^{-\imath s L + \imath s' L}}{\imath s - \imath s'} - \frac{e^{\imath s L + \imath s' L} - e^{-\imath s L - \imath s' L}}{\imath s + \imath s'} = \frac{2 \sin[(s - s')L]}{s - s'} - \frac{2 \sin[(s + s')L]}{s + s'},$$

and in the last line we used the eigenvalue equation $\sqrt{|E_k|} \sin(s_k L) = -s_k \cos(s_k L)$. We conclude that, as advertised, $\langle\psi_k|\psi_l\rangle \sim \delta_{kl}$.

Diffusion states. For positive energy states $E \geq 0$ we go back to the general solution:

$$\psi(x) = A(e^{\imath\sqrt{E+V_0}x} - e^{-\imath\sqrt{E+V_0}x}) \mathbf{1}_{[0,L]}(x) + (C e^{\imath\sqrt{E}x} + D e^{-\imath\sqrt{E}x}) \mathbf{1}_{[L,\infty)}(x),$$

and this time the continuity conditions read $Ce^{\imath\sqrt{E}L} + De^{-\imath\sqrt{E}L} = A(e^{\imath\sqrt{E+V_0}L} - e^{-\imath\sqrt{E+V_0}L})$ and $\sqrt{E}(Ce^{\imath\sqrt{E}L} - De^{-\imath\sqrt{E}L}) = \sqrt{E+V_0}A(e^{\imath\sqrt{E+V_0}L} + e^{-\imath\sqrt{E+V_0}L})$ which can be solved in terms of A :

$$C = \frac{e^{-\imath\sqrt{E}L}}{2} A \left[\left(1 + \frac{\sqrt{E+V_0}}{\sqrt{E}} \right) e^{\imath\sqrt{E+V_0}L} - \left(1 - \frac{\sqrt{E+V_0}}{\sqrt{E}} \right) e^{-\imath\sqrt{E+V_0}L} \right] = Ae^{-\imath\sqrt{E}L} \left[\imath \sin(L\sqrt{E+V_0}) + \frac{\sqrt{E+V_0}}{\sqrt{E}} \cos(L\sqrt{E+V_0}) \right],$$

$$D = \frac{e^{\imath\sqrt{E}L}}{2} A \left[\left(1 - \frac{\sqrt{E+V_0}}{\sqrt{E}} \right) e^{\imath\sqrt{E+V_0}L} - \left(1 + \frac{\sqrt{E+V_0}}{\sqrt{E}} \right) e^{-\imath\sqrt{E+V_0}L} \right] = Ae^{\imath\sqrt{E}L} \left[\imath \sin(L\sqrt{E+V_0}) - \frac{\sqrt{E+V_0}}{\sqrt{E}} \cos(L\sqrt{E+V_0}) \right].$$

Denoting $c_E = \imath \sin(L\sqrt{E+V_0}) + \frac{\sqrt{E+V_0}}{\sqrt{E}} \cos(L\sqrt{E+V_0})$ we have $\psi^I I(x) = Ac_E e^{\imath\sqrt{E}(x-L)} - A\bar{c}_E e^{-\imath\sqrt{E}(x-L)}$. Note that the condition which suppresses the divergent term for the bounded states and ultimately leads to a discrete spectrum of eigenvalues writes $\bar{c}_E|_{E=-|E|} = 0$ and it is possible to impose this without having at the same time $c_E = 0$ because for negative energy states we have:

$$\bar{c}_E|_{E=-|E|} = \frac{\sqrt{V_0-|E|}}{\imath\sqrt{|E|}} \cos(L\sqrt{V_0-|E|}) - \imath \sin(L\sqrt{V_0-|E|}), \quad c_E|_{E=-|E|} = \frac{\sqrt{V_0-|E|}}{\imath\sqrt{|E|}} \cos(L\sqrt{V_0-|E|}) + \imath \sin(L\sqrt{V_0-|E|}),$$

and the two are not complex conjugate to each other.

The positive energy states are clearly not normalizable. In order to evaluate the scalar product between two such states we split the integration interval into regions:

$$\begin{aligned} \frac{\langle \psi_E | \psi_{E'} \rangle}{A_E A_{E'}} &= \int_0^L dx \left(e^{-\imath\sqrt{E+V_0}x} - e^{\imath\sqrt{E+V_0}x} \right) \left(e^{\imath\sqrt{E'+V_0}x} - e^{-\imath\sqrt{E'+V_0}x} \right) + \int_L^\infty dx \left(\bar{c}_E e^{-\imath\sqrt{E}(x-L)} - c_E e^{\imath\sqrt{E}(x-L)} \right) \left(c_{E'} e^{\imath\sqrt{E'}(x-L)} - \bar{c}_{E'} e^{-\imath\sqrt{E'}(x-L)} \right) \\ &= \int_0^L dx \left(e^{-\imath\sqrt{E+V_0}x} - e^{\imath\sqrt{E+V_0}x} \right) \left(e^{\imath\sqrt{E'+V_0}x} - e^{-\imath\sqrt{E'+V_0}x} \right) + \int_0^\infty dx \left(\bar{c}_E e^{-\imath\sqrt{E}x} - c_E e^{\imath\sqrt{E}x} \right) \left(c_{E'} e^{\imath\sqrt{E'}x} - \bar{c}_{E'} e^{-\imath\sqrt{E'}x} \right), \end{aligned}$$

where we shifted the argument in the second integral. We stress that this scalar product should be seen as a distribution acting on test functions dependent on the energy, $f(E')$. Denoting $\sqrt{E+V_0} = s$ and $\sqrt{E'+V_0} = s'$ hence $c_E = \imath \sin(sL) + \frac{s}{\sqrt{E}} \cos(sL)$ the definite integral computes to:

$$\frac{2 \sin[(s-s')L]}{s-s'} - \frac{2 \sin[(s+s')L]}{s+s'} = 4 \frac{s' \sin(sL) \cos(s'L) - s \cos(sL) \sin(s'L)}{s^2 - (s')^2} = 4 \frac{s' \sin(sL) \cos(s'L) - s \cos(sL) \sin(s'L)}{E - E'}.$$

For the indefinite integral, we add an exponential regulator and use the Sokhotski–Plemelj formula. Using principal value when needed we get:

$$\begin{aligned} \int_0^\infty dx e^{-\epsilon x} \left(\bar{c}_E e^{-\imath\sqrt{E}x} - c_E e^{\imath\sqrt{E}x} \right) \left(c_{E'} e^{\imath\sqrt{E'}x} - \bar{c}_{E'} e^{-\imath\sqrt{E'}x} \right) &= \frac{\bar{c}_E c_{E'}}{\epsilon + \imath(\sqrt{E} - \sqrt{E'})} - \frac{\bar{c}_E \bar{c}_{E'}}{\epsilon + \imath(\sqrt{E} + \sqrt{E'})} - \frac{c_E c_{E'}}{\epsilon + \imath(-\sqrt{E} - \sqrt{E'})} + \frac{c_E \bar{c}_{E'}}{\epsilon + \imath(-\sqrt{E} + \sqrt{E'})} \\ &= (-\imath) \frac{\bar{c}_E c_{E'}}{(\sqrt{E} - \sqrt{E'}) - \imath\epsilon} - (-\imath) \frac{\bar{c}_E \bar{c}_{E'}}{\imath(\sqrt{E} + \sqrt{E'}) - \imath\epsilon} - (-\imath) \frac{c_E c_{E'}}{(-\sqrt{E} - \sqrt{E'}) + \imath\epsilon} + (-\imath) \frac{c_E \bar{c}_{E'}}{(-\sqrt{E} + \sqrt{E'}) - \imath\epsilon} \\ &= (-\imath \bar{c}_E c_{E'}) \left[\imath \pi \delta(\sqrt{E} - \sqrt{E'}) + \frac{1}{\sqrt{E} - \sqrt{E'}} \right] - (-\imath \bar{c}_E \bar{c}_{E'}) \left[\imath \pi \delta(\sqrt{E} + \sqrt{E'}) + \frac{1}{\sqrt{E} + \sqrt{E'}} \right] - (-\imath c_E c_{E'}) \left[\imath \pi \delta(-\sqrt{E} - \sqrt{E'}) + \frac{1}{-\sqrt{E} - \sqrt{E'}} \right] + (-\imath c_E \bar{c}_{E'}) \left[\imath \pi \delta(-\sqrt{E} + \sqrt{E'}) + \frac{1}{-\sqrt{E} + \sqrt{E'}} \right]. \end{aligned}$$

Observing that $\sqrt{E} + \sqrt{E'} > 0$ the indefinite integral computes to $2\pi|c_E|^2 \delta(\sqrt{E} - \sqrt{E'}) + \frac{-\imath \bar{c}_E c_{E'} + \imath c_E \bar{c}_{E'}}{\sqrt{E} - \sqrt{E'}} + \frac{\imath \bar{c}_E \bar{c}_{E'} - \imath c_E c_{E'}}{\sqrt{E} + \sqrt{E'}}$. The second term in this expression is:

$$\begin{aligned} &\frac{(\sqrt{E} + \sqrt{E'}) 2 \operatorname{Re} \left[(-\imath) \left(-\imath \sin(sL) + \frac{s}{\sqrt{E}} \cos(sL) \right) \left(\imath \sin(s'L) + \frac{s'}{\sqrt{E'}} \cos(s'L) \right) \right]}{E - E'} + \frac{(\sqrt{E} - \sqrt{E'}) 2 \operatorname{Re} \left[\imath \left(-\imath \sin(sL) + \frac{s}{\sqrt{E}} \cos(sL) \right) \left(-\imath \sin(s'L) + \frac{s'}{\sqrt{E'}} \cos(s'L) \right) \right]}{E - E'} \\ &= \frac{(\sqrt{E} + \sqrt{E'}) 2 \left[-\sin(sL) \frac{s'}{\sqrt{E'}} \cos(s'L) + \frac{s}{\sqrt{E}} \cos(sL) \sin(s'L) \right]}{E - E'} + \frac{(\sqrt{E} - \sqrt{E'}) 2 \left[\sin(sL) \frac{s'}{\sqrt{E'}} \cos(s'L) + \frac{s}{\sqrt{E}} \cos(sL) \sin(s'L) \right]}{E - E'} \\ &= \frac{1}{E - E'} \sin(sL) \frac{s'}{\sqrt{E'}} \cos(s'L) (-4\sqrt{E'}) + \frac{1}{E - E'} \frac{s}{\sqrt{E}} \cos(sL) \sin(s'L) (4\sqrt{|E|}), \end{aligned}$$

and precisely cancels the contribution of the definite integral thus $\frac{\langle \psi_E | \psi_{E'} \rangle}{A_E A_{E'}} = 2\pi|c_E|^2 \delta(\sqrt{E} - \sqrt{E'}) = 4\pi|c_E|^2 \sqrt{E} \delta(E - E')$ where $|c_E|^2 = \frac{s^2}{E} \cos^2(sL) + \sin^2(sL) = 1 + \frac{V_0}{E} \cos^2(sL)$.

It follows that a complete set of eigenvectors consists in (we fix $A = \frac{1}{2\imath}$):

$$\begin{aligned} -V_0 \leq E_k < 0, \quad \psi_k(x) &= \sin(\sqrt{V_0 - |E_k|}x) \mathbf{1}_{[0, L]}(x) + \sin(\sqrt{V_0 - |E_k|}L) e^{-\sqrt{|E_k|}(x-L)} \mathbf{1}_{[L, \infty)}(x), \quad \tan(\sqrt{V_0 - |E_k|}L) = -\frac{\sqrt{V_0 - |E_k|}}{\sqrt{|E_k|}} \\ E \geq 0, \quad \psi_E(x) &= \sin(\sqrt{E+V_0}x) \mathbf{1}_{[0, L]}(x) + \left[\sin(L\sqrt{E+V_0}) \cos(\sqrt{E}(x-L)) + \frac{\sqrt{E+V_0}}{\sqrt{E}} \cos(L\sqrt{E+V_0}) \sin(\sqrt{E}(x-L)) \right] \mathbf{1}_{[L, \infty)}(x). \end{aligned}$$

The orthogonality and completeness write:

$$\langle \psi_k | \psi_l \rangle = \|\psi_k\|^2 \delta_{kl}, \quad \langle \psi_E | \psi_{E'} \rangle = \pi \left(1 + \frac{V_0}{E} \cos^2(\sqrt{E+V_0}L) \right) \sqrt{E} \delta(E - E'), \quad 1 = \sum_k \frac{1}{\|\psi_k\|} \psi_k(x) \psi_k(x') + \int_0^\infty dE \frac{1}{\left[\pi \sqrt{E} \left(1 + \frac{V_0}{E} \cos^2(\sqrt{E+V_0}L) \right) \right]^{1/2}} \psi_E(x) \psi_E(x').$$

Chapter 19

Differential equations

We first study ordinary differential equations (ODEs) which concern functions of one variable. We will discuss partial differential equations at the end.

19.1 Initial value problems

An initial value problem specifies how a system evolves in time. Specifically, one is interested in determining the unknown function $x : \mathbb{R} \rightarrow \mathbb{R}^N$ which respects the differential equation with initial condition:

$$\dot{x}(t) = \frac{dx}{dt}(t) = Y(t, x(t)) , \quad x(t_0) = x_0 .$$

where $Y : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}^N$, $(t, x) \mapsto Y(t, x)$ is some known function.

Theorem 19.1 (**Picard–Lindelöf theorem**). *Let $I \subset \mathbb{R}$ an open interval, $U \subset \mathbb{R}^N$ an open set and $Y : I \times U \rightarrow \mathbb{R}^N$ a (jointly in its arguments) continuous function. Assume that there exists a constant $L > 0$ such that $\forall t \in I$ and $\forall x_1, x_2 \in U$:*

$$\|Y(t, x_1) - Y(t, x_2)\| \leq L \|x_1 - x_2\| .$$

Then $\forall t_0 \in I$ and $\forall x_0 \in U$, $\exists \delta$ with $t_0 + \delta \in I$ such that $\forall t_1 \in (t_0, t_0 + \delta]$ there exists a unique solution $x : [t_0, t_1] \rightarrow U$ of the initial value problem:

$$\dot{x}(t) = \frac{dx}{dt}(t) = Y(t, x(t)) , \quad x(t_0) = x_0 .$$

Proof. The proof relies on Banach fixed point theorem which states that if T is a contraction in a complete metric space $d(T(x), T(y)) < q d(x, y)$ with $q < 1$ then it admits a fixed point $T(x^*) = x^*$ and the iteration $x_{n+1} = T(x_n)$ converges to x^* for any initial condition x_0 .

The initial value problem can be reformulated as an integral equation:

$$x(t) = x_0 + \int_{t_0}^t ds Y(s, x(s)) .$$

If this integral equation admits a continuous solution $x(t)$, then $x(t)$ is differentiable and satisfies the initial value problem. Thus we search for a fixed point of the map:

$$T : \mathcal{C}^0(I, U) \rightarrow \mathcal{C}^0(I, U) , \quad T(x(\cdot))(t) = x_0 + \int_{t_0}^t ds Y(s, x(s)) ,$$

for some sub interval of I containing t_0 , which we relabel abusively by I .

Let $R > 0$ be such that $\overline{B_R(x_0)} \subset U$, and take $t_1 \in I$, $t_1 > t_0$. Then the space:

$$X = \left\{ x : [t_0, t_1] \rightarrow \overline{B_R(x_0)} \mid x \text{ continuous, } x(t_0) = x_0 \right\}, \quad d(x_1, x_2) = \sup_{t \in [t_0, t_1]} |x_1(t) - x_2(t)|,$$

is a metric space (but not a vector space!). Consider a sequence of functions x_n which is Cauchy¹ in (X, d) . For any ϵ , for m, n large enough $|x_m(t) - x_n(t)| < \epsilon$, $\forall t$. Then $x_n(t)$ converges pointwise to some $x(t)$ and, as any distance is continuous $|x(t) - x_n(t)| < \epsilon$, $\forall t$, that is x_n converges to x in d . It follows that $x(t)$ is the uniform limit of continuous functions, hence it is continuous and (X, d) is Cauchy complete as $|x_n(t) - x_0| \leq R \Rightarrow |x(t) - x_0| \leq R$. Note that X is neither empty (it contains at least the constant function), nor too large (it does not contain the continuous functions that vary by more than R in the interval $[t_0, t_1]$).

It remains to chose R and δ such that for $t_1 \leq t_0 + \delta$ we have $T(X) \subset X$ and $d(T(x), T(y)) < q d(x, y)$ for some $q < 1$. We fix some $R > 0$ with $\overline{B_R(x_0)} \subset U$ and some $t_* \in I$ with $t_* > t_0$. We set $M = \sup_{t \in [t_0, t_*], x \in \overline{B_R(x_0)}} \|Y(t, x)\|$, which is finite as Y is continuous with compact support. Then for $t_1 < t_*, t_0 + \frac{R}{M}$ we have:

$$\forall x \in X, \quad \forall t \in [t_0, t_1], \quad x(t) \in B_R(x_0) \Rightarrow \forall t \in [t_0, t_1], \quad \|T(x(\cdot))(t) - x_0\| \leq |t - t_0| \sup_{s \in [t_0, t]} \|Y(s, x(s))\| \leq R \Rightarrow T(x) \in X.$$

On the other hand:

$$\begin{aligned} \|T(x_1(\cdot))(t) - T(x_2(\cdot))(t)\| &\leq |t - t_0| \sup_{s \in [t_0, t]} \|Y(s, x_1(s)) - Y(s, x_2(s))\| \leq |t - t_0| L \sup_{s \in [t_0, t]} \|x_1(s) - x_2(s)\| \\ &\leq |t - t_0| L \sup_{s \in [t_0, t_1]} \|x_1(s) - x_2(s)\|, \end{aligned}$$

hence T is a contracting map as long as $|t_1 - t_0| L < 1$.

We conclude that if $\delta < t - t_*, R/M, 1/L$, then T has a fixed point and the initial value problem a unique solution. $\square$

This theorem guarantees the local existence of solutions of the initial value problem of arbitrary order:

$$\frac{d^n x}{dt^n}(t) = Y\left(t, x(t), \frac{dx}{dt}(t), \dots, \frac{d^{n-1}x}{dt^{n-1}}(t)\right), \quad x(t_0) = x_0, \quad \frac{dx}{dt}(t_0) = x_0^{(1)}, \dots, \quad \frac{d^{n-1}x}{dt^{n-1}}(t_0) = x_0^{(n-1)}.$$

Indeed, one can write any such equation in vector form by organizing the derivatives of x in a column vector:

$$\mathbf{x} = \begin{pmatrix} x \\ x^{(1)} \\ \vdots \\ x^{(n-1)} \end{pmatrix}, \quad \dot{\mathbf{x}} = \mathbf{Y}(t, \mathbf{x}(t)), \quad \mathbf{x}(t_0) = \begin{pmatrix} x_0 \\ x_0^{(1)} \\ \vdots \\ x_0^{(n-1)} \end{pmatrix}, \quad \mathbf{Y}(t, \mathbf{x}) = \begin{pmatrix} x^{(1)} \\ \vdots \\ x^{(n-1)} \\ Y(t, x, x^{(1)} \dots x^{(n-1)}) \end{pmatrix},$$

and note that one needs to specify n initial conditions in order to fix the solution of an initial value problem of order n .

We stress that the uniform Lipschitz continuity in x is not a mere technicality, but a required condition. Indeed, the initial value problem

$$\frac{dx}{dt} = \frac{3}{2}x^{1/3}, \quad x(0) = 0,$$

has three solutions $x(t) = 0$ and $x(t) = \pm t^{3/2}$.

¹We have used a similar argument to prove that the space of bounded functions with sup norm $\|\cdot\|_\infty$ is a Banach space.

Exercise 19.1. Solve the initial value problem $\frac{dx}{dt}(t) = Y(t, x(t))$, $x(0) = x_0$ for $Y(t, x) = t$ and $Y(t, x) = x$ by taking explicitly the limit of the sequence $x_{n+1}(t) = x_0 + \int_0^t ds Y(s, x_n(s))$. We have in the first case $x_{n+1} = x_0 + \int_0^t ds s = x_0 + \frac{t^2}{2}$ and in the second:

$$x_1 = x_0 + \int_0^t ds x_0 = x_0(1+t), \quad x_2 = x_0 + \int_0^t ds x_0(1+s) = x_0(1+t + \frac{t^2}{2}), \dots x_n = x_0(1+t + \dots + \frac{t^n}{n!}) \Rightarrow x_0 e^t.$$

19.1.1 Flows of vector fields

A particularly intuitive picture for initial value problems is given by the notion of flow. A flow describes the motion of a particle subjected with a give velocity field in $\mathbb{R}^N$.

Definition 19.1 (**Flows on $\mathbb{R}^N$**). A flow on $\mathbb{R}^N$ is a map $\varphi: I \times X \rightarrow X$ with $I \subset \mathbb{R}$ a real interval and $X \subset \mathbb{R}^N$ such that for any $x \in X$ and any $s, t \in I$:

- $\varphi(0, x) = x$,
- $\varphi(s, \varphi(t, x)) = \varphi(s+t, x)$.

We sometimes denote $\varphi(t, x) = \varphi^t(x)$. The orbit (flow line) of a flow is the set $\{\varphi(t, x) | t \in I, x \text{ fixed}\}$. Note that we can set the initial condition at any time $t_0 \in I$. We consider here only continuous (actually differentiable) flows.

An initial value problem $\dot{x}(t) = Y(t, x(t))$, $x(t_0) = x_0$ is called *autonomous* if Y does not depend on t , that is $Y = \sum_{i=1}^N Y^i(x) e_i$ is a vector field in $\mathbb{R}^N$. To any such initial value problem one can associate a flow which transports the initial condition:

$$\dot{x}(t) = Y(x(t)), \quad x(t_0) = x_0 \quad \Rightarrow \quad \varphi(t, x_0) = x(t),$$

where $x(t)$ is the solution of the initial value problem with initial condition set at x_0 . Conversely, for any vector field $Y(x)$, one calls the solution of the differential equation $\dot{x}(t) = Y(x(t))$ with $x(t_0) = x_0$ an integral curve or a trajectory. The flow plot of $Y(x)$ is the plot of the integral curves.

Given a vector field, of particular interests are the *fixed points* of the flow, that is points such that $\varphi(t, x_*) = x_*$. Such fixed points correspond to x_* solutions of $Y(x_*) = 0$ which are so that the differential equation $\dot{x}(t) = Y(x(t))$, $x(t_0) = x_*$ admits a stationary (fixed point) solution $x(t) = x_*$. One can perform a systematic study of the flow in the neighborhood of fixed points. A linearization of the ODE for $x(t) = x_* + h(t)$ yields at first order in h (in components):

$$\dot{h}^j(t) = Y^j(x_* + h(t)) \approx (\partial_i Y^j)(x_*) h^i(t), \quad h^i(t_0) = h_0^i,$$

where $Y' = (\partial_i Y^j)(x_*)$ is called the *stability matrix* at x_* . The evolution of the perturbation is then in matrix notation:²

$$\varphi(t, h_0) = h(t) = e^{(t-t_0)Y'} h_0, \quad e^{(t-t_0)Y'} = \sum_{n \geq 0} \frac{(t-t_0)^n}{n!} (Y')^n.$$

If the matrix Y' can be diagonalized³ $Y' = S^{-1}DS$ with $D = \text{diag}(\nu_1, \dots, \nu_N)$ then:

$$(Sh(t))^i = e^{(t-t_0)\nu_i} (Sh_0)^i.$$

We call the exponents ν_i the critical exponents of the fixed point, and the eigenvectors of Y' the associated eigen-perturbations. An eigen-perturbation is called relevant if $\text{Re}(\nu_i) > 0$ (it increases with t), marginal if $\text{Re}(\nu_i) = 0$ and irrelevant if $\text{Re}(\nu_i) < 0$ (it goes to zero for large t).⁴ Note that for real vector fields, if the critical exponents are complex

²Check this is a solution of the linearized equation.

³This is not a very restrictive assumption as diagonalizable matrices are dense in arbitrary matrices in any norm.

⁴Somewhat confusingly, we are often interested in physics in the limit $t \rightarrow -\infty$. In this case we switch terminology and talk about *infrared relevant* perturbations with $\text{Re}(\nu_i) < 0$ which grow for $t \rightarrow -\infty$ and *infrared irrelevant* perturbations with $\text{Re}(\nu_i) > 0$ which wash out for $t \rightarrow -\infty$.

they come in complex conjugated pairs and the associated eigenvectors are also complex conjugated to each other. Such a complex conjugated pair describes a rotation accompanied by a scale transformation and describes spiralling flows with vorticity.

In order to draw a flow plot one first identifies fixed points, solves for the critical exponents and eigen-directions and then finds the flow lines (integral curves).

Exercise 19.2. *Analyze the flow of the system of differential equations:*

$$\dot{x} = -x^3, \quad \dot{y} = y^2 - x^2.$$

If Y' can not be diagonalized, then it can be brought in Jordan normal form $Y' = S^{-1}JS$ with ν_i the diagonal entries of J . The exponential of a Jordan block of size p , denoting B the matrix with 1 entries over the diagonal, hence $B^p = 0$, and using $J^k = \sum_{q=0}^k \binom{k}{q} \lambda^{k-q} B^q$ computes to:

$$e^{(t-t_0)J} = \sum_{k \geq 0} \frac{(t-t_0)^k}{k!} \sum_{q \geq 0} \binom{k}{q} \lambda^{k-q} B^q = e^{(t-t_0)\lambda} \sum_{q=0}^p \frac{(t-t_0)^q}{q!} B^q.$$

19.1.2 First integrals

In some cases one can identify constants of motion, that is quantities that are conserved along the trajectories of a system. They typically help in reducing the order of the differential equation. Consider a differential equation of order p :

$$\frac{d^p x}{dt^p}(t) = Y(x^{(p-1)}(t), \dots, x(t), t), \quad x^{(q)}(t_0) = x_0^q.$$

A first integral of the equation is a function $E(y_{p-1}, \dots, y_0, t)$ such that:

$$\frac{d}{dt} \left(E(y_{p-1}, \dots, y_0, t) \Big|_{y_i = \frac{d^i x}{dt^i}(t)} \right) = \sum_{q=0}^{p-1} \left(\frac{\partial E}{\partial y_q} \right) \Big|_{y_i = \frac{d^i x}{dt^i}(t)} \frac{d^{q+1} x(t)}{dt^{q+1}} + \partial_t E \Big|_{y_i = \frac{d^i x}{dt^i}(t)} = 0.$$

If such a first integral exists, the differential equation is equivalent to the differential equation of order $p-1$:

$$E(y_{p-1}, \dots, y_0, t) \Big|_{y_i = \frac{d^i x}{dt^i}(t)} = E(x_0^{p-1}, \dots, x_0, t_0),$$

and given enough first integral the differential equation can be solved.

The prototypical example is the one dimensional motion of a particle in a potential:

$$\ddot{x}(t) = -\nabla U(x(t)),$$

this can be turned into an initial value problem in terms of the velocity $\dot{x}(t) = v(t)$, $\dot{v}(t) = -\nabla U(x(t))$ with $x(t_0) = x_0$ and $v(t_0) = v_0$. The total energy $E(x, v) = \frac{1}{2}v^2 + U(x)$ is conserved along any trajectory:

$$E(x, v) \Big|_{x=x(t), v=\dot{x}(t)} = \frac{1}{2}[\dot{x}(t)]^2 + U(x(t)), \quad \frac{d}{dt} E(x, v) \Big|_{x=x(t), v=\dot{x}(t)} = \dot{x}(t) \ddot{x}(t) + \nabla U(x(t)) \dot{x}(t) = 0.$$

Our ODE is then equivalent to:

$$\dot{x}(t) = \sqrt{2(E - U(x(t)))} \Rightarrow t - t_0 = \int_{x_0}^{x(t)} \frac{du}{\sqrt{2(E - U(s))}}, \quad E = \frac{1}{2}v_0^2 + U(x_0),$$

which defines $x(t)$ implicitly.

Exercise 19.3 (Separation of variables). *Prove that the equation $\frac{dx}{dt} = \frac{T(t)}{F(x(t))}$, $x(t_0) = x_0$ admits a first integral $V(x, t) = \int^x dx' F(x') - \int^{t_0} dt' T(t')$ hence it is solved implicitly by $\int_{x_0}^{x(t)} dx' F(x') - \int_{t_0}^t dt' T(t') = 0$. We have $\frac{d}{dt} V(x(t), t) = F(x(t)) \frac{dx}{dt} - T(t) = 0$. We say that the variables are separated and write also:*

$$\frac{dx}{dt} = \frac{T(t)}{F(x(t))} \Rightarrow F(x)dx = T(t)dt \Rightarrow \int_{x_0}^{x(t)} dx F(x) - \int_{t_0}^t dt T(t) = 0.$$

This can be interpreted geometrically as follows: $x(t)$ is a submanifold embedded in the plane (x, t) defined by the constraint $V(x, t) = V(x_0, t_0)$. The form $dV = F(x)dx - T(t)dt$ is the normal form to this submanifold. The tangent space to this submanifold is one dimensional and consists in the vectors $X = X^x \partial_x + X^t \partial_t$ such that $dV(X) = F(x)X^x - T(t)X^t = 0$. Choosing some coordinate s , the submanifold is embedded as $x(s), t(s)$ and the tangent space is spanned by $\phi_(\partial_s) = \frac{dx}{ds} \partial_x + \frac{dt}{ds} \partial_t$. Fixing $s = t$, $x(t)$ must satisfy $F(x) \frac{dx}{dt} = T(t)$.*

19.1.3 The Kepler problem

Sometimes first integrals can completely solve a differential equation. The aim of the Kepler problem is to identify all the possible trajectories of a body of mass m (planet) moving in a central force field with potential $U(r) = -\frac{k}{r}$ in $\mathbb{R}^3$. We denote in bold $\mathbf{r}$ the position vector of the planet with respect to the center of force, and $\mathbf{p} = m \dot{\mathbf{r}}$ its momentum. We have:

$$\nabla \left(\frac{1}{r} \right) = -\frac{\mathbf{r}}{r^3} = -\frac{1}{r^2} \mathbf{e}_r, \quad \frac{d}{dt} \left(\frac{1}{r} \right) = -\frac{\mathbf{r} \cdot \dot{\mathbf{r}}}{r^3} = -\frac{1}{r^2} \mathbf{e}_r \cdot \dot{\mathbf{r}},$$

where $\mathbf{e}_r = \frac{\mathbf{r}}{r}$ is the unit vector pointing along $\mathbf{r}$. Furthermore, we denote $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ the standard basis in $\mathbb{R}^3$, $\mathbf{a} \times \mathbf{b} = \epsilon_{ijk} a_j b_k$ the vector product in $\mathbb{R}^3$ and $\mathbf{L} = \mathbf{r} \times \mathbf{p} = m \mathbf{r} \times \dot{\mathbf{r}}$ the angular momentum with respect to the center of force.

The trajectories of a planet of m are the solution of Newton's first law $m \ddot{\mathbf{r}} = -\nabla U(r) = -k \frac{\mathbf{r}}{r^3} = -\frac{k}{r^2} \mathbf{e}_r$. This equation has three first integrals:

- the energy $E = \frac{m}{2} \dot{\mathbf{r}}^2 - \frac{k}{r}$, as $\frac{d}{dt} E = (m \ddot{\mathbf{r}} + \frac{k}{r^2} \mathbf{e}_r) \cdot \dot{\mathbf{r}} = 0$,
- the angular momentum $\mathbf{L}$, as $\frac{d}{dt} \mathbf{L} = m \dot{\mathbf{r}} \times \dot{\mathbf{r}} + m \mathbf{r} \times \ddot{\mathbf{r}} = \mathbf{r} \times (-\frac{k}{r^2} \mathbf{e}_r) = 0$. This implies that the orbits are restricted to the plane orthogonal to $\mathbf{L}$.
- the Runge-Lenz vector $\mathbf{A} = \mathbf{p} \times \mathbf{L} - mk \mathbf{e}_r = \mathbf{p} \times \mathbf{L} - mk \frac{\mathbf{r}}{r}$, as:

$$\frac{d}{dt} \mathbf{A} = \dot{\mathbf{p}} \times \mathbf{L} - mk \left(-\frac{\mathbf{r} \cdot \dot{\mathbf{r}}}{r^3} \right) \mathbf{r} - mk \frac{1}{r} \dot{\mathbf{r}} = -\frac{mk}{r^3} \mathbf{r} \times (\mathbf{r} \times \dot{\mathbf{r}}) + \frac{mk}{r^3} [\mathbf{r}(\mathbf{r} \cdot \dot{\mathbf{r}}) - \dot{\mathbf{r}}(\mathbf{r} \cdot \mathbf{r})] = 0,$$

as the double vector product is $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$. Observe that:

$$\begin{aligned} A^2 &= [\mathbf{r} \mathbf{p}^2 - \mathbf{p}(\mathbf{p} \cdot \mathbf{r}) - mk \mathbf{e}_r]^2 = r^2 p^4 + p^2 (\mathbf{p} \cdot \mathbf{r})^2 + m^2 k^2 - 2p^2 (\mathbf{p} \cdot \mathbf{r})^2 - 2mk p^2 r + 2 \frac{mk}{r} (\mathbf{p} \cdot \mathbf{r})^2 \\ &= m^2 k^2 + p^2 [r^2 p^2 - (\mathbf{p} \cdot \mathbf{r})^2] - 2 \frac{mk}{r} [r^2 p^2 - (\mathbf{p} \cdot \mathbf{r})^2] = m^2 k^2 + 2m E L^2, \end{aligned}$$

and we remark that the total energy is bounded below by $E \geq -\frac{mk^2}{2L^2}$.

Now, we note that $\mathbf{A} \cdot \mathbf{r} = (\mathbf{p} \times \mathbf{L}) \cdot \mathbf{r} - mk r = (\mathbf{r} \times \mathbf{p}) \cdot \mathbf{L} - mk r = L^2 - mk r = Ar \cos \theta$, therefore the trajectories are the curves:

$$r = \frac{p}{1 + e \cos \theta}, \quad e = \frac{A}{mk} = \sqrt{1 + \frac{2L^2}{mk^2} E}, \quad p = \frac{L^2}{mk}.$$

This is conic with eccentricity $e = \frac{A}{mk}$. For $e < 1$, that is $E \leq 0$ the orbits are ellipses, for $e = 0$ it is a parabola while for $e \geq 1$ they are hyperbolas. In all cases $\mathbf{A}$ points to the perihelion of the conic, that is the point closest to the focal point.

For $e < 1$, the ellipse is described by the equation:

$$\frac{(x+f)^2}{a^2} + \frac{y^2}{b^2} = 1,$$

with semiaxes $a = \frac{p}{1-e^2}$ and $b = \frac{pe}{\sqrt{1-e^2}}$, focal distance $f = \frac{pe}{1-e^2}$, and center at $(-f, 0)$. Indeed:

$$\begin{aligned} \left(r \cos \theta + \frac{pe}{1-e^2}\right)^2 + \frac{r^2 \sin^2 \theta}{1-e^2} &= r^2 \cos^2 \theta + \frac{p^2 e^2}{(1-e^2)^2} + \frac{2per \cos \theta}{1-e^2} + \frac{r^2 \sin^2 \theta}{1-e^2} = \frac{r^2}{1-e^2} - \frac{e^2 r^2 \cos^2 \theta}{1-e^2} + \frac{p^2 e^2}{(1-e^2)^2} + \frac{2per \cos \theta}{1-e^2} \\ &= \frac{p^2}{(1-e^2)^2} + \frac{1}{1-e^2} \left[r^2 - e^2 r^2 \cos^2 \theta + 2per \cos \theta - p^2 \right], \end{aligned}$$

and $r^2 - e^2 r^2 \cos^2 \theta + 2per \cos \theta - p^2 = r^2 - (p-r)^2 + 2p(p-r) - p^2 = 0$.

19.2 Linear ordinary differential equations

A *linear* ordinary differential equation (ODE) of order n is an equation of the form:

$$\frac{d^n x}{dt^n}(t) + a_{n-1}(t) \frac{d^{n-1} x}{dt^{n-1}}(t) \cdots + a_0(t) x(t) = f(t), \quad \frac{d^i x}{dt^i}(t_0) = x^{(i)}(t_0) = v^i, \quad i = 0, \dots, n-1,$$

with v^i some fixed initial values.

We denote this $Dx = f$ with D the linear differential operator $D = \sum_{i=1}^n a_i(t) \frac{d^i}{dt^i}$, where $a_n(t) = 1$, which acts on a suitable space of functions. If $f = 0$ the ODE is called homogeneous and if $f \neq 0$ it is called inhomogeneous. We assume that the coefficients $a_i(t)$ and the function $f(t)$ are sufficiently regular such that the Picard–Lindelöf theorem holds.⁵ As function space we choose $\mathcal{C}^n(I, \mathbb{R})$ for some closed interval $I \ni t_0$. The hypothesis of the Picard–Lindelöf theorem imply that a_i are continuous hence $D : (\mathcal{C}^n(I, \mathbb{R}), \|\cdot\|_{\mathcal{C}^n}) \rightarrow (\mathcal{C}^0(I, \mathbb{R}), \|\cdot\|_{\mathcal{C}^0})$ is a bounded linear operator between Banach spaces with $\|D\|_{\text{op}} \leq \max_i \sup_{t \in I} |a_i(t)|$.⁶

Homogenous equations. Solving a homogeneous differential equation $Dx = 0$ comes to identifying the kernel of the operator D , which is a vector space.⁷

On the other hand, $Dx = 0$ with n initial conditions $\{x^{(i)}(t_0) = v^i\}_{i=0, n-1}$ admits a unique solution. We generate n different solutions $x_\nu(t)$, one for each choice of initial conditions $v^i = \delta_{i\nu}$ with ν fixed. In particular the function $x_\nu(t)$ is such that $\frac{d^i x_\nu}{dt^i}(t_0) = \delta_{i\nu}$ hence, by linearity, $x(t) = \sum_{\nu=0}^{n-1} v^\nu x_\nu(t)$ is the unique solution of $Dx = 0$ which obeys the initial condition $\frac{d^i x}{dt^i}(t_0) = v^i$. The functions $x_\nu(t)$ provide a Hamel basis in the kernel,⁸ hence the latter has dimension n . Any other basis in the kernel consists in n linearly independent solutions. We denote somewhat abusively some arbitrary basis

⁵Writing the ODE as:

$$\begin{pmatrix} \frac{dy}{dt} \\ \frac{dy_1}{dt} \\ \vdots \\ \frac{dy_{n-2}}{dt} \\ \frac{dy_{n-1}}{dt} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0(t) & -a_1(t) & -a_2(t) & \dots & -a_{n-1}(t) \end{pmatrix} \begin{pmatrix} y \\ y_1 \\ \vdots \\ y_{n-2} \\ y_{n-1} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ f(t) \end{pmatrix},$$

a sufficient condition is that the norm of the matrix is finite hence, using the Hilbert Schmidt norm, $\sup_t (\sum_i |a_i(t)|^2) < \infty$.

⁶We can extend by density D to an operator $D : L^2(I, \mathbb{R}) \rightarrow L^2(I, \mathbb{R})$, but then D is an unbounded densely defined operator because for any c , there exists $f \in L^2(I, \mathbb{R})$ with $\|Df\|_{L^2} \geq c\|f\|_{L^2}$.

⁷As $D : (\mathcal{C}^n(I, \mathbb{R}), \|\cdot\|_{\mathcal{C}^n}) \rightarrow (\mathcal{C}^0(I, \mathbb{R}), \|\cdot\|_{\mathcal{C}^0})$ is bounded, this vector space is closed in $\|\cdot\|_{\mathcal{C}^n}$.

⁸They are linearly independent as, by Picard–Lindelöf, $x(t) = 0$ for all t if and only if all the initial conditions are $\frac{d^i x}{dt^i}(t_0) = 0$.

in the kernel by $x_\nu(t)$ with $\nu = 1, \dots, n$ and we have:

$$Dx = 0 \Rightarrow x = \sum_{\nu=1}^n c_\nu x_\nu(t), \quad v^k = \sum_{\nu=1}^n c_\nu x_\nu^{(k)}(t_0), \quad k = 0, \dots, n-1,$$

and the initial conditions fix the constants c_ν . The *Wronskian* of the independent solutions is the determinant:

$$W(x_1, \dots, x_n)(t) = \begin{vmatrix} x_1(t) & \dots & x_n(t) \\ x_1^{(1)}(t) & \dots & x_n^{(1)}(t) \\ \vdots & & \vdots \\ x_1^{(n-1)}(t) & \dots & x_n^{(n-1)}(t) \end{vmatrix},$$

and is non zero (that is it does not vanish identically: we will see below the Wronskian is in fact nowhere zero) as the solutions are linearly independent.

Inhomogenous equations. A solution of the inhomogeneous equation $Dx = f$ is also uniquely determined by n initial conditions, which is exactly the dimension of the kernel of D . The general solution of the inhomogeneous equation is determined up to an element in the kernel:

$$Dx = f \Rightarrow x(t) = h(t) + \sum_{\nu=1}^n c_\nu x_\nu(t),$$

where $h(t)$ is a particular solution of $Dh = f$ and c_ν are n parameters which are fixed by imposing the initial conditions $x^{(i)}(t_0) = v^i$.

Once the general solution $\sum_{\nu=1}^n c_\nu x_\nu(t)$ of the homogeneous equation is known (i.e. a basis is identified in the kernel of D), a particular solution of the inhomogeneous equation can be found by the method of *variation of constants*. This goes as follows:

- we posit that a solution of the inhomogeneous equation can be written as a linear combination of solutions of the homogeneous one where the coefficients are promoted to functions:

$$h(t) = \sum_{\nu=1}^n c_\nu(t) x_\nu(t),$$

with $c_\nu(t)$ some functions to be determined.

- we require that (we denote $c'_\nu = \frac{dc_\nu}{dt}$):

$$\sum_{i=1}^n c'_\nu(t) x_\nu^{(i)}(t) = 0, \quad k = 0, \dots, n-2,$$

which is always possible as it imposes $n-1$ linear constraints among n arbitrary functions.

- we impose that $x(t)$ is a solution of the inhomogeneous equation; taking into account that:

$$x^{(k)}(t) = \sum_{\nu=1}^n c_\nu(t) x_\nu^{(k)}(t), \quad k = 0, \dots, n-1, \quad \text{and} \quad x^{(n)}(t) = \sum_{\nu=1}^n c_\nu(t) x_\nu^{(n)}(t) + \sum_{\nu=1}^n c'_\nu(t) x_\nu^{(n-1)}(t),$$

and recalling that $x_\nu(t)$ are solutions of the homogeneous equation this comes to:

$$D \left(\sum_{\nu=1}^n c_\nu(t) x_\nu(t) \right) = \sum_{\nu=1}^n c_\nu(t) Dx_\nu(t) + \sum_{\nu=1}^n c'_\nu(t) x_\nu^{(n-1)}(t) = f(t) \Rightarrow \sum_{\nu=1}^n c'_\nu(t) x_\nu^{(n-1)}(t) = f,$$

hence the unknown functions $c'_\nu(t)$ must obey then the linear system of equations:

$$\begin{pmatrix} x_1(t) & \dots & x_n(t) \\ x_1^{(1)}(t) & \dots & x_n^{(1)}(t) \\ \vdots & & \vdots \\ x_1^{(n-1)}(t) & \dots & x_n^{(n-1)}(t) \end{pmatrix} \begin{pmatrix} c'_1(t) \\ c'_2(t) \\ \vdots \\ c'_n(t) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ f(t) \end{pmatrix},$$

which always admits a unique solution as the Wronskian matrix has non zero determinant. Indeed, denoting $W_\nu(t) = (-1)^{n+\nu} W_{\hat{n}, \hat{\nu}}(t) f(t)$ the Wronskian where the ν 'th column has been replaced by the column vector $\begin{pmatrix} 0 & 0 & \dots & f(t) \end{pmatrix}^T$, that is $W_{\hat{n}, \hat{\nu}}(t)$ is the minor of the Wronskian where the row n and column ν have been deleted, we have:

$$c'_\nu(t) = \frac{W_\nu(t)}{W(t)} = \frac{1}{W(t)} (-1)^{n+\nu} W_{\hat{n}, \hat{\nu}}(t) f(t) \Rightarrow h(t) = \sum_{\nu=1}^n \left(\int_{t_0}^t ds \frac{(-1)^{n+\nu} W_{\hat{n}, \hat{\nu}}(s)}{W(s)} f(s) \right) x_\nu(t),$$

hence a Green function of the differential operator D is $G(t, s) = \sum_{\nu=1}^n x_\nu(t) \Theta(t-s) \frac{(-1)^{n+\nu} W_{\hat{n}, \hat{\nu}}(s)}{W(s)}$.

The solution of the initial value problem $Dx = f$, $x^{(i)}(t_0) = v^i$, $i = 0, \dots, n-1$ is then:

$$x(t) = \sum_{\nu=1}^n x_\nu(t) \left(c_\nu + \int_{t_0}^t ds \frac{(-1)^{n+\nu} W_{\hat{n}, \hat{\nu}}(s)}{W(s)} f(s) \right) = \int_{t_0}^t ds G(t, s) f(s) + \sum_{\nu=1}^n c_\nu x_\nu(t), \quad v^i = \sum_{\nu=1}^n c_\nu x_\nu^{(i)}(t_0), \quad i = 0, \dots, n-1,$$

where the initial conditions serve to fix the constants c_ν by solving the linear system of equations. We note that⁹ $W(t) = W(t_0) e^{-\int_{t_0}^t ds a_{n-1}(s)}$.

Exercise 19.4. Consider the operator $D = \frac{d^3}{dt^3}$. Solve the homogeneous ODE $Dx = 0$ and compute the Wronskian of the solutions. Solve the inhomogeneous ODE $Dx = f$. The generic solution of the homogeneous equation is any polynomial of degree 2, $Dx = 0 \Rightarrow x = c_2 t^2 + c_1 t + c_0$. The Wronskian and W_ν are:

$$W = \begin{vmatrix} 1 & t & t^2 \\ 0 & 1 & 2t \\ 0 & 0 & 2 \end{vmatrix} = 2, \quad W_1(t) = \begin{vmatrix} 0 & t & t^2 \\ 0 & 1 & 2t \\ f(t) & 0 & 2 \end{vmatrix} = f(t)t^2, \quad W_2(t) = \begin{vmatrix} 1 & 0 & t^2 \\ 0 & 0 & 2t \\ 0 & f(t) & 2 \end{vmatrix} = -2t f(t), \quad W_3(s) = \begin{vmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & f(t) \end{vmatrix} = f(t).$$

We obtain a solution of the inhomogeneous equation as $h(t) = \sum_{\nu} \left(\int_{t_0}^t ds \frac{W_\nu}{W} \right) x_\nu(t) = \int_{t_0}^t ds \frac{1}{2} (s^2 - 2st + t^2) f(s)$. This coincides with the fundamental solution computation $h(t) = G * f$ for the operator $D = \frac{d^{n+1}}{dt^{n+1}}$ with Green function $G(t-s) = \Theta(t-s) \frac{(t-s)^n}{n!}$.

19.2.1 Linear equations with constant coefficients

The simplest linear ODE is the one whose coefficients are constants independent on t :

$$\frac{d^n x}{dt^n}(t) + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}}(t) + \dots + a_0 x(t) = f(t), \quad x^{(i)}(t_0) = v^i, \quad i = 0, \dots, n-1.$$

This equation can be solved by taking the Fourier transform, as this diagonalizes the action of the derivative. Alternatively, we posit that a solution of the homogeneous equation has the form of an exponential:

$$x(t) = e^{\lambda t}, \quad \left(\sum_{i=0}^n a_i \lambda^i \right) e^{\lambda t} = 0,$$

⁹We have:

$$W'(t) = \begin{vmatrix} x_1^{(1)}(t) & \dots & x_n^{(1)}(t) \\ x_1^{(1)}(t) & \dots & x_n^{(1)}(t) \\ \vdots & & \vdots \\ x_1^{(n-1)}(t) & \dots & x_n^{(n-1)}(t) \end{vmatrix} + \begin{vmatrix} x_1(t) & \dots & x_n(t) \\ x_1^{(2)}(t) & \dots & x_n^{(2)}(t) \\ \vdots & & \vdots \\ x_1^{(n-1)}(t) & \dots & x_n^{(n-1)}(t) \end{vmatrix} + \dots + \begin{vmatrix} x_1(t) & \dots & x_n(t) \\ x_1^{(1)}(t) & \dots & x_n^{(1)}(t) \\ \vdots & & \vdots \\ x_1^{(n)}(t) & \dots & x_n^{(n)}(t) \end{vmatrix},$$

and only the last term contributes yielding $W'(t) = -a_{n-1}(t)W(t)$.

where $a_n = 1$. As the exponential is not zero, this is satisfied only for λ one of the roots of the characteristic polynomial:

$$\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_0 = 0.$$

We denote λ_ν with $\nu = 1, \dots, n$ the n complex roots. We have two cases:

All the roots are different. In this case the general solution of the homogeneous equation is $x(t) = \sum_{\nu=1}^n c_\nu e^{\lambda_\nu t}$. The Wronskian of these solutions is proportional to a Vandermonde determinant:

$$W = e^{\sum_{\nu} \lambda_\nu t} \begin{vmatrix} 1 & 1 & \dots & 1 \\ \lambda_1 & \lambda_2 & \dots & \lambda_n \\ \vdots & & & \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \dots & \lambda_n^{n-1} \end{vmatrix} = e^{\sum_{\nu} \lambda_\nu t} \prod_{1 \leq \mu < \nu \leq n} (\lambda_\nu - \lambda_\mu) \neq 0, \quad \frac{W_\nu}{W} = \frac{(-1)^{n+\nu} e^{-\lambda_\nu t} f(t)}{\prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu) \prod_{\mu > \nu} (\lambda_\mu - \lambda_\nu)},$$

and the Green function is:

$$G(t, s) = \sum_{\nu=1}^n e^{\lambda_\nu t} \Theta(t-s) \frac{e^{-\lambda_\nu s}}{\prod_{\mu \neq \nu} (\lambda_\nu - \lambda_\mu)}.$$

Exercise 19.5. Prove directly that the Green function of $D = \left(\frac{d}{dt} - \lambda_2\right)\left(\frac{d}{dt} - \lambda_1\right)$ is $G(t, s) = \frac{1}{\lambda_1 - \lambda_2} \Theta(t-s) e^{\lambda_1(t-s)} + \frac{1}{\lambda_2 - \lambda_1} \Theta(t-s) e^{\lambda_2(t-s)}$. We have:

$$D_t \int ds G(t, s) f(s) = \left(\frac{d}{dt} - \lambda_2\right)\left(\frac{d}{dt} - \lambda_1\right) \int ds \left(\frac{1}{\lambda_1 - \lambda_2} e^{\lambda_1(t-s)} + \frac{1}{\lambda_2 - \lambda_1} e^{\lambda_2(t-s)} \right) f(s) = \left(\frac{d}{dt} - \lambda_2\right) \left[\frac{1}{\lambda_1 - \lambda_2} f(t) + \frac{1}{\lambda_2 - \lambda_1} f(t) + \int ds e^{\lambda_2(t-s)} f(s) \right] = f(t).$$

Some roots are degenerate. Assuming that the characteristic polynomial has m distinct roots λ_i each with multiplicity d_i , the differential operator writes as $D = \prod_{i=1}^m \left(\frac{d}{dt} - \lambda_i\right)^{d_i}$. The kernel of D splits into m subspaces, each corresponding to one root $\ker D = \bigoplus_{i=1}^m \ker \left(\frac{d}{dt} - \lambda_i\right)^{d_i}$. The subspace corresponding to λ_i is spanned by d_i linearly independent functions solutions of:

$$\left(\frac{d}{dt} - \lambda_i\right)^{d_i} v = 0.$$

A system of such independent functions is given by $v = t^d e^{\lambda_i t}$ for $d = 0, \dots, d_i - 1$ and a full list of independent eigen-functions spanning the kernel of D is:

$$e^{\lambda_1 t}, t e^{\lambda_1 t}, t^{d_1-1} e^{\lambda_1 t}, e^{\lambda_2 t}, t e^{\lambda_2 t}, \dots, t^{d_m-1} e^{\lambda_m t}.$$

Alternatively, one can rewrite the homogenous differential equation as a first order linear ODE for a n dimensional vector:

$$\begin{pmatrix} \frac{dy}{dt} \\ \frac{dy_1}{dt} \\ \vdots \\ \frac{dy_{n-2}}{dt} \\ \frac{dy_{n-1}}{dt} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{pmatrix} \begin{pmatrix} y \\ y_1 \\ \vdots \\ y_{n-2} \\ y_{n-1} \end{pmatrix},$$

or in matrix notations $\frac{dv}{dt} = Av$ with general solution $v = e^{At} v_0$. By expanding the determinant along the last line, we note that the characteristic polynomial of the A is exactly the characteristic polynomial of the ODE:

$$(-1)^n \det(A - \lambda) = (a_{n-1} + \lambda) \lambda^{n-1} + \sum_{p=0}^{n-1} a_p \lambda^p$$

Now, the matrix exponential is easy to compute up to a caveat: any eigenvalue of A has geometrical multiplicity 1 that is if $Av = \lambda v$ then $v \sim (1, \lambda, \lambda^2, \dots, \lambda^n)$, hence if there are degenerate roots, A is not diagonalizable. However, it can be brought in Jordan form and the independent solutions of the order n homogeneous ODE can be identified by noting that for any function f and J a Jordan block of size m corresponding to the eigenvalue λ we have:

$$f(J) = \begin{pmatrix} f(\lambda) & \frac{df}{d\lambda}(\lambda) & \frac{1}{2!} \frac{d^2 f}{d\lambda^2}(\lambda) & \cdots & \frac{1}{(m-1)!} \frac{d^{m-1} f}{d\lambda^{m-1}}(\lambda) \\ 0 & f(\lambda) & \frac{df}{d\lambda} & \cdots & \frac{1}{(m-2)!} \frac{d^{m-2} f}{d\lambda^{m-2}}(\lambda) \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{df}{d\lambda} \\ 0 & 0 & 0 & \cdots & f(\lambda) \end{pmatrix},$$

which reproduces the solutions listed previously.

Exercise 19.6. Prove the the Vandermonde determinant equals $\prod_{1 \leq \mu < \nu \leq n} (\lambda_\nu - \lambda_\mu)$.

Exercise 19.7. Derive a formula for the Wronskian in the case when the roots are degenerate.

Exercise 19.8. Why is this similar to recursive sequences defined by linear recursions with constant coefficients?

19.2.2 Generic linear first order ODE

A generic linear first order ODE can always be written as:

$$x'(t) + a(t)x(t) = f(t), \quad x(t_0) = x_0.$$

The homogeneous equation is solved by “separating the variables” x and t :

$$\frac{dx}{dt} + a(t)x(t) = 0 \Rightarrow \frac{dx}{x} = -a(t)dt \Rightarrow \ln\left(\frac{x}{x_0}\right) = -\int_{t_0}^t ds a(s) \Rightarrow x(t) = x_0 e^{-\int_{t_0}^t ds a(s)},$$

and the general solution of the inhomogeneous equation is obtained by varying the constant:

$$x(t) = c(t) e^{-\int_{t_0}^t ds a(s)}, \quad c'(t) e^{-\int_{t_0}^t ds a(s)} = f(t) \Rightarrow c(t) = \int_{t_0}^t du f(u) e^{\int_{t_0}^u ds a(s)},$$

hence imposing the initial condition we find:

$$x(t) = \left(\int_{t_0}^t du f(u) e^{\int_{t_0}^u ds a(s)} + x_0 \right) e^{-\int_{t_0}^t ds a(s)}.$$

The time ordered exponential

In many instances one needs to solve a linear ODE:

$$\frac{dx}{dt} = H(t)x, \quad x(t_0) = x_0$$

where x is a vector in some vector space and $H(t)$ is a possibly time dependent linear operator. If H is time independent the equation is immediately solved by taking its exponential $x(t) = e^{(t-t_0)H} x_0$. If H is time dependent, one can not talk about the simple exponential as the $H(t)$ at different times do not commute. The solution of this equation is written in terms of the “time ordered exponential” of H :

$$x(t) = \left[\mathcal{T} e^{\int_{t_0}^t ds H(s)} \right] x_0,$$

where

$$\mathcal{T}e^{\int_{t_0}^t ds H(s)} = \sum_{n \geq 0} \frac{1}{n!} \int ds^n \dots ds^1 T(s_n, \dots, s_1) H(s_n) \dots H(s_1), \quad T(s_n, \dots, s_1) = \begin{cases} 1, & \text{if } s_n \geq s_{n-1} \geq \dots \geq 1 \\ 0, & \text{if not} \end{cases}.$$

19.3 Sturm–Liouville theory

We are interested in the second order differential equation:

$$\frac{d}{dt} \left(p(t) \frac{dx}{dt}(t) \right) + q(t)x(t) = -\lambda w(t)x(t),$$

subjected to boundary conditions on an interval $[a, b]$. The functions p, q and w are known functions and we search for solutions $x : [a, b] \rightarrow \mathbb{C}$. The equation admits solutions only for some values of λ , which are called the eigenvalues of the problem and the associated solutions are called its eigenfunctions.

Exercise 19.9. *Prove that any second order linear differential operator can be brought into Sturm-Liouville form. $P_y'' + Q_y' + Ry = 0$ is equivalent to $[e^{\int^t Q/P} y']' + R/P e^{\int^t Q/P} y = 0$.*

Theorem 19.2 (Sturm–Liouville). *If the functions $p(t), p'(t), q(t)$ and $w(t)$ are real valued, continuous on the interval $[a, b]$, $p(t), w(t) > 0$ on $[a, b]$ and $x(t)$ obeys separated boundary conditions:*

$$\alpha_1 x(a) + \alpha_2 x'(a) = 0, \quad \beta_1 x(b) + \beta_2 x'(b) = 0, \quad \alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}, \quad |\alpha_1| + |\alpha_2| > 0, \quad |\beta_1| + |\beta_2| > 0,$$

with $\alpha_{1,2}$ and $\beta_{1,2}$ fixed the problem is called regular. In this case:

- the eigenvalues of the problem are real and form a countable set bounded below $\lambda_1 < \lambda_2 < \dots < \lambda_n < \dots \rightarrow \infty$.
- for each n there is a unique (up to rescaling) eigenfunction $x_n : [a, b] \rightarrow \mathbb{R}$ (the eigenfunctions can be chosen to be real valued), called the n 'th fundamental solution, verifying:

$$\frac{d}{dt} \left[p(t) \frac{dx_n}{dt}(t) \right] + q(t)x_n(t) = -\lambda_n w(t)x_n(t),$$

and the boundary conditions. The function $x_n(t)$ has exactly $n - 1$ zeros in (a, b) .

- the normalized eigenfunctions form an orthonormal basis in the Hilbert space $L^2([a, b], w(x)dx)$:

$$\langle f, g \rangle = \int_a^b [w(t)dt] \overline{f(t)} g(t).$$

Proof. The Sturm-Liouville equation is the eigenvalue equation of the operator:

$$Lx(t) = -\frac{1}{w(t)} \left[\frac{d}{dt} \left(p(t) \frac{dx}{dt}(t) \right) + q(t)x(t) \right], \quad Lx = \lambda x.$$

which is an unbounded, densely defined operator on $L^2([a, b], w(x)dx)$. The operator is symmetric if the functions in its domain respect the separated boundary conditions, as:

$$\begin{aligned} \int_a^b [w(t)dt] \left\{ -\frac{1}{w(t)} \left[\frac{d}{dt} \left(p(t) \frac{dx}{dt}(t) \right) + q(t)x(t) \right] \right\} y(t) &= \\ = -[p \overline{x'} y]_a^b + [\overline{x} p y']_a^b + \int_a^b [w(t)dt] \overline{x(t)} \left\{ -\frac{1}{w(t)} \left[\frac{d}{dt} \left(p(t) \frac{dy}{dt}(t) \right) + q(t)y(t) \right] \right\}, \end{aligned}$$

and the boundary term $p(b)[\overline{x(b)}y'(b) - \overline{x'(b)}y(b)] - p(a)[\overline{x(a)}y'(a) - \overline{x'(a)}y(a)]$ cancel as both x and y obey the boundary conditions. For instance, if $\beta_1 \neq 0$ then:

$$\overline{x(b)}y'(b) - \overline{x'(b)}y(b) = [-\frac{\beta_2}{\beta_1}\overline{x'(b)}]y'(b) - \overline{x'(b)}[-\frac{\beta_2}{\beta_1}y'(b)] = 0.$$

For z not belonging to the spectrum of L , the resolvent $(L - z)^{-1}$ can be computed by solving $(L - z)y = f$ with $\alpha_1 y(a) + \alpha_2 y'(a) = 0$ and $\beta_1 y(b) + \beta_2 y'(b) = 0$. This is a second order ODE and can be solved by the variation of constants. We set $w(t) = 1$ (the general $w(t)$ follows the same steps). The operator L is bounded from below¹⁰ hence z real and small enough belongs to the resolvent set. We chose such a z and we note that:

$$(L - z)y = -f \Leftrightarrow y'' + \frac{p'}{p}y' + \frac{q+z}{p}y = Dy = \frac{f}{p}, \quad D \equiv \frac{d^2}{dt^2} + \frac{p'}{p} \frac{d}{dt} + \frac{q+z}{p}.$$

As D is a second order differential operator its kernel is two dimensional. We denote $\phi(t)$ and $\psi(t)$ a basis in the kernel such that

$$\alpha_1 \phi(a) + \alpha_2 \phi'(a) = 0, \quad \beta_1 \psi(b) + \beta_2 \psi'(b) = 0.$$

Note that ϕ (and ψ) is determined only up to a multiplicative constant, as it respects only one initial condition: in order to have a initial value problem with a unique solution, one would need to impose a second condition relating $\phi(a)$ and $\phi'(a)$. Furthermore, ϕ and ψ are linearly independent, that is the corresponding Wronskian $W(s) = \phi(s)\psi'(s) - \psi(s)\phi'(s)$ is non zero: indeed, if $\psi = \phi$ then $(L - z)\phi = 0$ and ϕ respects both boundary conditions, hence $L - z$ has a non trivial kernel and z is in the spectrum of L . We solve the inhomogeneous equation $Dy = \frac{f}{p}$ using the symmetric Green function $G(t, s) = \frac{\Theta(t-s)\psi(t)\phi(s) + \Theta(s-t)\phi(t)\psi(s)}{W(s)}$:

$$y(t) = \int_a^b ds \frac{\Theta(t-s)\psi(t)\phi(s) + \Theta(s-t)\phi(t)\psi(s)}{p(s)W(s)} f(s) = \int_a^t ds \frac{-\phi(t)\psi(s) + \psi(t)\phi(s)}{p(s)W(s)} f(s) + \phi(t) \int_a^b ds \frac{\psi(s)}{p(s)W(s)} f(s),$$

where in the last equality we have explicitly written the symmetric Green function in terms of the one sided Green function from the previous section plus a piece in the kernel of D . Incidentally, as $W(s) = W(a)e^{-\int_a^s ds' \frac{p'}{p}}$, $p(s)W(s)$ is a constant. Taking into account that:

$$\begin{aligned} y(a) &= \phi(a) \int_a^b ds \frac{\psi(s)}{p(s)W(s)} f(s), \quad y'(a) = \phi'(a) \int_a^b ds \frac{\psi(s)}{p(s)W(s)} f(s), \\ y(b) &= \psi(b) \int_a^b ds \frac{\phi(s)}{p(s)W(s)} f(s), \quad y'(b) = \psi'(b) \int_a^b ds \frac{\phi(s)}{p(s)W(s)} f(s), \end{aligned}$$

we conclude that $y(t)$ respects both boundary conditions.

We have thus written the operator inverse of $L - z$ with boundary conditions as an explicit integral operator which, as integral operators are Hilbert-Schmidt, is compact. For z real outside the spectrum of L , as $G(t, s)$ is symmetric, this operator is also self-adjoint. By the spectral theorem for compact self adjoint operators we have that $(L - z)^{-1}$ has a sequence of real eigenvalues $\mu_i \rightarrow 0$ with associated eigenfunctions x_i :

$$(L - z)^{-1}(t, s) = \sum_i \mu_i x_i(t)x_i(s) \Rightarrow L = \sum_i (z + \frac{1}{\mu_i}) x_i(t)x_i(s).$$

As L is bounded from below, the set $\{\lambda_n\}$ is bounded below and as $|\lambda_n| \rightarrow \infty$ we conclude that we can order the eigenvalue as $\lambda_1 < \lambda_2 < \dots \rightarrow \infty$. The fact that x_n is unique up to an overall factor follows from observing that, as the kernel

¹⁰We have $\langle x, Lx \rangle = -[p\overline{x}x']_a^b + \int_a^b p(t) \left| \frac{dx}{dt}(t) \right|^2 - \int_a^b dt q(t) |x(t)|^2 \geq -\sup_{t \in [a, b]} |q(t)| \|x\|_{L^2}$ and the boundary term drops for boundary conditions $x(a) = x(b) = 0$. The case with generic boundary conditions follows from this one.

of $L - \lambda$ is two dimensional, imposing one boundary condition projects on a one dimensional subspace. Concerning that fact that x_n can be taken real, we observe that the Picard-Lindelöf iteration for an equation with real coefficients and real initial conditions gives a real solution, hence the same holds up to a multiplicative constant for a real boundary condition. For the proof that x_n has exactly $n - 1$ zeros, the reader is invited to search the literature. Note that L is selfadjoint as its eigenvectors form a basis and all its eigenvalues are real. $\square$

After bringing a differential equation in Sturm Liouville form, one can wonder whether it admits eigenvalues and eigenvectors. This can be the case even if the problem is not regular: for instance for infinite intervals, replacing the boundary conditions by conditions on the growth at infinity suffices in order to fix the eigenvalues and eigenvectors; alternatively, using periodic boundary conditions one can also find eigenvalues and eigenvectors (contrary to the regular problems, the eigenvalues can be degenerated in this case). The point is that whatever boundary conditions one uses (even if they are not regular) one needs to ensure that the Sturm Liouville operator $Lx = -\frac{1}{w} \left[\frac{d}{dt} \left(p \frac{dx}{dt} \right) + q x \right]$ is symmetric. As a thumb rule, a set of boundary conditions for a differential operator L is “good”, in the sense that L with the boundary conditions is a self adjoint operator if, in order to ensure that L is symmetric, $\langle x, Ly \rangle = \langle Lx, y \rangle$, one needs to use the same boundary conditions *both* for x and for y to cancel the boundary terms. Morally, this ensures that the boundary conditions are “just right” (not too strong, not too weak) and $D(L^\dagger) = D(L)$.

Consider the Sturm-Liouville problem $Lx = -\frac{1}{w} \left[\frac{d}{dt} \left(p \frac{dx}{dt} \right) + q x \right] = \lambda x$, $\alpha_1 x(a) + \alpha_2 x'(a) = \beta_1 x(b) + \beta_2 x'(b) = 0$, and denote λ_n, x_n its eigenvalues and fundamental solutions. Assuming 0 is not an eigenvalue, the symmetric Green function of L writes as:

$$G(t, s) = \sum_n \frac{1}{\lambda_n} x_n(t) x_n(s), \quad L_t G(t, s) = \sum_n x_n(t) x_n(s), \quad \alpha_1 G(a, s) + \alpha_2 \partial_t G(a, s) = \beta_1 G(b, s) + \beta_2 \partial_t G(b, s) = 0,$$

As x_n is a complete basis, we have:

$$f(t) = \sum_n \left(\int_a^b [w(s) ds] f(s) x_n(s) \right) x_n(t), \quad \sum_n x_n(t) x_n(s) = \delta^w(t - s),$$

where the last equation is to be interpreted in the sense of distributions and $\delta^w(t - s) = \frac{1}{w(t)} \delta(t - s)$ is the Dirac δ distribution for the measure $w(t)dt$. The unique solution of the inhomogeneous ODE $Ly = f$ which respects the boundary conditions $\alpha_1 y(a) + \alpha_2 y'(a) = 0$ and $\beta_1 y(b) + \beta_2 y'(b) = 0$ is $y(t) = \int_a^b [w(s) ds] G(t, s) f(s)$. For instance, if L with Dirichlet boundary conditions $x(a) = x(b) = 0$ is invertible then one can solve the equation $Ly = f$ for arbitrary boundary conditions, $y(a) = y_a, y(b) = y_b$. Denoting ϕ and ψ a basis in the kernel of L with $\phi(a) = 0$ and $\psi(b) = 0$ (ϕ and ψ are linearly independent, as we assumed L with Dirichlet boundary conditions is invertible) we have:

$$y(t) = \int_a^b ds \frac{\Theta(t-s)\psi(t)\phi(s) + \Theta(s-t)\phi(t)\psi(s)}{p(s)W(s)} f(s) + y_a \frac{\psi(t)}{\psi(a)} + y_b \frac{\phi(t)}{\phi(b)},$$

and the symmetric Green function is known as the Dirichlet Green function in this case.

19.3.1 The role of the boundary / initial conditions

Consider the operator $L = -\frac{d^2}{dt^2}$ and consider the equation $Lx = \lambda x$ with $\lambda \neq 0$ (a similar discussion goes through for $\lambda = 0$, but the eigenfunctions below must be replaced with $y_1(t) = 1$ and $y_2(t) = t$).

ODE with initial conditions. As ODE with initial conditions we aim to solve $Lx = \lambda x \Rightarrow x'' + \lambda x = 0$, $x(a) = x_0$, $x'(a) = x'_0$. As the ODE is second order, the kernel of $L - \lambda$ is two dimensional, spanned by two functions $y_1(t) = e^{i\sqrt{\lambda}t}$ and $y_2(t) = e^{-i\sqrt{\lambda}t}$. The general solution of the homogeneous equation writes $x(t) = c_1 e^{i\sqrt{\lambda}t} + c_2 e^{-i\sqrt{\lambda}t}$ and the solution of the

equation with initial conditions is:

$$x'' + \lambda x = 0, \quad x(a) = x_0, \quad x'(a) = x'_0 \quad \Rightarrow \quad x(t) = \frac{x'_0}{\sqrt{\lambda}} \sin(\sqrt{\lambda}(t-a)) + x_0 \cos(\sqrt{\lambda}(t-a)).$$

Note that the ODE has a solution for any λ and any initial condition.¹¹

Regular Sturm Liouville problem. On the other hand we have an associated Sturm Liouville problem $Lx = \lambda x \Rightarrow x'' + \lambda x = 0$, $\alpha_1 x(a) + \alpha_2 x'(a) = 0$, $\beta_1 x(b) + \beta_2 x'(b) = 0$. The boundary conditions *do not fix a unique solution*, but they fix a set of acceptable eigenvalues λ_n and associated eigenfunctions $x_n(t)$, that is they fix the self adjoint extension of L . Note that due to the boundary condition of the Sturm Liouville problem, that is for the Sturm Liouville self adjoint extension, the eigenvalues are non degenerate. This is not true for other self adjoint extensions (see below).

Let us consider for simplicity Dirichlet boundary conditions $x(a) = x(b) = 0$. At fixed λ , the ODE with one boundary condition is under determined (the solution is fixed up to an overall scaling):

$$x'' + \lambda x = 0, \quad x(a) = 0 \quad \Rightarrow \quad x(t) = A \sin(\sqrt{\lambda}(t-a)), \quad A \in \mathbb{C} \setminus \{0\}.$$

If the solution happens to respect the second boundary condition $x(b) = 0$, then λ is an eigenvalue. That is $x(b) = 0 \Rightarrow \sqrt{\lambda} = \pm \frac{\pi}{b-a} n$ with $n \in \mathbb{N}_{>0}$ therefore $\lambda = \lambda_n = \left(\frac{\pi}{b-a} n\right)^2$ with $n \in \mathbb{N}_{>0}$ and the associated eigenfunction is $x_n = \sin\left(\frac{\pi}{b-a} n(t-a)\right)$. Note that both $\sqrt{\lambda} \sim \pm n$ lead to the same eigenvalue and eigenfunction, that is the eigenvalues of the Sturm Liouville self adjoint extension are non degenerate.

If $x(b) \neq 0$, then λ does not belong to the spectrum of L , $\lambda \neq \left(\frac{\pi}{b-a} n\right)^2$, and one has two linearly independent solutions of $Lx = \lambda x$ such that each of them respects one of the boundary conditions:

$$y_1(t) = \sin(\sqrt{\lambda}(t-a)), \quad y_1(a) = 0 \quad ; \quad y_2(t) = \sin(\sqrt{\lambda}(b-t)), \quad y_2(b) = 0.$$

In this case λ is in the resolvent set, $L - \lambda$ can be explicitly inverted using the symmetric Green's function and $(L - \lambda)^{-1}$ is a compact self adjoint operator.

Regular Sturm Liouville problem reloaded. Another regular Sturm Liouville problem is defined by the boundary conditions $x(a) = 0$, $x'(b) = 0$. We have:

$$x(a) = 0 \Rightarrow x(t) = A \sin(\sqrt{\lambda}(t-a)), \quad x'(b) = 0 \Rightarrow \sqrt{\lambda}(b-a) = \pm n\pi + \frac{\pi}{2} \Rightarrow \lambda_n = \left(\frac{n\pi + \frac{\pi}{2}}{b-a}\right)^2, \quad n \in \mathbb{N},$$

that is a very different set of eigenvalues with respect to what we obtained for the Dirichlet boundary conditions.

Other self adjoint extensions: periodic boundary conditions. There exist other self adjoint extensions of the operator $L = -\frac{d^2}{dt^2}$. For instance one can consider $Lx = \lambda x$ with periodic boundary conditions $x(a) = x(b)$ and $x'(a) = x'(b)$, that is:

$$x(t) = c_1 e^{i\sqrt{\lambda}(t-a)} + c_2 e^{-i\sqrt{\lambda}(t-a)}, \quad c_1 + c_2 = c_1 e^{i\sqrt{\lambda}(b-a)} + c_2 e^{-i\sqrt{\lambda}(b-a)}, \quad c_1 - c_2 = c_1 e^{i\sqrt{\lambda}(b-a)} - c_2 e^{-i\sqrt{\lambda}(b-a)}.$$

The periodic boundary conditions can be reorganized in the homogeneous linear system of equations:

$$\begin{pmatrix} 1 - e^{i\sqrt{\lambda}(b-a)} & 1 - e^{-i\sqrt{\lambda}(b-a)} \\ 1 - e^{i\sqrt{\lambda}(b-a)} & -1 + e^{-i\sqrt{\lambda}(b-a)} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \begin{vmatrix} 1 - e^{i\sqrt{\lambda}(b-a)} & 1 - e^{-i\sqrt{\lambda}(b-a)} \\ 1 - e^{i\sqrt{\lambda}(b-a)} & -1 + e^{-i\sqrt{\lambda}(b-a)} \end{vmatrix} = 2(e^{i\sqrt{\lambda}(b-a)} + e^{-i\sqrt{\lambda}(b-a)} - 2),$$

¹¹One can also solve the inhomogeneous ODE $(L - \lambda)x = f$. The general solution writes as:

$$x(t) = \int_a^b G(t, s) f(s) ds + c_1 y_1(t) + c_2 y_2(t) \quad G(t, s) = \Theta(t-s) \frac{-y_1(t)y_2(s) + y_2(t)y_1(s)}{y_1(s)y_2'(s) - y_2(s)y_1'(s)},$$

and an initial condition fixes c_1 and c_2 .

which admits a non trivial solution only if the matrix is singular, that is $\cos(\sqrt{\lambda}(b-a)) = 1$, leading to $\sqrt{\lambda} = \pm \frac{2\pi}{b-a}n$, $n \in \mathbb{N}$ and eigenvalues $\lambda_n = \left(\frac{2\pi}{b-a}n\right)^2$, $n \in \mathbb{N}$. Of course, as L with periodic boundary conditions is self adjoint, its spectrum is real, but it is again different from the previous two cases. Moreover, now the eigenvalues are degenerate, that is for each λ_n with $n \neq 0$ there exist two linearly independent periodic eigenfunctions:

$$x_{n,1}(t) = e^{i\frac{2\pi}{b-a}n(t-a)}, \quad x_{n,2}(t) = e^{-i\frac{2\pi}{b-a}n(t-a)}.$$

In practice we always start from the general solution of the homogeneous ODE $Lx = \lambda x$, but we use it in different ways. For the initial value problem, fixing both x and x' at the same point fixes one solution for any λ . For the Sturm Liouville problem, fixing x at two different points yields a set of acceptable solutions (λ_n, x_n) with x_n fixed up to overall scaling. Periodic boundary conditions fix the eigenvalues λ_n , but the latter are degenerate.

19.3.2 Hermite functions

The classical harmonic oscillator respects the equation of motion $\ddot{x} = -U'$ with $U(x) = \omega^2 \frac{x^2}{2}$ and has an integral of motion $E = \frac{1}{2}(p^2 + \omega^2 x^2)$. In quantum mechanics the energy is promoted to the Hamiltonian operator $p \rightarrow -i\frac{d}{dx}$ and the study of the quantum harmonic oscillator is the study of the eigenvalues and the eigenfunctions of the differential operator $H_x = \frac{1}{2}(-\frac{d^2}{dx^2} + x^2)$. We thus aim to solve the equation:

$$-\frac{1}{2}\psi'' + \frac{x^2}{2}\psi = \lambda\psi, \quad \psi(x) \in L^2(\mathbb{R}),$$

or in physics notation $H|\psi\rangle = \lambda|\psi\rangle$. Although this is not a regular Sturm Liouville problem, as the interval of definition is unbounded, it is nevertheless solvable in an eigenbasis. One solution is simple: $\psi_0(x) = \frac{1}{\pi^{1/4}}e^{-\frac{x^2}{2}}$ is such that $H\psi_0 = \frac{1}{2}\psi_0$ and $\int dx |\psi_0(x)|^2 = 1$. In physics notations this becomes $\psi_0 \sim |0\rangle$ and $\langle 0|0\rangle = 1$. Other are slightly more involved. Let us define for $n \geq 1$:

$$\psi_n = \frac{1}{\sqrt{2n}} \left(x - \frac{d}{dx}\right) \psi_{n-1}(x) = \frac{1}{\sqrt{n!}} (A^\dagger)^n \psi_0, \quad |n\rangle = \frac{1}{\sqrt{n!}} (A^\dagger)^n |0\rangle,$$

where we introduced the raising and lowering (differential) operators $A^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx}\right)$ and $A = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx}\right)$ with:

$$[A, A^\dagger] = 1, \quad [A, (A^\dagger)^n] = n(A^\dagger)^{n-1} \quad A\psi_0 = A|0\rangle = 0, \quad H = A^\dagger A + \frac{1}{2}.$$

The functions ψ_n form an orthonormal system (assume $m \geq n$):

$$\langle n|m\rangle = \int dx \psi_n(x) \psi_m(x) = \frac{1}{\sqrt{n!m!}} \int dx [A^m (A^\dagger)^n \psi_0(x)] \psi_0(x) = \frac{n!}{\sqrt{n!m!}} \int dx [A^{m-n} \psi_0(x)] \psi_0(x) = \delta_{mn},$$

and are eigenfunctions of H :

$$H\psi_n = \left(A^\dagger A + \frac{1}{2}\right) \frac{1}{\sqrt{n!}} (A^\dagger)^n \psi_0 = n \frac{1}{\sqrt{n!}} (A^\dagger)^n \psi_0 + \frac{1}{2} \psi_n = \left(n + \frac{1}{2}\right) \psi_n, \quad H|n\rangle = \left(n + \frac{1}{2}\right) |n\rangle.$$

The family $\{\psi_n\}_{n \in \mathbb{N}}$ forms a complete orthonormal system, that is this set consists in all the normalized eigenfunctions of H_x . In order to prove the completeness of $\{\psi_n\}_{n \in \mathbb{N}}$, let us first remark that:

$$e^{-\frac{x^2}{2}} \left(x - \frac{d}{dx}\right)^n e^{-\frac{x^2}{2}} = (-1)^n \frac{d^n}{dx^n} e^{-x^2}.$$

Indeed, the $n = 1$ case is straightforward¹² and for $n \geq 1$ we have:

$$e^{-\frac{x^2}{2}} \left(x - \frac{d}{dx}\right)^n e^{-\frac{x^2}{2}} = e^{-\frac{x^2}{2}} \left(x - \frac{d}{dx}\right) \left[e^{\frac{x^2}{2}} (-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} e^{-x^2} \right] = (-1)^n \frac{d^n}{dx^n} e^{-x^2}.$$

We now consider the auxiliary object:

$$E(u; x, y) = \sum_{n \geq 0} u^n \psi_n(x) \psi_n(y) = \sum_{n \geq 0} \frac{u^n}{\sqrt{\pi} 2^n n!} e^{\frac{x^2+y^2}{2}} \left(\frac{d^n}{dx^n} e^{-x^2} \right) \left(\frac{d^n}{dy^n} e^{-y^2} \right),$$

and substituting the Fourier transform $e^{-x^2} = \int \frac{dp}{(2\pi)^{1/2}} e^{-\frac{p^2}{4} + ipx}$ we get:

$$\begin{aligned} E(u; x, y) &= \frac{e^{\frac{x^2+y^2}{2}}}{2\pi^{3/2}} \int dp dq e^{-\frac{p^2}{4} - \frac{q^2}{4} + ipx - i q y + \frac{u p q}{2}} = \frac{e^{\frac{x^2+y^2}{2}}}{2\pi^{3/2}} \int dp dq e^{-\frac{1}{4} \begin{pmatrix} p & q \end{pmatrix} \begin{pmatrix} 1 & -u \\ -u & 1 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} + i \begin{pmatrix} p & q \end{pmatrix} \begin{pmatrix} x \\ -y \end{pmatrix}} \\ &= \frac{1}{\sqrt{\pi(1-u^2)}} \exp \left\{ -\frac{1-u}{1+u} \frac{(x+y)^2}{4} - \frac{1+u}{1-u} \frac{(x-y)^2}{4} \right\}, \end{aligned}$$

and in the sense of distributions $\lim_{u \rightarrow 1} E(u; x, y) = \delta(x - y)$.

Using the coherent states:¹³

$$\psi_{\alpha, \bar{\beta}} = e^{\alpha A^\dagger - \bar{\beta} A} \psi_0 = e^{-\frac{\alpha \bar{\beta}}{2}} e^{\alpha A^\dagger} e^{-\bar{\beta} A} \psi_0 = e^{-\frac{\alpha \bar{\beta}}{2}} \sum_{n \geq 0} \frac{\alpha^n}{\sqrt{n!}} \psi_n,$$

for $\alpha = -\beta = \sqrt{2}t$ we obtain the generating function of Hermite functions:

$$\frac{1}{\pi^{1/4}} e^{-\frac{x^2}{2} + 2xt - t^2} = \sum_{n \geq 0} \frac{2^{\frac{n}{2}} t^n}{\sqrt{n!}} \psi_n(x)$$

19.3.3 The Legendre polynomials

The following points are all equivalent and one can take any of them as definition.

Rodrigues' formula. Let us consider the family of polynomials P_n of degree n defined by Rodrigues' formula:

$$P_n(x) = \frac{(-1)^n}{2^n n!} \frac{d^n}{dx^n} (1-x^2)^n = \frac{(2n)!}{2^n n! n!} x^n + O(x^{n-1}).$$

Orthogonality and completeness. This is a family of orthogonal polynomials with trivial measure on $[-1, 1]$: if $m > n$ then (note that $\frac{d^{m-q}}{dx^{m-q}} (x^2 - 1)^m = (x^2 - 1)^q Q(x)$ for some polynomial $Q(x)$, hence all the boundary terms are zero):

$$\begin{aligned} \int_{-1}^1 dx P_m(x) P_n(x) &= \frac{1}{2^{m+n} m! n!} \int_{-1}^1 dx \left[\frac{d^m}{dx^m} (x^2 - 1)^m \right] \left[\frac{d^n}{dx^n} (x^2 - 1)^n \right] \\ &= \frac{(-1)^{n+1}}{2^{m+n} m! n!} \int_{-1}^1 dx \left[\frac{d^{m-n-1}}{dx^{m-n-1}} (x^2 - 1)^m \right] \left[\frac{d^{2n+1}}{dx^{2n+1}} (x^2 - 1)^n \right] = 0, \end{aligned}$$

¹²Note that for any function $f(x)$ we have $e^{-\frac{x^2}{2}} \left(x - \frac{d}{dx}\right) [f(x) e^{-\frac{x^2}{2}}] = e^{-\frac{x^2}{2}} (2x - f') e^{-\frac{x^2}{2}} = -\frac{d}{dx} [f(x) e^{-x^2}]$.

¹³We use the Baker-Campbell-Hausdorff formula $e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]}$ if $[X, Y]$ commutes with X and Y .

and they are normalized as:¹⁴

$$\int_{-1}^1 dx P_n(x) P_n(x) = \frac{(-1)^n}{2^{2n} n! n!} \int_{-1}^1 dx (1-x^2)^n \left[\frac{d^{2n}}{dx^{2n}} (1-x^2)^n \right] = \frac{(2n)!}{2^{2n} n! n!} \int_{-1}^1 dx (1-x^2)^n = \frac{2}{2n+1}.$$

As the polynomials are dense in L^2 and we have one polynomial by degree, this is a complete basis in $L^2([-1, 1], \mathbb{R})$. One can obtain P_n alternatively by applying Gram Schmidt orthonormalization of the monomials $1, x, x^2, \dots$, and requiring that $P_n(1) = 1$.¹⁵ The completeness of the set $\{P_n\}_{n \in \mathbb{N}}$ writes as:

$$\sum_{n \geq 0} \frac{2n+1}{2} P_n(x) P_n(y) = \delta(x-y).$$

Recursion relation. The Legendre polynomials respect the recursion relation

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

Indeed, $xP_n(x)$ is a polynomial of degree $n+1$ hence expands uniquely as:

$$xP_n(x) = \sum_{q=0}^{n+1} c_q P_q(x), \quad c_q = \frac{2q+1}{2} \int_{-1}^1 dy y P_n(y) P_q(y) = \frac{2q+1}{2^{n+q+1} n! q!} \int_{-1}^1 dy y [(y^2-1)^q]^{(q)} [(y^2-1)^n]^{(n)},$$

and the recursion follows.

Exercise 19.10. *Finish the proof.* If $n = q$, taking into account that $[(y^2-1)^n]$ has only even degrees, integrating by parts we get:

$$c_n \sim \int_{-1}^1 dy (y^2-1)^n \left[y[(y^2-1)^n]^{(n)} \right] \sim \int_{-1}^1 dy [(y^2-1)^n] \left[y[y^{2n}]^{(n)} \right] \sim \int_{-1}^1 dy (y^2-1)^n y \sim \int_{-1}^1 dy \frac{d}{dy} [(y^2-1)^{n+1}].$$

Taking $q > n$ and integrating by parts we get:

$$(-1)^{n+1} \int_{-1}^1 dy [(y^2-1)^q]^{(q-n-1)} \left[y[(y^2-1)^n]^{(n)} \right]^{(n+1)} = (-1)^{n+1} \frac{(2n)!}{n!} (n+1)! \delta_{n,q-1} \int_{-1}^1 dy (y^2-1)^q = (2n)! (n+1) \delta_{n,q-1} \frac{2^{2n+3} (n+1)! (n+1)!}{(2n+3)!} = \frac{2^{2n+2} (n+1)! (n+1)!}{(2n+1)(2n+3)} \delta_{n,q-1},$$

$$\text{hence } c_{n+1} = \frac{2^{2n+3}}{2^{2n+2} n! (n+1)!} \frac{2^{2n+2} (n+1)! (n+1)!}{(2n+1)(2n+3)} = \frac{n+1}{2n+1} \text{ and } c_{n-1} = \frac{2^{2n-1}}{2^{2n} n! (n-1)!} \frac{2^{2n} n! n!}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

Generating function. The Legendre polynomials can be repackaged together in generating function which converges for $x, t < 1$:

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n \geq 0} t^n P_n(x).$$

Exercise 19.11. *Show that the Taylor series in t of $(1-2xt+t^2)^{-1/2}$ is $\sum_{n \geq 0} t^n Q_n(x)$ for some polynomials $Q_n(x)$.*

Prove that $Q_n(x)$ respect the Legendre recursion. Deriving with respect to t yields $\frac{x-t}{\sqrt{1-2xt+t^2}} = (1-2xt+t^2) \sum_{n \geq 0} n t^{n-1} Q_n$, and the recursion follows by rearranging the terms in powers of t .

Sturm Liouville problem. The Legendre polynomials are a complete set of solutions of the Legendre equation, which is the Sturm-Liouville problem:

$$[(1-x^2)P'(x)]' + \lambda P(x) = 0.$$

¹⁴Prove that $\int_{-1}^1 dx (1-x^2)^n = \frac{\sqrt{\pi} n!}{\Gamma(n+\frac{3}{2})} = \frac{2^{2n+1} n! n!}{(2n+1)!}$.

¹⁵We have $\frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2-1)^n|_{x=1} = \frac{1}{2^n n!} [n!(x+1)^n]_{x=1}$.

on the interval $[-1, 1]$ with boundary conditions that P is regular (not divergent) at the end points. While this is not a regular Sturm Liouville problem, as $p(\pm 1) = 0$, requiring that P is not divergent at the end points is enough to fix the eigenvalues to $\lambda_n = n(n+1)$.

A direct proof of the fact that $n(n+1)$ are eigenvalues of the Legendre equation with eigenfunctions the Legendre polynomials goes as follows. On the one hand we have:

$$\sum_{n \geq 0} t^n [(1-x^2)P'_n(x)]' = \left[\frac{(1-x^2)t}{(1-2xt+t^2)^{3/2}} \right]' = \frac{-2xt}{(1-2xt+t^2)^{3/2}} + \frac{3(1-x^2)t^2}{(1-2xt+t^2)^{5/2}},$$

and on the other:¹⁶

$$\begin{aligned} \sum_{n \geq 0} n(n+1)t^n P_n(x) &= t \frac{d^2}{dt^2} \left(\frac{t}{(1-2xt+t^2)^{1/2}} \right) = t \frac{d}{dt} \left(\frac{1}{(1-2xt+t^2)^{1/2}} + \frac{t(-\frac{1}{2})(-2x+2t)}{(1-2xt+t^2)^{3/2}} \right) \\ &= \frac{t(x-t)}{(1-2xt+t^2)^{3/2}} + t \frac{x-2t}{(1-2xt+t^2)^{3/2}} + t^2(x-t) \frac{(-\frac{3}{2})(-2x+2t)}{(1-2xt+t^2)^{5/2}}, \end{aligned}$$

therefore collecting all the terms we get $\sum_{n \geq 0} t^n \left[[(1-x^2)P'_n(x)]' + n(n+1)P_n(x) \right] = 0$, and as t^n are linearly independent we conclude.

The associated Legendre polynomials. Closely associated with the Legendre polynomials we have the associated Legendre polynomials (which are sometimes not polynomials):

$$P_l^m(x) = (-1)^m (1-x^2)^{\frac{m}{2}} \frac{d^m}{dx^m} P_l(x) = \frac{(-1)^{m+l}}{2^l l!} (1-x^2)^{\frac{m}{2}} \frac{d^{l+m}}{dx^{l+m}} (1-x^2)^l,$$

where the second definition holds in the whole range $-l \leq m \leq l$ and $P_l^{-m} = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m$. These functions obey the orthogonality relations:

$$\int_{-1}^1 dx P_l^m(x) P_k^m(x) = \frac{2(l+m)!}{(2l+1)(l-m)!} \delta_{kl}, \quad \int_{-1}^1 dx \frac{P_l^m(x) P_l^n(x)}{1-x^2} = \frac{(l+m)!}{m(l-m)!} \delta_{mn},$$

where the last expression is ∞ for $m = 0$.

Derivating the Legendre equation m times¹⁷ it follows that the associated Legendre polynomials are the fundamental solutions of the general Legendre equation:

$$[(1-x^2)P']' + \lambda P - \frac{m^2}{1-x^2} P = 0,$$

with $m \in \mathbb{N}$. This is again not a regular Sturm Liouville problem, as $p(\pm 1) = 0$, but demanding regularity of the solutions fixes the eigenvalues to $\lambda = l(l+1)$ with $l \in \mathbb{N}$, $l \geq |m|$:

$$[(1-x^2)(P_l^m)']' + l(l+1)P_l^m - \frac{m^2}{1-x^2} P_l^m = 0,$$

in particular the set $\{P_l^m\}_{l \geq |m|}$ at fixed m is a basis in $L^2([-1, 1], dx)$.

¹⁶Observe that $-2xt(1-2xt+t^2) + 3(1-x^2)t^2 + t(2x-3t)(1-2xt+t^2) + 3t^2(x-t)^2 = 0$.

¹⁷One gets $(1-x^2)P^{(m+2)} - 2xmP^{(m+1)} - 2\frac{m(m-1)}{2}P^{(m)} - 2xP^{(m+1)} - 2mP^{(m)} + \lambda P^{(m)} = 0$. Then using $[(1-x^2)^{m/2}P^{(m)}]' = -mx(1-x^2)^{m/2-1}P^{(m)} + (1-x^2)^{m/2}P^{(m+1)}$ and $[(1-x^2)^{m/2}P^{(m)}]'' = m(m-2)x^2(1-x^2)^{m/2-2}P^{(m)} - 2mx(1-x^2)^{m/2-1}P^{(m+1)} + (1-x^2)^{m/2}P^{(m+2)}$ we conclude.

Exercise 19.12. *Prove the stated properties of the associated Legendre polynomials.*

19.3.4 The multipole expansion

We denote in bold vectors in $\mathbb{R}^3$ and without bold their modulus $R = \sqrt{\mathbf{R} \cdot \mathbf{R}}$. We have seen that in dimension 3 the solution of the Poisson equation in $\mathbb{R}^3$ writes using fundamental solutions as:

$$-\Delta\varphi = \rho \Rightarrow \varphi(\mathbf{R}) = \frac{1}{4\pi} \int d^3\mathbf{r} \frac{\rho(\mathbf{r})}{|\mathbf{R} - \mathbf{r}|},$$

where $d^3\mathbf{r}$ is the volume element in dimension 3. Using the generating function of Legendre polynomials, we have:

$$\frac{1}{|\mathbf{R} - \mathbf{r}|} = \frac{1}{\sqrt{R^2 + r^2 - 2Rr \cos \theta}} = \sum_{n \geq 0} \frac{r^n}{R^{n+1}} P_n(\cos \theta),$$

therefore:

$$\varphi(\mathbf{R}) = \frac{1}{4\pi} \sum_{n \geq 0} \frac{1}{R^{n+1}} \int d^3\mathbf{r} r^n P_n(\cos \theta) \rho(\mathbf{r}).$$

This is the multipole expansion. As $P_0(\cos \theta) = 1$, if the total charge on the system is 0 then the first terms in this expansion drops and dipole term dominates:

$$\varphi(\mathbf{R}) \sim \frac{1}{4\pi} \frac{1}{R^2} \int_0^\infty r^2 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi r P_1(\cos \theta) \rho(\mathbf{r}),$$

that is the potential drops like R^{-2} . If the dipole moment $\int d^3\mathbf{r} r P_1(\cos \theta) \rho(\mathbf{r})$ is zero, then the series starts at the quadrupole term $n = 2$, if this one is also zero then it starts at the octopole $n = 3$ and so on. An example of configurations of charges in which the series starts exactly at the term $n = 1$ consists two equal point charges $+$ and $-$; for $n = 2$ one takes 4 charges alternatig in the corner of a square; for $n = 3$ one takes 8 charges at the corners of a cube and so on.

Exercise 19.13 (Classical orthogonal polynomials). *With minimal regularity assumptions, prove that Sturm Liouville problems $Qf'' + Lf + \lambda f = 0$ for Q and L a second (respectively first) degree polynomial have eigenvalues $\lambda_n = -n(\frac{n-1}{2}Q'' + L')$ and polynomial eigenfucntions $f_n = W^{-1}[WQ^n]^{(n)}$, which are orthogonal with the weight $W(x) = e^{\int^x ds L(s)/Q(s)}/Q(x)$.*

19.4 Power series solutions

Given a generic homogeneous ODE (and sometimes even for non linear differential equations):

$$Df = \sum_{k=0}^n a_k(x) \frac{d^k f}{dx^k}(x) = 0,$$

one can attempt to solve for f as a power series in x , that is one makes the ansatz $f(x) = \sum_n f_n x^n$, and translates the differential equation in a recursion relation for the coefficients f_n . For instance for the Legendre equation we get:

$$(1-x^2)f'' - 2xf' + \lambda f = 0 \Rightarrow \sum_{n \geq 0} x^n \left[(n+2)(n+1)f_{n+2} - n(n-1)f_n - 2nf_n + \lambda f_n \right] = 0,$$

and f_0 and f_1 are free (they are fixed by initial conditions), hence:

$$f_{n+2} = \frac{n(n+1) - \lambda}{(n+1)(n+2)} f_n,$$

which in particular implies $f_{-1} = f_{-2} = 0$. The general solution of the Legendre equation for arbitrary λ is then:

$$c_1 y_1(x) + c_2 y_2(x), \quad y_1 = x + \sum_{n \geq 1} \left[\prod_{i=0}^{n-1} \frac{(2i+1)(2i+2) - \lambda}{(2i+2)(2i+3)} \right] x^{2n+1}, \quad y_2(x) = 1 + \sum_{n \geq 1} \left[\prod_{i=0}^{n-1} \frac{(2i)(2i+1) - \lambda}{(2i+1)(2i+2)} \right] x^{2n}.$$

Although we have solved the Legendre differential equation for arbitrary λ , there is an issue: the two series have radius of convergence 1 and diverge in $x \rightarrow \pm 1$ like $\ln(1-x^2)$, hence in general y_1 and y_2 are not in L^2 . The only way to solve the equation in L^2 is for the series to truncate, that is for λ to take one of the possible values $l(l+1)$ for $l \geq 0$ (and for l even, $c_1 = 0$ while for l odd $c_2 = 0$). We thus recover directly that the eigenvalues of the Sturm Liouville Legendre problem are $l(l+1)$ and that the associated functions are polynomials. Fixing one eigenvalue one can directly check that the explicit polynomials above can be rewritten as a Rodrigues formula.

19.4.1 The hypergeometric series

An important linear ODE is the hypergeometric equation (here $z \in \mathbb{C}$):

$$z(1-z)f''(z) + [c - (a+b+1)z]f'(z) - abf(z) = 0.$$

This equation has three singular points $z = 0, 1, \infty$ at which the coefficients have singularities. For z a regular point, we propose a power series ansatz $f = \sum_{n \geq 0} F_n z^n$ and we obtain:

$$(n+1)n F_{n+1} - n(n-1)F_n + c(n+1)F_{n+1} - (a+b+1)n F_n - ab F_n = 0 \Rightarrow F_{n+1} = \frac{(n+a)(n+b)}{(n+1)(n+c)} F_n,$$

that is for $c \neq -n$ the equation is solved by the ${}_2F_1$ *hypergeometric series*:

$$F(a, b | c; z) = {}_2F_1\left(\begin{smallmatrix} a, b \\ c \end{smallmatrix}; z\right) = \sum_{n \geq 0} \frac{\Gamma(a+n)}{\Gamma(a)} \frac{\Gamma(b+n)}{\Gamma(b)} \frac{\Gamma(c)}{\Gamma(c+n)} \frac{z^n}{n!},$$

which has radius of convergence 1. Note that the recursion implies $F_{-1} = 0$. The hypergeometric function can be continued on simply connected domains that avoid its branch point in 1 with cut $[1, \infty)$. The general solution of the hypergeometric equation is then:¹⁸

$$f = c_1 F(a, b | c; z) + c_2 z^{1-c} F(a-c+1, b-c+1 | 2-c; z).$$

An important degenerate case of the hypergeometric function is the confluent hypergeometric function which solves the Kummer equation:

$$zf''(z) + (b-z)f'(z) - af(z) = 0.$$

Taking again a power series ansatz $f = \sum_n F_n z^n$ we get:

$$(n+1)n F_{n+1} + b(n+1) F_{n+1} - n F_n - a F_n = 0 \Rightarrow F_{n+1} = \frac{n+a}{(n+1)(n+b)} F_n,$$

¹⁸We note that:

$$\begin{aligned} & z^{-1+c} \left\{ z(1-z)[z^{1-c}f]'' + [c - (a+b+1)z][z^{1-c}f]' - ab z^{1-c}f \right\} \\ &= z(1-z)[-c(1-c)z^{-2}f + 2(1-c)z^{-1}f' + f''] + [c - (a+b+1)z][(1-c)z^{-1}f + f'] - ab f \\ &= z(1-z)f'' + [2(1-z)(1-c) + c - (a+b+1)z]f' - \left[\frac{c(1-c)(1-z)}{z} - \frac{c(1-c)}{z} + (a+b+1)(1-c) + ab \right] f \\ &= z(1-z)f'' - [2-c - (2-2c+a+b+1)z]f' - [-c(1-c) + (a+b+1)(1-c) + ab]f. \end{aligned}$$

which implies $F_{-1} = 0$ and, as long as $b \neq -n$ the equation admits as power series solution the ${}_1F_1$ hypergeometric series:

$$M(a, b; z) = {}_1F_1\left(\begin{matrix} a \\ b \end{matrix}; z\right) = \sum_{n \geq 0} \frac{\Gamma(a+n)}{\Gamma(a)} \frac{\Gamma(b)}{\Gamma(b+n)} \frac{z^n}{n!},$$

and the general solution writes:

$$f(z) = c_1 M(a, b; z) + c_2 z^{1-b} M(a+1-b, 2-b; z).$$

Note that M is an entire function and if a is a non positive integer, $a \in \mathbb{Z}_{\leq 0}$, then $M(a, b; z)$ truncates to a polynomial.¹⁹

19.4.2 General algorithm

For an equation like:

$$a_k(z)f^{(k)}(z) + a_{k-1}(z)f^{(k-1)}(z) + \cdots + a_1(z)f'(z) + a_0(z)f(z) = 0,$$

with $a_q(z)$ polynomials propose the ansatz $f = \sum_{n \geq 0} F_n(z)z^{n+r}$ and obtain a recursion for F_n . The recursion can be solved forwards and backwards. Requiring that the series stops at $n = 0$ gives an equation $F_{-1} = 0$ which fixes r , and the forward solution of the recursion gives the formula for F_n ,

19.5 Partial differential equations

A partial differential equation is an equation involving the partial derivatives of a function. An existence theorem for solutions of PDEs with boundary conditions is the Cauchy–Kovalevskaya theorem which states that if the coefficients of a linear PDE are analytic functions then the PDE admits a solution which is analytic.

Theorem 19.3 (Cauchy–Kovalevskaya). *Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$, $V = \mathbb{K}^m$ and $W = \mathbb{K}^n$. Let $A_1, \dots, A_{n-1}$ be analytic functions in some neighborhood of $(0, 0) \subset W \times V$ (i.e. defined by power series convergent in this neighborhood) taking values in the $m \times m$ matrices and let b be an analytic function defined on the same neighborhood of $(0, 0)$ with values in V . Then there exists a neighborhood of 0 in W in which the Cauchy problem:*

$$\frac{\partial f}{\partial x_n}(x) = A_1(x, f) \frac{\partial f}{\partial x_1}(x) + \cdots + A_{n-1}(x, f) \frac{\partial f}{\partial x_{n-1}}(x) + b(x, f), \quad f(x)|_{x_n=0} = 0,$$

has a unique analytic solution $f : W \rightarrow V$ near 0. If F and h_j are analytic near 0 then the Cauchy problem:

$$\partial_t^k f = F(x, t, \partial_t^j \partial_x^\alpha f) \text{ for } j < k, \quad |\alpha| + j \leq k, \quad \text{with initial conditions} \quad \partial_t^j f(x, 0) = h_j(x), \quad 0 \leq j < k,$$

has a unique solution in some neighborhood of 0.

Proof. We only sketch here the proof. Expanding both sides of the PDE in Taylor series around $(0, 0)$, for instance:

$$f(x) = \sum_{p_1, \dots, p_n \geq 0} \frac{1}{p_1! \cdots p_n!} x_1^{p_1} \cdots x_n^{p_n} \partial_{x_1}^{p_1} \cdots \partial_{x_n}^{p_n} f \Big|_{x=0},$$

we obtain recurrence relations for the coefficients $\partial_{x_1}^{p_1} \cdots \partial_{x_n}^{p_n} f \Big|_{x=0}$ in terms of the Taylor coefficients of the A s and the b . Using the analyticity of the A s and the b , one can show that this formal power series converges. $\square$

¹⁹The generalized Laguerre polynomial $L_n^{(\alpha)}(z) = \binom{n+\alpha}{n} M(-n, \alpha+1, z)$.

19.5.1 The Poisson equation

Partial differential equations can be solved in terms of Green's functions and boundary conditions. We consider the problem of determining the electric potential φ generated by a charge density $-\rho$ inside a domain $D \subset \mathbb{R}^d$ with boundary conditions at $\Sigma = \partial D$ fixed.

We fix some notation. We denote $x^1, \dots, x^d$ coordinates on $\mathbb{R}^d$ and we denote a metric on $\mathbb{R}^d$ by g_{ij} . We assume that the surface $\Sigma = \partial D$ is parametrized by $d-1$ coordinates s , that is $\Sigma = \phi(S)$ where ϕ is the embedding map $(s^1, \dots, s^{d-1}) \mapsto (x^1, \dots, x^d)$. The exterior normal to $\partial\Sigma$ is a vector such that (repeated indices are summed) $\phi^*(\nu_i) = 0$, that is $\nu^i g_{ij} \frac{\partial x^j}{\partial s^\mu} = 0$ and $g_{ij} \nu^i \nu^j = 1$. The volume and area forms on D and Σ are:

$$dV = \omega = \sqrt{\det g} \, dx^1 \wedge \dots \wedge dx^d, \quad d\sigma = \phi^*(i_\nu \omega) = \sqrt{\det g} \sum_i (-1)^{i+1} \nu^i \frac{\partial(x^1 \dots \widehat{x^i} \dots x^d)}{\partial(s^1, \dots, s^{d-1})} ds^1 \wedge \dots \wedge ds^{d-1},$$

and recalling that $\phi^*(i_Y \omega) = g(Y, \nu) \phi^*(i_\nu \omega)$ and $d(i_Y \omega) = \operatorname{div} Y \, \omega$ the Stokes' theorem writes:

$$\int_D d(i_Y \omega) = \int_{\phi(S)} i_Y \omega = \int_S \phi^*(i_Y \omega) \Rightarrow \int_D dV \operatorname{div} Y = \int_S d\sigma \, g(\nu, Y) \Rightarrow \int_D dV \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} Y^i) = \int_S d\sigma \, g_{ij} \nu^i Y^j,$$

and we denote $\partial_\nu f = \nu(f) = \nu^i \partial_i f = g(\nu, \nabla f)$ the Gateaux derivative of f along ν . The Laplacian in arbitrary coordinates is the divergence of the gradient:

$$\Delta \varphi = \operatorname{div} \nabla \varphi = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} \, g^{ij} \partial_j \varphi).$$

Exercise 19.14. Under a changing of variables $x \rightarrow x(y)$ the Laplacian with standard metric δ_{ij} on $\mathbb{R}^d$ changes like:

$$\Delta = \delta^{ij} \partial_{x^i} \partial_{x^j} = \delta^{ij} \frac{\partial y^k}{\partial x^i} \frac{\partial}{\partial y^k} \left(\frac{\partial y^l}{\partial x^j} \frac{\partial}{\partial y^l} \right) = \delta^{ij} \frac{\partial y^k}{\partial x^i} \frac{\partial y^l}{\partial x^j} \frac{\partial}{\partial y^k \partial y^l} + \delta^{ij} \frac{\partial^2 y^k}{\partial x^i \partial x^j} \frac{\partial}{\partial y^k},$$

and the metric like $g_{kl} = \delta_{ij} \frac{\partial x^i}{\partial y^k} \frac{\partial x^j}{\partial y^l}$. Prove that this reproduces the above formula.

Applying the Stokes theorem for the vector $Y = \phi \nabla \psi$ and taking into account that $g_{ij} \nu^i Y^j = \phi \nu^i \partial_i \psi = \phi \partial_\nu \psi$, we get:

$$\int_S d\sigma \, \varphi \partial_\nu \psi = \int_D dV \frac{1}{\sqrt{\det g}} \partial_i [\sqrt{\det g} \, g^{ij} (\phi \partial_j \psi)] = \int_D dV \, \varphi \Delta \psi + \int_D dV \, g^{ij} \partial_i \varphi \partial_j \psi,$$

therefore:

$$\int_S d\sigma (\varphi \partial_\nu \psi - \psi \partial_\nu \varphi) = \int_D dV (\varphi \Delta \psi - \psi \Delta \varphi),$$

and we conclude that the operator Δ can be extended to a self-adjoint operator by restricting its domain to functions respecting either Dirichlet $\varphi|_\Sigma = 0$, or Neumann $\partial_\nu \varphi|_\Sigma = 0$, boundary conditions. It can be shown that the resolvent of the Dirichlet (or Neumann) Laplacian is compact, hence the eigenvalues of these operators are real, positive, discrete and such that $\lambda_n \rightarrow \infty$. This parallels Sturm Liouville problems in higher dimensions.

Note that if we substitute $\psi(x) = G(x, y)$ with $G(x, y) = G(x - y)$ the fundamental solution of the Laplacian $-\Delta G = \frac{1}{\sqrt{\det g}} \delta^{20}$ then we obtain the following integral representation valid for arbitrary φ :

$$\begin{cases} \forall y \in D & \varphi(y) = -\int_D dV_x \, G(x, y) \Delta \varphi(x) - \int_S d\sigma_x [\varphi(x) \partial_{\nu_x} G(x, y) - G(x, y) \partial_\nu \varphi(x)] \\ \forall y \notin D & 0 = -\int_D dV_x \, G(x, y) \Delta \varphi(x) - \int_S d\sigma_x [\varphi(x) \partial_{\nu_x} G(x, y) - G(x, y) \partial_\nu \varphi(x)] \end{cases}.$$

²⁰With the standard metric $g_{ij} = \delta_{ij}$ on $\mathbb{R}^d$ we have $G(x, y) = \frac{\Gamma(d/2+1)}{d(d-2)\pi^{d/2}} \frac{1}{|x-y|^{d-2}}$ for $d \neq 2$ and $G(x, y) = -\frac{1}{2\pi} \ln |x-y|$ for $d = 2$.

Our aim is to solve the inhomogenous Poisson equation $-\Delta\varphi = \rho$ inside the domain D . In order for the solution to be unique, one needs to specify additional boundary conditions:

- Dirichlet boundary conditions $\varphi|_{\Sigma} = f$ form some given $f : \Sigma \rightarrow \mathbb{R}$.
- Neumann boundary conditions $\partial_{\nu}\varphi|_{\Sigma} = d\varphi(\nu)|_{\Sigma} = h$ form some given $h : \Sigma \rightarrow \mathbb{R}$.

One even also consider mixed boundary conditions $\alpha\varphi + \beta\partial_{\nu}\varphi = g$ for some constants α and β .

Observe that imposing either Dirichlet or Neumann conditions will completely determine the solution of the inhomogeneous Poisson equation (trying to impose both at the same time is not possible in general, that is for any solution of the Poisson equation φ and $\nabla\varphi$ are not independent). Indeed, if $\Delta\varphi_1 = \Delta\varphi_2$ and φ_1, φ_2 respect the same boundary condition, then $\varphi = \varphi_1 - \varphi_2$ is such that:

$$\int_S d\sigma \varphi \partial_{\nu}\varphi = \int_D dV \varphi \Delta\varphi + \int_D dV \partial_i \varphi \partial^i \varphi \Rightarrow \int_D dV |\nabla\varphi|^2 = 0 ,$$

which implies $\nabla\varphi = 0$ in D , hence $\varphi = 0$ for Dirichlet boundary conditions, or a constant for Neumann.

Let us focus on the Dirichlet problem. We solve for the Dirichlet eigenfunctions of the Laplacian:

$$\Delta u(x) + \lambda u(x) = 0 , \quad u|_{x \in \Sigma} = 0 .$$

As already mentioned, on general grounds one can prove that the Dirichlet eigenvalues are $0 < \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_n \rightarrow \infty$, have finite multiplicity, and that the corresponding eigenfunctions form a basis $\sum_{n \geq 0} u_n(x)u_n(y) = \frac{1}{\sqrt{\det g}}\delta(x-y)$. We define the Dirichlet Green function:

$$G_D(x, y) = \sum_{n \geq 0} \frac{1}{\lambda_n} u_n(x)u_n(y) , \quad -\Delta_x G_D(x, y) = \frac{1}{\sqrt{\det g}}\delta(x-y) , \quad G_D(x, y)|_{x \in \Sigma} = 0 ,$$

which respects the Dirichlet boundary conditions $G_D(x, y)|_{x \in \Sigma} = 0$ and substituting $\psi(x) = G_D(x, y)$ the inhomogeneous Dirichlet problem $\Delta\varphi|_D = -\rho$, $\varphi|_{\Sigma} = f$ is solved by:

$$\forall y \in D , \quad \varphi(y) = \int_D dV_x G_D(x, y)\rho(x) - \int_S d\sigma_x f(x)\partial_{\nu_x} G_D(x, y) .$$

The Neumann problem is solved similarly.

19.5.2 The spherical harmonics

We aim to identify *harmonic functions*, that is the solutions of the Laplace equation $\Delta f = 0$ in $d = 3$ dimensions. In spherical coordinates in $\mathbb{R}^3$, $x^1 = r \sin \theta \cos \varphi$, $x^2 = r \sin \theta \sin \varphi$, $x^3 = r \cos \theta$ with $0 \leq r < \infty$, $0 \leq \theta \leq \pi$, $0 \leq \varphi \leq 2\pi$ the standard metric writes $ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\varphi^2$ and the Laplacian is:

$$\Delta = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} g^{ij} \partial_j) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\mathbf{L}^2}{r^2} , \quad -\mathbf{L}^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} ,$$

where $\mathbf{L}^2$ is the *orbital* angular momentum operator in quantum mechanics.

Exercise 19.15. *Derive the expressions of the orbital angular momentum operator in spherical coordinates starting from $L_i = \epsilon_{ijk} x^j (-i\partial_k)$. We have $L_z = -i\frac{\partial}{\partial \varphi}$ and $L_{\pm} = L_x \pm iL_y = e^{\pm i\varphi} \left(\pm \frac{\partial}{\partial \theta} + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \right)$.*

As the angular and radial part of the Laplacian do not mix, we will search for solutions $f(r, \theta, \varphi)$ that *separate variables* in spherical coordinates, that is it write as a product $f(r, \theta, \varphi) = R(r)Y(\theta, \varphi)$. As $f(r, \theta, \varphi)$ is single valued we have $Y(\theta, \varphi) = Y(\theta, \varphi + 2\pi)$ and rearranging the terms, the Laplace equation becomes:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{d}{dr} R \right) = \frac{1}{Y} \mathbf{L}^2 Y = \lambda.$$

As this is valid for all r, θ, φ and the left hand side is a function of r only while the right hand side a function of θ, φ only it follows that λ must be a constant.

For any²¹ λ the radial equation is solved by $R(r) = r^a$ with $a(a+1) = \lambda$, that is the general solution is:

$$R(r) = Ar^{\frac{-1+\sqrt{1+4\lambda}}{2}} + Br^{\frac{-1-\sqrt{1+4\lambda}}{2}}.$$

It remains to solve:

$$\mathbf{L}^2 Y = \lambda Y, \quad |Y(\theta, \varphi)| < \infty, \quad Y(\theta, \varphi) = Y(\theta, \varphi + 2\pi).$$

The operator $\mathbf{L}^2$ with these boundary conditions is self adjoint on $L^2([0, \pi] \times [0, 2\pi])$ and this is a well posed Sturm Liouville problem. Assuming again separation of variables $Y(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$ we get:

$$\lambda \sin^2 \theta + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -\frac{1}{\Phi} \frac{d^2 \Phi}{d\varphi^2} = \mu^2, \quad |\Theta(\theta)| < \infty, \quad \Phi(\varphi) = \Phi(\varphi + 2\pi).$$

The Φ equation is solved by $\Phi(\varphi) = e^{i\mu\varphi}$ and the periodicity imposes $\mu \in \mathbb{Z}$. We denote $\mu = m$. Denoting $P(\cos \theta) = \Theta(\theta)$ the resulting equation for P is:

$$[(1-x^2)P']' + \lambda P - \frac{m^2}{1-x^2} P = 0, \quad |P(x)|_{x \in [-1, 1]} < \infty,$$

that is the general Legendre equation which has eigenvalues $\lambda = l(l+1)$, $l \geq |m|$ and eigenfunctions the associated Legendre polynomial $P_l^m(x)$. It follows that the eigenvalues and eigenfuctions of the orbital momentum, know as the spherical harmonics, are:

$$\mathbf{L}^2 Y_l^m = l(l+1)Y_l^m, \quad Y_l^m(\theta, \varphi) = N_l^m P_l^m(\cos \theta) e^{im\varphi}, \quad l \geq 0, \quad m = -l, \dots, l.$$

The spherical harmonics are orthogonal and the normalization N_l^m is chosen such that they form an orthonormal set:

$$\int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta \overline{Y_l^m(\theta, \varphi)} Y_{l'}^{m'}(\theta, \varphi) = \delta_{ll'} \delta^{mm'},$$

hence a Hilbert basis in $L^2([0, \pi] \times [0, 2\pi], d\Omega = \sin \theta d\theta d\varphi)$.

Combining the spherical harmonics with the radial solution of the Laplace equation $R(r) = Ar^l + Br^{-l-1}$ we obtain the regular and irregular solid harmonics $r^l Y_l^m$ and $r^{-l-1} Y_l^m$ and harmonic functions that are regular at the origin expand on the regular solid harmonics:

$$f(r, \theta, \varphi) = \sum_{l \geq 0} \sum_{m=-l}^l f_l^m r^l Y_l^m(\theta, \varphi).$$

This expansion is convergent for $r \sim 0$. For $r \rightarrow \infty$ one uses the irregular harmonics and expands f in a Laurent series. In Cartesian coordinates ($x = r \sin \theta \cos \varphi$, $y = r \sin \theta \sin \varphi$, $z = r \cos \theta$) the regular solid harmonics $R_l^m = r^l Y_l^m$ are homogeneous polynomials:

$$R_l^{\pm m} = \sum_{k \geq 0} \frac{(-1)^k z^{2k+\eta}}{(2k+\eta)!} (\partial_x^2 + \partial_y^2)^k \left[(x^2 + y^2)^{\frac{l-m-\eta}{2}} (x \pm iy)^m \right],$$

²¹To be precise for $\lambda = 1/4$ one would use $Ar^{-1/2} + B(\ln r)r^{-1/2}$, but this case does not arise.

with $0 \leq m \leq l$ and $\eta = 1$ if $l - m$ is odd and zero if it is even.

The spherical multipole expansion. We consider two unit vectors with spherical coordinates $x = (\theta, \varphi)$ and $x' = (\theta', \varphi')$. Denoting γ the angle between these two vectors, $\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi')$, we have the so called spherical harmonics addition theorem:

$$P_l(\cos \gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^l \overline{Y_l^m(\theta', \varphi')} Y_l^m(\theta, \varphi).$$

In order to prove this, let us denote somewhat abusively $Y_l^m(\theta, \varphi) = Y_l^m(x)$ and so on. The coordinate system in which x' is aligned to the z axis is obtained by a solid rotation R which is such that $e_z = Rx'$ and we denote $y = Rx$. In this coordinates system we have $x = (\gamma, \beta)$ and $P_l(\cos \gamma) = \sqrt{\frac{4\pi}{2l+1}} Y_l^0(\gamma, \beta)$, that is $P_l(x \cdot x') = \sqrt{\frac{4\pi}{2l+1}} Y_l^0(Rx)$. As the angular momentum is rotation invariant we have²² $\mathbf{L}_x^2 Y_l^0(Rx) = l(l+1) Y_l^0(Rx)$ and it follows that:

$$Y_l^0(Rx) = \sum_{m=-l}^l b_l^m Y_l^m(x), \quad b_l^m = \int d\Omega_u \overline{Y_l^m(u)} Y_l^0(Ru) = \int d\Omega_v \overline{Y_l^m(R^{-1}v)} Y_l^0(v),$$

where we denoted $d\Omega_u = \sin \theta d\theta d\varphi$ the solid angle in the coordinates u and we noticed that the solid angle does not change under solid rotations. On the other hand, being harmonic $\mathbf{L}_y^2 \overline{Y_l^m(R^{-1}y)} = l(l+1) \overline{Y_l^m(R^{-1}y)}$, any spherical harmonic expands as:

$$\overline{Y_l^m(R^{-1}y)} = \sum_{m'=-l}^l B_{mm'} Y_l^{m'}(y), \quad B_{mm'} = \int d\Omega_v \overline{Y_l^m(R^{-1}v)} \overline{Y_l^{m'}(v)} \Rightarrow B_{m0} = b_l^m.$$

Taking into account $Y_l^m(e_z) = \delta_{m0} \sqrt{\frac{2l+1}{4\pi}}$ we conclude that $\overline{Y_l^m(x')} = \overline{Y_l^m(R^{-1}e_z)} = \sqrt{\frac{2l+1}{4\pi}} B_{m0} = \sqrt{\frac{2l+1}{4\pi}} b_l^m$ hence:

$$P_l(x \cdot x') = \sqrt{\frac{4\pi}{2l+1}} Y_l^0(Rx) = \frac{4\pi}{2l+1} \sum_{m=-l}^l \overline{Y_l^m(x')} Y_l^m(x).$$

Using the generating function of the Legendre polynomials, the Green function of the Laplacian in dimension 3 expands for $|y| > |x|$ as:

$$\frac{1}{|y-x|} = \sum_{l \geq 0} \frac{|x|^l}{|y|^{l+1}} P_l(\cos \gamma) = \sum_{l \geq 0} \frac{|x|^l}{|y|^{l+1}} \frac{4\pi}{2l+1} \sum_{m=-l}^l \overline{Y_l^m(\theta', \varphi')} Y_l^m(\theta, \varphi),$$

and the electric potential generated by a charge density $\rho(x)$ localized at $|x| < |y|$ is:

$$\varphi(R, \theta', \varphi') = \frac{1}{4\pi} \int dV_x \frac{\rho(x)}{|y-x|} = \sum_{l \geq 0} \frac{1}{R^{l+1}} \frac{1}{2l+1} \sum_{m=-l}^l \overline{Y_l^m(\theta', \varphi')} \int_0^\infty r^2 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi \rho(r, \theta, \varphi) r^l Y_l^m(\theta, \varphi),$$

which is known as the spherical multipole expansion.

19.5.3 Separation of variables

We have seen that one can search for solutions of the Laplace equation assuming separation of variables. The reader might wonder why does this strategy work. The idea is that, from Fubini's theorem, if a function is integrable with respect to a tensor product measure, then it is integrable iteratively.

To keep things concrete, consider $w(x)dx$ a measure on I_1 and $\rho(y)dy$ a measure on I_2 . Any $F \in L^2(I_1 \times I_2, w(x)\rho(y)dx dy)$ with $\int_{I_1 \times I_2} w(x)\rho(y)dx dy |F(x, y)| < \infty$ is square integrable in y at fixed x ,²³ that is $F(x, \cdot)$ is such

²²This is most easily seen in Cartesian coordinates in which $L_i = -i\epsilon_{ijk} x_j \partial_{x_k}$.

²³To be precise this holds only almost everywhere in x , but we will ignore such subtleties.

that $\forall x, \int_{I_2} \rho(y) dy |F(x, y)|^2 < \infty$, hence $F(x, \cdot) \in L^2(I_2, \rho(y) dy)$. Denoting $f_m(y)$ some basis in $L^2(I_2, \rho(y) dy)$, $F(x, y)$ expands on this basis as:

$$F(x, y) = \sum_k f_m(y) C_m(x), \quad C_m(x) = \int_{I_2} \rho(y') dy' \overline{f_m(y')} F(x, y').$$

From Bessel's inequality $|C_m(x)|^2 \leq \|F(x, \cdot)\|_{L^2}^2 = \int_{I_2} \rho(y) dy |F(x, y)|^2$ hence $\int_{I_1} w(x) dx |C_m(x)|^2 \leq \int_{I_1} w(x) dx \int_{I_2} \rho(y) dy |F(x, y)|^2 = \int_{I_1 \times I_2} w(x) \rho(y) dx dy |F(x, y)|^2 < \infty$ and $C_m \in L^2(I_1, w(x) dx)$. If one identifies at fixed m a basis $e_l^m(x)$ in $L^2(I_1, w(x) dx)$, $C_m(x)$ expands on this basis as:

$$C_m(x) = \sum_l e_l^m(x) C_{ml}, \quad C_{ml} = \int_{I_1} w(x') dx' \overline{e_l^m(x')} C_m(x') \Rightarrow F(x, y) = \sum_m f_m(y) \sum_l e_l^m(x) C_{ml}.$$

If f_m and e_l^m at fixed m are orthonormal, the spanning set $e_l^m(x) f_m(y)$ is orthonormal, hence linearly independent, therefore it is a basis:

$$\int_{I_1} w(x) dx e_l^m(x) e_{l'}^{m'}(x) \int_{I_2} \rho(y) dy f_m(y) f_{m'}(y) = \int_{I_1} w(x) dx e_l^m(x) e_{l'}^{m'}(x) \delta^{mm'} = \delta_{ll'} \delta^{mm'}.$$

The procedure can be iterated for more variables. In the context of PDEs one identifies iterated bases by formulating iterated well posed Sturm Liouville problems for each coordinate.

19.5.4 The Hydrogen atom

The hydrogen atom is described by the Hamiltonian operator $H = -\frac{1}{2}\Delta - \frac{e^2}{r}$. We are interested in finding its eigenvalues, degeneracies and eigenfunction:

$$H\psi = -\frac{1}{2} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{\mathbf{L}^2 \psi}{2r^2} - \frac{e^2}{r} \psi = E\psi, \quad |\psi(r, \theta, \varphi)| < \infty, \quad \psi(r, \theta, \varphi + 2\pi) = \psi(r, \theta, \varphi),$$

where we used spherical coordinates. The problem is solved (again) by separating variables, or equivalently by expanding on regular solid harmonics $\psi = R(r)Y(\theta, \varphi)$:

$$\frac{\mathbf{L}^2 Y}{Y} = 2r^2 \left(E + \frac{e^2}{r} \right) + \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right).$$

The angular part is solved by the spherical harmonics $\mathbf{L}^2 Y_l^m = l(l+1)Y_l^m$ and the radial part writes:

$$\frac{1}{2r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{l(l+1)}{2r^2} R + \frac{e^2}{r} R + ER = 0,$$

and $\int_0 r^2 dr |R(r)|^2 < \infty$ as ψ should be square integrable close to the origin. Denoting $R = r^{-1}P$ the radial equation becomes²⁴:

$$P'' - \frac{l(l+1)}{r^2} P + \frac{2e^2}{r} P + 2EP = 0.$$

²⁴We have $[r^2(r^{-1}P)']' = [-P + rP']' = rP''$

This is a second order ODE which one can solve. Defining $a = (-2E)^{-1/2}$ and substituting $P(r) = f(r/a)e^{-r/a}$, $f(y)$ respects:²⁵

$$f'' - 2f' - \frac{l(l+1)}{y^2}f + \frac{2e^2a}{y}f = 0,$$

and $\int_0 dy |f(y)|^2 < \infty$. Assuming a (Laurent) power series solution $f = \sum_n F_n y^n$, where in principle we allow n to take negative values we obtain the recursion:²⁶

$$F_{n+1}[n(n+1) - l(l+1)] = F_n[2n - 2e^2a].$$

Now, the power series can not have arbitrary negative powers (otherwise f is not square integrable at the origin), it follows that $F_n = 0$ for $n \leq l$ and the first non trivial coefficient²⁷ is F_{l+1} (which we set to one up to a rescaling):

$$\begin{aligned} f(y) &= y^{l+1} \left(1 + \sum_{n \geq 1} y^n \prod_{k=1}^n \frac{2[k+l-e^2a]}{(k+l)(k+l+1)-l(l+1)} \right) = y^{l+1} \left(1 + \sum_{n \geq 1} y^n \prod_{k=1}^n \frac{2[k+l-e^2a]}{k(k+2l+1)} \right) \\ &= y^{l+1} \sum_{n \geq 0} \frac{\Gamma(n+l+1-e^2a)}{\Gamma(l+1-e^2a)} \frac{\Gamma(2l+2)}{\Gamma(n+2l+2)} \frac{2^n y^n}{n!} = y^{l+1} M(l+1-e^2a, 2l+2, 2y), \end{aligned}$$

where we recognize the confluent hypergeometric series ${}_1F_1$. Gathering everything, we find the eigenfunctions:

$$\psi(r, \theta, \varphi) = r^l \exp\left\{-\frac{r}{a}\right\} M(l+1-e^2a, 2l+2, \frac{2r}{a}) Y_l^m(\theta, \varphi), \quad a = (-2E)^{-1/2}.$$

As it will be relevant below we note that the large order behavior of the hypergeometric series at non exceptional arguments yields the large argument asymptotic behavior:

$$\frac{\Gamma(a+n)}{\Gamma(a)} \frac{\Gamma(b)}{\Gamma(b+n)} \frac{z^n}{n!} \sim n^{a-b} \frac{z^n}{n!} \sim z^{a-b} \frac{z^{n-(a-b)}}{[n-(a-b)]!} \Rightarrow M(a, b, z) \sim_{|z| \rightarrow \infty} z^{a-b} e^z.$$

As the Hamiltonian is self adjoint, E is real.²⁸ The analysis proceeds differently according to the sign of E :

- **For $E < 0$.** In this case $a = (2|E|)^{-1/2}$ is a positive real number and the large argument behavior of the wave function is:

$$\psi \sim \exp\left\{-\frac{r}{a} + \frac{2r}{a}\right\}.$$

This diverges badly at infinity: such a ψ is not a tempered distribution. One obtains an L^2 function only if the hypergeometric series truncates to a polynomial, that is if:

$$e^2a = n \Rightarrow |E| = \frac{e^4}{2} \frac{1}{n^2}, \quad n = 1, 2, \dots; \quad l = 0, \dots, n-1; \quad m = -l \dots l,$$

where we used $F_{n+1}[n(n+1) - l(l+1)] = F_n[2n - 2e^2a]$ and we remark that in order for the solution to exist we need that the series truncates at an order higher than the order at which it starts, $n \geq l+1$. We thus obtain the

²⁵We have $P' = a^{-1}[f'(r/a) - f(r/a)]e^{-r/a}$ and $P'' = a^{-2}[f''(r/a) - 2f'(r/a) + f(r/a)]e^{-r/a}$ hence denoting $y = r/a$ we get:

$$P'' - \frac{l(l+1)}{r^2}P + \frac{2e^2}{r}P - a^{-2}P = 0 \Rightarrow f'' - 2f' + f - \frac{l(l+1)}{y^2}f + \frac{2e^2a}{y}f - f = 0.$$

²⁶The ODE rewrites as $\sum y^{n-1}[(n+1)nF_{n+1} - 2nF_n - l(l+1)F_{n+1} + 2e^2aF_n]$ and y^n are linearly independent.

²⁷The series can only truncate at $n = l$ or $n = -l-1$, but the second choice leads to a function that is not square integrable at the origin.

²⁸The reader should identify the domain of the Hamiltonian.

corresponding eigenfunctions:

$$\psi_{n,l,m}(r, \theta, \varphi) = r^l \exp \left\{ -\frac{r}{a} \right\} M(-(n-l-1), 2l+2, \frac{2r}{a}) Y_l^m(\theta, \varphi), \quad a = (-2E)^{-1/2}.$$

Proceeding more carefully, one can formulate a Sturm Liouville problem for the radial equation allowing us to conclude that $\psi_{n,l,m}(r, \theta, \varphi)$ form a basis in $L^2([0, \infty] \times [0, \pi] \times [0, 2\pi], r^2 dr \sin \theta d\theta d\varphi)$.

The electrons in an atom have spin, hence up to two electrons can occupy the same state. The orbital quantum number $l = 0, 1, 2, 3$ identifies the orbitals s, p, d, f .

Exercise 19.16. *Neglecting any other effects, discuss the ideal Mendeleev table. In practice the electron-electron interactions in heavy atoms lift the degeneracy of the energy levels and change the spectrum of energies. The levels are occupied in the order:*

/
1s / /
2s 2p / /
3s 3p 3d / /
4s 4p 4d 4f /
5s 5p 5d 5f ...
6s 6p 6d 6f ...
7s 7p

that is $1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p, \dots$. Discuss the physical Mendeleev table. What is the orbital occupation of the element 92 in the table? What is this element?

- **For $E > 0$.** In this case $a = i(2|E|)^{-1/2} = i|a|$, the hypergeometric series does not truncate and we obtain the Coulomb wave function:

$$\psi_{E,l,m}(r, \theta, \varphi) = r^l \exp \left\{ i \frac{r}{|a|} \right\} M(l+1 - ie^2|a|, 2l+2, -i \frac{2r}{|a|}) Y_l^m(\theta, \varphi).$$

This describes an electron that comes from infinity, scatters on the nucleus and departs for infinity, as in the Rutherford experiment. It is a non normalizable state which does not belong to L^2 but is a distribution.²⁹

Note that, with respect to the Coulomb waves one finds in the literature, we considered here the scattering of a negatively charged electron on a positively charged nucleus: this leads to a $-e^2$ instead of an e^2 in this formula.

The orthogonality and completeness write:

$$\begin{aligned} \langle \psi_{n,l,m} | \psi_{n',l',m'} \rangle &= \|\psi_{n,l,m}\|^2 \delta_{nn'} \delta_{ll'} \delta_{mm'}, & \langle \psi_{E,l,m} | \psi_{E',l',m'} \rangle &= \frac{1}{[\rho(E)]^2} \delta(E-E') \delta_{ll'} \delta_{mm'} \\ &\frac{1}{r^2} \delta(r-r') \frac{1}{\sin \theta} \delta(\theta-\theta') \delta(\varphi-\varphi') \\ &= \sum_{n \geq 1, l \leq n-1, |m| \leq l} \frac{1}{\|\psi_{n,l,m}\|} \psi_{n,l,m}(r, \theta, \varphi) \psi_{n,l,m}(r', \theta', \varphi') + \sum_{l \geq 0, |m| \leq l} \int_0^\infty \rho(E) dE \psi_{E,l,m}(r, \theta, \varphi) \psi_{E,l,m}(r', \theta', \varphi'). \end{aligned}$$

²⁹Using the asymptotic behavior of the hypergeometric function we have at large argument:

$$\psi_{E,l,m}(r, \theta, \varphi) \sim r^{-1-ie^2|a|} e^{-i \frac{r}{|a|}} Y_l^m(\theta, \varphi), \quad \int dV \|\psi_{E,l,m}\|^2 \sim \int_0^\infty (r^2 dr) r^{-2}.$$

Appendices

Appendix A

Lie groups and representations, a primer

A group $(G, \cdot, e)$ is a set endowed with an operation $\cdot$ which is associative $(g \cdot h) \cdot f = g \cdot (h \cdot f)$ such that e is the identity $g \cdot e = e \cdot g = g$ and any element has an inverse $g \cdot g^{-1} = g^{-1} \cdot g = e$. Symmetry transformations in physics are elements of groups acting on states. We sometimes denote the identity by 1.

An example of a group is the group S_n of permutations of n elements.

A.1 Classical Lie groups

In many instances one deals with Lie groups, that is groups that are at the same time differentiable manifolds. Many Lie groups are matrix groups and examples include:

- the general linear group $GL(N, \mathbb{R})$ is the group of invertible $N \times N$ matrices with real entries. $SL(N, \mathbb{R})$ is the subgroup of $GL(N, \mathbb{R})$ consisting in matrices with determinant 1. Versions with complex entries can be defined.
- the unitary group $U(N)$ is the group of unitary $N \times N$ matrices $UU^\dagger = U^\dagger U = I$ (hence with complex entries). The special unitary group $SU(N)$ is the subgroup of $U(N)$ of unitary matrices with determinant 1.
- $O(N)$ is the group of orthogonal $N \times N$ matrices with real entries $OO^T = O^T O = 1$. The group $O(N)$ is the disjoint union of two connected components: the orthogonal matrices with determinant +1, which form the special orthogonal group $SO(N)$, respectively those with determinant -1.¹

A one parameter subgroup of G is a map $g : \mathbb{R} \rightarrow G$ respecting $g(t+s) = g(t)g(s)$. The one parameter subgroups are the curves on G seen as a manifold. The tangent space at the identity in a Lie group, denoted $\mathfrak{g}$, consists in all the tangent vectors $\frac{d}{dt}g(t)$ at curves $g(t)$ with $g(0) = e$. The exponential map $e : \mathfrak{g} \rightarrow G$ is defined as $e^V = g(1)$, where $g(t)$ is the unique one parameter subgroup of G whose tangent vector at the identity is V .² Using the group multiplication, any element in the connected component containing e of G will write as the exponential of some element in the tangent space. The tangent space $\mathfrak{g}$ is a Lie algebra, that is a vector space endowed with an alternating bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, $[U, V] \mapsto X$ which satisfies the Jacobi identity:

$$[U, V] = -[V, U] \quad [W, [U, V]] + [U, [V, W]] + [V, [W, U]] = 0.$$

The Lie algebras of matrix groups are matrix algebras, and the Lie bracket is the commutator. In this case the exponential map is trivial $e^X = \sum_{n \geq 0} \frac{1}{n!} X^n$. Close to the origin, any element in a Lie group writes as $g \approx e + X$ with X in

¹Prove the two components are disconnected.

²This is straightforward for matrix groups we are concerned with here. While less obvious, it also holds on general Lie groups.

the Lie algebra. For instance, an infinitesimal orthogonal transformation writes as $O = e^A$ with:

$$I = OO^T \approx (I + A)(I + A^T) \Rightarrow A = -A^T ,$$

that is the Lie algebra $\mathfrak{o}(N)$ is the vector space of antisymmetric $N \times N$ matrices with Lie bracket:

$$[A_1, A_2] = A_1 A_2 - A_2 A_1 , \quad [A_1, A_2]^T = [A_2^T, A_1^T] = -[A_1, A_2] .$$

Note that the exponential map e^A with $A^T = -A$ generates only the connected component containing the identity I in $O(N)$, that is the subgroup $SO(N)$.

A.1.1 The $SU(N)$ Lie algebra

As another example, any element in the group $SU(N)$ writes as e^{iH} with H a Hermitian traceless matrix,³ that is the Lie algebra $\mathfrak{su}(N)$ consists in Hermitian traceless $N \times N$ matrices. The traceless condition comes from requiring $\det U = e^{i \text{Tr} H} = 1$. The space $\mathfrak{su}(N)$ is a real vector space with inner product⁴ $\langle A | B \rangle = \text{Tr}(AB)$ and with real dimension $N^2 - 1$. Let $T^a, a = 1 \dots N^2 - 1$ be an orthogonal but not normal basis in this space:

$$T^a = (T^a)^\dagger , \quad \text{Tr}(T^a) = 0 , \quad \text{Tr}(T^a T^b) = \frac{1}{2} \delta^{ab} = \langle T^a | T^b \rangle ,$$

where the last condition is a choice of normalization. Any H can be expanded on this basis as $H = x_a T^a$ with x_a real. In particular $[T^a, T^b]$ is anti hermitian traceless hence expands as:

$$[T^a, T^b] = i f^{abc} T^c , \quad \frac{i}{2} f^{abc} = \text{Tr}(T^c [T^a, T^b]) ,$$

for some real structure constants f^{abc} . We have

- due to the Jacobi identity $[T^a, [T^b, T^c]] + [T^b, [T^c, T^a]] + [T^c, [T^a, T^b]] = 0$ the structure constants respect $f^{ade} f^{bcd} + f^{bde} f^{cad} + f^{cde} f^{abd} = 0$.
- the structure constants are totally anti symmetric $0 = \text{Tr}([T^c T^a, T^b]) \Rightarrow f^{abc} + f^{cba} = 0$.
- the Casimir operator $C = T^a T^a$ is such that $[C, T^b] = T^a i f^{abc} T^c + i f^{abc} T^c T^a = 0$, that is it is a central element (commutes with everybody) and therefore it is proportional to the identity in any irreducible representation (see below). In particular, in the defining representation as $N \times N$ Hermitian traceless matrices, $(T^a T^a)_{ij} = C \delta_{ij}$ with $C = \frac{N^2 - 1}{2N}$.
- as any Hermitian matrix writes as $M = \frac{1}{N} \text{Tr}(M) \mathbb{I} + 2 \text{Tr}(M T^a) T^a$, writing the right hand side as an operator acting on hermitian $N \times N$ matrices we get:

$$T_{ij}^a T_{kl}^a = \frac{1}{2} \left(\delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl} \right) .$$

³The unitary matrices are normal $UU^\dagger = U^\dagger U$ hence diagonalizable. Their eigenvalues have absolute value 1 hence write as $e^{i\lambda_i}$ for some real λ_i . The Lie bracket in this case is $(-i)[H_1, H_2]$: the awkward $(-i)$ is needed if one insists to work with hermitian generators. Strictly speaking the Lie algebra consists in antihermitian matrices which close under commutator.

⁴Any matrix space can be endowed with the inner product $\langle A | B \rangle = \text{Tr}(A^\dagger B)$.

- the Killing form is defined by $B(X, Y) = \text{tr}([X, [Y, \cdot]]) = \sum_a \frac{1}{\|T^a\|^2} \langle T^a | [X, [Y, T^a]] \rangle$, where tr is the trace in the sense of operators from $\mathfrak{su}(N)$ to $\mathfrak{su}(N)$ is:

$$\begin{aligned} B(T_p, T_q) &= 2 \text{Tr}(T^a [T^p, [T^q, T^a]]) = f^{pam} f^{qam} = 2 \text{Tr}(T^a T^p T^q T^a - T^a T^p T^a T^q - T^a T^q T^a T^p + T^a T^a T^q T^p) \\ &= 4C \text{Tr}(T^p T^q) - 4 \left(T_{li}^p T_{jk}^q \frac{1}{2} \left(\delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl} \right) \right) = N \delta^{pq} \equiv C_2 \delta^{pq}, \end{aligned}$$

is non degenerate.

A.1.2 The adjoint representation

A group G can be mapped into the automorphisms of G (bijections from G to G preserving the group structure) via:

$$g \mapsto \Psi_g, \quad \Psi_g(h) = ghg^{-1},$$

which respects $\Psi_{g_1 g_2} = \Psi_{g_1} \circ \Psi_{g_2}$. We define the *adjoint action* of g as the derivative of Ψ_g at the origin:

$$\text{Ad}_g = (d\Psi_g)_e : T_e G \rightarrow T_e G,$$

where Ad_g is a linear transformation of $\mathfrak{g}$ to itself that preserves the Lie bracket. The map $g \mapsto \text{Ad}_g$ is a mapping of G onto the linear operators on $\mathfrak{g}$ and it is called the adjoint representation of G .

This is particularly transparent for matrix groups. Close to the origin any group element writes as $e + X$ with X in the Lie algebra hence $\Psi_g(h) \approx \Psi_g(e + X)$ and:⁵

$$(d\Psi_g)(X) \approx \Psi_g(e + X) - \Psi_g(e) = gXg^{-1} = \text{Ad}_g(X).$$

The adjoint representation induces by differentiating at the origin a representation of the Lie algebra $\text{ad} : \mathfrak{g} \rightarrow \text{Lie}(\text{Aut})(\mathfrak{g})$, the Lie algebra of the automorphisms of $\mathfrak{g}$ (bijections preserving the Lie bracket):

$$\text{ad}_Y : \mathfrak{g} \rightarrow \mathfrak{g}, \quad \text{ad}_Y(X) \approx (\text{Ad}_{e+Y} - \text{Ad}_e)(X) \approx (e + Y)X(e - Y) - X \approx [Y, X]$$

Note that the matrix elements of the Lie algebra generators in the adjoint representations are the structure constants:

$$[T^a, T^b] = c^{abc} T^c \Rightarrow (\text{ad}_{T^a})^{bc} = c^{acb},$$

where we used non Hermitian generators.

Exercise 1.1. *Prove that*

$$[\text{ad}_X, \text{ad}_Y](Z) = \text{ad}_{[X, Y]}(Z).$$

⁵More rigorously, the Lie algebra provides coordinates X on G with $e = e^X|_{X=0}$. In this coordinate system $\Psi_g^\varphi(X) = \ln(ge^X g^{-1})$ and from the definition of the differential $\Psi_g^\varphi(Y) - \Psi_g^\varphi(0) = (d\Psi_g^\varphi)_e(Y) + O(Y^2)$ thus:

$$\ln(ge^Y g^{-1}) - \ln(ge^0 g^{-1}) = \ln(g(1 + Y)g^{-1}) + O(Y^2) = \ln(1 + gYg^{-1}) + O(Y^2) = gYg^{-1} + O(Y^2).$$

A.2 $SO(3)$ versus $SU(2)$

The group $SO(3)$ is the group of spatial rotations in 3 dimensional Euclidean space. It is the group of 3×3 matrices that preserve the standard metric and the orientation:

$$\delta_{aa'} O_{ab} O_{a'c} = \delta_{bc} \quad \det O = 1$$

In Cartesian coordinates, rotations around the axes are described by (we take positive angles when rotating counterclockwise around the axes):

$$R_1(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix}, \quad R_2(\alpha) = \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix}, \quad R_3(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

A particular basis in $\mathfrak{so}(3)$ is obtained as: $L_i = \frac{dR_i}{d\alpha}(0)$,

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

or in short $(L_i)_{jk} = -\epsilon_{ijk}$ where ϵ_{ijk} is the totally antisymmetric Levi-Civita symbol. Clearly the matrices L_1, L_2, L_3 form a basis in the space of antisymmetric traceless matrices. One can check that:

$$[L_i, L_j] = \epsilon_{ijk} L_k, \quad \text{Tr}(L_i^T L_j) = 2\delta_{ij}.$$

or switch to Hermitian generators $J_i = \imath L_i$ such that $[J_i, J_j] = \imath \epsilon_{ijk} J_k$, $\text{Tr}(J_i J_j) = 2\delta_{ij}$. A rotation of angle ψ around the direction $\vec{n} = (n_1, n_2, n_3)$ is then represented by a matrix:

$$O_n(\psi) = \exp \left\{ -\imath \psi \vec{n} \cdot \vec{J} \right\}, \quad \vec{n} \cdot \vec{J} = \sum_{i=1}^3 n_i J_i.$$

Topologically the group $SO(3)$ can be seen as a solid ball of radius π in $\mathbb{R}^3$ with antipodal points identified. A point at distance ψ in the direction $\vec{n}$ represents the rotation of angle ψ around the direction $\vec{n}$ and the rotations of π around $\vec{n}$ and $-\pi$ around $-\vec{n}$ are identified. $SO(3)$ is connected but not simply connected as there exist non contractible closed paths: go from the center to the north pole, out through the south pole and back to the center. However, going twice along such a path leads to a contractible loop.

The universal covering group of $SO(3)$ is the smallest simply connected group that is generated by the same Lie algebra. This universal covering group is isomorphic with the group $SU(2)$ which we now describe.

Any element in $SU(2)$ can be parametrized as:

$$U = \begin{pmatrix} u_4 + \imath u_3 & u_2 + \imath u_1 \\ -u_2 + \imath u_1 & u_4 - \imath u_3 \end{pmatrix} = u_4 I + \imath(u_1 \sigma_1 + u_2 \sigma_2 + u_3 \sigma_3), \quad u_i \in \mathbb{R}, \quad \sum u_i^2 = 1,$$

where the Pauli matrices are:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\imath \\ \imath & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_i \sigma_j = \delta_{ij} + \imath \epsilon_{ijk} \sigma_k.$$

It follows that $SU(2)$ is in bijection with the sphere $S^3 \subset \mathbb{R}^4$, which is simply connected. We can parametrize any element in $SU(2)$ by a unit three dimensional vector and an angle:

$$n_i = \frac{-u_i}{\sqrt{\sum_{j=1}^3 u_j^2}}, \quad i = 1, 2, 3, \quad \theta \in [0, 2\pi], \quad \cos \frac{\theta}{2} = u_4, \quad \sin \frac{\theta}{2} = \sqrt{\sum_{j=1}^3 u_j^2}, \quad U = \exp \left\{ -i\theta \frac{\vec{n} \cdot \vec{\sigma}}{2} \right\} = \cos \frac{\theta}{2} - i \sin \frac{\theta}{2} \vec{n} \cdot \vec{\sigma}.$$

Exercise 1.2. *Prove the statements above concerning $SU(2)$.*

The generators of $\mathfrak{su}(2)$ can be chosen $T_i = \frac{\sigma_i}{2}$ and:

$$[T_i, T_j] = i\epsilon_{ijk} T_k, \quad \text{Tr}(T_i T_j) = \frac{1}{2} \delta_{ij},$$

that is the Lie algebras of $O(3)$ and $SU(2)$ are identical.

However, although locally they two groups coincide, they differ globally: one is simply connected and one is not. In fact there exists a two to one homomorphism between $SU(2)$ and $O(3)$, that is $SU(2)$ is the double cover of $O(3)$.

Let us denote $X = \vec{x} \cdot \vec{\sigma}$, with $\vec{x} = (x^1, x^2, x^3)$ and note that $\det(X) = -\vec{x}^2$. Pick any $U \in SU(2)$, and denote $X' = U^\dagger X U$. Then X' is hermitian traceless, $(X')^\dagger = X'$ and $\text{Tr}(X') = 0$. Expanding X' back on the basis $\vec{\sigma}$ we get $X' = \vec{x}' \cdot \vec{\sigma}$ and $\det(X') = -(\vec{x}')^2$. It follows that $\vec{x}' = R\vec{x}$ for some rotation $R \in SO(3)$. Clearly the map $f : SU(2) \rightarrow SO(3)$, $f(U) = R$ is two to one: $f(U) = f(-U)$ because $U^\dagger X U = (-U)^\dagger X (-U)$

Exercise 1.3. *Prove that for $U = e^{-i\theta \frac{\vec{n} \cdot \vec{\sigma}}{2}}$:*

$$X' = U^\dagger X U = \vec{x}' \cdot \vec{\sigma} \quad \Rightarrow \quad \vec{x}' = e^{-i\theta \vec{n} \cdot \vec{J}} \vec{x}.$$

Exercise 1.4. *Prove that the defining representation of $SO(3)$ is the adjoint representation of $SU(2)$. Think of the matrix elements of the generators of $SO(3)$ in the defining representation.*

A.3 The Haar measure

Although the Haar measure exists for more general groups, we focus here on matrix Lie groups consisting in $N \times N$ matrices. Let us consider the matrix 1-form dU with base point U , written in some parametrization $U(y)$ with $y \in \mathbb{R}^n$ of the group G . This is a $N \times N$ matrix with entries the 1-forms dU_{ij} :

$$dU = \sum_{i,j=1}^N \sum_{\mu} e_{ij} \frac{\partial U_{ij}}{\partial y^\mu} dy^\mu, \quad (e_{ij})_{kl} = \delta_{ik} \delta_{jl},$$

that is $dU = \sum_{\mu} T_{\mu} dy^{\mu}$ with the matrix generators $T_{\mu} = \frac{\partial U}{\partial y^{\mu}} = \sum_{i,j=1}^N e_{ij} \frac{\partial U_{ij}}{\partial y^{\mu}} = dU(\partial_{y^{\mu}})$. A one parameter subgroup of G is a curve $(U \circ y)(t)$ and its velocity vector $\frac{d(U \circ y)}{dt} = \sum_{\mu} \dot{y}^{\mu} T_{\mu}$ belongs to the tangent space $T_U G$, that is $T_U G$ is spanned by the matrices T_{μ} . We define the metric (symmetric non degenerate bilinear form):

$$\text{Tr} \left[dU^\dagger (U^{-1})^\dagger U^{-1} dU \right] = \sum_{\mu\nu} g_{\mu\nu} dy^\mu dy^\nu, \quad g_{\mu\nu} = \text{Tr} \left(\frac{\partial U^\dagger}{\partial y^\mu} \bigg|_y (U^{-1})^\dagger \bigg|_y U^{-1} \frac{\partial U}{\partial y^\nu} \bigg|_y \right).$$

The associated volume form $\sqrt{\det g} dy^1 \wedge \cdots \wedge dy^n$ is called the Haar measure and it is a left invariant measure on the group: changing variables to $U \rightarrow VU$ with V fixed clearly does not change g .

For instance, the elements of $SU(2)$ can be represented by Euler angles in the $z - y - z$ convention:

$$U(\alpha, \beta, \gamma) = e^{-i\alpha T_3} e^{-i\beta T_2} e^{-i\gamma T_3} = \begin{pmatrix} e^{-i\frac{\alpha}{2}} & 0 \\ 0 & e^{i\frac{\alpha}{2}} \end{pmatrix} \begin{pmatrix} \cos \frac{\beta}{2} & -\sin \frac{\beta}{2} \\ \sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix} \begin{pmatrix} e^{-i\frac{\gamma}{2}} & 0 \\ 0 & e^{i\frac{\gamma}{2}} \end{pmatrix} = \begin{pmatrix} e^{-i\frac{\alpha}{2}} \cos \frac{\beta}{2} e^{-i\frac{\gamma}{2}} & -e^{-i\frac{\alpha}{2}} \sin \frac{\beta}{2} e^{i\frac{\gamma}{2}} \\ e^{i\frac{\alpha}{2}} \sin \frac{\beta}{2} e^{-i\frac{\gamma}{2}} & e^{i\frac{\alpha}{2}} \cos \frac{\beta}{2} e^{i\frac{\gamma}{2}} \end{pmatrix},$$

with $\alpha, \gamma \in [0, 2\pi]$ and $\beta \in [0, \pi]$. Using:

$$dU = \begin{pmatrix} e^{-i\frac{\alpha}{2}} \cos \frac{\beta}{2} e^{-i\frac{\gamma}{2}} \left(-i \frac{d\alpha}{2} - \frac{d\beta}{2} t \frac{\beta}{2} - i \frac{d\gamma}{2} \right) & -e^{-i\frac{\alpha}{2}} \sin \frac{\beta}{2} e^{i\frac{\gamma}{2}} \left(-i \frac{d\alpha}{2} + \frac{d\beta}{2} ct \frac{\beta}{2} + i \frac{d\gamma}{2} \right) \\ e^{i\frac{\alpha}{2}} \sin \frac{\beta}{2} e^{-i\frac{\gamma}{2}} \left(i \frac{d\alpha}{2} + \frac{d\beta}{2} ct \frac{\beta}{2} - i \frac{d\gamma}{2} \right) & e^{i\frac{\alpha}{2}} \cos \frac{\beta}{2} e^{i\frac{\gamma}{2}} \left(i \frac{d\alpha}{2} - \frac{d\beta}{2} t \frac{\beta}{2} + i \frac{d\gamma}{2} \right) \end{pmatrix},$$

the metric is:

$$\begin{aligned} \text{Tr}[dU^\dagger dU] &\sim \cos^2 \frac{\beta}{2} \left(-i \frac{d\alpha}{2} - \frac{d\beta}{2} t \frac{\beta}{2} - i \frac{d\gamma}{2} \right) \left(i \frac{d\alpha}{2} - \frac{d\beta}{2} t \frac{\beta}{2} + i \frac{d\gamma}{2} \right) + \sin^2 \frac{\beta}{2} \left(-i \frac{d\alpha}{2} + \frac{d\beta}{2} ct \frac{\beta}{2} + i \frac{d\gamma}{2} \right) \left(i \frac{d\alpha}{2} + \frac{d\beta}{2} ct \frac{\beta}{2} - i \frac{d\gamma}{2} \right) \\ &= \frac{1}{4} d\beta^2 + \cos^2 \frac{\beta}{2} \left(\frac{d\alpha + d\gamma}{2} \right)^2 + \sin^2 \frac{\beta}{2} \left(\frac{d\alpha - d\gamma}{2} \right)^2 \sim d\beta^2 + d\alpha^2 + d\gamma^2 + 2 \cos \beta d\alpha d\gamma. \end{aligned}$$

Using $\det g = 1 - \cos^2 \beta = \sin^2 \beta$ the Haar measure on $SU(2)$ is $[dU] = \sin \beta d\alpha d\beta d\gamma$. For compact groups the Haar measure is usually normalized to $\int [dU] = 1$.

The Haar measure is important as it allows one to discuss integrable functions over the group.

Exercise 1.5. Determine the normalization of the $SU(2)$ Haar measure.

Exercise 1.6. The group $SL(2, \mathbb{R})$ can be parametrized in the Iwasawa decomposition as:

$$U = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

with $\theta \in [0, 2\pi]$, $x \in \mathbb{R}$ and $y \in \mathbb{R}_{\geq 0}$. Prove that up to normalization the Haar measure is $\frac{1}{y^2} d\theta dx dy$. We have

$$\begin{aligned} U^{-1} dU &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} y^{-1/2} & 0 \\ 0 & y^{1/2} \end{pmatrix} \begin{pmatrix} 1 & -x \\ 0 & 1 \end{pmatrix} \\ &= \left\{ \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} dx \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} + \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} y^{-1/2} & 0 \\ 0 & -\frac{1}{2} y^{-3/2} \end{pmatrix} dy \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} + \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} -\sin \theta & -\cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix} d\theta \right\} \\ &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} y^{-1/2} & 0 \\ 0 & y^{1/2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} dx + \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \frac{dy}{2y} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} d\theta \\ &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 0 & y^{-1} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} dx + \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix} \frac{dy}{2y} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} d\theta \\ &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \sin \theta & \cos \theta \\ 0 & 0 \end{pmatrix} \frac{dx}{y} + \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \frac{dy}{2y} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} d\theta = \begin{pmatrix} \frac{\sin(2\theta)}{2} & \cos^2 \theta \\ -\sin^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \frac{dx}{y} + \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \frac{dy}{2y} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} d\theta \end{aligned}$$

We collect the terms

$$\begin{aligned} \frac{dx^2}{y^2} \text{Tr} \left[\begin{pmatrix} \frac{\sin(2\theta)}{2} & -\sin^2 \theta \\ \cos^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \begin{pmatrix} \frac{\sin(2\theta)}{2} & \cos^2 \theta \\ -\sin^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \right] &= \frac{dx^2}{y^2} \left[\frac{\sin^2(2\theta)}{2} + \sin^4 \theta + \cos^4 \theta \right] = \frac{dx^2}{y^2}, \\ \frac{dy^2}{4y^2} \text{Tr} \left[\begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \right] &= \frac{dy^2}{4y^2} [1 + 1] = \frac{dy^2}{2y^2}, \\ d\theta^2 \text{Tr} \left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] &= 2d\theta^2, \\ \frac{dx dy}{2y^2} \text{Tr} \left[\begin{pmatrix} \frac{\sin(2\theta)}{2} & -\sin^2 \theta \\ \cos^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} + \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \begin{pmatrix} \frac{\sin(2\theta)}{2} & \cos^2 \theta \\ -\sin^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \right] &= \frac{dx dy}{2y^2} \left[\frac{\sin(4\theta)}{2} + \sin^2 \theta \sin(2\theta) - \cos^2 \theta \sin(2\theta) \right] = 0, \\ \frac{dx d\theta}{y} \text{Tr} \left[\begin{pmatrix} \frac{\sin(2\theta)}{2} & -\sin^2 \theta \\ \cos^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\sin(2\theta)}{2} & \cos^2 \theta \\ -\sin^2 \theta & -\frac{\sin(2\theta)}{2} \end{pmatrix} \right] &= -2 \frac{dx d\theta}{y}, \\ \frac{dy d\theta}{2y} \text{Tr} \left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} + \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ -\sin(2\theta) & -\cos(2\theta) \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] &= 0 \end{aligned}$$

for a metric:

$$\frac{dx^2}{y^2} + \frac{dy^2}{2y^2} + 2d\theta^2 - 2\frac{dx d\theta}{y} \quad g = \begin{pmatrix} \frac{1}{y^2} & 0 & -\frac{1}{y} \\ 0 & \frac{1}{2y^2} & 0 \\ -\frac{1}{y} & 0 & 2 \end{pmatrix}, \quad \det g = \frac{1}{2y^2} \left(\frac{2}{y^2} - \frac{1}{y^2} \right) = \frac{1}{2y^4}$$

A.4 Finite dimensional unitary representations of $SU(2)$

Definition 1.1 (**Finite dimensional representations**). A finite dimensional representation of a group G is a group homomorphism $\rho : G \rightarrow GL(V)$, the space of invertible linear operators on some finite dimensional Hilbert space V :

$$\rho(g_1 g_2) = \rho(g_1) \rho(g_2) .$$

If ρ is injective, then the representation is said to be faithful. We often denote abusively a representation as $\rho : G \rightarrow V$, which is to be understood as $\rho : G \rightarrow GL(V)$.

A representation is called unitary if it maps G to the unitary operations on V .⁶ Two unitary representations $\rho_i : G \rightarrow U(V_i)$ are said to be unitarily equivalent if there exists a unitary operator $A : V_1 \rightarrow V_2$, $\langle Av | Aw \rangle = \langle v | w \rangle$ such that $\rho_1(g) = A^{-1} \circ \rho_2(g) \circ A$. Such an A is called an intertwining operator. A finite dimensional representation is unitary if and only if the associated Lie algebra representation $d\rho : \mathfrak{g} \rightarrow GL(V)$ maps into the space of anti Hermitian operators on H (equivalently it maps the hermitian generators $J = \imath L$ to Hermitian operators on H).

A subspace $X \subset V$ is an invariant subspace of ρ if $\rho(G)X \subset X$. A representation $\rho : G \rightarrow V$ is called irreducible if its only invariant subspaces are 0 and V itself.

Finite dimensional unitary representations are significantly simpler than generic representations. A finite dimensional unitary representation is completely reducible, in the sense that if $\rho(G)X \subset X$ for some subspace $X \subset V$, then $\rho(G)X^\perp \subset X^\perp$ where $X^\perp$ is the orthogonal complement of X . Any unitary representation can be decomposed into block diagonal irreducible representations by choosing suitable bases in all the invariant subspaces:

$$\rho(g) = \begin{pmatrix} \rho^{(1)}(g) & & & \\ & \rho^{(2)}(g) & & \\ & & \ddots & \\ & & & \rho^{(k)}(g) \end{pmatrix} = \rho^{(1)}(g) \oplus \rho^{(2)}(g) \oplus \dots \oplus \rho^{(k)}(g) .$$

As $SU(2)$ is simply connected and compact, its representations can be obtained by exponentiation of the representation of its Lie algebra. In order to identify the finite dimensional unitary irreducible representations of $SU(2)$ it suffices to classify the finite dimensional representations of $\mathfrak{su}(2)$ into the hermitian operators on finite dimensional vector spaces.

Theorem 1.1 (**Schur's lemma**). Let ρ and ρ' be two irreducible representations of a group G on V respectively V' and let $T : V \rightarrow V'$ linear such that $T\rho(g) = \rho'(g)T$, $\forall g \in G$. Then either T is zero or T is an isomorphism (a bijection).

If ρ is a complex representation of G on a finite dimensional vector space V and $T : V \rightarrow V$ is a linear operator such that $T\rho(g) = \rho(g)T$, $\forall g \in G$ then T is proportional to the identity, $T = \lambda I$ for some $\lambda \in \mathbb{C}$.

Proof. We have that if $x \in \ker T$ then for all $g \in G$, $T\rho(g)x = \rho'(g)Tx = 0$, that is $\rho(g)x \in \ker T$. If $x \in \text{Im } T$ then for all $g \in G$, $\rho'(g)x = \rho'(g)Ty = T\rho(g)y$, that is $\rho'(g)x \in \text{Im } T$.

Thus $\ker T$ (resp. $\text{Im } T$) is an invariant subspace of $\rho(G)$ (resp. $\rho'(G)$) and as ρ (resp. ρ') is an irreducible representation, $\ker T$ (resp. $\text{Im } T$) is either 0 or the whole of T (resp. V'). If $\ker T = V$ or $\text{Im } T = 0$ then $T = 0$. If not, then T is injective (as $\ker T = 0$) and surjective (as $\text{Im } T = V'$).

⁶For infinite dimensional unitary representations the map ρ is required to be continuous in the strong topology on V .

For the second part, the characteristic polynomial $\det(T - \lambda I)$ admits a root in $\mathbb{C}$, which is a non trivial eigenvalue with associated non zero eigenvector $Tx = \lambda x$. We have for all $g \in G$:

$$(T - \lambda)\rho(g) = \rho(g)(T - \lambda) ,$$

hence $\ker(T - \lambda)$ is a non trivial invariant subspace for $\rho(G)$. It follows that $\ker(T - \lambda)$ is V , hence $Tx = \lambda x$ for all $x \in V$. $\square$

A.4.1 Classification of the irreducible finite dimension unitary representations of $SU(2)$

Let us denote $J_i = \frac{\sigma_i}{2}$ and $J^2 = \sum_{i=1}^3 J_i^2$ the generators and Casimir element in $\mathfrak{su}(2)$ with structure constants $[J_i, J_j] = \epsilon_{ijk} J_k$. We define the ladder operators $J_{\pm} = J_1 \pm iJ_2$:

$$[J_3, J_{\pm}] = iJ_2 \pm i(-iJ_1) = \pm J_{\pm} , \quad [J_+, J_-] = 2J_3 , \quad J^2 = J_3^2 + J_- J_+ + J_3 = J_3^2 + J_+ J_- - J_3 .$$

As J^2 commutes with all the generators, it commutes with $\rho(SU(2))$ for any finite dimensional irreducible representation $\rho : SU(2) \rightarrow GL(V)$, thus $J^2 = \lambda I$ in V . If furthermore ρ is unitary, $d\rho$ maps on hermitian operators and a basis in V consists in the orthonormal eigenbasis of the operator J_3 . We label this basis $|\lambda \mu\rangle$:

$$J^2 |\lambda \mu\rangle = \lambda |\lambda \mu\rangle , \quad J_3 |\lambda \mu\rangle = \mu |\lambda \mu\rangle .$$

The raising and lowering operators shift between the basis elements:

$$J_3 J_{\pm} |\lambda \mu\rangle = (\mu \pm 1) J_{\pm} |\lambda \mu\rangle \quad \Rightarrow \quad J_{\pm} |\lambda \mu\rangle = C_{\pm}(\lambda, \mu) |\lambda \mu \pm 1\rangle$$

and as eigenvectors corresponding to distinct eigenvalues of J_3 are orthogonal and V has finite dimension, there must exist a maximal element $|\lambda \mu_0\rangle$ such that:

$$J_3 |\lambda \mu_0\rangle = \mu_0 |\lambda \mu_0\rangle , \quad J_+ |\lambda \mu_0\rangle = 0 , \quad J^2 |\lambda \mu_0\rangle = \lambda |\lambda \mu_0\rangle = \mu_0(\mu_0 + 1) |\lambda \mu_0\rangle .$$

Again as the representation is finite dimensional, there must exist an $d \in \mathbb{N}$ such that $|\lambda \mu_0 - k\rangle \neq 0$ for $k = 0, \dots, d$ but $J_- |\lambda \mu_0 - d\rangle = 0$. But then:

$$J^2 |\lambda \mu_0 - d\rangle = \lambda |\lambda \mu_0 - d\rangle = (\mu_0 - d)(\mu_0 - d - 1) |\lambda \mu_0 - d\rangle ,$$

therefore

$$\mu_0^2 + \mu_0 = \mu_0^2 - 2d\mu_0 + d^2 - \mu_0 + d \Rightarrow \mu_0 = \frac{d}{2} , \quad \lambda = \frac{d}{2} \left(\frac{d}{2} + 1 \right) .$$

It follows that the unitary irreducible representations of $SU(2)$ are indexed by a non negative half integer $j \in \frac{\mathbb{N}}{2}$, $\rho^j : SU(2) \rightarrow V^j$ when V^j is the $d_j = 2j + 1$ dimensional Hilbert space spanned by the vectors $|j m\rangle$ where $m = j, j - 1, \dots, -j$ and

$$J_3 |j m\rangle = m |j m\rangle , \quad J^2 |j m\rangle = j(j + 1) |j m\rangle .$$

We call this the spin j representation of $SU(2)$.

Note that $J_{\pm} |j m\rangle = C_{\pm}(j, m) |j m \pm 1\rangle$ and as $J_{\pm}^{\dagger} = J_{\mp}$ we have $C_+(j, m) = \langle j m + 1 | J_+ |j m\rangle = \langle j m + 1 | J_- |j m\rangle = C_-(j, m + 1)$. As furthermore $j(j + 1) - m(m + 1) = C_-(j, m + 1)C_+(j, m)$, we fix up to a phase:

$$C_{\pm}(j, m) = \sqrt{j(j + 1) - m(m \pm 1)} .$$

A.4.2 $SU(2)$ recoupling theory

Composite quantum systems are described by tensor product of Hilbert spaces. The tensor product of two vector spaces V and W is the linear span of vectors of the form $v \otimes w$ with $v \in V$ and $w \in W$. This is particularly transparent if one picks a basis $\{e_i\}$ in V and a basis $\{f_j\}$ in W . Then:

$$V \otimes W = \left\{ \sum_{ij} \alpha_{ij} e_i \otimes f_j, \alpha_{ij} \in \mathbb{C} \right\}.$$

Clearly there exist elements in the tensor product $V \otimes W$ that do not write as the product of a $v \in V$ and $w \in W$. In finite dimension $\dim(V \otimes W) = \dim(V) \dim(W)$.

For any linear operators $A : V \rightarrow V$ and $B : W \rightarrow W$ we define the linear tensor product operator:

$$(A \otimes B)(v \otimes w) = (Av) \otimes (Bw).$$

A way to fabricate new representations of a group is to take tensor products of representations.

Definition 1.2 (**Tensor product representation**). If $\rho_V : G \rightarrow V$ and $\rho_W : G \rightarrow W$ are two representations of a group, we call the representation $\rho_{V \otimes W} : G \rightarrow V \otimes W$:

$$\rho_{V \otimes W}(g) = \rho_V(g) \otimes \rho_W(g),$$

tensor product representation. *The tensor product of two finite dimensional unitary representations is a finite dimensional unitary representation.*

However, when taking tensor products of representations one usually obtains reducible representations. For unitary representations of $SU(2)$ for instance one obtains in general:

$$\rho^{j_1} \otimes \rho^{j_2} = \bigoplus_{J \in \frac{\mathbb{N}}{2}} \rho^J,$$

and in all generality some representations might appear more than once on the right hand side.

Consider a quantum system made of two sub systems. If the system 1 leaves in the spin j_1 representation of $\vec{J}^{(1)}$ and the system 2 in the spin j_2 representation of $\vec{J}^{(2)}$, the total system will leave in the tensor product representation $\rho^{j_1} \otimes \rho^{j_2}$ spanned by:

$$|j_1 m_1\rangle \otimes |j_2 m_2\rangle = |j_1 j_2 m_1 m_2\rangle,$$

and total angular momentum of the systems will be $\vec{J} = \vec{J}^{(1)} \otimes 1 + 1 \otimes \vec{J}^{(2)}$.

It is easy to check that $\vec{J}$ respects the $\mathfrak{su}(2)$ Lie algebra, hence the Hilbert space can be decomposed on irreducible representations of the total spin $\rho^J : SU(2) \rightarrow V^J$ whose basis is $|J M\rangle$ for some J and $M = -J, \dots, J$ with the constraint that $\sum_J (2J+1) = (2j_1+1)(2j_2+1)$ (in principle the same representation can appear several times):

$$|J M\rangle = \sum_{j_1, j_2, m_1, m_2} |j_1 j_2 m_1 m_2\rangle \langle j_1 j_2 m_1 m_2 | J M \rangle,$$

where the coefficients $\langle j_1 j_2 m_1 m_2 | J M \rangle$ are called the Clebsch–Gordan coefficients. Looking at the matrix element $\langle j_1 j_2 m_1 m_2 | J_3 | J M \rangle$ we note that the Clebsch–Gordan coefficient is zero unless $M = m_1 + m_2$. Acting with $J_{\pm}$ we get:

$$C_{\pm}(J, M) |J M \pm 1\rangle = \sum_{j'_1, j'_2, m'_1, m'_2} \left[C_{\pm}(j'_1, m'_1) |j'_1 j'_2 m'_1 \pm 1 m'_2\rangle + C_{\pm}(j'_2, m'_2) |j'_1 j'_2 m'_1 m'_2 \pm 1\rangle \right] \langle j'_1 j'_2 m'_1 m'_2 | J M \rangle,$$

and we conclude that:

$$C_{\pm}(J, M) \langle j_1 j_2 \ m_1 m_2 | JM \pm 1 \rangle = C_{\pm}(j_1, m_1 \mp 1) \langle j_1 j_2 \ m_1 \mp 1 m_2 | JM \rangle + C_{\pm}(j_2, m_2 \mp 1) \langle j_1 j_2 \ m_1 m_2 \mp 1 | JM \rangle .$$

We initiate the recursion at $M = J$ with the convention that $\langle j_1 j_2 j_1 (J - j_1) | JJ \rangle \geq 0$ if $j_1 \geq j_2$. Note that the recursion can be initiated non trivially only if $j_1 - j_2 \leq J \leq j_1 + j_2$: otherwise the unique solution of the recursion is that all the Clebsch–Gordan coefficients are zero. The first level of the recursion reads:

$$0 = C_+(j_1, m_1 - 1) \langle j_1 j_2 \ m_1 - 1 m_2 | JJ \rangle + C_+(j_2, m_2 - 1) \langle j_1 j_2 \ m_1 m_2 - 1 | JJ \rangle .$$

We first fix $m_2 = J - j_1 + 1$ and solve for $m_1 = j_1, \dots, j_1$ we then proceed to $m_2 = J - j_1$ and iterate. Once the first level recursion is solved, we go to $|J \ J - 1\rangle$ and so on. We get:

$$\rho^{j_1} \otimes \rho^{j_2} = \bigoplus_{|j_1 - j_2| \leq J \leq j_1 + j_2} \rho^J , \quad \sum_{|j_1 - j_2| \leq J \leq j_1 + j_2} (2J + 1) = (2j_1 + 1)(2j_2 + 1) .$$

Closely related to the Clebsch–Gordan we define the Wigner 3-j symbol:

$$\begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & -M \end{pmatrix} = \frac{(-1)^{-j_1 + j_2 + M}}{\sqrt{2J + 1}} \langle j_1 j_2 \ m_1 m_2 | JM \rangle$$

One sometimes needs to deal with iterated tensor products of representations. For $\rho^i \otimes \rho^j \otimes \rho^k$ one has three possible decompositions $(\rho^i \otimes \rho^j) \otimes \rho^k$, $(\rho^i \otimes \rho^k) \otimes \rho^j$, $(\rho^j \otimes \rho^k) \otimes \rho^i$. There exist canonical mappings between the various decompositions.

A.4.3 Wigner D -matrices

As already mentioned, the elements of $SU(2)$ can be represented by Euler angles in the $z - y - z$ convention:

$$U(\alpha, \beta, \gamma) = e^{-i\alpha J_3} e^{-i\beta J_2} e^{-i\gamma J_3} = \begin{pmatrix} e^{-i\frac{\alpha}{2}} & 0 \\ 0 & e^{i\frac{\alpha}{2}} \end{pmatrix} \begin{pmatrix} \cos \frac{\beta}{2} & -\sin \frac{\beta}{2} \\ \sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix} \begin{pmatrix} e^{-i\frac{\gamma}{2}} & 0 \\ 0 & e^{i\frac{\gamma}{2}} \end{pmatrix} .$$

In the representation j this group element is represented by:

$$D_{mk}^j(\alpha, \beta, \gamma) = \langle jm | e^{-i\alpha J_3} e^{-i\beta J_2} e^{-i\gamma J_3} | jk \rangle = e^{i(\alpha m - \gamma k)} d_{mk}^j(\beta) ,$$

and a tedious computations shows that:

$$d_{mk}^j(\beta) = \sqrt{(j+m)!(j-m)!(j+k)!(j-k)!} \sum_s \frac{(-1)^{m-k+s} (\cos \frac{\beta}{2})^{2j+k-m-2s} (\sin \frac{\beta}{2})^{m-k+2s}}{s!(j+k-s)!(j-m-s)!(m-k+s)!} ,$$

where the sum runs over the integer s such that the factorials are well defined (i.e. have non negative argument).

The recoupling theory of $SU(2)$ implies in particular that the product of two matrix elements can be re decomposed as:

$$D_{m_1 k_1}^{j_1}(\alpha, \beta, \gamma) D_{m_2 k_2}^{j_2}(\alpha, \beta, \gamma) = \sum_{J=j_1+j_2} \sum_{K=|j_1-j_2|}^{j_1+j_2} \langle j_1 j_2 \ m_1 m_2 | J \ M \rangle D_{MK}^J(\alpha, \beta, \gamma) \langle J \ K | j_1 j_2 \ k_1 k_2 \rangle ,$$

and note that the sum over MK is superfluous as the selection rules fix $M = m_1 + m_2$ and $K = k_1 + k_2$. Integrating over $SU(2)$ with the Haar measure and using $\overline{D_{m_1 k_1}^{j_1}(\alpha, \beta, \gamma)} = (-1)^{m_1 - k_1} D_{-m_1 -k_1}^{j_1}(\alpha, \beta, \gamma)$ we obtain that the matrix elements

are orthogonal:

$$\int_0^{2\pi} d\alpha \int_0^\pi \sin \beta d\beta \int_0^{2\pi} d\gamma \overline{D_{m_1 k_1}^{j_1}(\alpha, \beta, \gamma)} D_{m_2 k_2}^{j_2}(\alpha, \beta, \gamma) = \frac{8\pi^2}{2j+1} \delta^{j_1 j_2} \delta_{m_1 m_2} \delta_{k_1 k_2} .$$

Theorem 1.2 (Peter–Weyl Theorem). *The set of matrix coefficients D_{mk}^j is dense in the space of continuous complex functions on $SU(2)$ with the sup norm and dense in L^2 norm in the space L^2 with respect to the Haar measure. The set of characters $\chi^j(g) = \text{Tr } D^j(g)$ for $g \in SU(2)$ of the irreducible representations j form a Hilbert basis in the L^2 space of class functions $f(\text{gh}^{-1}) = f(g)$. The same holds for general finite dimensional representations of arbitrary Lie groups.*

Let ρ be a unitary representation of a compact group G on a complex Hilbert space V (of infinite dimension). Then V splits into an orthogonal direct sum of irreducible finite-dimensional unitary representations of G .

Proof. The reader should consult the literature. □

Exercise 1.7. *Compute the character of an arbitrary element in $SU(2)$.* $\chi^j(g) = \frac{\sin((2j+1)\frac{\theta}{2})}{\sin(\frac{\theta}{2})}$ with $\cos \frac{\theta}{2} = \cos \frac{\beta}{2} \cos \frac{\alpha+\gamma}{2}$

Exercise 1.8. *Prove that*

$$\frac{1}{8\pi^2} \int_0^{2\pi} d\alpha \int_0^\pi \sin \beta d\beta \int_0^{2\pi} d\gamma D_{m_1 k_1}^{j_1}(\alpha, \beta, \gamma) D_{m_2 k_2}^{j_2}(\alpha, \beta, \gamma) D_{m_3 k_3}^{j_3}(\alpha, \beta, \gamma) = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ k_1 & k_2 & k_3 \end{pmatrix} .$$

Exercise 1.9. *Prove that on class functions we have:*

$$\delta(g) = \sum_{j \geq 0} (2j+1) \chi^j(g) .$$

A.5 $SU(N)$ Yang Mills theory

Many theories in physics (electromagnetism, electroweak, quantum chromodynamics) are gauge theories. We will write below the classical action of such a gauge theory.

Gauge field, covariant derivative. We consider a gauge field (gauge connection), that is a Lie algebra valued covector field:

$$A_\mu = A_\mu^a T^a ,$$

where A_μ behaves under rotations of the space as $\mathbb{R}^d$, $A \rightarrow A_\mu^R(Rx) = R_{\mu\nu} A_\nu(x)$. The functions $A_\mu^a(x)$ are functions of space time points, but we suppress the x dependence in order to simplify the notation. This field represents the gauge bosons, for instance the gluons.

We define the covariant derivative $D_\mu \psi = \partial_\mu \psi - \imath A_\mu \psi$ on vectors on which the defining representation of the Lie algebra acts $\psi = (\psi_i)_{i=1, N}$. Under conjugation by the gauge group the covariant derivative behaves like:

$$D \rightarrow D^U = U D U^\dagger = U (\partial_\mu - \imath A_\mu) U^\dagger = \partial_\mu - \imath (U A_\mu U^\dagger + \imath U \partial_\mu U^\dagger) , \quad \text{that is} \quad A_\mu \rightarrow A_\mu^U = U A_\mu U^\dagger + \imath U \partial_\mu U^\dagger .$$

In particular $\bar{\psi} D_\mu \psi$ is invariant under a gauge transformation $\psi \rightarrow \psi^U = U \psi$, $D \rightarrow D^U$ and the covariant derivative is used to couple in the gauge invariant manner $\bar{\psi} \gamma^\mu D_\mu \psi$ matter fields with the gauge field. Note that the matter fields $\psi_{\alpha i}$ are vectors in a tensor product space with $\alpha = 1, \dots, 4$ Dirac spinor indices and $i = 1, \dots, N$ color indices.

In a representation ρ of the Lie algebra we have generators $(T^a)_j^i$ which are $\dim \rho \times \dim \rho$ matrices and $D_\mu \psi^i = \partial_\mu \psi^i - \imath A_\mu^a (T^a)_j^i \psi^j$. A representation which will appear frequently is the adjoint representation with $\dim(\rho) = N^2 - 1$. It is useful

to package the fields valued in the adjoint representation into Lie algebra valued objects $H = h_a T^a$. The action of the covariant derivative in this representation is represented by the adjoint action in the Lie algebra:

$$D_\mu H \equiv [D_\mu, H] = \partial_\mu H - \imath[A_\mu, H], \quad D_\mu(h_a T^a) = (\delta^{ab} \partial_\mu h_a + f^{pab} A_\mu^p h_a) T^b.$$

We denote $D_\mu X$ with the understanding that D_μ acts on X in the appropriate representation. Note that $D_\mu X$ will live in the same representation as X .

Field strength, action. The field strength (curvature of the gauge connection) is the Lie algebra element (Lie algebra valued two form):

$$-\imath F_{\mu\nu} = [D_\mu, D_\nu] = [\partial_\mu - \imath A_\mu, \partial_\nu - \imath A_\nu] = -\imath \left(\partial_\mu A_\nu - \partial_\nu A_\mu - \imath [A_\mu, A_\nu] \right) = -\imath \left(\partial_\mu A_\nu^c - \partial_\nu A_\mu^c + A_\mu^a A_\nu^b f^{abc} \right) T^c.$$

Note that in the fundamental representation we have $[D_\mu, D_\nu]\psi = -\imath F_{\mu\nu}\psi$ and respectively in the adjoint we get $[D_\mu, D_\nu]H = [D_\mu, [D_\nu, H]] - [D_\nu, [D_\mu, H]] = [[D_\mu, D_\nu], H] = [-\imath F_{\mu\nu}, H]$. The field strength transforms covariantly under a gauge transformation $F_{\mu\nu} \rightarrow F_{\mu\nu}^U = U F_{\mu\nu} U^\dagger$.

The Yang Mills action is the gauge invariant action describing the propagation and self interactions of the gauge bosons:

$$S_{\text{YM}}(A) = \frac{1}{2g^2} \int_x \text{tr}(F_{\mu\nu} F^{\mu\nu}) = \frac{1}{4g^2} \int_x (F_{\mu\nu})^c (F^{\mu\nu})^c,$$

where space time indices are raised with the Euclidean metric $\delta^{\mu\nu}$. The classical equations of motion are obtained by varying the action with respect to A_μ :

$$\begin{aligned} (F_{\mu\nu})^c \delta_{A_\rho^d} (F^{\mu\nu})^c &= (F_{\mu\nu})^c \left(\partial_\mu \delta_{\nu\rho} \delta^{dc} - \partial_\nu \delta_{\mu\rho} \delta^{dc} + \delta_{\rho\mu} \delta^{ad} A_\nu^b f^{abc} + \delta_{\rho\nu} \delta^{bd} A_\mu^a f^{abc} \right) \\ &= -\partial_\mu (F_{\mu\rho})^d + \partial_\nu (F_{\nu\rho})^d + (F_{\rho\nu})^c A_\nu^b f^{dbc} + (F_{\mu\rho})^c A_\mu^a f^{adc} \propto \partial_\mu (F_{\mu\rho})^d + A_\mu^a (F_{\mu\rho})^c f^{acd}, \end{aligned}$$

thus read $D_\mu F^{\mu\nu} = 0$, as D acts on F in the adjoint representation. Moreover, defining the Hodge dual tensor $*F^{\rho\sigma} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}$ with $\epsilon^{\mu\nu\rho\sigma}$ totally anti symmetric we have the Bianchi identity

$$D_\mu (*F^{\mu\nu}) \propto \epsilon^{\nu\mu\rho\sigma} [D_\mu, [D_\rho, D_\sigma]] \propto [D_\mu, [D_\rho, D_\sigma]] + [D_\rho, [D_\sigma, D_\mu]] + [D_\sigma, [D_\mu, D_\rho]] = 0,$$

by the Jacobi identity. The equations of motion and Bianchi identity of Yang Mills theory write in short as:

$$D_\mu F^{\mu\nu} = 0, \quad D_\mu (*F^{\mu\nu}) = 0.$$

The theta term. One can build a second Lorentz and gauge invariant term and consider the more general action:

$$S_{YM} - \imath S_\theta = \frac{1}{2g^2} \int_x \text{tr}(F_{\mu\nu} F^{\mu\nu}) - \imath \frac{\theta}{16\pi^2} \int_x \text{tr}((*F^{\mu\nu}) F_{\mu\nu}).$$

The theta term is a total derivative $S_\theta = \frac{\theta}{8\pi^2} \int_x \partial_\mu K^\mu$ with $K^\mu = \epsilon^{\mu\nu\rho\sigma} (A_\nu \partial_\rho A_\sigma - \frac{2i}{3} A_\nu A_\rho A_\sigma)$, hence this term is topological. The boundary action obtained by using Stokes theorem is the Chern–Simons action.

Appendix B

Exams

B.1 Sample exams HöMa II

B.1.1 Sample 1

Exercise 1 (Elementary exercises, 30 points). *Solve the questions below.*

- (5 points) Consider the change of variables $u = x^2 + x + y$, $v = x^2 + y$. Compute the Jacobian of the change of variables and the inverse transformation. Comment on the result.

We have $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x+1 & 1 \\ 2x & 1 \end{vmatrix} = 1$, and $x = u - v$, $y = v - (u - v)^2 = v - u^2 - v^2 + 2uv$. This is change of variables with Jacobian 1 hence always invertible. Both the change of variables and its inverse are polynomial.

- (10 points) Derive the Taylor series representation around 0 of $\frac{1}{(1-x)^{k+1}}$, $k \in \mathbb{N}$ and discuss its convergence properties.

For any power series $P(x) = \sum_{n \geq 0} c_n x^n$ with radius of convergence $r^{-1} = \limsup |c_n|^{1/n}$, for any ϵ , for all but a finite number of n , $|c_n| \leq (r^{-1} + \epsilon)^n$. We denote $P'(x) = \sum_{n \geq 1} c_n n x^{n-1}$ and we observe that for $|x|, |x_0| < r$ we have $\left| \frac{P(x) - P(x_0)}{x - x_0} - P'(x_0) \right| = \left| \sum_{n \geq 2} a_n \sum_{p=0}^{n-1} (x^p - x_0^p) x_0^{n-p} \right| \leq \sum_{n \geq 2} \frac{(1+\epsilon r)^n}{r^n} |x - x_0| \sum_{p=1}^{n-1} |x|^{p-1} |x_0|^{n-p} = |x - x_0| \sum_{n \geq 2} (1 + \epsilon r)^n \frac{n(n-1)}{r^n} \rho^{n-2}$, hence $\lim_{x \rightarrow x_0} \frac{P(x) - P(x_0)}{x - x_0} = P'(x_0)$, that is inside the disk of convergence we can derivate a power series term by term. We then have $\frac{1}{(1-x)^{k+1}} = \frac{1}{k!} \frac{d^k}{dx^k} \left(\frac{1}{1-x} \right) = \frac{1}{k!} \frac{d^k}{dx^k} \left[\sum_{p=0}^{k-1} x^p + \sum_{p \geq 0} x^{p+k} \right] = \sum_{p \geq 0} \frac{(p+k)!}{k!p!} x^p$, convergent for $|x| < 1$.

- (15 points) Assuming $\int_{-\infty}^{\infty} dx e^{-\frac{x^2}{2}} = \sqrt{2\pi}$, compute the moments of the normalized Gaußian integral $\int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} e^{-\frac{x^2}{2\alpha}} x^n$ for $\alpha \in (0, \infty)$.

We define $\int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} e^{-\frac{x^2}{2\alpha} + Jx} = e^{\frac{1}{2}\alpha J^2}$. Commuting the derivatives and the integrals (the integrand is continuous in both variables and continuously partially differentiable in J and we use the "differentiation under the integral sign" theorem in the script) $\frac{d^n}{dJ^n} e^{\frac{1}{2}\alpha J^2} = \frac{d^n}{dJ^n} \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} e^{-\frac{x^2}{2\alpha} + Jx} = \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} \frac{d^n}{dJ^n} e^{-\frac{x^2}{2\alpha} + Jx} = \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi\alpha}} e^{-\frac{x^2}{2\alpha} + Jx} x^n$, and setting $J = 0$ we get $\langle x^{2p+1} \rangle = 0$, $\langle x^{2p} \rangle = \alpha^p \frac{(2p)!}{2^p p!}$.

Exercise 2 (Lagrange multipliers, 30 points). *Find the extrema of the function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x, y, z) = x^2 + y^2 + z^2$ subjected to the constraints $z^2 = x^2 + y^2$ and $x + y - z + 1 = 0$.*

We have $L = x^2 + y^2 + z^2 + \lambda(x^2 + y^2 - z^2) + \mu(x + y - z + 1)$ and requiring criticality in all the arguments leads to $z^2 = x^2 + y^2$, $z = x + y + 1$, $2x + 2\lambda y + \mu = 0$, $2y + 2\lambda x + \mu = 0$, $2z - 2\lambda z - \mu = 0$. Observe that if $\lambda = 1$ then $\mu = 0$, $x, y = 0$ and $z = 0, 1$ hence contradiction; if $\lambda = -1$ then $\mu = 0$, $z = 0$ hence both $x^2 + y^2 = 0$, $x + y + 1 = 0$ and x, y real which is impossible. It follows that $z^2 = x^2 + y^2$, $z = x + y + 1$, $x = y = -\frac{\mu}{2(1+\lambda)}$, $z = \frac{\mu}{2(1-\lambda)}$. If $\mu = 0$ then $x = y = z = 0$ but $z = x + y + 1$, hence $\mu \neq 0$. Then: $\frac{\mu^2}{4(1-\lambda)^2} = 2 - \frac{\mu^2}{4(1+\lambda)^2} \Rightarrow (1+\lambda)^2 = 2(1-\lambda)^2 \Rightarrow 1 + 2\lambda + \lambda^2 = 2 - 4\lambda + 2\lambda^2 \Rightarrow \lambda^2 - 6\lambda + 1 = 0 \Rightarrow \lambda = \frac{6 \pm \sqrt{36-4}}{2} = \boxed{3 \pm 2\sqrt{2} = \lambda}$. Furthermore $\frac{\mu}{2(1-\lambda)} = 1 - \frac{\mu}{(1+\lambda)} \Rightarrow \mu[(1+\lambda) + 2(1-\lambda)] = 2(1-\lambda^2) \Rightarrow \mu = \frac{2(1-\lambda^2)}{3-\lambda} = \frac{2[1-(9+8\pm 12\sqrt{2})]}{\mp 2\sqrt{2}} = \frac{1-17 \mp 12\sqrt{2}}{\mp \sqrt{2}} = \boxed{12 \pm 8\sqrt{2} = \mu}$. also $z = \frac{4(3 \pm 2\sqrt{2})}{2(-2 \mp 2\sqrt{2})} = \frac{3 \pm 2\sqrt{2}}{-1 \mp \sqrt{2}} = (3 \pm 2\sqrt{2})(1 \mp \sqrt{2}) = \boxed{-1 \mp \sqrt{2} = z}$, and $x = y = -\frac{4(3 \pm 2\sqrt{2})}{2(4 \pm 2\sqrt{2})} = -\frac{3 \pm 2\sqrt{2}}{2 \pm \sqrt{2}} = -\frac{1}{2}(3 \pm 2\sqrt{2})(2 \mp \sqrt{2}) = -\frac{1}{2}(2 \pm \sqrt{2}) = \boxed{-1 \mp \sqrt{2} = x = y}$.

Exercise 3 (Euler-Maclaurin summation, 40 points). *We define the Bernoulli numbers as the coefficients B_n in $\frac{x}{e^x - 1} = \sum_{n \geq 0} B_n \frac{x^n}{n!}$.*

- (10 points) Compute B_0, B_1, B_2 and B_3 . It can be shown that $\forall n \geq 1$, $|B_{2n}| \leq (2n)! K^{2n}$ and $B_{2n+1} = 0$ for some constant K . In the rest of this exercise you can use this result without proof.

We have $\frac{x}{e^x - 1} = \frac{\frac{x}{2} + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots}{1 + \frac{x}{2} + \frac{x^2}{3!} + \frac{x^3}{4!} + \dots}$, hence $\sum_{n \geq 0} B_n \frac{x^n}{n!} = \frac{\frac{1}{2} + \frac{x}{2} + \frac{x^2}{3!} + \frac{x^3}{4!} + \dots}{1 + \frac{x}{2} + \frac{x^2}{3!} + \frac{x^3}{4!} + \dots} = 1 - \frac{x}{2} - \frac{x^2}{3!} - \frac{x^3}{4!} + \left(\frac{x}{2} + \frac{x^2}{3!} \right)^2 - \left(\frac{x}{2} \right)^3 + \dots = 1 - \frac{x}{2} + \frac{x^2}{2!} \left(-\frac{1}{3} + \frac{1}{2} \right) + \frac{x^3}{3!} \left(-\frac{1}{4} + 1 - \frac{3}{4} \right)$, leading to $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$ and $B_3 = 0$.

- (10 points) We denote the differential operator $D = \frac{d}{dx}$. Assuming that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is indefinitely differentiable:

– show that in the sense of formal series (that is do not worry about convergence yet), $\forall a \in \mathbb{R}$, $(e^{aD} f)(x) = f(x + a)$. We recognize the

Taylor expansion $f(x + a) = \sum_{k \geq 0} \frac{a^k}{k!} \frac{d^k}{dx^k} f(x) = e^{a \frac{d}{dx}} f(x)$.

B. EXAMS

- compute in the sense of formal series $\left(\frac{1}{1-e^D}f\right)(x)$.

We recognize the geometric series $\frac{1}{1-e^D} = \sum_{k \geq 0} e^{kD} \Rightarrow \left(\frac{1}{1-e^D}f\right)(x) = \sum_{n \geq 0} f(x+n)$.

- propose sufficient conditions such that these expressions converge.

The Taylor expansion converges absolutely and uniformly if for instance $f^{(k)} < A^k$; it has some finite radius of convergence for $f^{(k)} < k! A^k k^\beta$ for any fixed β . The sum converges if $f(x+n)$ decreases fast enough at infinity, e.g. $f(x) < |x|^{-1-\alpha}$ for some $\alpha > 1$ for x large enough.

- (10 points) Prove the Euler-Maclaurin summation formula:

$$\sum_{n \geq 0} f(x+n) = \int_x^\infty dt f(t) + \frac{1}{2}f(x) - \sum_{n \geq 1} \frac{B_{2n}}{(2n)!} \frac{d^{2n-1}f}{dx^{2n-1}}(x).$$

Propose sufficient conditions such that the sum converges.

Applying D once more we get $\left(\frac{D}{1-e^D}f\right)(x) = \frac{d}{dx}(\sum_{n \geq 0} f(x+n)) = -f(x) + \frac{1}{2}f'(x) - \sum_{n \geq 1} \frac{B_{2n}}{(2n)!} \frac{d^{2n}f}{dx^{2n}}(x)$. It follows by integrating once that the series $\sum_{n \geq 0} f(x+n)$ computes in terms of the integral of f and corrections $\sum_{n \geq 0} f(x+n) = \int_x^\infty dt f(t) + \frac{1}{2}f(x) - \sum_{n \geq 1} \frac{B_{2n}}{(2n)!} \frac{d^{2n-1}f}{dx^{2n-1}}(x) + C$ with C some integration constant. In fact $C = 0$, as it can be checked by considering f with compact support. For instance $f^{(2n-1)}(x) \leq (aK)^{-2n}$ with $a > 1$ ensures convergence.

- (5 points) Consider the sum $S = 1 + 2 + 3 + \dots = \sum_{n \geq 0} n$. We introduce some cutoff function h such that $h(0) = 1$, $vh(v)$ is integrable at infinity and h together with its derivatives is suitably bounded. Prove that:

$$S_N = \sum_{n \geq 0} nh\left(\frac{n}{N}\right) = N^2 I(h) - \frac{1}{12} + O(N^{-2}),$$

with $I(h)$ some cutoff dependent term. Propose a cutoff such that the $O(N^{-2})$ term is a convergent series. We remark that S_N consists in a cutoff dependent divergent part, a universal finite part $-\frac{1}{12}$ and terms which go to zero when lifting the cutoff $N \rightarrow \infty$. We call the finite part, $-\frac{1}{12}$, the renormalized value of the sum S .

$S_N = \sum_{n \geq 0} nh\left(\frac{n}{N}\right) = \int_0^\infty du \, u h\left(\frac{u}{N}\right) - \frac{1}{12} \left[h\left(\frac{0}{N}\right) + \frac{1}{N} x h'\left(\frac{x}{N}\right)\right]_{x=0} + O(N^{-2}) = N^2 \int_0^\infty dv \, v h(v) - \frac{1}{12} + O(N^{-2})$. A good cutoff is e^{-u} : the rest term is bounded by $|\sum_{n \geq 2} K^{2n} [xe^{-x/N}]_{x=0}^{(2n-1)}| = |\sum_{n \geq 0} K^{2n+4} [e^{-x/N}]_{x=0}^{(2n+2)}| \leq \sum_{n \geq 0} K^{2n+4} \frac{1}{N^{2n+2}} = \frac{K^4}{N^2} \frac{1}{1-K^2/N^2}$.

- (5 points) What are the renormalized values of the sums $\sum_{n \geq 0} n^2$ and $\sum_{n \geq 0} 1$?

For $\sum_{n \geq 0} n^2$ the constant drops and we only have odd number of derivatives acting on $x^2 h(x/N)$, hence $\sum_{n \geq 0} n^2 = 0$. Furthermore, $\sum_{n \geq 0} 1 = \frac{1}{2} h(x/N) = 1/2$.

Exercise 4 (Classical mechanics, 20 points). We denote Ω the phase space of a point particle in classical mechanics. For the purpose of this exercise, Ω is considered as a two dimensional manifold parametrized by a generalized coordinate and a generalized momentum (p, q) . We define the differential 2-form $\omega = dq \wedge dp$, which is a volume form on Ω .

- (5 points) On what does ω act? How?

ω acts on pairs of tangent vectors $X^P \partial_P + X^Q \partial_Q \in T\Omega$ like $\omega(X, Y) = X^Q Y^P - X^P Y^Q$.

The Hamiltonian of the point particle is a function $H : \Omega \rightarrow \mathbb{R}$, for instance: $H(p, q) = \frac{1}{2}p^2 + V(q)$. All the questions below refer to a general Hamiltonian. We define the Hamiltonian vector field associated to H as the vector field $X_H \in T\Omega$ such that $i_{X_H} \omega = dH$.

- (5 points) Write the Hamiltonian vector field in the natural basis of $T\Omega$.

We have $X_H = X^P \partial_P + X^Q \partial_Q$ and $\omega(X_H, Y) = dH(Y)$ rewrites as $(\partial_P H)Y^P + (\partial_Q H)Y^Q = dq(X_H)Y^P - dp(X_H)Y^Q \Rightarrow X_H^Q = \partial_P H$, $X_H^P = -\partial_Q H$.

- (5 points) Consider the integral curves of X_H , that is the curves $\gamma : \mathbb{R} \rightarrow \Omega$, $\gamma(t) = (q(t), p(t))$ such that $\frac{d\gamma}{dt} \equiv \dot{\gamma} = X_H$. The variable t is called “time”. What do these curves represent? A function $f(p, q)$ is called an observable. Compute the time derivative of an observable $f(p, q)$ along an integral curve.

The integral curves $(q(t), p(t))$ respect $\dot{q} = X_H^Q = \partial_P H$, $\dot{p} = X_H^P = -\partial_Q H$, hence are the (classical) trajectories of the particle; $\frac{d}{dt} f \circ \gamma = \partial_P f \dot{\gamma}^P + \partial_Q f \dot{\gamma}^Q = \partial_P f (\partial_P H) + \partial_Q f (-\partial_Q H) \equiv \{f, H\}$.

We define the Lie derivative of ω as $\mathcal{L}_X \omega = i_X d\omega + d(i_X \omega)$.

- (5 points) Solve the following:

- Compute for any vector field X the Lie derivative $\mathcal{L}_X \omega$ in components.

We have $d\omega = 0$ hence $\mathcal{L}_X \omega = d(i_X \omega) = d(X^Q dp - X^P dq) = (\partial_Q X^Q - \partial_P X^P) dq \wedge dp$.

- Compute $\mathcal{L}_{X_H} \omega$ for the Hamiltonian vector field. Comment on the result.

We have $\mathcal{L}_{X_H} \omega = d(i_{X_H} \omega) = d(dH) = 0$, known as Liouville’s theorem.

- For any $f : \Omega \rightarrow \mathbb{R}$, we define X_f as the vector field such that $df = i_{X_f} \omega$. Compute in coordinates $\omega(X_f, X_g)$. Compare $df(X_H)$ with $dH(X_f)$, compute them in coordinates and revisit the time derivative of an observable along an integral curve of the Hamiltonian vector field.

We have $\omega(X_f, X_g) = X_f^Q X_g^P - X_f^P X_g^Q = \partial_P f (-\partial_Q g) - (\partial_P f)(\partial_P g) = \{f, g\}$. Also $\frac{d}{dt} f \circ \gamma = \{f, H\} = \omega(X_f, X_H) = -i_{X_H} \omega(X_f) = -dH(X_f) = -X_f(H) = -\partial_P f \frac{\partial H}{\partial q} + \partial_Q f \frac{\partial H}{\partial p}$. Finally $df(X_H) = \omega(X_f, X_H) = -\omega(X_H, X_f) = -dH(X_f)$.

B.1.2 Sample 2

Exercise 1 (Elementary exercises, 30 points). *Solve the following:*

- (10 points) For $z \in \mathbb{C}$, compute $s_N = \sum_{n=0}^{N-1} z^n$. Prove that the series $\sum_{n \geq 0} z^n$ converges for $|z| < 1$ and find its sum.

As $(1-z) \sum_{n=0}^{N-1} z^n = 1 - z + z - z^2 + z^2 - \dots - z^N$ we conclude that $\sum_{n=0}^{N-1} z^n = \frac{1-z^N}{1-z}$. The series converges if z^N converges for $N \rightarrow \infty$. If $|z| < 1$, $\forall \epsilon$ if $N > \ln(1/\epsilon)/\ln(1/|z|)$ then $|z|^N < \epsilon$, hence $\lim_{N \rightarrow \infty} z^N = 0$ and $\sum_{n=0}^{\infty} z^n = 1/(1-z)$. If $|z| \geq 1$, z^N does not converge to zero and the series does not converge (following the script, if a series converges its general term must converge to zero).

- (10 points) Determine for what values of $x \in \mathbb{R}$ the series $\zeta(x) = \sum_{n \geq 1} \frac{1}{n^x}$ converges.

By Cauchy's condensation criterion, $\zeta(x)$ has the same nature as the series $\sum_{n \geq 1} 2^n a_{2^n}$, that is $\sum_{n \geq 1} 2^n \frac{1}{2^{nx}} = \sum_{n \geq 1} \frac{1}{2^{n(x-1)}}$ hence converges if and only if $\frac{1}{2^{x-1}} < 1 \Rightarrow x > 1$.

- (10 points) Find the Taylor series of $f(x, y) = (2-x)e^{x+2y}$, around the point $(-2, 1)$ up to order 2 (included).

We have up to order 2 $f(x_0 + a, y_0 + b) = f(x_0, y_0 + b) + a \partial_x f(x_0, y_0 + b) + \frac{1}{2} a^2 \partial_x^2 f(x_0, y_0 + b) + R = f(x_0, y_0) + b \partial_y f(x_0, y_0) + \frac{1}{2} b^2 \partial_y^2 f(x_0, y_0) + a \partial_x f(x_0, y_0) + ab \partial_x \partial_y f(x_0, y_0) + \frac{1}{2} a^2 \partial_x^2 f(x_0, y_0) + R'$. The derivatives are $f(-2, 1) = 4$, $\partial_x f = (-1 + 2 - x)e^{x+2y}|_{-2,1} = (1-x)e^{x+2y}|_{-2,1} = 3$, $\partial_x \partial_x f = (-1 + 1 - x)e^{x+2y}|_{-2,1} = 2$, $\partial_y f = (2-x)2e^{x+2y}|_{-2,1} = 8$, $\partial_y \partial_y f = (2-x)4e^{x+2y}|_{-2,1} = 16$, $\partial_x \partial_y f = [-2 + (2-x)2]e^{x+2y}|_{-2,1} = 2(1-x)e^{x+2y}|_{-2,1} = 6$.

Exercise 2 (Critical points, 30 points). *Find and classify all the critical points of:*

- (15 points) the function $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, $f(x, y) = x^2 y + 2x^2 + y^2$.

The critical points are solutions of $\partial_x f = 2xy + 4x = 2x(y+2) = 0$, $\partial_y f = x^2 + 2y = 0$, hence $(0, 0)$ and $(\pm 2, -2)$. The Hessians is $\begin{pmatrix} 2y+4 & 2x \\ 2x & 2 \end{pmatrix}$, hence $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}_{(0,0)}$, $\begin{pmatrix} 0 & 4 \\ 4 & 2 \end{pmatrix}_{(2,-2)}$, $\begin{pmatrix} 0 & -4 \\ -4 & 2 \end{pmatrix}_{(-2,-2)}$. As all the eigenvalues are strictly positive $(0, 0)$ is a local minimum. The characteristic polynomial for the other two critical points is $-\lambda(2-\lambda)-16 = \lambda^2 - 2\lambda - 16$ hence $\lambda_{\pm} = 1 \pm \sqrt{1+16}$, that is they are both saddle points.

- (15 points) the function $f: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, $f(x, y, z) = x^2 + xy + y^2 + 3yz^2 + 6z^2$.

We have $\partial_x f = 2x + y$, $\partial_y f = x + 2y + 3z^2$, $\partial_z f = 6yz + 12z$, hence at critical points $y = -2x$ and $x - 4x + 3z^2 = 0$ hence $x = z^2$ and $6(-2z^2)z + 12z = 0$ hence $z = 0, 1, -1$ and the critical points are $(0, 0, 0)$, $(1, -2, 1)$ and $(1, -2, -1)$. The Hessian is: $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 6z \\ 0 & 6z & 6y+12 \end{pmatrix}$ that is $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 12 \end{pmatrix}_{(0,0,0)}$, $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 6 \\ 0 & 6 & 0 \end{pmatrix}_{(1,-2,1)}$ and $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & -6 \\ 0 & -6 & 0 \end{pmatrix}_{(1,-2,-1)}$.

The characteristic polynomial and eigenvalues at the critical point $(0, 0, 0)$ is $(2-\lambda)[(2-\lambda)^2 - 1] = 0$ hence $\lambda = 2$ and $\lambda^2 - 4\lambda + 3 = 0$ that is $\lambda_{\pm} = 2 \pm \sqrt{4-3} = 2 \pm 1$, minimum.

For the other two critical points, the characteristic polynomial is $(2-\lambda)(2-\lambda)(-\lambda) - 36(2-\lambda) + \lambda = -\lambda^3 + 4\lambda^2 + 33\lambda - 72 = g(\lambda)$. As $g(-10) = 1000 + 400 - 330 - 72 > 0$, $g(0) = -72 < 0$, $g(4) = 33 \times 4 - 72 > 0$ and $g(10) = -1000 + 400 - 330 - 72 < 0$, we have one negative and two positive roots, hence both are saddle points.

Exercise 3 (Differential geometry, 40 points). *Consider the 2 dimensional manifold Σ embedded in $\mathbb{R}^3$ with standard metric $g_{ij} = \delta_{ij}$ and volume form $\omega = dx^1 \wedge dx^2 \wedge dx^3$ defined by the embedding map $\phi(s, t) = (s, t, s^2 t + 2s^2 + t^2)$.*

- (10 points) Compute the pushforward map $d\phi: T\Sigma \rightarrow T\mathbb{R}^3$, $d\phi = \phi_* = \sum \frac{\partial x^i}{\partial s^\mu} ds^\mu$.

We have $dx^1 = ds$, $dx^2 = dt$, $dx^3 = (2st + 4s)ds + (s^2 + 2t)dt$ hence $d\phi = \phi_* = \partial_{x^1} ds + \partial_{x^2} dt + \partial_{x^3} [(2st + 4s)ds + (s^2 + 2t)dt] = \begin{pmatrix} \partial_{x^1} & \partial_{x^2} & \partial_{x^3} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2st+4s & s^2+2t \end{pmatrix} \begin{pmatrix} ds \\ dt \end{pmatrix}$.

- (10 points) Compute the pulled back metric $G = \phi^*(g)$.

This is $G_{\mu\nu} ds^\mu ds^\nu = g_{ij} \frac{\partial x^i}{\partial s^\mu} \frac{\partial x^j}{\partial s^\nu} ds^\mu ds^\nu = ds^2 + dt^2 + [(2st + 4s)ds + (s^2 + 2t)dt]^2 = [1 + (2st + 4s)^2]ds^2 + [1 + (s^2 + 2t)^2]dt^2 + 2(2st + 4s)(s^2 + 2t)dsdt$

- (10 points) Compute the exterior normal ν . We fix the orientation by requiring that $\nu^3 > 0$ for $x = y = 0$.

We have $g(\nu, d\phi) = 0$ for all $V \in T_{(s,t)}\Sigma$. In components this means that $\forall \mu$, $g_{ij} \nu^i \frac{\partial x^j}{\partial s^\mu} = 0$ and $g_{ij} \nu^i \nu^j = 1$, that is $\nu^1 + \nu^3(2st + 4s) = 0$, $\nu^2 + \nu^3(s^2 + 2t) = 0$, $\nu_i \nu^i = 1$, thus $\nu = \frac{1}{\sqrt{1+(2st+4s)^2+(s^2+2t)^2}} \left[-(2st+4s)\partial_{x^1} - (s^2+2t)\partial_{x^2} + \partial_{x^3} \right]$.

- (10 points) Compute the induced volume form: compare $\phi^*(i_\nu \omega)$ with $\omega_\Sigma = \sqrt{\det G} ds \wedge dt$.

We have on the one hand $i_\nu \omega = \frac{-(2st+4s)}{\sqrt{\dots}} dx^2 \wedge dx^3 - \frac{-(s^2+2t)}{\sqrt{\dots}} dx^1 \wedge dx^3 + dx^1 \wedge dx^2$ hence $\phi^*(i_\nu \omega) = \frac{-(2st+4s)}{\sqrt{\dots}} dt \wedge ds - \frac{-(s^2+2t)}{\sqrt{\dots}} ds \wedge dt + ds \wedge dt = \frac{\sqrt{1+(2st+4s)^2+(s^2+2t)^2}}{\sqrt{\dots}} ds \wedge dt$. On the other hand $\omega_\Sigma = \left| \begin{matrix} 1+(2st+4s)^2 & (2st+4s)(s^2+2t) \\ (2st+4s)(s^2+2t) & 1+(s^2+2t)^2 \end{matrix} \right|^{1/2} ds \wedge dt$ and we note that $\left| \begin{matrix} 1+(2st+4s)^2 & (2st+4s)(s^2+2t) \\ (2st+4s)(s^2+2t) & 1+(s^2+2t)^2 \end{matrix} \right| = 1+(2st+4s)^2+(s^2+2t)^2$ hence, as expected, the two coincide.

Exercise 4 (Stirling's approximation, 20 points). *We define the Euler Gamma function $\Gamma(x) = \int_0^\infty dt t^{x-1} e^{-t}$ for $x \in (0, \infty)$.*

- (5 points) Prove that, for $n \in \mathbb{N}$ we have $\Gamma(n+1) = n!$.

We have, by integration by parts $\Gamma(1) = \int_0^\infty dt e^{-t} = (-e^{-t})|_0^\infty = 1$ and $\Gamma(n+1) = \int_0^\infty dt t^n e^{-t} = (-t^n e^{-t})|_0^\infty + n \int_0^\infty dt t^{n-1} e^{-t} = n\Gamma(n)$.

For any function $f: \mathbb{R} \rightarrow \mathbb{R}$ having a unique minimum t_0 , that is $f'(t_0) = 0$ and $f''(t_0) > 0$, we have the asymptotic Laplace approximation formula: if $a < t_0 < b$ then $\int_a^b dt e^{-f(t)} \approx \sqrt{\frac{2\pi}{f''(t_0)}} e^{-f(t_0)}$.

- (10 points) Using the above formula, deduce an approximation of $n!$ for $n > 1$.

We have for $x > 1$, $\Gamma(x) = \int_0^\infty dt e^{-f(t)}$, $f(t) = -(x-1)\ln t + t$, $f'(t) = -\frac{x-1}{t} + 1$ hence $t_0 = x-1$, $f''(t_0) = \frac{x-1}{t_0^2} = \frac{1}{x-1}$, thus $\Gamma(x) = \sqrt{2\pi(x-1)} e^{-(x-1)\ln(x-1)-(x-1)} = \sqrt{2\pi(x-1)} \left(\frac{x-1}{e}\right)^{x-1}$ and $n! = \Gamma(n+1) \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$.

- (5 points) Approximating f by a second order Taylor expansion around its minimum, justify the Laplace approximation formula.

$\int_a^b dt e^{-f(t)} = \int_{a-t_0}^{b-t_0} dt e^{-f(t_0+t)} \approx \int_{-\infty}^{\infty} dt e^{-f(t_0)-\frac{1}{2}t^2 f''(t_0)+\dots} \approx \sqrt{\frac{2\pi}{f''(t_0)}} e^{-f(t_0)}$.

B.1.3 Sample 3

Exercise 1 (Elementary exercises, 30 points). *Solve the following tasks:*

- (10 points) Compute the definite integral:

$$\int_0^\infty dt e^{-t} \sin(t) .$$

Any correct way to obtain: $\int_0^\infty dt e^{-t} \sin(t) = \frac{1}{2i} \int_0^\infty dt (e^{-(1-i)t} - e^{-(1+i)t}) = \frac{1}{2i} \left[\frac{1}{1-i} - \frac{1}{1+i} \right] = \frac{1}{2} .$

- (10 points) Find the primitive of:

$$\int dx \sqrt{1 - \frac{1}{x}} ,$$

in terms of simple functions.

Any correct way to obtain: $\int dx \sqrt{1 - \frac{1}{x}} = \sqrt{x^2 - x} - \ln(\sqrt{x} + \sqrt{x-1}) + C$ as $\left[\sqrt{x^2 - x} - \ln(\sqrt{x} + \sqrt{x-1}) \right]' = \frac{2x-1}{2\sqrt{x^2-x}} - \frac{\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{x-1}}}{\sqrt{x} + \sqrt{x-1}} = \frac{2x-2}{2\sqrt{x^2-x}} = \sqrt{\frac{x-1}{x^2-x}} .$ One way to do it $\int dx \frac{1}{\sqrt{x}} \sqrt{x-1} = 2 \int d(\sqrt{x}) \sqrt{x-1} \xrightarrow{\sqrt{x}=t} 2 \int dt \sqrt{t^2-1} \xrightarrow{t=\cosh u} 2 \int du (\sinh u)^2 = \int du \frac{e^{2u} + e^{-2u} - 2}{2} = \frac{1}{4} e^{2u} - \frac{1}{4} e^{-2u} - u$, and observe that $2\sqrt{x} = e^u + e^{-u} \Rightarrow e^{2u} - 2\sqrt{x}e^u + 1 = 0 \Rightarrow e^u = \sqrt{x} \pm \sqrt{x-1} .$

- (10 points) We denote $\mathcal{C}([0,1], \mathbb{R})$ the space of real valued continuous functions on $[0,1]$. This is Banach space with norm $\|x\|_\infty = \sup_{x \in [0,1]} |x(t)|$. Compute the Fréchet differential of the function $F : \mathcal{C}([0,1], \mathbb{R}) \rightarrow \mathbb{R}$:

$$F(x) = x(0) \int_0^1 dt \sin(x(t)) .$$

We have: $dF|_x(h) = h(0) \int_0^1 dt \sin(x(t)) + x(0) \int_0^1 dt \cos(x(t)) h(t)$. In order to prove this: $|F(x+h) - f(x) - dF|_x(h)| = \left| (x(0) + h(0)) \int_0^1 dt \sin(x(t) + h(t)) - x(0) \int_0^1 dt \sin(x(t)) - h(0) \int_0^1 dt \sin(x(t)) - x(0) \int_0^1 dt \cos(x(t)) h(t) \right| \leq \left| x(0) \int_0^1 dt [\sin(x(t) + h(t)) - \sin(x(t)) - \cos(x(t)) h(t)] \right| + \left| h(0) \int_0^1 dt [\sin(x(t) + h(t)) - \sin(x(t))] \right| \leq |x(0)| \int_0^1 dt \left| \sin(x(t)) [\cos(h(t)) - 1] + \cos(x(t)) [\sin(h(t)) - h(t)] \right| + |h(0)| \int_0^1 dt \left| \sin(x(t)) [\cos(h(t)) - 1] + \cos(x(t)) \sin(h(t)) \right| . Now, $|h(0)| \leq \|h\|_\infty$ and $|\sin(h(t))| \leq |h(t)| \leq \|h\|_\infty$. Also $|\cos(h(t)) - 1| = \left| -2 \left(\sin \frac{h(t)}{2} \right)^2 \right| \leq \frac{1}{2} \|h\|_\infty^2$. From the mean value theorem, there exist $\beta \in [0, \alpha]$ such that $\sin(\alpha) = \alpha \cos(\beta)$, hence $|\sin(\alpha) - \alpha| = |\alpha| |\cos(\beta) - 1| = 2|\alpha| \left| \left(\sin \frac{\beta}{2} \right)^2 \right| \leq \frac{1}{2} \|\alpha\| \|\beta\|^2 \leq \frac{1}{2} \|\alpha\|^3$, hence $\|\sin(h(t)) - h(t)\| \leq \frac{1}{2} \|h(t)\|^3 \leq \frac{1}{2} \|h\|_\infty^3$. Gathering everything we have $|F(x+h) - f(x) - dF|_x(h)| \leq \|x\|_\infty \left(\frac{1}{2} \|h\|_\infty^2 + \frac{1}{2} \|h\|_\infty^3 \right) + \|h\|_\infty \left(\frac{1}{2} \|h\|_\infty^2 + \|h\|_\infty \right) \Rightarrow \lim_{\|h\|_\infty \rightarrow 0} \frac{|F(x+h) - f(x) - dF|_x(h)|}{\|h\|_\infty} = 0$.$

Exercise 2 (A sequence of integrals, 30 points). *Consider the sequence of integrals:*

$$I_n = \int_0^1 dx (1-x^2)^{n/2} .$$

- (5 points) Compute I_0 and I_1 .

We have $I_0 = 1$ and $I_1 = \int_0^1 dx (1-x^2)^{1/2} \xrightarrow{x=\sin \theta} \int_0^{\pi/2} d\theta \cos^2 \theta = \int_0^{\pi/2} d\theta \frac{1+\cos(2\theta)}{2} = \frac{\pi}{4} + \int_0^{\pi/2} \frac{d\varphi}{2} \cos \varphi = \frac{\pi}{4} + \frac{1}{2} \sin \varphi \Big|_0^{\pi/2} \Rightarrow I_1 = \frac{\pi}{4} .$

- (10 points) Prove that $I_{n+2} = \frac{n+2}{n+3} I_n$.

We have $I_{n+2} = \int_0^1 dx (1-x^2)^{1+\frac{n}{2}} = x(1-x^2)^{1+\frac{n}{2}} \Big|_0^1 - \int_0^1 dx x \frac{d}{dx} \left[(1-x^2)^{1+\frac{n}{2}} \right] = (n+2) \int_0^1 dx x^2 (1-x^2)^{\frac{n}{2}} = (n+2) [I_n - I_{n+2}]$, hence $I_{n+2} = \frac{n+2}{n+3} I_n$.

- (5 points) Prove that $I_{n-1} \geq I_n \geq I_{n+1}$. Prove that $\lim_{n \rightarrow \infty} \frac{I_n}{I_{n-1}} = 1$.

As $(1-x^2)^{(n-1)/2} \geq (1-x^2)^{n/2} \geq (1-x^2)^{(n+1)/2}$ the inequality follows. We then have $1 \geq \frac{I_n}{I_{n-1}} \geq \frac{I_{n+1}}{I_n} = \frac{n+1}{n+2} \xrightarrow{n \rightarrow \infty} 1$, and we conclude by the sandwich theorem.

- (10 points) Using the previous results, compute the limit $\lim_{n \rightarrow \infty} B_n$, where:

$$B_n = \prod_{k=1}^n \frac{(2k)^2}{(2k-1)(2k+1)} .$$

We have $\frac{2}{1} \cdot \frac{2}{3} = 2I_2$ and for $k \geq 2$, $\frac{2k}{2k-1} = \frac{I_{2k-3}}{I_{2k-1}}$, $\frac{2k}{2k+1} = \frac{I_{2k}}{I_{2k-2}}$ thus $B_n = 2I_2 \prod_{k=2}^n \frac{I_{2k-3}}{I_{2k-1}} \frac{I_{2k}}{I_{2k-2}} = 2I_2 \frac{I_1}{I_2} \frac{I_{2n}}{I_{2n-1}} = \frac{\pi}{2} \frac{I_{2n}}{I_{2n-1}}$ $\lim_{n \rightarrow \infty} B_n = \frac{\pi}{2}$.

Exercise 3 (Lagrange inversion, 30 points). *The inverse function theorem states that if $f : U \rightarrow \mathbb{R}$ is continuously differentiable and $f(x_0) = y_0$ for some $x_0 \in U$ with $f'(x_0) \neq 0$, then there exist open sets U_0 and V_0 with $x_0 \in U_0 \subset U$ and $y_0 \in V_0$ and a continuously differentiable function $g : V_0 \rightarrow U_0$ such that for $x \in U_0$ and $y \in V_0$, $f(x) = y \Leftrightarrow x = g(y)$. We consider the particular case when f is indefinitely differentiable and admits a convergent Taylor series representation in some neighbourhood of x_0 . Our aim is to prove that the inverse function g is the sum of the series:*

$$g(y) = x_0 + \sum_{k \geq 1} \frac{(y - y_0)^k}{k!} \left\{ \frac{d^{k-1}}{dx^{k-1}} \left[\frac{x - x_0}{f(x) - y_0} \right] \right\}_{x=x_0} , \quad (2.1)$$

whenever it converges.

- (10 points) Assuming Eq. (2.1) proven, write an explicit power series representation for the function $T_p(y)$, the solution of the equation $T_p(y) = 1 + y T_p(y)^p$ respecting $T_p(0) = 1$, for any $p \in \mathbb{N}$. What is the radius of convergence of this series?

We have for low values of p : $T_0(y) = 1$; $T_1(y) = \frac{1}{1-y} = \sum_{n \geq 0} y^n$ and $0 = 1 - T_2(y) + y T_2(y)^2 \Rightarrow T_2(y) = \frac{1 - \sqrt{1-4y}}{2y}$. For $p \geq 2$, we define $f(x) = \frac{x-1}{x^p}$, and we aim to invert $f(x) = y$ in the neighbourhood of $x_0 = 1$, $y_0 = 0$. By the inversion formula $T_p(y) = g(y) = 1 + \sum_{k \geq 1} \frac{y^k}{k!} \left\{ \frac{d^{k-1}}{dx^{k-1}} \left[\frac{x-1}{f(x)} \right]^k \right\}_{x=1} = 1 + \sum_{k \geq 1} \frac{y^k}{k!} [x^p]_{x=1}^{(k-1)} = 1 + y + \sum_{k \geq 2} \frac{y^k}{k!} (pk - 1) \dots (pk - k + 2) = 1 + \sum_{k \geq 1} \frac{y^k}{k!} \frac{(pk)!}{[(p-1)k+1]!} = \sum_{k \geq 0} \frac{1}{pk+1} \binom{pk+1}{k} y^k$. By the ratio criterion, $r = \lim_k \left| \frac{g_k}{g_{k+1}} \right|$, hence $r = \lim_k \left(\frac{(pk)!}{k![(p-1)k+1]!} \cdot \frac{(k+1)![(p-1)(k+1)+1]!}{[p(k+1)]!} \right) = \lim_k \frac{(k+1)[(p-1)k+2] \dots [(p-1)k+p]}{(pk+1)(pk+2) \dots (pk+p)} = \frac{(p-1)^{p-1}}{p^p}$. This can also be obtained by noting that $f' = -(p-1)x^{-p} + px^{-p-1}$ and $f' = 0 \Rightarrow x = \frac{p}{p-1}$ and $f\left(\frac{p}{p-1}\right) = \frac{\frac{p}{p-1}-1}{\left(\frac{p}{p-1}\right)^p} = \frac{(p-1)^{p-1}}{p^p}$, hence the inversion must fail at $y = \frac{(p-1)^{p-1}}{p^p}$.

In order to prove Eq. (2.1) we proceeded as follows. Without loss of generality we can always consider $x_0 = y_0 = 0$, that is invert $f(x) = y$, given that $f(0) = 0$ and $f'(0) \neq 0$.

- (5 points) We consider the convergent Taylor series expansion of f around zero, $f(x) = \sum_{n \geq 0} f_n x^n$ and we assume that the inverse function admits also a convergent Taylor expansion around 0, $g(y) = \sum_{n \geq 0} g_n y^n$. Compute directly g_0, g_1, g_2 and g_3 in terms of f_0, f_1, f_2, f_3 .

As $f(0) = 0$ and $f'(0) \neq 0$ we conclude that $f_0 = 0$ and $f_1 \neq 0$; as $g(0) = 0$ we conclude $g_0 = 0$. We solve $y = f(g(y))$ order by order in y : $y = f_1(g_1 y + g_2 y^2 + g_3 y^3) + f_2(g_1 y + g_2 y^2 + g_3 y^3)^2 + f_3(g_1 y + g_2 y^2 + g_3 y^3)^3$ hence $f_1 g_1 = 1$, $f_1 g_2 + f_2 g_1^2 = 0$, $f_1 g_3 + 2f_2 g_1 g_2 + f_3 g_1^3$ thus $g_1 = \frac{1}{f_1}$, $g_2 = -\frac{f_2}{f_1} g_1^2 = -\frac{f_2}{f_1^3}$, $g_3 = -\frac{f_3}{f_1} g_1^3 - 2\frac{f_2}{f_1} g_1 g_2 = -\frac{f_3}{f_1^4} + 2\frac{f_2^2}{f_1^5}$.

The derivative of a formal power series $h(x) = \sum_{n \in \mathbb{Z}} h_n x^n$ is $h'(x) = \sum_{n \in \mathbb{Z}} h_n n x^{n-1}$. We denote $[h(x)]_{x^n}$ the coefficient of the term x^n in the series, that is h_n , and we stress that n can be negative. For instance: $[\sum_{n > 0} h_n x^n + \sum_{n < 0} h_n x^n]_{x^{-1}} = 0$.

- (5 points) Prove that, if $h(x) = \sum_{n=-\infty}^{\infty} h_n x^n$ then $[h'(x)]_{x^{-1}} = 0$. Prove that, if $f(x) = \sum_{n \geq 1} f_n x^n$ with $f_1 \neq 0$ and $i \in \mathbb{Z}$, then:

$$[f(x)^i f'(x)]_{x^{-1}} = \begin{cases} 1, & \text{if } i = -1 \\ 0, & \text{if } i \neq -1 \end{cases}.$$

You can assume that $\sum_{n \geq 1} f_n x^n$ has non zero radius of convergence.

The first point is trivial as $h'(x) = \sum_{n=-\infty}^{\infty} h_n n x^{n-1}$, hence the coefficient of x^{-1} is 0. For the second point, if $i \neq -1$ then $f(x)^i f'(x) = \frac{1}{i+1} \frac{d}{dx} f^{i+1}(x)$ and $f(x)$ and all its powers, positive or negative, can be expanded as $\sum_{n=-\infty}^{\infty} h_n x^n$. For $i = -1$ we have $\frac{1}{f(x)} f'(x) = \frac{f_1 + f_2 2x + \dots}{f_1 x + f_2 x^2 + \dots} = \frac{1}{x} \frac{1 + 2\frac{f_2}{f_1}x + \dots}{1 + \frac{f_2}{f_1}x + \dots} = \frac{1}{x} + O(x^0)$.

- (5 points) Prove that if $f(x) = \sum_{n \geq 1} f_n x^n$ and $g(x) = \sum_{n \geq 1} g_n x^n$ are inverse to each other, $x = g(f(x))$, then:

$$[g(x)]_{x^n} = \frac{1}{n} \left[\frac{1}{f(x)^n} \right]_{x^{-1}},$$

and conclude. Hint: we observe that $\frac{d}{dx} x = 1$.

We have $x = g(f(x)) = \sum_{p \geq 1} g_p f(x)^p \xrightarrow{\frac{d}{dx}} 1 = \sum_{p \geq 1} g_p p f(x)^{p-1} f'(x) \xrightarrow{\frac{f(x)^n}{f(x)^n} \times} \frac{1}{f(x)^n} = \sum_{p \geq 1} g_p p f(x)^{p-1-n} f'(x) \Rightarrow \left[\frac{1}{f(x)^n} \right]_{x^{-1}} = n g_n$, therefore $[g(x)]_{x^n} = \frac{1}{n} \left[\frac{1}{f(x)^n} \right]_{x^{-1}} = \frac{1}{n} \left[\frac{x^n}{f(x)^n} \right]_{x^{-n-1}}$ hence $g(x) = \sum_{n \geq 1} x^n \frac{1}{n} \left[\frac{1}{(n-1)!} \frac{d^{n-1}}{dx^{n-1}} \left(\frac{x^n}{f(x)^n} \right) \right]_{x=0}$.

- (5 points) Rederive starting from $[g(x)]_{x^n} = \frac{1}{n} \left[\frac{1}{f(x)^n} \right]_{x^{-1}}$ the formulas for g_0, g_1, g_2 and g_3 obtained before.

We have: $g_1 = \left[\frac{1}{f_1 x + f_2 x^2 + f_3 x^3} \right]_{x^{-1}} = \left[\frac{1}{f_1 x} \frac{1}{1 + O(x)} \right]_{x^{-1}} = \frac{1}{f_1}$ and $g_2 = \frac{1}{2} \left[\frac{1}{(f_1 x + f_2 x^2 + f_3 x^3)^2} \right]_{x^{-1}} = \frac{1}{2} \left[\frac{1}{f_1^2 x^2} \frac{1}{(1 + \frac{f_2}{f_1}x + \frac{f_3}{f_1}x^2)^2} \right]_{x^{-1}} = \frac{1}{2} \left[\frac{1}{f_1^2 x^2} (1 - 2\frac{f_2}{f_1}x) \right]_{x^{-1}} = -\frac{f_2}{f_1^3}$ and finally $g_3 = \frac{1}{3} \left[\frac{1}{(f_1 x + f_2 x^2 + f_3 x^3)^3} \right]_{x^{-1}} = \frac{1}{3} \left[\frac{1}{f_1^3 x^3} \frac{1}{(1 + \frac{f_2}{f_1}x + \frac{f_3}{f_1}x^2)^3} \right]_{x^{-1}} = \frac{1}{3} \left[\frac{1}{f_1^3 x^3} \frac{1}{(1 + 3\frac{f_2}{f_1}x + 3\frac{f_2^2}{f_1^2}x^2 + 3\frac{f_3}{f_1}x^2)} \right]_{x^{-1}} = \frac{1}{3} \left[\frac{1}{f_1^3 x^3} (1 - 3\frac{f_2}{f_1}x - 3\frac{f_2^2}{f_1^2}x^2 - 3\frac{f_3}{f_1}x^2 + 9\frac{f_2^2}{f_1^2}x^2) \right]_{x^{-1}} = -\frac{f_3}{f_1^4} + 2\frac{f_2^2}{f_1^5}$, where we used $(a+b+c)^3 = \sum_{n_1, n_2, n_3 \geq 0}^{n_1+n_2+n_3=3} \frac{3!}{n_1!n_2!n_3!} a^{n_1} b^{n_2} c^{n_3}$.

Exercise 4 (Envelope of a family of curves, 30 points). A family of curves in $\mathbb{R}^2$ is defined implicitly by an equation $F_\tau(x, y) = 0$, where τ is a parameter. A curve in the family, parametrized either as $y_\tau(x)$ or $x_\tau(y)$, is obtained by solving the equation at fixed τ and we switch between different curves in the family by varying τ . For example the curves in the family $F_\tau(x, y) = x - y - \tau = 0$ can be parametrized either as $y_\tau(x) = x - \tau$ or as $x_\tau(y) = y + \tau$ and are straight lines. In order for the implicit function theorem to guarantee that the equation $F_\tau(x, y) = 0$ can be solved at fixed τ , we assume that either $\frac{\partial F_\tau}{\partial y} \neq 0$ or $\frac{\partial F_\tau}{\partial x} \neq 0$.

We can treat the parameter τ as an additional variable and relabel $F_\tau(x, y) \equiv F(x, y; \tau)$, and we assume F is smooth in all its arguments in some appropriate domain. The envelope of the family of curves, if it exists, is a curve $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$ in $\mathbb{R}^2$ which is tangent to all the curves in the family. Our aim is to prove that the envelope is the solution of the system of equations:

$$F(x, y; \tau) = 0, \quad \frac{\partial F}{\partial \tau}(x, y; \tau) = 0. \quad (2.2)$$

B. EXAMS

- (10 points) A cannon launches projectiles from the origin $(0,0) \in \mathbb{R}^2$ with initial speed v at an arbitrary angle α with the horizontal, in a constant gravitational field with acceleration $-g$ along the vertical axis. Assuming Eq. (2.3) proven, determine the “safety curve”, that is the curve which separates the points which can be reached by the projectile from those which can not.

Denoting t the time, we have $x = v \cos \alpha t$, $y = v \sin \alpha t - \frac{g}{2} t^2 = x \tan \alpha - \frac{g}{2} \frac{x^2}{v^2 \cos^2 \alpha}$ that is $y = x \tan \alpha - \frac{g}{2} \frac{x^2}{v^2} (1 + \tan^2 \alpha)$, hence the trajectories are the family of parabolas $F(x, y; \tau) = y - \tau x + \frac{g}{2} \frac{x^2}{v^2} (1 + \tau^2) = 0$. The “safety curve” (a.k.a. the parabola of safety) is the envelope of the trajectories and is determined by $y - \tau x + \frac{g}{2} \frac{x^2}{v^2} (1 + \tau^2) = 0$, $-x + g \frac{x^2}{v^2} \tau = 0 \Rightarrow \tau = \frac{v^2}{gx} \Rightarrow y = \frac{v^2}{g} - \frac{g}{2} \frac{x^2}{v^2} (1 + \frac{v^4}{g^2 x^2})$ hence $y = \frac{v^2}{2g} - \frac{g}{2} \frac{x^2}{v^2}$.

- (10 points) Assuming Eq. (2.3) proven, consider the family of curves defined by the equation:

$$F(x, y; \tau) = \frac{x^2}{\tau} + \frac{y^2}{1 - \tau} - 1 = 0, \quad 0 < \tau < 1.$$

Describe this family of curves and determine the envelope.

The curves are ellipses centered at the origin with semiaxes $a = \sqrt{\tau}$ and $b = \sqrt{1 - \tau}$, that is such that $a^2 + b^2 = 1$. The semiaxes are the projections of the points on a circle of radius 1 on the coordinate axes. The envelope is determined by the system of equations $F(x, y; \tau) = \frac{x^2}{\tau} + \frac{y^2}{1 - \tau} - 1 = 0$ and $\partial_\tau F(x, y; \tau) = -\frac{x^2}{\tau^2} + \frac{y^2}{(1 - \tau)^2} = 0$. The second equation leads to $y^2 \tau^2 - x^2 (1 - \tau)^2 = 0 \Rightarrow |y| \tau = |x| (1 - \tau) \Rightarrow \tau = \frac{|x|}{|x| + |y|}$, and substituting into the first we get $\frac{x^2}{|x|} (|x| + |y|) + \frac{y^2}{|y|} (|x| + |y|) - 1 = 0 \Rightarrow (|x| + |y|)^2 = 1 \Rightarrow |x| + |y| = \pm 1$. The envelope is a square centered at the origin, tilted by 45 degrees with respect to the axes.

- (10 points) In order to prove Eq. (2.3), consider the system of equations:

$$F(x, y; \tau) = 0, \quad \frac{\partial F}{\partial \tau}(x, y; \tau) = 0,$$

and assume (x_0, y_0, τ_0) is a solution. Give a sufficient condition such that this system admits locally around (x_0, y_0, τ_0) a solution $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$.

We denote $x_\tau(y)$ the solution of the equation $F_\tau(x, y) \equiv F(x, y; \tau) = 0$ corresponding to the fixed parameter τ and such that $x_{\tau_0}(y_0) = x_0$. This curve is an embedded submanifold defined by a constraint. Prove that the envelope $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$ is tangent to the curve $x_\tau(y)$ at the point $y = y^{\text{en}}(\tau)$.

The differential of the system of equations is: $dH|_{x_0, y_0, \tau_0} = \begin{bmatrix} \frac{\partial F}{\partial \tau} \\ \frac{\partial^2 F}{\partial \tau^2} \end{bmatrix} \begin{bmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial^2 F}{\partial x \partial \tau} \end{bmatrix} \begin{bmatrix} \frac{\partial F}{\partial y} \\ \frac{\partial^2 F}{\partial y \partial \tau} \end{bmatrix}$, $d_\perp H|_{x_0, y_0, \tau_0} = \begin{pmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \\ \frac{\partial^2 F}{\partial x \partial \tau} & \frac{\partial^2 F}{\partial y \partial \tau} \end{pmatrix}_{x_0, y_0, \tau_0}$, and a sufficient condition for the system to admit locally a solution is that $d_\perp H|_{x_0, y_0, \tau_0}$ is invertible, hence has rank 2. As $F(x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau) = 0$, we have $0 = \frac{d}{d\tau} F(x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau) \Big|_\tau = \frac{\partial F}{\partial x} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dx^{\text{en}}}{d\tau} + \frac{\partial F}{\partial y} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dy^{\text{en}}}{d\tau} + \frac{\partial F}{\partial \tau} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} = 0$. As both $F(x_\tau(y), y, \tau) = 0$ and $F(x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau) = 0$, we have $x_\tau(y) \Big|_{y=y^{\text{en}}(\tau)} = x^{\text{en}}(\tau)$, that is the envelope intersects the curve $x_\tau(y)$ at $y = y^{\text{en}}(\tau)$. The curve $x_\tau(y)$ is an embedded submanifold and the normal to this manifold is the form $dF_\tau|_{x_\tau(y), y} = \frac{\partial F_\tau}{\partial x} \Big|_{x_\tau(y), y} dx|_{x_\tau(y), y} + \frac{\partial F_\tau}{\partial y} \Big|_{x_\tau(y), y} dy|_{x_\tau(y), y}$, where we emphasised that this is an element of in $T^*_{x_\tau(y), y} \mathbb{R}^2$. Evaluating the normal form dF_τ at the point $(x_\tau(y), y)_{y=y^{\text{en}}(\tau)}$ on the vector tangent to the envelope at the same point $(x, y) = (x^{\text{en}}(\tau), y^{\text{en}}(\tau))$, which is $\frac{dx^{\text{en}}}{d\tau} \partial_x|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)} + \frac{dy^{\text{en}}}{d\tau} \partial_y|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)}$ and using $F_\tau(x, y) = F(x, y; \tau)$ we have $dF_\tau|_{x_\tau(y), y} \Big|_{y=y^{\text{en}}(\tau)} \left(\frac{dx^{\text{en}}}{d\tau} \partial_x|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)} + \frac{dy^{\text{en}}}{d\tau} \partial_y|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)} \right) = \frac{\partial F}{\partial x} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dx^{\text{en}}}{d\tau} + \frac{\partial F}{\partial y} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dy^{\text{en}}}{d\tau} = -\frac{\partial F}{\partial \tau} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau}$, and $\frac{\partial F}{\partial \tau} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} = 0$ as $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$ is a solution of $F(x, y; \tau) = 0$, $\partial_\tau F(x, y; \tau) = 0$.

B.1.4 Sample 4

Exercise 1 (Elementary exercises, 40 points). *Solve the following tasks:*

- (10 points) Compute the limit:

$$\lim_{x \rightarrow 2} \frac{x^3 - 6x^2 + 12x - 8}{\sin(\pi x) + 2\pi - x\pi}.$$

First observe it is an indeterminate 0/0. Method 1: Taylor expand for $x = 2 + \epsilon$, $\frac{\epsilon^3}{\sin(2\pi + \epsilon\pi) - \epsilon\pi} = -\frac{\epsilon^3}{\frac{1}{3!}\epsilon^3\pi^3 + O(\epsilon^4)} = -\frac{6}{\pi^3}$. Method 2: Iterated L'Hôpital rule $\lim_{x \rightarrow 2} \frac{x^3 - 6x^2 + 12x - 8}{\sin(\pi x) + 2\pi - x\pi} = \frac{3x^2 - 12x + 12}{\pi \cos \pi x - \pi} = \lim_{x \rightarrow 2} \frac{6x - 12}{-\pi^2 \sin \pi x} = \lim_{x \rightarrow 2} \frac{6}{-\pi^3 \cos \pi x} = -\frac{6}{\pi^3}$.

- (10 points) Find the general term of the sequence x_n defined by the recursion:

$$x_{n+2} = 4x_{n+1} - 4x_n, \quad x_0 = 5, \quad x_1 = 14.$$

The characteristic equation is $r^2 - 4r + 4$ hence the root is $r = 2$ with multiplicity 2. The general term is $\alpha 2^n + \beta n 2^n$, and the initial conditions fix $\alpha = 5, \beta = 2$, hence $x_n = 5 \cdot 2^n + n \cdot 2^{n+1}$.

- (10 points) Let $f: [0, 1] \rightarrow [0, 1]$ be a function such that $f(f(x) + y) = f(x) + f(y)$ whenever both sides are defined. Denote $f(0) = \alpha$ and, considering the sequence $f(0), f \circ f(0), f \circ f \circ f(0)$ and so on, prove that $f(f(x)) = f(x)$ for any $x \in [0, 1]$. Give an example of a function with this property.

If $\alpha = 0$, then set $y = 0$ and conclude that we have $f(f(x)) = f(f(x) + 0) = f(x) + f(0) = f(x)$. Now, assume $0 < \alpha$ then $(f \circ f)(0) = f(f(0) + 0) = f(0) + f(0) = 2\alpha$ and $(f \circ f \circ f)(0) = f[f(f(0)) + 0] = f(f(0)) + f(0) = 3\alpha$. By induction $\underbrace{f \circ \dots \circ f(0)}_{n \text{ times}} = n\alpha$ as $f[\underbrace{f \circ \dots \circ f(0)}_{n-1 \text{ times}}] = f[\underbrace{f \circ \dots \circ f(0)}_{n-1 \text{ times}} + 0] = \underbrace{f \circ \dots \circ f(0)}_{n-1 \text{ times}} + f(0) = (n-1)\alpha + \alpha$. But for n large enough $n\alpha > 1$, which contradicts the hypothesis that f takes values in $[0, 1]$. Examples: $f(x) = x$, $f(x) = 0$ or $f(x) = 0$ for $0 \leq x \leq \beta$ and $f(x) = 1$ for $\beta < x \leq 1$.

- (10 points) We denote $M_{N \times N}(\mathbb{C})$ the set of $N \times N$ matrices with complex entries. Find the Fréchet differentials of the functions $F_n: M_{N \times N}(\mathbb{C}) \rightarrow M_{N \times N}(\mathbb{C})$, $F_n(A) = A^n$ for $n = 1, 2$ and arbitrary $n \in \mathbb{N} \setminus \{0\}$. Note: You do not need to check all the conditions of the definition, it is enough to find the correct differentials.

The differentials $dF|_A(\cdot)$ are linear operators which takes matrices to matrices. We use $F(A+h) = F(A) + dF|_A(h) + O(h^2)$ to find them. $(A+h) - A = h \Rightarrow dF|_A(h) = h$, $(A+h)^2 - A^2 = Ah + hA + h^2 \Rightarrow dF|_A(h) = Ah + hA$ and the general result is $(A+h)^n - A^n = \sum_{p=0}^{n-1} A^p h A^{n-1-p} + O(h^2) \Rightarrow dF_n|_A(h) = \sum_{p=0}^{n-1} A^p h A^{n-1-p}$.

Exercise 2 (Intersecting surfaces, 20 points). *The cylinder $x^2 + y^2 = 1$ and the surface $2x^2 + 4y^4 + z = 3$ intersect along a curve. Find the points along this curve furthest from and closest to the origin.*

We want to find the extrema of the function $x^2 + y^2 + z^2$ with the given constraints. We consider the Lagrange function $x^2 + y^2 + z^2 - \lambda(x^2 + y^2 - 1) - \mu(2x^2 + 4y^4 + z - 3)$. We identify critical points subjected to constraints by solving the system: (i): $x^2 + y^2 - 1 = 0$, (ii): $2x^2 + 4y^4 + z - 3 = 0$, (iii): $x - \lambda x - 2\mu x = 0$, (iv): $y - \lambda y - 8\mu y^3 = 0$, (v): $2z - \mu = 0$ and changing variable to $1 - \lambda = \lambda'$ and dropping the prime we have (i): $x^2 + y^2 - 1 = 0$, (ii): $2x^2 + 4y^4 + z - 3 = 0$, (iii): $(\lambda - 2\mu)x = 0$, (iv): $(\lambda - 8\mu y^2)y = 0$, (v): $2z - \mu = 0$. Looking at (iii):

- $x = 0$ then (i): $y = \pm 1$ hence (ii): $z = -1$ hence (v): $\mu = -2$ and (iv): $\lambda = -16$.

- else (*) : $\lambda = 2\mu$ hence (iv): $(\lambda - 8\mu y^2)y = 0$ then:

- $y = 0$ then (i): $x = \pm 1$ hence (ii): $z = 1$ and (v): $\mu = 2$ finally $\lambda = 4$

- else (a): $\lambda = 8\mu y^2$ which together with (*) leads to (b): $\mu = 4\mu y^2$ then:

$$* \quad \mu = 0 \text{ and } (*) : \lambda = 0 \text{ and } (v) : z = 0 \text{ and } (ii) - 2(i) : 4y^4 - 2y^2 - 1 = 0 \text{ solved by } y^2 = \frac{2 + \sqrt{20}}{2} > 1 \text{ and can not solve } x^2 + y^2 = 1$$

$$* \quad y = \pm \frac{1}{2} \text{ and } (i) : x = \pm \sqrt{\frac{3}{4}} \text{ hence } (ii) : z = \frac{5}{4} \text{ thus } (v) : \mu = \frac{5}{2} \text{ and } (*) : \lambda = 5$$

We have 2 points at minimal distance $x^2 + y^2 + z^2 = 2$ and 4 points at maximal distance $x^2 + y^2 + z^2 = 1 + \frac{25}{16}$.

Exercise 3 (A sequence of integrals, 30 points). *Consider the sequence of integrals:*

$$I_n = \int_0^1 dx (1 - x^2)^{n/2}.$$

- (5 points) Compute I_0 and I_1 .

We have $I_0 = 1$ and $I_1 = \int_0^1 dx (1 - x^2)^{1/2} \xrightarrow{x=\sin \theta} \int_0^{\pi/2} d\theta \cos^2 \theta = \int_0^{\pi/2} d\theta \frac{1 + \cos(2\theta)}{2} = \frac{\pi}{4} + \int_0^{\pi/2} \frac{d\varphi}{2} \cos \varphi = \frac{\pi}{4} + \frac{1}{2} \sin \varphi|_0^{\pi/2} \Rightarrow I_1 = \frac{\pi}{4}$.

- (10 points) Prove that $I_{n+2} = \frac{n+2}{n+3} I_n$.

We have $I_{n+2} = \int_0^1 dx (1 - x^2)^{1+\frac{n}{2}} = x(1 - x^2)^{1+\frac{n}{2}} \Big|_0^1 - \int_0^1 dx x \frac{d}{dx} \left[(1 - x^2)^{1+\frac{n}{2}} \right] = (n+2) \int_0^1 dx x^2 (1 - x^2)^{\frac{n}{2}} = (n+2)[I_n - I_{n+2}]$, hence $I_{n+2} = \frac{n+2}{n+3} I_n$.

- (5 points) Prove that $I_{n-1} \geq I_n \geq I_{n+1}$. Prove that $\lim_{n \rightarrow \infty} \frac{I_n}{I_{n-1}} = 1$.

As $(1 - x^2)^{(n-1)/2} \geq (1 - x^2)^{n/2} \geq (1 - x^2)^{(n+1)/2}$ the inequality follows. We then have $1 \geq \frac{I_n}{I_{n-1}} \geq \frac{I_{n+1}}{I_n} = \frac{n+1}{n+2} \xrightarrow{n \rightarrow \infty} 1$, and we conclude by the sandwich theorem.

B. EXAMS

- (10 points) Using the previous results, compute the limit $\lim_{n \rightarrow \infty} B_n$, where:

$$B_n = \prod_{k=1}^n \frac{(2k)^2}{(2k-1)(2k+1)}.$$

We have $\frac{2}{1} \cdot \frac{2}{3} = 2I_2$ and for $k \geq 2$, $\frac{2k}{2k-1} = \frac{I_{2k-3}}{I_{2k-1}}$, $\frac{2k}{2k+1} = \frac{I_{2k}}{I_{2k-2}}$ thus $B_n = 2I_2 \prod_{k=2}^n \frac{I_{2k-3}}{I_{2k-1}} \frac{I_{2k}}{I_{2k-2}} = 2I_2 \frac{I_1}{I_2} \frac{I_{2n}}{I_{2n-1}} = \frac{\pi}{2} \frac{I_{2n}}{I_{2n-1}}$ $\lim_{n \rightarrow \infty} B_n = \frac{\pi}{2}$.

Exercise 4 (Envelope of a family of curves, 30 points). A family of curves in $\mathbb{R}^2$ is defined implicitly by an equation $F_\tau(x, y) = 0$, where τ is a parameter. A curve in the family, parametrized either as $y_\tau(x)$ or $x_\tau(y)$, is obtained by solving the equation at fixed τ and we switch between different curves in the family by varying τ . For example the curves in the family $F_\tau(x, y) = x - y - \tau = 0$ can be parametrized either as $y_\tau(x) = x - \tau$ or as $x_\tau(y) = y + \tau$ and are straight lines. In order for the implicit function theorem to guarantee that the equation $F_\tau(x, y) = 0$ can be solved at fixed τ , we assume that either $\frac{\partial F_\tau}{\partial y} \neq 0$ or $\frac{\partial F_\tau}{\partial x} \neq 0$.

We can treat the parameter τ as an additional variable and relabel $F_\tau(x, y) \equiv F(x, y, \tau)$, and we assume F is smooth in all its arguments in some appropriate domain. The envelope of the family of curves, if it exists, is a curve $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$ in $\mathbb{R}^2$ which is tangent to all the curves in the family. Our aim is to prove that the envelope is the solution of the system of equations:

$$F(x, y, \tau) = 0, \quad \frac{\partial F}{\partial \tau}(x, y, \tau) = 0. \quad (2.3)$$

- (10 points) A cannon launches projectiles from the origin $(0, 0) \in \mathbb{R}^2$ with initial speed v at an arbitrary angle α with the horizontal, in a constant gravitational field with acceleration $-g$ along the vertical axis. Assuming Eq. (2.3) proven, determine the “safety curve”, that is the curve which separates the points which can be reached by the projectile from those which can not.

Denoting t the time, we have $x = v \cos \alpha t$, $y = v \sin \alpha t - \frac{gt^2}{2} = x \tan \alpha - \frac{g}{2} \frac{x^2}{v^2 \cos^2 \alpha}$ that is $y = x \tan \alpha - \frac{g}{2} \frac{x^2}{v^2} (1 + \tan^2 \alpha)$, hence the trajectories are the family of parabolas $F(x, y, \tau) = y - \tau x + \frac{g}{2} \frac{x^2}{v^2} (1 + \tau^2) = 0$. The “safety curve” (a.k.a. the parabola of safety) is the envelope of the trajectories and is determined by $y - \tau x + \frac{g}{2} \frac{x^2}{v^2} (1 + \tau^2) = 0$, $-x + g \frac{x^2}{v^2} \tau = 0 \Rightarrow \tau = \frac{v^2}{gx} \Rightarrow y = \frac{v^2}{g} - \frac{g}{2} \frac{x^2}{v^2} (1 + \frac{v^4}{g^2 x^2})$ hence $y = \frac{v^2}{2g} - \frac{g}{2} \frac{x^2}{v^2}$.

- (10 points) Assuming Eq. (2.3) proven, consider the family of curves defined by the equation:

$$F(x, y, \tau) = \frac{x^2}{\tau} + \frac{y^2}{1-\tau} - 1 = 0, \quad 0 < \tau < 1.$$

Describe this family of curves and determine the envelope.

The curves are ellipses centered at the origin with semiaxes $a = \sqrt{\tau}$ and $b = \sqrt{1-\tau}$, that is such that $a^2 + b^2 = 1$. The semiaxes are the projections of the points on a circle of radius 1 on the coordinate axes. The envelope is determined by the system of equations $F(x, y, \tau) = \frac{x^2}{\tau} + \frac{y^2}{1-\tau} - 1 = 0$ and $\partial_\tau F(x, y, \tau) = -\frac{x^2}{\tau^2} + \frac{y^2}{(1-\tau)^2} = 0$. The second equation leads to $y^2 \tau^2 - x^2 (1-\tau)^2 = 0 \Rightarrow |y|\tau = |x|(1-\tau) \Rightarrow \tau = \frac{|x|}{|x|+|y|}$, and substituting into the first we get $\frac{x^2}{|x|} (|x|+|y|) + \frac{y^2}{|y|} (|x|+|y|) - 1 = 0 \Rightarrow (|x|+|y|)^2 = 1 \Rightarrow |x|+|y| = \pm 1$. The envelope is a square centered at the origin, tilted by 45 degrees with respect to the axes.

- (10 points) In order to prove Eq. (2.3), consider the system of equations:

$$F(x, y, \tau) = 0, \quad \frac{\partial F}{\partial \tau}(x, y, \tau) = 0,$$

and assume (x_0, y_0, τ_0) is a solution. Give a sufficient condition such that this system admits locally around (x_0, y_0, τ_0) a solution $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$.

We denote $x_\tau(y)$ the solution of the equation $F_\tau(x, y) \equiv F(x, y, \tau) = 0$ corresponding to the fixed parameter τ and such that $x_{\tau_0}(y_0) = x_0$. This curve is an embedded submanifold defined by a constraint. Prove that the envelope $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$ is tangent to the curve $x_\tau(y)$ at the point $y = y^{\text{en}}(\tau)$.

The differential of the system of equations is: $dH|_{x_0, y_0, \tau_0} = \begin{bmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial F}{\partial y} \\ \frac{\partial F}{\partial \tau} \end{bmatrix}_{x_0, y_0, \tau_0} = \begin{bmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial F}{\partial y} \\ \frac{\partial F}{\partial \tau} \end{bmatrix}_{x_0, y_0, \tau_0}$, and a sufficient condition for the system to admit locally a solution is that $d_\perp H|_{x_0, y_0, \tau_0}$ is invertible, hence has rank 2. As $F(x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau) = 0$, we have $0 = \frac{d}{d\tau} F(x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau) \Big|_\tau = \frac{\partial F}{\partial x} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dx^{\text{en}}}{d\tau} + \frac{\partial F}{\partial y} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dy^{\text{en}}}{d\tau} + \frac{\partial F}{\partial \tau} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} = 0$. As both $F(x_\tau(y), y, \tau) = 0$ and $F(x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau) = 0$, we have $x_\tau(y)|_{y=y^{\text{en}}(\tau)} = x^{\text{en}}(\tau)$, that is the envelope intersects the curve $x_\tau(y)$ at $y = y^{\text{en}}(\tau)$. The curve $x_\tau(y)$ is an embedded submanifold and the normal to this manifold is the form $dF_\tau|_{x_\tau(y), y} = \frac{\partial F_\tau}{\partial x} \Big|_{x_\tau(y), y} dx|_{x_\tau(y), y} + \frac{\partial F_\tau}{\partial y} \Big|_{x_\tau(y), y} dy|_{x_\tau(y), y}$, where we emphasised that this is an element of $T^*_{x_\tau(y), y} \mathbb{R}^2$. Evaluating the normal form dF_τ at the point $(x_\tau(y), y)_{y=y^{\text{en}}(\tau)}$ on the vector tangent to the envelope at the same point $(x, y) = (x^{\text{en}}(\tau), y^{\text{en}}(\tau))$, which is $\frac{dx^{\text{en}}}{d\tau} \partial_x|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)} + \frac{dy^{\text{en}}}{d\tau} \partial_y|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)}$ and using $F_\tau(x, y) = F(x, y, \tau)$ we have $dF_\tau|_{x_\tau(y), y}|_{y=y^{\text{en}}(\tau)} \left(\frac{dx^{\text{en}}}{d\tau} \partial_x|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)} + \frac{dy^{\text{en}}}{d\tau} \partial_y|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau)} \right) = \frac{\partial F}{\partial x} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dx^{\text{en}}}{d\tau} + \frac{\partial F}{\partial y} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} \frac{dy^{\text{en}}}{d\tau} + \frac{\partial F}{\partial \tau} \Big|_{x^{\text{en}}(\tau), y^{\text{en}}(\tau), \tau} = 0$ as $(x^{\text{en}}(\tau), y^{\text{en}}(\tau))$ is a solution of $F(x, y, \tau) = 0$, $\partial_\tau F(x, y, \tau) = 0$.

B.1.5 Sample 5

Exercise 1 (Exercises, 20 points). *Solve the following tasks:*

- (10 points, The Heidelberg Integration Bee, simplified) Compute the definite integral:

$$I = \int_0^\pi dx \ln(\sin(x)) .$$

Hint: recall that $\sin(x) = 2\sin(\frac{x}{2})\cos(\frac{x}{2})$.

We have:

$$I = \int_0^\pi dx \ln(\sin(x)) \stackrel{\sin(x)=2\sin\frac{x}{2}\cos\frac{x}{2}}{=} \pi \ln(2) + \int_0^\pi dx \left[\ln(\sin\frac{x}{2}) + \ln(\cos\frac{x}{2}) \right] .$$

Now:

$$\int_0^\pi dx \left[\ln(\sin\frac{x}{2}) + \ln(\cos\frac{x}{2}) \right] = 2 \int_0^{\frac{\pi}{2}} dx \left[\ln(\sin(x)) + \ln(\cos(x)) \right] \stackrel{\cos(x)=\sin(x+\frac{\pi}{2})}{=} 2 \int_0^{\frac{\pi}{2}} dx \left[\ln(\sin(x)) + \ln(\sin(x+\frac{\pi}{2})) \right] = 2 \int_0^\pi dx \ln(\sin(x)) ,$$

hence $I = \pi \ln(2) + 2I$ therefore $I = -\pi \ln(2)$.

- (10 points) We denote $\mathcal{C}([0, 1], \mathbb{R}_{\geq 0})$ the space of real positive continuous functions on $[0, 1]$. This is a Banach space with norm $\|x\|_\infty = \sup_{t \in [0, 1]} |x(t)|$ (the functions $f \in \mathcal{C}([0, 1], \mathbb{R}_{\geq 0})$ are bounded as they are continuous with compact domain). Compute (rigorously) the Fréchet differential of the functional:

$$F : \mathcal{C}([0, 1], \mathbb{R}_{\geq 0}) \rightarrow \mathbb{R} , \quad F(x) = \int_0^1 dt e^{x(t)} \sin(x(t)) .$$

Hint: the Taylor theorem states that for any $f : \mathbb{R} \rightarrow \mathbb{R}$ twice continuously differentiable and for any $h \geq 0$, there exists $0 \leq \xi \leq h$ such that $f(h) = f(0) + hf'(0) + \frac{h^2}{2!} f''(\xi)$.

We have: $\mathrm{d}F|_x(h) = \int_0^1 dt e^{x(t)} [\sin(x(t) + \cos(x(t))) h(t)]$. Obviously linear and $\|\mathrm{d}F|_x\|_{\text{op}} \leq 2e^{\|x\|_\infty}$.

In order to prove this (we suppress arguments for simplicity):

$$\begin{aligned} |F(x+h) - f(x) - \mathrm{d}F|_x(h)| &= \left| \int_0^1 dt [e^{x+h} \sin(x+h) - e^x \sin(x) - e^x (\sin(x) + \cos(x))h] \right| = \left| \int_0^1 dt e^x [e^h \sin(x) \cos(h) + \cos(x) \sin(h) - \sin(x) - [\sin(x) + \cos(x)]h] \right| \\ &\leq \int_0^1 dt e^x \sin(x) |e^h \cos(h) - 1 - h| + \int_0^1 dt e^x \cos(x) |e^h \sin(h) - h| \end{aligned}$$

As $h \geq 0$, from Taylor theorem with rest term, and using $|\sin(\xi)|, |\cos(\xi)| \leq 1$ and the fact that the exponential is increasing we get:

- we have $[e^h \cos(h)]' = e^h \cos(h) - e^h \sin(h)$ and $[e^h \cos(h)]'' = -2e^h \sin(h)$ hence $|e^h \cos(h) - 1 - h| = \left| \frac{h^2}{2} (-2)e^\xi \sin(\xi) \right| \leq h^2 e^h$,
- we have $[e^h \sin(h) - h]' = e^h \sin(h) + e^h \cos(h) - 1$ and $[e^h \sin(h) - h]'' = 2e^h \cos(h)$ hence $|e^h \sin(h) - h| \leq \left| \frac{h^2}{2} 2e^\xi \cos(\xi) \right| \leq h^2 e^h$.

Bounding by the sup and using again $|\sin(x)|, |\cos(x)| < 1$ we have:

$$\frac{|F(x+h) - f(x) - \mathrm{d}F|_x(h)|}{\|h\|_\infty} \leq \|h\|_\infty \int_0^1 dt e^{\|x\|_\infty} 2e^{\|h\|_\infty} \frac{\|h\|_\infty \rightarrow 0}{0} ,$$

Exercise 2 (Matrix diagonalization, 30 points). *Consider the matrix:*

$$A = \begin{pmatrix} 4 & 1 & -3 \\ -2 & 1 & 3 \\ 0 & 0 & 2 \end{pmatrix} .$$

- (5 points) Compute the eigenvalues of A and their algebraic multiplicities. We have the characteristic polynomial:

$$\begin{vmatrix} 4-\lambda & 1 & -3 \\ -2 & 1-\lambda & 3 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 4-\lambda & 1 \\ -2 & 1-\lambda \end{vmatrix} = (2-\lambda) [(4-\lambda)(1-\lambda) + 2] = (2-\lambda) [6 - 5\lambda + \lambda^2] = (2-\lambda)(2-\lambda)(3-\lambda) ,$$

hence the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$ with algebraic multiplicities $m_1 = 2$ and $m_2 = 1$.

- (10 points) Determine the geometric multiplicities of the eigenvalues and identify bases in the corresponding eigenspaces. We have:

$$A - 2I = \begin{pmatrix} 2 & 1 & -3 \\ -2 & -1 & 3 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & \frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} , \quad A - 3I = \begin{pmatrix} 1 & 1 & -3 \\ -2 & -2 & 3 \\ 0 & 0 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} ,$$

hence we have:

$$\ker(A - 2I) = \text{Span}\{f_1 = e_2 - \frac{1}{2}e_1, f_2 = e_3 + \frac{3}{2}e_1\} , \quad \ker(A - 3I) = \text{Span}\{f_3 = e_2 - e_1\} ,$$

with geometric multiplicities $\gamma_1 = 2, \gamma_2 = 1$ hence A is diagonalizable.

- (15 points) Identify the projectors on the eigenspaces, write the spectral decomposition of A and compute $\ln(A - I)$, where I is the identity matrix.

Note that if $f_1, \dots, f_N$ are a basis in $\mathbb{R}^n$, denoting $S = \begin{pmatrix} f_1 & \dots & f_N \end{pmatrix}$ the matrix formed by this column vectors expressed in the standard basis, and regarding S^{-1} as a matrix made by the row vectors L^i , then the projector along f_i is $P_i = f_i L^i$ (the index i is not summed!).

B. EXAMS

In order to find the projectors we need to expand an arbitrary vector on the basis of eigenvectors, $x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 = \alpha\mathbf{f}_1 + \beta\mathbf{f}_2 + \gamma\mathbf{f}_3$. This is a linear system with matrix the column vectors $\mathbf{f}_i$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{3}{2} & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}, \quad \mathbf{x} = S\boldsymbol{\alpha},$$

which is solved by inverting the matrix $S^{-1}\mathbf{x} = \boldsymbol{\alpha}$. Regarding S^{-1} as a matrix of row vectors L_i , the projector along $\mathbf{f}_i$ is $P_i = \mathbf{f}_i L_i$.

Instead of inverting the matrix mechanically, we can note that $z = \beta$ and the remaining two equations write $x = -\frac{1}{2}\alpha + \frac{3}{2}z - \gamma$ respectively $y = \alpha + \gamma$ or equivalently:

$$2x - 3z = -\alpha - 2\gamma, \quad y = \alpha + \gamma \Rightarrow \quad 2x + 2y - 3z = \alpha, \quad 2x + y - 3z = -\gamma,$$

and we get $S^{-1} = \begin{pmatrix} 2 & 2 & -3 \\ 0 & 0 & 1 \\ -2 & -1 & 3 \end{pmatrix}$. Hence:

$$P_1 = \begin{pmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 2 & 2 & -3 \end{pmatrix} = \begin{pmatrix} -1 & -1 & \frac{3}{2} \\ 2 & 2 & -3 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} \frac{3}{2} \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & \frac{3}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P_3 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} -2 & -1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 & -3 \\ -2 & -1 & 3 \\ 0 & 0 & 0 \end{pmatrix},$$

and the projector on $\ker(A - 2I)$ is $\boxed{P_1 + P_2}$ while the projector on $\ker(A - 3I)$ is $\boxed{P_3}$.

Consequently:

$$A = \begin{pmatrix} 4 & 1 & -3 \\ -2 & 1 & 3 \\ 0 & 0 & 2 \end{pmatrix} = 2 \begin{pmatrix} -1 & -1 & 3 \\ 2 & 2 & -3 \\ 0 & 0 & 1 \end{pmatrix} + 3 \begin{pmatrix} 2 & 1 & -3 \\ -2 & -1 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \quad \ln(A - I) = \ln(2)P_3,$$

where we used $I = P_1 + P_2 + P_3$, hence $A - I = (2 - 1)(P_1 + P_2) + (3 - 1)P_3$.

Exercise 3 (Bipolar coordinates, 30 points). The bipolar coordinate system (v, ξ) with $v \in (-\infty, \infty)$ and $\xi \in [0, 2\pi)$ in $\mathbb{R}^2$ is defined by:

$$x = a \frac{\sinh(v)}{\cosh(v) - \cos(\xi)}, \quad z = a \frac{\sin(\xi)}{\cosh(v) - \cos(\xi)} \quad \text{where} \quad \sinh(v) = \frac{e^v - e^{-v}}{2}, \quad \cosh(v) = \frac{e^v + e^{-v}}{2}, \quad \coth(v) = \frac{\cosh(v)}{\sinh(v)}.$$

We denote the natural embedding of $\mathbb{R}^2$ in bipolar coordinates into $\mathbb{R}^2$ in Cartesian coordinates by $\phi(v, \xi) = (x(v, \xi), z(v, \xi))$ and $\eta = dx^2 + dz^2$ the Euclidean metric in Cartesian coordinates.

- (15 points) Compute the Frechét differential $d\phi \equiv \phi_* = \sum_{i,\mu} \frac{\partial}{\partial x^i} \frac{\partial x^i}{\partial s^\mu} ds^\mu$, the pullback of the Euclidean metric $g = \phi^*(\eta)$, the volume element ω in bipolar coordinates and the push forwards $\phi_*(\partial_\xi)$ and $\phi_*(\partial_v)$.

Hint: Note that $[1 - \cosh(v) \cos(\xi)]^2 + \sinh^2(v) \sin^2(\xi) = [\cosh(v) - \cos(\xi)]^2$.

We have $d\phi \equiv \phi_* = \begin{pmatrix} \frac{\partial}{\partial x^i} & \frac{\partial}{\partial s^\mu} \end{pmatrix} ds^\mu = \begin{pmatrix} \partial_x & \partial_z \end{pmatrix} \begin{pmatrix} a \frac{1 - \cosh(v) \cos(\xi)}{[\cosh(v) - \cos(\xi)]^2} & -a \frac{\sinh(v) \sin(\xi)}{[\cosh(v) - \cos(\xi)]^2} \\ -a \frac{\sin(\xi) \sinh(v)}{[\cosh(v) - \cos(\xi)]^2} & -a \frac{1 - \cosh(v) \cos(\xi)}{[\cosh(v) - \cos(\xi)]^2} \end{pmatrix} \begin{pmatrix} dv \\ d\xi \end{pmatrix}$. We simplify the notation: $C_v = \cosh(v)$, $S_v = \sinh(v)$ and $C_\xi = \cos(\xi)$, $S_\xi = \sin(\xi)$, hence $dx = a \frac{1 - C_v C_\xi}{(C_v - C_\xi)^2} dv - a \frac{S_v S_\xi}{(C_v - C_\xi)^2} d\xi$ and $dz = -a \frac{S_v S_\xi}{(C_v - C_\xi)^2} dv - a \frac{1 - C_v C_\xi}{(C_v - C_\xi)^2} d\xi$. We have:

$$\phi^*(\eta) = a^2 \frac{(1 - C_v C_\xi)^2 + S_v^2 S_\xi^2}{(C_v - C_\xi)^4} (dv^2 + d\xi^2) = \frac{a^2}{(C_v - C_\xi)^2} [dv^2 + d\xi^2] = \phi^*(\eta), \quad \omega = \frac{a^2}{(C_v - C_\xi)^2} dv \wedge d\xi,$$

$$\text{and } \phi_*(\partial_v) = \frac{a}{(C_v - C_\xi)^2} [(1 - C_v C_\xi) \partial_x - S_v S_\xi \partial_z], \quad \phi_*(\partial_\xi) = \frac{a}{(C_v - C_\xi)^2} [-S_v S_\xi \partial_x - (1 - C_v C_\xi) \partial_z].$$

- (10 points) Prove that $x^2 + z^2 + a^2 = 2ax \coth(v)$. The curve Σ with constant $v = v_0$ can also be defined as the solution of the constraint $F(x, z) = x^2 + z^2 + a^2 - 2ax \coth(v_0) = 0$. Draw Σ , compute $dF|_\Sigma$ and $dF|_\Sigma(\phi_*(\partial_\xi))$.

$$\text{We have } x^2 + z^2 + a^2 - 2ax \coth(v) = a^2 \frac{S_v^2 + S_\xi^2}{(C_v - C_\xi)^2} + a^2 - 2a \frac{a S_v}{C_v - C_\xi} \frac{C_v}{S_v} = \frac{a^2}{(C_v - C_\xi)^2} \times \underbrace{[S_v^2 + S_\xi^2 + (C_v - C_\xi)^2 - 2C_v(C_v - C_\xi)]}_{S_v^2 + S_\xi^2 + C_v^2 + C_\xi^2 - 2C_v C_\xi - 2C_v^2 + 2C_v C_\xi = S_\xi^2 + C_\xi^2 + S_v^2 - C_v^2 = 1 - 1 = 0}.$$

The equation defining Σ is also $(x - a \coth(v_0))^2 + z^2 + a^2(1 - \coth^2(v_0)) \Rightarrow (x - a \coth(v_0))^2 + z^2 = \frac{a^2}{\sinh^2(v_0)}$. This is a circle with center $(x_c = a \coth(v_0), z_c = 0)$ and radius $\frac{a}{\sinh(v_0)}$.

We have $dF = 2x dx + 2z dz - 2a \coth(v_0) dx$. Writing the points in $\Sigma \subset \mathbb{R}^2$ in terms of their bipolar coordinates (v_0, ξ) we have:

$$dF|_\Sigma = 2a \frac{S_\xi}{C_{v_0} - C_\xi} dz + (2a \frac{S_{v_0}}{C_{v_0} - C_\xi} - 2a \frac{C_{v_0}}{S_{v_0}}) dx = \boxed{dF|_\Sigma = -2a \frac{1 - C_{v_0} C_\xi}{S_{v_0}(C_{v_0} - C_\xi)} dx + 2a \frac{S_\xi}{C_{v_0} - C_\xi} dz}, \quad \boxed{dF|_\Sigma(\phi_*(\partial_\xi)) = 0}.$$

- (5 points) Propose an embedding ψ of the curve Σ in $\mathbb{R}^2$, compute the differential $d\psi = \psi_*$, the pullback metric $\psi^*(\eta)$, the volume form for this pullback metric and compare it with $\psi^*(i_\nu dx \wedge dz)$, where ν is a unit vector parallel to $\phi_*(\partial_v)$. Evaluate $\psi^*(dF)$. We propose

$$\psi(\xi) = \phi(v_0, \xi), \text{ then } d\psi \equiv \psi_* = \begin{pmatrix} \frac{\partial}{\partial x^i} & \frac{\partial}{\partial s^\mu} \end{pmatrix} ds^\mu = \begin{pmatrix} \partial_x & \partial_z \end{pmatrix} \begin{pmatrix} -a \frac{S_{v_0} S_\xi}{[C_{v_0} - C_\xi]^2} \\ -a \frac{1 - C_{v_0} C_\xi}{[C_{v_0} - C_\xi]^2} \end{pmatrix} (d\xi) \text{ and } dx = -a \frac{S_{v_0} S_\xi}{(C_{v_0} - C_\xi)^2} d\xi, \quad dz = -a \frac{1 - C_{v_0} C_\xi}{(C_{v_0} - C_\xi)^2} d\xi. \text{ Hence:}$$

$$\psi^*(\eta) = a^2 \frac{(1 - C_{v_0} C_\xi)^2 + S_{v_0}^2 S_\xi^2}{(C_{v_0} - C_\xi)^4} d\xi^2 = \frac{a^2}{(C_{v_0} - C_\xi)^2} d\xi^2 = \psi^*(\eta), \quad \omega_\Sigma = \frac{a}{(C_{v_0} - C_\xi)} d\xi,$$

On the other hand $\nu = \frac{1}{(C_v - C_\xi)} \left[(1 - C_v C_\xi) \partial x - S_v S_\xi \partial z \right]$, $i_\nu dx \wedge dz = \frac{(1 - C_v C_\xi)}{(C_v - C_\xi)} dz - \frac{(-S_v S_\xi)}{(C_v - C_\xi)} dx$ and:

$$\psi^*(i_\nu dx \wedge dz) = \frac{(1 - C_{v0} C_\xi)}{(C_{v0} - C_\xi)} \left(-a \frac{1 - C_{v0} C_\xi}{(C_{v0} - C_\xi)^2} d\xi \right) - \frac{(-S_{v0} S_\xi)}{(C_{v0} - C_\xi)} \left(-a \frac{S_{v0} S_\xi}{(C_{v0} - C_\xi)^2} d\xi \right) = -d\xi \frac{a}{(C_{v0} - C_\xi)^3} \left[(1 - C_{v0} C_\xi)^2 + S_{v0}^2 S_\xi^2 \right] = \boxed{-\omega_\Sigma}.$$

Finally $dF = [2x - 2a \coth(v_0)] dx + 2z dz$ and:

$$\psi^*(dF) = \left[2a \frac{S_{v0}}{C_{v0} - C_\xi} - 2a \frac{C_{v0}}{S_{v0}} \right] \left(-a \frac{S_{v0} S_\xi}{(C_{v0} - C_\xi)^2} d\xi \right) + 2a \frac{S_\xi}{C_{v0} - C_\xi} \left(-a \frac{1 - C_{v0} C_\xi}{(C_{v0} - C_\xi)^2} d\xi \right) = -2a \frac{1 - C_{v0} C_\xi}{S_{v0} (C_{v0} - C_\xi)} \left(-a \frac{S_{v0} S_\xi}{(C_{v0} - C_\xi)^2} d\xi \right) + 2a \frac{S_\xi}{C_{v0} - C_\xi} \left(-a \frac{1 - C_{v0} C_\xi}{(C_{v0} - C_\xi)^2} d\xi \right) = \boxed{0}.$$

Exercise 4 (Particle on the cycloid, 40 points). The cycloid is the curve embedded in $\mathbb{R}^2$ by $\phi : [0, 2\pi] \rightarrow \mathbb{R}^2$, $\phi(u) = (x(u), y(u))$ with:

$$x(u) = u - \sin(u), \quad y(u) = 1 - \cos(u) = 2 \sin^2\left(\frac{u}{2}\right).$$

- (5 points) Compute the pullback of the Euclidean metric in $\mathbb{R}^2$ on the cycloid (seen as an embedded manifold) and write the Lagrangian of a free particle $L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2)$ constrained to move on the cycloid.

We have: $g_{uu} du^2 \equiv ds^2 = dx^2 + dy^2 = (1 - \cos(u))^2 du^2 + \sin^2(u) du^2 = 2(1 - \cos(u)) du^2 = \boxed{4 \sin^2\left(\frac{u}{2}\right) du^2}$. The Lagrangian is $L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) = \boxed{2m \sin^2\left(\frac{u}{2}\right) \dot{u}^2}$.

- (10 points) Starting from the Euler Lagrange equations $-\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \Big|_{q(t), \frac{dq}{dt}(t)} \right) + \frac{\partial L}{\partial q} \Big|_{q(t), \frac{dq}{dt}(t)} = 0$, prove that for any Lagrangian $L(q, \dot{q})$, the energy function $E(q, \dot{q}) = \dot{q} \frac{\partial L}{\partial \dot{q}} - L(q, \dot{q})$ is conserved along the classical trajectories. For a particle on the cycloid, assuming $\dot{u} \geq 0$, determine $\dot{u}$ as a function of u , the total energy $E > 0$ and the mass m .

Along the classical trajectories $\frac{d}{dt} E \Big|_{q(t), \frac{dq}{dt}(t)} = \frac{d^2 q}{dt^2} \frac{\partial L}{\partial \dot{q}} \Big|_{q(t), \frac{dq}{dt}(t)} + \frac{dq}{dt} \left(\frac{\partial}{\partial t} \frac{\partial L}{\partial \dot{q}} \Big|_{q(t), \frac{dq}{dt}(t)} \right) - \frac{\partial L}{\partial q} \Big|_{q(t), \frac{dq}{dt}(t)} \frac{dq}{dt} - \frac{\partial L}{\partial \dot{q}} \Big|_{q(t), \frac{dq}{dt}(t)} \frac{d^2 q}{dt^2} = 0$

We have $E = 2m \sin^2\left(\frac{u}{2}\right) \dot{u}^2 \Rightarrow \dot{u} = \frac{\sqrt{E/2m}}{\sin\left(\frac{u}{2}\right)}$.

- (10 points) Assuming $u(0) = 0$, determine $u(t)$ and $x(t)$, $y(t)$. It can be useful to denote $\frac{E}{2m} = A^2$.

Hint: Note that if $\frac{du}{dt}(t') = f(u(t'))$ then $1 = \frac{1}{f(u(t'))} \frac{du}{dt}(t')$ and consequently $t = \int_0^t \frac{1}{f(u(t'))} \frac{du}{dt}(t') dt' = \int_{u(0)}^{u(t)} \frac{1}{f(u)} du$.

We have:

$$\sin\left(\frac{u}{2}\right) \frac{du}{dt} = A \Rightarrow \int_0^t \sin\left(\frac{u}{2}\right) \frac{du}{dt} dt = A t \Rightarrow \int_0^{u(t)} \sin\left(\frac{u}{2}\right) du = A t \Rightarrow -2 \cos\left(\frac{u}{2}\right) \Big|_0^{u(t)} = A t \Rightarrow 2(1 - \cos\left(\frac{u(t)}{2}\right)) = A t,$$

that is $\cos\left(\frac{u(t)}{2}\right) = 1 - \frac{At}{2}$ and $\sin\left(\frac{u(t)}{2}\right) = \sqrt{\frac{At}{2} \left(2 - \frac{At}{2}\right)}$.

As $x(u) = u - 2 \sin\left(\frac{u}{2}\right) \cos\left(\frac{u}{2}\right)$ and $y(u) = 2 \sin^2\left(\frac{u}{2}\right)$ it follows that $x(t) = 2 \arccos\left(1 - \frac{At}{2}\right) - 2 \sqrt{\frac{At}{2} \left(2 - \frac{At}{2}\right)} \left(1 - \frac{At}{2}\right)$, $y(t) = 2 \frac{At}{2} \left(2 - \frac{At}{2}\right)$.

- (15 points) Assuming $u(0) = 0$, determine $u(t)$ for a particle on a cycloid in a gravitational potential $V(x, y) = mgy$, assuming $0 < \frac{E - 2mg}{2m} = \bar{A}^2$. Check that for $g \rightarrow 0$ we recover the result we obtained previously.

The energy conservation writes $E = 2m \sin^2\left(\frac{u}{2}\right) \dot{u}^2 + 2mg \sin^2\left(\frac{u}{2}\right) \Rightarrow \dot{u} = \sqrt{\frac{E/2m}{\sin^2\left(\frac{u}{2}\right)} - g}$. Hence:

$$t = \int_0^{u(t)} du \frac{\sin\left(\frac{u}{2}\right)}{\sqrt{\frac{E}{2m} - g \sin^2\left(\frac{u}{2}\right)}} = \frac{v = \cos\left(\frac{u}{2}\right)}{-2} \int_1^{\cos\left(\frac{u(t)}{2}\right)} dv \frac{1}{\sqrt{\frac{E - 2mg}{2m} + gv^2}},$$

and we denote $\frac{E - 2mg}{2m} = \bar{A}^2$. Either changing variables to $v = \frac{\bar{A}}{\sqrt{g}} \sinh(s)$ or directly we have $\int dv \frac{1}{\sqrt{\bar{A}^2 + gv^2}} = \frac{1}{\sqrt{g}} \ln(v + \sqrt{\bar{A}^2 + v^2})$. Hence:

$$-\frac{t}{2} = \frac{1}{\sqrt{g}} \ln \frac{\cos\left(\frac{u(t)}{2}\right) + \sqrt{\frac{\bar{A}^2}{g} + \cos^2\left(\frac{u(t)}{2}\right)}}{1 + \sqrt{\frac{\bar{A}^2}{g} + 1}} = \frac{1}{\sqrt{g}} \ln \frac{\sqrt{1 + \frac{g}{\bar{A}^2} \cos^2\left(\frac{u(t)}{2}\right)} + \frac{\sqrt{g}}{\bar{A}} \cos\left(\frac{u(t)}{2}\right)}{\sqrt{1 + \frac{g}{\bar{A}^2}} + \frac{\sqrt{g}}{\bar{A}}} \Rightarrow \sqrt{1 + \frac{g}{\bar{A}^2} \cos^2\left(\frac{u(t)}{2}\right)} + \frac{\sqrt{g}}{\bar{A}} \cos\left(\frac{u(t)}{2}\right) = \left(\sqrt{1 + \frac{g}{\bar{A}^2}} + \frac{\sqrt{g}}{\bar{A}}\right) e^{-\frac{t}{2} \sqrt{g}},$$

Which rewrites as:

$$1 = \left(\sqrt{1 + \frac{g}{\bar{A}^2}} + \frac{\sqrt{g}}{\bar{A}}\right)^2 e^{-t\sqrt{g}} - 2 \frac{\sqrt{g}}{\bar{A}} \cos\left(\frac{u(t)}{2}\right) \left(\sqrt{1 + \frac{g}{\bar{A}^2}} + \frac{\sqrt{g}}{\bar{A}}\right) e^{-\frac{t}{2} \sqrt{g}} \Rightarrow \cos\left(\frac{u(t)}{2}\right) = \frac{\left(\sqrt{1 + \frac{g}{\bar{A}^2}} + \frac{\sqrt{g}}{\bar{A}}\right)^2 e^{-t\sqrt{g}} - 1}{2 \frac{\sqrt{g}}{\bar{A}} \left(\sqrt{1 + \frac{g}{\bar{A}^2}} + \frac{\sqrt{g}}{\bar{A}}\right) e^{-\frac{t}{2} \sqrt{g}}}.$$

At first order in $\sqrt{g}$ we have $\bar{A} = A$ hence:

$$\frac{\left(\sqrt{1 + \frac{g}{A^2}} + \frac{\sqrt{g}}{A}\right)^2 e^{-t\sqrt{g}} - 1}{2 \frac{\sqrt{g}}{A} \left(\sqrt{1 + \frac{g}{A^2}} + \frac{\sqrt{g}}{A}\right) e^{-\frac{t}{2} \sqrt{g}}} = \frac{(1 + 2 \frac{\sqrt{g}}{A} + O(g))(1 - t\sqrt{g} + O(g)) - 1}{2 \frac{\sqrt{g}}{A} (1 + O(\sqrt{g}))} = \frac{1 + 2 \frac{\sqrt{g}}{A} - t\sqrt{g} - 1}{2 \frac{\sqrt{g}}{A}} = \boxed{1 - \frac{At}{2}}.$$

B.1.6 Sample 6

Exercise 1 (Exercises, 30 points). *Solve the following tasks:*

- (15 points, The Heidelberg Integration Bee) Compute the primitive:

$$I = \int dx \frac{1}{\cos^3(x)}.$$

We have $I = \int dx \frac{1}{\cos^3(x)} = \int dx [\tan(x)]' \frac{1}{\cos(x)} = \frac{\tan(x)}{\cos(x)} - \int dx \tan(x) \frac{\sin(x)}{\cos^2(x)} = \frac{\tan(x)}{\cos(x)} - \int dx \left(\frac{1}{\cos^3(x)} - \frac{1}{\cos(x)} \right) \Rightarrow 2I = \frac{\tan(x)}{\cos(x)} + \int dx \frac{1}{\cos(x)}.$

Now $\int dx \frac{1}{\cos(x)} = \int dx \frac{\cos(x)}{1-\sin^2(x)} = \int dx \cos(x) \frac{1}{2} \left(\frac{1}{1+\sin(x)} + \frac{1}{1-\sin(x)} \right) = \frac{1}{2} [\ln(1+\sin(x)) - \ln(1-\sin(x))]$, hence:

$$I = \frac{1}{2} \frac{\tan(x)}{\cos(x)} + \frac{1}{4} \ln \frac{1+\sin(x)}{1-\sin(x)} = \frac{1}{2} \frac{\tan(x)}{\cos(x)} + \frac{1}{4} \ln \frac{[1+\sin(x)]^2}{1-\sin^2(x)} = \frac{1}{2} \frac{\tan(x)}{\cos(x)} + \frac{1}{2} \ln \left| \frac{1+\sin(x)}{\cos(x)} \right|.$$

- (10 points) Compute rigorously:

$$\lim_{x \rightarrow 0} \frac{x(\sin(x) - x)}{\cos(x) - 1 + \frac{x^2}{2}}.$$

Either L'Hospital:

$$\lim_{x \rightarrow 0} \frac{x(\sin(x) - x)}{\cos(x) - 1 + \frac{x^2}{2}} = \lim_{x \rightarrow 0} \frac{\sin(x) - x + x \cos(x) - x}{-\sin(x) + x} = \lim_{x \rightarrow 0} \frac{\cos(x) - 2 + \cos(x) - x \sin(x)}{-\cos(x) + 1} = \lim_{x \rightarrow 0} \frac{-2 \sin(x) - \sin(x) - x \cos(x)}{\sin(x)} = \lim_{x \rightarrow 0} \frac{-3 \cos(x) - \cos(x) + x \sin(x)}{\cos(x)} = -4$$

Or from Taylor with rest term there exist $\xi, \xi' \in (-|x|, |x|)$ so that $\frac{x(\sin(x)-x)}{\cos(x)-1+\frac{x^2}{2}} = \frac{x\left(-\frac{x^3}{3!} + \frac{x^4}{4!} \sin^{(4)}(\xi)\right)}{\frac{x^4}{4!} + \frac{x^5}{5!} \cos^{(5)}(\xi')} = \frac{-\frac{1}{3!} + \frac{x}{4!} \sin^{(4)}(\xi)}{\frac{1}{4!} + \frac{x}{5!} \cos^{(5)}(\xi')} \xrightarrow{x \rightarrow 0} -\frac{4!}{3!} = -4.$

- (5 points) For any $N \times N$ matrices X and Y we have:

$$e^{\epsilon X} e^{\epsilon Y} = e^{\epsilon X + \epsilon Y + \frac{\epsilon^2}{2} A + O(\epsilon^3)},$$

where $O(\epsilon^3)$ denotes terms of order at least ϵ^3 . Determine A in terms of X and Y .

We have $\left(1 + \epsilon X + \frac{1}{2} \epsilon^2 X^2 + O(\epsilon^3)\right) \left(1 + \epsilon Y + \frac{1}{2} \epsilon^2 Y^2 + O(\epsilon^3)\right) = 1 + \epsilon(X + Y) + \frac{\epsilon^2}{2} A + \frac{1}{2} (\epsilon X + \epsilon Y)^2 + O(\epsilon^3)$, and expanding the left hand side we get:

$$1 + \epsilon X + \frac{1}{2} \epsilon^2 X^2 + \epsilon Y + \epsilon^2 XY + \frac{1}{2} \epsilon^2 Y^2 + O(\epsilon^3) = 1 + \epsilon(X + Y) + \frac{\epsilon^2}{2} A + \frac{1}{2} (\epsilon X + \epsilon Y)^2 + O(\epsilon^3),$$

Collecting the ϵ^2 terms we have $X^2 + Y^2 + 2XY = A + X^2 + XY + YX + Y^2$ hence $A = XY - YX = [X, Y]$.

Exercise 2 (Rotations, 30 points). A proper rotation of a vector in $\mathbb{R}^3$ is a real linear transformation $x \rightarrow x' = Rx$ which preserves the norm and the orientation. It follows that R is a 3×3 matrix with real entries such that $RR^T = R^T R = I$, where R^T is the transposed of R and I the 3×3 identity matrix, and $\det R = 1$.

- (5 points) Prove that the compositions of two rotations is a rotation. Check that the composition of rotations is associative and that every rotation has an inverse.

We have $(R_1 R_2)^T (R_1 R_2) = R_2^T R_1^T R_1 R_2 = I$ and $\det(R_1 R_2) = \det(R_1) \det(R_2) = 1$. Associativity follows from the associativity of the matrix product, existence of inverse follows from observing that $R^T = R^{-1}$.

- (10 points) We can consider rotation matrices acting on complex vectors. Prove that any rotation R has an invariant vector $u \in \mathbb{C}^3$ that is such that $Ru = u$.

Hint: This means any rotation must admit 1 as an eigenvalue.

We first prove that R has an eigenvalue 1, $\det(R - I) = \det(R - RR^T) = \det(R) \det(I - R^T) = \underbrace{\det(R)}_1 \underbrace{\det(I - R)^T}_{\det(I - R)} = \det(I - R) = -\det(R - I) \Rightarrow \det(R - I) = 0$. The invariant vector

$u \in \mathbb{C}^3$ with $Ru = u$ is the eigenvector associated to this eigenvalue.

- (10 points) As R has real entries, $R^T = R^\dagger$, the adjoint of R . It follows that $RR^\dagger = R^\dagger R = I$ that is, seen as a complex matrix acting on $\mathbb{C}^3$, R is unitary hence diagonalizable in an orthonormal basis in $\mathbb{C}^3$. Prove that there exist two linearly independent complex vectors $u_1, u_2 \in \mathbb{C}^3$ and a $\phi \in [0, 2\pi)$ such that:

$$Ru_1 = e^{i\phi} u_1, \quad Ru_2 = e^{-i\phi} u_2.$$

We denote $\lambda_{1,2,3}$ the three eigenvalues of R , we have $|\lambda_i| = 1$ (as R is unitary), $\lambda_3 = 1$ (already proven) and $\det R = \lambda_1 \lambda_2 \lambda_3 = 1$ hence $\lambda_1 = e^{i\phi}$ and $\lambda_2 = e^{-i\phi}$. It follows that there exist two eigenvectors $u_1, u_2 \in \mathbb{C}^3$ such that $Ru_1 = e^{i\phi} u_1$, $Ru_2 = e^{-i\phi} u_2$ and $\{u_1, u_2, u\}$ is an orthonormal basis in $\mathbb{C}^3$.

- (5 points) Prove that there exist a basis of real vectors n_1, n_2, n in $\mathbb{C}^3$, $\bar{n}_1 = n_1, \bar{n}_2 = n_2$ and $\bar{n} = n$, such that in this basis a rotation matrix takes the form:

$$\begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Prove that n_1, n_2, n are also a basis in $\mathbb{R}^3$.

Hint: Note that if $z = x + iy \in \mathbb{C} \setminus \{0\}$ respects $z = \alpha \bar{z}$ for some $\alpha \in \mathbb{C}$ then either $y = 0$ and $\alpha = 1$ or $x = 0$ and $\alpha = -1$.

If $R = I$ we take $\phi = 0$ and the standard basis and we conclude. If not, $\phi \neq 0$ and the eigenvalues $e^{i\phi}, e^{-i\phi}, 1$ of R are all distinct. It follows that the geometric multiplicity of each eigenvalue is 1. As R has real entries:

- as $Ru = u$, $R\bar{u} = \bar{u}$ and $\lambda_3 = 1$ has geometric multiplicity 1 it follows that $\bar{u}$ and u are parallel, $\bar{u} = \alpha u$. Then, if $\alpha = 1$ then $n = u$ is real and is such that $Rn = n$. If $\alpha = -1$ then $n = iu$ is real (as u is purely imaginary) and such that $Rn = n$.
- $Ru_1 = e^{i\phi} u_1 \Rightarrow R\bar{u}_1 = e^{-i\phi} \bar{u}_1$ and as the geometric multiplicity of $e^{-i\phi}$ is 1 we must have $\bar{u}_1 = \alpha u_2$. We set $n_1 = u_1 + \alpha u_2$ and $n_2 = -i(u_1 - \alpha u_2)$ which both are real vectors and:

$$\begin{aligned} Rn_1 &= e^{i\phi} u_1 + \alpha e^{-i\phi} u_2 = \cos \phi (u_1 + \alpha u_2) + i \sin \phi (u_1 - \alpha u_2) = \begin{bmatrix} \cos \phi n_1 - \sin \phi n_2 \end{bmatrix}, \\ Rn_2 &= (-i)(e^{i\phi} u_1 - \alpha e^{-i\phi} u_2) = (-i) \cos \phi (u_1 - \alpha u_2) + (-i) i \sin \phi (u_1 + \alpha u_2) = \begin{bmatrix} \sin \phi n_1 + \cos \phi n_2 \end{bmatrix}. \end{aligned}$$

As u_1, u_2, u are a basis in $\mathbb{C}^3$, any $w \in \mathbb{C}^3$ decomposes uniquely as $w = w_1 u_1 + w_2 u_2 + w_3 u = w_1 \frac{n_1 + i n_2}{2} + w_2 \alpha \frac{n_1 - i n_2}{2} + w_3 \begin{cases} n & \text{if } n = u \\ -in & \text{if } n = iu \end{cases}$ hence n_1, n_2, n are a basis in $\mathbb{C}^3$. As n_1, n_2, n are linearly independent over $\mathbb{C}$, they are linearly independent over $\mathbb{R}$, hence they are three linearly independent vectors in $\mathbb{R}^3$, therefore a basis.

Exercise 3 (Stokes theorm and the half torus, 40 points). We embed the half torus T with radii $0 < r < R$ in $\mathbb{R}^3$ via:

$$\phi : [0, 2\pi] \times [0, \pi] \rightarrow \mathbb{R}^3, \quad \phi(\theta, \varphi) = (x = (R + r \cos \varphi) \cos \theta, y = (R + r \cos \varphi) \sin \theta, z = r \sin \varphi),$$

with positively oriented volume form $\omega = d\theta \wedge d\varphi$.

- (5 points) The boundary ∂T of the half torus has two boundary components, $\gamma_+(\theta) = \phi(\theta, \varphi = 0)$ and $\gamma_-(\theta) = \phi(\theta, \varphi = \pi)$. Compute the volume form induced by the exterior normal on each of the two boundary components.

Hint: Note that the exterior normal at $\gamma_+(\theta)$ is $\nu = -\partial_\varphi$ while at $\gamma_-(\theta)$ it is $\nu = \partial_\varphi$.

We have $\gamma_+(\theta) = ((R + r) \cos \theta, (R + r) \sin \theta, 0)$, $\nu = -\partial_\varphi$, $i_\nu \omega = d\theta$ and $\gamma_-(\theta) = ((R - r) \cos \theta, (R - r) \sin \theta, 0)$, $\nu = \partial_\varphi$, $i_\nu \omega = -d\theta$.

- (10 points) Compute the pullbacks $\phi^*(dx)$, $\phi^*(dy)$, $\phi^*(dz)$ and the pullback $\phi^*(\alpha)$ of the 2-form $\alpha = (x^2 + y^2 + z^2)dx \wedge dy + 2xz \, dz \wedge dy$ on the half torus.

We have $\phi^*(dx) = -r \sin \varphi \cos \theta d\varphi - (R + r \cos \varphi) \sin \theta d\theta$, $\phi^*(dy) = -r \sin \varphi \sin \theta d\varphi + (R + r \cos \varphi) \cos \theta d\theta$, $\phi^*(dz) = r \cos \varphi d\varphi$. Hence:

$$\phi^*(dx \wedge dy) = [(R + r \cos \varphi) \sin \theta r \sin \varphi \sin \theta + r \sin \varphi \cos \theta (R + r \cos \varphi) \cos \theta] d\theta \wedge d\varphi = (R + r \cos \varphi) r \sin \varphi d\theta \wedge d\varphi, \quad \phi^*(dz \wedge dy) = -r \cos \varphi (R + r \cos \varphi) \cos \theta d\theta \wedge d\varphi,$$

and using $x^2 + y^2 + z^2 = (R + r \cos \varphi)^2 + r^2 \sin^2 \varphi = R^2 + r^2 + 2Rr \cos \varphi$:

$$\begin{aligned} \phi^* \alpha &= (R^2 + r^2 + 2Rr \cos \varphi) (R + r \cos \varphi) r \sin \varphi d\theta \wedge d\varphi - 2(R + r \cos \varphi)^2 \cos^2 \theta r^2 \sin \varphi \cos \varphi d\theta \wedge d\varphi \\ &= [R^2 + r^2 + 2Rr \cos \varphi - 2(R + r \cos \varphi) \cos^2 \theta r \cos \varphi] (R + r \cos \varphi) r \sin \varphi d\theta \wedge d\varphi = [R^2 + r^2 + 2Rr \sin^2 \theta \cos \varphi - 2r^2 \cos^2 \theta \cos^2 \varphi] (R + r \cos \varphi) r \sin \varphi d\theta \wedge d\varphi \end{aligned}$$

- (10 points) Consider the 1-form $\beta = (\frac{x^3}{3} + xz^2)dy - \frac{y^3}{3}dx$. Compute $d\beta$, and the pullbacks $\gamma_+^* \beta$ and $\gamma_-^* \beta$.

We have $d\beta = x^2 dx \wedge dy + z^2 dx \wedge dy + 2xz dz \wedge dy - y^2 dy \wedge dx = \alpha$. And:

$$\gamma_\pm^*(dx) = -(R \pm r) \sin \theta d\theta, \quad \gamma_\pm^*(dy) = (R \pm r) \cos \theta d\theta, \quad \gamma_\pm^* \beta = \frac{(R \pm r)^3 \cos^3 \theta}{3} (R \pm r) \cos \theta d\theta + \frac{(R \pm r)^3 \sin^3 \theta}{3} (R \pm r) \sin \theta d\theta = \frac{(R \pm r)^4}{3} (\cos^4 \theta + \sin^4 \theta) d\theta$$

- (15 points) Compute $\int_T \alpha$.

Hint: We have $\int d\theta (\cos^4 \theta + \sin^4 \theta) = \frac{\sin(2\theta) \cos(2\theta) + 6\theta}{8}$.

By the Stokes theorem $\int_T \alpha = \int_T d\beta = \int_{\partial T} \beta = \int_{\gamma_+} \beta + \int_{\gamma_-} \beta = \int_{[0, 2\pi]} \gamma_+^*(\beta) + \int_{[0, 2\pi]} \gamma_-^*(\beta) = \int_0^{2\pi} d\theta \frac{(R+r)^4}{3} (\cos^4 \theta + \sin^4 \theta) - \int_0^{2\pi} d\theta \frac{(R-r)^4}{3} (\cos^4 \theta + \sin^4 \theta)$, which is $\frac{1}{3} ((R+r)^4 - (R-r)^4) \frac{6 \cdot 2\pi}{8} = \frac{\pi}{2} ((R+r)^2 - (R-r)^2) ((R+r)^2 + (R-r)^2) = 4\pi r R (R^2 + r^2)$.

Exercise 4 (Cycloidal pendulum, 20 points). The reversed cycloid is the curve embedded in $\mathbb{R}^2$ by $\phi : [-\pi, \pi] \rightarrow \mathbb{R}^2$, $\phi(u) = (x(u), y(u))$ with:

$$x(u) = a[u + \sin(u)], \quad y(u) = a[1 - \cos(u)].$$

We have proven in the Hauptklausur that for any Lagrangian $L(q, \dot{q})$, the energy function $E(q, \dot{q}) = \dot{q} \frac{\partial L}{\partial \dot{q}} - L(q, \dot{q})$ is conserved along the classical trajectories.

- (5 points) Write the Lagrangian and the energy function of a particle constrained to move on the cycloid subjected to a gravitational potential $V(x, y) = mgy$.

The free Lagrangian is $L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) = \frac{m}{2}a^2[(\dot{u} + \cos(u)\dot{u})^2 + (-\sin(u)\dot{u})^2] = ma^2(1 + \cos(u))\dot{u}^2 = 2ma^2 \cos^2(\frac{u}{2}) \dot{u}^2$ leading to:

$$L = 2ma^2 \cos^2(\frac{u}{2}) \dot{u}^2 - mga[1 - \cos(u)] \quad E = 2ma^2 \cos^2(\frac{u}{2}) \dot{u}^2 + mga[1 - \cos(u)] = 2ma^2 \cos^2(\frac{u}{2}) \dot{u}^2 + 2mga \sin^2(\frac{u}{2}).$$

B. EXAMS

- (10 points) Assuming the particle is dropped from the point with coordinate u_0 with zero initial velocity at time $t = 0$, determine $u(t)$ up to the first passage of the particle by the minimum of the curve, $u = 0$. Before the first passage by $u = 0$, if $u_0 \in [0, \pi]$, then $\dot{u} < 0$ and if $u_0 \in [-\pi, 0]$, then $\dot{u} > 0$.

Hint: Note that if $\frac{du}{dt}(t') = f(u(t'))$ then $1 = \frac{1}{f(u(t'))} \frac{du}{dt}(t')$ and consequently $t = \int_0^t \frac{1}{f(u(t'))} \frac{du}{dt}(t') dt' = \int_{u(0)}^{u(t)} du \frac{1}{f(u)}$.

The conservation of the energy writes $E = 2mga \sin^2(\frac{u_0}{2}) = 2ma^2 \cos^2(\frac{u}{2}) \dot{u}^2 + 2mga \sin^2(\frac{u}{2}) \Rightarrow \dot{u} = \pm \sqrt{g/a \frac{\sin^2(\frac{u_0}{2}) - \sin^2(\frac{u}{2})}{\cos^2(\frac{u}{2})}}$. We take $u_0 \in [0, \pi]$, hence $\dot{u} < 0$:

$$-\sqrt{g/a}t = \int_{u_0}^{u(t)} \frac{\cos(\frac{u}{2}) du}{\sqrt{\sin^2(\frac{u_0}{2}) - \sin^2(\frac{u}{2})}} = 2 \int_{\sin(\frac{u_0}{2})}^{\sin(\frac{u(t)}{2})} \frac{dx}{\sqrt{\sin^2(\frac{u_0}{2}) - x^2}} = 2 \arcsin\left(\frac{x}{\sin(\frac{u_0}{2})}\right) \Big|_{\sin(\frac{u_0}{2})}^{\sin(\frac{u(t)}{2})} = 2 \arcsin\left(\frac{\sin(\frac{u(t)}{2})}{\sin(\frac{u_0}{2})}\right) - 2 \frac{\pi}{2} \Rightarrow \boxed{\sin\left(\frac{u_0}{2}\right) \sin\left(\frac{\pi - \sqrt{g/a}t}{2}\right) = \sin\left(\frac{u(t)}{2}\right)}.$$

- (5 points) Determine the time t_c at which the particle reaches for the first time the coordinate $u = 0$. Comment on the result.

We have $\boxed{t_c = \frac{\pi}{\sqrt{g/a}}}$ irrespective of u_0 , that is the curve is isochronal – the particle reaches the minimum in the same amount of time, not matter from what height the particle is launched.

B.2 Sample exams HöMa III

B.2.1 Sample 1

Exercise 1 (Complex analysis, 20 points). *Solve the following:*

- (10 points) We denote C the positively oriented circle of center 0 and radius 1 in the complex plane. Compute the integrals:

$$\frac{1}{2\pi i} \int_C dz z^n \bar{z}^m \quad \forall n, m \in \mathbb{Z}; \quad I = \frac{1}{2\pi i} \int_C dz z^3 [\cos(\bar{z})]^2.$$

We have $\frac{1}{2\pi i} \int_C dz z^n \bar{z}^m = \delta_{n+1, m}$ and $\frac{1}{2\pi i} \int_C dz z^3 [\cos(\bar{z})]^2 = \frac{1}{2\pi i} \int_C dz z^3 (1 - \frac{1}{2!} \bar{z}^2 + \frac{1}{4!} \bar{z}^4)^2 = \frac{1}{2\pi i} \int_C dz z^3 (\frac{1}{2!2!} \bar{z}^4 + \frac{2}{4!} \bar{z}^4) = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$.

- (10 points) Compute the integral:

$$J = \int_{-\infty}^{\infty} dx \frac{1}{(x^2 + 1)(x^2 + 4)}.$$

We close the contour by a semicircle at infinity in the upper half plane and we get $J = \int_C dz \frac{1}{(z-i)(z+i)(z-i2)(z+i2)}$. By the residue theorem we have $2\pi i \lim_{z \rightarrow i} \frac{1}{(z+i)(z-i2)(z+i2)} + 2\pi i \lim_{z \rightarrow 2i} \frac{1}{(z-i)(z+i)(z-i2)} = 2\pi i \left(\frac{1}{(2i)(-i)(3i)} + \frac{1}{(i)(3i)(4i)} \right) = 2\pi i \left(\frac{1}{6} - \frac{1}{12} \right) = \frac{\pi}{6}$.

Exercise 2 (Differential equations, 20 points). *Solve the following:*

- (10 points) Find the general solution of the linear ordinary differential equation:

$$y''' + 3y'' - 9y' + 5y = 0.$$

The characteristic polynomial is $\lambda^3 + 3\lambda^2 - 9\lambda + 5 = 0$. The polynomial factors as $\lambda^3 + 3\lambda^2 - 9\lambda + 5 = (\lambda + 5)(\lambda - 1)^2$, hence -5 is a simple root and 1 is a double root, leading to the general solution $c_1 e^{-5x} + c_2 e^x + c_3 x e^x$.

- (10 points) Consider $j_l : [0, \infty] \rightarrow \mathbb{R}$ and $l \in \mathbb{N}$. Find a solution of the differential equation:

$$x^2 j_l'' + 2x j_l' + [x^2 - l(l+1)] j_l = 0,$$

such that $j_l(0) < \infty$ using the power series method. Prove that this solution can also be written as $j_l(x) = (-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\sin x}{x}$. Note that this differential equation admits a second linearly independent solution, but this diverges at the origin.

We substitute $j_l(x) = \sum a_k x^k$ and we obtain the recursion $\sum_k x^k \left[[k(k-1) + 2k - l(l+1)] a_k + a_{k-2} \right] = 0$ hence $a_k = -\frac{1}{k(k+1) - l(l+1)} a_{k-2}$. It follows that the series truncates at $k = l$ (the alternative $k = -l - 1$ diverges at the origin). $j_l(x) < \infty$ implies $a_{l+2q+1} = 0$ for all $q \in \mathbb{N}$ and (with the convention that empty products are 1):

$$\forall k \in \mathbb{N}, \quad a_{l+2k} = a_l (-1)^k \prod_{q=1}^k \frac{1}{(l+2q)(l+2q+1) - l(l+1)} = a_l (-1)^k \prod_{q=1}^k \frac{1}{4q^2 + 2q(2l+1)} = a_l \frac{(-1)^k}{2^k k!} \prod_{q=1}^k \frac{1}{(2q+2l+1)} = a_l \frac{(-1)^k}{2^k k!} \frac{2^{k+l} (k+l)!}{(2k+2l+1)!} \frac{(2l)!}{2^l l!}.$$

Hence $j_l = \left(a_l \frac{(2l)!}{2^l l!} \right) 2^l x^l \sum_{k \geq 0} \frac{(-1)^k (k+l)!}{k! (2k+2l+1)!} x^{2k}$, and by convention the prefactor is set to 1. We check directly on the series that:

$$j_0(x) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)!} x^{2k} = \frac{\sin x}{x}, \quad \frac{(-1)^{k+l}}{(2k+2l+1)!} (-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l x^{2k+2l} = \frac{(-1)^{k+l}}{(2k+2l+1)!} (-1)^l (2k+2l)(2k+2l-2) \dots (2k+2) x^{2k+l} = \frac{(-1)^k (k+l)!}{k! (2k+2l+1)!} 2^l x^{2k+l}.$$

Exercise 3 (Fourier series, 30 points). *Consider the function $f : [-\pi, \pi] \rightarrow \mathbb{R}$, $f(x) = x^2$.*

- (10 points) Write the Fourier series of f and compute its Fourier coefficients.

$$f(x) = \sum_{n \in \mathbb{Z}} c_n e^{inx}, \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} dx e^{-inx} x^2 = \begin{cases} n=0, & \frac{1}{2\pi} \left[\frac{1}{3} x^3 \right]_{-\pi}^{\pi} = \frac{\pi^2}{3} \\ n \neq 0, & \frac{1}{2\pi} \left[\underbrace{\frac{e^{-inx}}{-in} x^2}_{=0} \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} dx \underbrace{\frac{e^{-inx}}{-in}}_{=0} 2x \right] = \frac{1}{in\pi} \left[\frac{e^{-inx}}{-in} x \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} dx \underbrace{\frac{e^{-inx}}{-in}}_{=0} = \frac{2(-1)^n}{n^2} \right], \end{cases}$$

Hence $x^2 = \frac{\pi^2}{3} + \sum_{n \neq 0} 2 \frac{(-1)^n}{n^2} e^{inx} = \frac{\pi^2}{3} + \sum_{n \geq 1} \frac{(-1)^n}{n^2} 4 \cos(nx)$.

- (10 points) Prove that:

$$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}, \quad \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

As $f(x) = x^2$ is continuous and continuously differentiable on $(-\pi, \pi)$, the Fourier series converges pointwise in $(-\pi, \pi)$, see the script, Lemma 13.1. We have setting $x = 0$ that $-\frac{\pi^2}{3} = \sum_{n \geq 1} \frac{(-1)^n}{n^2} 4 \Rightarrow \sum_{n \geq 1} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$ and setting $x = \pi$, $\pi^2 = \frac{\pi^2}{3} + \sum_{n \geq 1} \frac{(-1)^n}{n^2} 4(-1)^n \Rightarrow \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$.

THE FOLLOWING IS NOT REQUIRED: Note that in the script we only showed pointwise convergence on the open interval. The series converges also at the end points, as the function is continuous when continued by periodicity. To see this directly, note that the Fourier series converges pointwise for $x = \pi - \delta$, hence:

$$(\pi - \delta)^2 = \frac{\pi^2}{3} + \sum_{n \geq 1} \frac{(-1)^n}{n^2} 4 \cos(n\pi - n\delta) = \pi^2 - 2\delta\pi + \delta^2 = \frac{\pi^2}{3} + \sum_{n \geq 1} \frac{(-1)^n}{n^2} 4[(-1)^n \cos(n\delta)] \Rightarrow \frac{\pi^2}{6} = \sum_{n \geq 1} \frac{1}{n^2} + \frac{\delta\pi}{4} - \frac{\delta^2}{4} - \sum_{n \geq 1} \frac{1}{n^2} \left[1 - \cos(n\delta) \right].$$

B. EXAMS

The $\sum_{n \geq 1} \frac{1}{n^2}$ is monotonically increasing and converges to some M . Then $\forall \epsilon, \exists N_\epsilon$ such that $\sum_{n > N_\epsilon} \frac{1}{n^2} < \frac{\epsilon}{4}$. Chose δ such that $1 - \cos(N_\epsilon \delta) < \frac{\epsilon}{2M}$ (which always exists as the cosine is continuous at 0). Thus (using the monotonicity of the cosine close enough to zero) we get:

$$\sum_{n \geq 1} \frac{1}{n^2} [1 - \cos(n\delta)] = \sum_{n=1}^{N_\epsilon} \frac{1}{n^2} [1 - \cos(n\delta)] + \sum_{n > N_\epsilon} \frac{1}{n^2} [1 - \cos(n\delta)] \leq \sum_{n=1}^{N_\epsilon} \frac{1}{n^2} [1 - \cos(N_\epsilon \delta)] + \sum_{n > N_\epsilon} \frac{1}{n^2} 2 \leq \frac{\epsilon}{2M} \sum_{n=1}^{N_\epsilon} \frac{1}{n^2} + 2 \frac{\epsilon}{4} = \epsilon.$$

- (10 points) Prove that:

$$\sum_{n \geq 1} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

From Parseval's theorem (be carefull with the normalizations): $\frac{x^5}{5} \Big|_{-\pi}^{\pi} = \int_{-\pi}^{\pi} dx \ x^4 = 2\pi \sum_{n \in \mathbb{Z}} |c_n|^2 = 2\pi \left(\left(\frac{\pi^2}{3} \right)^2 + \sum_{n \neq 0} \frac{4}{n^4} \right) \Rightarrow 2 \frac{\pi^5}{5} = 2 \frac{\pi^5}{9} + 16\pi \sum_{n \geq 1} \frac{1}{n^4}.$

Exercise 4 (Infinite spherical well, 50 points). A particle in a spherically symmetric potential is described by the Hamiltonian $H = -\frac{1}{2}\Delta + V(r)$. We aim to find the eigenvalues E and eigenfunctions ψ , that is solve $H\psi = E\psi$ in the case of an infinite potential well $V(r) = 0$ for $r \leq a$ and $V(r) = \infty$, $r > a$. As $V(r) = \infty$ for $r > a$, we have $\psi(r) = 0$ for $r > a$. This is a good model for the nucleus.

We aim to solve in the domain $r \in [0, a], \theta \in [0, \pi], \varphi \in [0, 2\pi]$ the PDE:

$$-\frac{1}{2} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{L^2 \psi}{2r^2} = E\psi, \quad -L^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}.$$

with boundary and regularity conditions $|\psi(r, \theta, \varphi)| < \infty$, $\psi(a, \theta, \varphi) = 0$, $\psi(r, \theta, \varphi + 2\pi) = \psi(r, \theta, \varphi)$.

- (5 points) Assuming separation of variables $\psi(r, \theta, \varphi) = R(r)Y(\theta, \varphi)$, derive the corresponding equations for the angular and the radial part.

We posit $\psi(r, \theta, \varphi) = R(r)Y(\theta, \varphi)$ and we get

$$\frac{2r^2}{R Y} \left(\frac{1}{2r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R Y}{\partial r} \right) + E R Y - \frac{L^2 R Y}{2r^2} \right) = 0 \Rightarrow \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + 2E r^2 = \frac{L^2 Y}{Y} = \lambda.$$

- (15 points) The angular part can be solved in terms of spherical harmonics $Y_l^m(\theta, \varphi)$. Formulate an appropriate Sturm Liouville problem for the radial equation. Prove that $E \geq 0$. A useful notation is $2E = \kappa^2$.

We have $L^2 Y = l(l+1)Y_l^m(\theta, \varphi)$, see script. The radial Sturm Liouville problem is:

$$R: [0, a] \rightarrow \mathbb{R}, \quad [r^2 R']' + \kappa^2 r^2 R - l(l+1)R = 0, \quad R(0), R'(0) < \infty, \quad R(a) = 0, R'(a) < \infty.$$

The Sturm Liouville operator is $LR = -\frac{1}{r^2} [r^2 R']' - l(l+1)R$ and the eigenvalue equation $LR = \kappa^2 R$ corresponding to $p(r) = r^2$, $q(r) = -l(l+1)$ and the density $w(r) = r^2$. L is self andjoint as:

$$\int_0^a r^2 dr R_1 L R_2 = \int_0^a dr (-1) R_1 [r^2 R_2'] - l(l+1) R_2 = -[R_1 r^2 R_2']_0^a + [R_1' r^2 R_2]_0^a + \int_0^a dr (-1) [r^2 R_1'] - l(l+1) R_1 R_2,$$

and the boundary terms are zero as $R_1(a) = R_2(a) = 0$.

We have:

$$\kappa^2 \int_0^a dr r^2 R^2 = \int_0^a dr r^2 R L R = \int_0^a dr R (-1) [r^2 R'] - l(l+1) R = \kappa^2 = \frac{-[R r^2 R']_0^a + \int_0^a dr r^2 (R')^2 + \int_0^a dr l(l+1) R^2}{\int_0^a dr r^2 R^2} \geq 0.$$

We conclude that the eigenfunctions of the radial equation at fixed l are a basis in the Hilbert space $L^2([0, a], r^2 dr)$.

- (15 points) Using the results in Exercise 2 (even without having proven any of them!), conclude that the radial eigenfuctions must be of the form $j_l(\kappa r)$ with $j_l(x) = (-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\sin x}{x}$.

Denoting $x_{l,k} > 0$ the k 'th non trivial zero of j_l , $j_l(x_{l,k}) = 0$, find the eigenfuctions of the Hamiltonian, deduce the energy levels $E_{l,k}$ of the Hamiltonian, their degeneracies, and compute $E_{0,k}$.

For any κ real, we have $j_l(0) < \infty$ and:

$$\frac{d}{dr} \left[r^2 \frac{d}{dr} j_l(\kappa r) \right] + \kappa^2 r^2 j_l(\kappa r) - l(l+1) j_l(\kappa r) = \left(\frac{d}{dx} \left[x^2 \frac{d}{dx} j_l(x) \right] + [x^2 - l(l+1)] j_l(x) \right)_{x=\kappa r},$$

From the boundary condition $j_l(\kappa a) = 0$ we conclude that the only allowed values are $\kappa_{l,k} = \frac{x_{l,k}}{a} \Rightarrow E_{l,k} = \frac{x_{l,k}^2}{2a^2}$ with degeneracies $2l+1$. The eigenfuctions are $j_l(\frac{x_{l,k}}{a} r) Y_l^m(\theta, \varphi) = j_l(\frac{x_{l,k}}{a} r) P_l^m(\cos \theta) e^{im\varphi}$ with $k \geq 1$, $l \geq 0$ and $m = -l, \dots, l$. As $x_{0,k}$ are the zeroes of the sinus they are located at $x_{0,k} = k\pi$, hence $E_{0,k} = \frac{\pi^2}{2a^2} k^2$.

- (15 points) Using the differential equation, prove directly the orthogonality of $j_l(\kappa_{l,k} x)$ and $j_l(\kappa_{l,q} x)$.

We aim to prove $\int_0^a r^2 dr j_l(\kappa_{l,k} r) j_l(\kappa_{l,q} r) = 0$. We have for any κ_1, κ_2 :

$$(\kappa_1^2 - \kappa_2^2) r^2 j_l(\kappa_1 r) j_l(\kappa_2 r) = j_l(\kappa_2 r) \left[l(l+1) j_l(\kappa_1 r) - \frac{d}{dr} \left[r^2 \frac{d}{dr} j_l(\kappa_1 r) \right] \right] - j_l(\kappa_1 r) \left[l(l+1) j_l(\kappa_2 r) - \frac{d}{dr} \left[r^2 \frac{d}{dr} j_l(\kappa_2 r) \right] \right] \Rightarrow$$

$$(\kappa_1^2 - \kappa_2^2) \int_0^a dr r^2 j_l(\kappa_1 r) j_l(\kappa_2 r) = j_l(\kappa_1 r) r^2 \frac{d}{dr} j_l(\kappa_2 r) \Big|_{r=0}^{r=a} - j_l(\kappa_2 r) r^2 \frac{d}{dr} j_l(\kappa_1 r) \Big|_{r=0}^{r=a},$$

and using $j_l(\kappa_{l,k} a) = 0$ we conclude that $(\kappa_{l,k}^2 - \kappa_{l,q}^2) \int_0^a dr r^2 j_l(\kappa_{l,k} r) j_l(\kappa_{l,q} r) = 0$.

B.2.2 Sample 2

Exercise 1 (Complex analysis, 30 points). We denote C the positively oriented circle of center 0 and radius 1 in the complex plane.

- (10 points) Compute the integral:

$$\frac{1}{2\pi i} \int_C dz \frac{1}{\cos(z) - 1}.$$

- (10 points) Compute the integral:

$$\frac{1}{2\pi i} \int_C dz \frac{z}{\cos(z) - 1}.$$

- (10 points) Compute the integral:

$$\frac{1}{2\pi i} \int_C dz \frac{z^2}{\cos(z) - 1}.$$

The functions $\cos(z) - 1$, $\frac{\cos(z)-1}{z}$ and $\frac{\cos(z)-1}{z^2}$ are entire, hence their inverses are meromorphic in $\mathbb{C}$. Inside the circle of radius 1, their inverses display a singularity in $z = 0$.
For $\frac{1}{\cos(z)-1}$, $z = 0$ is a pole of order 2.

$$\frac{1}{2\pi i} \int_C dz \frac{1}{\cos(z) - 1} = \text{Res}\left(\frac{1}{\cos(z) - 1}, 0\right) = \lim_{z \rightarrow 0} \left[z^2 \frac{1}{\cos(z) - 1} \right]' = \lim_{z \rightarrow 0} \frac{2z[\cos(z) - 1] + z^2 \sin(z)}{[\cos(z) - 1]^2} \sim \frac{2z(-\frac{z^2}{2!} + \frac{z^4}{4!} - \dots) + z^2(z - \frac{z^3}{3!})}{(-\frac{z^2}{2} + \dots)^2} = 0,$$

or alternatively:

$$\frac{1}{\cos(z) - 1} = \frac{1}{-\frac{z^2}{2!} + \frac{z^4}{4!} - O(z^6)} = -\frac{2!}{z^2} \frac{1}{1 - 2! \frac{z^2}{4!} + O(z^4)} = -\frac{2!}{z^2} [1 + 2! \frac{z^2}{4!} + O(z^4)] = -\frac{2!}{z^2} - \frac{2!2!}{4!} + O(z^2),$$

hence the residue is zero at the pole.

For $\frac{z}{\cos(z)-1}$, $z = 0$ is a pole of order 1, hence:

$$\frac{1}{2\pi i} \int_C dz \frac{z}{\cos(z) - 1} = \text{Res}\left(\frac{z}{\cos(z) - 1}, 0\right) = \lim_{z \rightarrow 0} z \frac{z}{\cos(z) - 1} = -2.$$

For $\frac{z^2}{\cos(z)-1}$, $z = 0$ is a removable singularity, the function is holomorphic and by Cauchy's integral theorem:

$$\frac{1}{2\pi i} \int_C dz \frac{z^2}{\cos(z) - 1} = 0.$$

Exercise 2 (Differential equations, 20 points). Solve the following:

- (5 points) Solve the differential equation with boundary conditions:

$$Z : [0, L] \rightarrow \mathbb{C} : \quad Z'' + \alpha Z = 0, \quad Z(0) = 0, \quad Z(L) = 0.$$

The general solution is $Z(z) = C e^{i\sqrt{\alpha}z} + D e^{-i\sqrt{\alpha}z}$. The boundary conditions impose $C + D = 0$ and $e^{2i\sqrt{\alpha}L} = 1$ that is α must take one of the allowed values $\left(\frac{\pi}{L}l\right)^2$ with corresponding solution $Z_l(z) = \sin\left(\frac{\pi}{L}lz\right)$, $\alpha_l = \left(\frac{\pi}{L}l\right)^2$, $l \in \mathbb{N}_{\geq 1}$. For each α_l we have one solution Z_l .

- (5 points) Solve for a periodic function $\Phi : [0, 2\pi] \rightarrow \mathbb{C}$, $\Phi(\varphi + 2\pi) = \Phi(\varphi)$ the differential equation:

$$\Phi : [0, 2\pi] \rightarrow \mathbb{C} : \quad \Phi'' + \gamma \Phi = 0, \quad \Phi(\varphi + 2\pi) = \Phi(\varphi).$$

The two independent solutions of the ODE are $e^{\pm i\sqrt{\gamma}\varphi}$, and the periodicity imposes $\sqrt{\gamma} \in \mathbb{N}$, hence the solutions are $\Phi_{\pm m}(\varphi) = e^{\pm im\varphi}$, $\gamma_m = m^2$, $m \in \mathbb{N}$. Note that for each allowed γ_m we have two independent solutions, $\Phi_{\pm m}$, and only one for $\gamma_m = 0$.

- (10 points) Denote $J_n : [0, \infty] \rightarrow \mathbb{R}$. Find for $n \in \mathbb{N}$ a solution of the differential equation:

$$x^2 J_n'' + x J_n' + (x^2 - n^2) J_n = 0,$$

such that $J_n(0) < \infty$ using the power series method. Note that this differential equation admits a second linearly independent solution, but this diverges at the origin.

We substitute $J_n(x) = \sum a_k x^k$ and we obtain the recursion:

$$\sum_k x^k [k(k-1) + k - n^2] a_k + a_{k-2} = 0 \Rightarrow a_k = -\frac{1}{k^2 - n^2} a_{k-2}.$$

It follows that the series truncates at $k = n$ (the alternative $k = -n$ diverges at the origin). $J_n(0) < \infty$ implies $a_{n+2q+1} = 0$ for all $q \in \mathbb{N}$ and (with the convention that empty products are 1):

$$\forall k \in \mathbb{N}, \quad a_{n+2k} = a_n (-1)^k \prod_{q=1}^k \frac{1}{(n+2q)^2 - n^2} = a_n (-1)^k \prod_{q=1}^k \frac{1}{4q^2 + 4qn} = a_n \frac{(-1)^k}{4^k k!} \prod_{q=1}^k \frac{1}{q+n} = a_n \frac{(-1)^k}{4^k k!} \frac{n!}{(n+k)!}.$$

Hence:

$$J_n = a_n n! 2^n \sum_{k \geq 0} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{2k+n},$$

and by convention the prefactor is set to 1.

Exercise 3 (Fourier transform, 20 points). We recall that the Fourier transform and its inverse in one dimension are:

$$\hat{f}(p) = \int_{-\infty}^{\infty} dx e^{-ipx} f(x), \quad f(x) = \int_{-\infty}^{\infty} \frac{dp}{2\pi} e^{ipx} \hat{f}(p).$$

B. EXAMS

- (10 points) Compute the Fourier transform of the function $f(x) = \Gamma(1 + ix)$, where $\Gamma(z) = \int_0^\infty dt t^{z-1} e^{-t}$ is the Euler gamma function.

Using $\delta(f(x)) = \frac{1}{|f'(x_0)|} \delta(x - x_0)$ if $f(x_0) = 0$ (note that $\frac{1}{|f'(x_0)|} \delta(x - x_0) = \frac{1}{|f'(x)|} \delta(x - x_0)$), we have:

$$f(p) = \int_{-\infty}^{\infty} dx e^{-ixp} \int_0^\infty dt t^{ix} e^{-t} = \int_0^\infty dt e^{-t} \int_{-\infty}^{\infty} dx e^{ix(\ln t - p)} = 2\pi \int_0^\infty dt e^{-t} \delta(\ln t - p) = 2\pi \int_0^\infty dt e^{-t} \frac{\delta(t - e^p)}{\frac{1}{t}} = 2\pi e^{p-e^p},$$

where the commutation of integrals makes sense in distributions.

- (10 points) Let $p \in \mathbb{R}^3$ and denote $p^2 = p_1^2 + p_2^2 + p_3^2$. Compute the inverse Fourier transform of $\hat{f}(p) = \frac{1}{p^2 + m^2}$.

We recall Jordan's Lemma: Let $C_R = Re^{i\varphi}$, $\varphi \in [0, \pi]$ and $a > 0$. Then:

$$\left| \int_{C_R} dz g(z) e^{iaz} \right| \leq \frac{\pi}{a} \sup_{\varphi \in [0, \pi]} |g(Re^{i\varphi})|.$$

THIS IS NOT REQUIRED. Jordan's Lemma proof:

$$\left| \int_{C_R} dz g(z) e^{iaz} \right| \leq 2R \sup_{\varphi \in [0, \pi]} |g(Re^{i\varphi})| \int_0^{\pi/2} d\varphi e^{-R \sin \varphi} \Big|_{\sin \varphi \geq \frac{2}{\pi} \varphi} \leq 2R \sup_{\varphi \in [0, \pi]} |g(Re^{i\varphi})| \int_0^{\pi/2} d\varphi e^{-\frac{2}{\pi} Ra \varphi},$$

We have:

$$\begin{aligned} \int \frac{d^3 p}{(2\pi)^3} e^{ipx} \frac{1}{p^2 + m^2} &= \frac{1}{(2\pi)^3} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \int_0^\infty dp p^2 e^{ip|x| \cos \theta} \frac{1}{p^2 + m^2} = \frac{1}{(2\pi)^2} \int_{-1}^1 du \int_0^\infty dp p^2 \frac{e^{ip|x|u}}{p^2 + m^2} \\ &= \frac{1}{(2\pi)^2} \int_0^\infty dp p^2 \frac{1}{p^2 + m^2} \left(\frac{e^{ip|x|}}{ip|x|} - \frac{e^{-ip|x|}}{ip|x|} \right) = \frac{1}{(2\pi)^2} \frac{1}{i|x|} \int_{-\infty}^\infty dp \frac{p}{p^2 + m^2} e^{ip|x|}. \end{aligned}$$

The poles are located at $p = \pm im$, we close with a circle at infinity in the upper half plane, and the integral over the circle goes to zero from Jordan's Lemma. We have from the residue theorem:

$$f(x) = \frac{1}{(2\pi)^2} \frac{1}{i|x|} 2\pi i \lim_{p \rightarrow im} \frac{p}{(p + im)} e^{ip|x|} = \frac{1}{4\pi} \frac{e^{-m|x|}}{|x|}$$

Exercise 4 (Cylindrical potential well, 50 points). A particle in a cylindrically symmetric potential is described by the Hamiltonian $H = -\frac{1}{2}\Delta + V(r, z)$. We aim to find the eigenvalues E and eigenfunctions ψ , that is solve $H\psi = E\psi$ in the case of an infinite potential well $V(r, z) = 0$ for $r \leq a$ and $0 \leq z \leq L$ and $V(r, z) = \infty$ if not. As $V(r, z) = \infty$ for $r > a$, $z \leq 0$ or $z \geq L$ we have $\psi(r, \varphi, z) = 0$ for $r > a$, $z \leq 0$ or $z \geq L$.

- (5 points) Recalling that the cylindrical coordinates (r, φ, z) are related to the Cartesian coordinates by: $x = r \cos \varphi$, $y = r \sin \varphi$, $z = z$, and using the formula for the Laplacian in an arbitrary metric $g_{\mu\nu}$:

$$\Delta = \frac{1}{\sqrt{\det g}} \partial_\mu \left(\sqrt{\det g} g^{\mu\nu} \partial_\nu \right),$$

show that the Laplacian in cylindrical coordinates is:

$$\Delta = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2}.$$

We have the line element (metric two form) in cylindrical coordinates $g_{\mu\nu} ds^\mu ds^\nu = dr^2 + r^2 d\varphi^2 + dz^2$ hence:

$$\Delta = \frac{1}{\sqrt{\det g}} \partial_\mu \left(\sqrt{\det g} g^{\mu\nu} \partial_\nu \right) = \frac{1}{r} \left[\partial_r (r \partial_r) + \partial_\varphi \left(r \frac{1}{r^2} \partial_\varphi \right) + \partial_z (r \partial_z) \right] = \frac{1}{r} \partial_r (r \partial_r) + \frac{1}{r^2} \partial_\varphi^2 + \partial_z^2.$$

Alternatively, $r = \sqrt{x^2 + y^2}$, $\varphi = \arctan(y/x)$ hence:

$$\begin{aligned} \partial_x r &= \frac{x}{\sqrt{x^2 + y^2}} = \cos \varphi, \quad \partial_y r = \sin \varphi, \quad \partial_x \varphi = \frac{-\frac{y}{x^2}}{1 + \frac{y^2}{x^2}} = -\frac{\sin \varphi}{r}, \quad \partial_y \varphi = \frac{\cos \varphi}{r}, \\ \partial_x &= \cos \varphi \partial_r - \frac{\sin \varphi}{r} \partial_\varphi, \quad \partial_y = \sin \varphi \partial_r + \frac{\cos \varphi}{r} \partial_\varphi, \end{aligned}$$

Leading to:

$$\begin{aligned} \partial_x^2 + \partial_y^2 &= \left(\cos \varphi \partial_r - \frac{\sin \varphi}{r} \partial_\varphi \right) \left(\cos \varphi \partial_r - \frac{\sin \varphi}{r} \partial_\varphi \right) + \left(\sin \varphi \partial_r + \frac{\cos \varphi}{r} \partial_\varphi \right) \left(\sin \varphi \partial_r + \frac{\cos \varphi}{r} \partial_\varphi \right) \\ &= \cos^2 \varphi \partial_r^2 - 2 \frac{\cos \varphi \sin \varphi}{r} \partial_r \partial_\varphi + \frac{\sin^2 \varphi}{r^2} \partial_r^2 + \frac{\sin \varphi \cos \varphi}{r^2} \partial_r \partial_\varphi + \frac{\sin^2 \varphi}{r^2} \partial_\varphi^2 \\ &\quad + \sin^2 \varphi \partial_r^2 + 2 \frac{\cos \varphi \sin \varphi}{r} \partial_r \partial_\varphi + \frac{\cos^2 \varphi}{r^2} \partial_r^2 - \frac{\sin \varphi \cos \varphi}{r^2} \partial_r \partial_\varphi + \frac{\cos^2 \varphi}{r^2} \partial_\varphi^2 = \partial_r^2 + \frac{1}{r} \partial_r + \frac{1}{r^2} \partial_\varphi^2 \end{aligned}$$

We aim to solve in the domain $r \in [0, a]$, $\varphi \in [0, 2\pi]$, $z \in [0, L]$ the PDE:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \varphi^2} + \frac{\partial^2 \psi}{\partial z^2} + 2E\psi = 0,$$

with boundary and regularity conditions $|\psi(r, \varphi, z)| < \infty$, $\psi(a, \varphi, z) = 0$; $\psi(r, \varphi, 0) = \psi(r, \varphi, L) = 0$; $\psi(r, \varphi + 2\pi, z) = \psi(r, \varphi, z)$.

- (10 points) Assuming separation of variables $\psi(r, \varphi, z) = f(r, \varphi)Z(z)$, derive the corresponding equations for $f(r, \varphi)$ and $Z(z)$. Assuming again separation of variables $f(r, \varphi) = R(r)\Phi(\varphi)$, derive the equations for $R(r)$ and $\Phi(\varphi)$.

We posit $\psi(r, \varphi, z) = f(r, \varphi)Z(z)$ and we get

$$\frac{1}{f} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \varphi^2} \right] + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} + 2E = 0 \Rightarrow \quad \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -\alpha, \quad \frac{1}{f} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \varphi^2} \right] = -\beta, \quad E = \frac{\alpha + \beta}{2}.$$

Furthermore,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \varphi^2} = -\beta f \xrightarrow{f=R\Phi} \frac{r}{R} \partial_r (r \partial_r R) + r^2 \beta + \frac{1}{\Phi} \partial_\varphi^2 \Phi = 0 \Rightarrow \quad \frac{1}{\Phi} \partial_\varphi^2 \Phi = -\gamma, \quad r \partial_r (r \partial_r R) + (r^2 \beta - \gamma) R = 0$$

- (20 points) Solve the equation for $\Phi(\varphi)$.

Recall that a Sturm Liouville problem is a differential equation for a function $x : [a, b] \rightarrow \mathbb{R}$ of the form $[px']' + qx = -\lambda w x$ for some functions p, q, w with boundary conditions $c_1 x(a) + c_2 x'(a) = d_1 x(b) + d_2 x'(b) = 0$ with $c_1^2 + c_2^2 \neq 0 \neq d_1^2 + d_2^2$. This is the eigenvalue equation $Lx = \lambda x$ for the self adjoint Sturm Liouville operator $Lx = -\frac{1}{w}[(px')' + qx]$ with boundary conditions. As they specify the domain of L , the boundary conditions are part of the definition of the operator.

Formulate the appropriate Sturm Liouville problem for $Z(z)$. Find the eigenfunctions and the corresponding eigenvalues.

Formulate the appropriate Sturm Liouville problem for $R(r)$ and prove that the eigenvalues λ of the corresponding Sturm Liouville operator are positive.

The angular differential equation:

$$\Phi : [0, 2\pi] \rightarrow \mathbb{C}, \quad \Phi(\varphi)'' + \gamma \Phi(\varphi) = 0, \quad \Phi(\varphi) = \Phi(\varphi + 2\pi),$$

is solved by $\Phi_{\pm m}(\varphi) = e^{\pm im\varphi}$ with $m \in \mathbb{N}$ and $\gamma = m^2$.

We have $Z : [0, L] \rightarrow \mathbb{C}$ with $Z'' + \alpha Z = 0$ with $Z(0) = Z(L) = 0$, hence $p = 1, q = 0, w = 1$. We have $Z_l(z) = \sin(\frac{\pi l}{L} z)$, $\alpha_l = (\frac{\pi l}{L})^2$.

Observing that $\gamma = m^2$, the radial equation for $R : [0, a] \rightarrow \mathbb{C}$ is:

$$r[rR']' + \beta r^2 R - m^2 R = 0, \quad R(0), R'(0) < \infty, \quad R(a) = 0, R'(a) < \infty.$$

The Sturm Liouville operator is $LR = -\frac{1}{w}[(pR')' + qR] = -\frac{1}{r}[rR']' - \frac{m^2}{r}R$ and the eigenvalue equation $LR = \beta R$ corresponds to $p(r) = r$, $q(r) = -\frac{m^2}{r}$, density $w(r) = r$ and β is exactly the eigenvalue λ . L is self adjoint as:

$$\int_0^a r dr R_1 L R_2 = \int_0^a dr R_1 (-1) \left[[rR_2']' - \frac{m^2}{r} R_2 \right] = -[R_1 r R_2']_0^a + [R_1' r R_2]_0^a + \int_0^a dr (-1) \left[[rR_1']' - \frac{m^2}{r} R_1 \right] R_2,$$

and the boundary terms are zero as $R_1(a) = R_2(a) = 0$.

We have:

$$\lambda \int_0^a dr r R^2 = \int_0^a dr r R L R = \int_0^a dr R (-1) \left[[rR']' - \frac{m^2}{r} R \right] \Rightarrow \lambda = \frac{-[RrR']_0^a + \int_0^a dr r (R')^2 + \int_0^a dr \frac{m^2}{r} R^2}{\int_0^a dr r R^2} \geq 0.$$

For fixed $\Phi(\varphi)$ and $Z(z)$ solutions of the angular and z equations, the eigenfunctions of the radial equation are a basis in the Hilbert space $L^2([0, a], r dr)$.

- (10 points) Using the results in Exercise 2 (even without having proven any of them), conclude that the radial eigenfunctions must be of the form $J_m(\sqrt{\lambda} r)$.

Denoting $x_{m,k} > 0$ the k 'th non trivial zero of J_m , $J_m(x_{m,k}) = 0$, find the eigenfunctions of the Hamiltonian, deduce the energy levels $E_{l,m,k}$ and their degeneracies.

For any $\sqrt{\lambda}$ real, we have $J_m(0) < \infty$ and:

$$r^2 \frac{d^2}{dr^2} J_m(\sqrt{\lambda} r) + r \frac{d}{dr} J_m(\sqrt{\lambda} r) + (\lambda r^2 - m^2) J_m(\sqrt{\lambda} r) = \left(x^2 \frac{d^2}{dx^2} J_m(x) + x \frac{d}{dx} J_m'(x) + (x^2 - m^2) J_m(x) \right)_{x=\sqrt{\lambda} r}.$$

From the boundary condition $J_m(\sqrt{\lambda} a) = 0$ we conclude that the only allowed values are $\sqrt{\lambda_{m,k}} = \frac{x_{m,k}}{a} \Rightarrow E_{l,m,k} = \frac{1}{2} \left(\frac{x_{m,k}^2}{a^2} + \left(\frac{\pi l}{L} \right)^2 \right)$ with degeneracy 2 for $m \neq 0$ and 1 for $m = 0$.

The eigenfunctions are $J_m(\frac{x_{m,k}}{a} r) e^{\pm im\varphi} \sin(\frac{\pi l}{L} z)$ with $k, l \in \mathbb{N}_{\geq 1}$ and $m \in \mathbb{N}$.

- (5 points) Using the differential equation, prove directly the orthogonality of $J_m(\sqrt{\lambda_{m,k}} x)$ and $J_m(\sqrt{\lambda_{m,q}} x)$.

We aim to prove $\int_0^a r dr J_m(\sqrt{\lambda_{m,k}} r) J_m(\sqrt{\lambda_{m,q}} r) = 0$. We have for any κ_1, κ_2 :

$$(\kappa_1^2 - \kappa_2^2) r J_m(\kappa_1 r) J_m(\kappa_2 r) = J_m(\kappa_2 r) r \left[\frac{m^2}{r^2} J_m(\kappa_1 r) - \frac{1}{r} \frac{d}{dr} \left[r \frac{d}{dr} J_m(\kappa_1 r) \right] \right] - J_m(\kappa_1 r) r \left[\frac{m^2}{r^2} J_m(\kappa_2 r) - \frac{1}{r} \frac{d}{dr} \left[r \frac{d}{dr} J_m(\kappa_2 r) \right] \right] \Rightarrow$$

$$(\kappa_1^2 - \kappa_2^2) \int_0^a dr r J_m(\kappa_1 r) J_m(\kappa_2 r) = -J_m(\kappa_2 r) r \frac{d}{dr} J_m(\kappa_1 r) \Big|_{r=0}^{r=a} + J_m(\kappa_1 r) r \frac{d}{dr} J_m(\kappa_2 r) \Big|_{r=0}^{r=a},$$

and using $J_m(\sqrt{\lambda_{m,k}} a) = J_m(\sqrt{\lambda_{m,q}} a) = 0$ we conclude.

B.2.3 Sample 3

Exercise 1 (The semicircles H_+ and H_- , 10 points). We denote $H_{+,R} = Re^{i\varphi}$ with $\varphi \in [0, \pi]$ and $H_{-,R} = Re^{-i\varphi}$ with $\varphi \in [0, \pi]$, the semicircles of radius R in the upper respectively lower half planes, oriented counterclockwise respectively clockwise. The semicircles of infinite radius H_+ and H_- are obtained as the limit $R \rightarrow \infty$ of $H_{+,R}$ and $H_{-,R}$. Prove that:

$$\int_{H_+} dz \frac{1}{z^2 + 1} = \int_{H_-} dz \frac{1}{z^2 + 1} = 0.$$

We have $\int_{H_+} dz \frac{1}{z^2 + 1} = \lim_{R \rightarrow \infty} \int_{H_{+,R}} dz \frac{1}{z^2 + 1} \leq \lim_{R \rightarrow \infty} \pi R \sup_{z \in H_{+,R}} \left| \frac{1}{z^2 + 1} \right| \leq \lim_{R \rightarrow \infty} \pi R \frac{1}{R^2 - 1} \rightarrow 0$, where we used $|z^2 + 1| \geq |z^2| - 1$.

Exercise 2 (Complex integrals, 30 points). Solve the following:

- (10 points) Compute the integral:

$$\int_{-\infty}^{\infty} dx \frac{1}{x^2 + 1}.$$

- (5 points) using the fundamental theorem of calculus.

The primitive of $1/(x^2 + 1)$ is $\arctan(x)$ hence $\int_{-\infty}^{\infty} dx \frac{1}{x^2 + 1} = \lim_{a \rightarrow -\infty, b \rightarrow +\infty} \arctan(x) \Big|_a^b = \frac{\pi}{2} - (-\frac{\pi}{2}) = \pi$.

- (5 points) using the residue theorem.

As $\int_{H_+} dz \frac{1}{z^2 + 1} = 0$, we have $\int_{-\infty}^{\infty} dx \frac{1}{x^2 + 1} = \int_{\mathbb{R} \cup H_+} dz \frac{1}{z^2 + 1} = 2\pi i \operatorname{Res}\left(\frac{1}{z^2 + 1}, i\right) = 2\pi i \lim_{z \rightarrow i} (z - i) \frac{1}{(z + i)(z - i)} = 2\pi i \frac{1}{2i} = \pi$.

- (10 points) Prove that, for $a \in \mathbb{R}$, we have:

$$I(a) = \int_{-\infty}^{\infty} dx \frac{e^{iax}}{x^2 + 1} = \pi e^{-|a|}.$$

Make sure that you justify all the steps in the computation, like proving that certain contributions are zero.

We have $|e^{iax}| = e^{-a \operatorname{Im}(z)}$ hence:

For $a > 0$, as $\int_{H_{+,R}} \leq \pi R \frac{1}{R^2 - 1} \rightarrow_{R \rightarrow \infty} 0$ we have $I(a) = \int_{\mathbb{R} \cup H_+} dz \frac{e^{iaz}}{z^2 + 1} = 2\pi i \operatorname{Res}\left(\frac{e^{iaz}}{z^2 + 1}, i\right) = 2\pi i \lim_{z \rightarrow i} (z - i) \frac{e^{iaz}}{(z - i)(z + i)} = \pi e^{-a}$

For $a < 0$, as $\int_{H_{-,R}} \leq \pi R \frac{1}{R^2 - 1} \rightarrow_{R \rightarrow \infty} 0$ we have $I(a) = \int_{\mathbb{R} \cup H_-} dz \frac{e^{iaz}}{z^2 + 1} = -2\pi i \operatorname{Res}\left(\frac{e^{iaz}}{z^2 + 1}, -i\right) = -2\pi i \lim_{z \rightarrow -i} (z + i) \frac{e^{iaz}}{(z - i)(z + i)} = \pi e^a$

- (10 points) Prove that for $a \in \mathbb{R}$ we have:

$$\int_{-\infty}^{\infty} dx \frac{e^{iax}}{(x^2 + 1)^2} = \frac{\pi}{2} (|a| + 1) e^{-|a|}.$$

Make sure that you justify all the steps in the computation, like proving that certain contributions are zero.

We have $|e^{iax}| = e^{-a \operatorname{Im}(z)}$ hence, as the poles are this times poles of order 2:

For $a > 0$, $\int_{H_{+,R}} \leq \pi R \frac{1}{R^4 - 1} \rightarrow_{R \rightarrow \infty} 0$ we have $I(a) = \int_{\mathbb{R} \cup H_+} dz \frac{e^{iaz}}{(z^2 + 1)^2} = 2\pi i \operatorname{Res}\left(\frac{e^{iaz}}{(z^2 + 1)^2}, i\right) = 2\pi i \lim_{z \rightarrow i} \frac{d}{dz} \left[(z - i)^2 \frac{e^{iaz}}{(z - i)^2 (z + i)^2} \right] = 2\pi i \lim_{z \rightarrow i} \left[\frac{ia e^{iaz}}{(z + i)^2} - 2 \frac{e^{iaz}}{(z + i)^3} \right] = 2\pi i \left[\frac{ia e^{-a}}{-4} - 2 \frac{e^{-a}}{-8i} \right] = \frac{\pi}{2} (a + 1) e^{-a}$

For $a < 0$, $\int_{H_{-,R}} \leq \pi R \frac{1}{R^4 - 1} \rightarrow_{R \rightarrow \infty} 0$ we have $I(a) = \int_{\mathbb{R} \cup H_-} dz \frac{e^{iaz}}{(z^2 + 1)^2} = -2\pi i \operatorname{Res}\left(\frac{e^{iaz}}{(z^2 + 1)^2}, -i\right) = -2\pi i \lim_{z \rightarrow -i} \frac{d}{dz} \left[(z + i)^2 \frac{e^{iaz}}{(z - i)^2 (z + i)^2} \right] = -2\pi i \lim_{z \rightarrow -i} \left[\frac{ia e^{iaz}}{(z - i)^2} - 2 \frac{e^{iaz}}{(z - i)^3} \right] = -2\pi i \left[\frac{ia e^a}{-4} - 2 \frac{e^a}{8i} \right] = \frac{\pi}{2} (-a + 1) e^a$

Exercise 3 (Resonance, 40 points). We recall that $G(t)$ and its Fourier transform $\hat{G}(p)$ are related by:

$$\hat{G}(p) = \int dt G(t) e^{-ipt}, \quad G(t) = \int \frac{dp}{2\pi} \hat{G}(p) e^{ipt}.$$

- (5 points) A point like body of mass m is attached to an elastic spring with elastic constant k and moves in an viscous environment with viscosity coefficient η . Use Newton's second law to describe the motion of the body when subjected to a driving external force $F(t)$. We denote $\omega_0^2 = \frac{k}{m}$, $\gamma = \frac{\eta}{2m}$ and $f = \frac{F}{m}$.

We denote x the displacement of the body with respect to the equilibrium position. The elastic force is $-kx$ and the drag is $-\eta \dot{x}$ (must justify the signs: the elastic force pulls towards the equilibrium point and is proportional to the displacement, the drag is proportional to the speed and opposed to it), hence Newton's first law reads $m\ddot{x} = -kx - \eta \dot{x} + F$, which is $\ddot{x} + 2\gamma \dot{x} + \omega_0^2 x = f(t)$.

We consider the differential operator $P(\frac{d}{dt}) = \frac{d^2}{dt^2} + 2\gamma \frac{d}{dt} + \omega_0^2$ with $\gamma \geq 0$. A fundamental solution of $P(\frac{d}{dt})$ is a distribution G such that $P(\frac{d}{dt})G = \delta$, and a solution of the inhomogenous equation $P(\frac{d}{dt})x = f$ is the convolution $x(t) = \int_{-\infty}^{\infty} ds G(t - s)f(s)$.

- (10 points) Prove that a fundamental solution of the operator $P(\frac{d}{dt})$ is:

$$G(t) = \Theta(t) e^{-\gamma t} \frac{\sin(\omega t)}{\omega},$$

where $\Theta(t)$ is the Heaviside theta function and $\omega = \sqrt{\omega_0^2 - \gamma^2}$. Hint: use the Fourier transform.

The equation $\ddot{G} + 2\gamma \dot{G} + \omega_0^2 G = \delta$ becomes in Fourier space $(-p^2 + 2ip\gamma + \omega_0^2)\hat{G}(p) = 1$ therefore:

$$G(t) = \int_{-\infty}^{\infty} \frac{dp}{2\pi} \frac{(-1)e^{ipt}}{[p - (i\gamma + \sqrt{\omega_0^2 - \gamma^2})][p - (i\gamma - \sqrt{\omega_0^2 - \gamma^2})]}.$$

For $t < 0$ we close the contour of integration with C_- : as the contour traps no poles, the integral is zero. For $t > 0$, we close the contour with C_+ and we trap both poles with positive orientation. Using the residue theorem we have:

$$G(t) = \Theta(t) \lim_{p \rightarrow i\gamma + \sqrt{\omega_0^2 - \gamma^2}} \frac{(-1)e^{ipt}}{p - (i\gamma - \sqrt{\omega_0^2 - \gamma^2})} + \lim_{p \rightarrow i\gamma - \sqrt{\omega_0^2 - \gamma^2}} \frac{(-1)e^{ipt}}{p - (i\gamma + \sqrt{\omega_0^2 - \gamma^2})} = \Theta(t)(-i)e^{-\gamma t} \left[\frac{e^{i\omega t}}{2\omega} + \frac{e^{-i\omega t}}{-2\omega} \right] = \Theta(t) e^{-\gamma t} \frac{\sin(\omega t)}{\omega}.$$

- (5 points) What is the physical interpretation of the Heaviside theta function in $G(t) = \Theta(t)e^{-\gamma t} \frac{\sin(\omega t)}{\omega}$?

Denoting $\tilde{G}(t) = e^{-\gamma t} \frac{\sin(\omega t)}{\omega}$, under the external driving force the system responds with $x(t) = \int_{-\infty}^{\infty} ds \Theta(t-s) \tilde{G}(t-s) f(s) = \int_{-\infty}^t ds \tilde{G}(t-s) f(s)$, that is the system responds only to the driving $f(s)$ with $s < t$, that is driving occurring in the past. The fundamental solution respects causality, that is the cause $f(s)$ precedes the effect $x(t)$.

- (10 points) Find a solution of the harmonic oscillator with periodic driving:

$$\left(\frac{d^2}{dt^2} + 2\gamma \frac{d}{dt} + \omega_0^2 \right) x = e^{i\alpha t},$$

and write it under the form $x(t) = A(\alpha)e^{i\alpha t + i\varphi(\alpha)}$, with $A(\alpha)$ and $\varphi(\alpha)$ real.

As $x(u) = \int_{-\infty}^{\infty} ds G(u-s) f(s) = \int_{-\infty}^{\infty} dt G(t) f(u-t)$ we have:

$$x(u) = \int_{-\infty}^{\infty} dt \Theta(t) e^{-\gamma t} \frac{\sin(\omega t)}{\omega} e^{i\alpha(u-t)} = e^{i\alpha u} \int_0^{\infty} dt \frac{e^{-(\gamma - i\omega + i\alpha)t} - e^{-(\gamma + i\omega + i\alpha)t}}{2i\omega} = \frac{e^{i\alpha u}}{2i\omega} \left[\frac{1}{\gamma - i\omega + i\alpha} - \frac{1}{\gamma + i\omega + i\alpha} \right] = \frac{e^{i\alpha u}}{(\gamma + i\alpha)^2 + \omega^2} = \frac{e^{i\alpha u}}{\omega_0^2 - \alpha^2 + 2i\alpha\gamma},$$

hence:

$$A(\alpha) = \left| \frac{1}{\omega_0^2 - \alpha^2 + 2i\alpha\gamma} \right| = \frac{1}{\sqrt{(\omega_0^2 - \alpha^2)^2 + 4\alpha^2\gamma^2}}, \quad \tan(\varphi(\alpha)) = \frac{-2\alpha\gamma}{\omega_0^2 - \alpha^2}$$

- (10 points) Discuss $A(\alpha)$ and $\varphi(\alpha)$. What happens for $\alpha \rightarrow \omega_0$? What happens if $\gamma = 0$?

$A(\alpha)$ has a maximum for $\alpha \sim \omega_0$, and diverges at this point if $\gamma = 0$. If $\gamma = 0$, $\varphi(\alpha) = 0$ that is the system is in phase with the driving. For $\gamma > 0$, the system lags by some amount $\varphi(\alpha)$ and $\varphi(\omega_0) = -\frac{\pi}{2}$, that is at resonance the system lags exactly a quarter of a period behind the driving.

Exercise 4 (The harmonic oscillator, 40 points). Recall that a Sturm Liouville problem is a differential equation for a function $H : [a, b] \rightarrow \mathbb{R}$ of the form $[pH']' + qH = -\lambda w H$ for some functions p, q, w with boundary conditions. This is the eigenvalue equation $LH = \lambda H$ for the self adjoint Sturm Liouville operator $LH = -\frac{1}{w} [(pH')' + qH]$, over a real vector space where the inner product is defined as $\langle H, F \rangle = \int dt w(t) H(t) F(t)$ with boundary conditions chosen such that L is symmetric.

We aim to identify the eigenvalues and the eigenfunctions of the differential operator $\frac{1}{2}(-\frac{d^2}{dx^2} + x^2)$, that is to solve the equation:

$$-\frac{1}{2}\psi'' + \frac{x^2}{2}\psi = \lambda\psi.$$

- (10 points) Substituting $\psi = e^{-\frac{x^2}{2}} H(x)$, write the eigenvalue equation of the Harmonic oscillator as a Sturm Liouville problem for H with corresponding differential equation:

$$\left[e^{-x^2} H' \right]' + (2\lambda - 1)e^{-x^2} H = 0.$$

Note that this is not a regular problem, as the interval of definition is unbounded, and the boundary conditions need to be replaced by conditions on the behaviour at infinity: we require that H grows at most polynomially at infinity.

Substituting $\psi = e^{-\frac{x^2}{2}} H$ and multiplying by $(-2)e^{-\frac{x^2}{2}}$ we have:

$$e^{-\frac{x^2}{2}} \left[e^{-\frac{x^2}{2}} H \right]'' - x^2 e^{-x^2} H = -2\lambda e^{-x^2} H, \quad \text{and} \quad \left[e^{-\frac{x^2}{2}} H \right]'' = \left[-x e^{-\frac{x^2}{2}} H + e^{-\frac{x^2}{2}} H' \right]' = -e^{-\frac{x^2}{2}} H + x^2 e^{-\frac{x^2}{2}} H - 2x e^{-\frac{x^2}{2}} H' + e^{-\frac{x^2}{2}} H'',$$

leading to:

$$e^{-x^2} H'' - 2x e^{-x^2} H' - e^{-x^2} H = -2\lambda e^{-x^2} H \quad \Rightarrow \quad \left[e^{-x^2} H' \right]' = -(2\lambda - 1)e^{-x^2} H, \quad w(x) = e^{-x^2}, \quad q(x) = 0, \quad p(x) = e^{-x^2}.$$

In order to ensure that L is symmetric we must have: $\int dx (pH')' F = [pH'F]_{-\infty}^{\infty} - [HpF']_{-\infty}^{\infty} + \int dx H(pF')'$, hence we need to require the solutions to satisfy $\lim_{x \rightarrow \pm\infty} e^{-x^2} (HF' - H'F) = 0$. This in particular is satisfied if F and H have at most polynomial growth at infinity.

- (20 points) Using the power series method to solve the equation for H . Using that fact that H must be a polynomial, find the eigenvalues of the harmonic oscillator. What is the problem if H is not a polynomial?

We assume $H = \sum_{k \geq 0} a_k x^k$ ($k \geq 0$ as we want H square integrable at origin) and we get:

$$H'' - 2xH' + (2\lambda - 1)H = 0 \Rightarrow \sum_{k \geq 0} x^k \left[(k+2)(k+1)a_{k+2} - 2ka_k + (2\lambda - 1)a_k \right] = 0 \Rightarrow a_{k+2} = \frac{2k - (2\lambda - 1)}{(k+2)(k+1)} a_k, \quad k \geq 0.$$

Iterating the recursion we have:

$$a_{2p+1} = a_1 \prod_{i=0}^{p-1} \frac{2(2i+1) - (2\lambda - 1)}{(2i+3)(2i+2)} \sim \frac{4^p p!}{(2p)!}, \quad a_{2p} = a_0 \prod_{i=0}^{p-1} \frac{4i - (2\lambda - 1)}{(2i+2)(2i+1)} \sim \frac{4^p p!}{(2p)!}$$

The series truncates to a polynomial only if:

$$2k = 2\lambda - 1 \Rightarrow \lambda = k + \frac{1}{2}, \quad k \geq 0.$$

B. EXAMS

The general solution is $H = c_0 y_0(x) + c_1 y_1(x)$ with $y_0(x) = \sum_{p \geq 0} a_{2p} x^{2p}$ and $y_1(x) = \sum_{p \geq 1} a_{2p+1} x^{2p+1}$, $y_0, y_1 \sim \sum_p \frac{1}{p!} e^p x^{2p} \sim e^{x^2}$. The solutions are square integrable (and L is symmetric) with measure e^{-x^2} if and only if one of c_1 or c_2 is 0 and the series truncates to a polynomial.

- (10 points) Denoting $\psi_n = e^{-\frac{x^2}{2}} H_n$ the solution corresponding to the n 'th eigenvalue λ_n of the harmonic oscillator, find the eigenvalues and eigenfunctions of the operator:

$$-\frac{1}{2} \frac{d^2}{dx^2} + V(x), \quad V(x) = \begin{cases} \frac{1}{2} x^2, & x \geq 0 \\ \infty, & x \leq 0 \end{cases}.$$

Everything goes through as before, except that the boundary condition of the Sturm Liouville problem which ensures Hermiticity of L is $H(0) = 0$ (one needs $\lim_{x \rightarrow 0} e^{-x^2} (H F' - H' F) = 0$). Alternatively, as $\psi(x) = 0$ for $x \leq 0$ and ψ is continuous at 0 we have $H(0) = 0$ (ψ is continuous at 0, ψ' has a finite discontinuity and ψ'' has an infinite discontinuity, like V). It follows that among the solutions of the previous SL problem we need to select the ones such that $H_n(0) = 0$. As:

$$\lambda = 2p + \frac{1}{2} : H_{2p} = \sum_{k=0}^p a_{2k} x^{2k}, \quad \lambda = 2p + 1 + \frac{1}{2} : H_{2p+1} = \sum_{k=0}^p a_{2k+1} x^{2k+1},$$

it follows that the eigenvalues and eigenfunctions which respect $\psi(0) = 0$ are $\lambda_{2p+1} = 2p + 1 + \frac{1}{2}$ and $\psi_{2p+1} = e^{-\frac{x^2}{2}} H_{2p+1}$.

B.2.4 Sample 4

Exercise 1 (Complex Analysis, 30 points). We consider the complex function:

$$f(z) = \frac{e^z \cos(\pi z)}{z^2 - 3z + 2}.$$

- (10 points) Find the poles of f and compute the residues of f at the poles.

We observe that $z^2 - 3z + 2 = (z - 2)(z - 1)$ therefore:

$$f(z) = e^z \cos(\pi z) \left(\frac{1}{z - 2} - \frac{1}{z - 1} \right).$$

As e^z and $\cos(\pi z)$ are entire functions, we conclude that has poles of order 1 at $z = 1, 2$. We have:

$$\text{Res}(f, 1) = \lim_{z \rightarrow 1} (z - 1)f(z) = -e \cos(\pi) = \boxed{e}, \quad \text{Res}(f, 2) = \lim_{z \rightarrow 2} (z - 2)f(z) = e^2 \cos(2\pi) = \boxed{e^2}.$$

- (10 points) We denote $C_r(0)$ the positively oriented circle of center zero and radius r in the complex plane. $C_r(0)$ can be parametrized as $re^{i\varphi}$ with $\varphi \in [0, 2\pi]$. Compute the integral:

$$\int_{C_{\frac{1}{2}}(0)} dz f(z),$$

and show that:

$$\int_{C_{\frac{3}{2}}(0)} dz f(z) = 2\pi i e, \quad \int_{C_{\frac{5}{2}}(0)} dz f(z) = 2\pi i e(1 + e).$$

The circle $C_{\frac{1}{2}}(0)$ encircles no poles hence $\int_{C_{\frac{1}{2}}(0)} dz f(z) = 0$. The circle $C_{\frac{3}{2}}(0)$ encircles only the pole at $z = 1$ hence $\int_{C_{\frac{3}{2}}(0)} dz f(z) = 2\pi i \text{Res}(f, 1) = \boxed{2\pi i e}$. The circle $C_{\frac{5}{2}}(0)$

encircles both poles hence $\int_{C_{\frac{5}{2}}(0)} dz f(z) = 2\pi i \text{Res}(f, 1) + 2\pi i \text{Res}(f, 2) = \boxed{2\pi i e(1 + e)}$.

- (10 points) Let β be a positive real number. Find the poles and the residues of the functions:

$$h_+(z) = \frac{\beta}{e^{\beta z} - 1} \quad \text{and} \quad h_-(z) = \frac{-\beta}{e^{\beta z} + 1}.$$

Hint: all the poles are of order 1.

We denote $\omega_k^B = \frac{2\pi}{\beta} k$ and $\omega_k^F = \frac{2\pi}{\beta} (k + \frac{1}{2})$ for $k \in \mathbb{Z}$. We have:

$$h_+ : \quad e^{\beta z} = 1 \Rightarrow \beta z = i 2\pi k \Rightarrow \boxed{z = i\omega_k^B}, \quad h_- : \quad e^{\beta z} = -1 \Rightarrow \beta z = i 2\pi (k + \frac{1}{2}) \Rightarrow \boxed{z = i\omega_k^F}.$$

As the poles are of order 1 we have:

$$\begin{aligned} \text{Res}(h_+(z), i\omega_k^B) &= \lim_{z \rightarrow i\omega_k^B} (z - i\omega_k^B) \frac{\beta}{e^{\beta z} - 1} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon \beta}{e^{i\beta \omega_k^B + \beta \epsilon} - 1} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon \beta}{e^{\beta \epsilon} - 1} = 1, \\ \text{Res}(h_-(z), i\omega_k^F) &= \lim_{z \rightarrow i\omega_k^F} (z - i\omega_k^F) \frac{-\beta}{e^{\beta z} + 1} = \lim_{\epsilon \rightarrow 0} \frac{-\epsilon \beta}{e^{i\beta \omega_k^F + \beta \epsilon} + 1} = \lim_{\epsilon \rightarrow 0} \frac{-\epsilon \beta}{-e^{\beta \epsilon} + 1} = 1. \end{aligned}$$

Exercise 2 (A differential equation, 20 points). Let z be a fixed complex number and consider the homogeneous differential equation with constant coefficients:

$$x'''(t) + 3z x''(t) + 3z^2 x'(t) + z^3 x(t) = 0.$$

- (10 points) Find three independent solutions $x_1(t)$, $x_2(t)$ and $x_3(t)$ of this equation. These form a basis for a general solution.

We posit $x = e^{\lambda t}$ and obtain the characteristic polynomial:

$$x''' + 3zx'' + 3z^2 x' + z^3 x|_{x=e^{\lambda t}} = P(\lambda)e^{\lambda t} \Rightarrow \quad P(\lambda) = \lambda^3 + 3z\lambda^2 + 3z^2\lambda + z^3 = (\lambda + z)^3$$

with triple root $\lambda = -z$, hence three independent solutions are:

$$\boxed{x_1(t) = e^{-zt}, \quad x_2(t) = te^{-zt}, \quad x_3(t) = t^2 e^{-zt}.$$

- (10 points) Compute their Wronskian:

$$W(t) = \begin{vmatrix} x_1 & x_2 & x_3 \\ x_1' & x_2' & x_3' \\ x_1'' & x_2'' & x_3'' \end{vmatrix}.$$

We have

$$\begin{vmatrix} e^{-zt} & te^{-zt} & t^2 e^{-zt} \\ -ze^{-zt} & e^{-zt} - zte^{-zt} & 2te^{-zt} - zt^2 e^{-zt} \\ z^2 e^{-zt} & -2ze^{-zt} + z^2 te^{-zt} & 2e^{-zt} - 4zte^{-zt} + z^2 t^2 e^{-zt} \end{vmatrix} \begin{vmatrix} c_2 - tc_1, c_3 - t^2 c_1 \\ e^{-zt} & 0 & 0 \\ -ze^{-zt} & e^{-zt} & 2te^{-zt} \\ z^2 e^{-zt} & -2ze^{-zt} & 2e^{-zt} - 4zte^{-zt} \end{vmatrix} \begin{vmatrix} c_3 - 2tc_2 \\ e^{-zt} & 0 & 0 \\ -ze^{-zt} & e^{-zt} & 0 \\ z^2 e^{-zt} & -2ze^{-zt} & 2e^{-zt} \end{vmatrix} = \boxed{2e^{-3zt}}.$$

B. EXAMS

Exercise 3 (The Airy function, 30 points). Consider the Airy differential equation:

$$\frac{d^2 y}{dx^2} - xy = 0 .$$

- (10 points) Find the general solution of this equation with $|y(0)| < \infty$ using the power series method. Hint: be careful with the boundary terms.

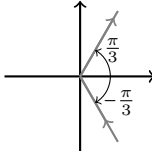
We substitute $y = \sum_{n \geq 0} a_n x^n$

$$0 = \sum_{n \geq 0} n(n-1)a_n x^{n-2} - \sum_{n \geq 0} a_n x^{n+1} = \sum_{n \geq 0} (n+2)(n+1)a_{n+2} x^n - \sum_{n \geq 1} a_{n-1} x^n = 2a_2 + \sum_{n \geq 1} x^n [(n+2)(n+1)a_{n+2} - a_{n-1}] \Rightarrow \quad a_2 = 0, \quad a_{n+3} = \frac{1}{(n+3)(n+2)} a_n .$$

The general solution is:

$$y(x) = a_0 \sum_{k \geq 0} \frac{1}{(2 \cdot 3)(5 \cdot 6) \cdots [(3k-1)(3k)]} x^{3k} + a_1 \sum_{k \geq 0} \frac{1}{(3 \cdot 4)(6 \cdot 7) \cdots [(3k)(3k+1)]} x^{3k+1} = \boxed{a_0 \sum_{k \geq 0} \frac{\prod_{i=1}^k (3i-2)}{(3k)!} x^{3k} + a_1 \sum_{k \geq 0} \frac{\prod_{i=1}^k (3i-1)}{(3k+1)!} x^{3k+1}} .$$

- (10 points) We denote:

$$\text{Ai}(x) = \frac{1}{2\pi i} \int_C dw e^{\frac{w^3}{3} - xw} ,$$


where the contour C depicted in the figure can be parametrized as $\{w = \rho e^{-i\frac{\pi}{3}} \mid \rho \in (\infty, 0)\} \cup \{w = \rho e^{i\frac{\pi}{3}} \mid \rho \in (0, \infty)\}$. Prove that $\text{Ai}(x)$ is a solution of the Airy equation. Hint: you can commute the derivative and integral as the integral is absolutely convergent along the contour C .

We have:

$$\frac{d^2 \text{Ai}}{dx^2}(x) - x \text{Ai}(x) = \frac{1}{2\pi i} \int_C dw e^{\frac{w^3}{3} - xw} (w^2 - x) = \frac{1}{2\pi i} \int_C dw \frac{d}{dw} \left(e^{\frac{w^3}{3} - xw} \right) = \lim_{\rho \rightarrow \infty} e^{\frac{w^3}{3} - xw} \Big|_{w=\rho e^{i\frac{\pi}{3}}} - \lim_{\rho \rightarrow \infty} e^{\frac{w^3}{3} - xw} \Big|_{w=\rho e^{-i\frac{\pi}{3}}} = \lim_{\rho \rightarrow \infty} e^{-\frac{\rho^3}{3} - x\rho e^{i\frac{\pi}{3}}} - \lim_{\rho \rightarrow \infty} e^{-\frac{\rho^3}{3} - x\rho e^{-i\frac{\pi}{3}}} = 0 .$$

- (10 points) Taking into account that for any function f :

$$\frac{1}{2\pi i} \int_C dw f(w) = \frac{1}{2\pi i} \int_0^\infty d\rho \left[e^{i\frac{\pi}{3}} f(\rho e^{i\frac{\pi}{3}}) - e^{-i\frac{\pi}{3}} f(\rho e^{-i\frac{\pi}{3}}) \right] ,$$

find the power series expansion of $\text{Ai}(x)$ around $x = 0$.

We have:

$$\text{Ai}(x) = \sum_{n \geq 0} \frac{(-1)^n}{n!} x^n \left(\frac{e^{i\frac{\pi}{3}} (n+1)}{2\pi i} - \frac{e^{-i\frac{\pi}{3}} (n+1)}{2\pi i} \right) \int_0^\infty d\rho e^{-\frac{\rho^3}{3}} \rho^n .$$

Using:

$$\frac{e^{i\frac{\pi}{3}} (n+1)}{2\pi i} - \frac{e^{-i\frac{\pi}{3}} (n+1)}{2\pi i} = \frac{\sin\left(\frac{\pi}{3}(n+1)\right)}{\pi} = \frac{1}{\pi} \begin{cases} \frac{\sqrt{3}}{2}, & n = 6k, 6k+1 \\ 0, & n = 6k+2, 6k+5 \\ -\frac{\sqrt{3}}{2}, & n = 6k+3, 6k+4 \end{cases} ,$$

respectively:

$$\int d\rho e^{-\frac{\rho^3}{3}} \rho^n \stackrel{u=\frac{\rho^3}{3}}{=} \int_0^\infty d(3u)^{1/3} e^{-u} (3u)^{n/3} = 3^{\frac{n}{3}} \int_0^\infty du e^{-u} u^{\frac{n-2}{3}} = 3^{\frac{n}{3}} \Gamma\left(-\frac{n+1}{3}\right) ,$$

we obtain:

$$\text{Ai}(x) = \frac{\sqrt{3}}{2\pi} \sum_{k \geq 0} \left(\frac{6k}{3^{\frac{6k}{3}}} \Gamma\left(\frac{6k+1}{3}\right) x^{6k} - \frac{3^{\frac{6k+1}{3}} \Gamma\left(\frac{6k+2}{3}\right)}{(6k+1)!} x^{6k+1} \right) - \frac{\sqrt{3}}{2\pi} \sum_{k \geq 0} \left(-\frac{3^{\frac{6k+3}{3}} \Gamma\left(\frac{6k+4}{3}\right)}{(6k+3)!} x^{6k+3} + \frac{3^{\frac{6k+4}{3}} \Gamma\left(\frac{6k+5}{3}\right)}{(6k+5)!} x^{6k+4} \right) = \frac{\sqrt{3}}{2\pi} \sum_{p \geq 0} \left(\frac{3^p \Gamma\left(p + \frac{1}{3}\right)}{(3p)!} x^{3p} - \frac{3^{p+\frac{1}{3}} \Gamma\left(p + \frac{2}{3}\right)}{(3p+1)!} x^{3p+1} \right) .$$

Observing that:

$$\Gamma\left(p + \frac{1}{3}\right) = \Gamma\left(\frac{1}{3}\right) \prod_{i=1}^p \left(p - \frac{2}{3}\right) , \quad \Gamma\left(p + \frac{2}{3}\right) = \Gamma\left(\frac{2}{3}\right) \prod_{i=1}^p \left(p - \frac{1}{3}\right) ,$$

we have

$$\boxed{\text{Ai}(x) = \frac{\sqrt{3}}{2\pi} \Gamma\left(\frac{1}{3}\right) \sum_{p \geq 0} \frac{\prod_{i=1}^p (3p-2)}{(3p)!} x^{3p} - \frac{\sqrt{3}}{2\pi} \frac{1}{3} \Gamma\left(\frac{2}{3}\right) \sum_{p \geq 0} \frac{\prod_{i=1}^p (3p-1)}{(3p+1)!} x^{3p+1}}$$

Exercise 4 (The δ potential well, 40 points). In a first approximation, an atom in one dimension is an infinite potential well $V(x) = -\lambda\delta(x)$. We are interested in solving the one dimensional eigenvalue equation $-\frac{1}{2}\psi'' + V(x)\psi = E\psi$. As $V(x) = 0$ for $x \neq 0$ we have for $x < 0$ and $x > 0$ that $-\psi'' = 2E\psi$ hence, denoting $k = \sqrt{2E}$, the general solution of the eigenvalue equation is:

$$\psi = \begin{cases} x < 0, & \psi_L(x) = A_r e^{ikx} + A_l e^{-ikx} \\ x > 0, & \psi_R(x) = B_r e^{ikx} + B_l e^{-ikx} \end{cases} .$$

- (Regularity, 10 points) Taking into account that ψ is continuous at 0, $\psi_R(0) = \psi_L(0)$, prove that $2\lambda\psi_{R,L}(0) = \psi'_L(0) - \psi'_R(0)$. Using:

$$\psi_R(0) = \psi_L(0), \quad 2\lambda\psi_{R,L}(0) = \psi'_L(0) - \psi'_R(0),$$

compute B_r and B_l in terms of A_r and A_l .

As ψ is continuous we have $\psi'' \sim \delta(x)$, that is $\psi' \sim \theta(x)$, that is ψ' has a finite discontinuity in 0. Integrating on a small interval around zero we have:

$$\lim_{\epsilon \rightarrow 0} \left(-\frac{1}{2} \int_{-\epsilon}^{\epsilon} dx \psi''(x) - \lambda \int_{-\epsilon}^{\epsilon} dx \delta(x) \psi(x) - E \int_{-\epsilon}^{\epsilon} dx \psi(x) \right) = 0 \Rightarrow -\frac{1}{2} (\psi'_R(0) - \psi'_L(0)) - \lambda \psi_{R,L}(0) = 0 \Rightarrow \boxed{2\lambda\psi_{R,L}(0) = \psi'_L(0) - \psi'_R(0)}.$$

As ψ is continuous we have $A_r + A_l = B_r + B_l$ and the discontinuity of ψ' yields $2\lambda(A_r + A_l) = ikA_r - ikA_l - ikB_r + ikB_l$. This is:

$$B_r + B_l = A_r + A_l, \quad B_r - B_l = A_r - A_l - \frac{2\lambda}{ik}(A_r + A_l) \Rightarrow \boxed{\begin{cases} B_r = A_r - \frac{\lambda}{ik}(A_r + A_l) \\ B_l = A_l + \frac{\lambda}{ik}(A_r + A_l) \end{cases}}.$$

- (Scattering, 10 points) For $E > 0$, e^{ikx} describes a particle propagating from left to right, which can either reflect off to well or be transmitted. This corresponds to $A_r = 1$ (the incident particle), $A_l = r$ (reflection), $B_r = t$ (transmission) and $B_l = 0$. Determine the probabilities of reflection $R = |r|^2$ respectively transmission $T = |t|^2$ and check that $R + T = 1$.

We have:

$$t = 1 - \frac{\lambda}{ik}(1 + r), \quad 0 = r + \frac{\lambda}{ik}(1 + r) \Rightarrow \begin{cases} r = \frac{-1}{1 + \frac{\lambda}{ik}} \\ t = 1 - \frac{\lambda}{ik}(1 + \frac{-1}{1 + \frac{\lambda}{ik}}) = \frac{1}{1 + \frac{\lambda}{ik}} \end{cases}.$$

The probability of reflection is $R = |r|^2 = \frac{1}{1 + \frac{2E}{\lambda^2}}$ and the one of transmission is $T = |t|^2 = \frac{1}{1 + \frac{\lambda^2}{2E}} = 1 - R$.

- (The bound state, 10 points) For $E < 0$ the system has a bound state. Compute its energy.

We denote $\kappa = \sqrt{2|E|}$ and $k = i\kappa$. The only square integrable solution is $\psi_L = A_r e^{\kappa x}$, $\psi_R = B_l e^{-\kappa x}$ and the regularity conditions are:

$$A_r = B_l, \quad \kappa A_r + \kappa B_l = 2\lambda A_r \Rightarrow \kappa = \lambda \Rightarrow \boxed{E = -\frac{\lambda^2}{2}}.$$

- (The molecule, 10 points) Show that the double well potential $V(x) = -\lambda \left[\delta(x + \frac{R}{2}) + \delta(x - \frac{R}{2}) \right]$, which describes a molecule, has two bound states. Derive self-consistent equations respected by the energy eigenvalues and identify the minimal energy. Hint: a self consistent equation for x is an equation of the form $x = f(x)$ for some function f . For example $x = e^{-x}$ is a self-consistent equation for x which admits a unique real solution.

Denoting $\kappa = \sqrt{2|E|}$, for $E < 0$ the wave function must be square integrable therefore:

$$\begin{cases} x < -\frac{R}{2}, & \psi_I(x) = A e^{\kappa(x + \frac{R}{2})} \\ -\frac{R}{2} < x < \frac{R}{2}, & \psi_{II}(x) = B_r e^{\kappa(x + \frac{R}{2})} + B_l e^{-\kappa(x - \frac{R}{2})} \\ x > \frac{R}{2}, & \psi_{III}(x) = C e^{-\kappa(x - \frac{R}{2})} \end{cases}.$$

The continuity of ψ and the finite discontinuities of ψ' lead to:

$$\begin{aligned} A = B_r + B_l e^{\kappa R}, \quad B_r e^{\kappa R} + B_l = C, \quad 2\lambda A = \kappa(A - B_r + B_l e^{\kappa R}), \quad 2\lambda C = \kappa(B_r e^{\kappa R} - B_l + C), \\ (2\lambda - \kappa)(B_r + e^{\kappa R} B_l) + \kappa B_r - \kappa e^{\kappa R} B_l = 0, \quad (2\lambda - \kappa)(e^{\kappa R} B_r + B_l) - \kappa e^{\kappa R} B_r + \kappa B_l = 0. \end{aligned}$$

The system is compatible undetermined only if:

$$\begin{vmatrix} 2\lambda & 2(\lambda - \kappa)e^{\kappa R} \\ 2(\lambda - \kappa)e^{\kappa R} & 2\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - (\lambda - \kappa)^2 e^{2\kappa R} = 0 \Rightarrow \kappa - \lambda = \pm \lambda e^{-\kappa R} \Rightarrow \boxed{\kappa_{\pm} = \lambda(1 \pm e^{-\kappa_{\pm} R})}.$$

As $\kappa_- < \lambda < \kappa_+$, the $+$ sign identifies the fundamental, $E_+ = -\frac{\kappa_+^2}{2}$.

